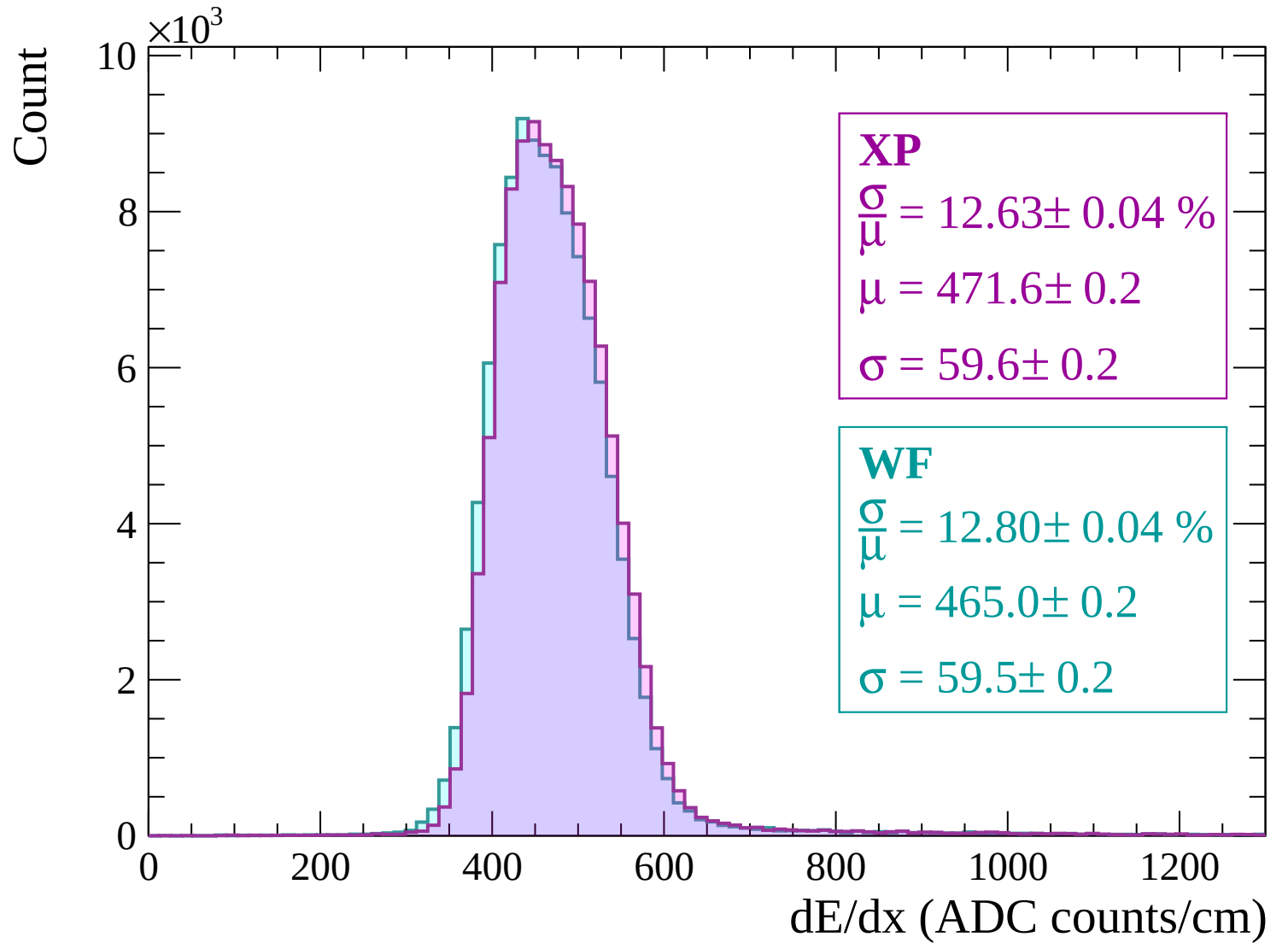
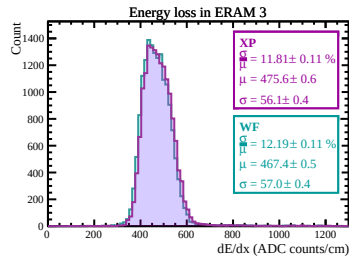
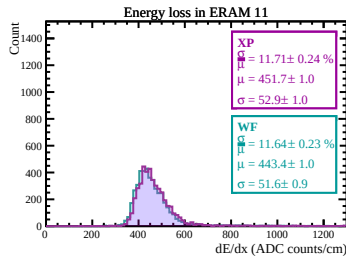
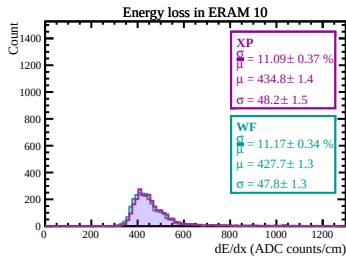
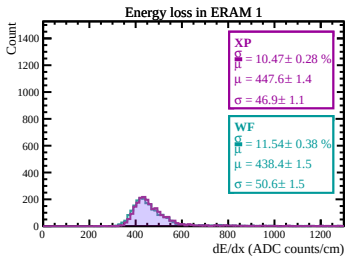
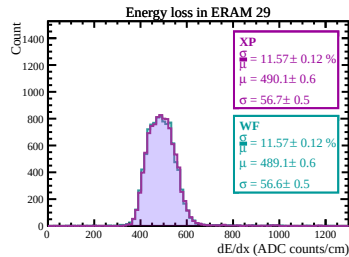
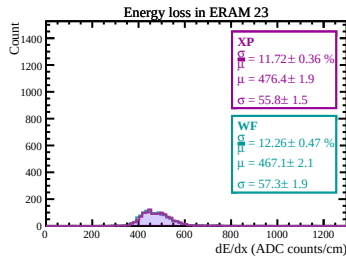
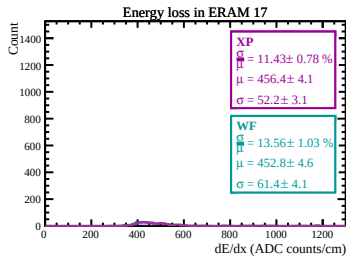
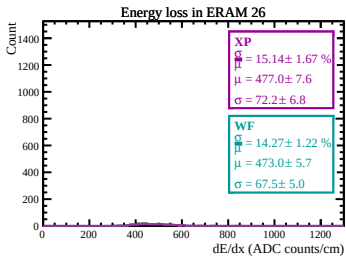
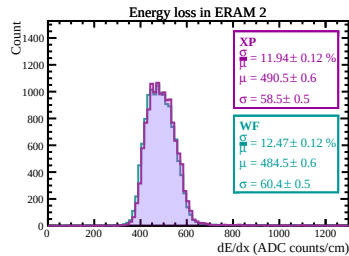
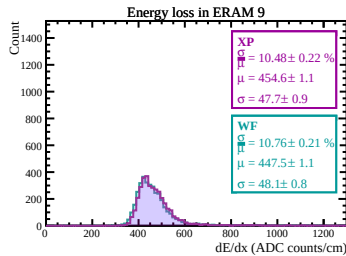
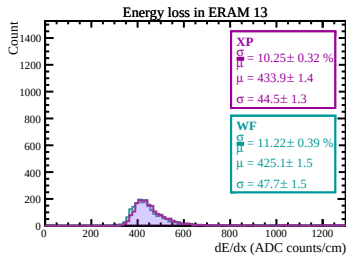
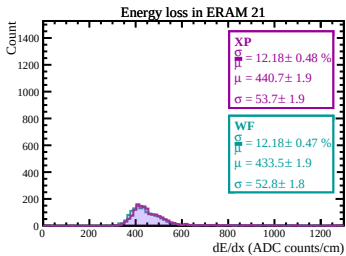
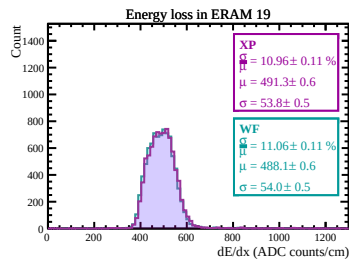
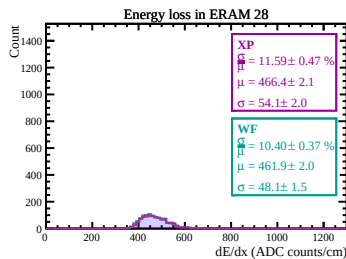
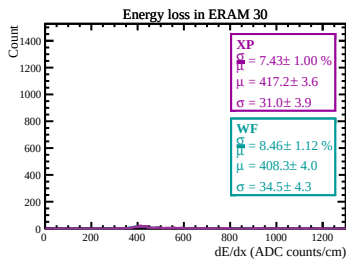
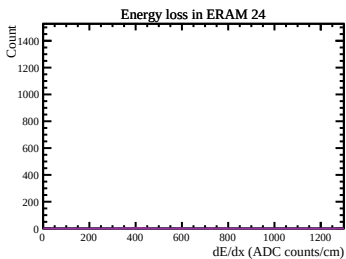
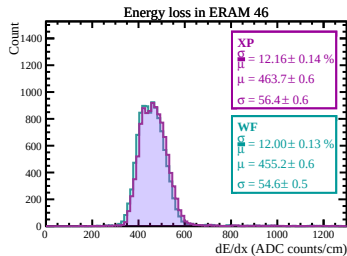
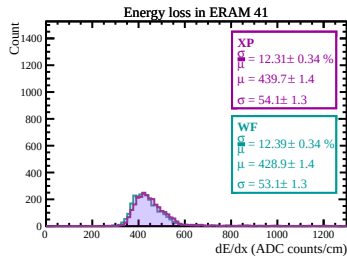
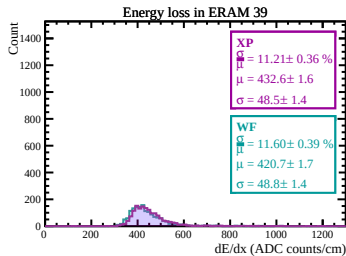
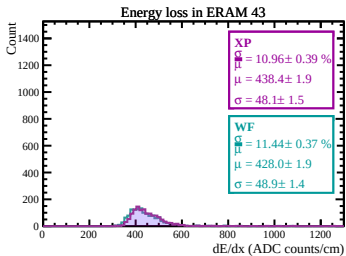
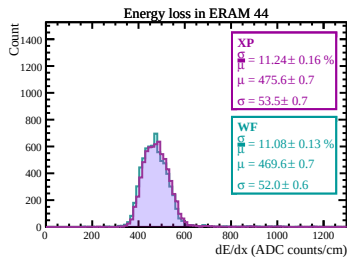
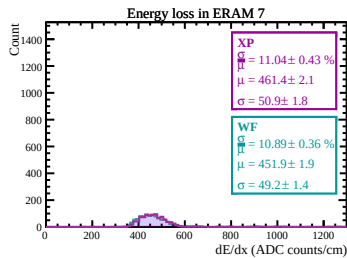
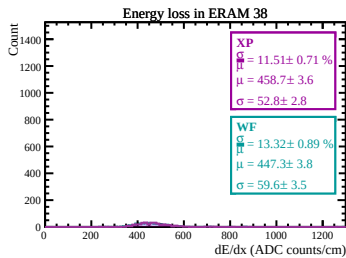
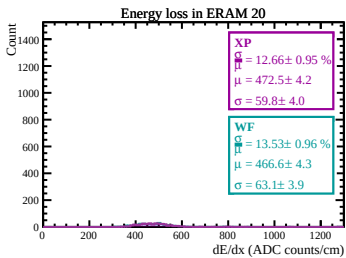
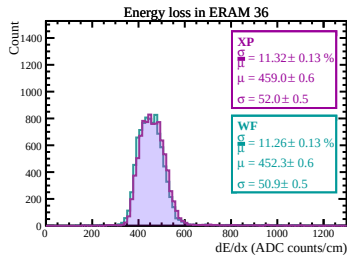
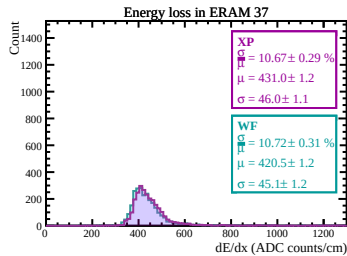
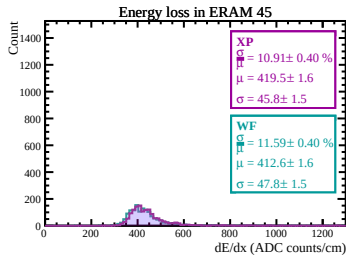
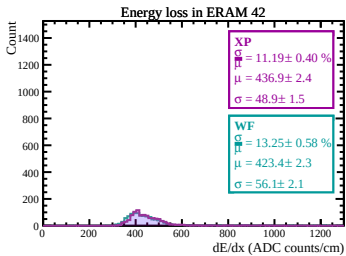
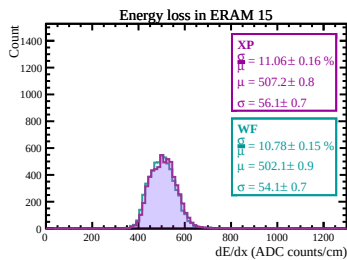
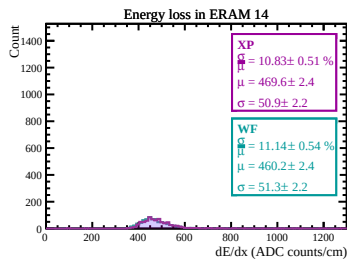
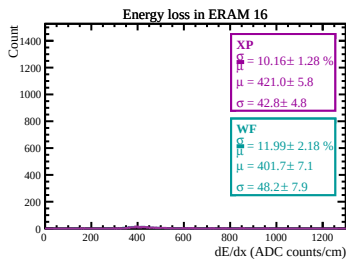
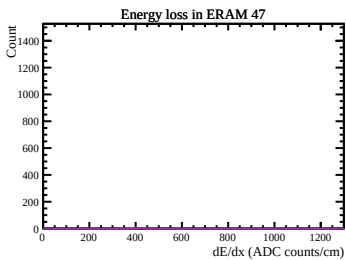
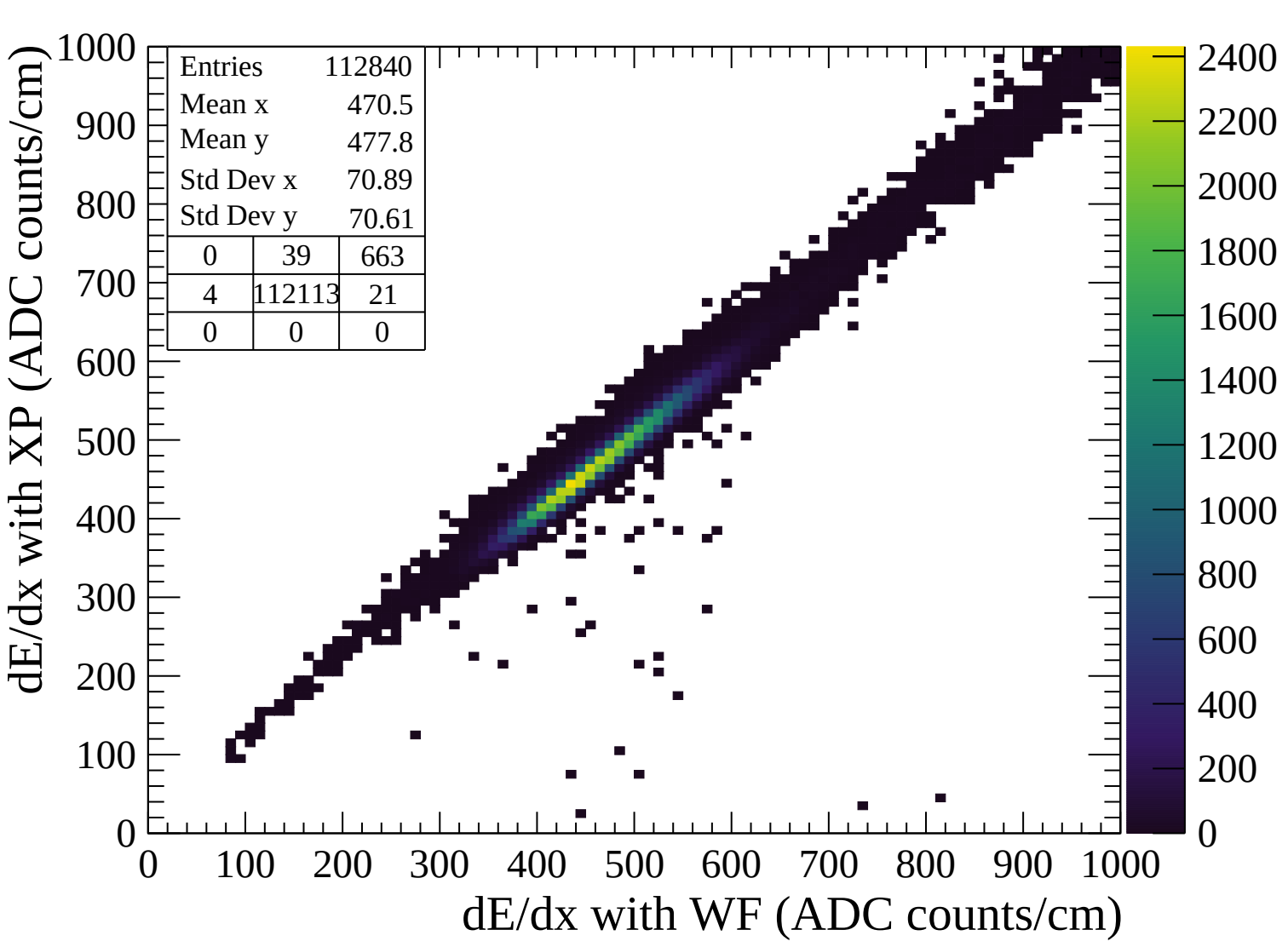
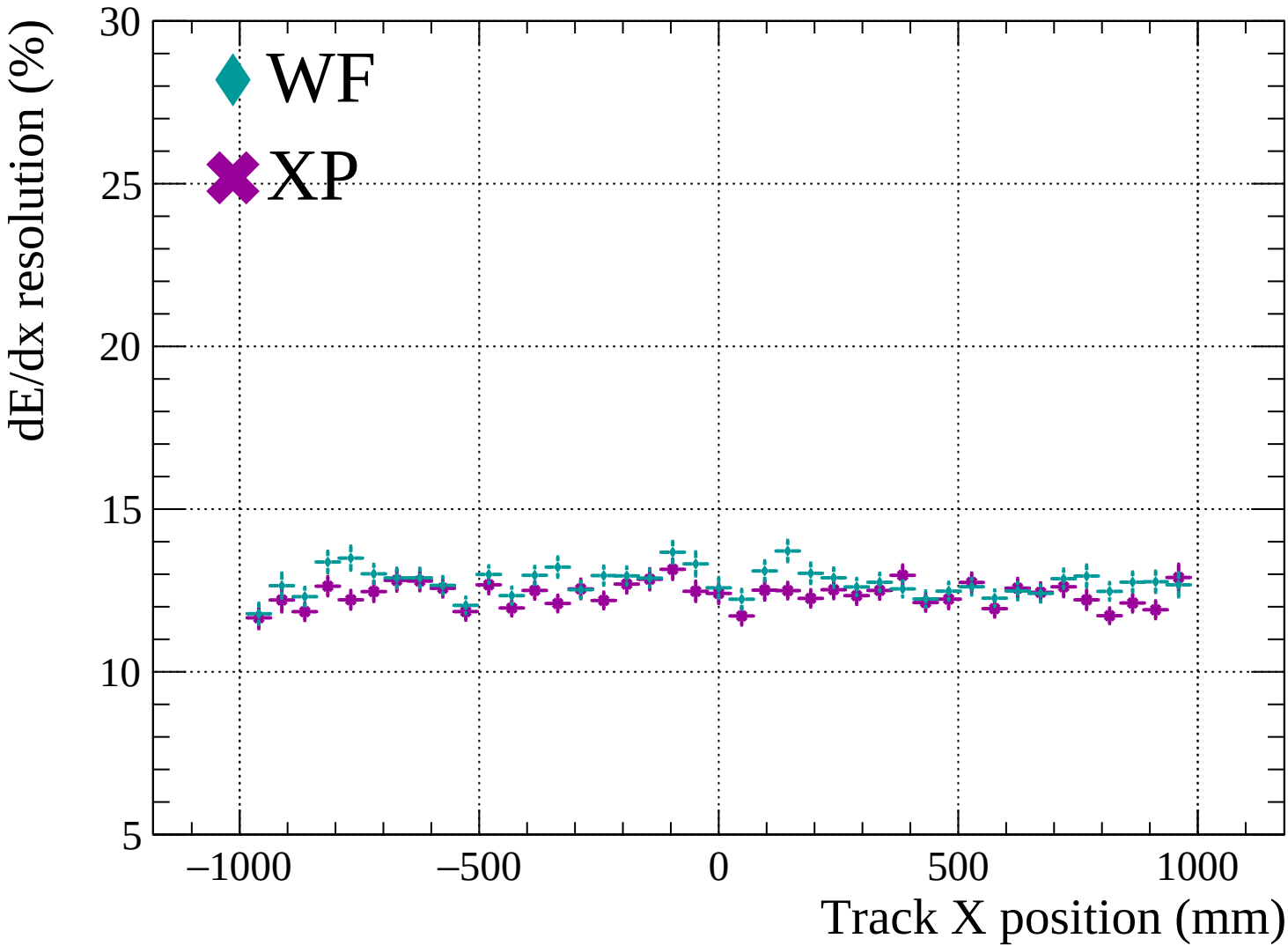


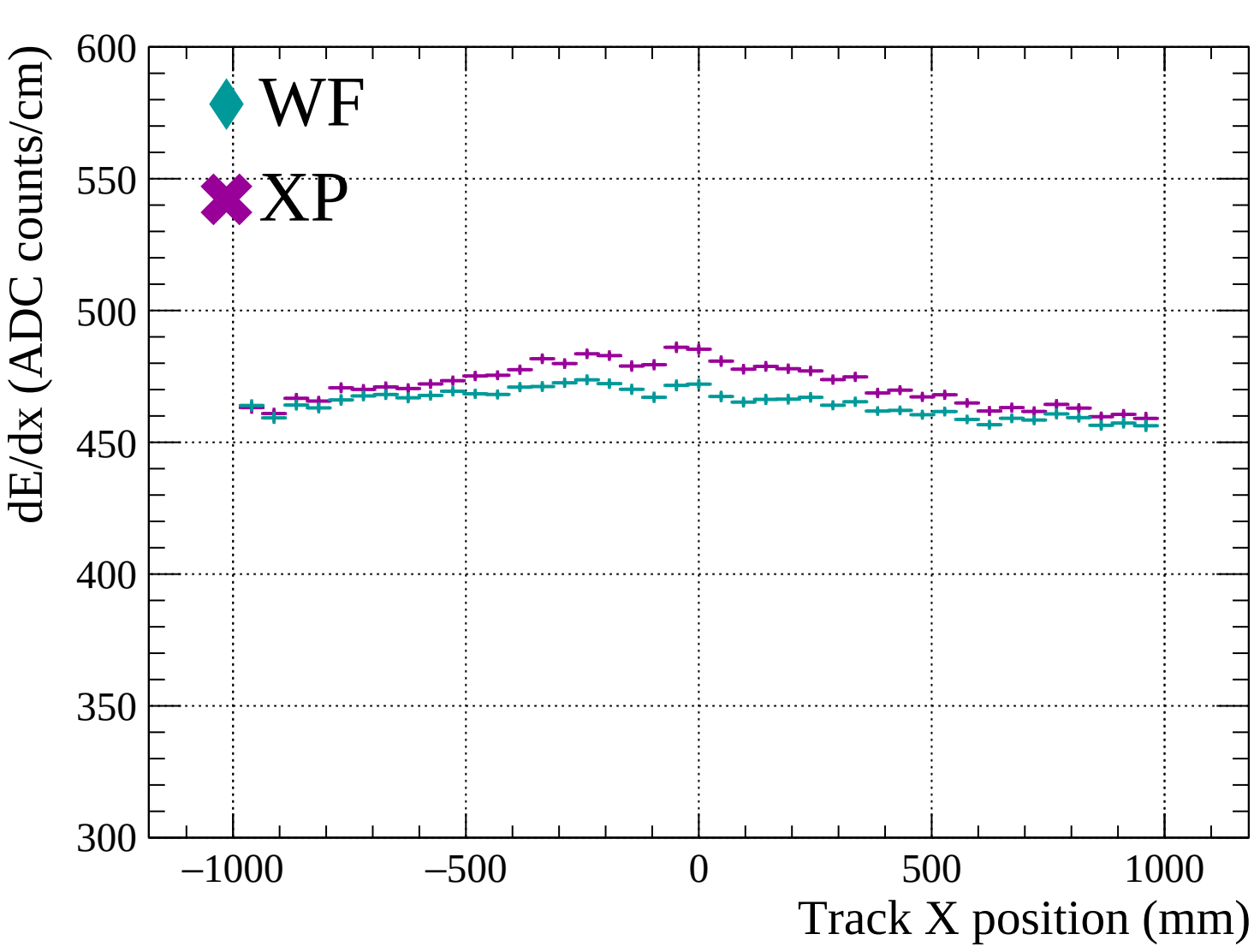


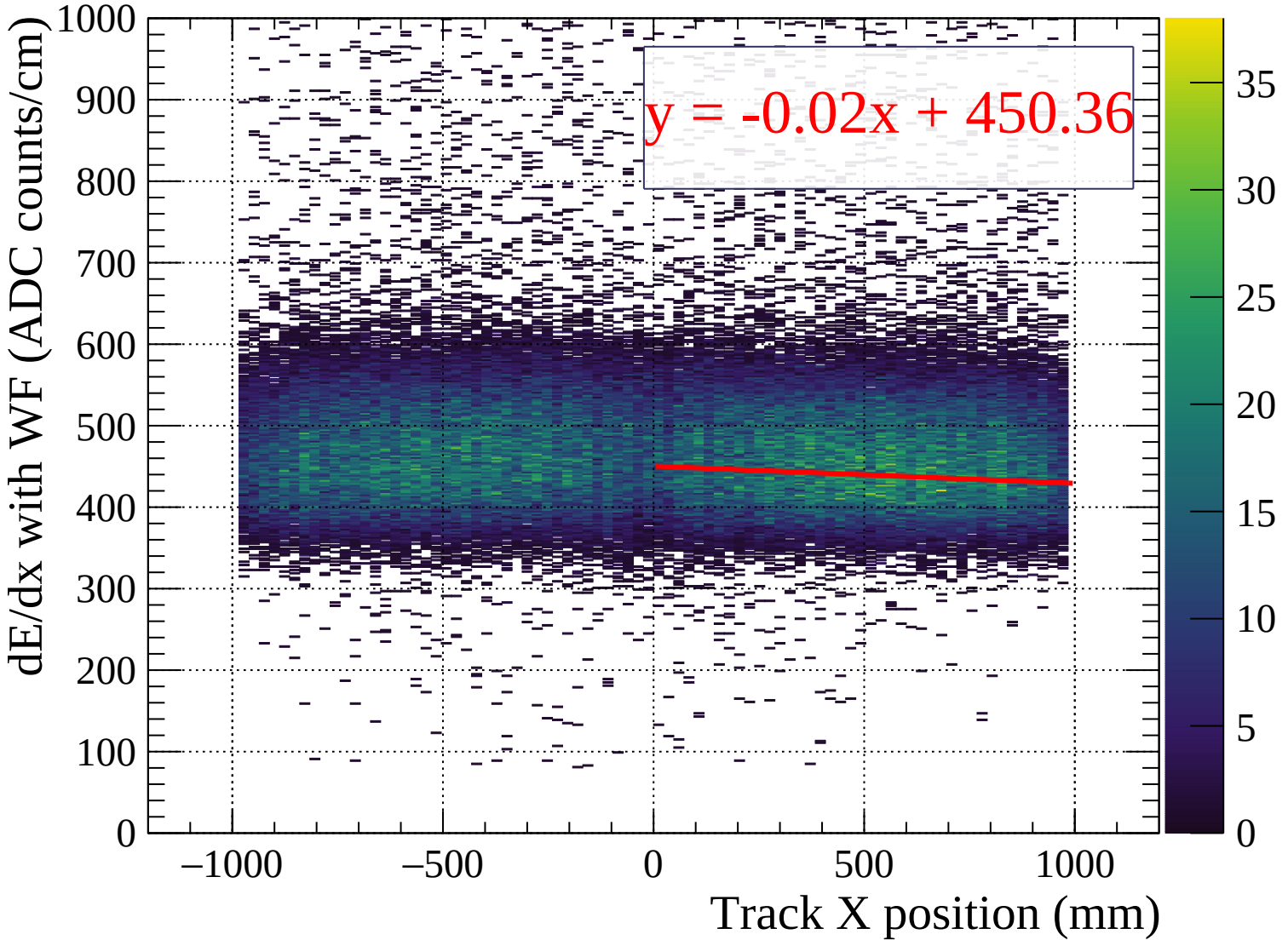


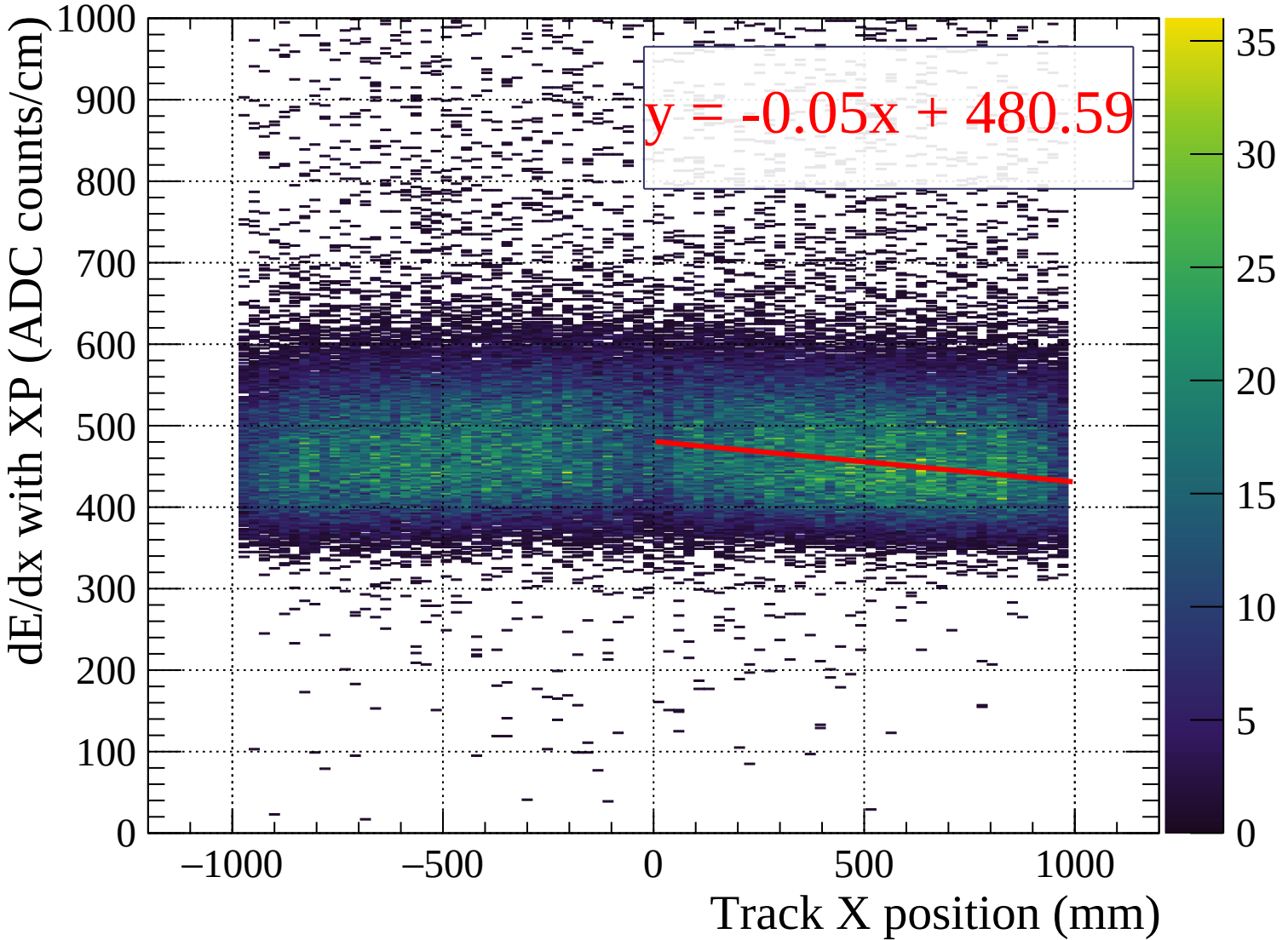


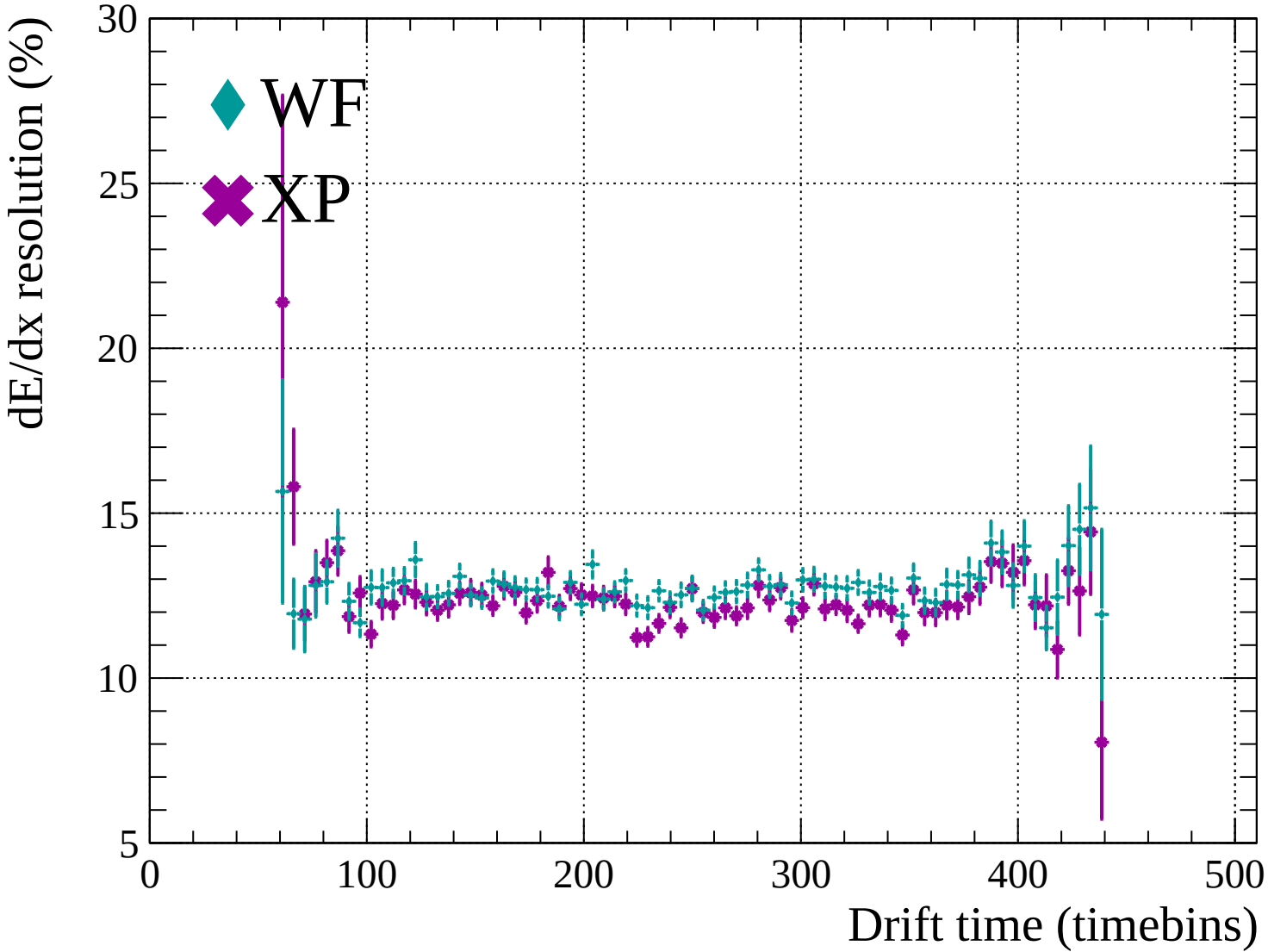


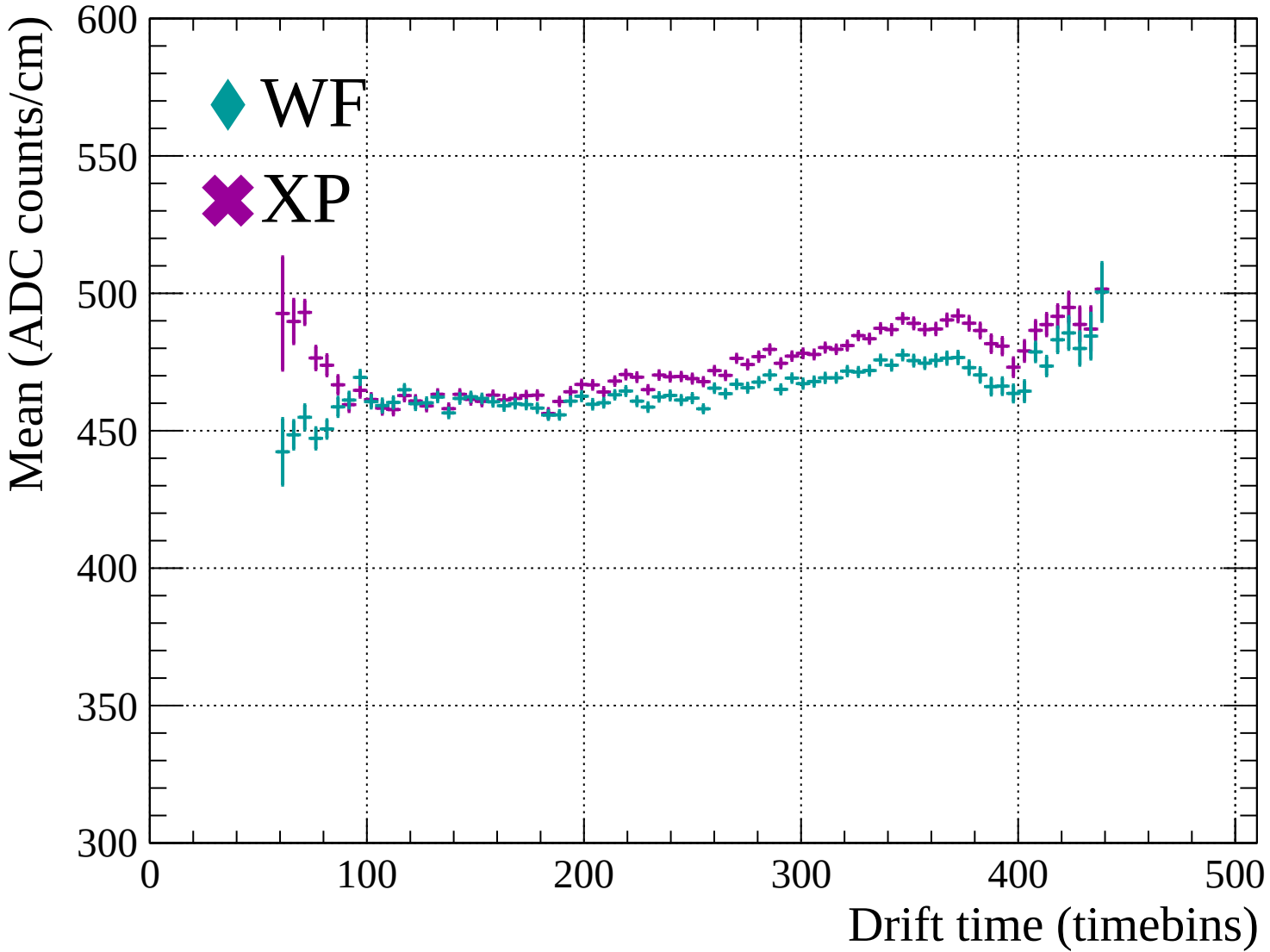


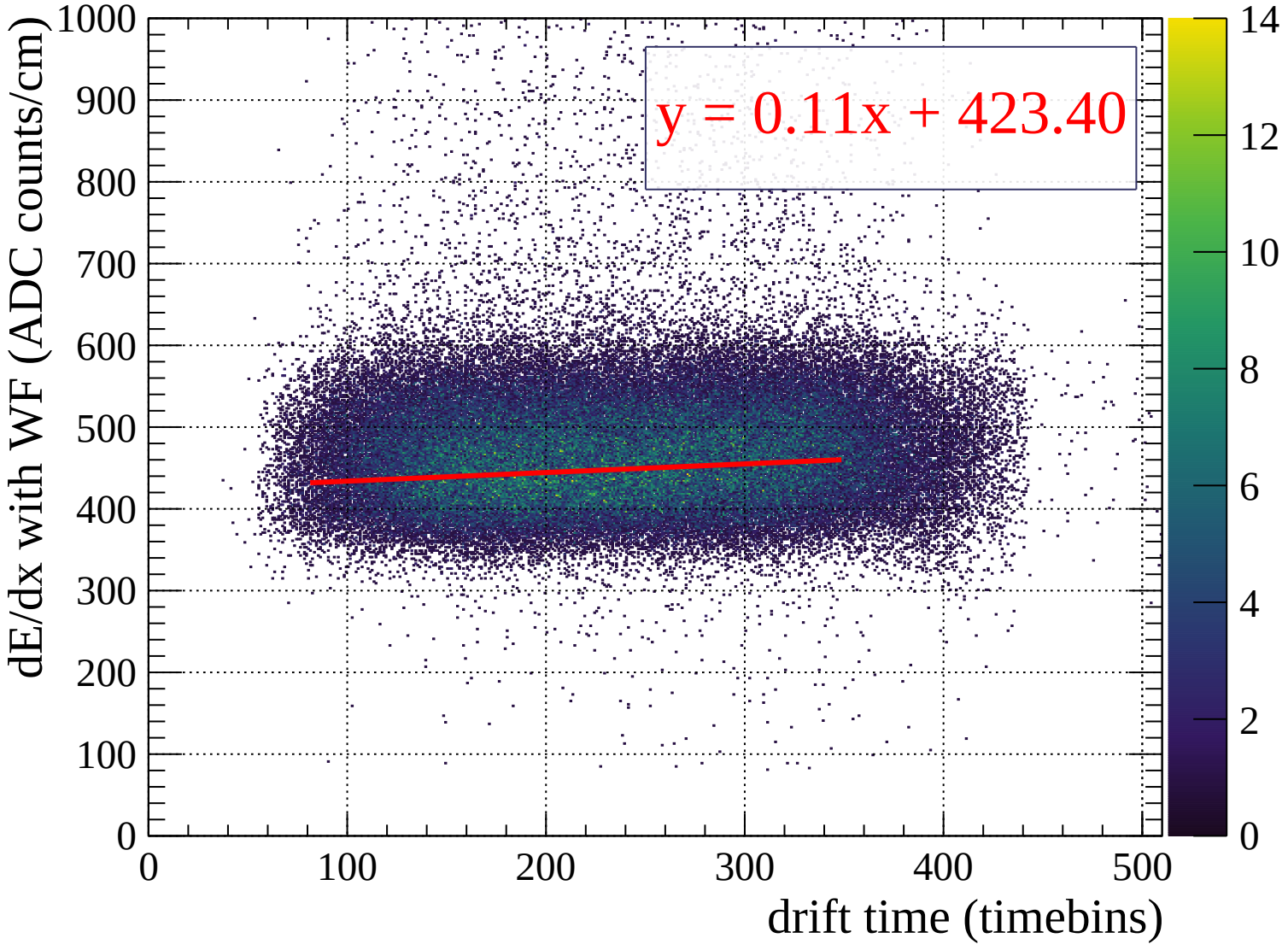


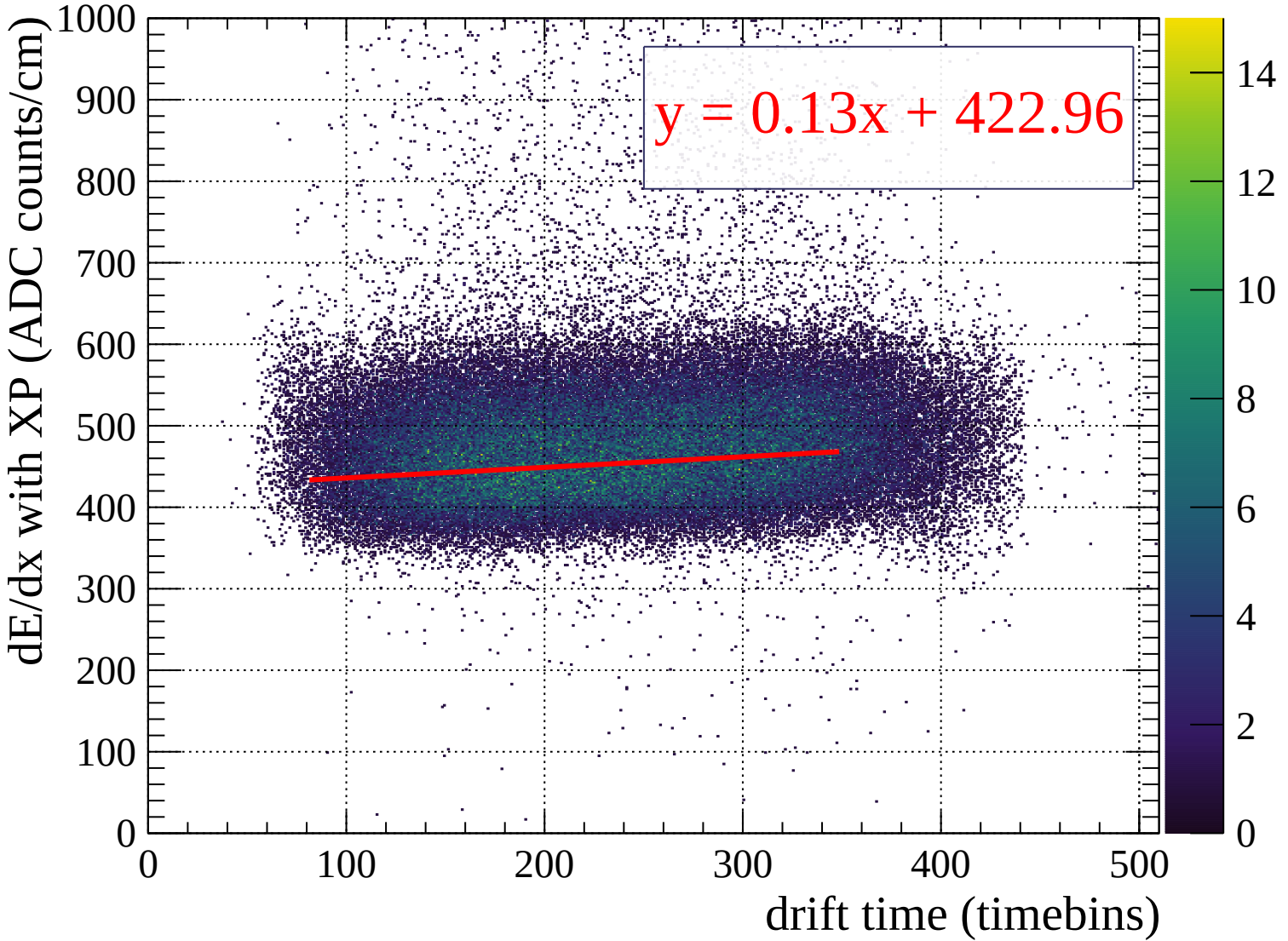


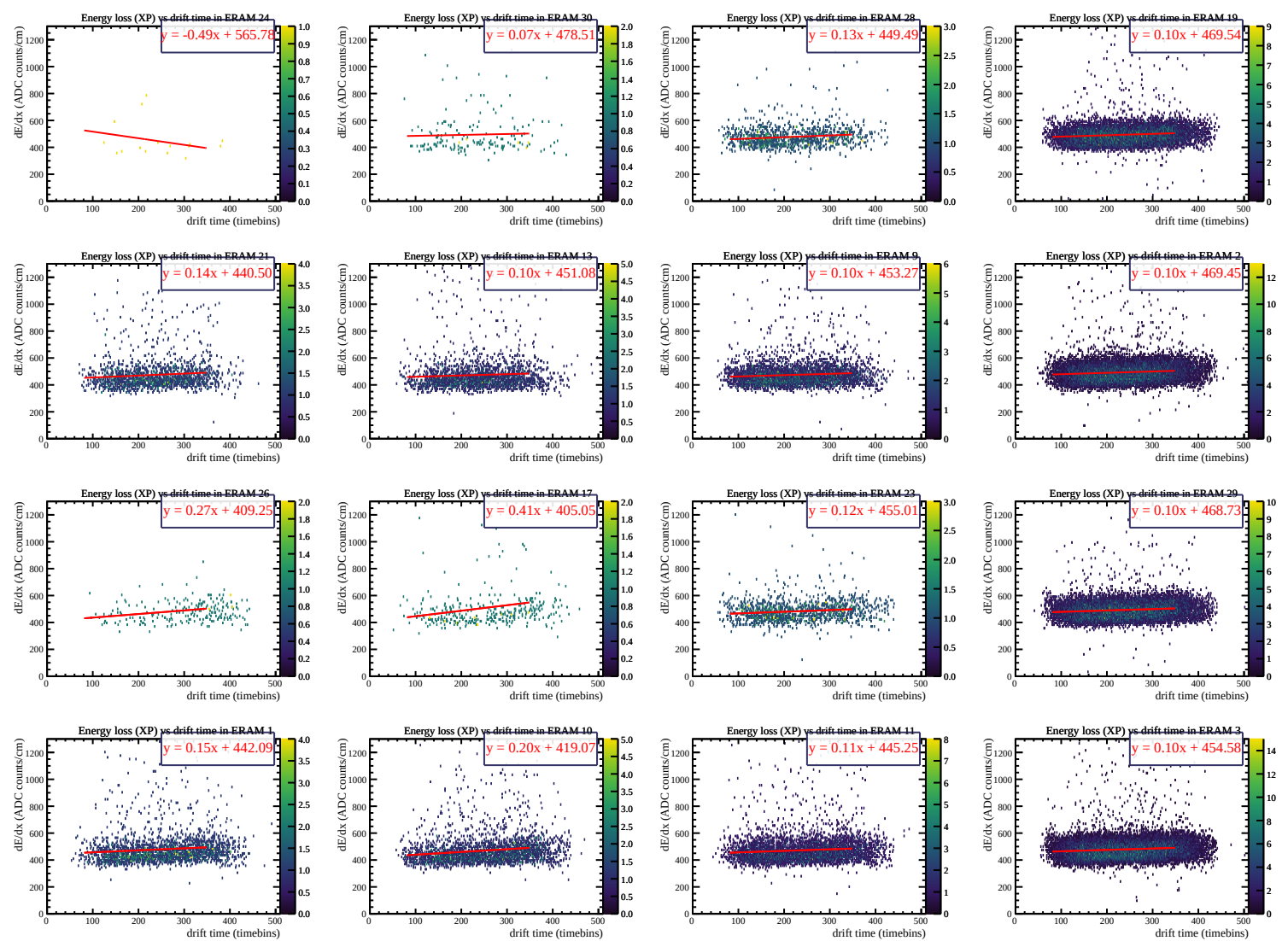


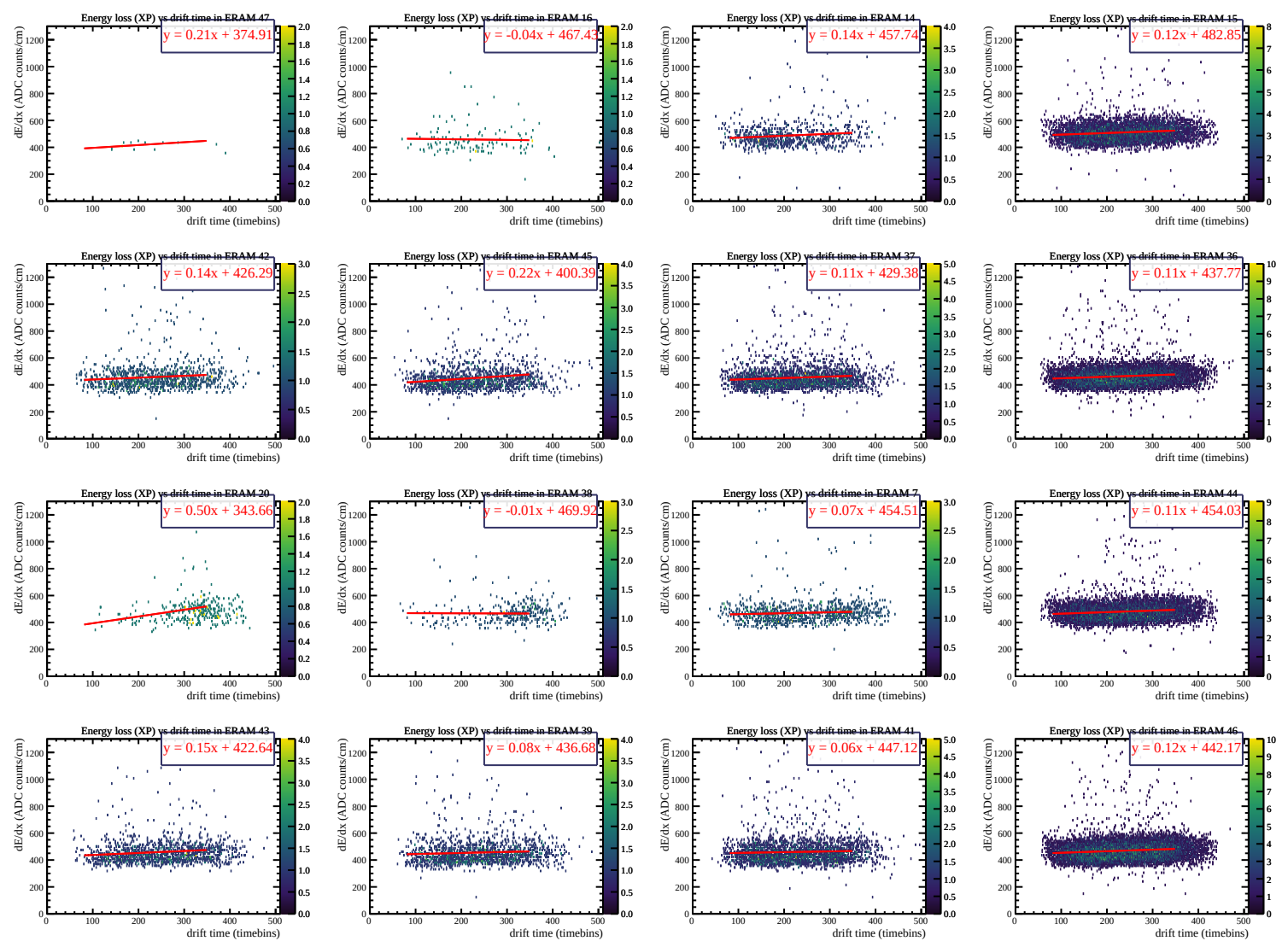


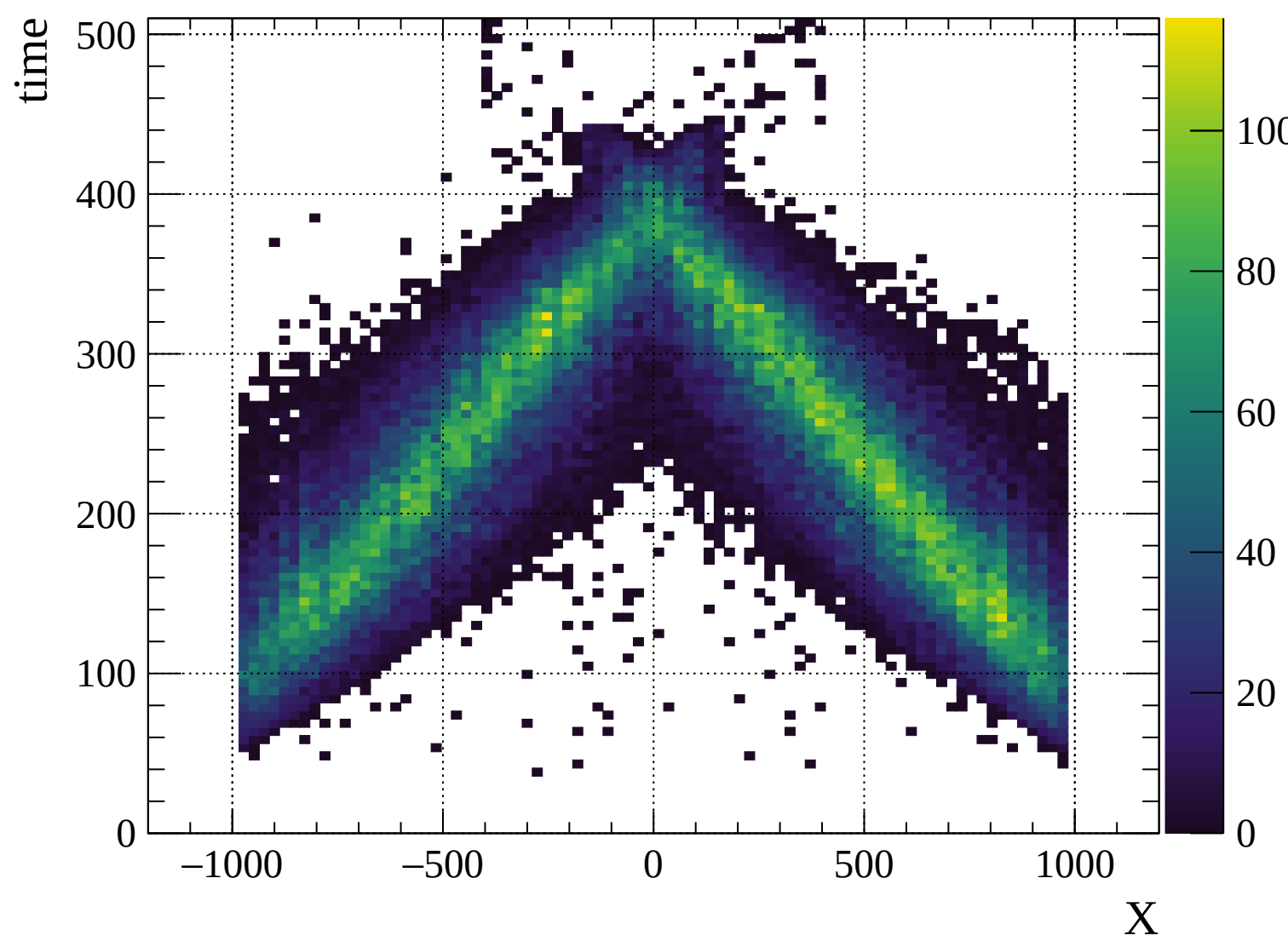


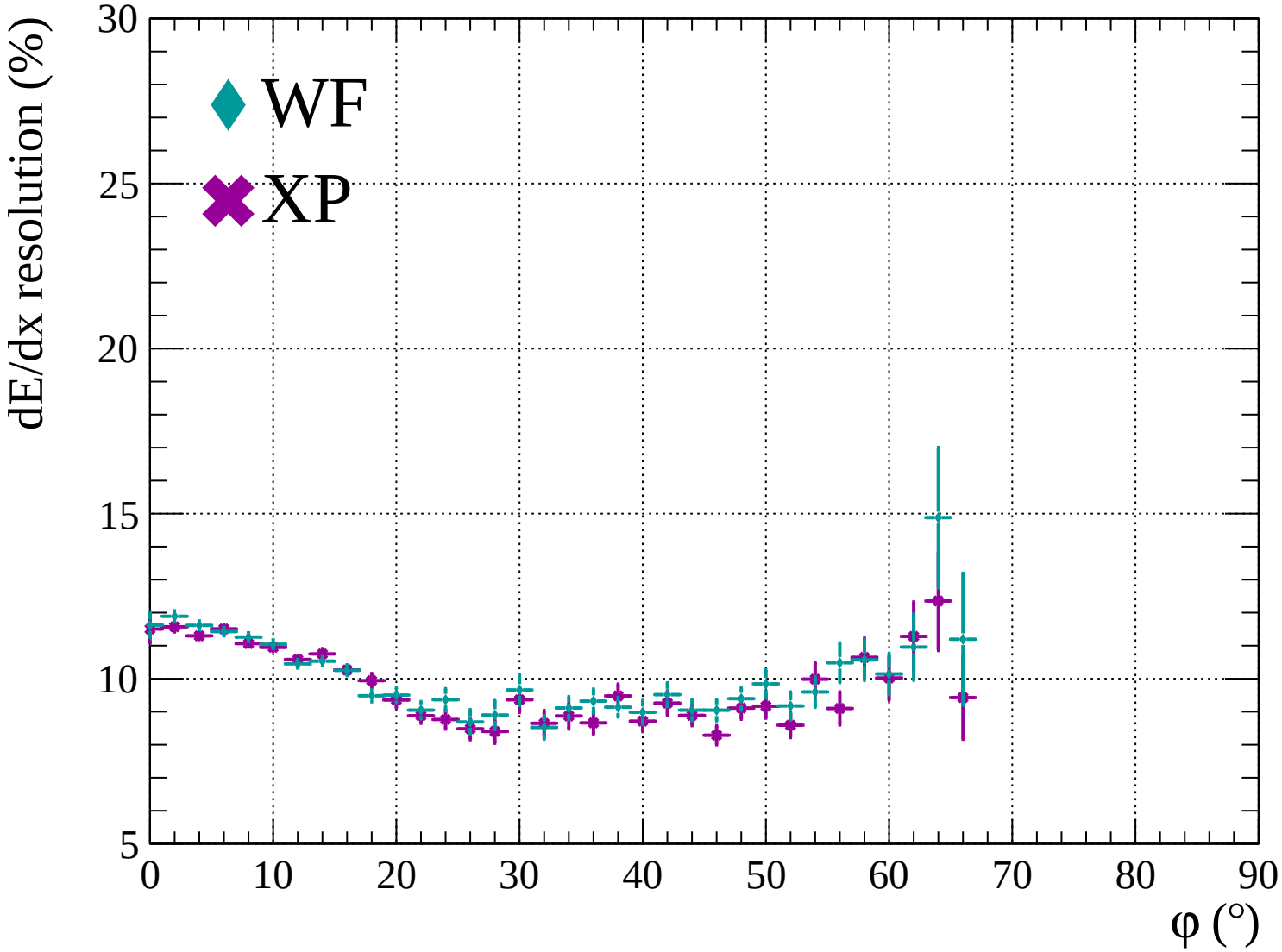


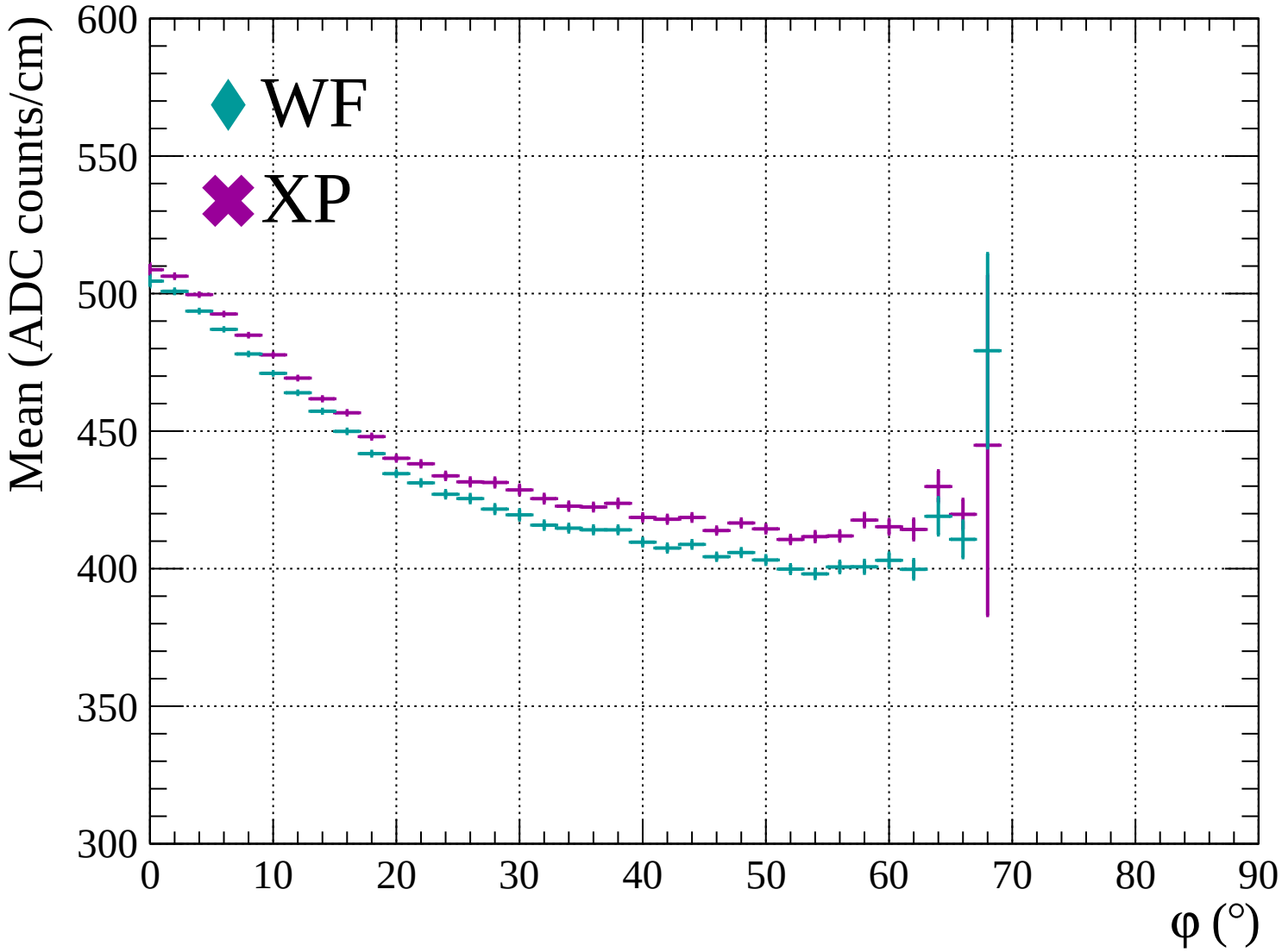


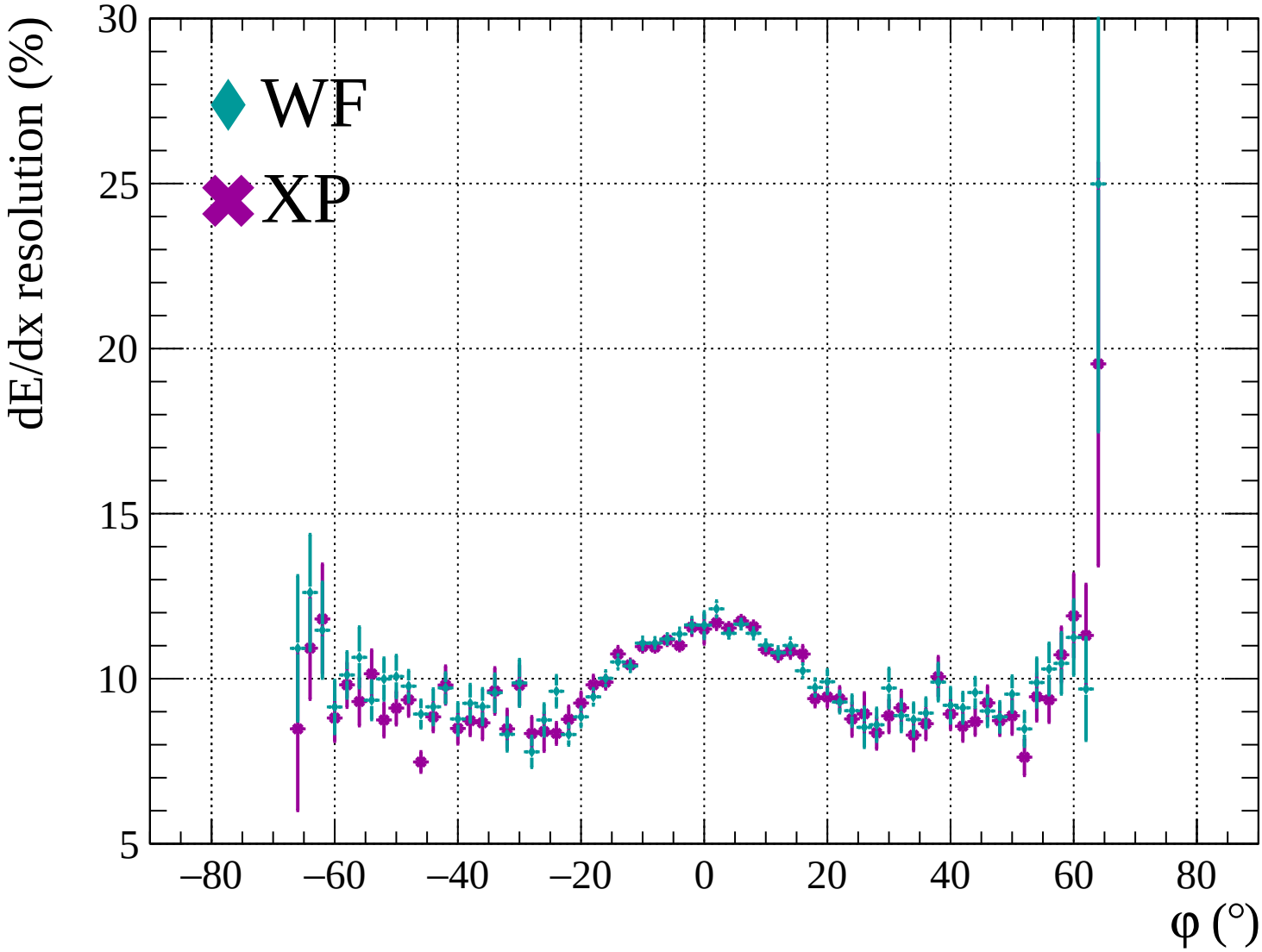


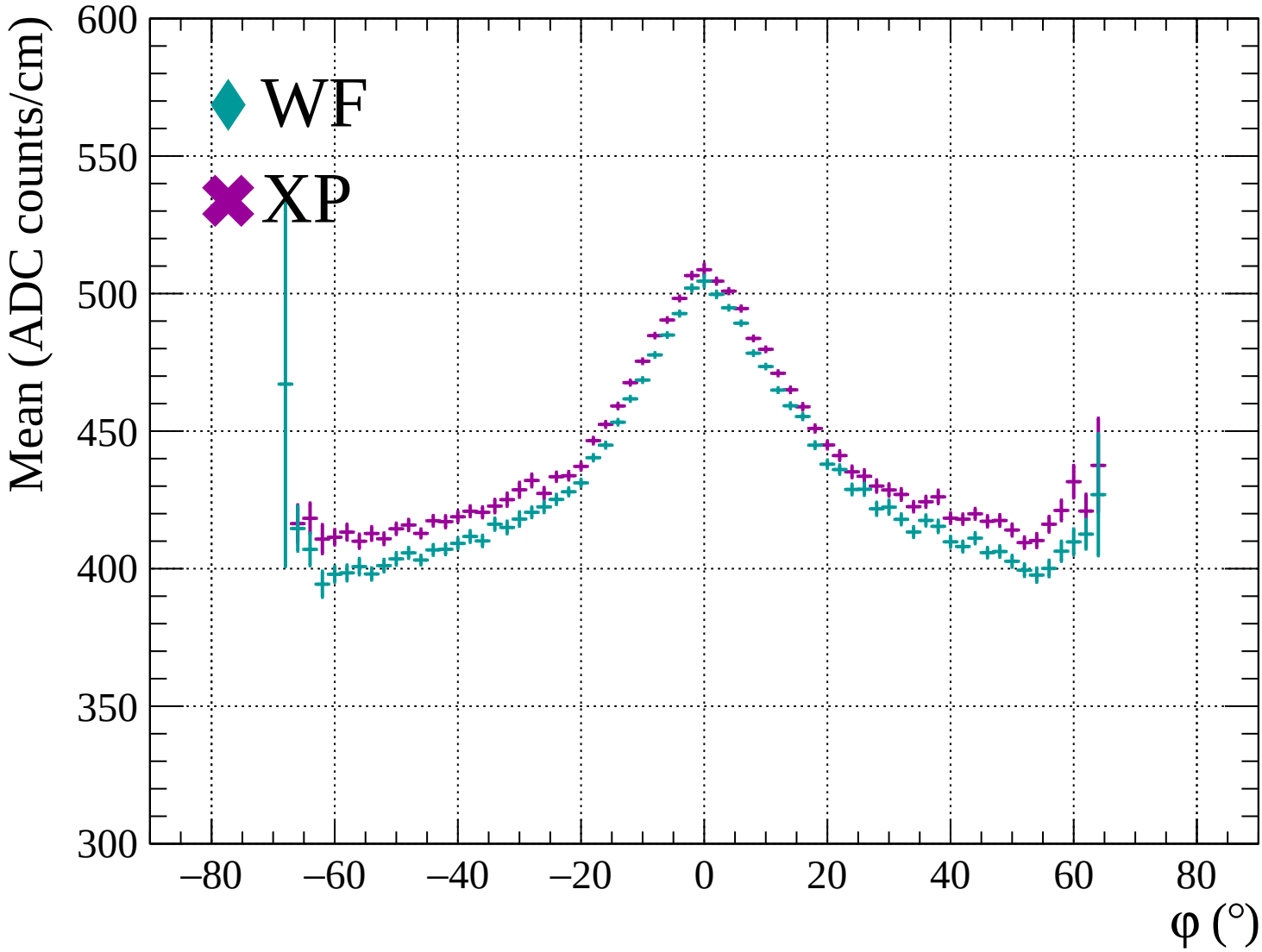


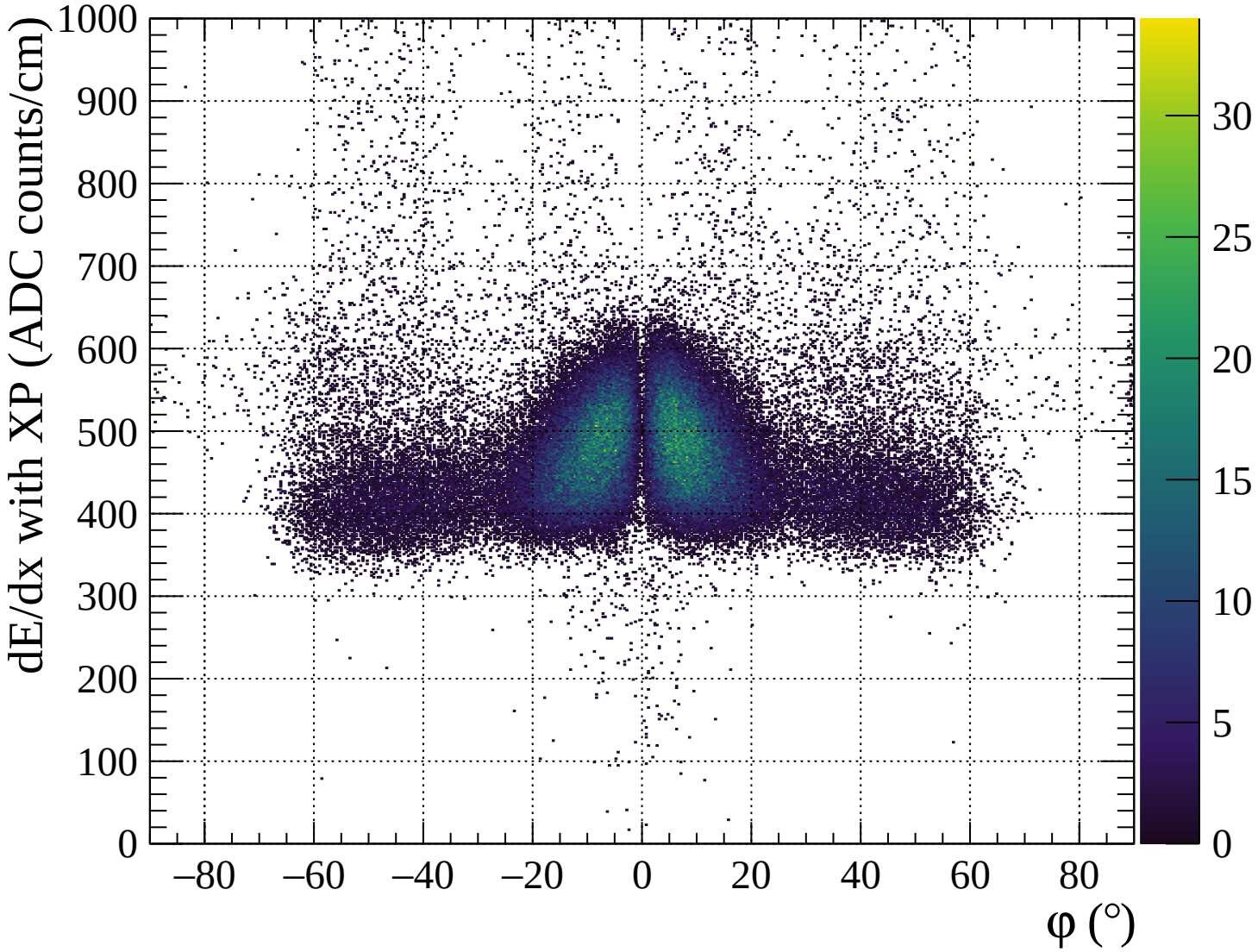


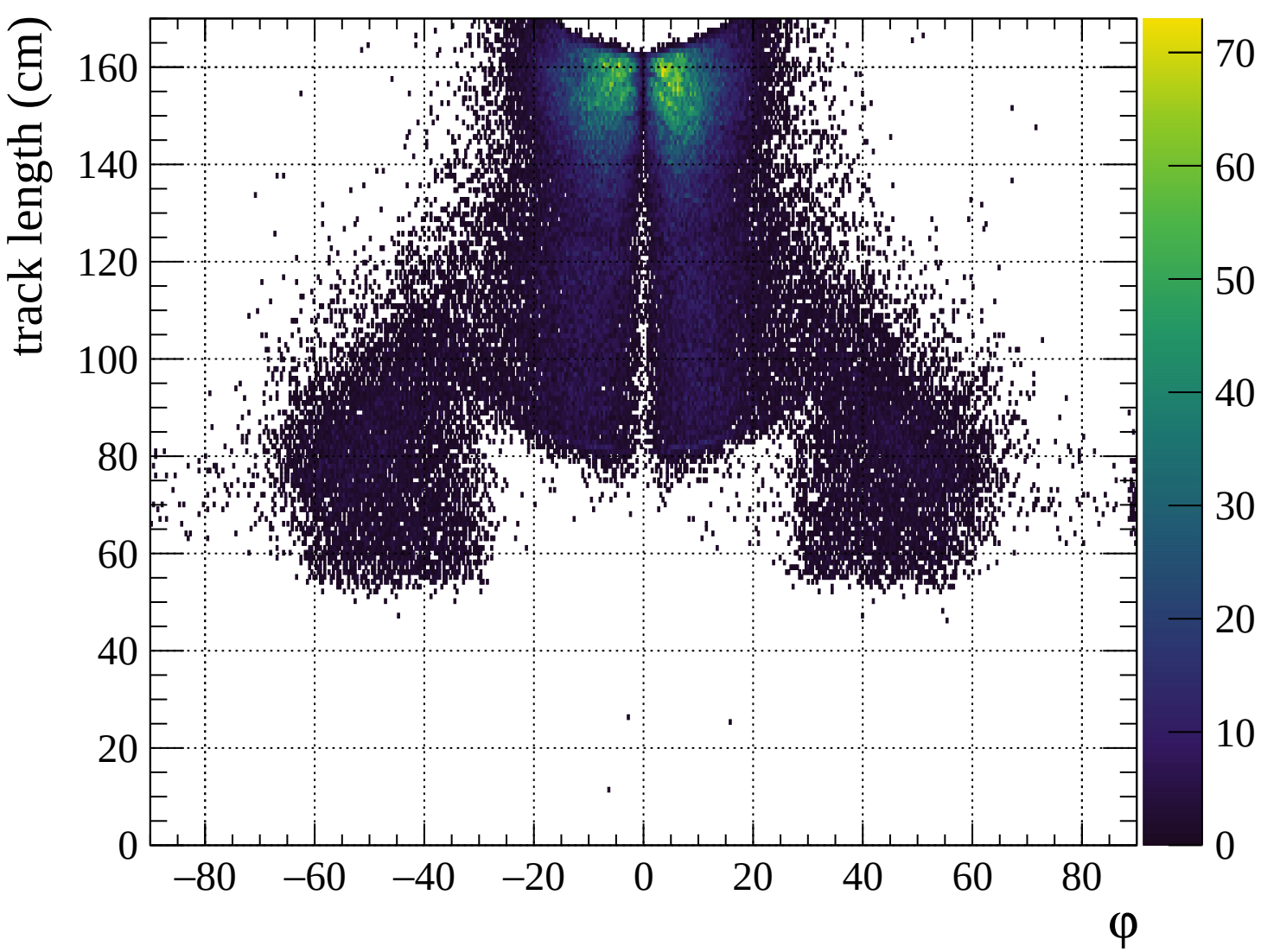


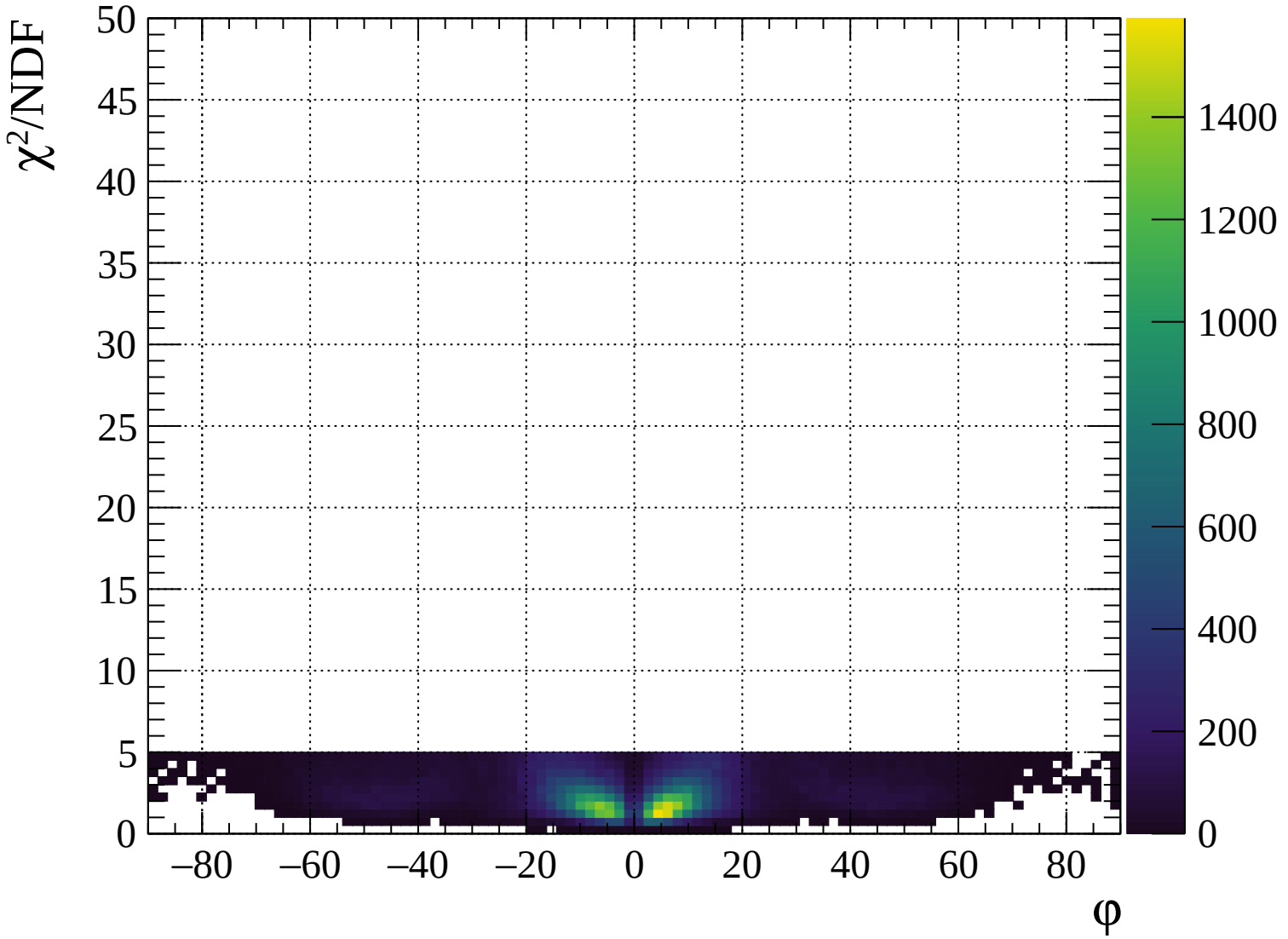


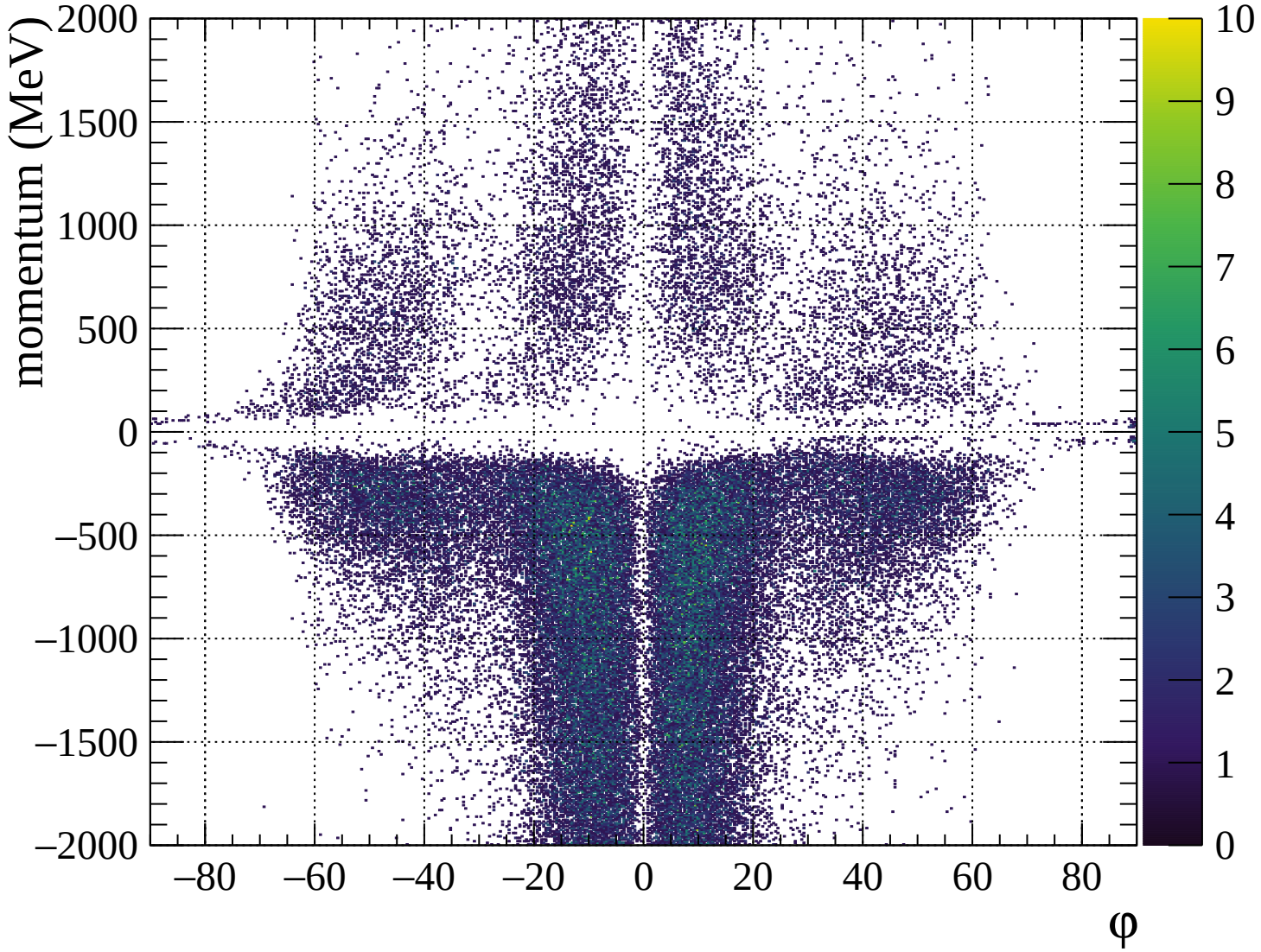


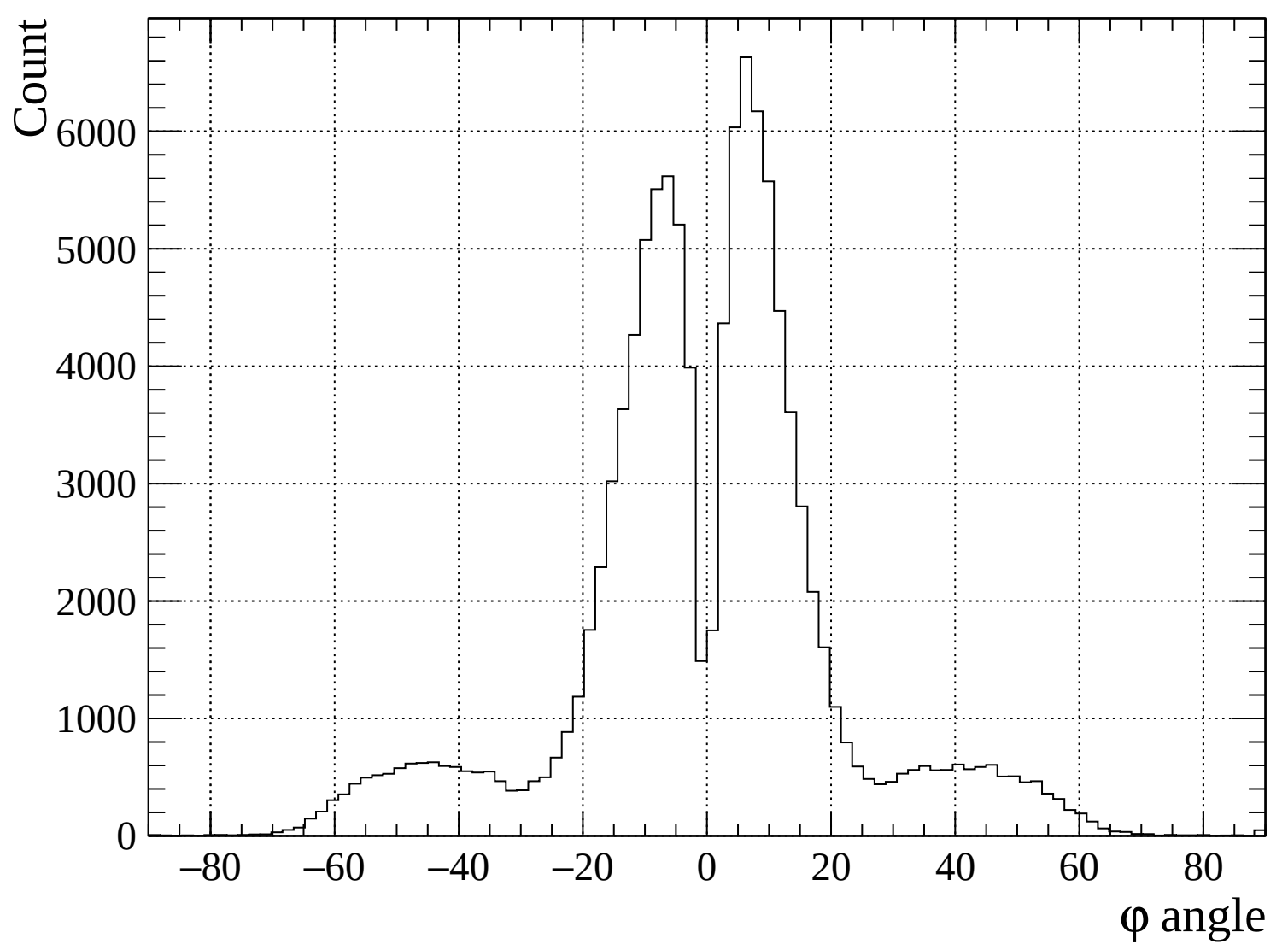


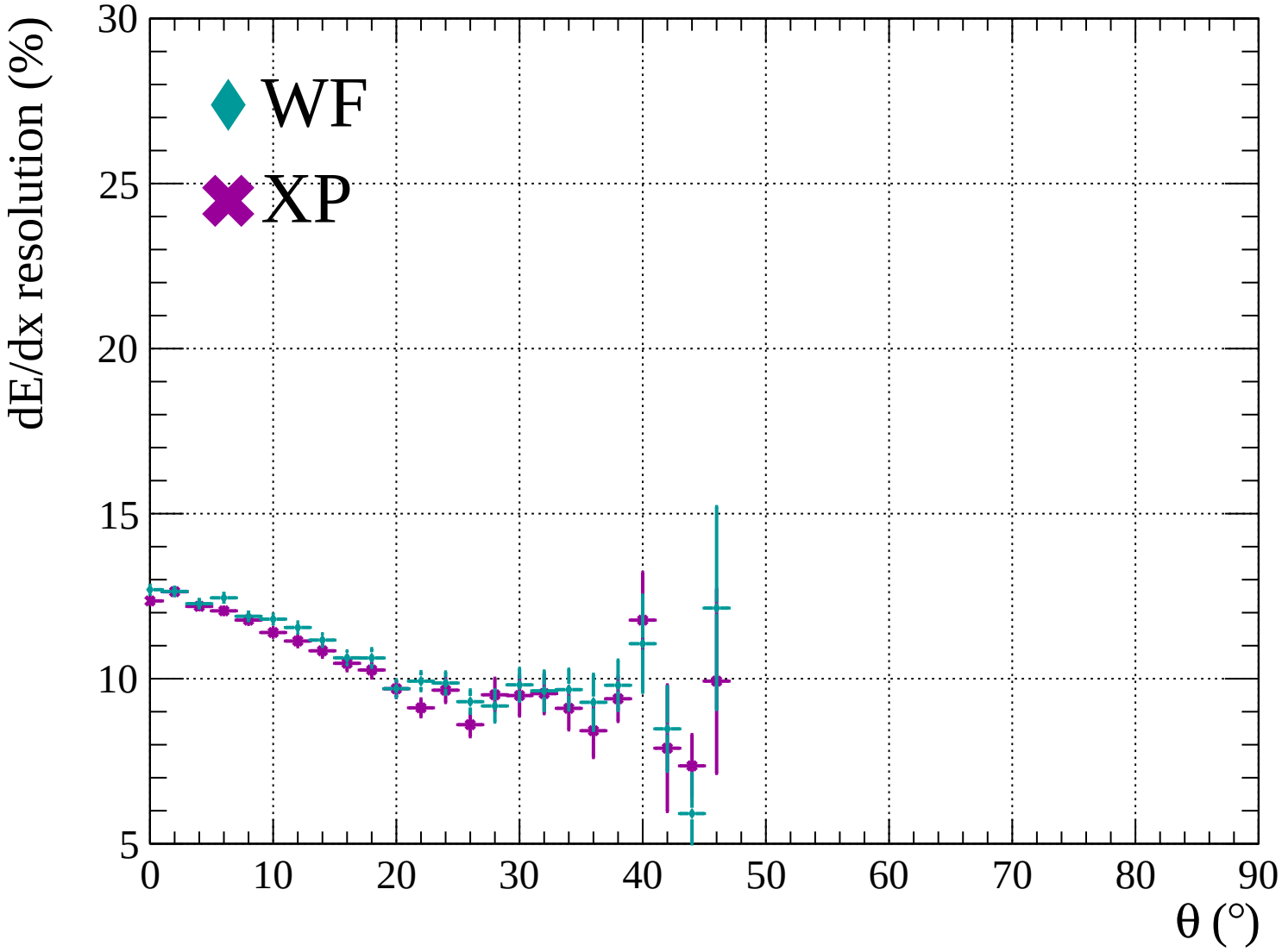


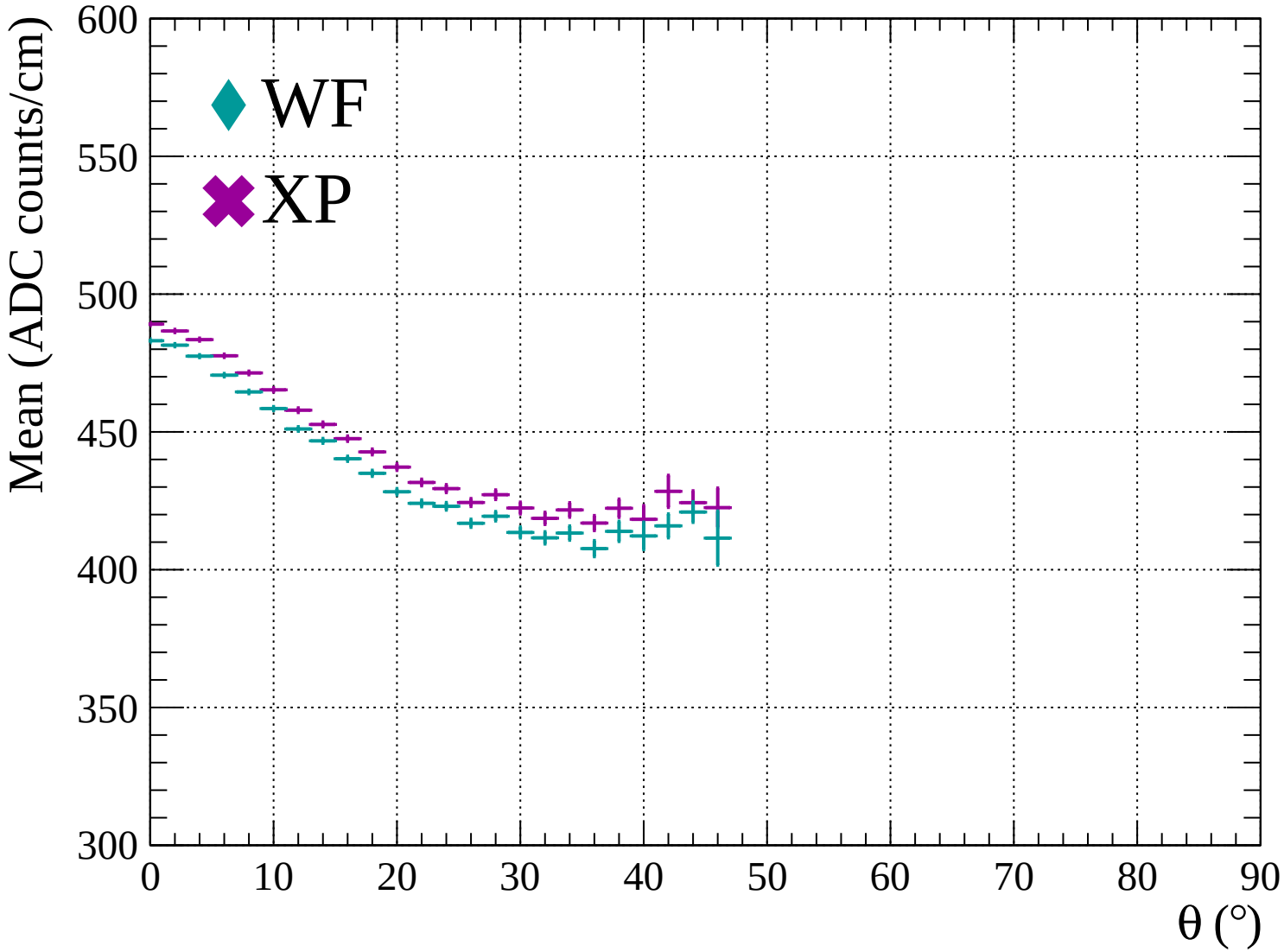


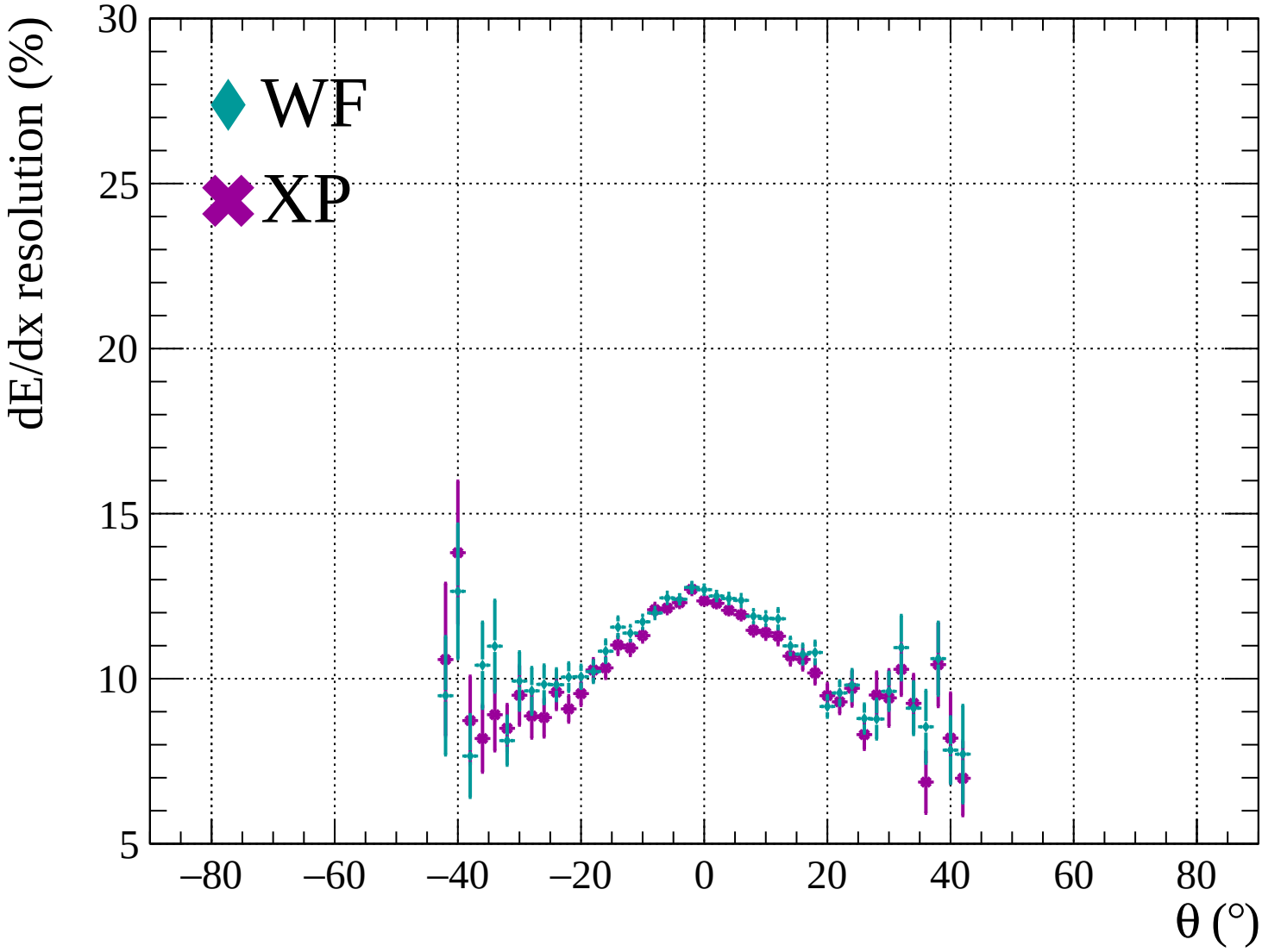


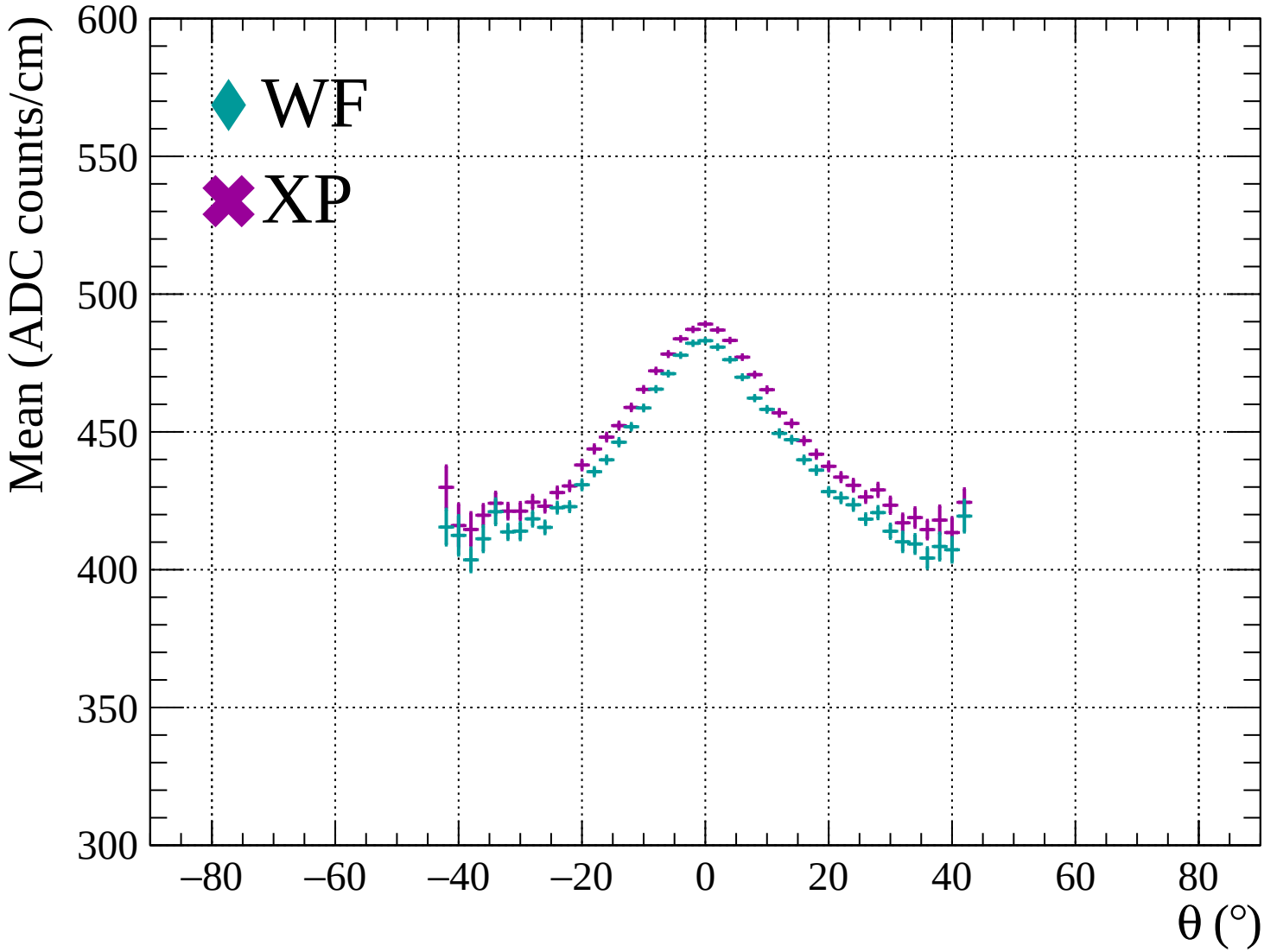


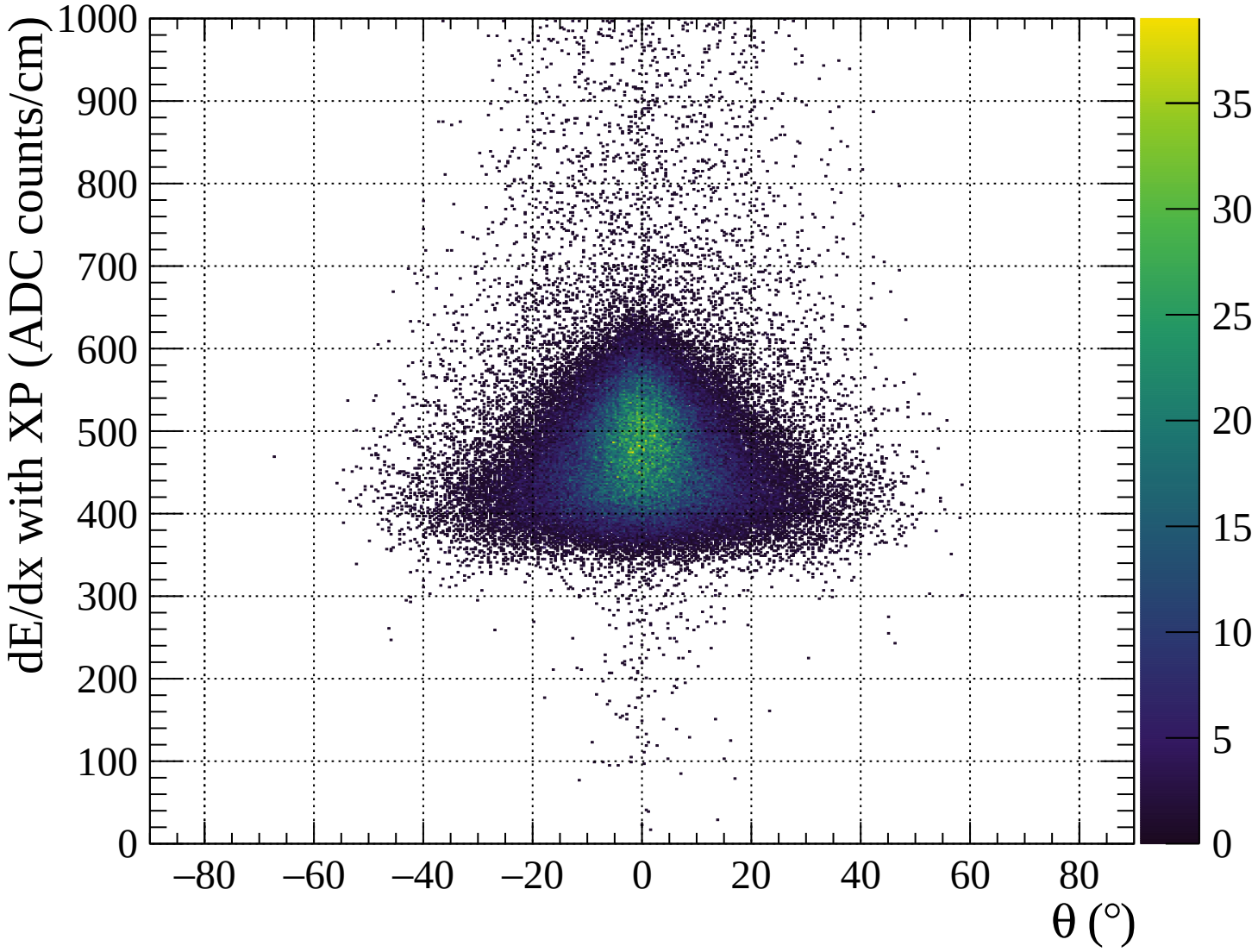


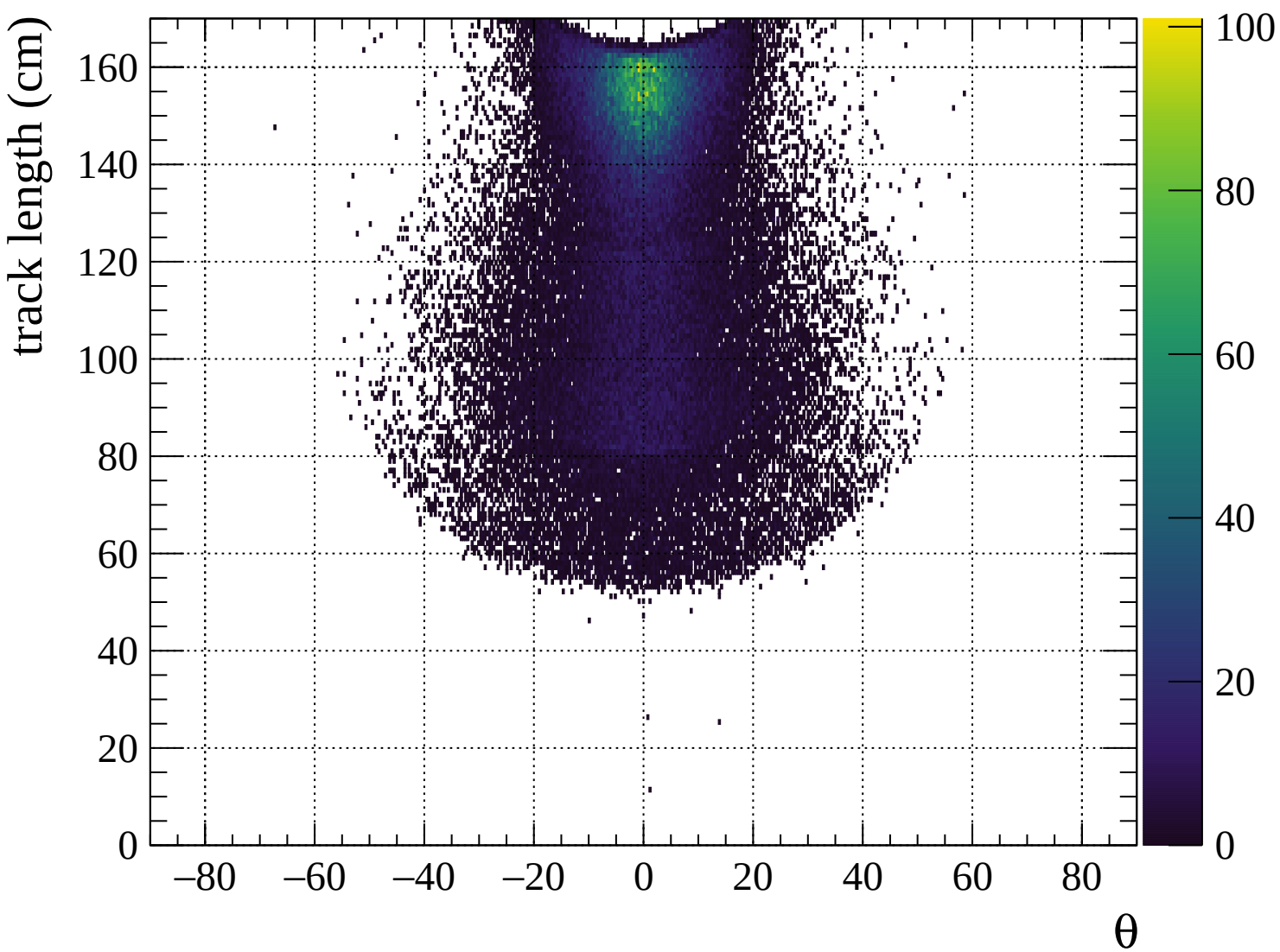


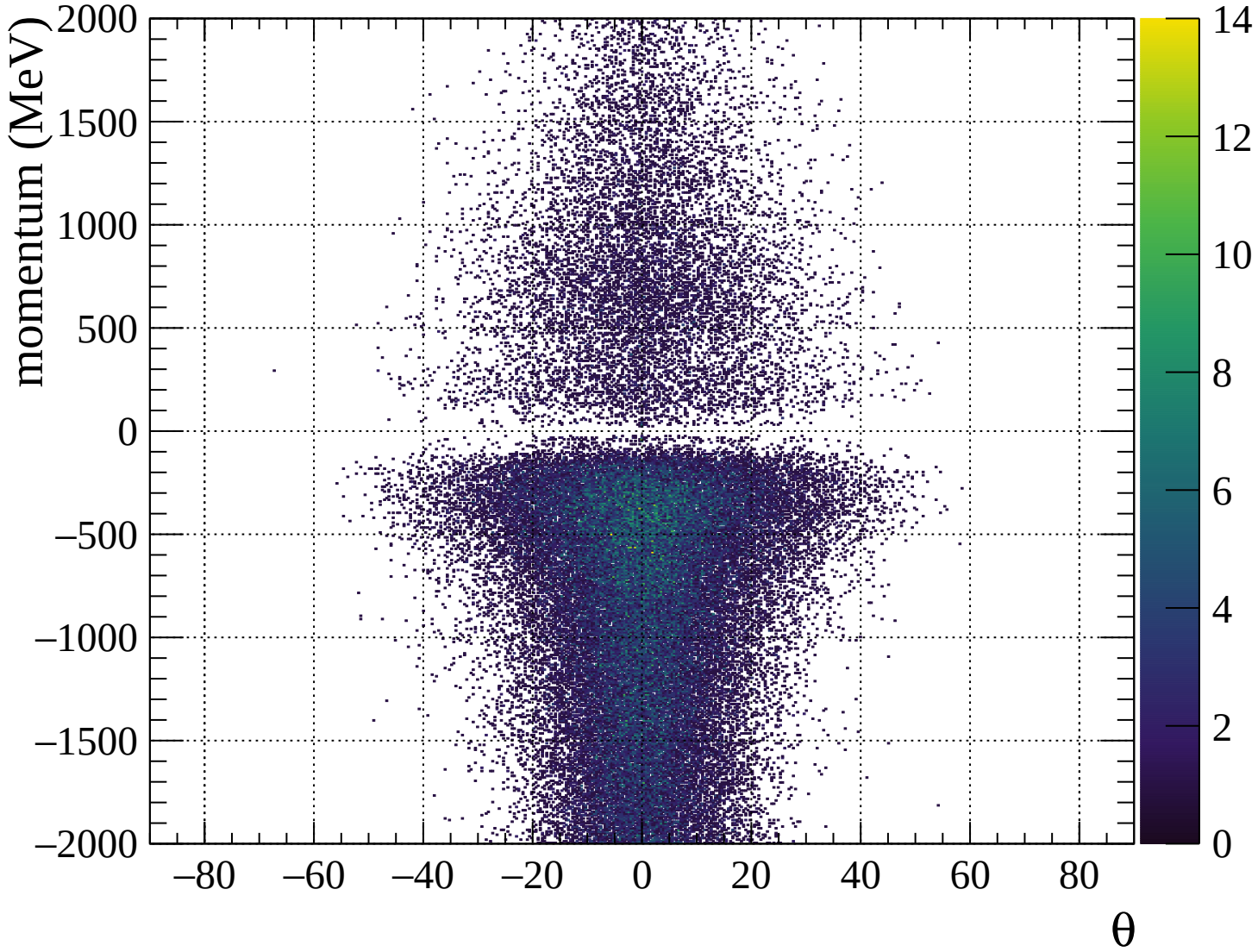


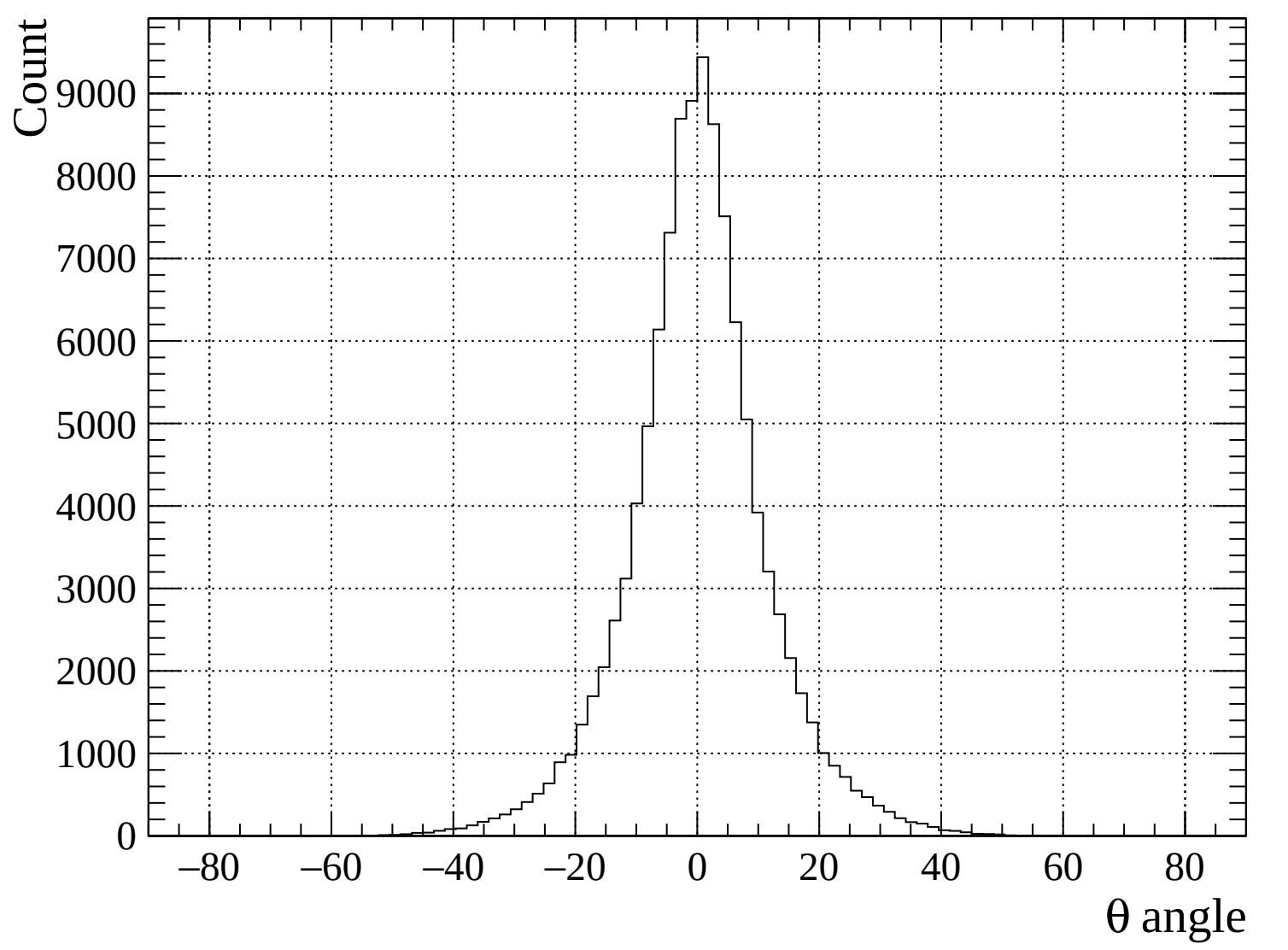


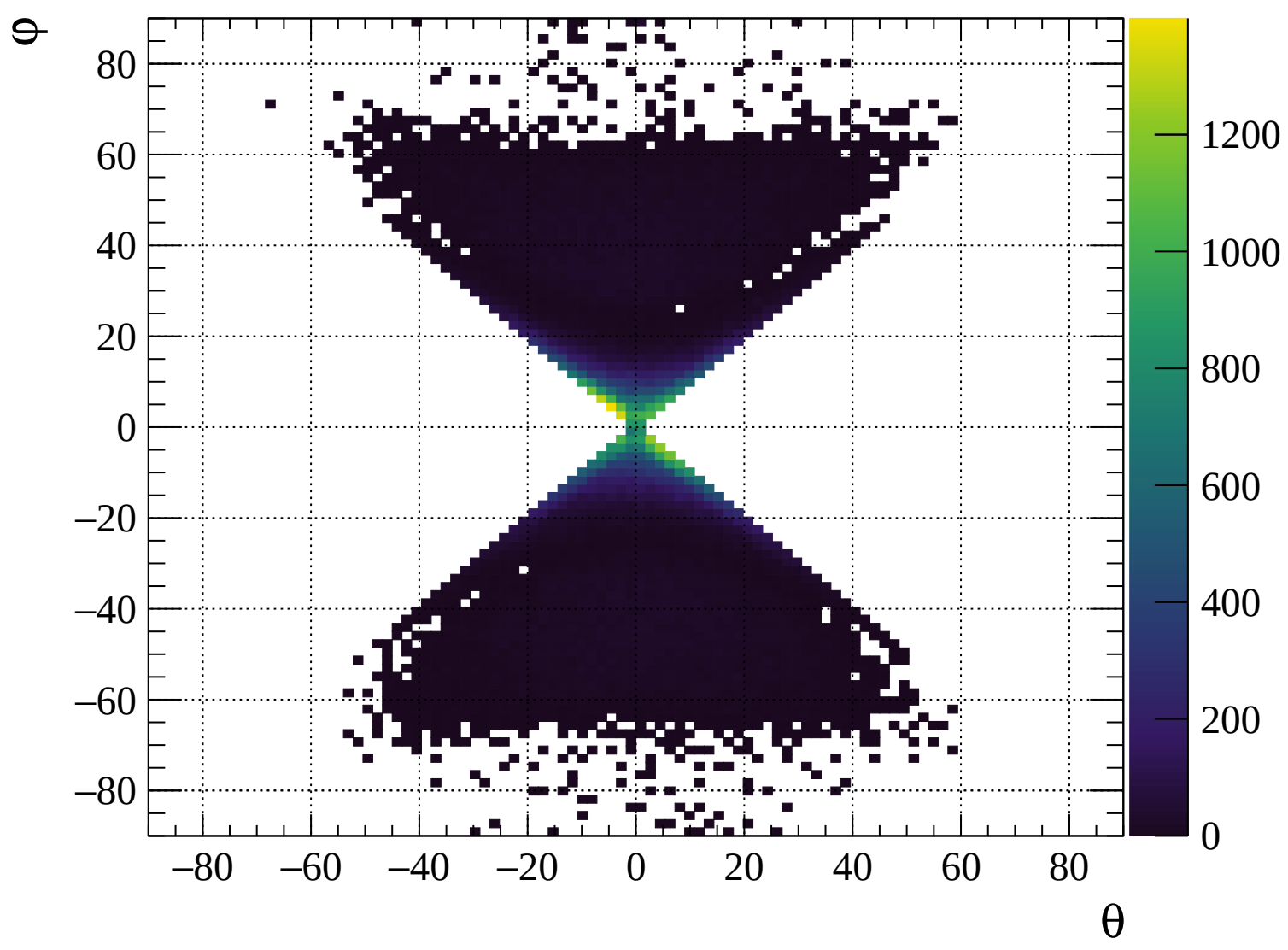


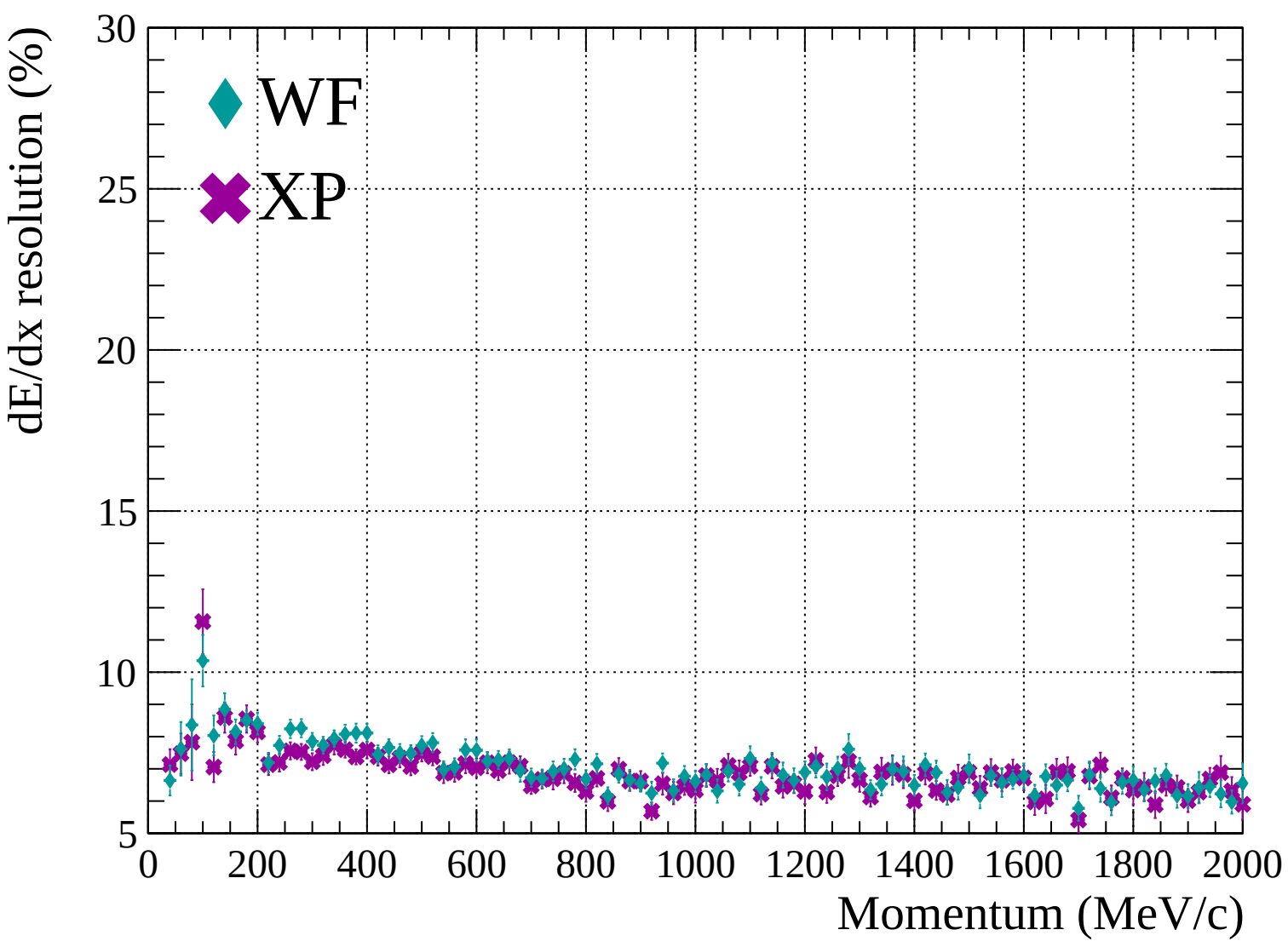


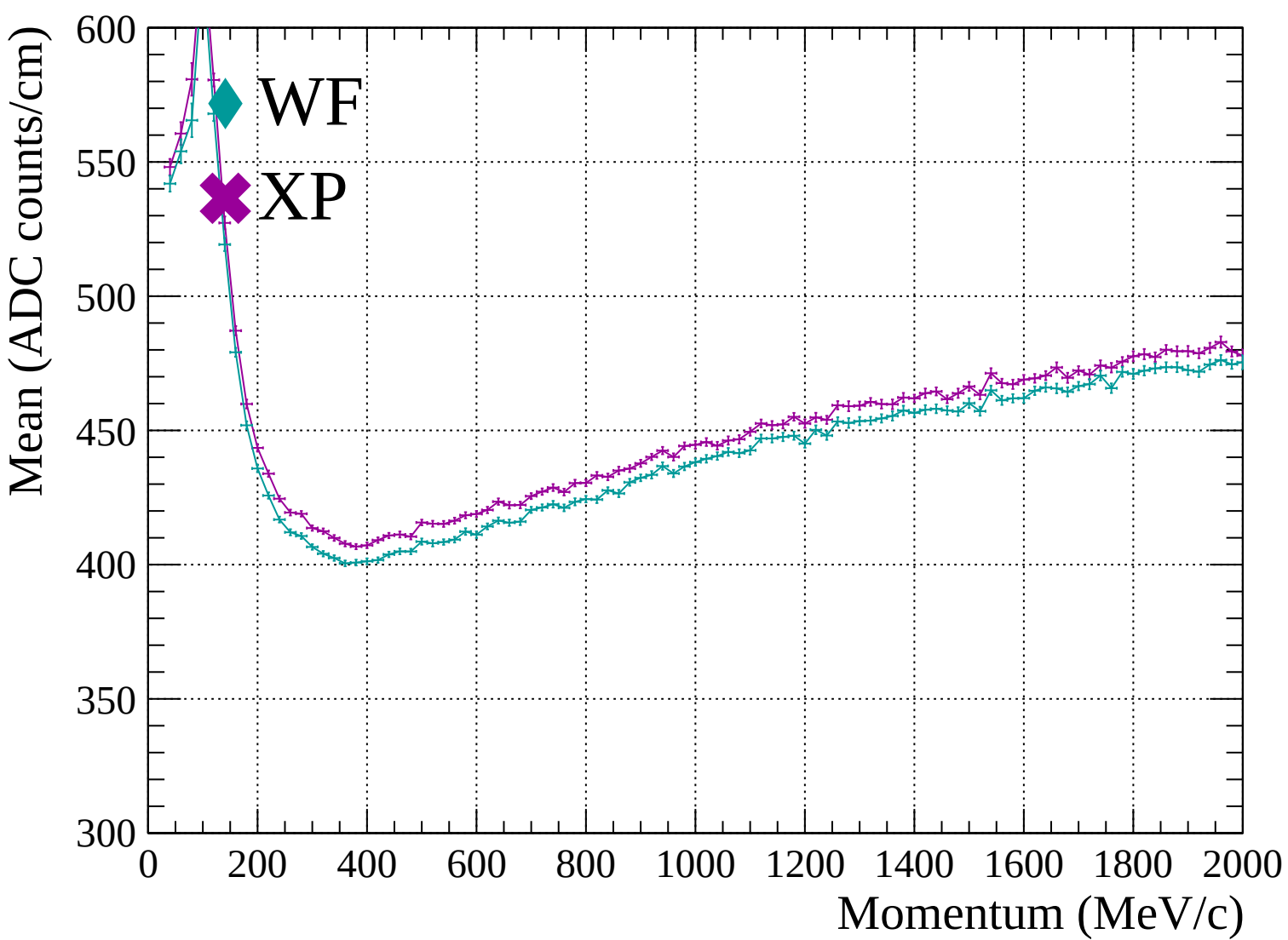


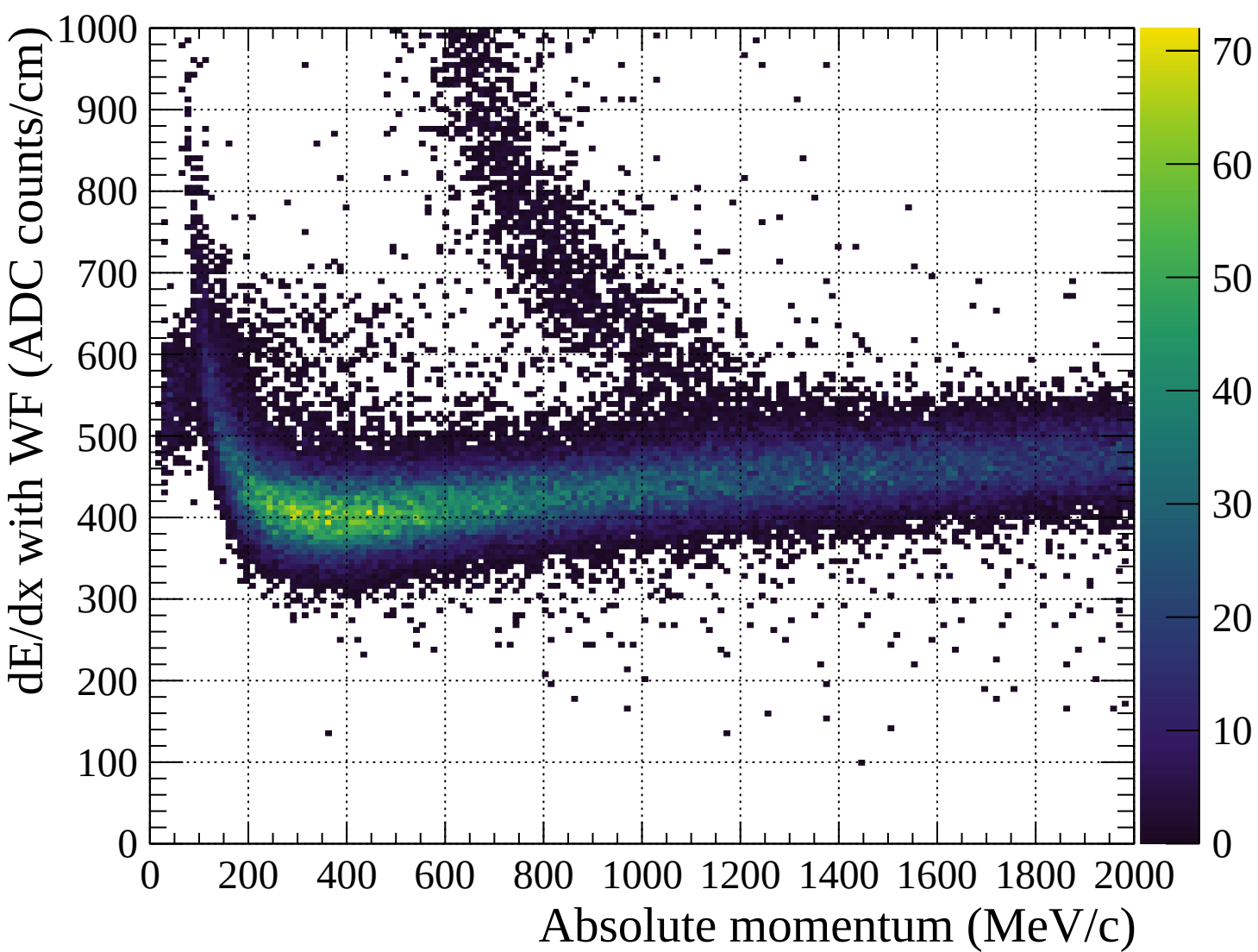


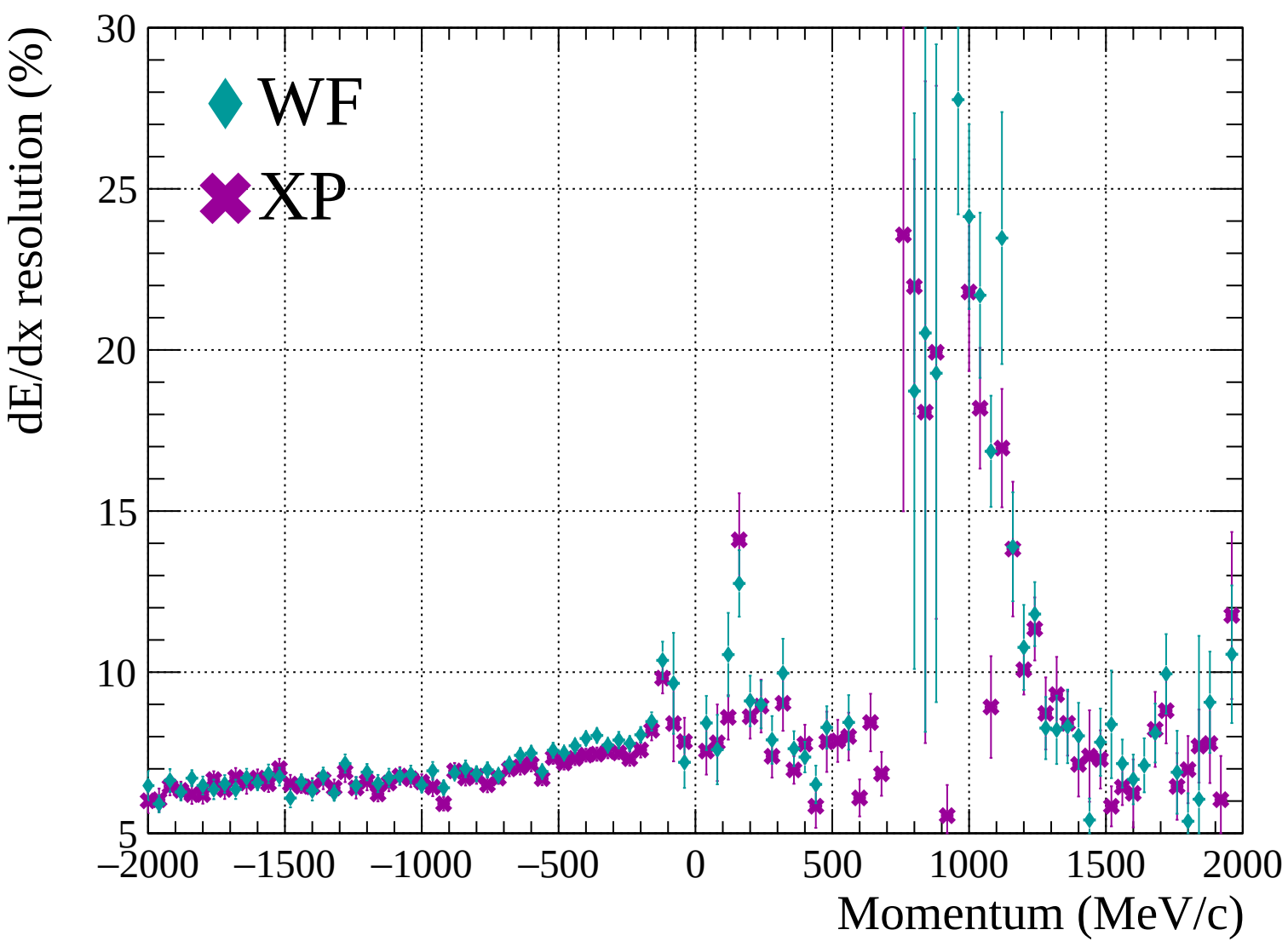


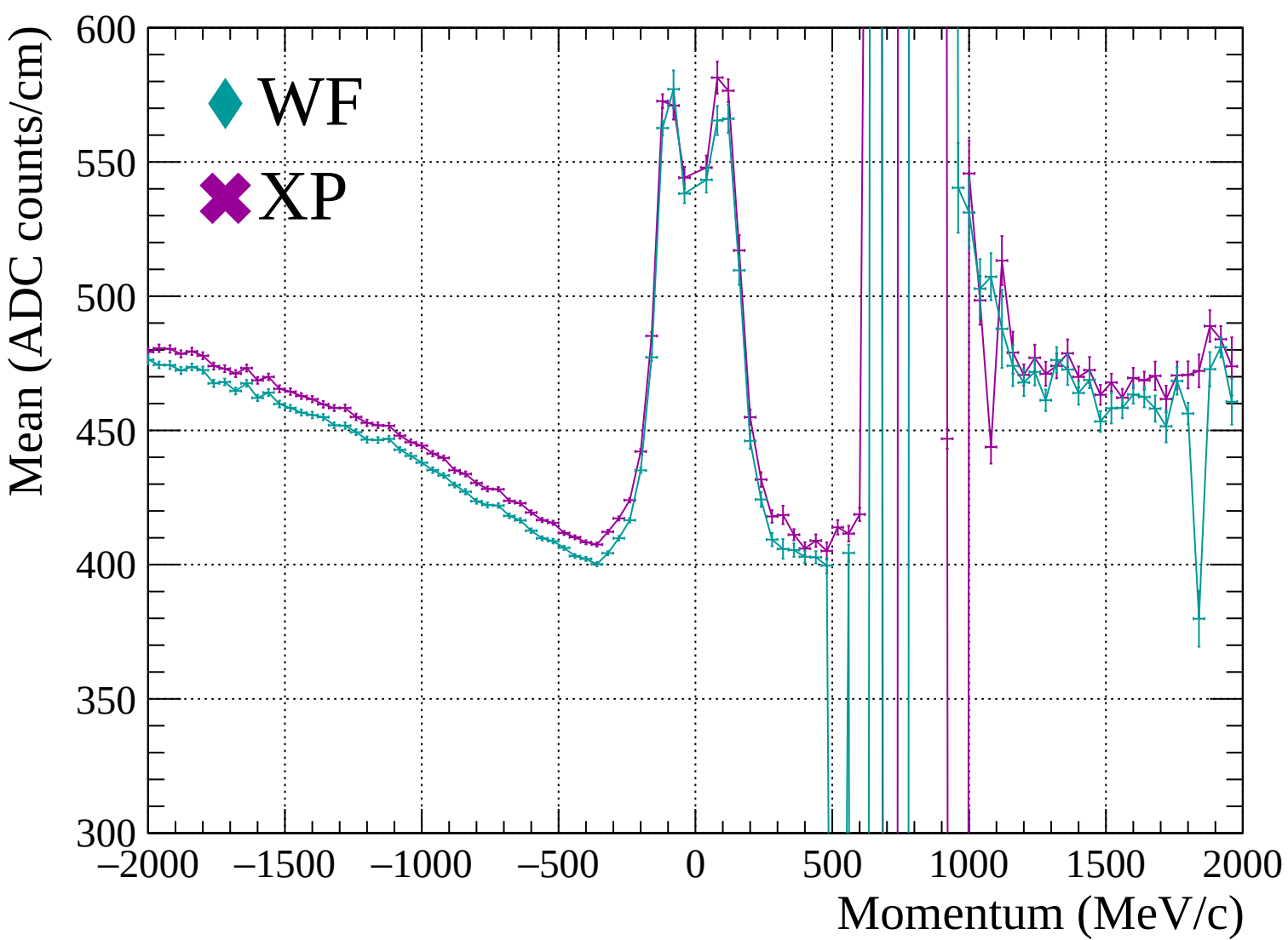


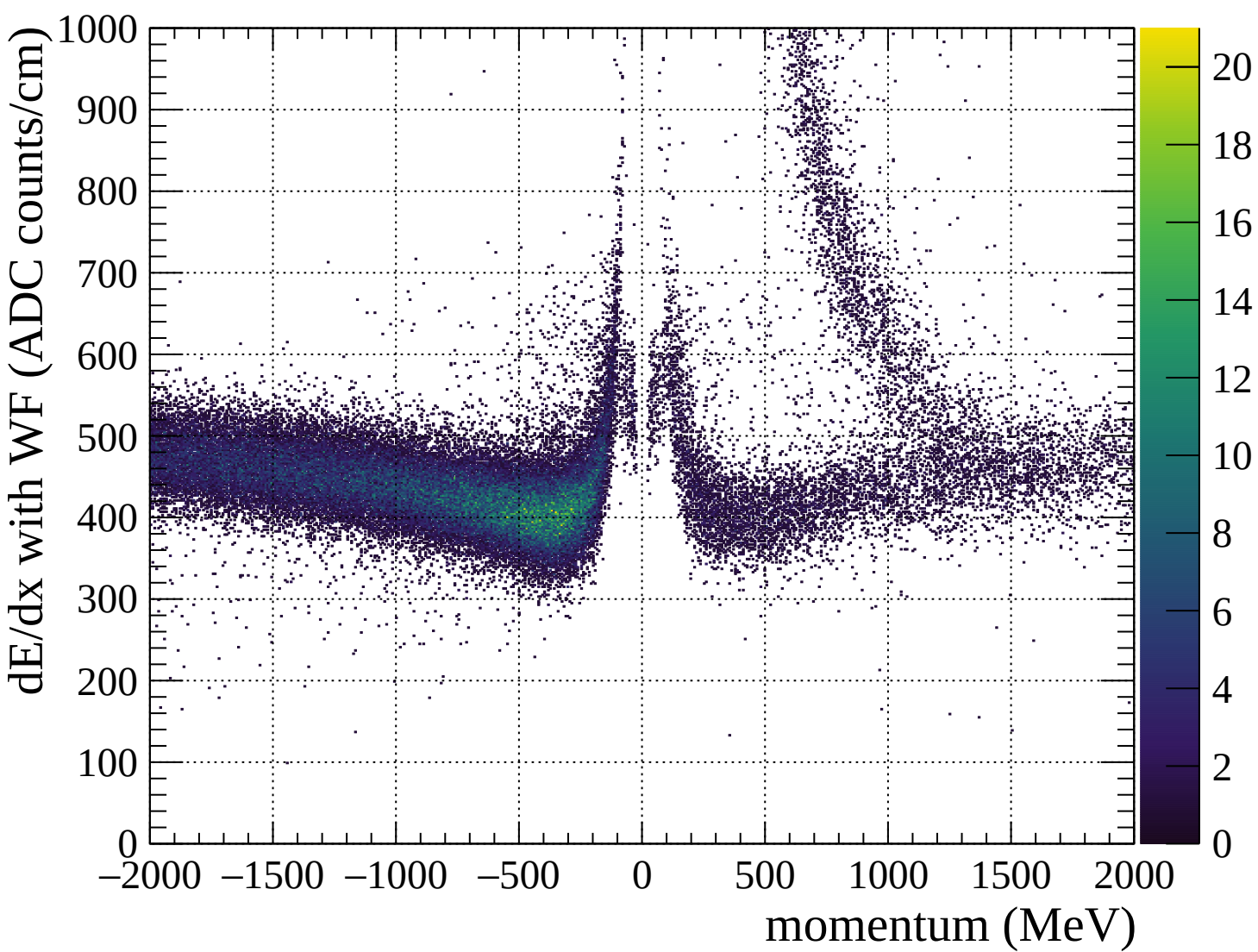


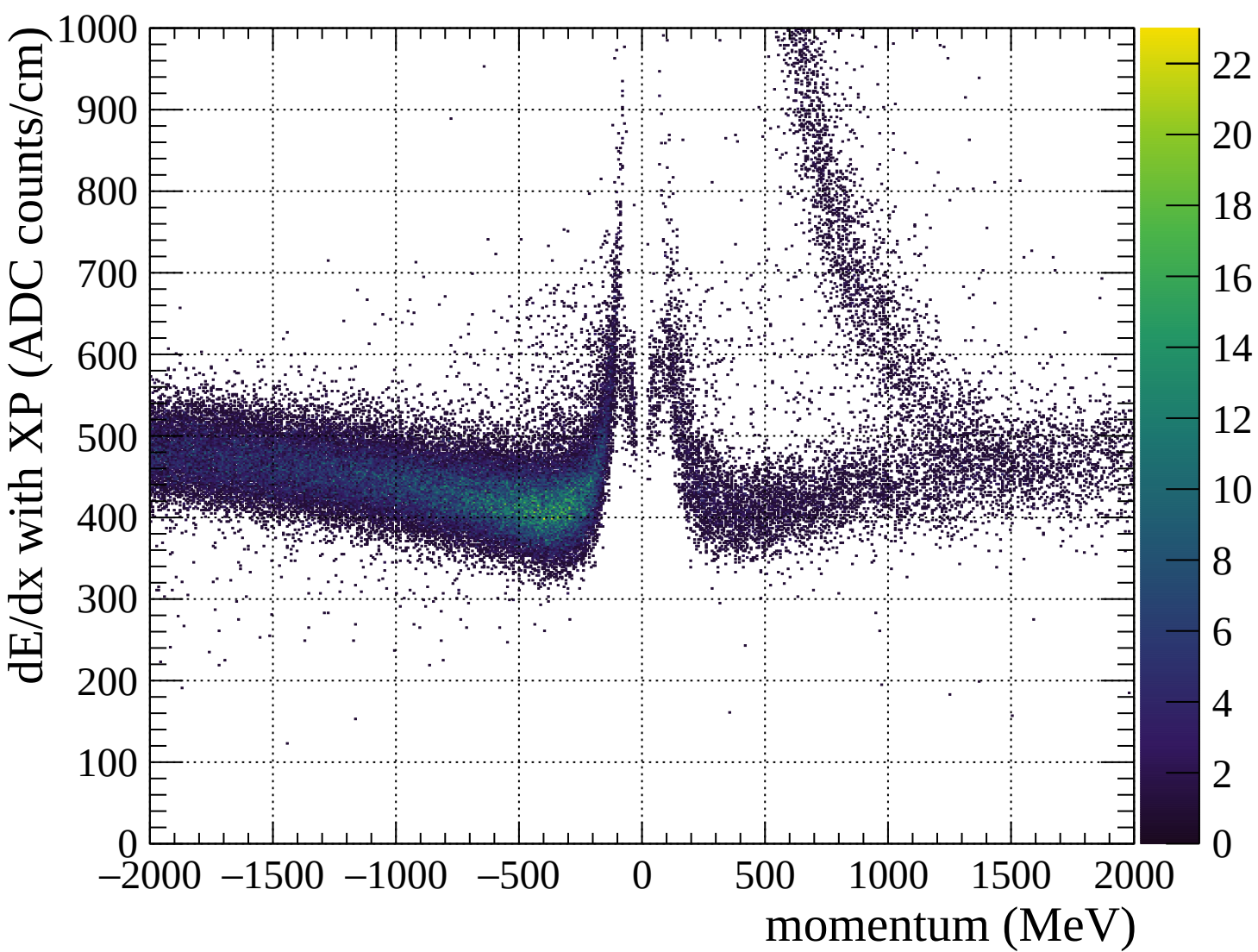


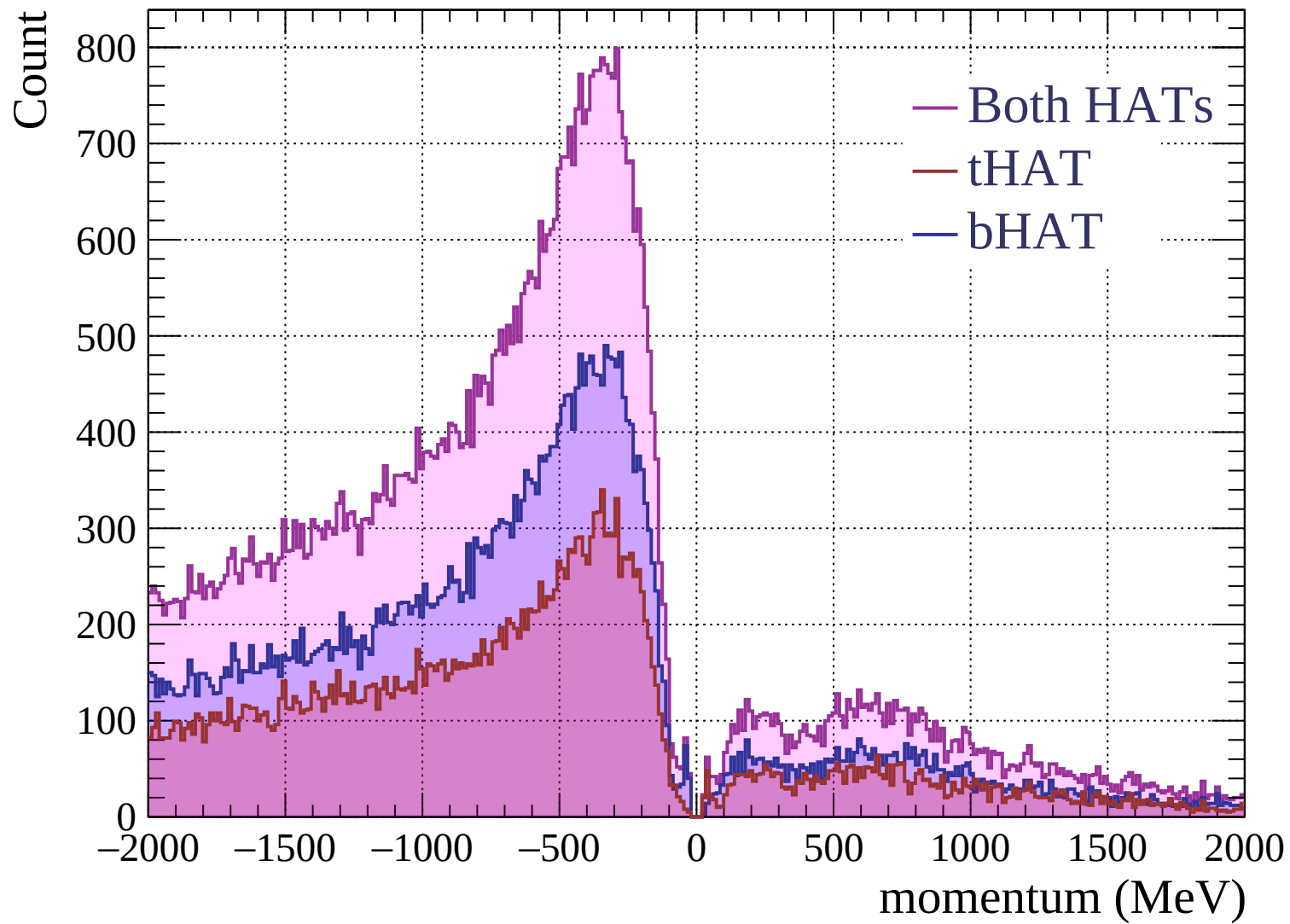


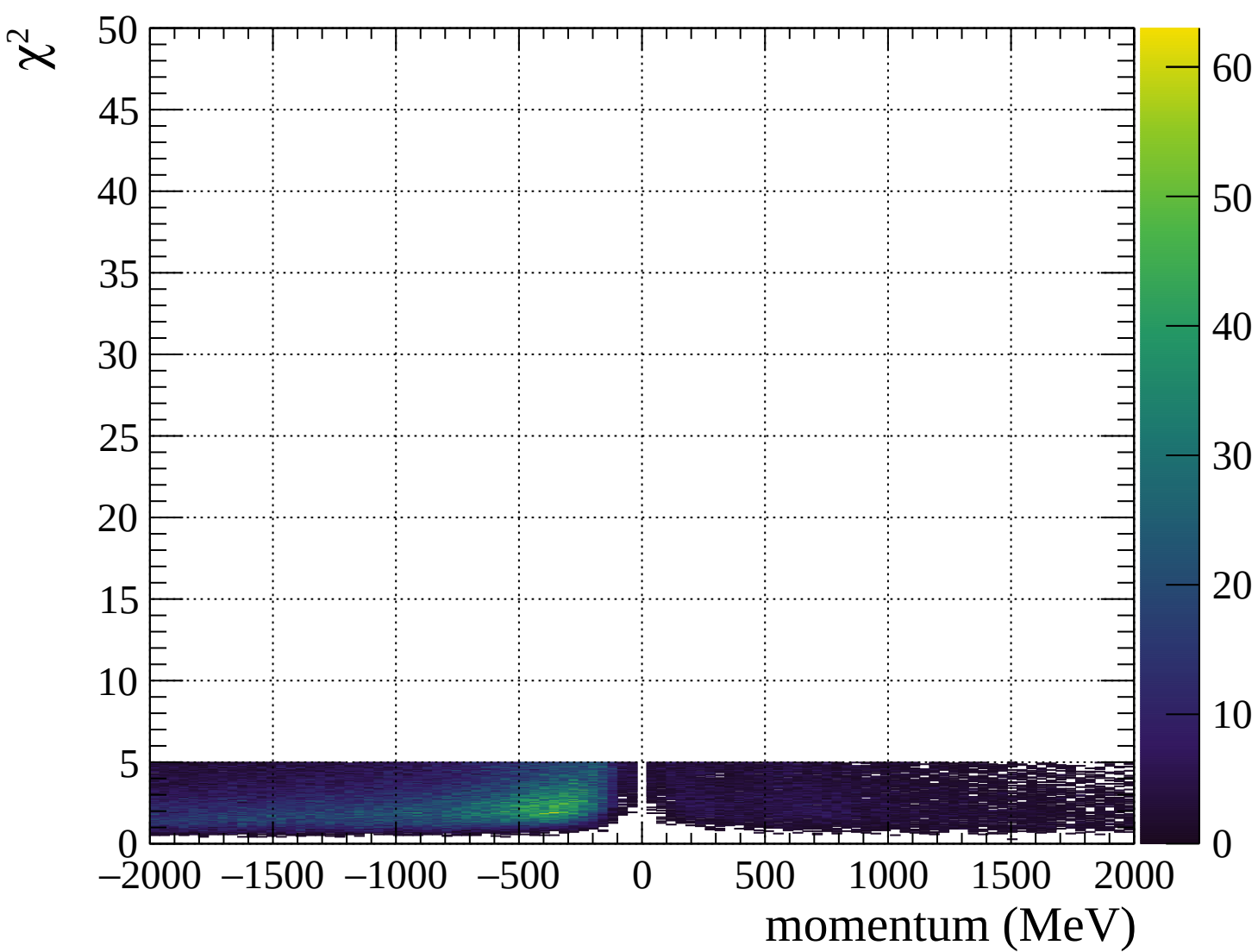




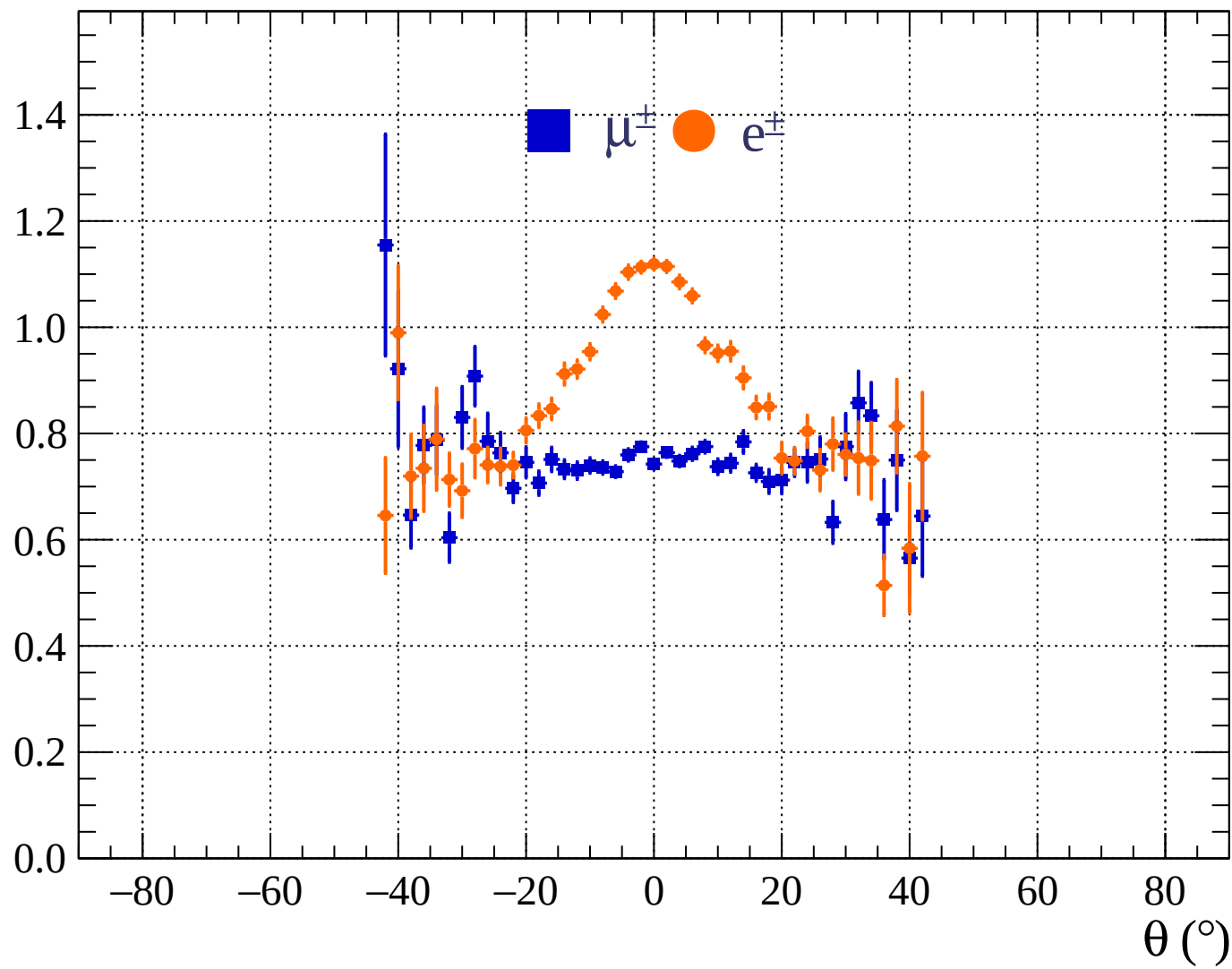


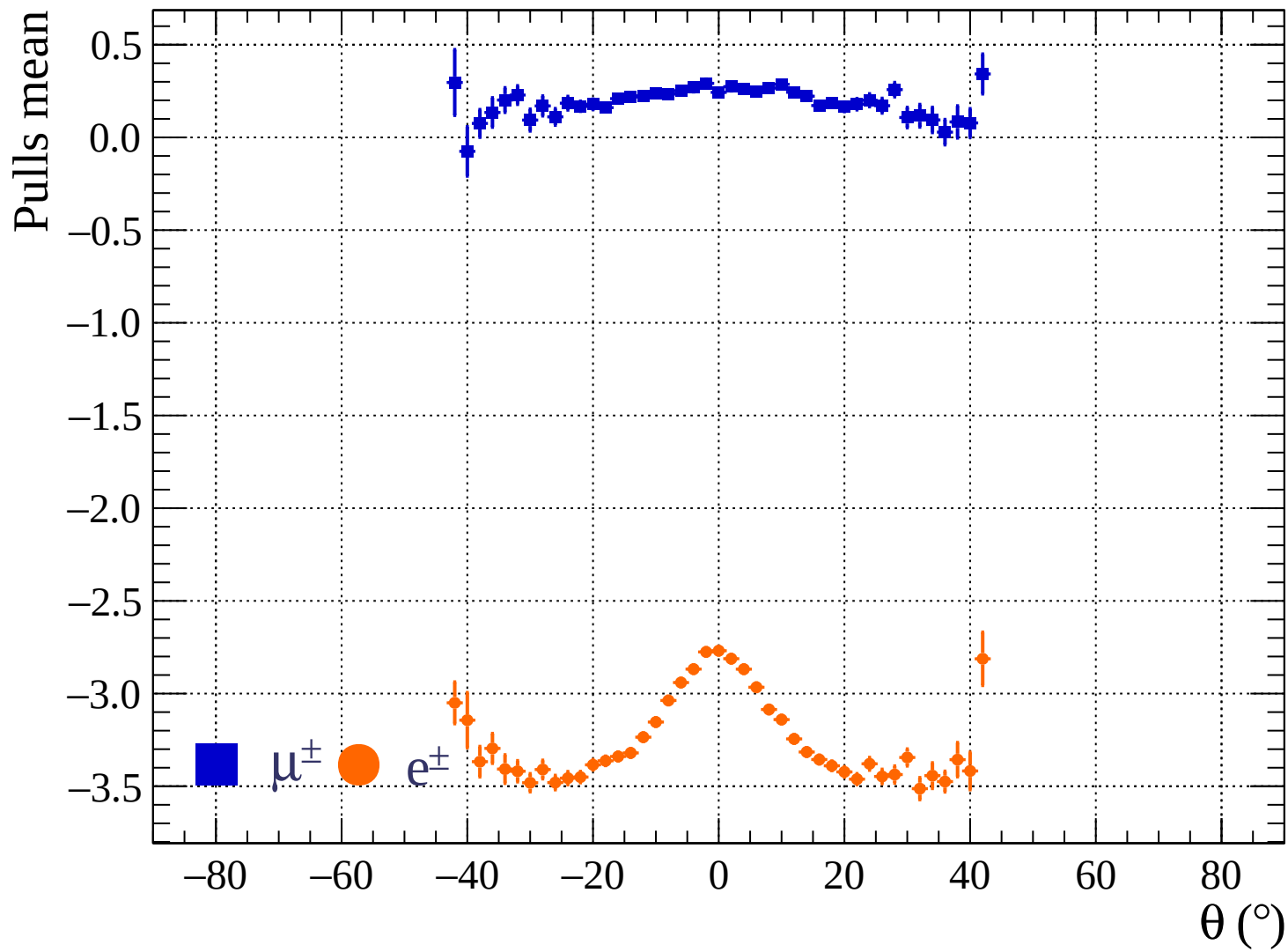




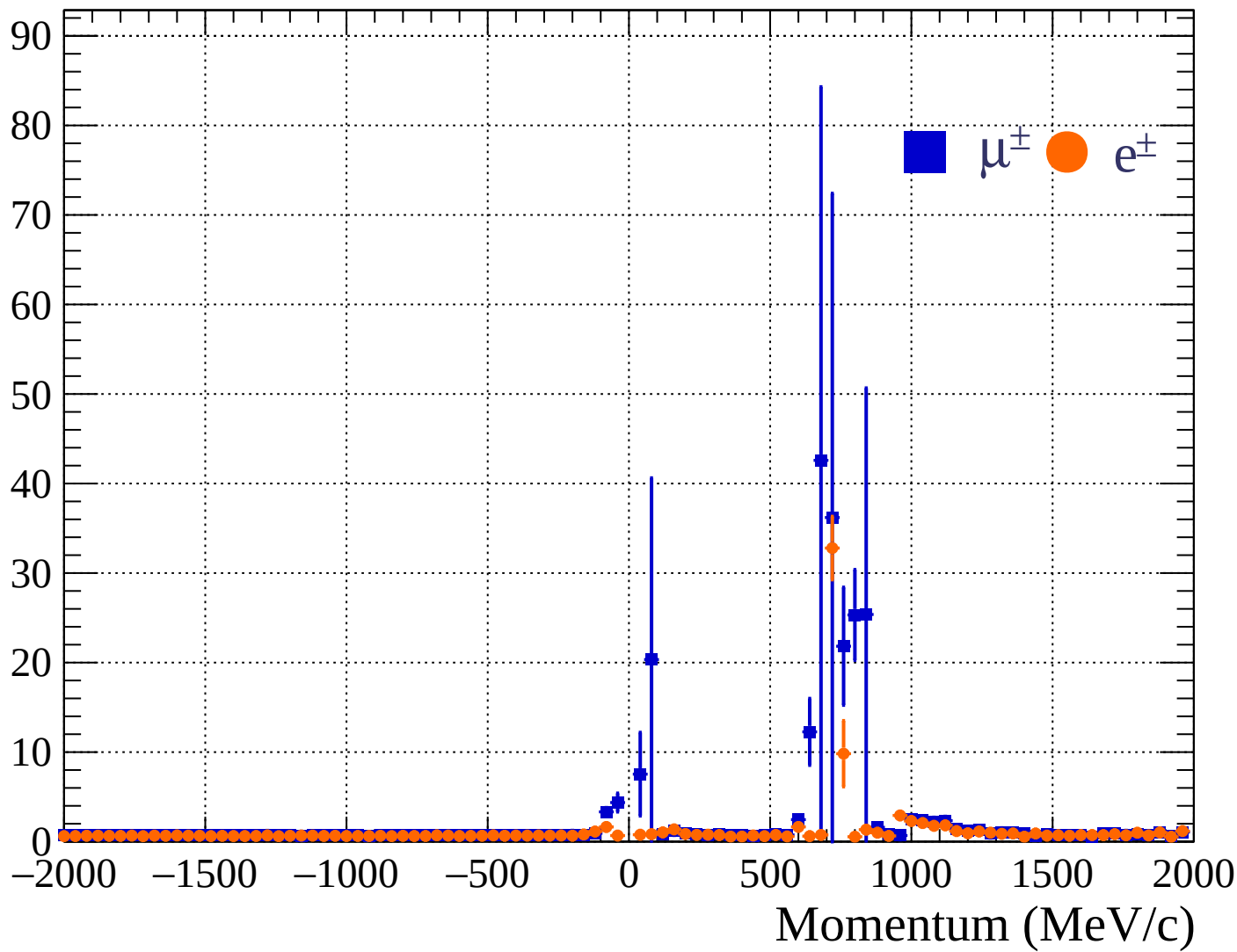


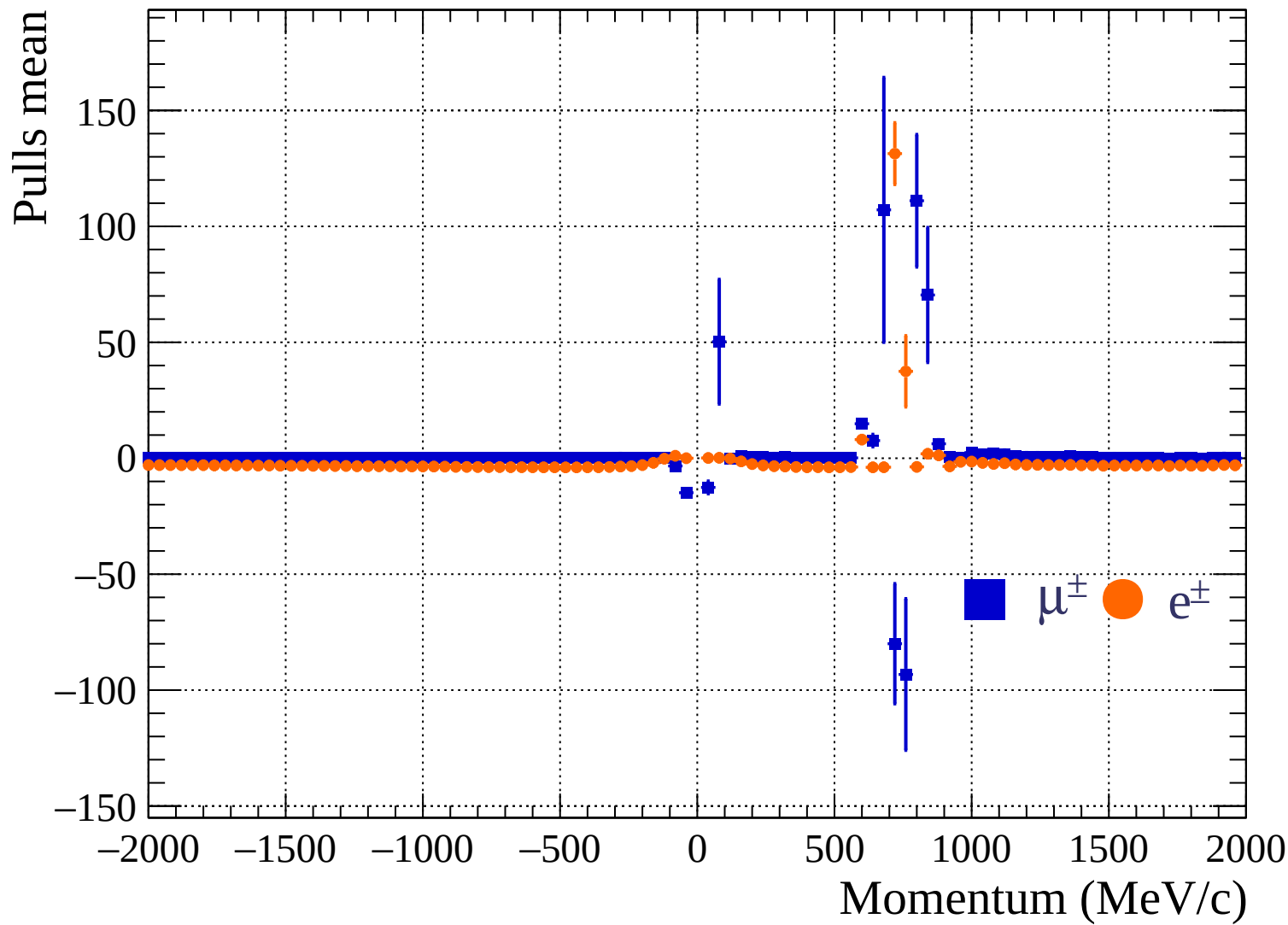
Pulls standard deviation

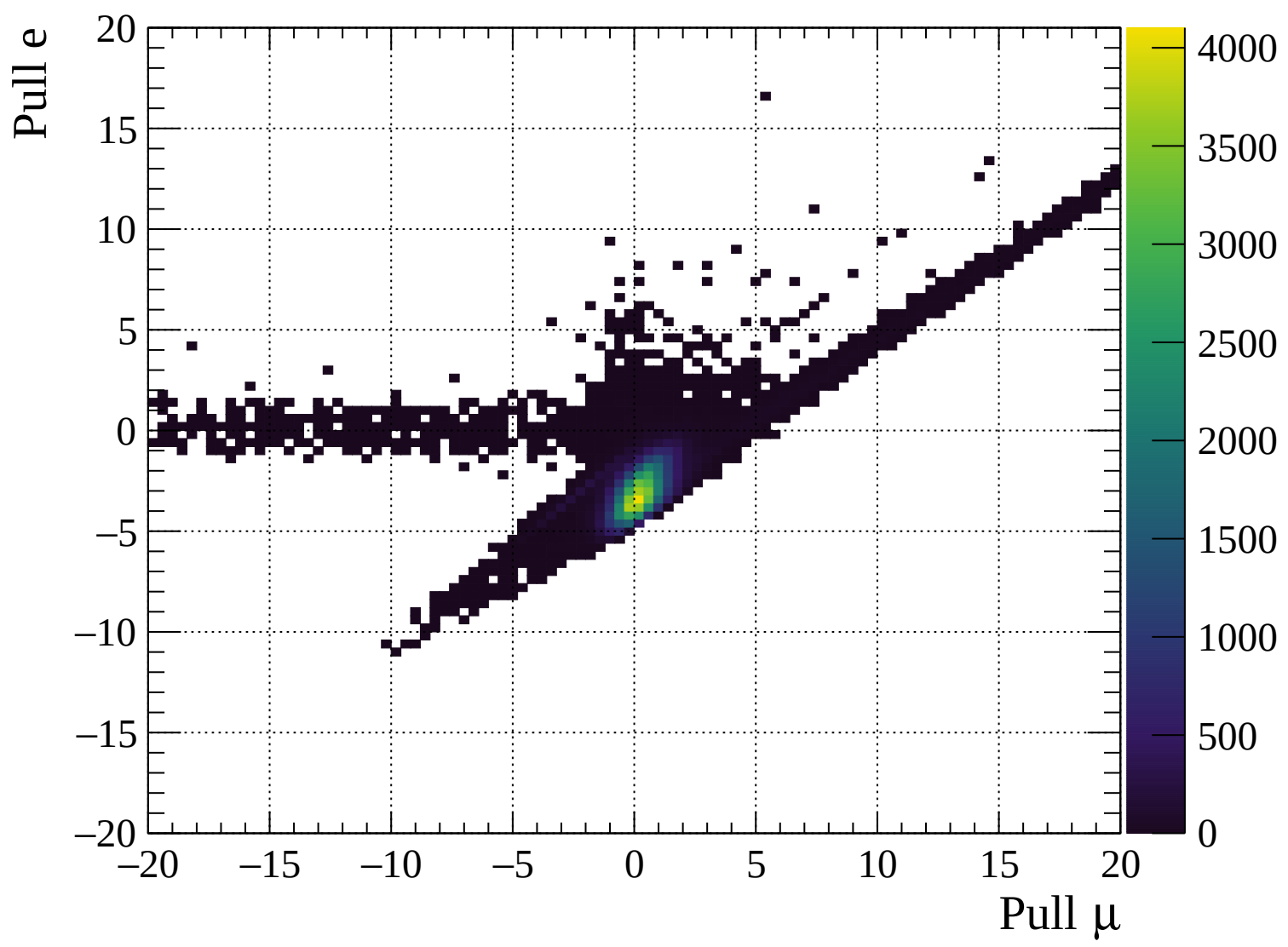


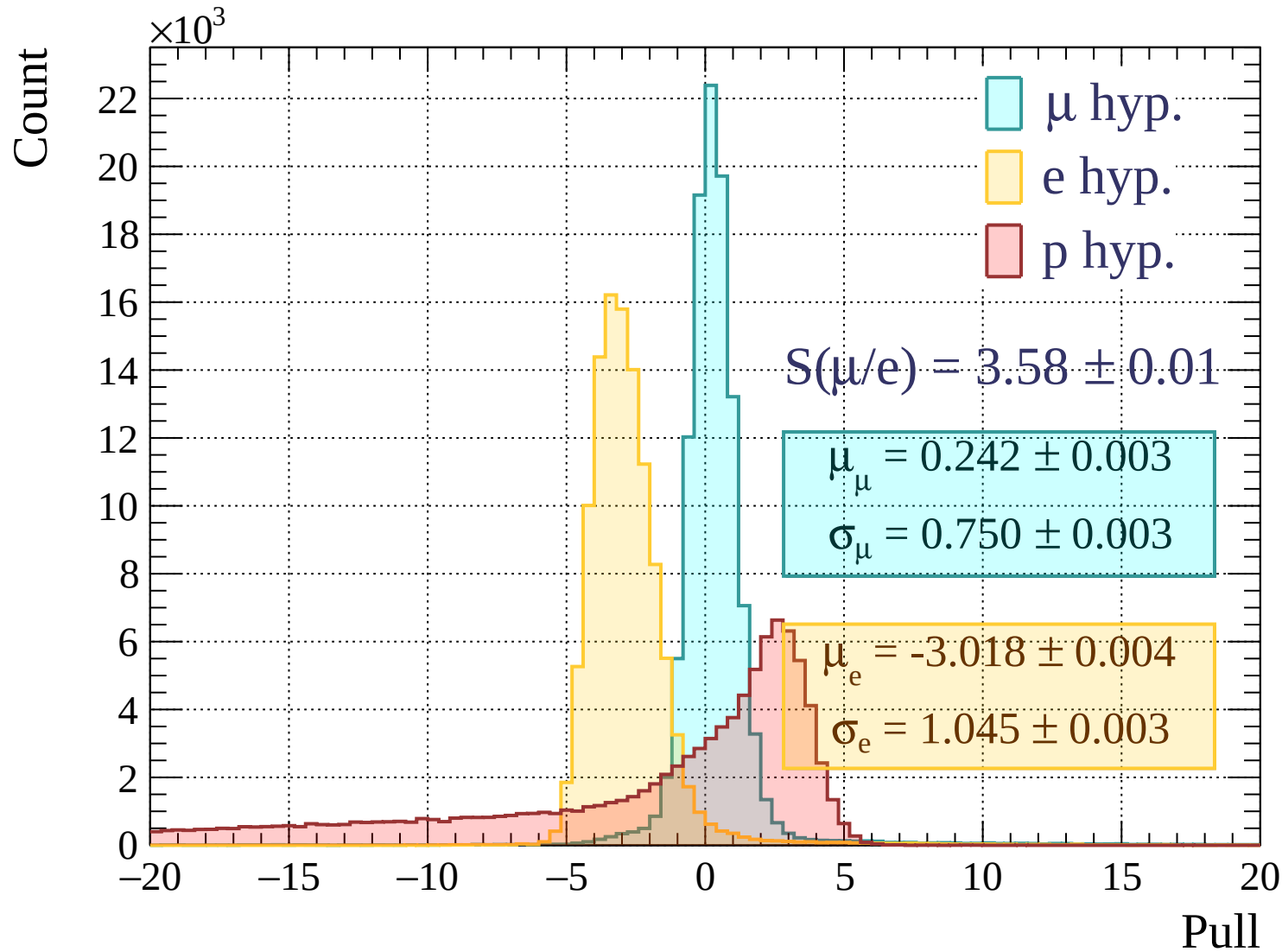


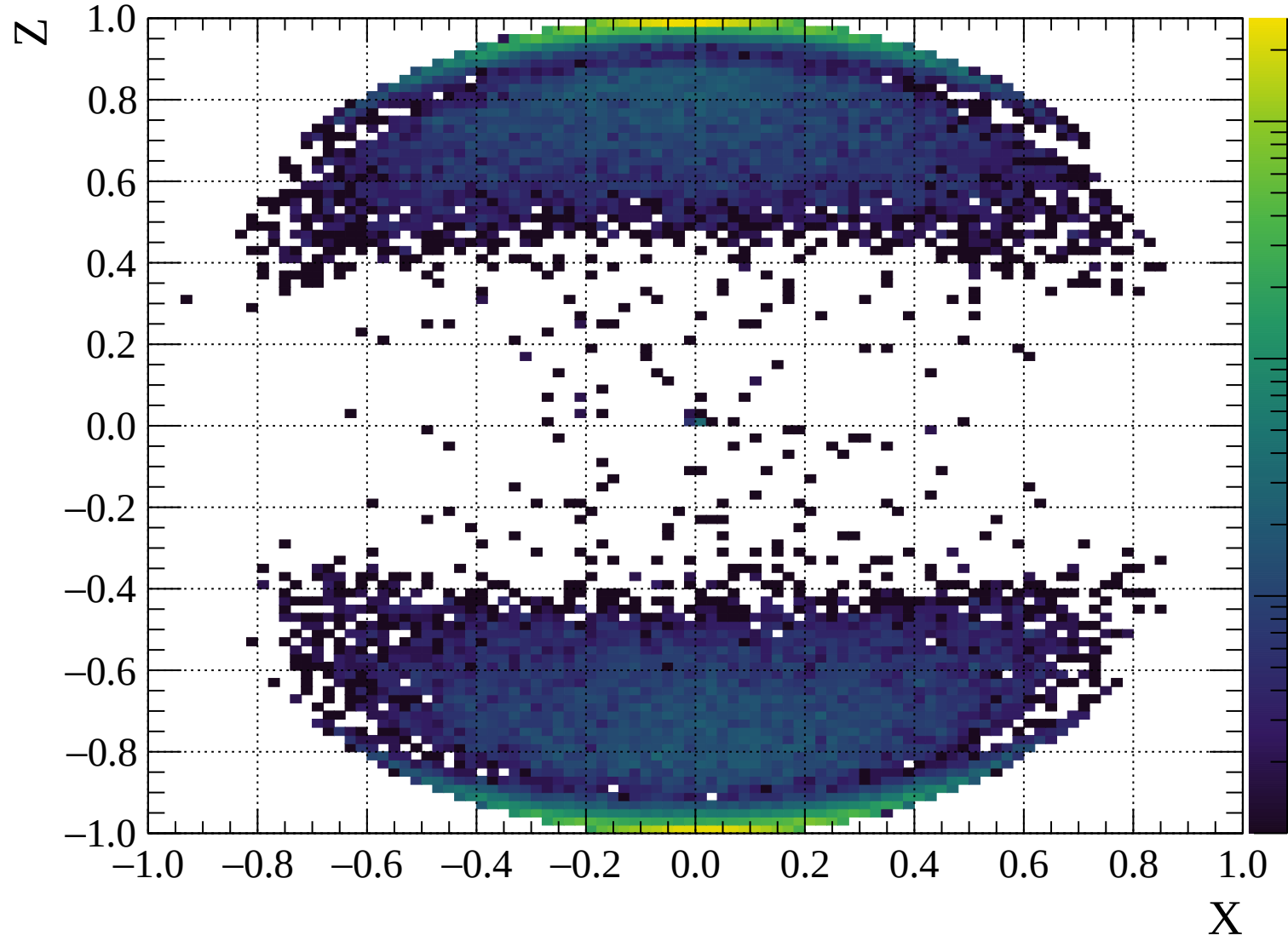
Pulls standard deviation

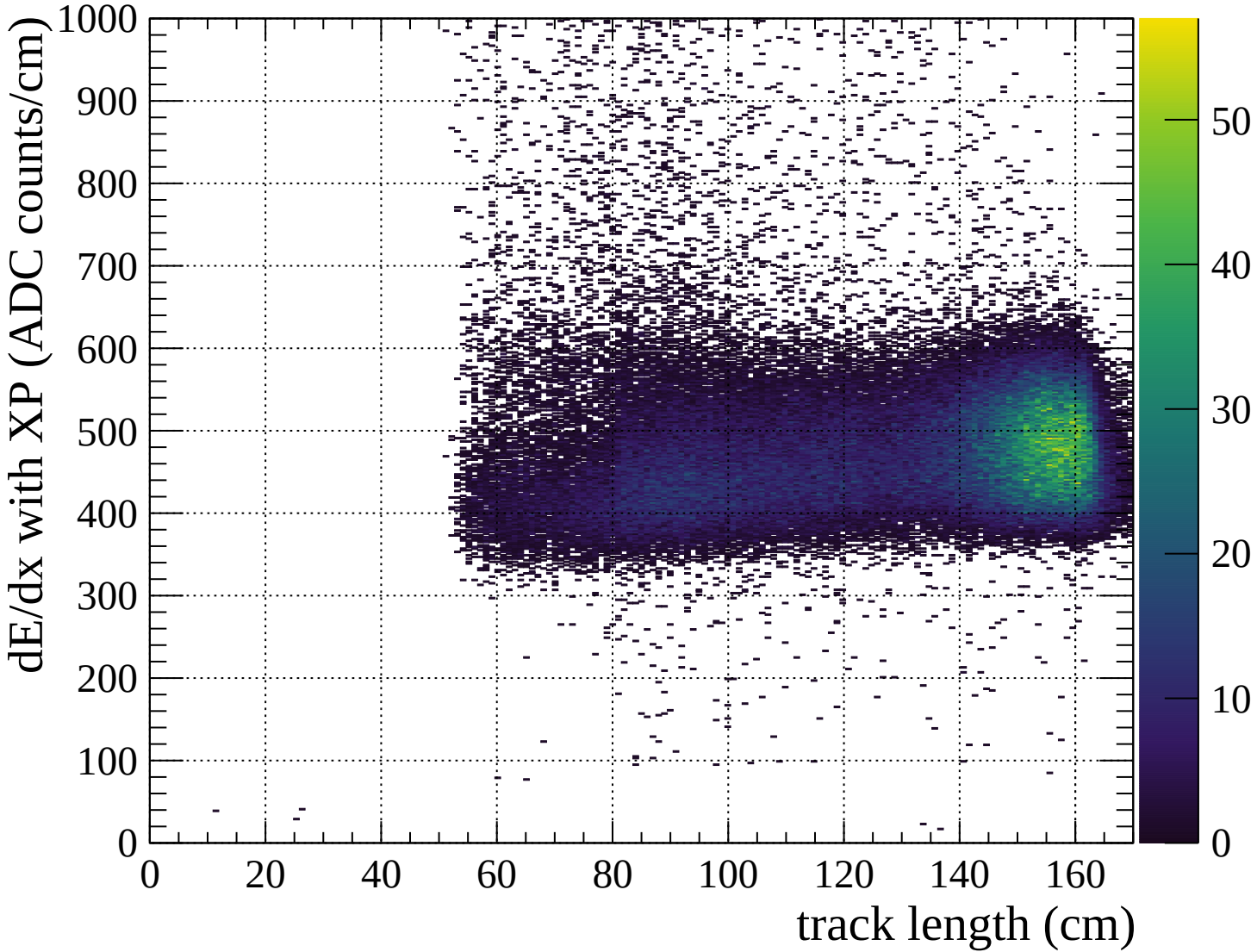


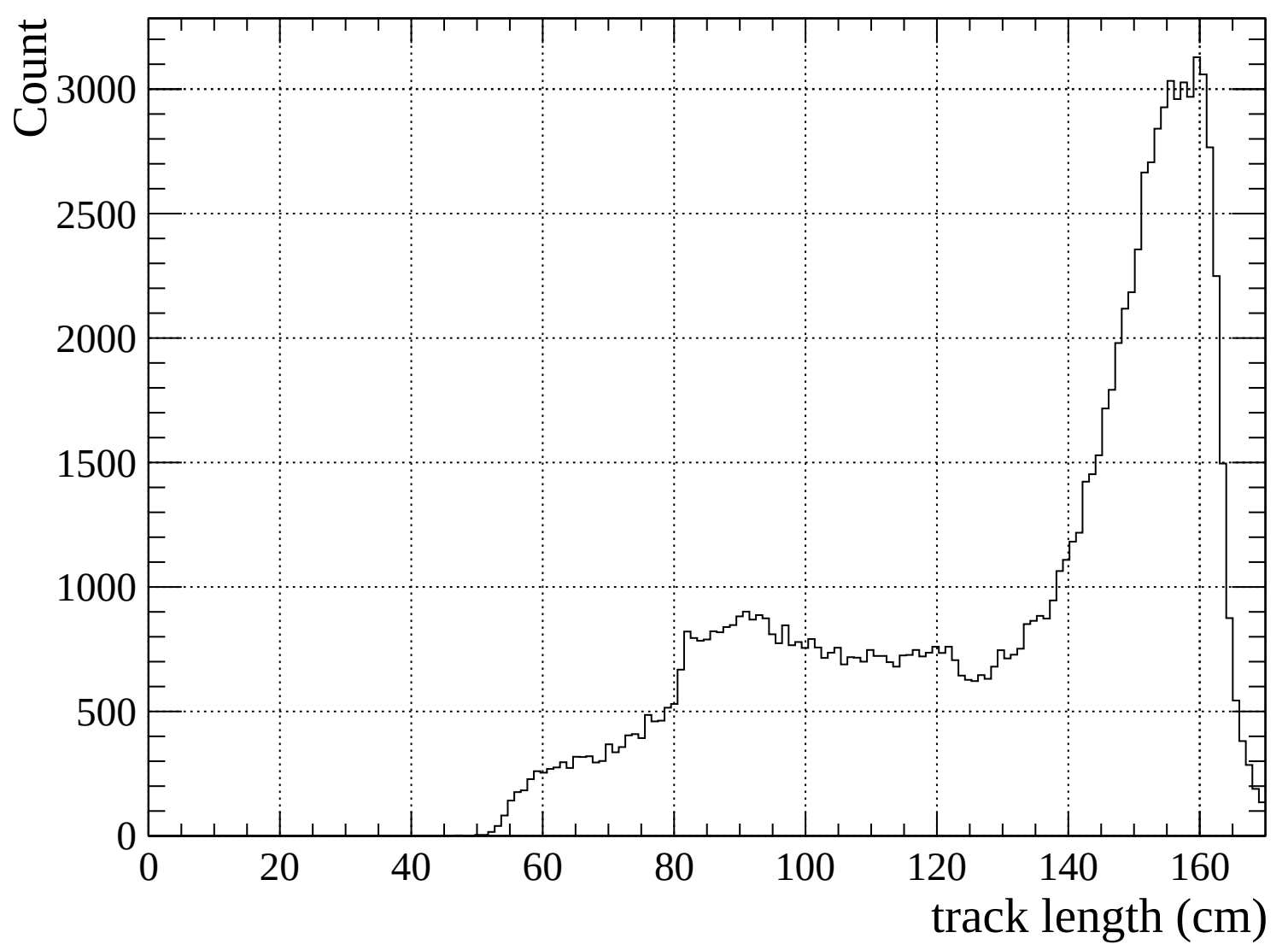


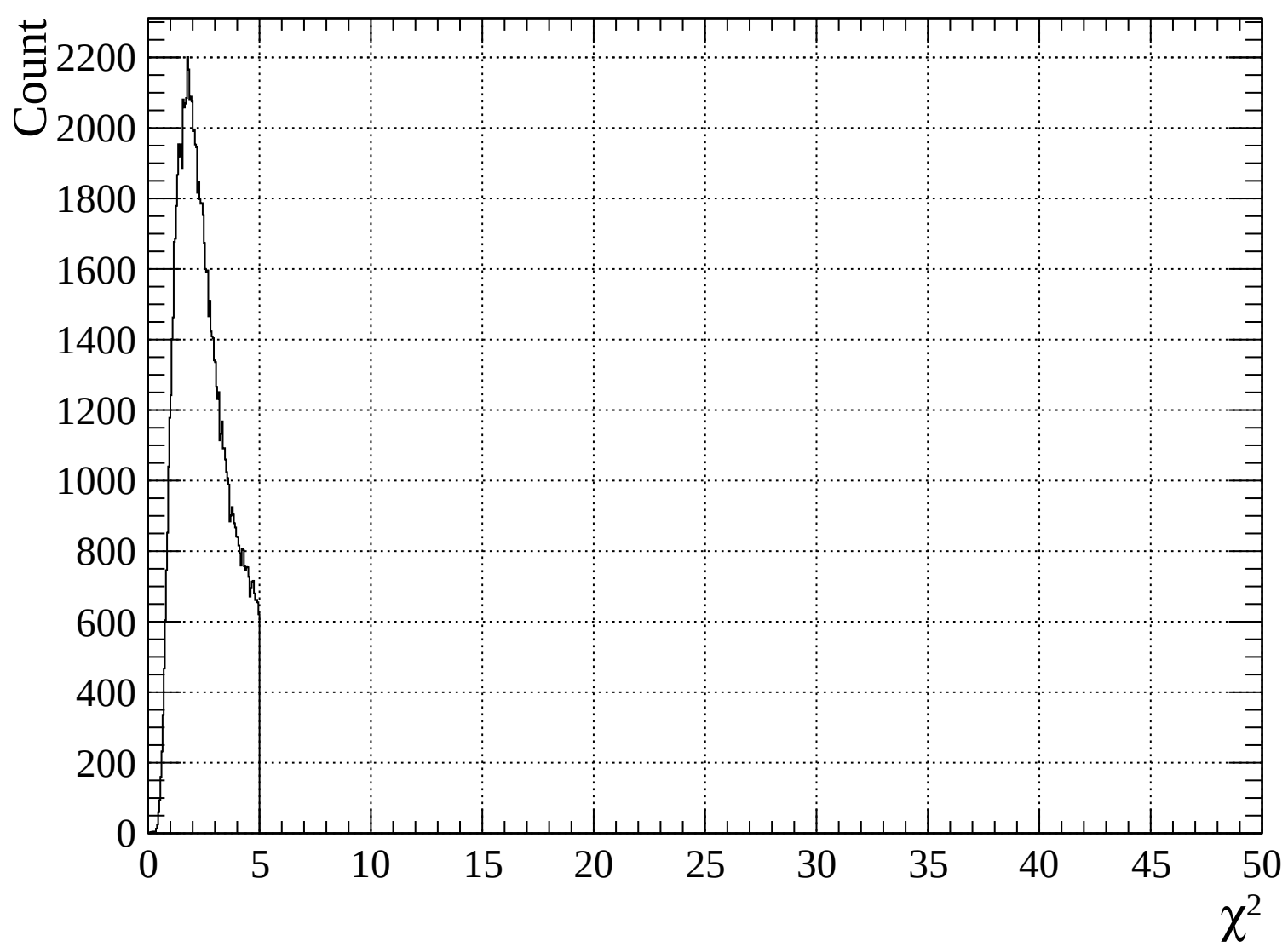


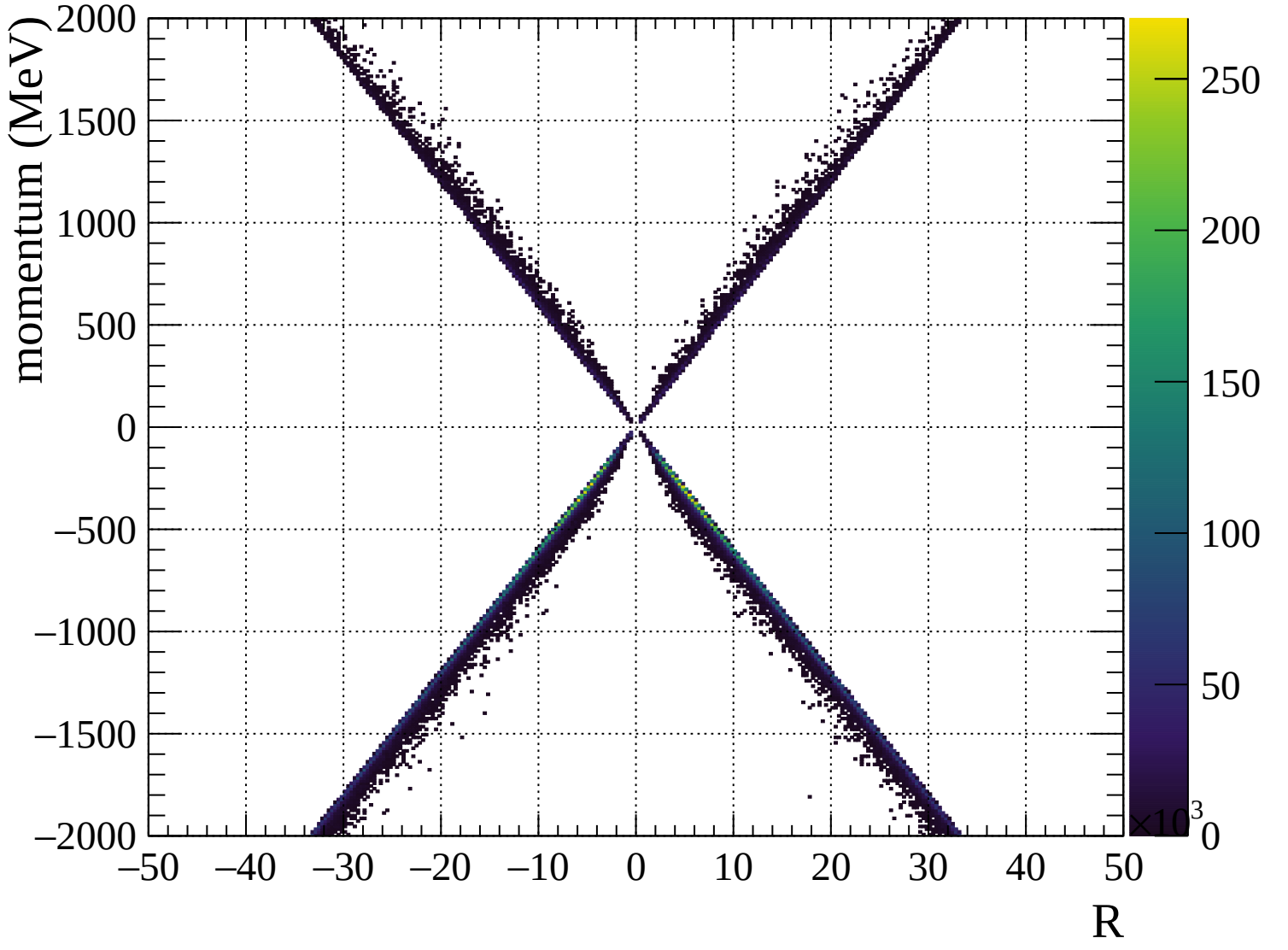


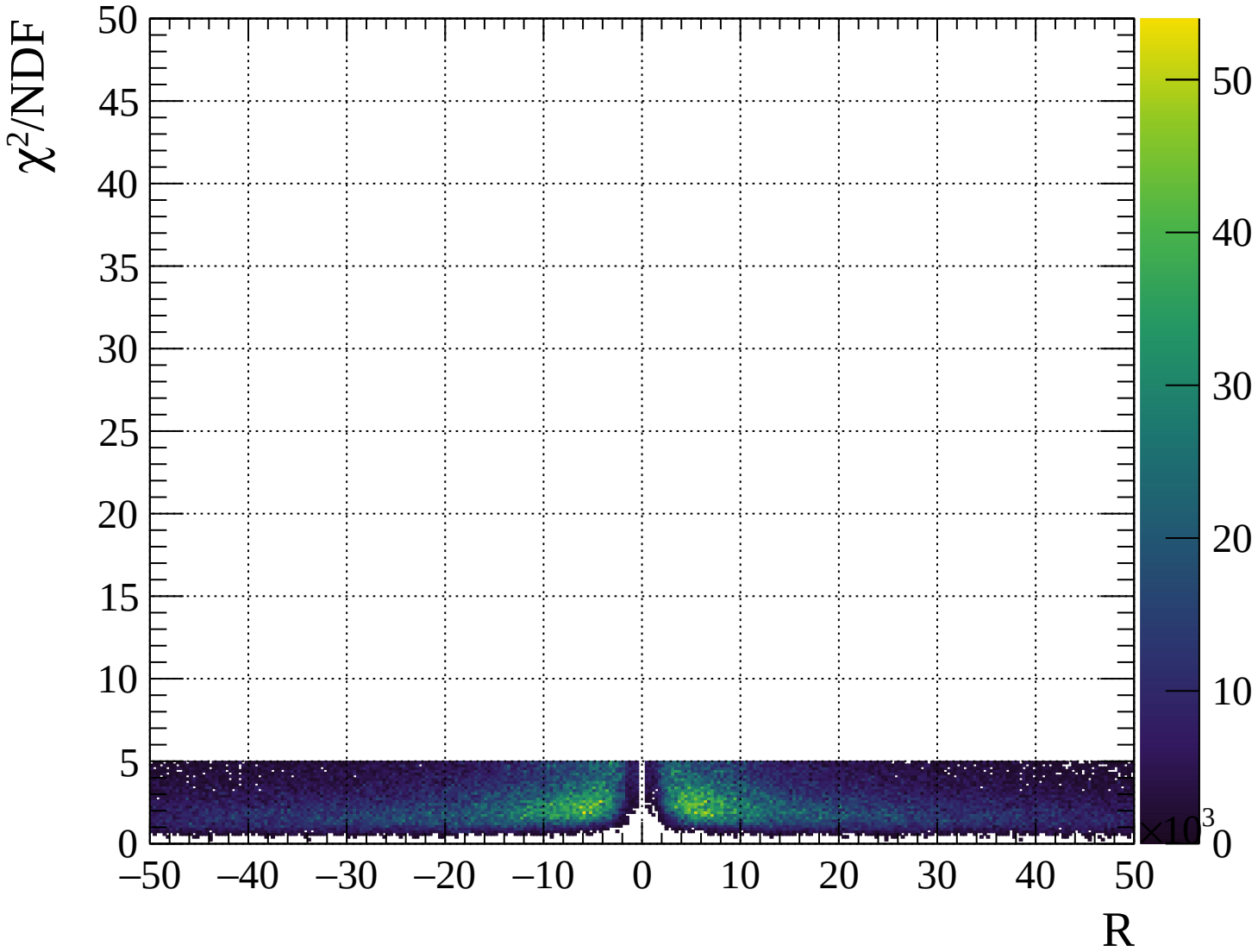






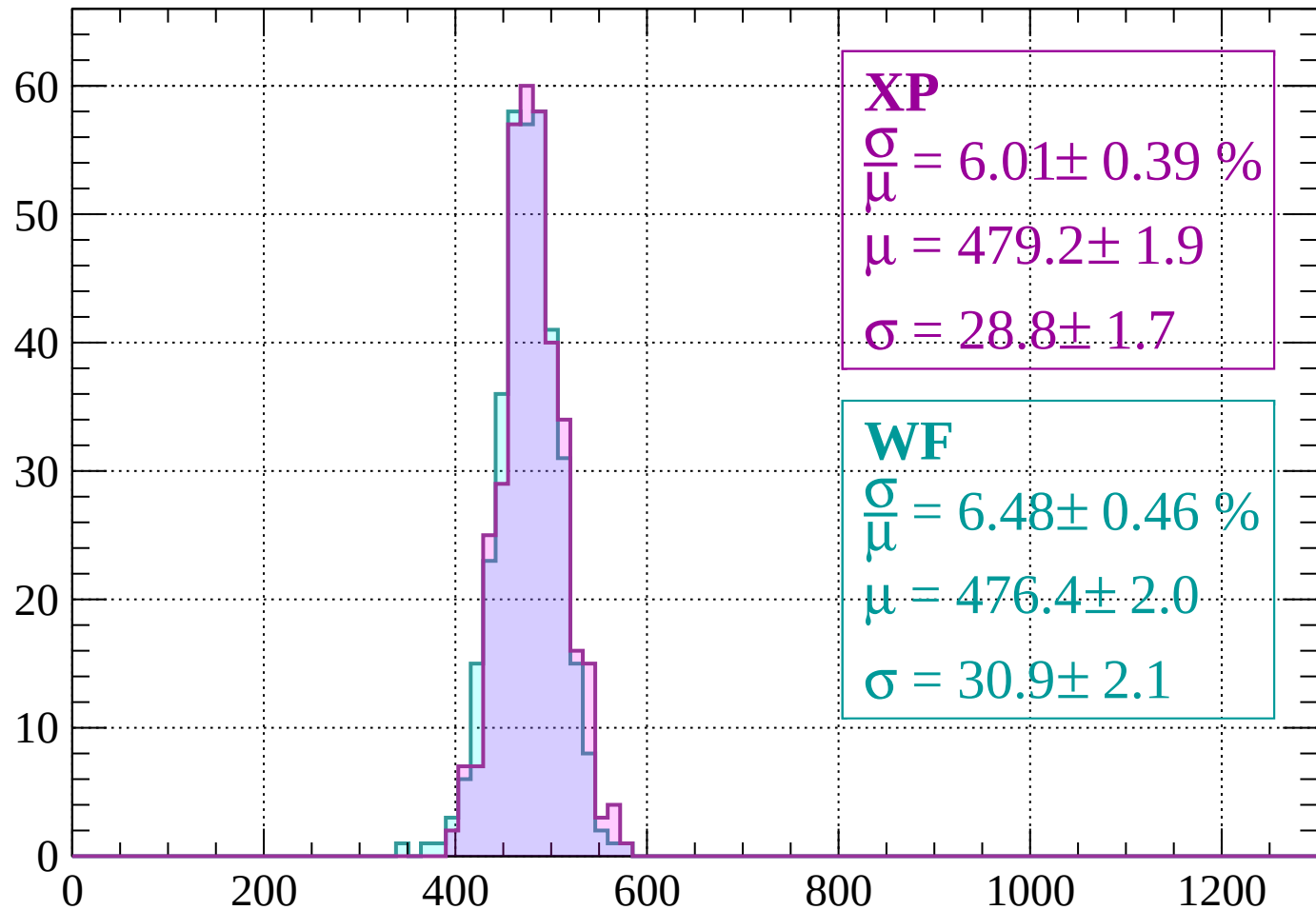






Energy loss | $-2000 < p < -1960$

Count



XP

$$\frac{\sigma}{\mu} = 6.01 \pm 0.39 \%$$

$$\mu = 479.2 \pm 1.9$$

$$\sigma = 28.8 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 6.48 \pm 0.46 \%$$

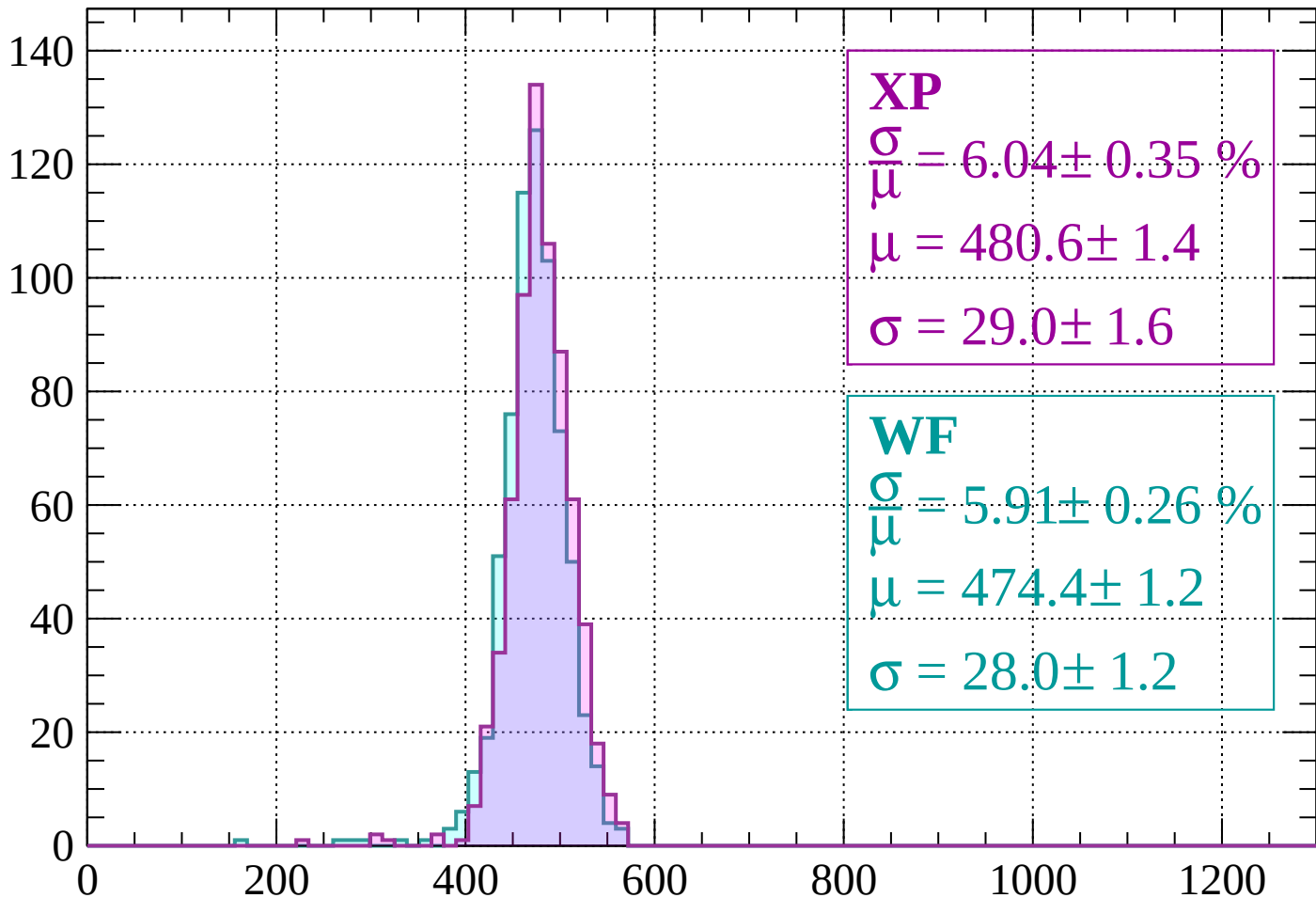
$$\mu = 476.4 \pm 2.0$$

$$\sigma = 30.9 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1960 < p < -1920$

Count



XP

$$\frac{\sigma}{\mu} = 6.04 \pm 0.35 \%$$

$$\mu = 480.6 \pm 1.4$$

$$\sigma = 29.0 \pm 1.6$$

WF

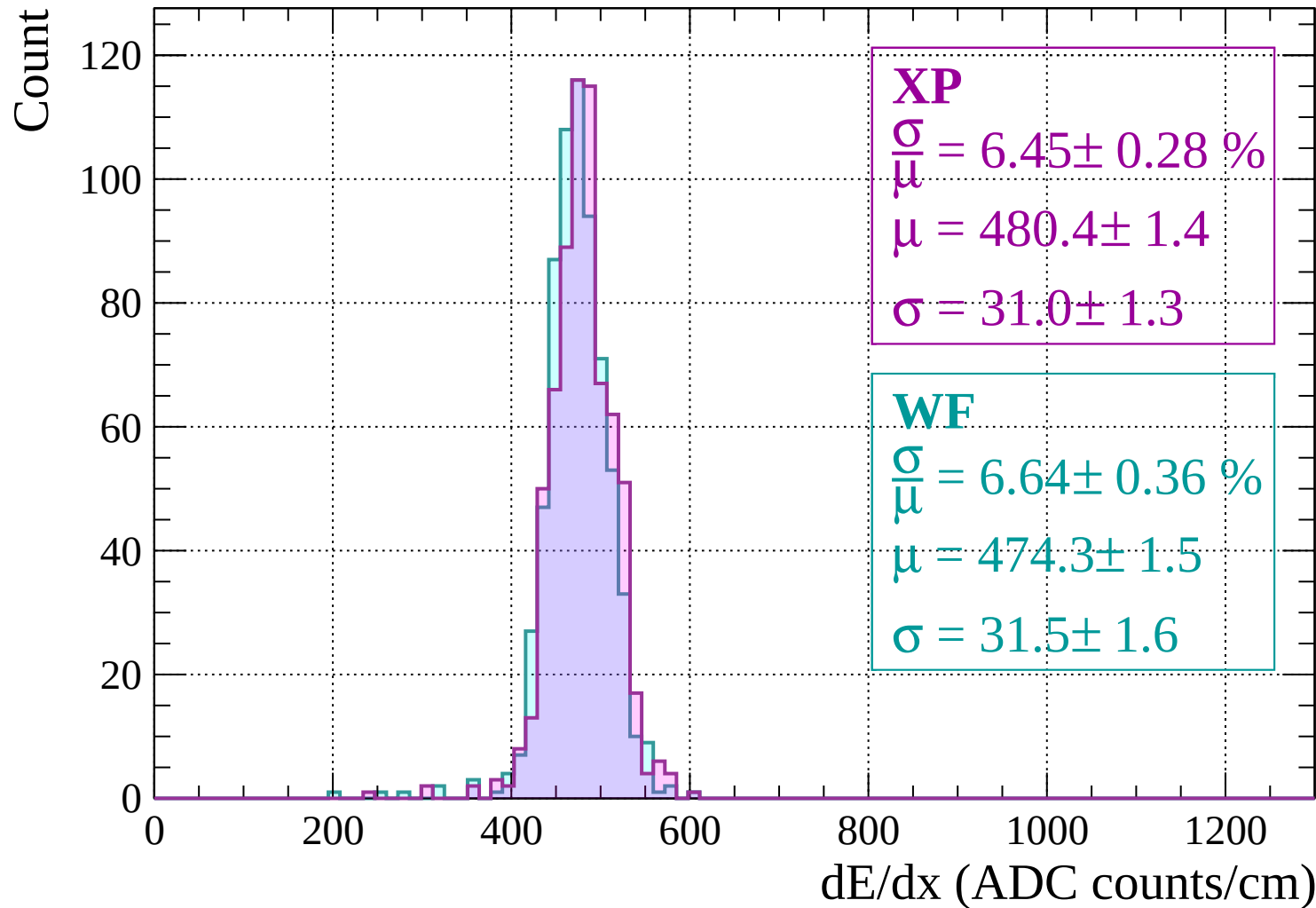
$$\frac{\sigma}{\mu} = 5.91 \pm 0.26 \%$$

$$\mu = 474.4 \pm 1.2$$

$$\sigma = 28.0 \pm 1.2$$

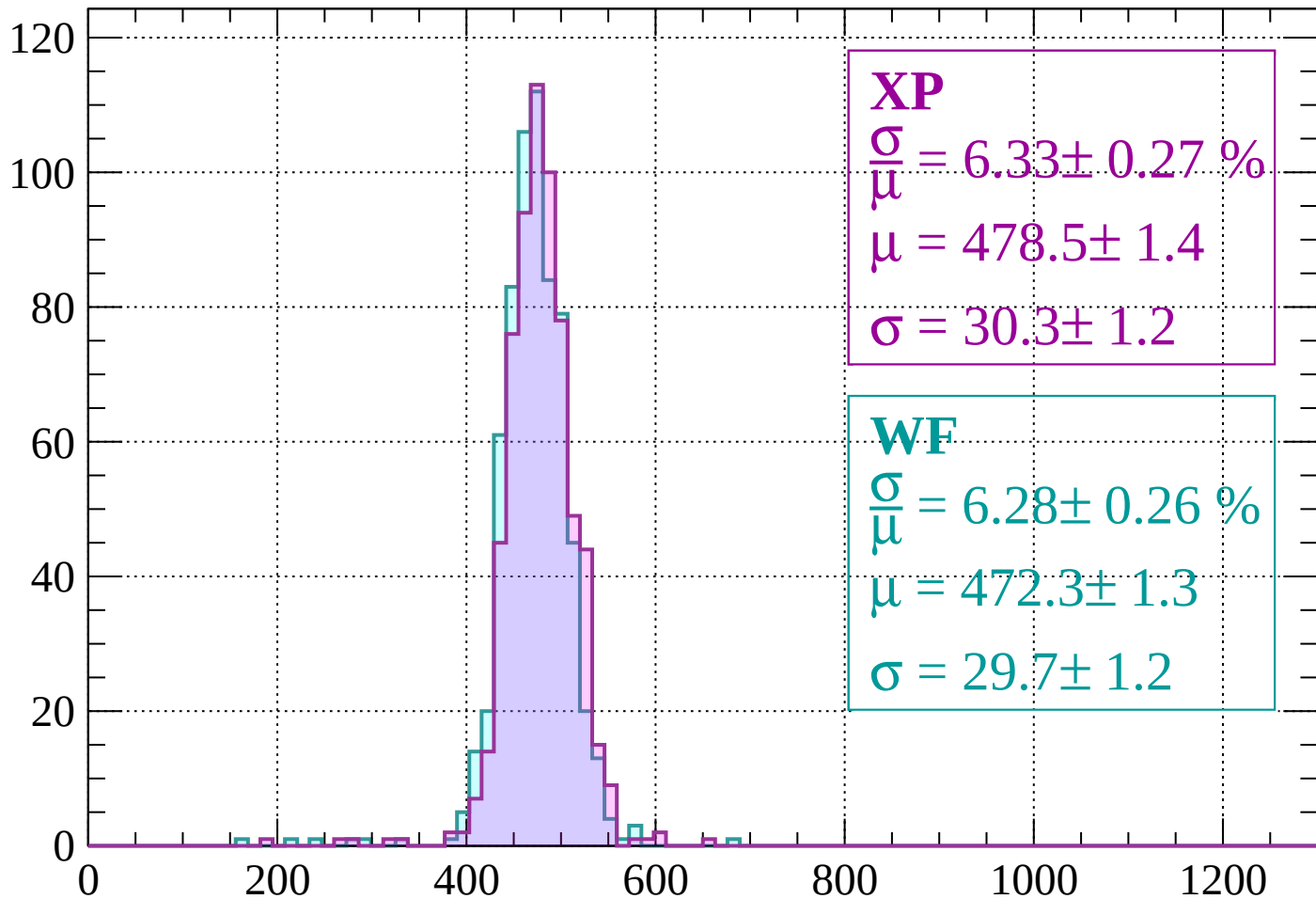
dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1880



Energy loss | $-1880 < p < -1840$

Count



XP

$$\frac{\sigma}{\mu} = 6.33 \pm 0.27 \%$$

$$\mu = 478.5 \pm 1.4$$

$$\sigma = 30.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.28 \pm 0.26 \%$$

$$\mu = 472.3 \pm 1.3$$

$$\sigma = 29.7 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1840 < p < -1800$

Count

120

100

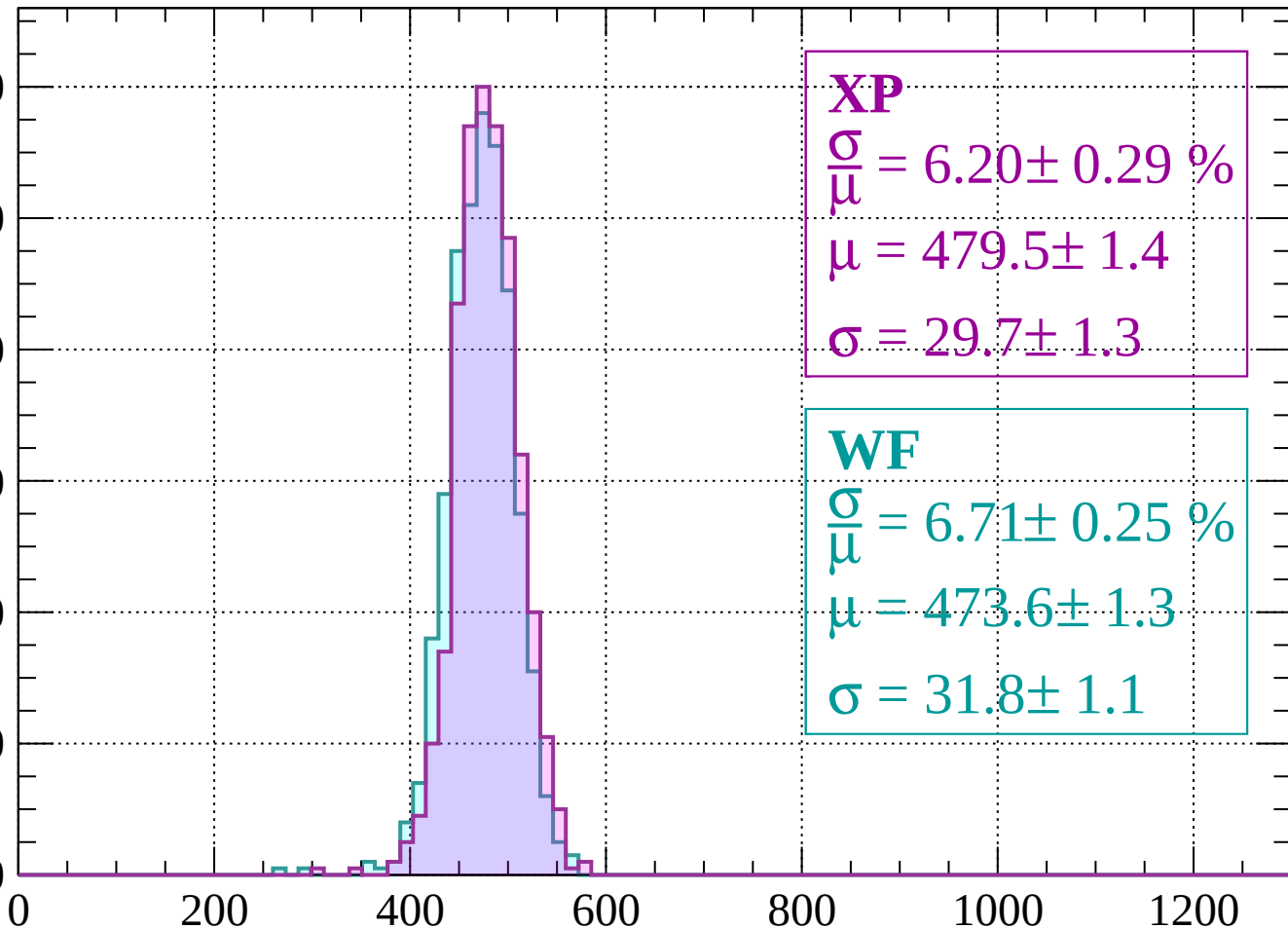
80

60

40

20

0



XP

$$\frac{\sigma}{\mu} = 6.20 \pm 0.29 \%$$

$$\mu = 479.5 \pm 1.4$$

$$\sigma = 29.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.71 \pm 0.25 \%$$

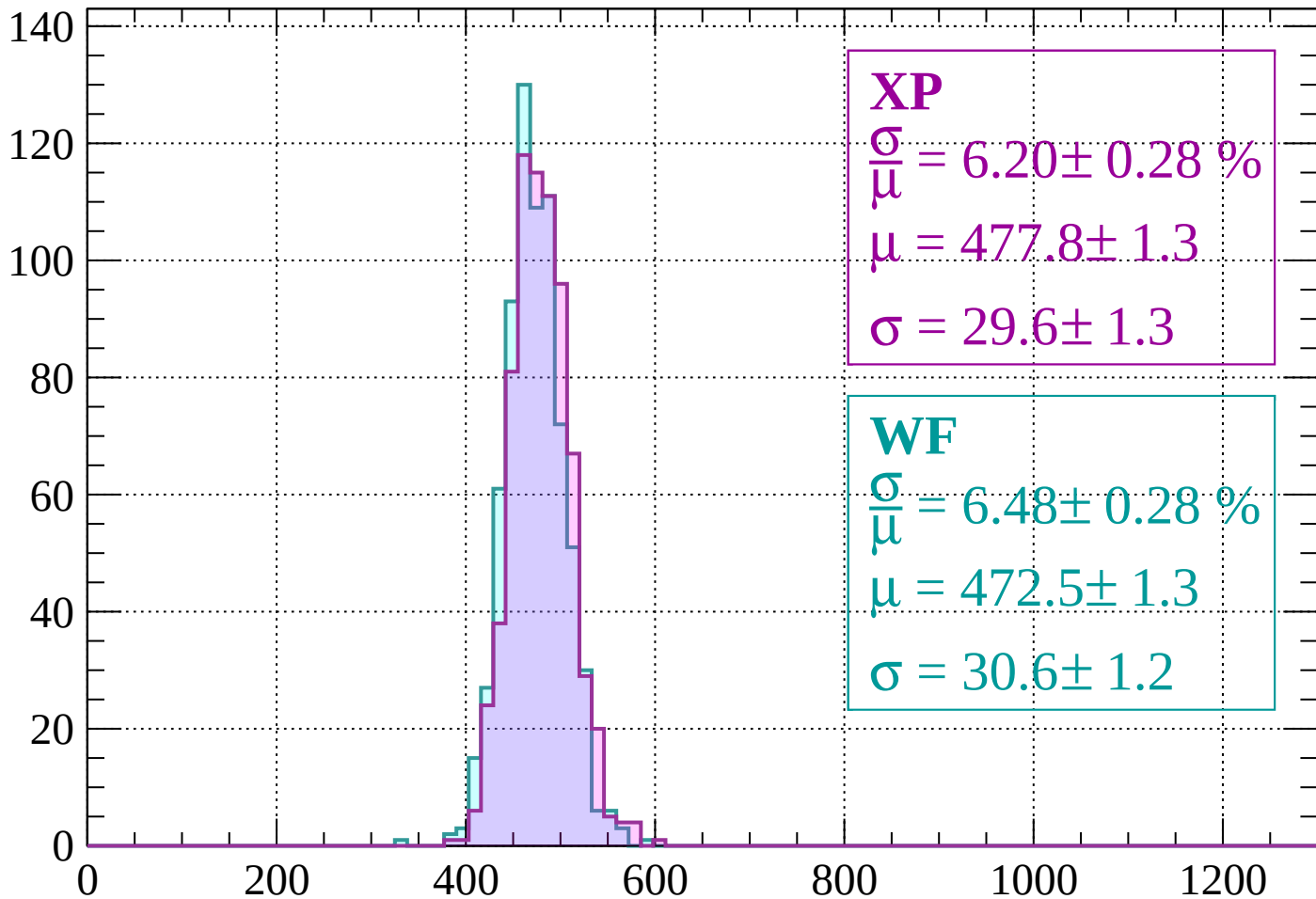
$$\mu = 473.6 \pm 1.3$$

$$\sigma = 31.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1760$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 6.20 \pm 0.28 \%$$

$$\mu = 477.8 \pm 1.3$$

$$\sigma = 29.6 \pm 1.3$$

WF

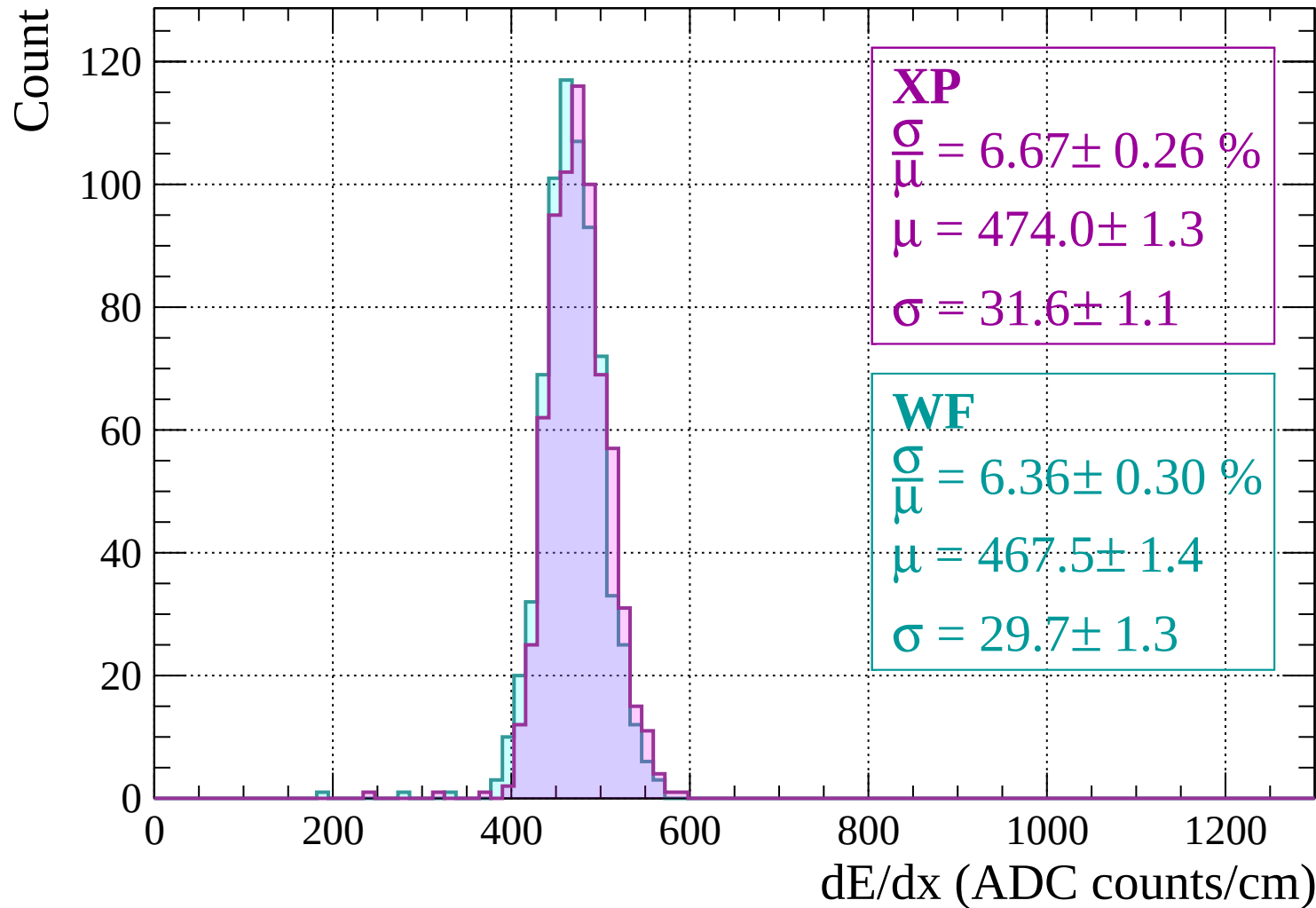
$$\frac{\sigma}{\bar{\mu}} = 6.48 \pm 0.28 \%$$

$$\mu = 472.5 \pm 1.3$$

$$\sigma = 30.6 \pm 1.2$$

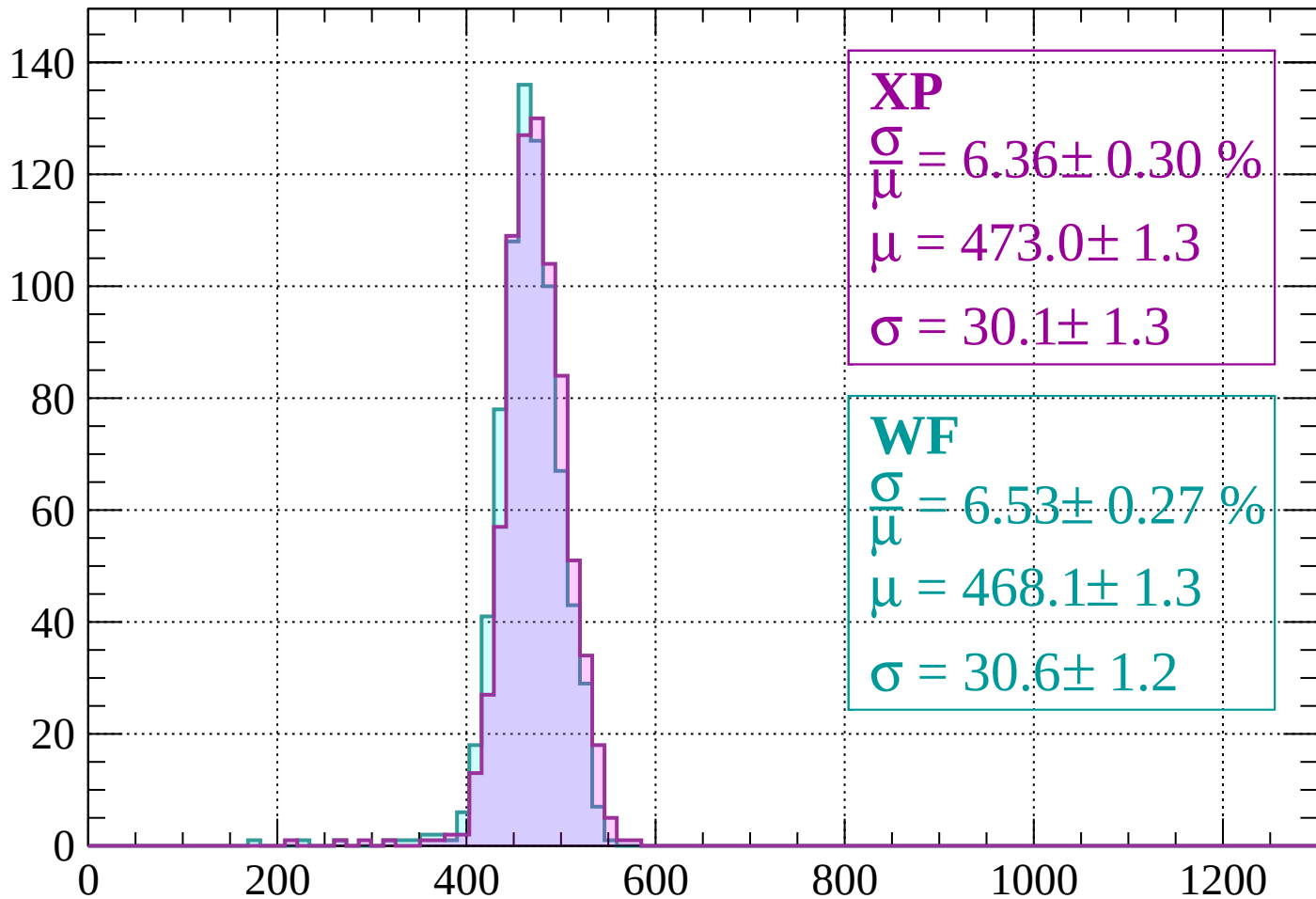
dE/dx (ADC counts/cm)

Energy loss | $-1760 < p < -1720$



Energy loss | $-1720 < p < -1680$

Count



XP

$$\frac{\sigma}{\mu} = 6.36 \pm 0.30 \%$$

$$\mu = 473.0 \pm 1.3$$

$$\sigma = 30.1 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.53 \pm 0.27 \%$$

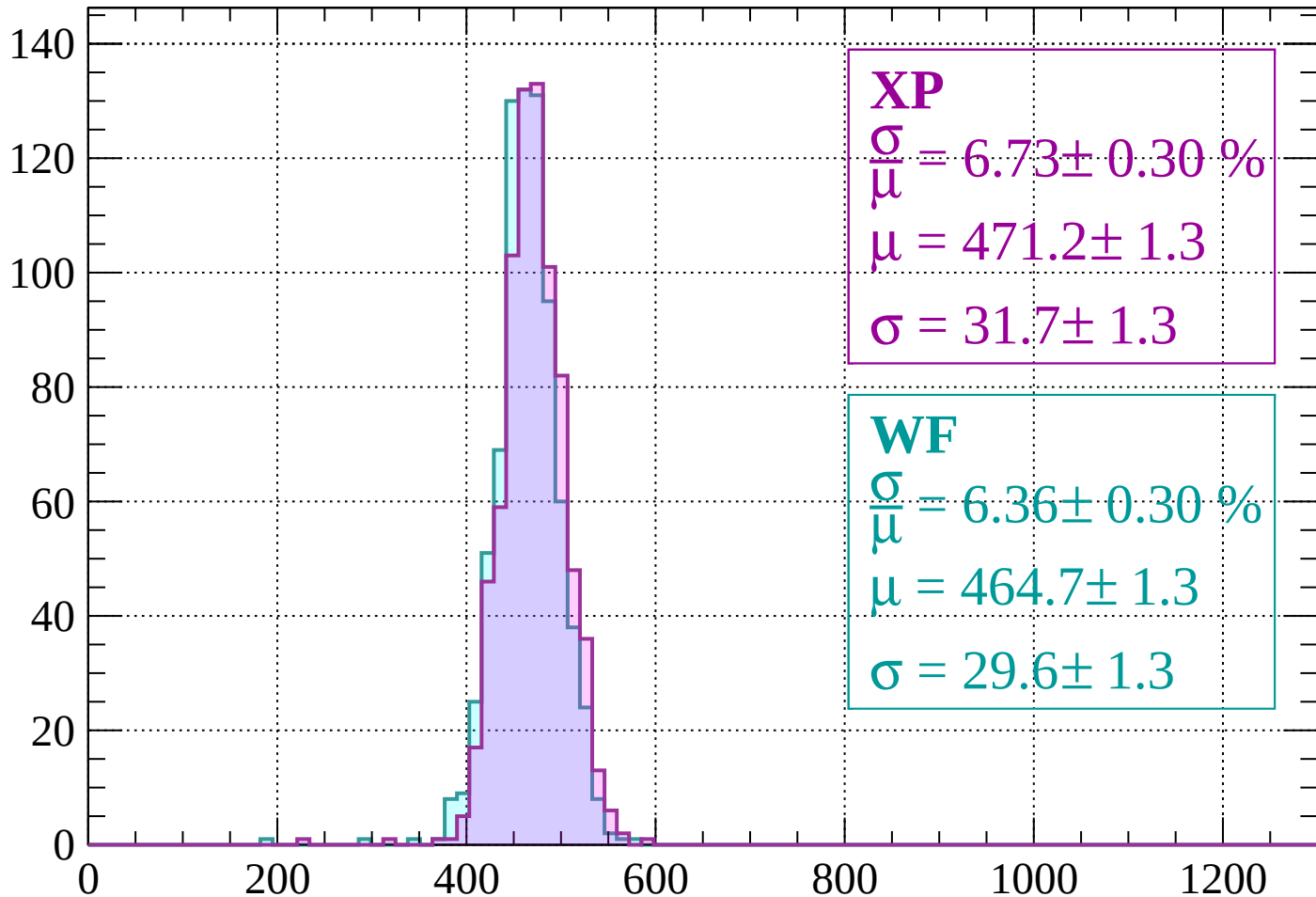
$$\mu = 468.1 \pm 1.3$$

$$\sigma = 30.6 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1640$

Count



XP

$$\frac{\sigma}{\mu} = 6.73 \pm 0.30 \%$$

$$\mu = 471.2 \pm 1.3$$

$$\sigma = 31.7 \pm 1.3$$

WF

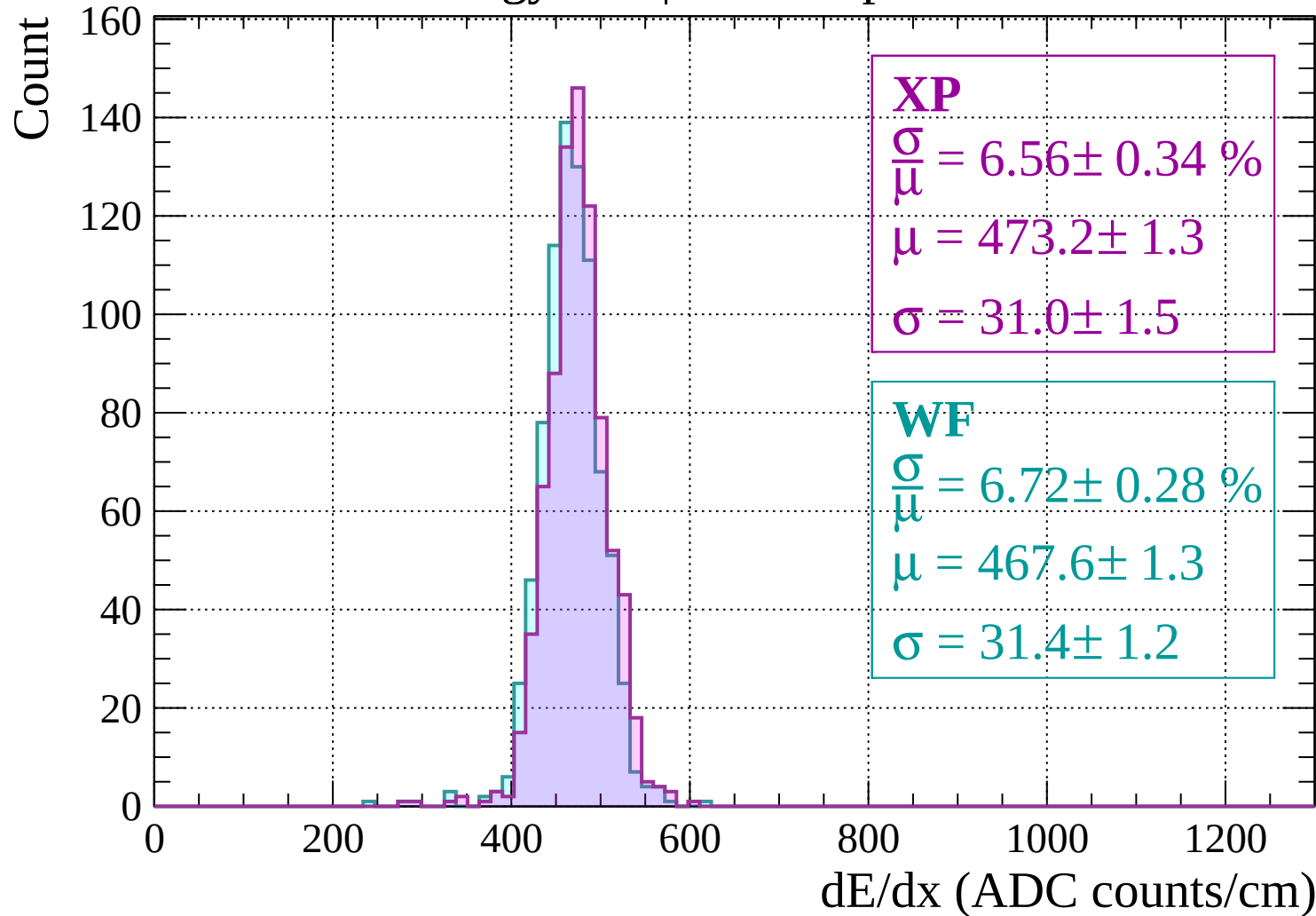
$$\frac{\sigma}{\mu} = 6.36 \pm 0.30 \%$$

$$\mu = 464.7 \pm 1.3$$

$$\sigma = 29.6 \pm 1.3$$

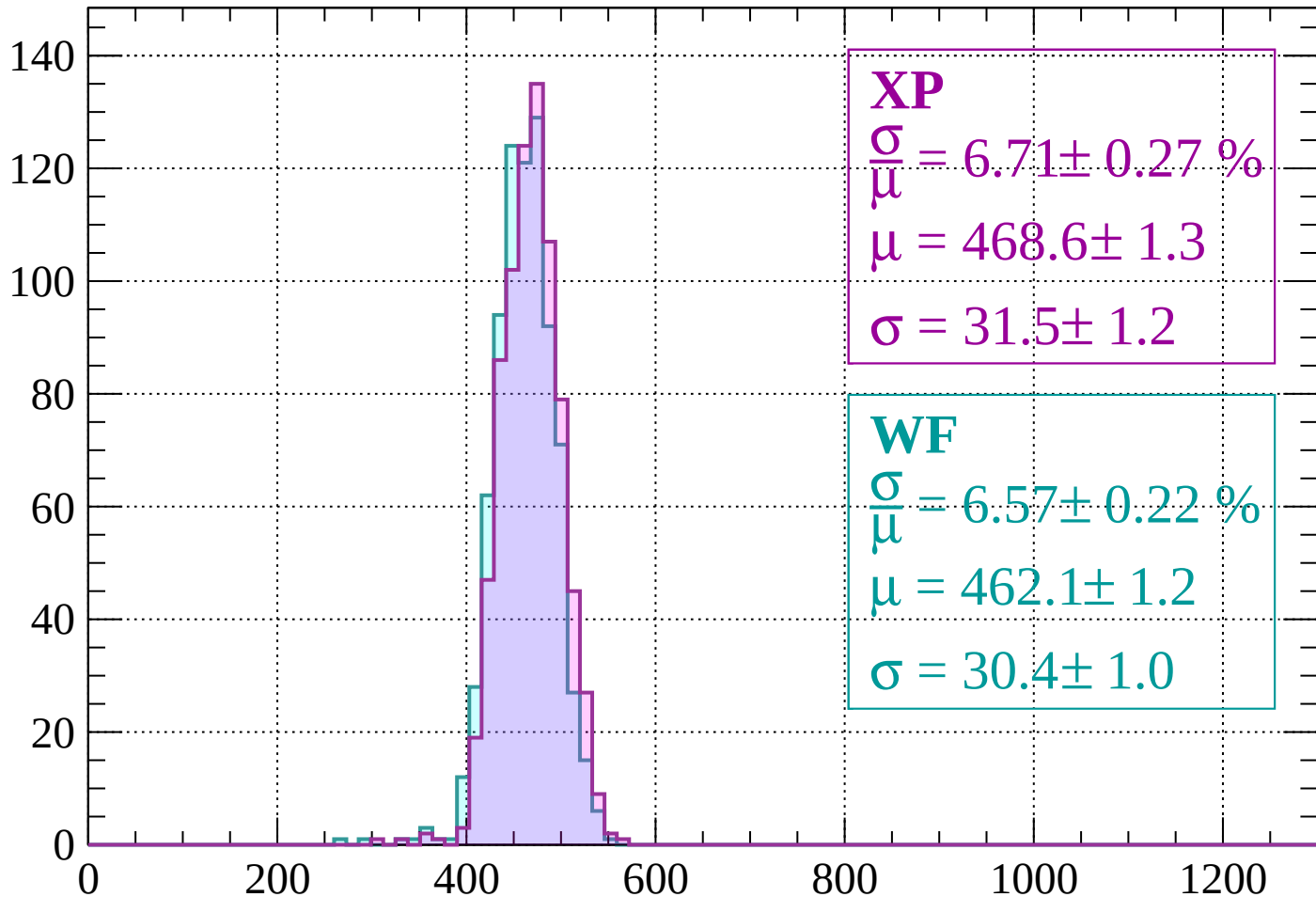
dE/dx (ADC counts/cm)

Energy loss | $-1640 < p < -1600$



Energy loss | $-1600 < p < -1560$

Count



XP

$$\frac{\sigma}{\mu} = 6.71 \pm 0.27 \%$$

$$\mu = 468.6 \pm 1.3$$

$$\sigma = 31.5 \pm 1.2$$

WF

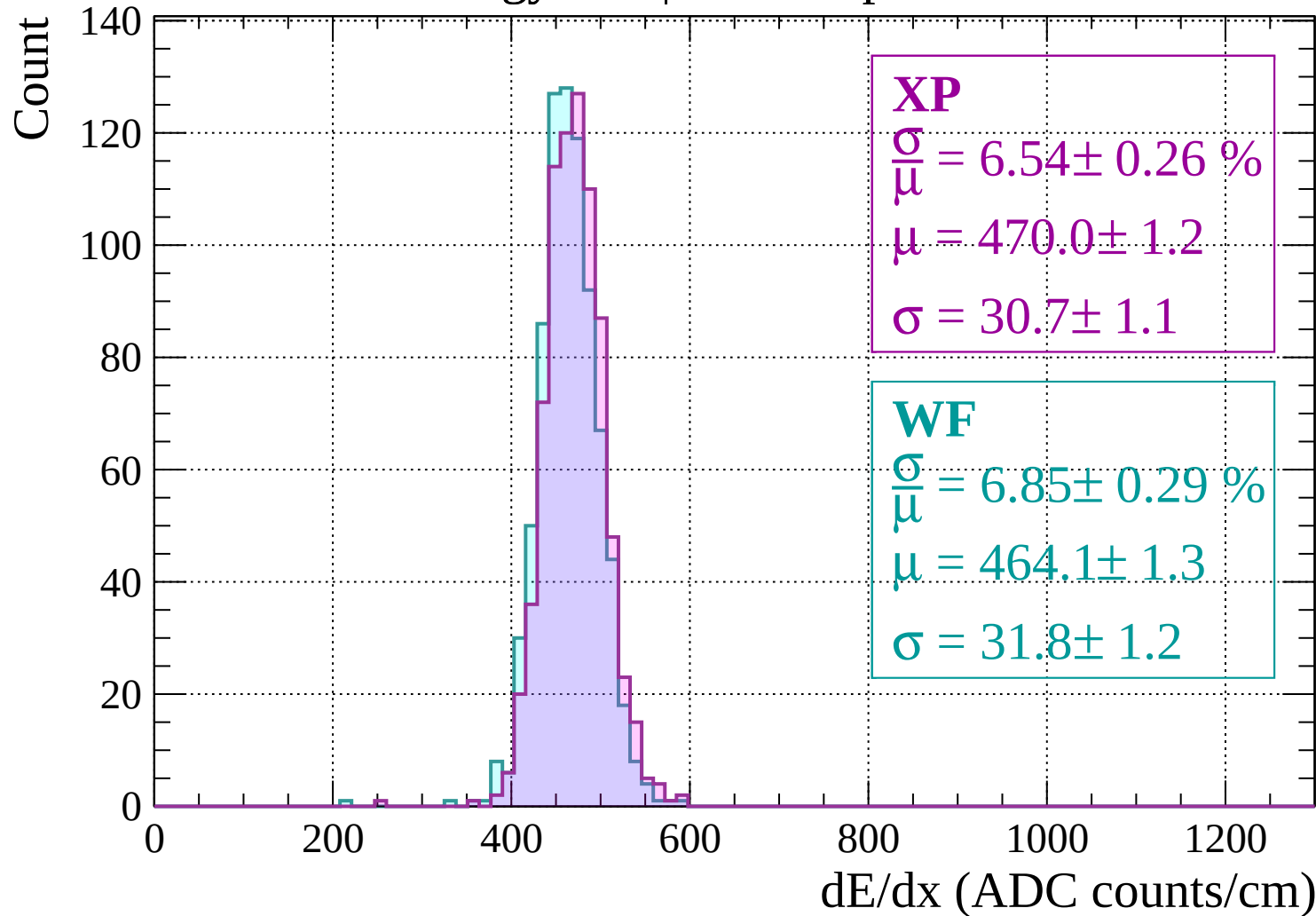
$$\frac{\sigma}{\mu} = 6.57 \pm 0.22 \%$$

$$\mu = 462.1 \pm 1.2$$

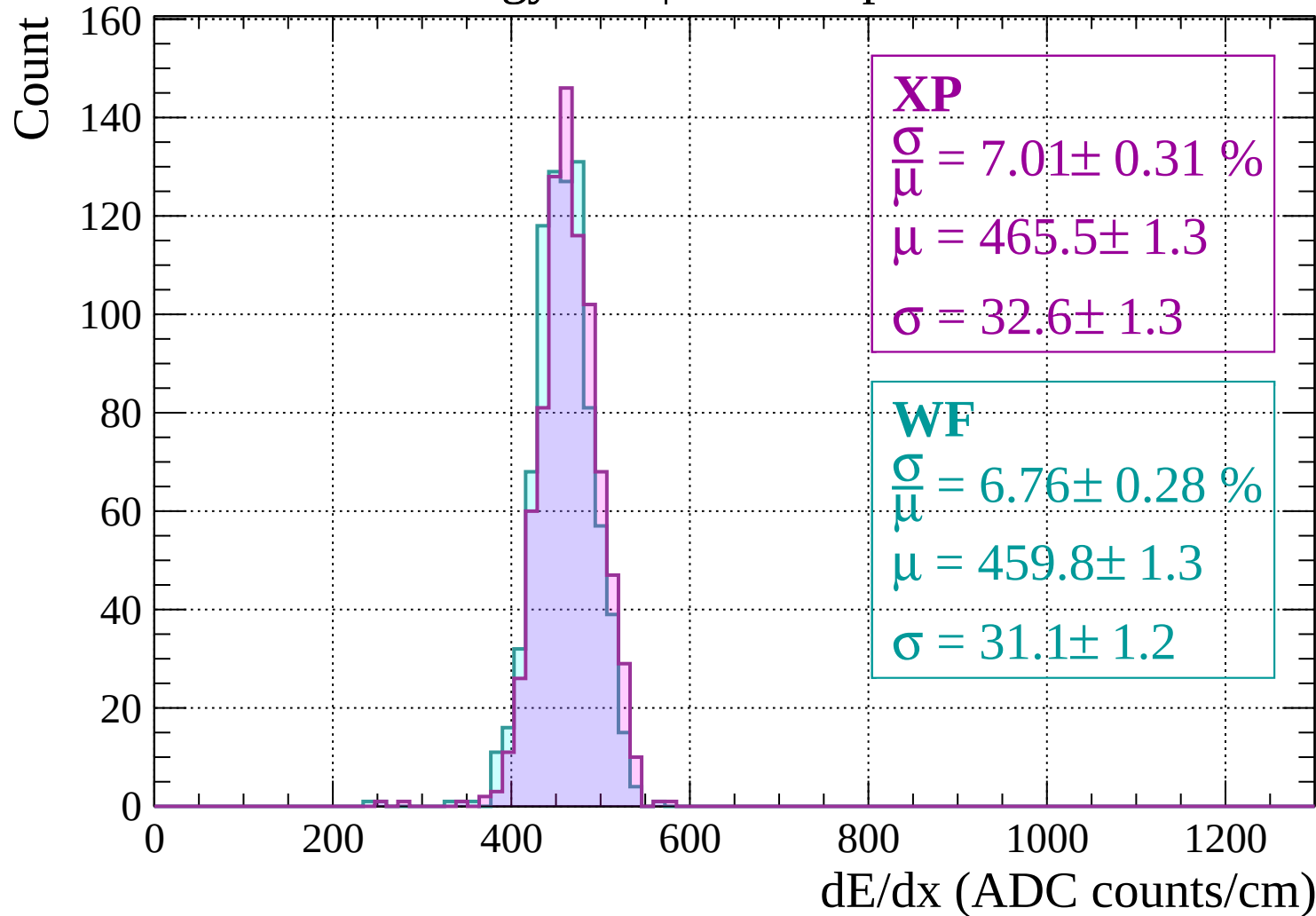
$$\sigma = 30.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1560 < p < -1520$

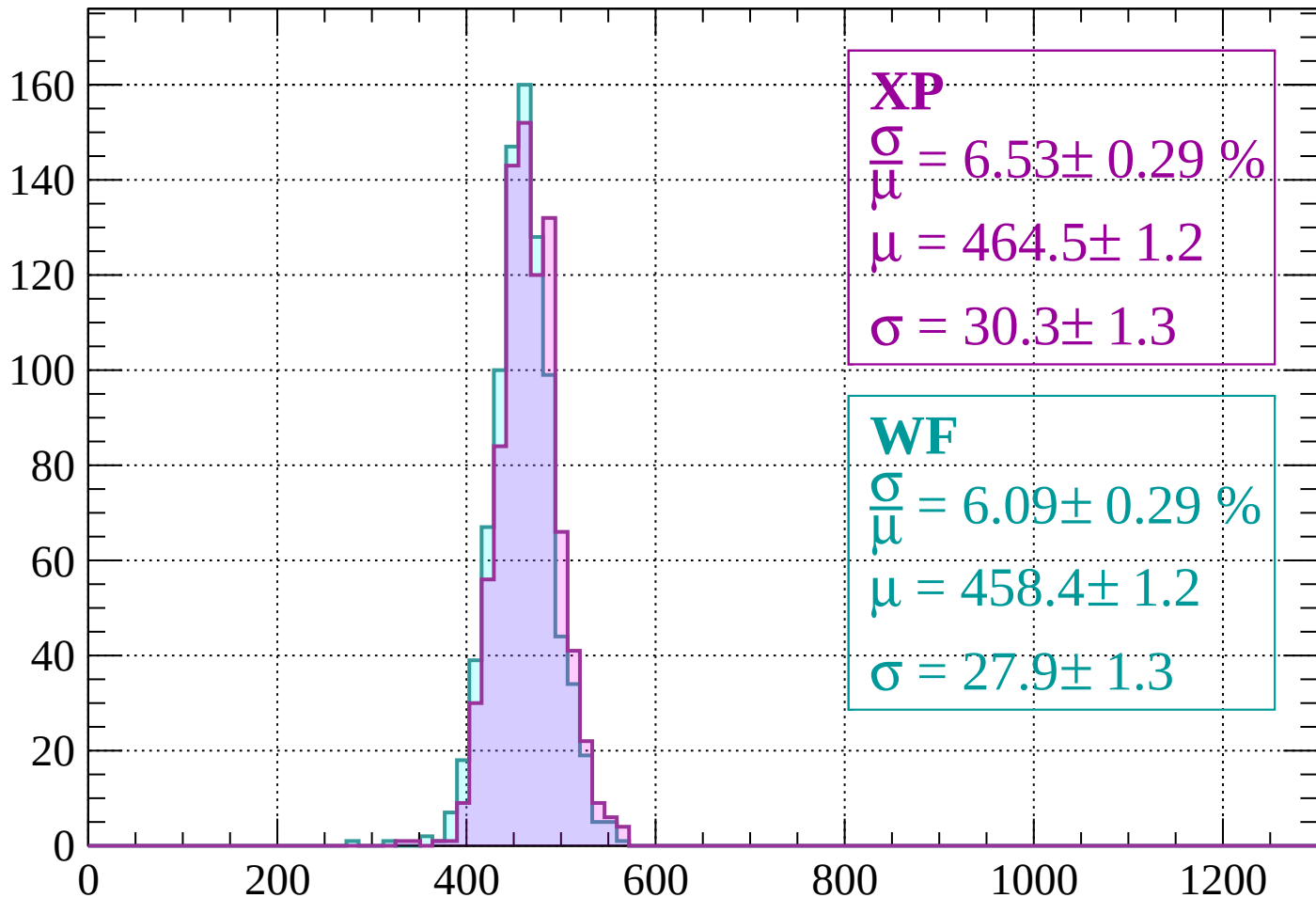


Energy loss | $-1520 < p < -1480$



Energy loss | $-1480 < p < -1440$

Count



XP

$$\frac{\sigma}{\mu} = 6.53 \pm 0.29 \%$$

$$\mu = 464.5 \pm 1.2$$

$$\sigma = 30.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.09 \pm 0.29 \%$$

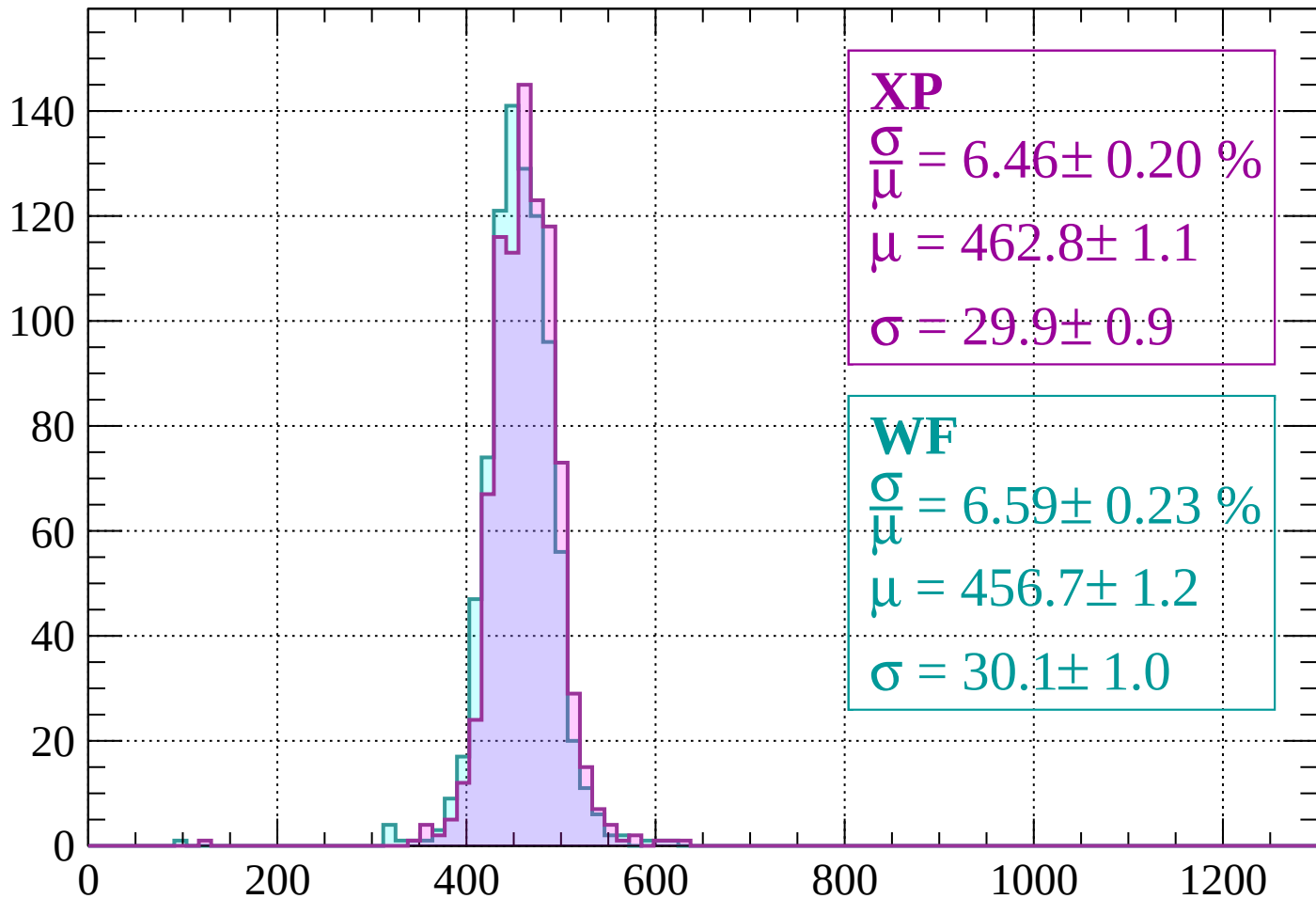
$$\mu = 458.4 \pm 1.2$$

$$\sigma = 27.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1400$

Count



XP

$$\frac{\sigma}{\mu} = 6.46 \pm 0.20 \%$$

$$\mu = 462.8 \pm 1.1$$

$$\sigma = 29.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.59 \pm 0.23 \%$$

$$\mu = 456.7 \pm 1.2$$

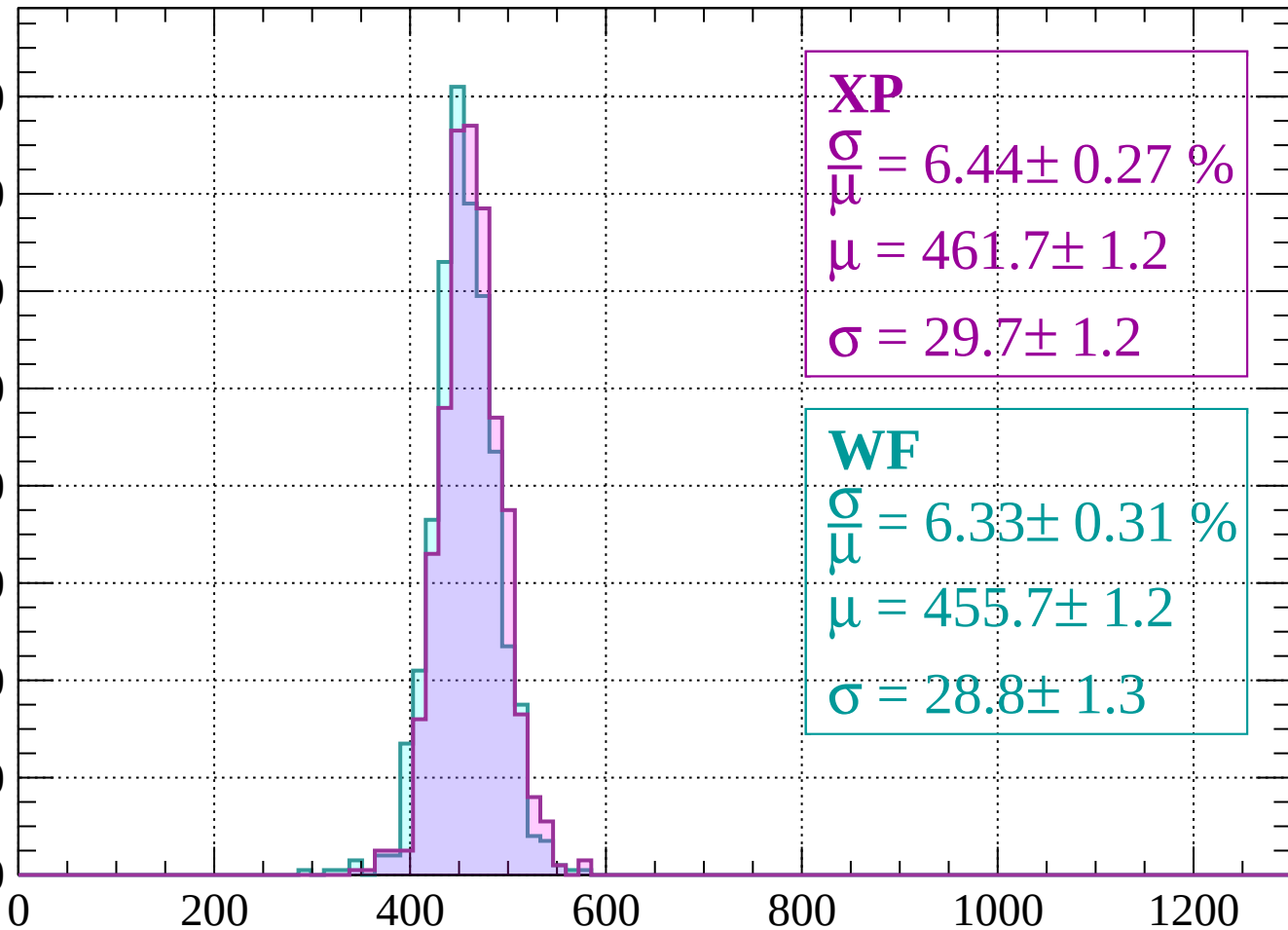
$$\sigma = 30.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1400 < p < -1360$

Count

160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 6.44 \pm 0.27 \%$$

$$\mu = 461.7 \pm 1.2$$

$$\sigma = 29.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.33 \pm 0.31 \%$$

$$\mu = 455.7 \pm 1.2$$

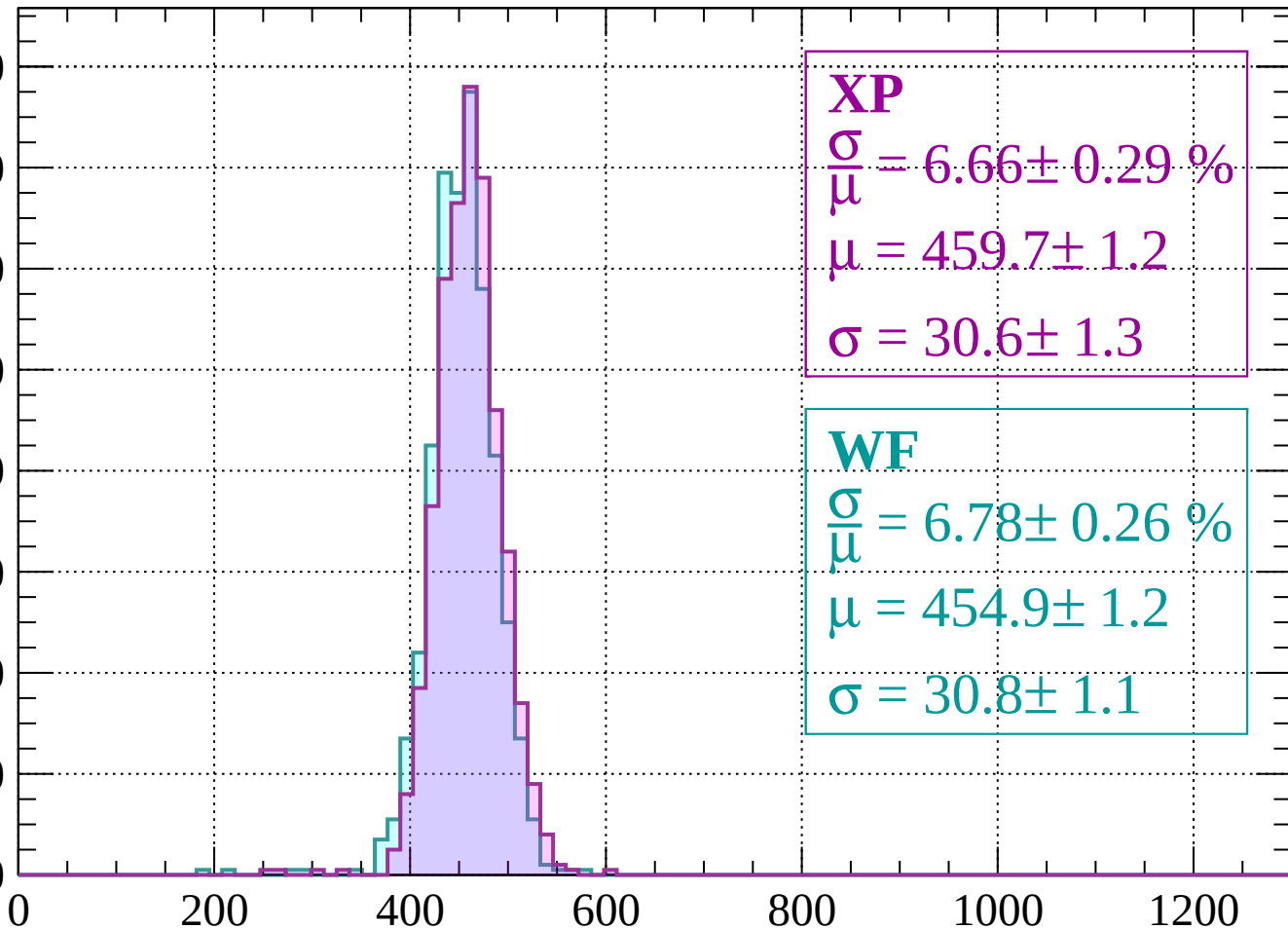
$$\sigma = 28.8 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-1360 < p < -1320$

Count

160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 6.66 \pm 0.29 \%$$

$$\mu = 459.7 \pm 1.2$$

$$\sigma = 30.6 \pm 1.3$$

WF

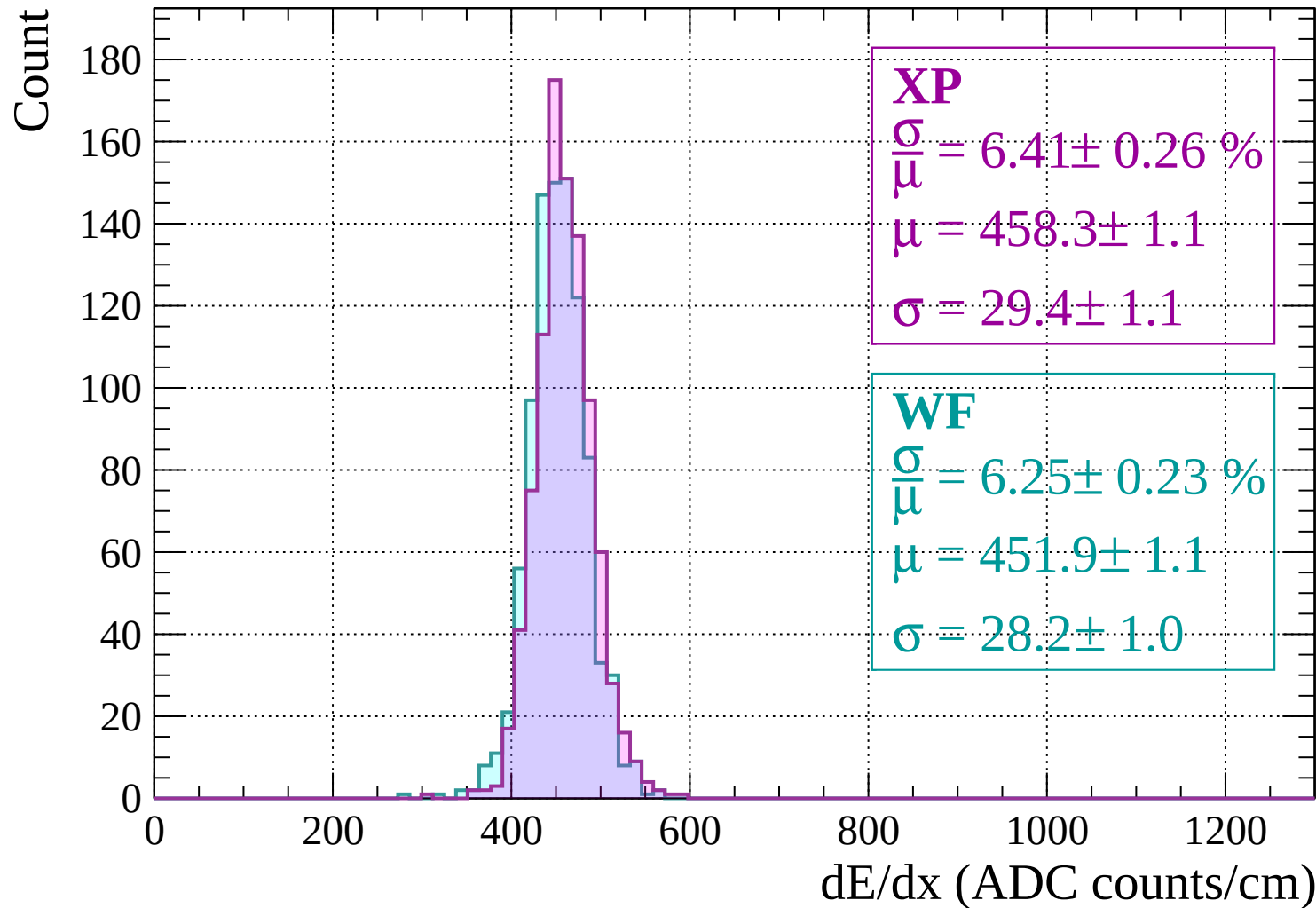
$$\frac{\sigma}{\mu} = 6.78 \pm 0.26 \%$$

$$\mu = 454.9 \pm 1.2$$

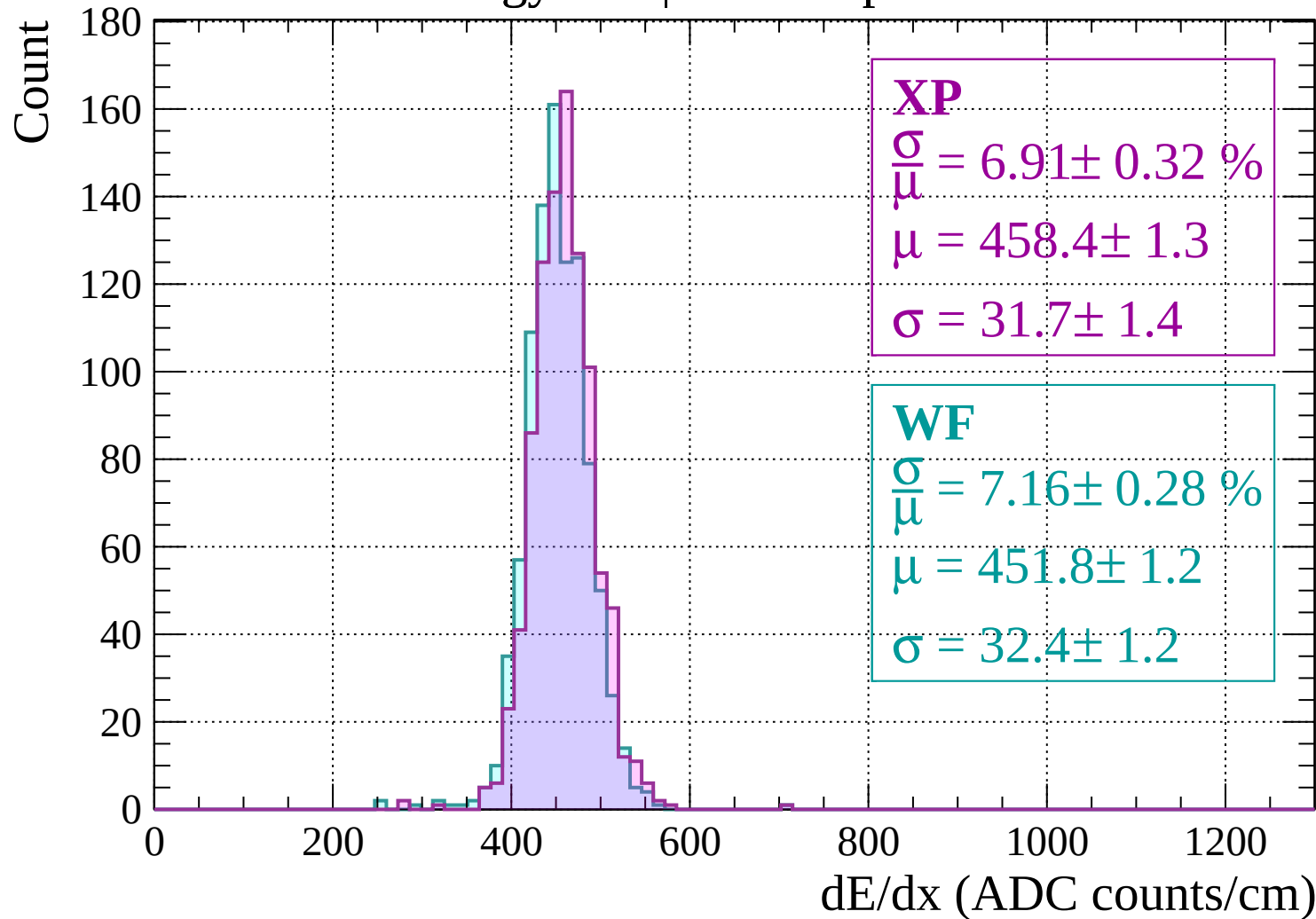
$$\sigma = 30.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1280

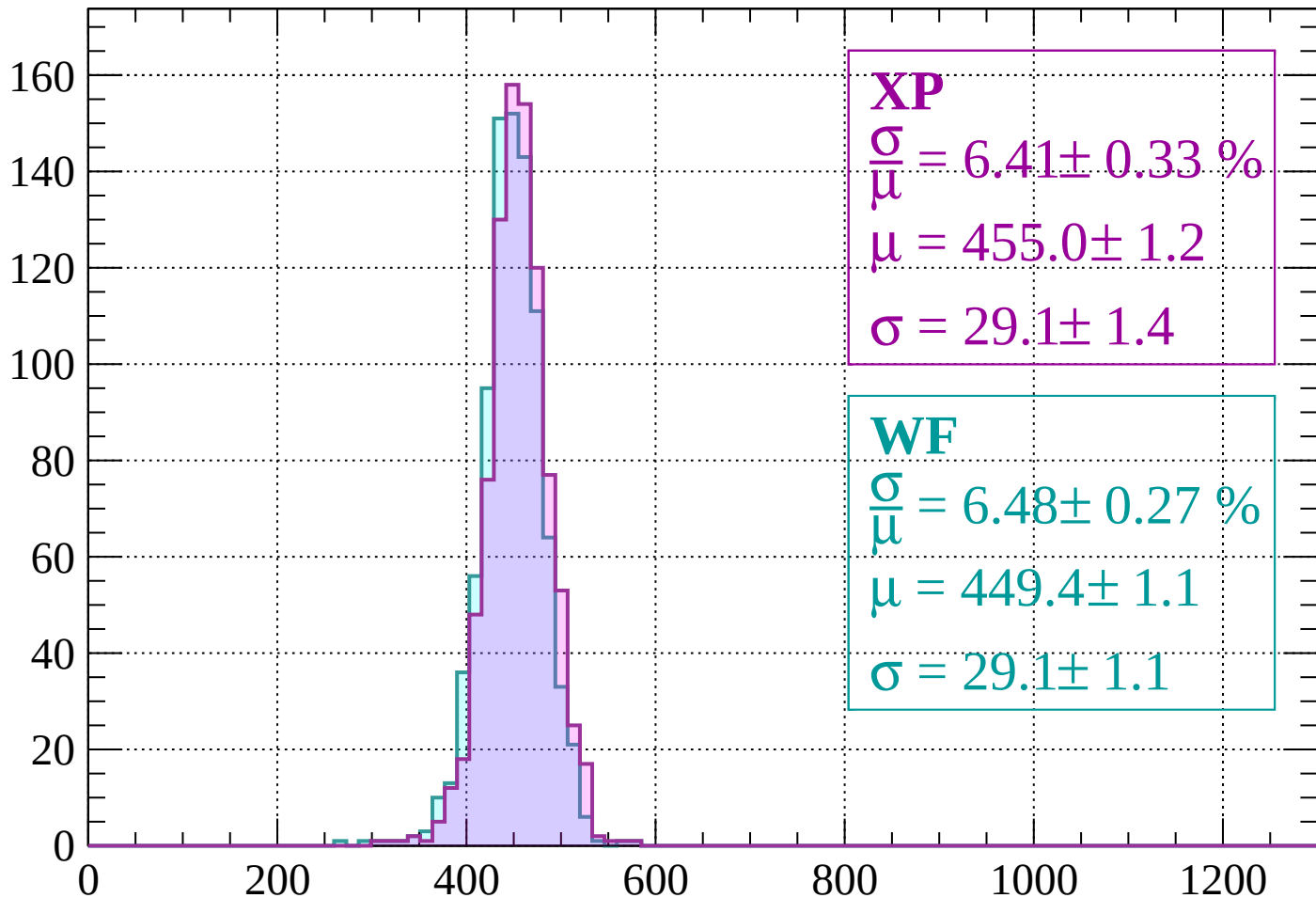


Energy loss | $-1280 < p < -1240$



Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\mu} = 6.41 \pm 0.33 \%$$

$$\mu = 455.0 \pm 1.2$$

$$\sigma = 29.1 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.48 \pm 0.27 \%$$

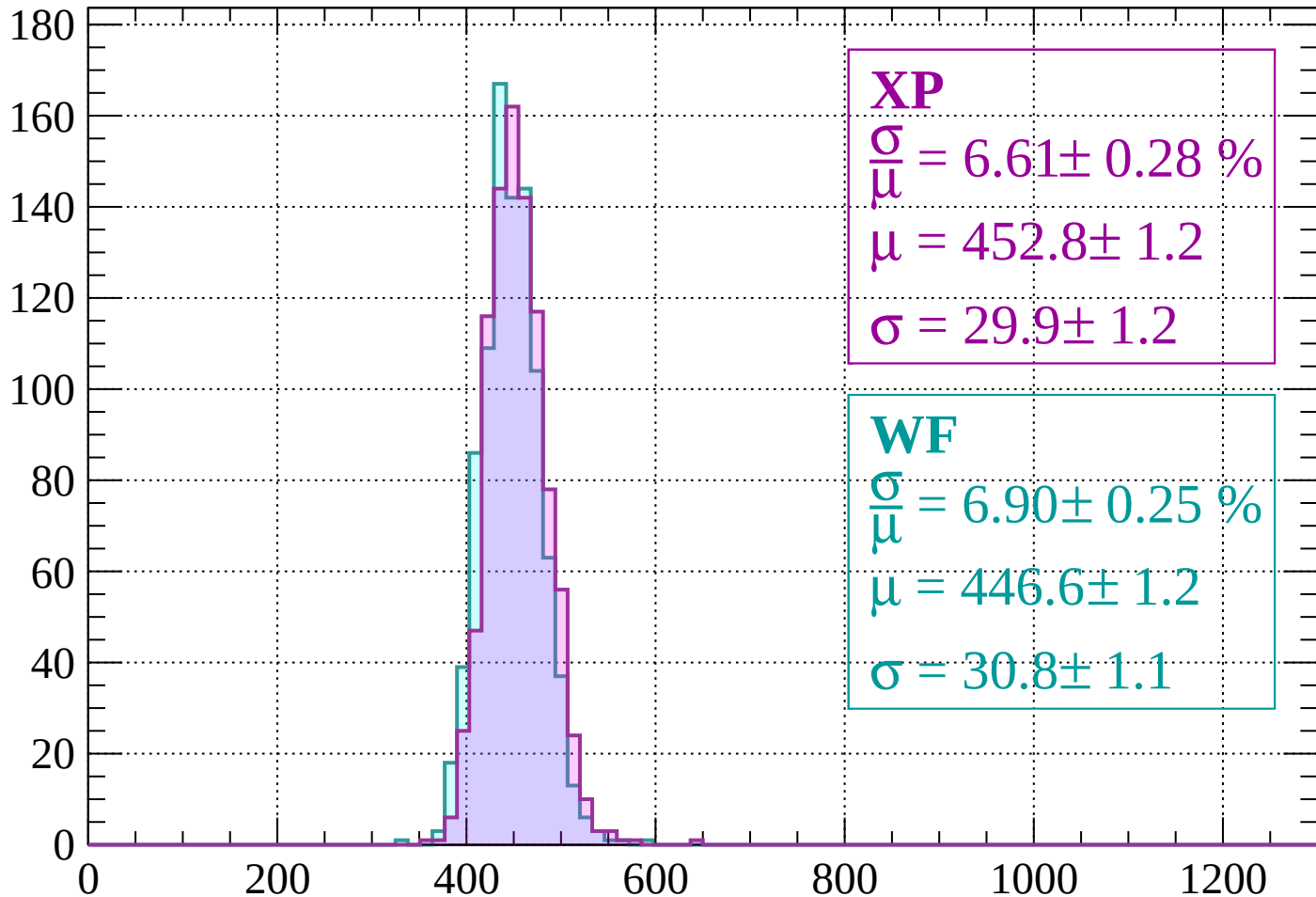
$$\mu = 449.4 \pm 1.1$$

$$\sigma = 29.1 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1160

Count



XP

$$\frac{\sigma}{\mu} = 6.61 \pm 0.28 \%$$

$$\mu = 452.8 \pm 1.2$$

$$\sigma = 29.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.90 \pm 0.25 \%$$

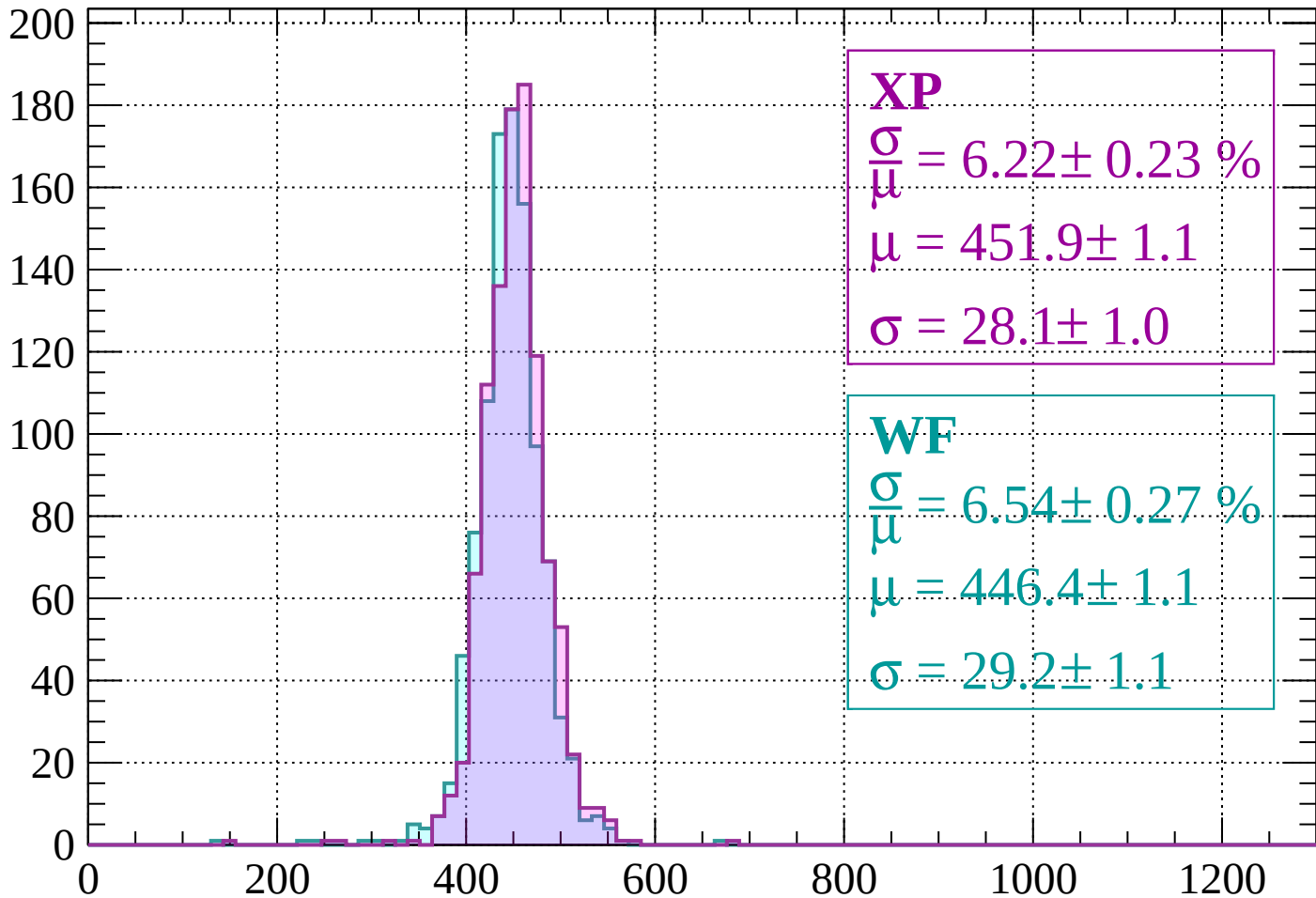
$$\mu = 446.6 \pm 1.2$$

$$\sigma = 30.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -1160 < p < -1120

Count



XP

$$\frac{\sigma}{\mu} = 6.22 \pm 0.23 \%$$

$$\mu = 451.9 \pm 1.1$$

$$\sigma = 28.1 \pm 1.0$$

WF

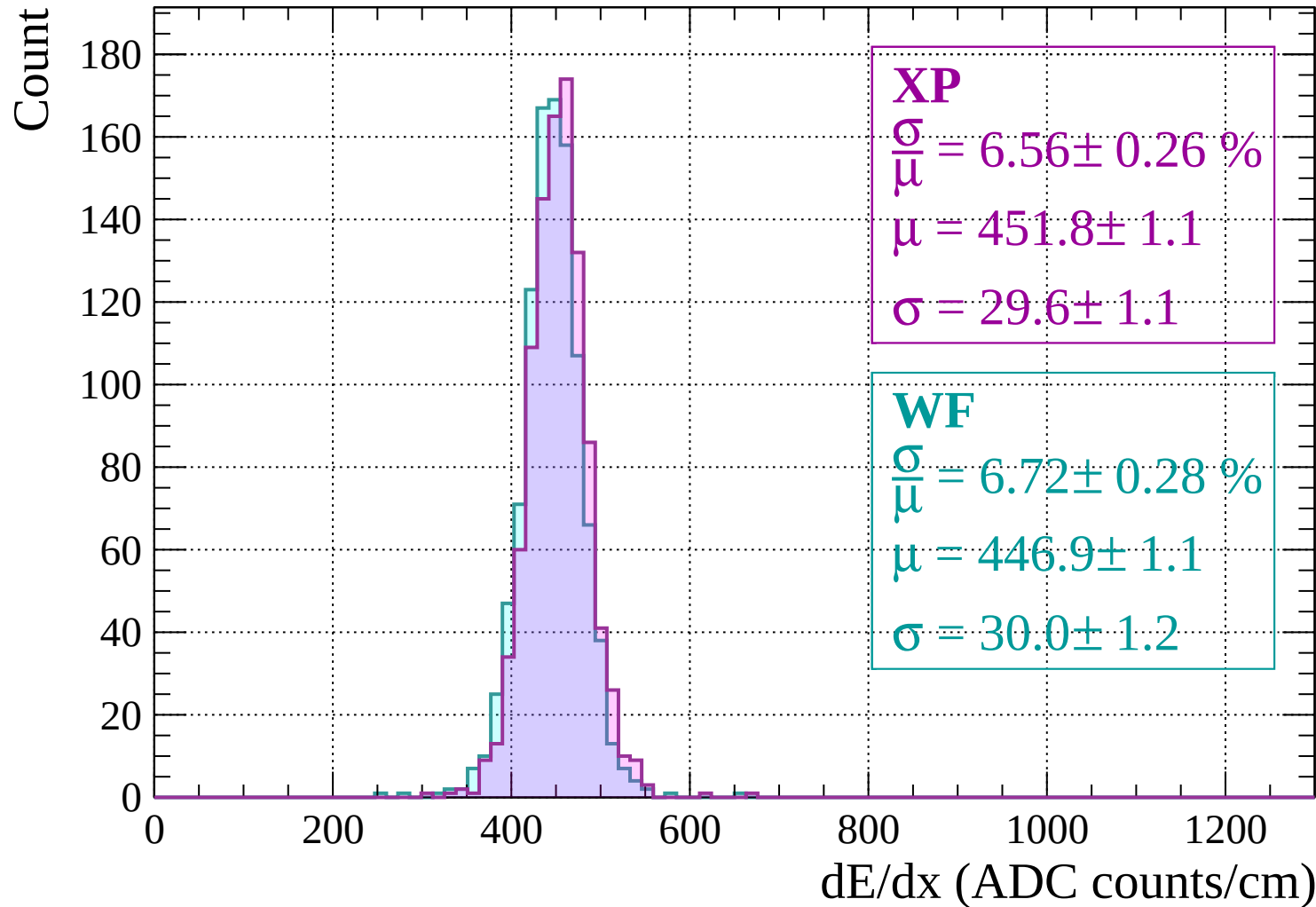
$$\frac{\sigma}{\mu} = 6.54 \pm 0.27 \%$$

$$\mu = 446.4 \pm 1.1$$

$$\sigma = 29.2 \pm 1.1$$

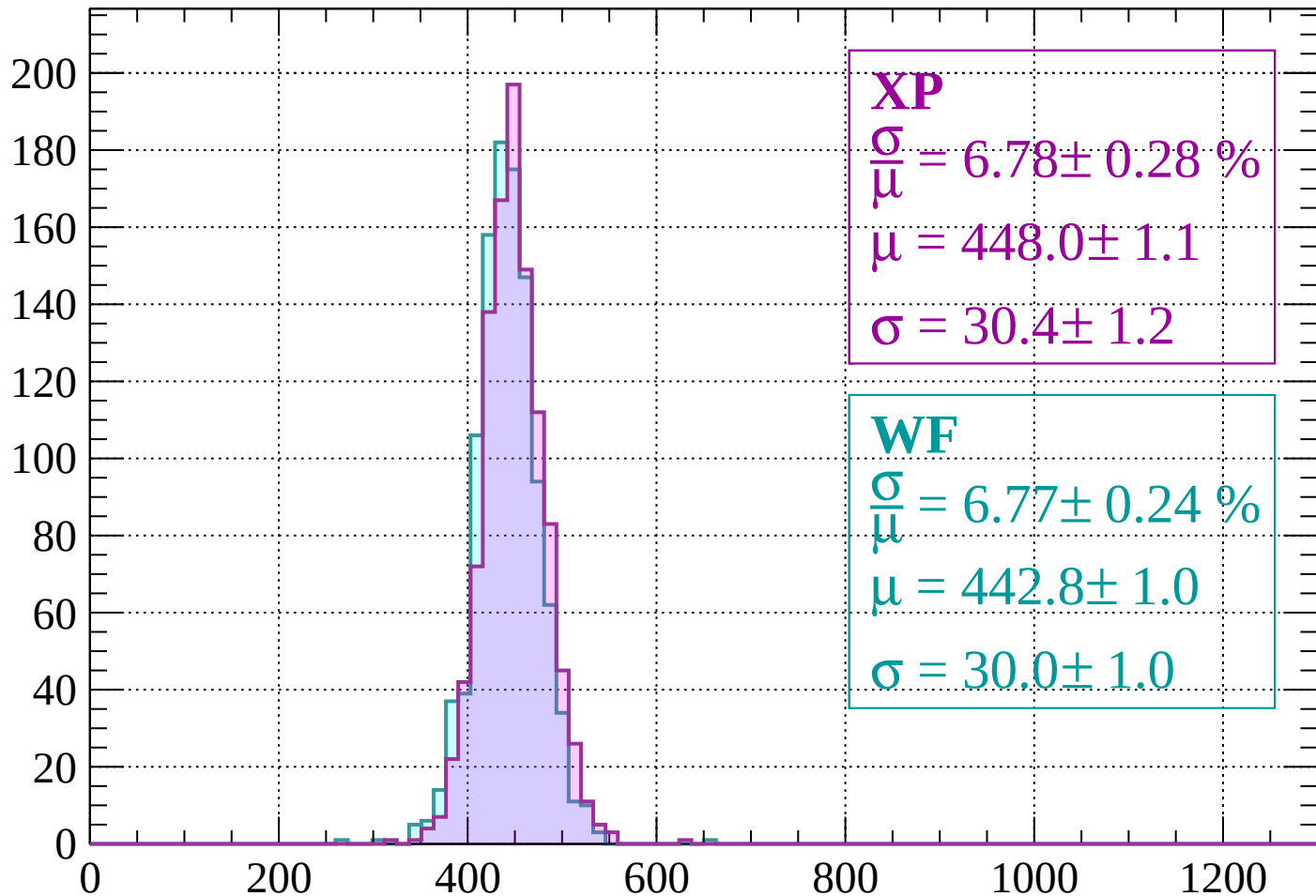
dE/dx (ADC counts/cm)

Energy loss | -1120 < p < -1080



Energy loss | -1080 < p < -1040

Count



XP

$$\frac{\sigma}{\mu} = 6.78 \pm 0.28 \%$$

$$\mu = 448.0 \pm 1.1$$

$$\sigma = 30.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.77 \pm 0.24 \%$$

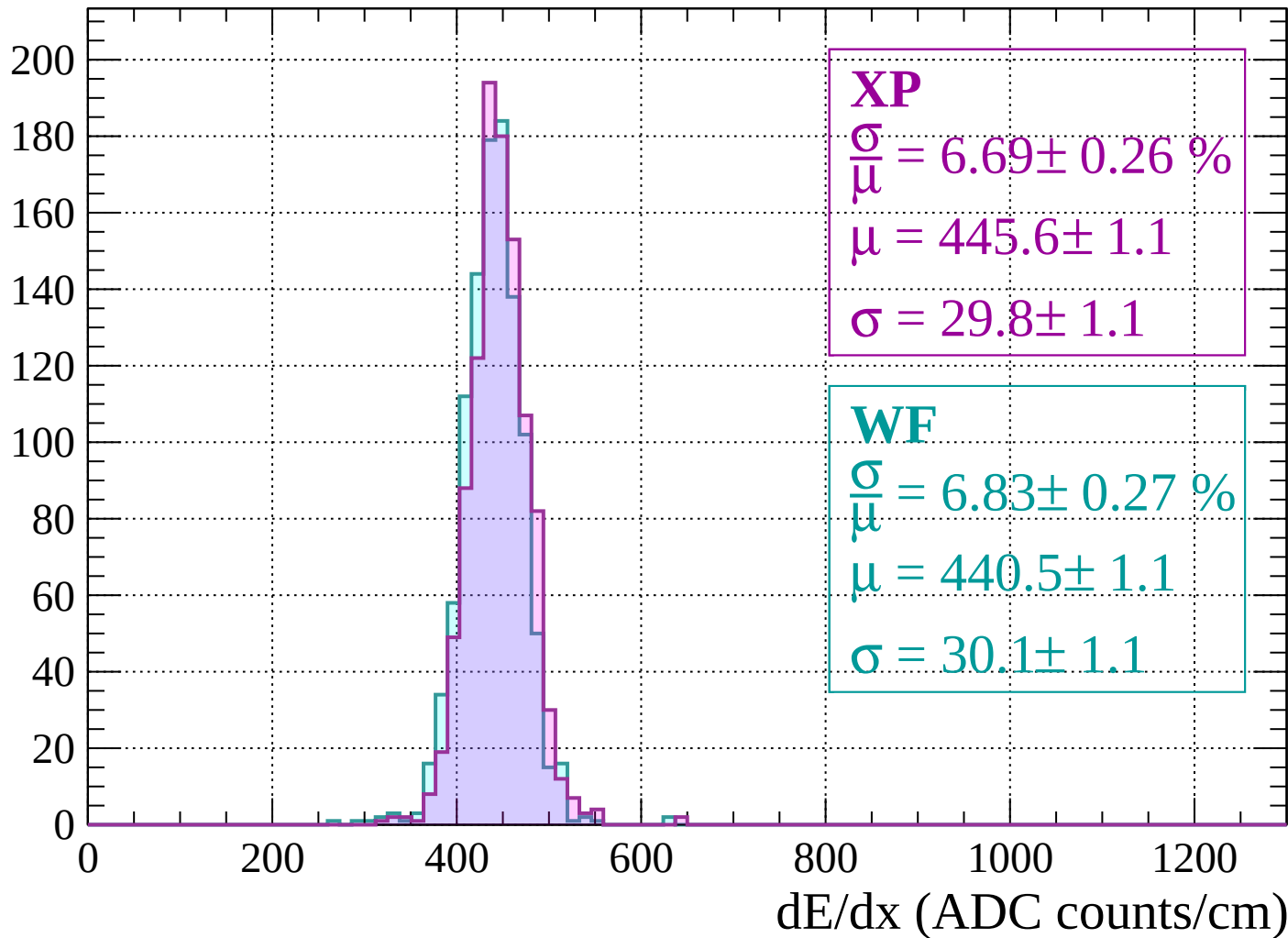
$$\mu = 442.8 \pm 1.0$$

$$\sigma = 30.0 \pm 1.0$$

dE/dx (ADC counts/cm)

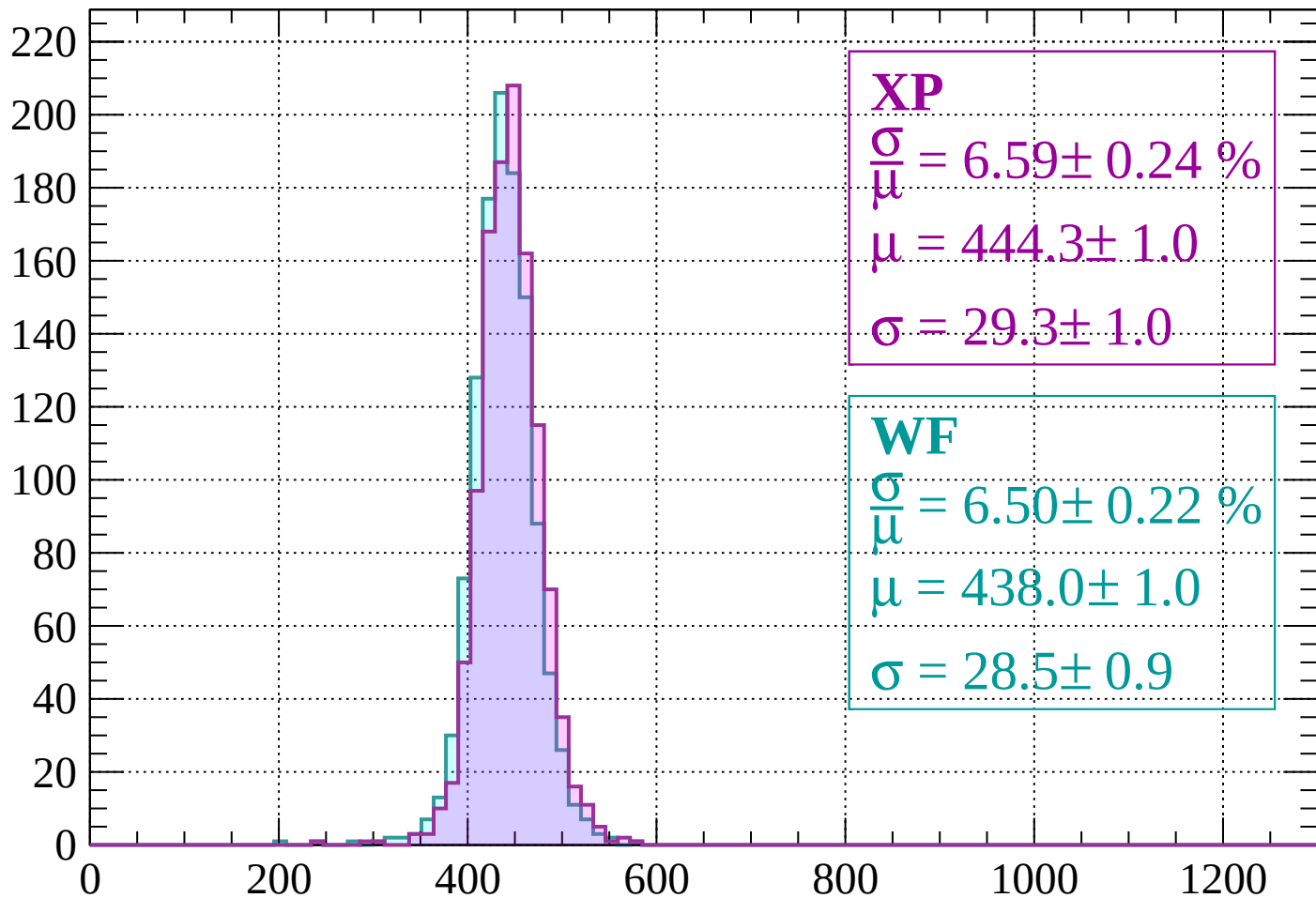
Energy loss | $-1040 < p < -1000$

Count



Energy loss | $-1000 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 6.59 \pm 0.24 \%$$

$$\mu = 444.3 \pm 1.0$$

$$\sigma = 29.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.50 \pm 0.22 \%$$

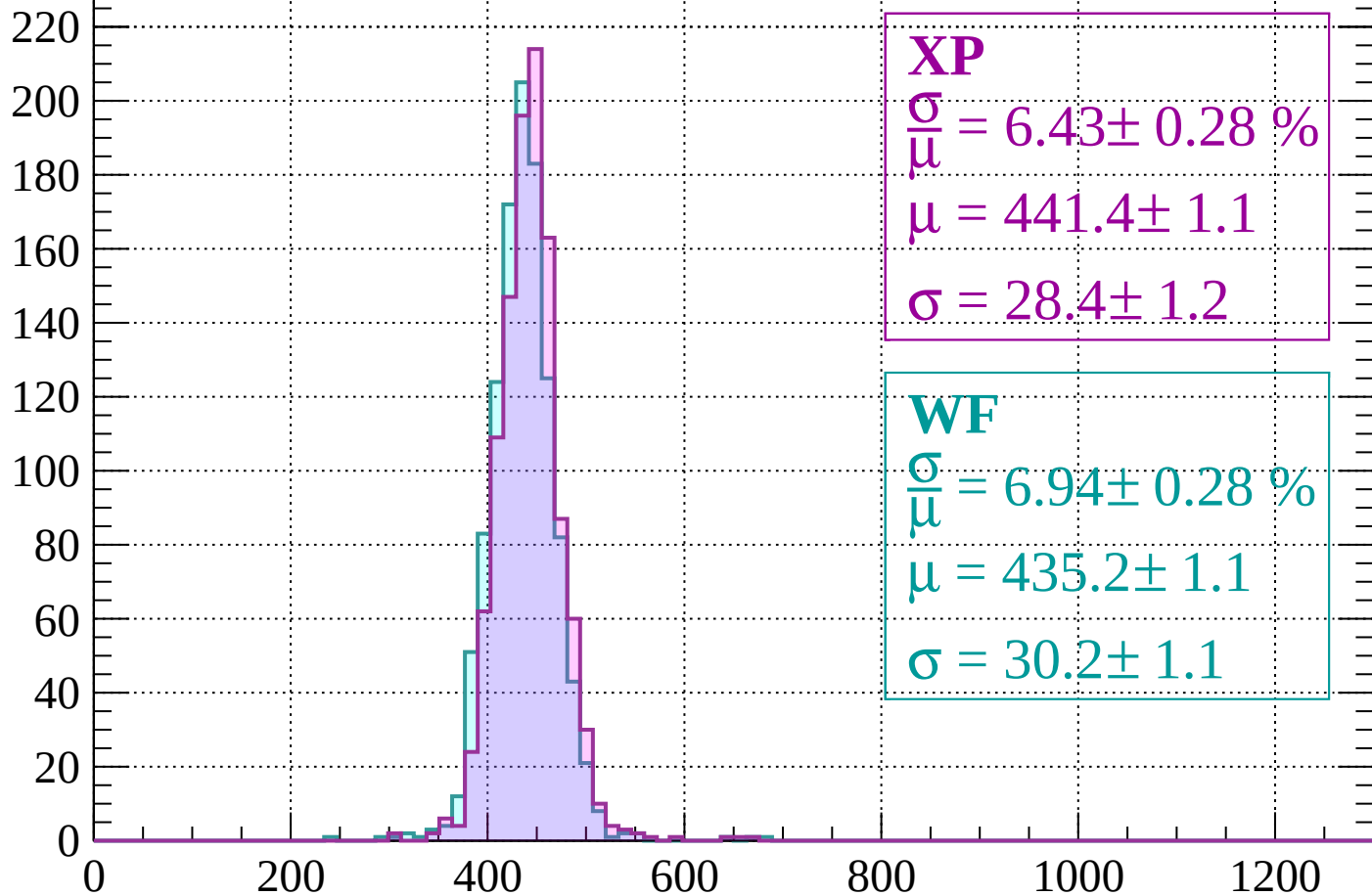
$$\mu = 438.0 \pm 1.0$$

$$\sigma = 28.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 6.43 \pm 0.28 \%$$

$$\mu = 441.4 \pm 1.1$$

$$\sigma = 28.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.94 \pm 0.28 \%$$

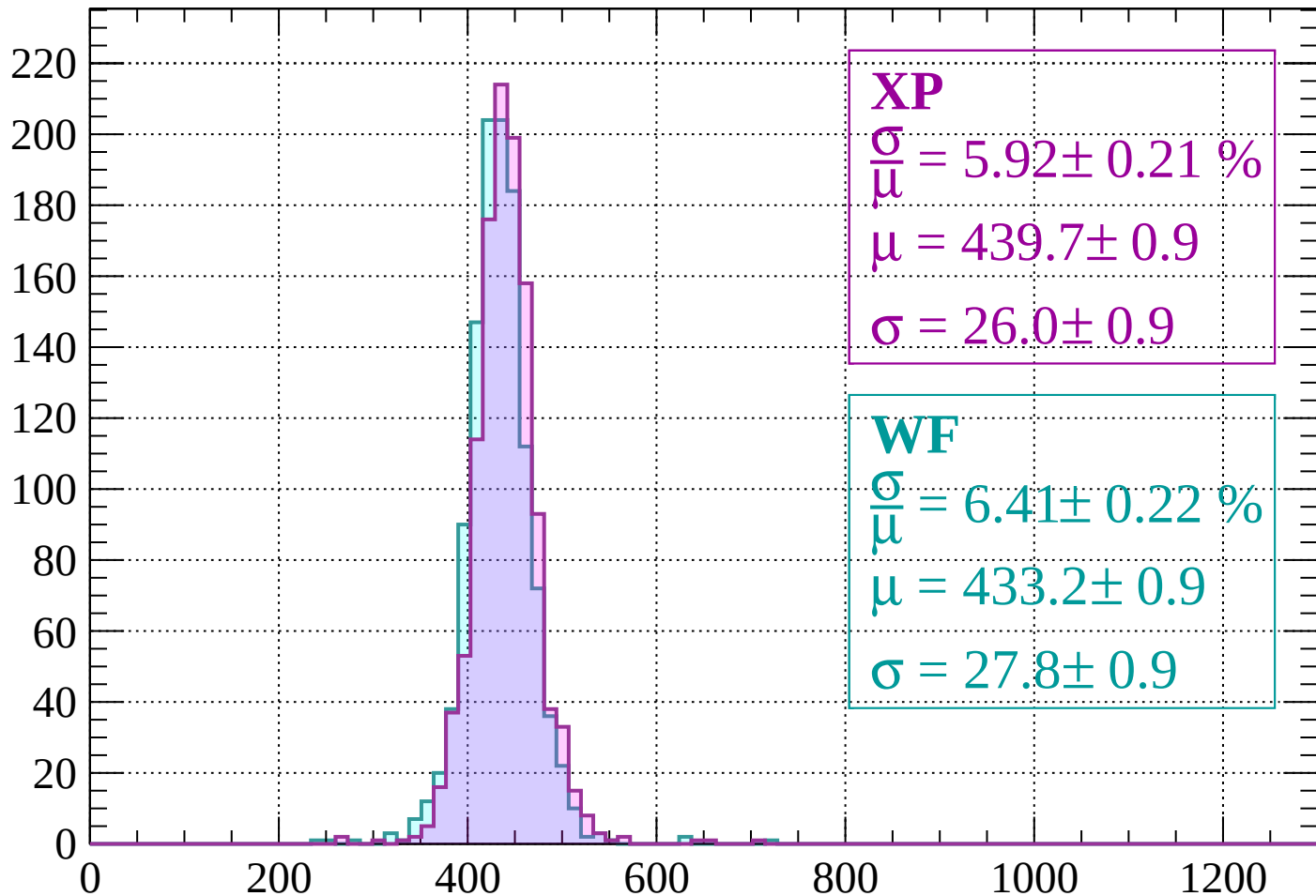
$$\mu = 435.2 \pm 1.1$$

$$\sigma = 30.2 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 5.92 \pm 0.21 \%$$

$$\mu = 439.7 \pm 0.9$$

$$\sigma = 26.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.41 \pm 0.22 \%$$

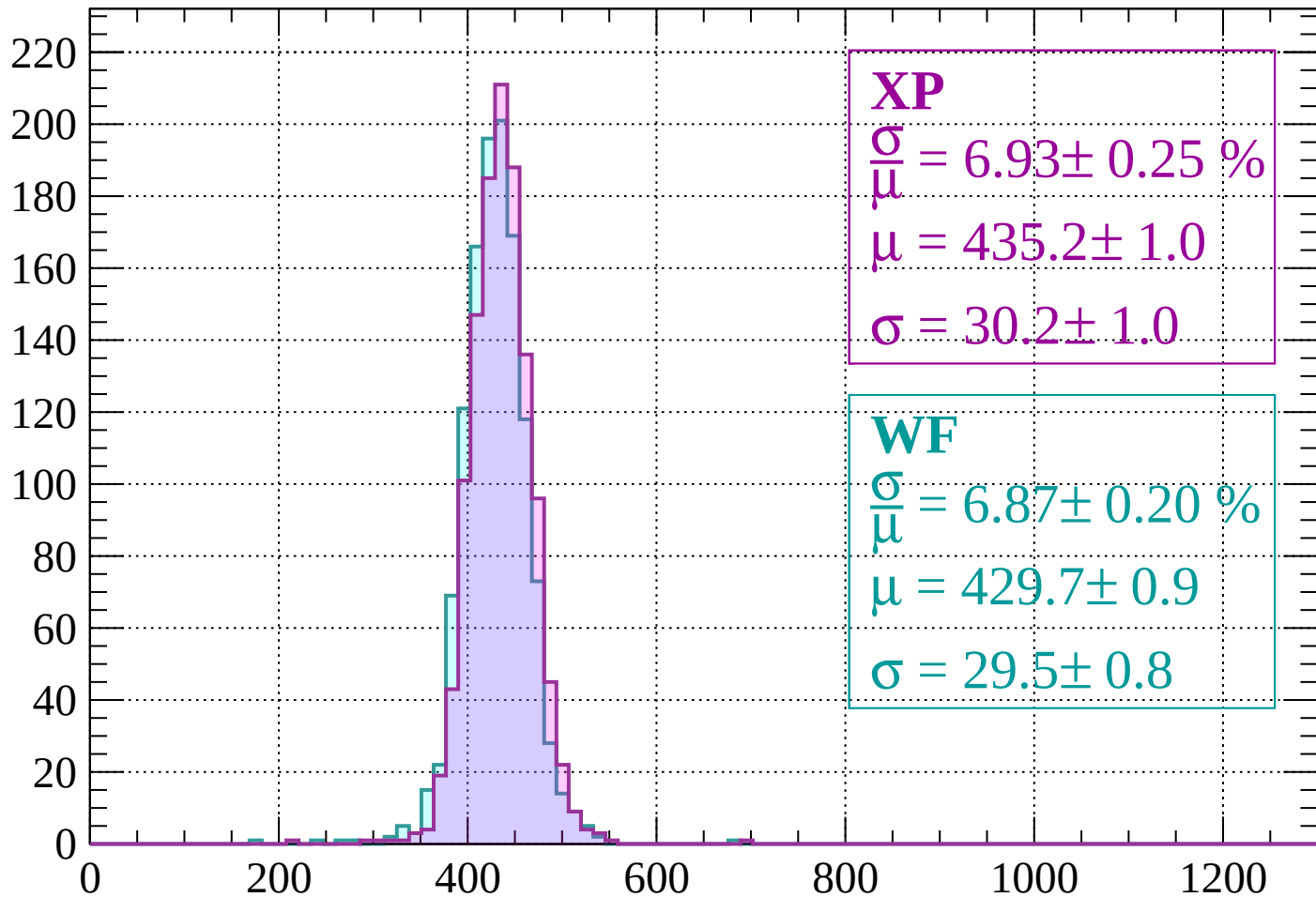
$$\mu = 433.2 \pm 0.9$$

$$\sigma = 27.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 6.93 \pm 0.25 \%$$

$$\mu = 435.2 \pm 1.0$$

$$\sigma = 30.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.87 \pm 0.20 \%$$

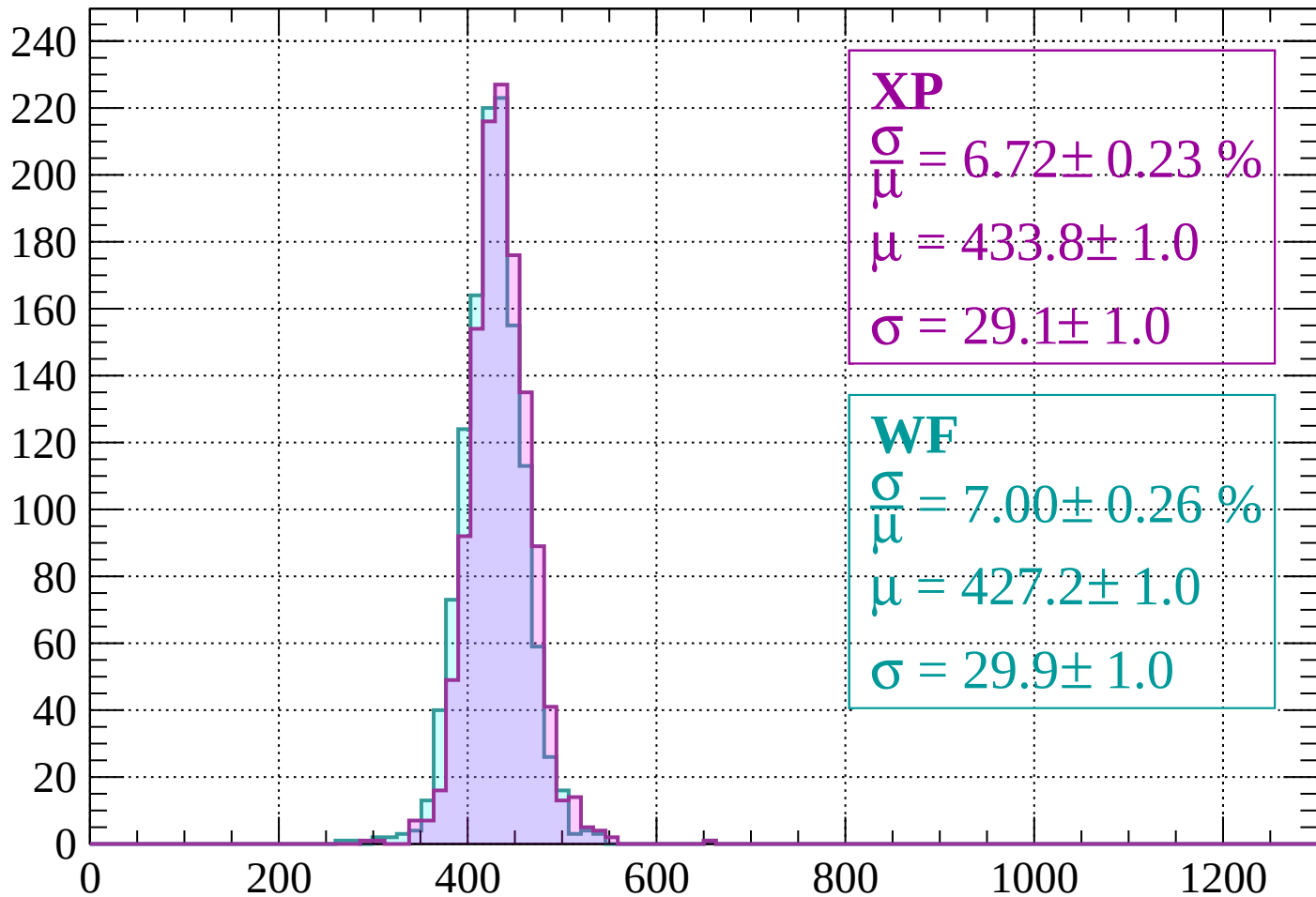
$$\mu = 429.7 \pm 0.9$$

$$\sigma = 29.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 6.72 \pm 0.23 \%$$

$$\mu = 433.8 \pm 1.0$$

$$\sigma = 29.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.00 \pm 0.26 \%$$

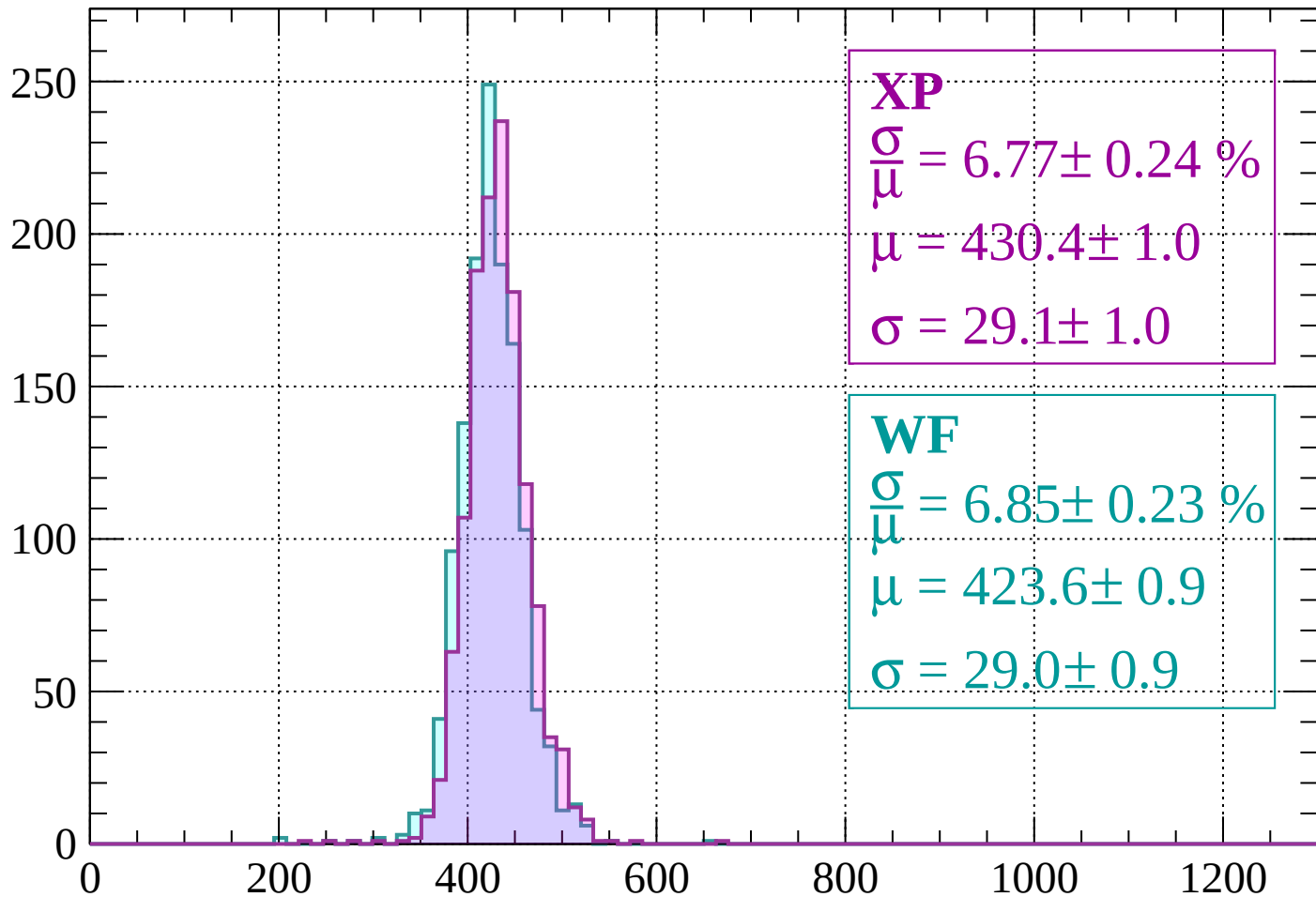
$$\mu = 427.2 \pm 1.0$$

$$\sigma = 29.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 6.77 \pm 0.24 \%$$

$$\mu = 430.4 \pm 1.0$$

$$\sigma = 29.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.85 \pm 0.23 \%$$

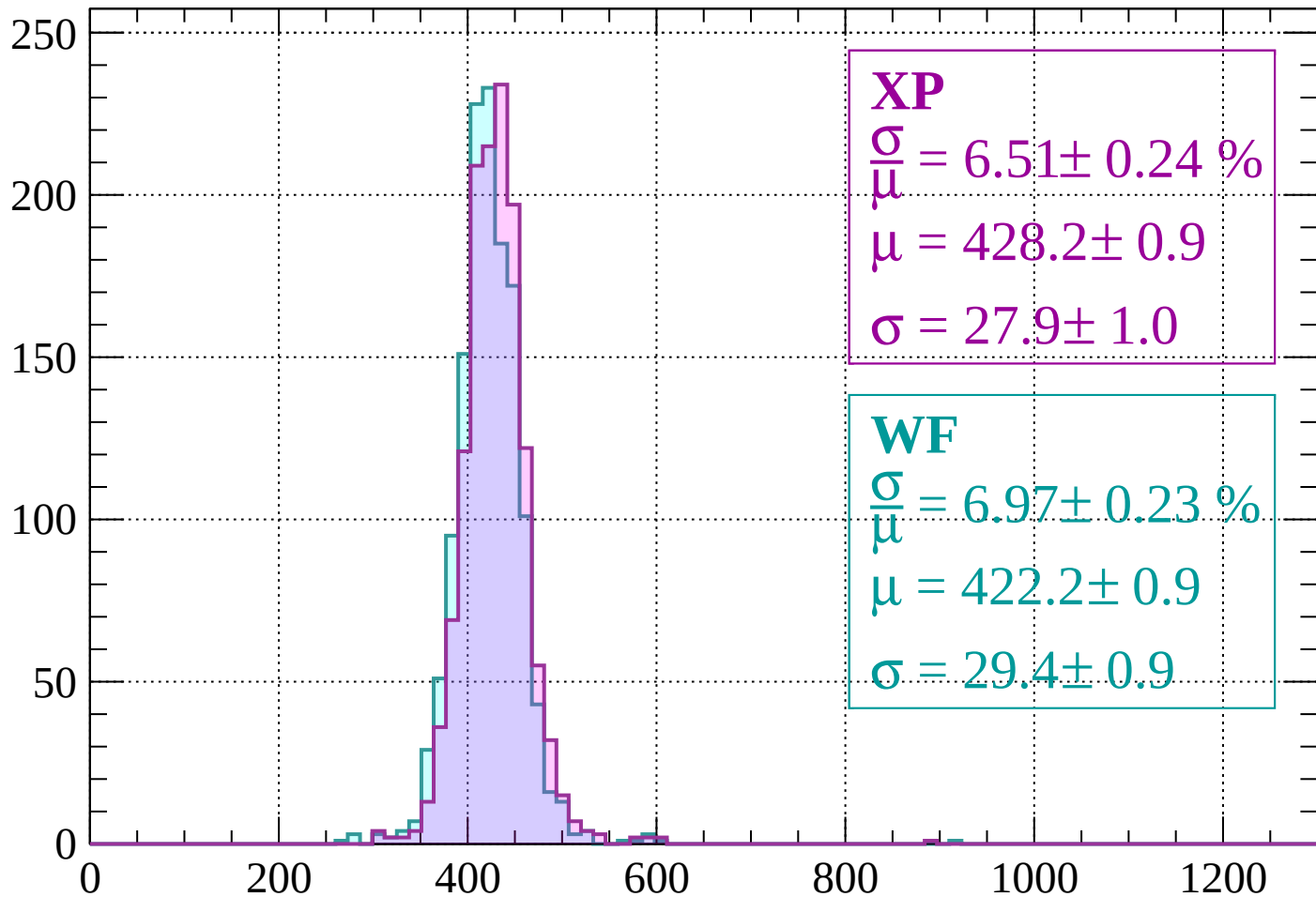
$$\mu = 423.6 \pm 0.9$$

$$\sigma = 29.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 6.51 \pm 0.24 \%$$

$$\mu = 428.2 \pm 0.9$$

$$\sigma = 27.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.97 \pm 0.23 \%$$

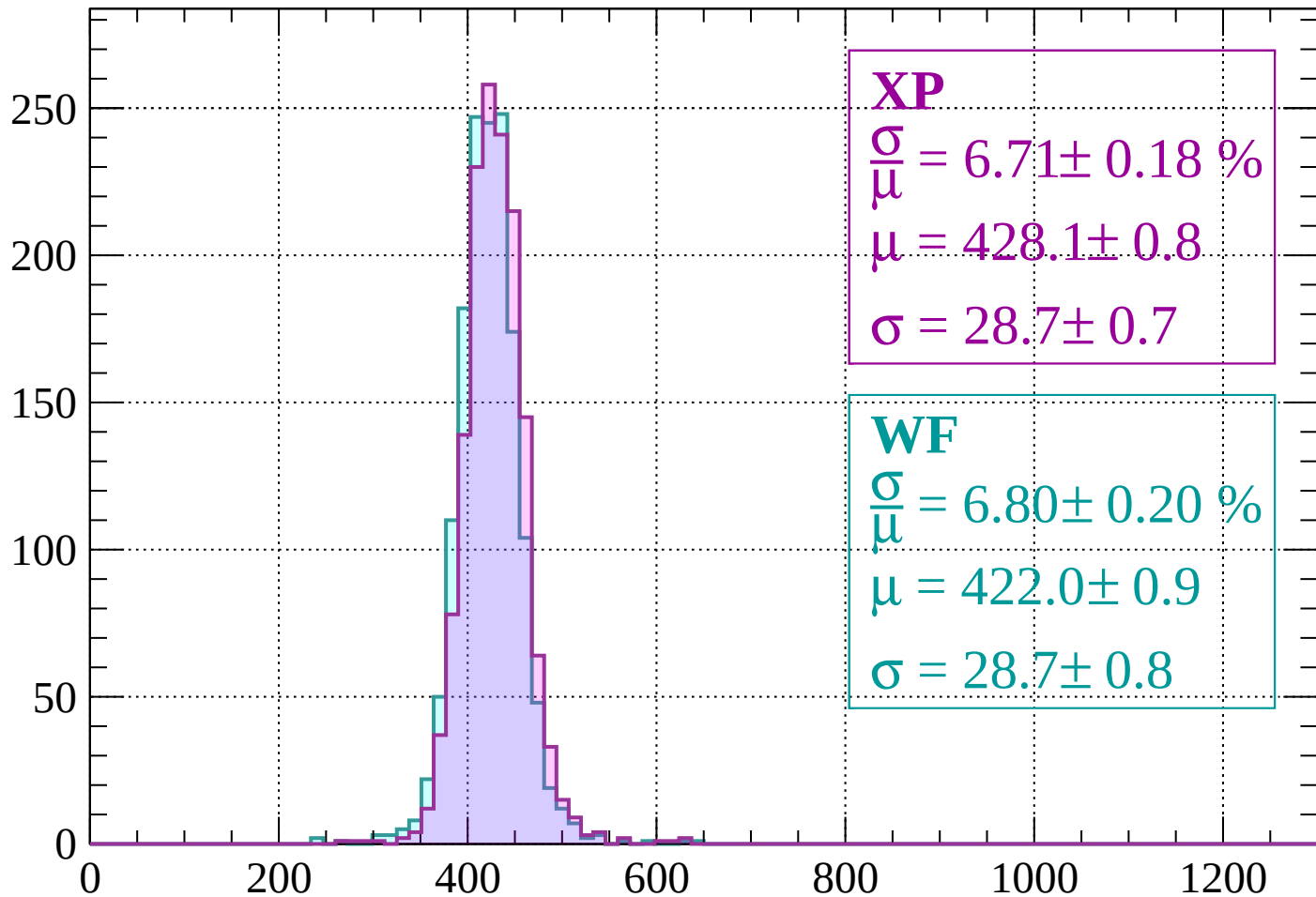
$$\mu = 422.2 \pm 0.9$$

$$\sigma = 29.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 6.71 \pm 0.18 \%$$

$$\mu = 428.1 \pm 0.8$$

$$\sigma = 28.7 \pm 0.7$$

WF

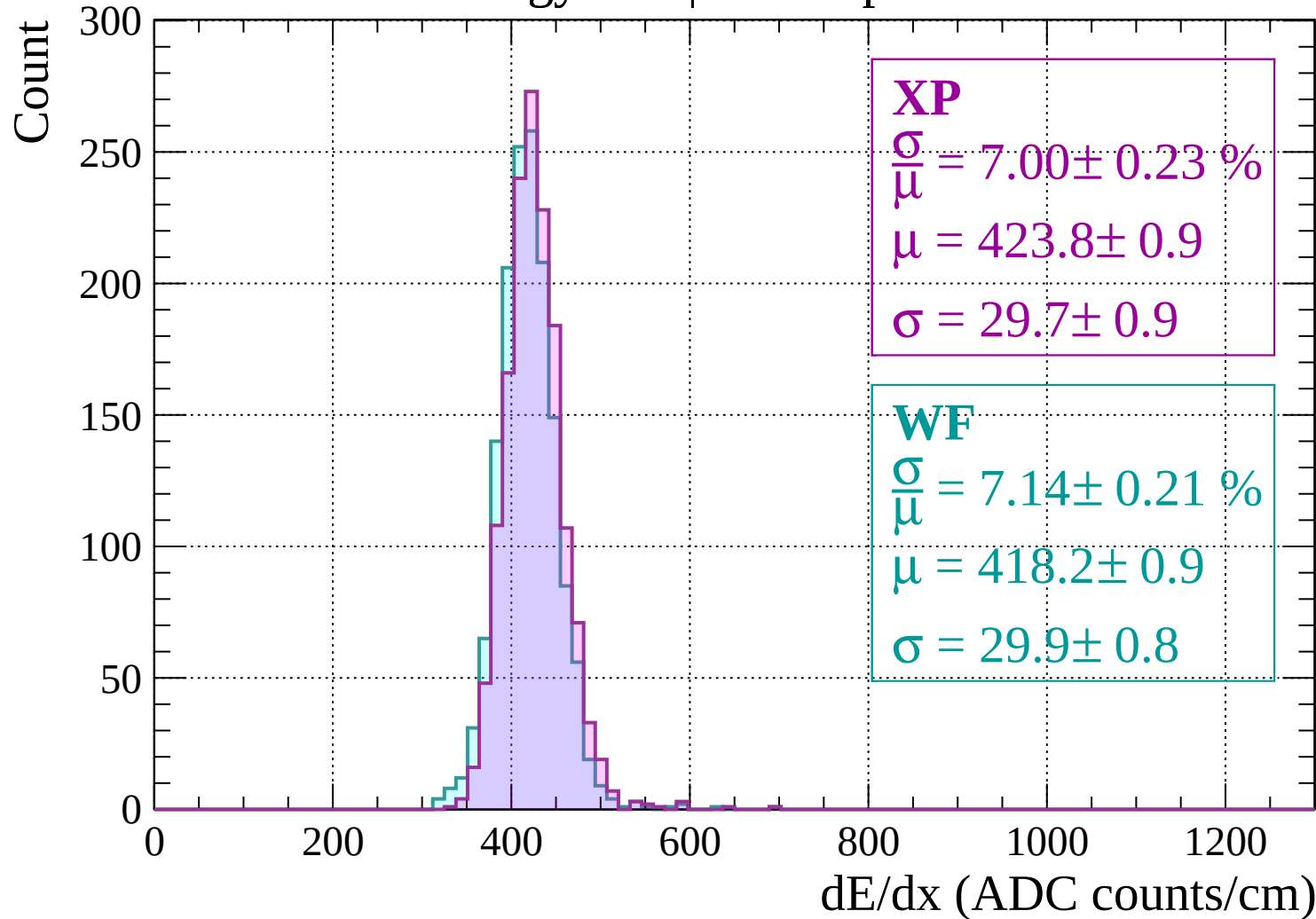
$$\frac{\sigma}{\mu} = 6.80 \pm 0.20 \%$$

$$\mu = 422.0 \pm 0.9$$

$$\sigma = 28.7 \pm 0.8$$

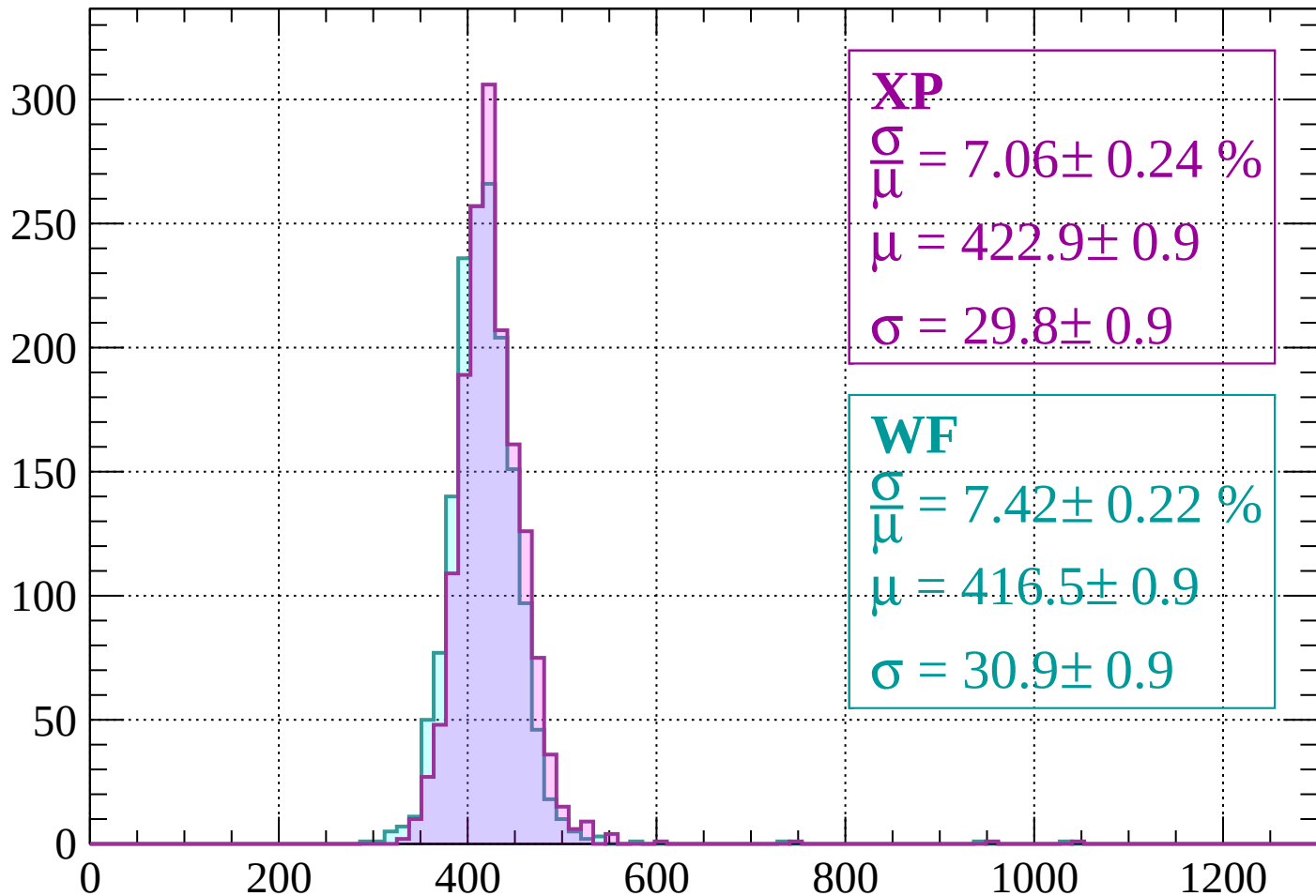
dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$



Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 7.06 \pm 0.24 \%$$

$$\mu = 422.9 \pm 0.9$$

$$\sigma = 29.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.42 \pm 0.22 \%$$

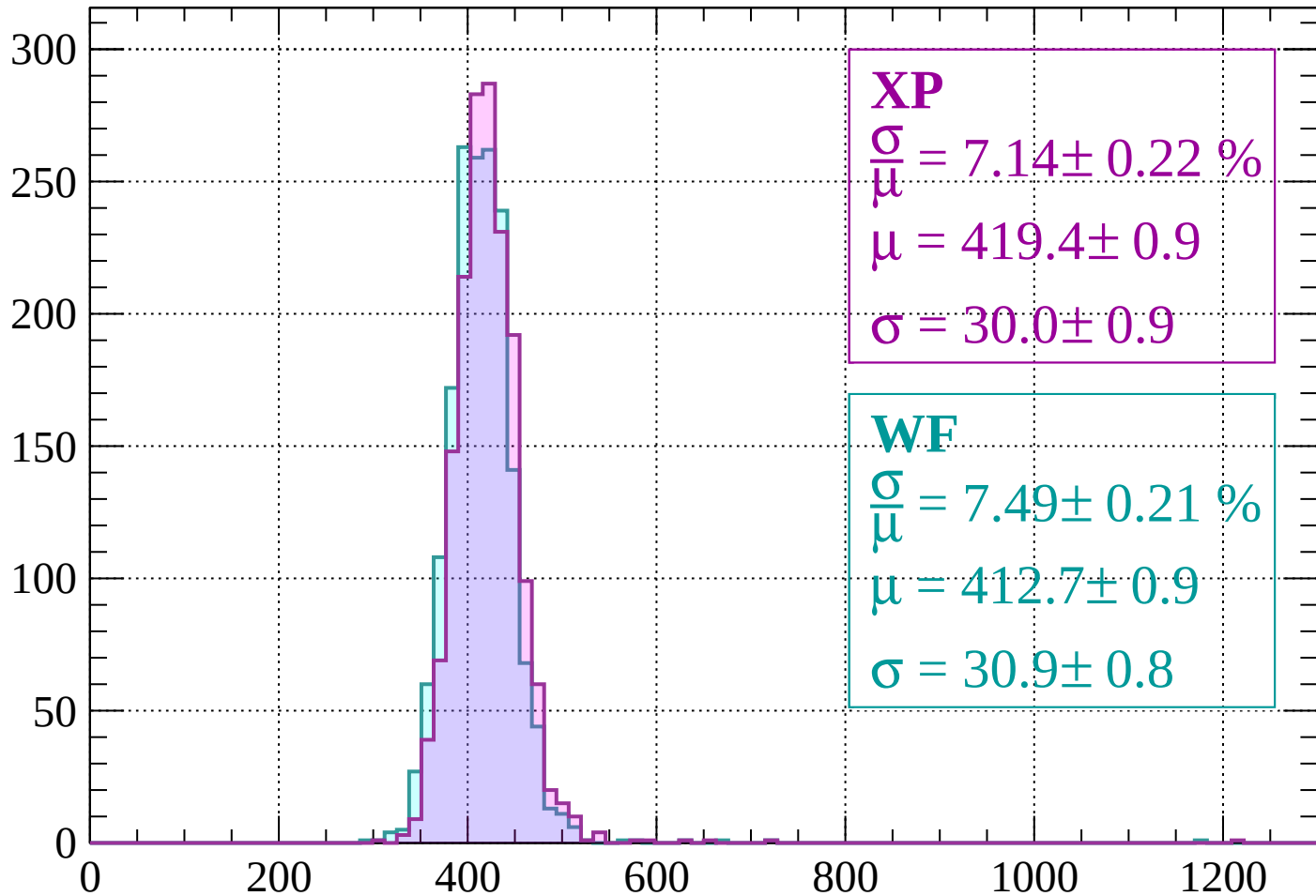
$$\mu = 416.5 \pm 0.9$$

$$\sigma = 30.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 7.14 \pm 0.22 \%$$

$$\mu = 419.4 \pm 0.9$$

$$\sigma = 30.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.49 \pm 0.21 \%$$

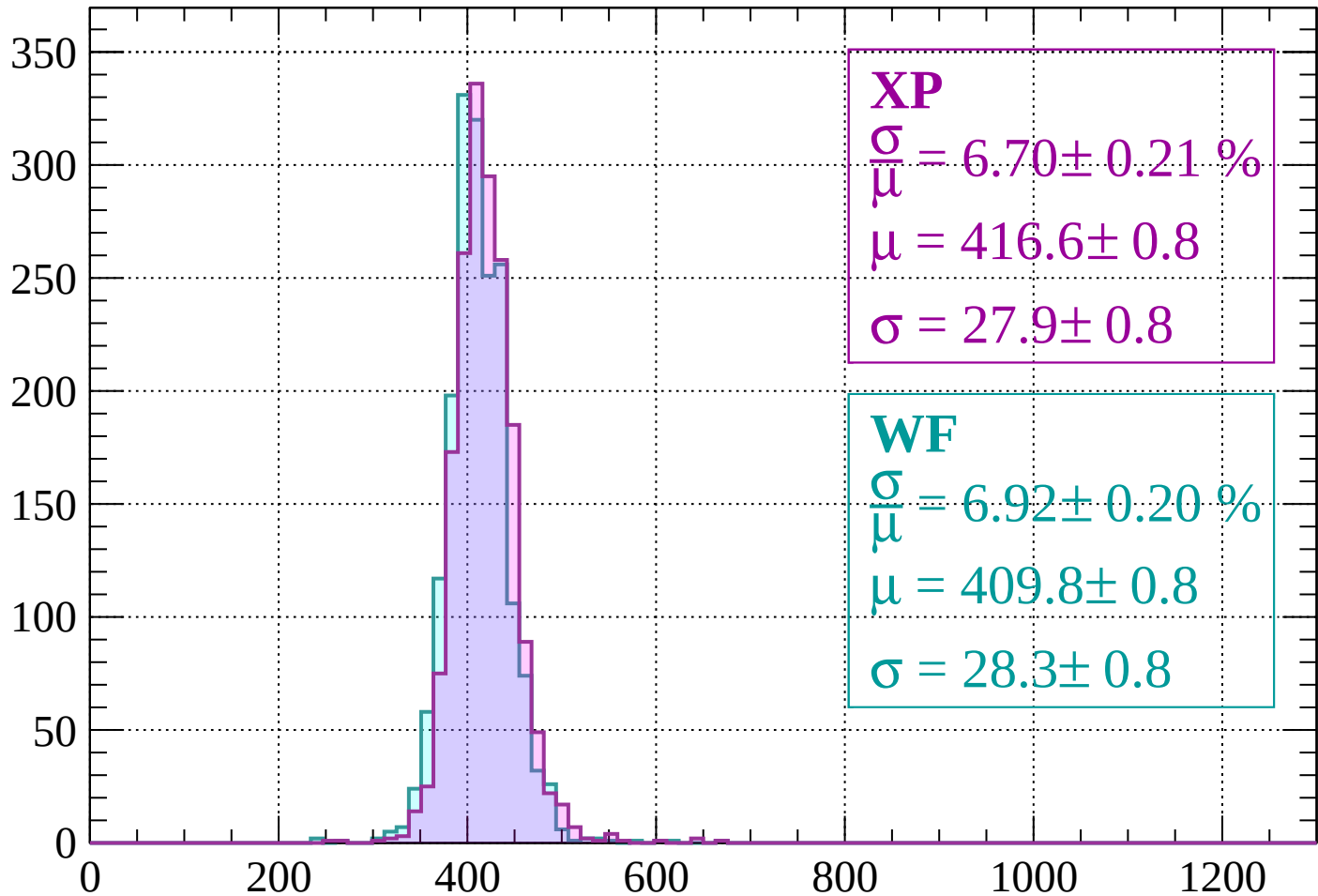
$$\mu = 412.7 \pm 0.9$$

$$\sigma = 30.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 6.70 \pm 0.21 \%$$

$$\mu = 416.6 \pm 0.8$$

$$\sigma = 27.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.92 \pm 0.20 \%$$

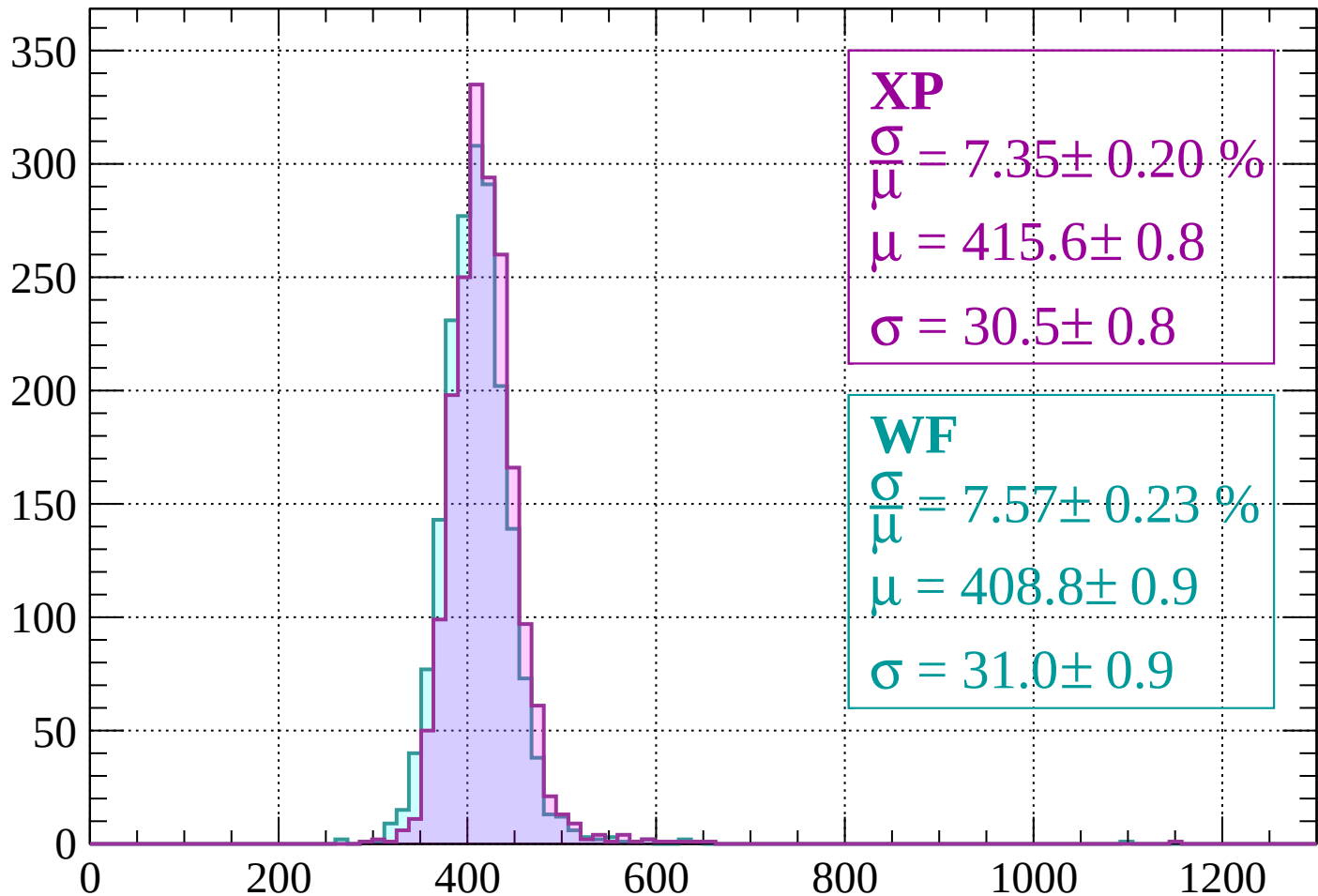
$$\mu = 409.8 \pm 0.8$$

$$\sigma = 28.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 7.35 \pm 0.20 \%$$

$$\mu = 415.6 \pm 0.8$$

$$\sigma = 30.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.57 \pm 0.23 \%$$

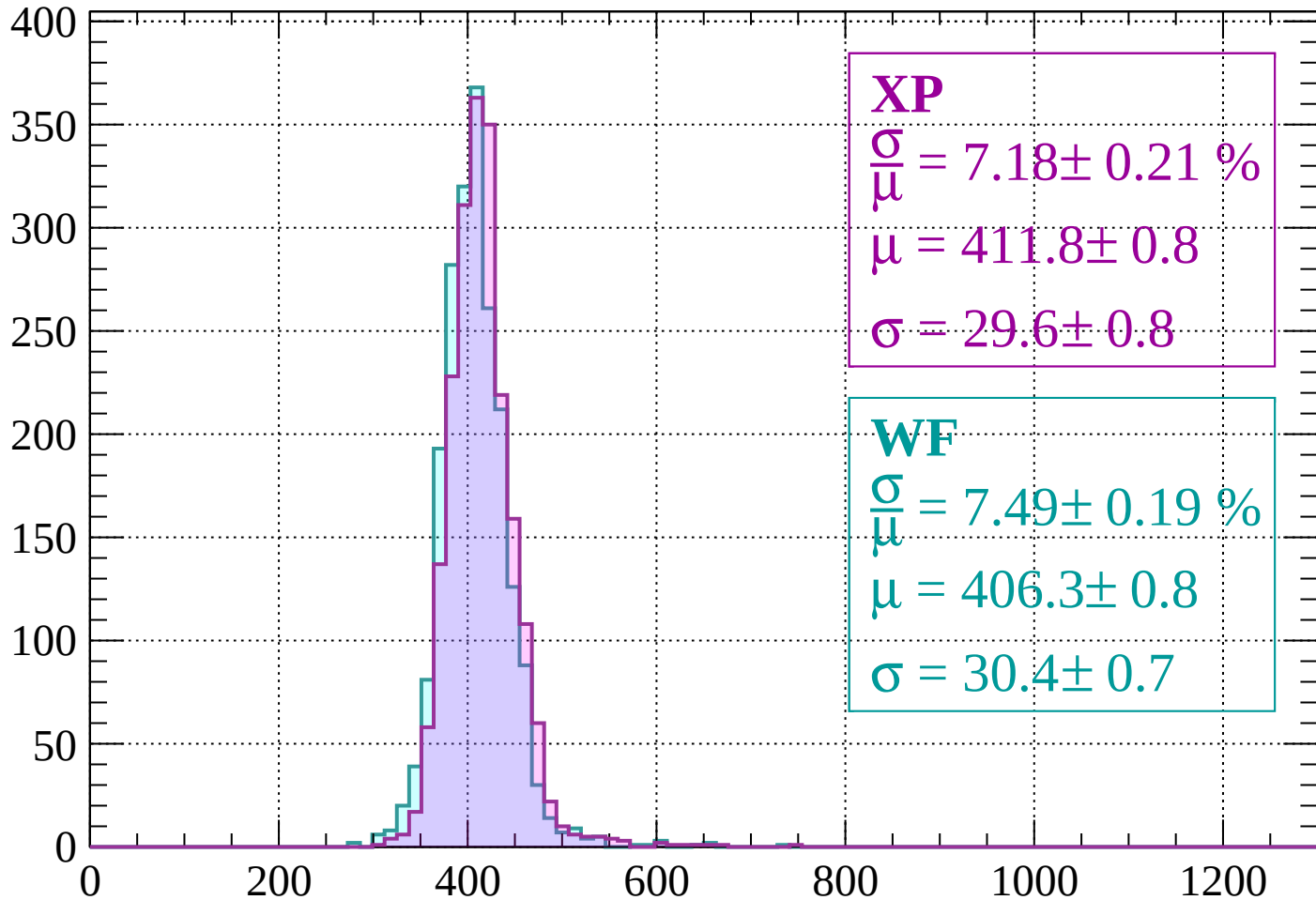
$$\mu = 408.8 \pm 0.9$$

$$\sigma = 31.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 7.18 \pm 0.21 \%$$

$$\mu = 411.8 \pm 0.8$$

$$\sigma = 29.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.49 \pm 0.19 \%$$

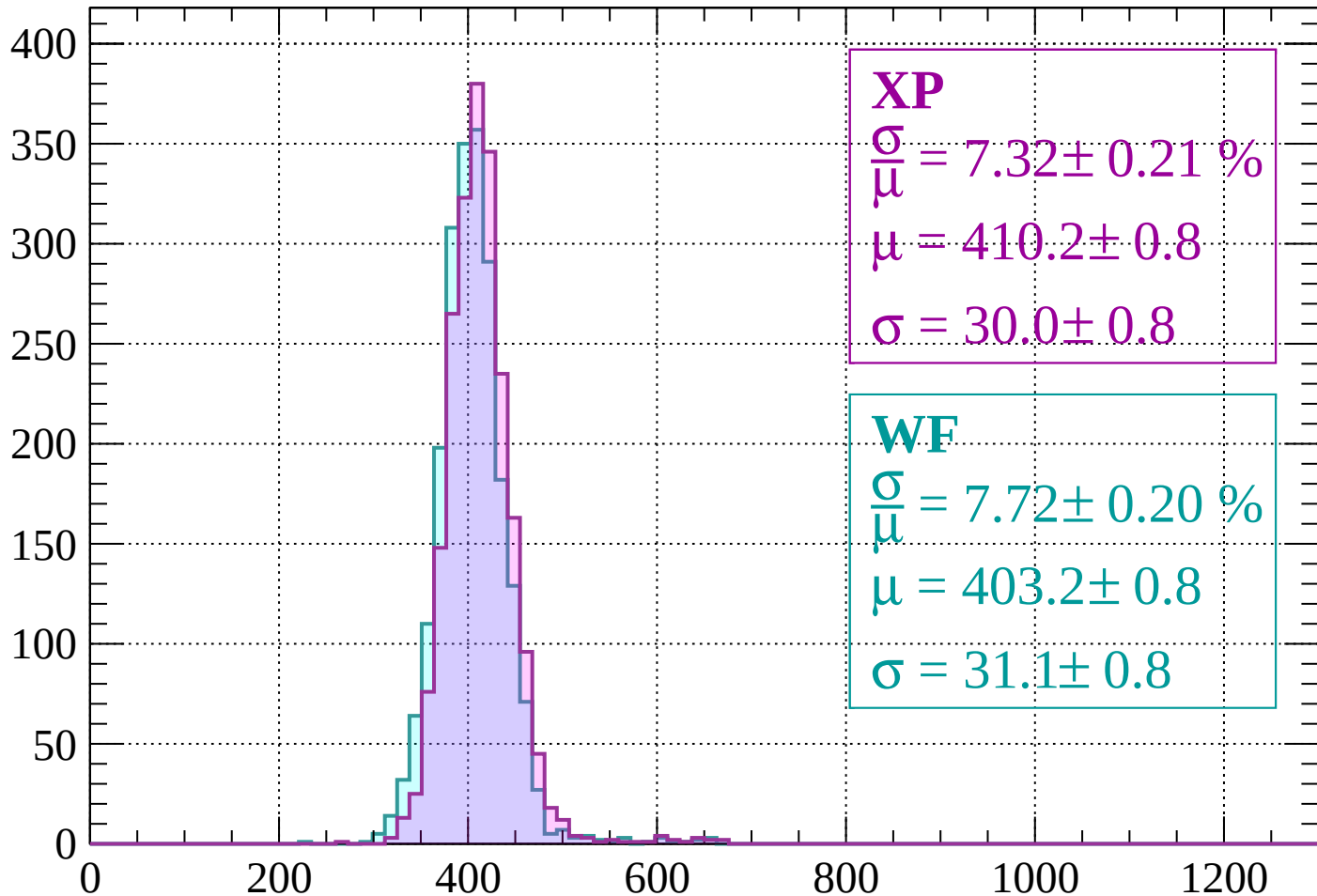
$$\mu = 406.3 \pm 0.8$$

$$\sigma = 30.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count



XP

$$\frac{\sigma}{\mu} = 7.32 \pm 0.21 \%$$

$$\mu = 410.2 \pm 0.8$$

$$\sigma = 30.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.72 \pm 0.20 \%$$

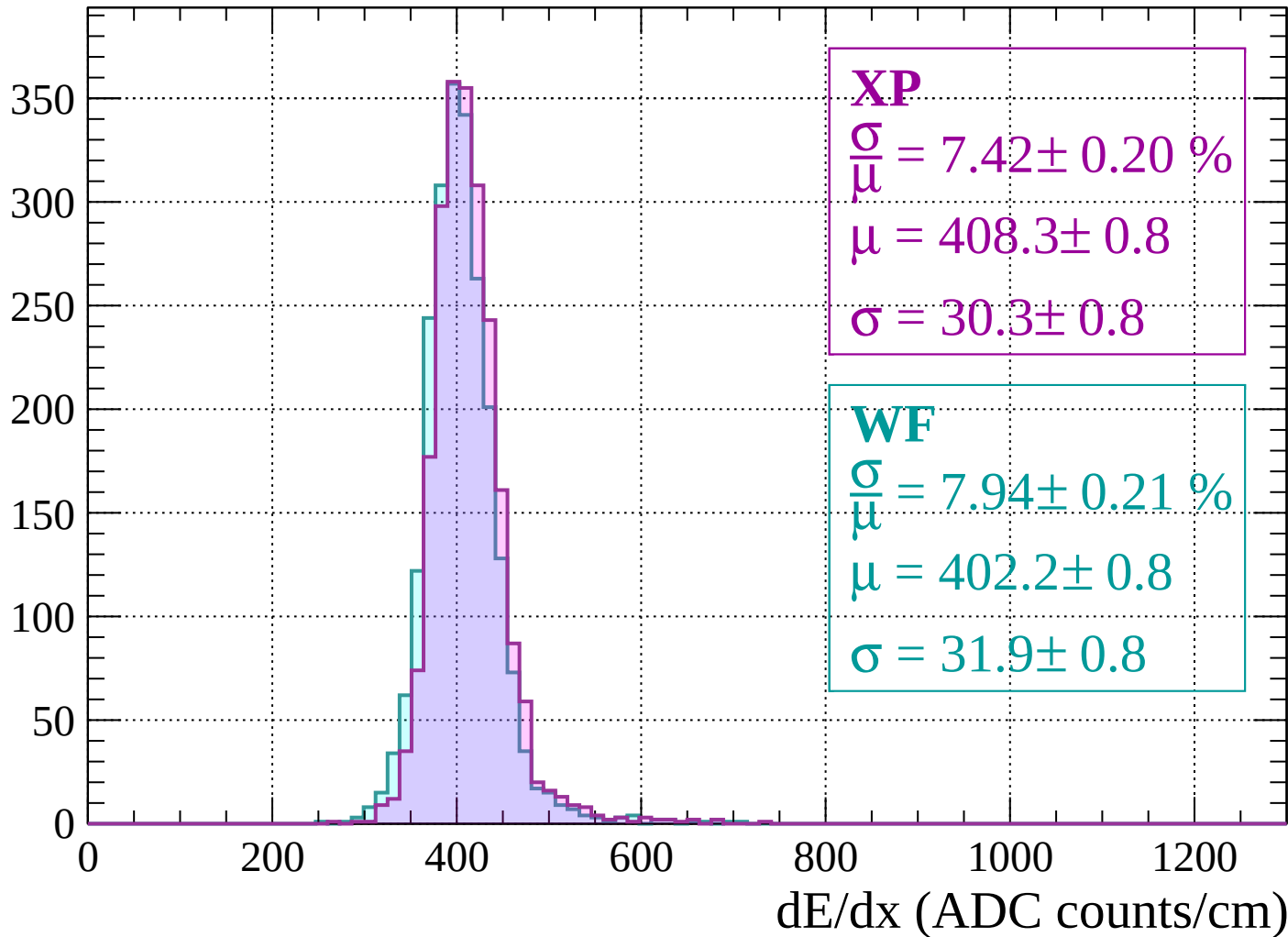
$$\mu = 403.2 \pm 0.8$$

$$\sigma = 31.1 \pm 0.8$$

dE/dx (ADC counts/cm)

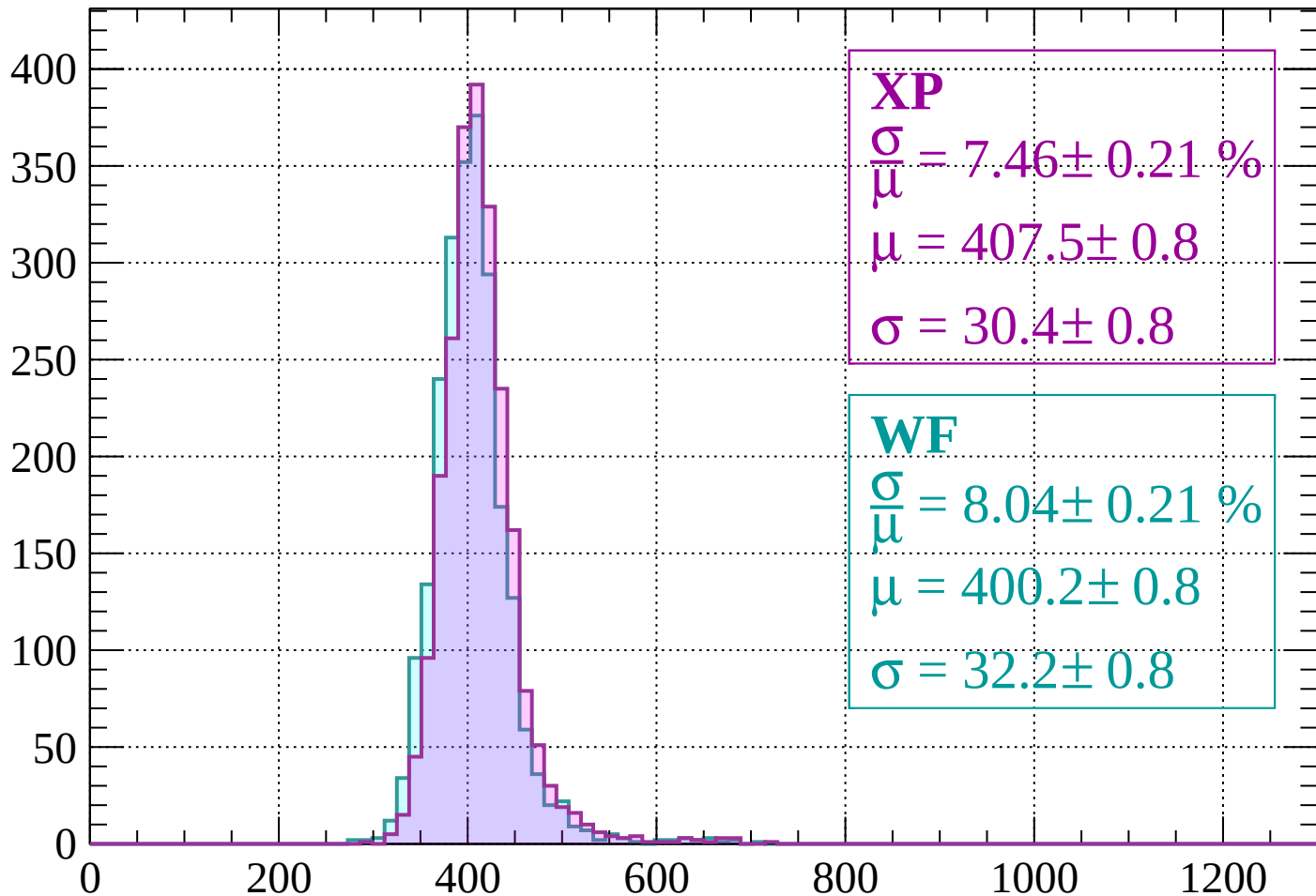
Energy loss | $-400 < p < -360$

Count



Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 7.46 \pm 0.21 \%$$

$$\mu = 407.5 \pm 0.8$$

$$\sigma = 30.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.04 \pm 0.21 \%$$

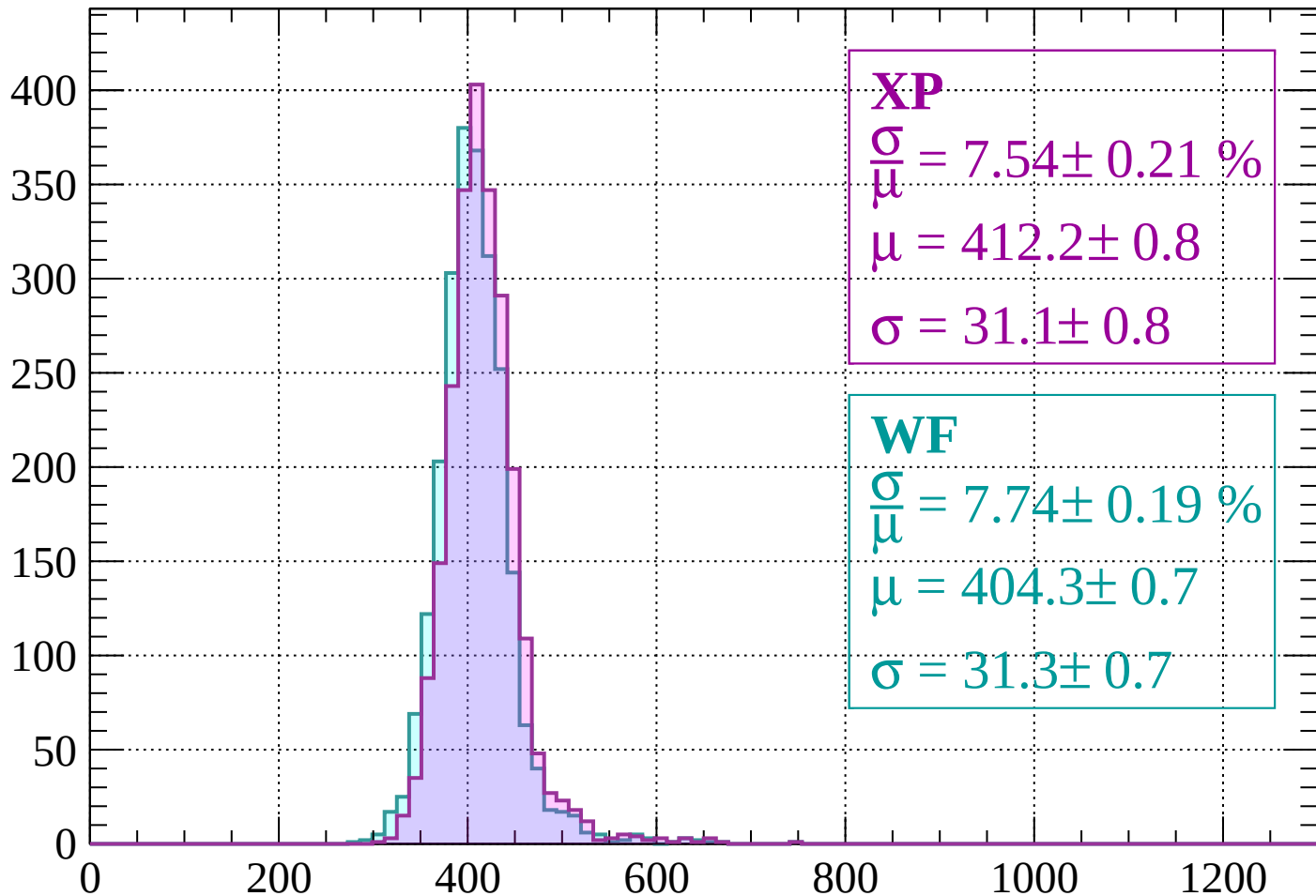
$$\mu = 400.2 \pm 0.8$$

$$\sigma = 32.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 7.54 \pm 0.21 \%$$

$$\mu = 412.2 \pm 0.8$$

$$\sigma = 31.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.74 \pm 0.19 \%$$

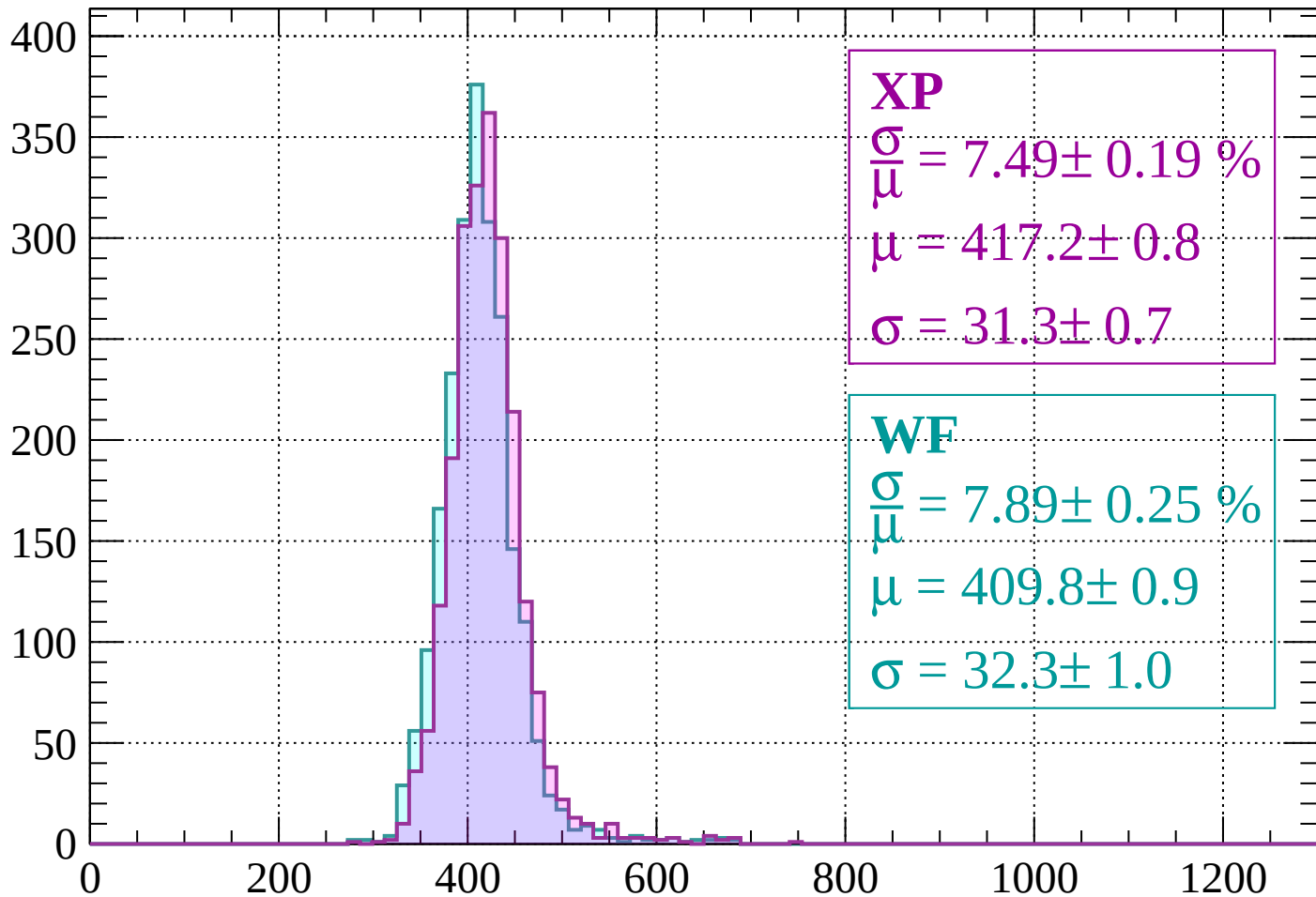
$$\mu = 404.3 \pm 0.7$$

$$\sigma = 31.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 7.49 \pm 0.19 \%$$

$$\mu = 417.2 \pm 0.8$$

$$\sigma = 31.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.89 \pm 0.25 \%$$

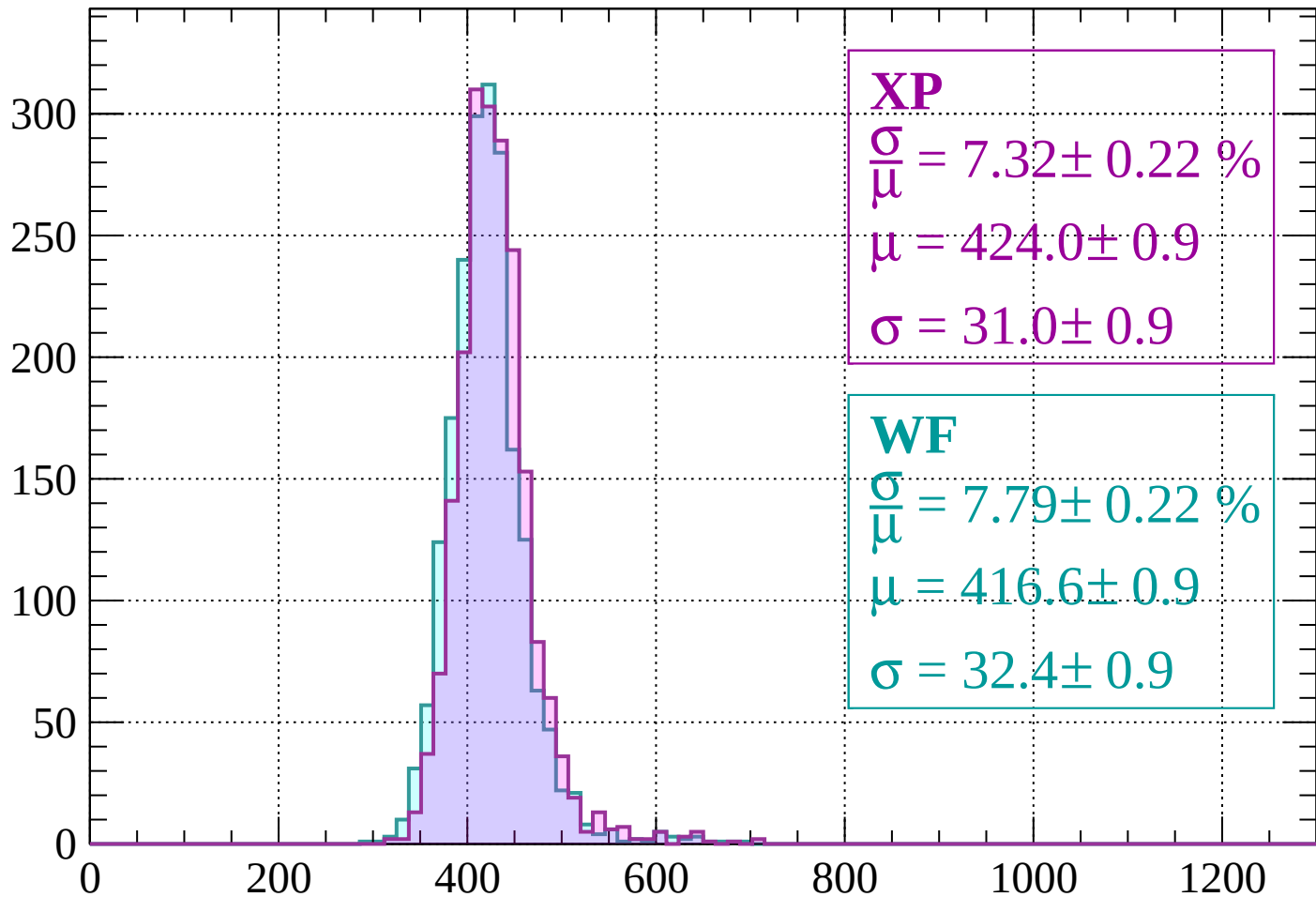
$$\mu = 409.8 \pm 0.9$$

$$\sigma = 32.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\mu} = 7.32 \pm 0.22 \%$$

$$\mu = 424.0 \pm 0.9$$

$$\sigma = 31.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.79 \pm 0.22 \%$$

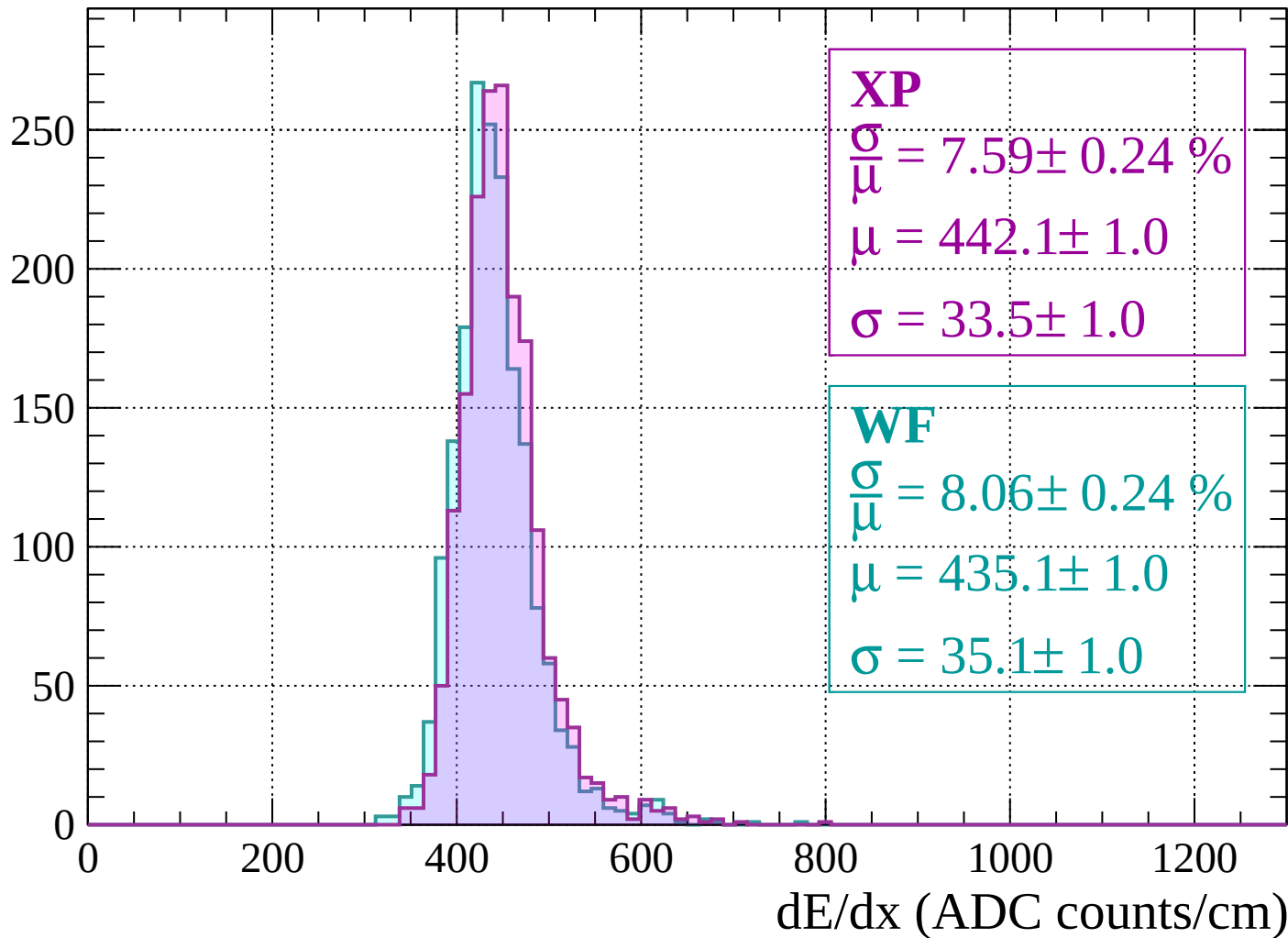
$$\mu = 416.6 \pm 0.9$$

$$\sigma = 32.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

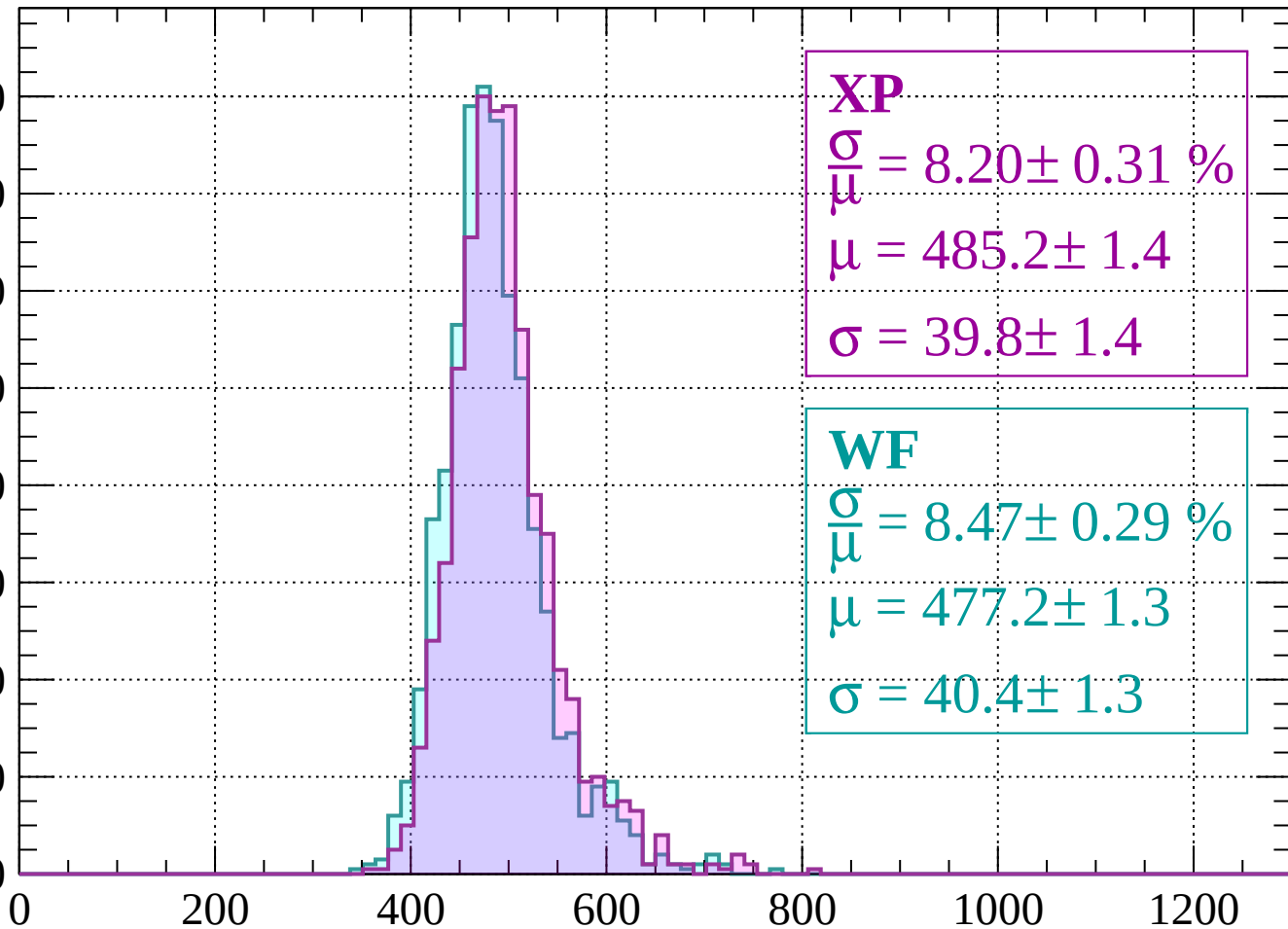
Count



Energy loss | $-160 < p < -120$

Count

160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.20 \pm 0.31 \%$$

$$\mu = 485.2 \pm 1.4$$

$$\sigma = 39.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.47 \pm 0.29 \%$$

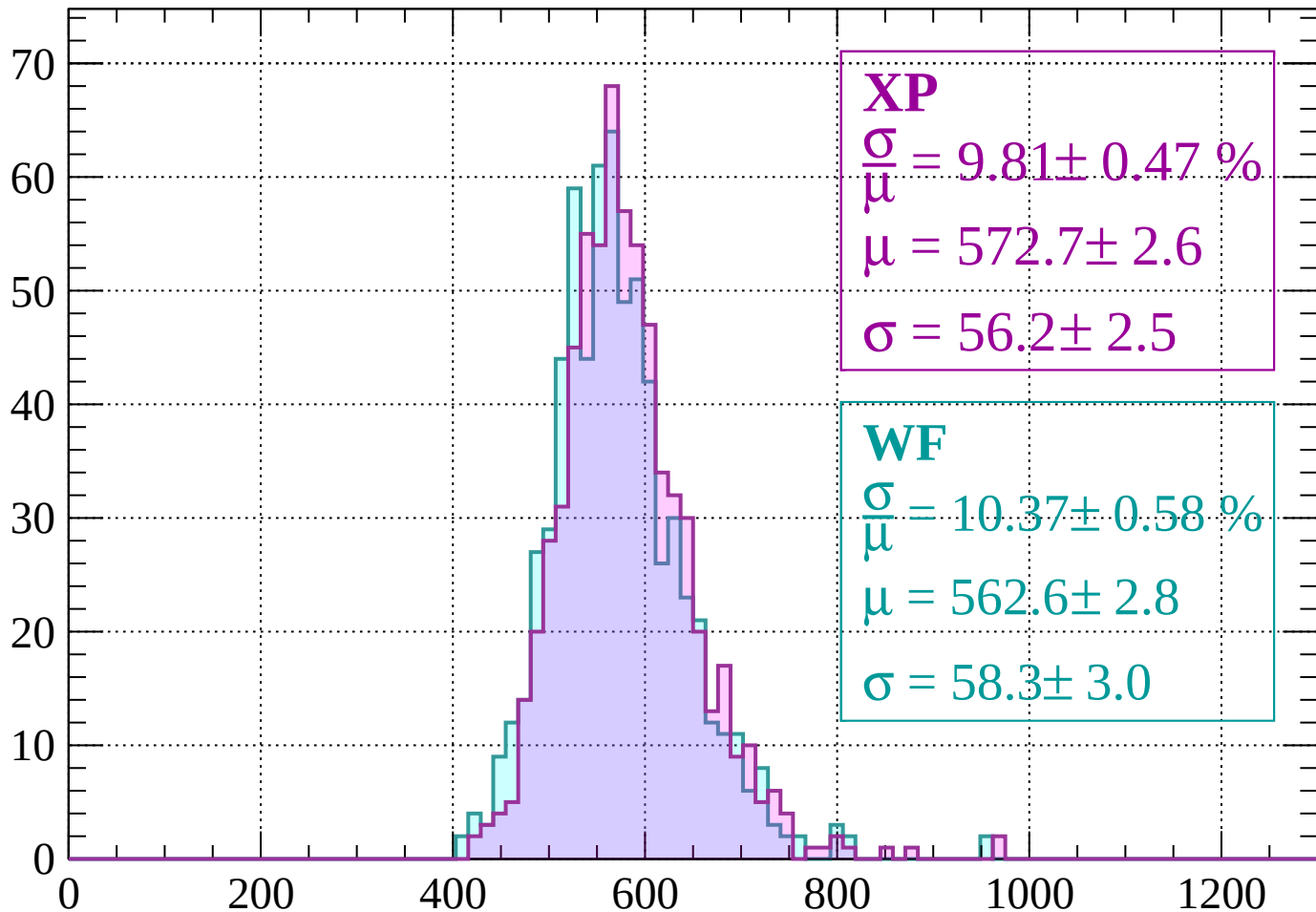
$$\mu = 477.2 \pm 1.3$$

$$\sigma = 40.4 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -80$

Count



XP

$$\frac{\sigma}{\mu} = 9.81 \pm 0.47 \%$$

$$\mu = 572.7 \pm 2.6$$

$$\sigma = 56.2 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 10.37 \pm 0.58 \%$$

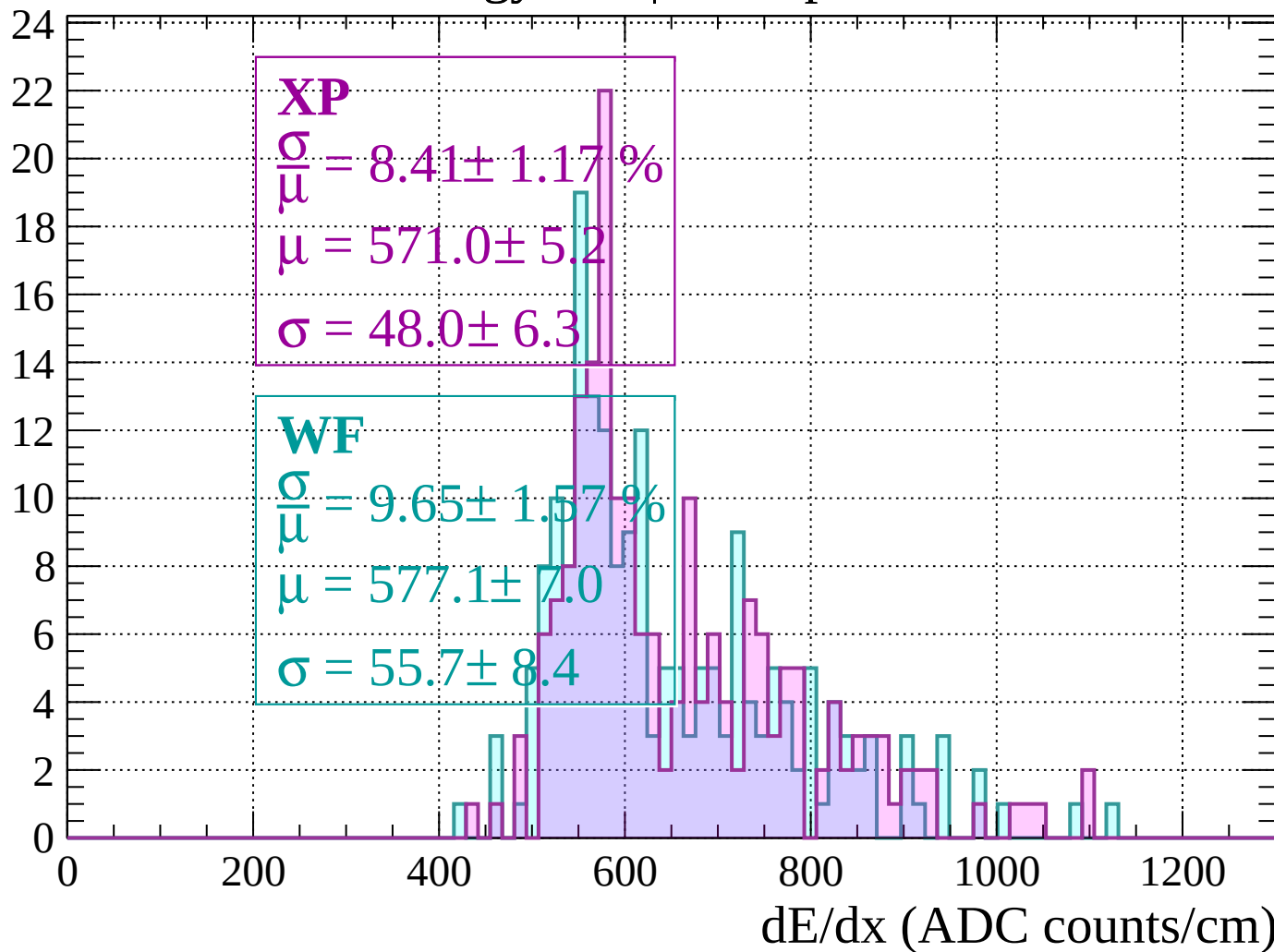
$$\mu = 562.6 \pm 2.8$$

$$\sigma = 58.3 \pm 3.0$$

dE/dx (ADC counts/cm)

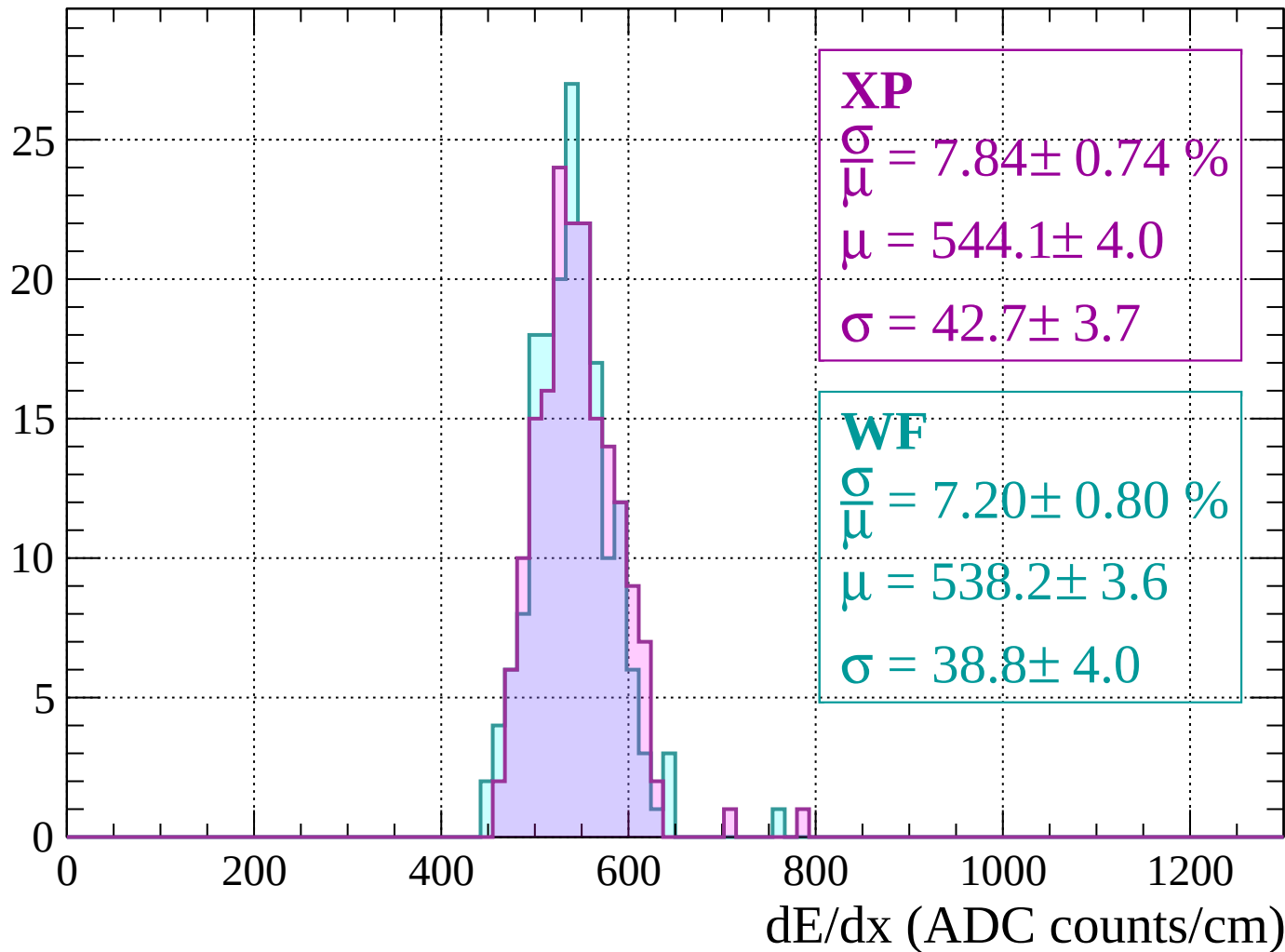
Energy loss | -80 < p < -40

Count



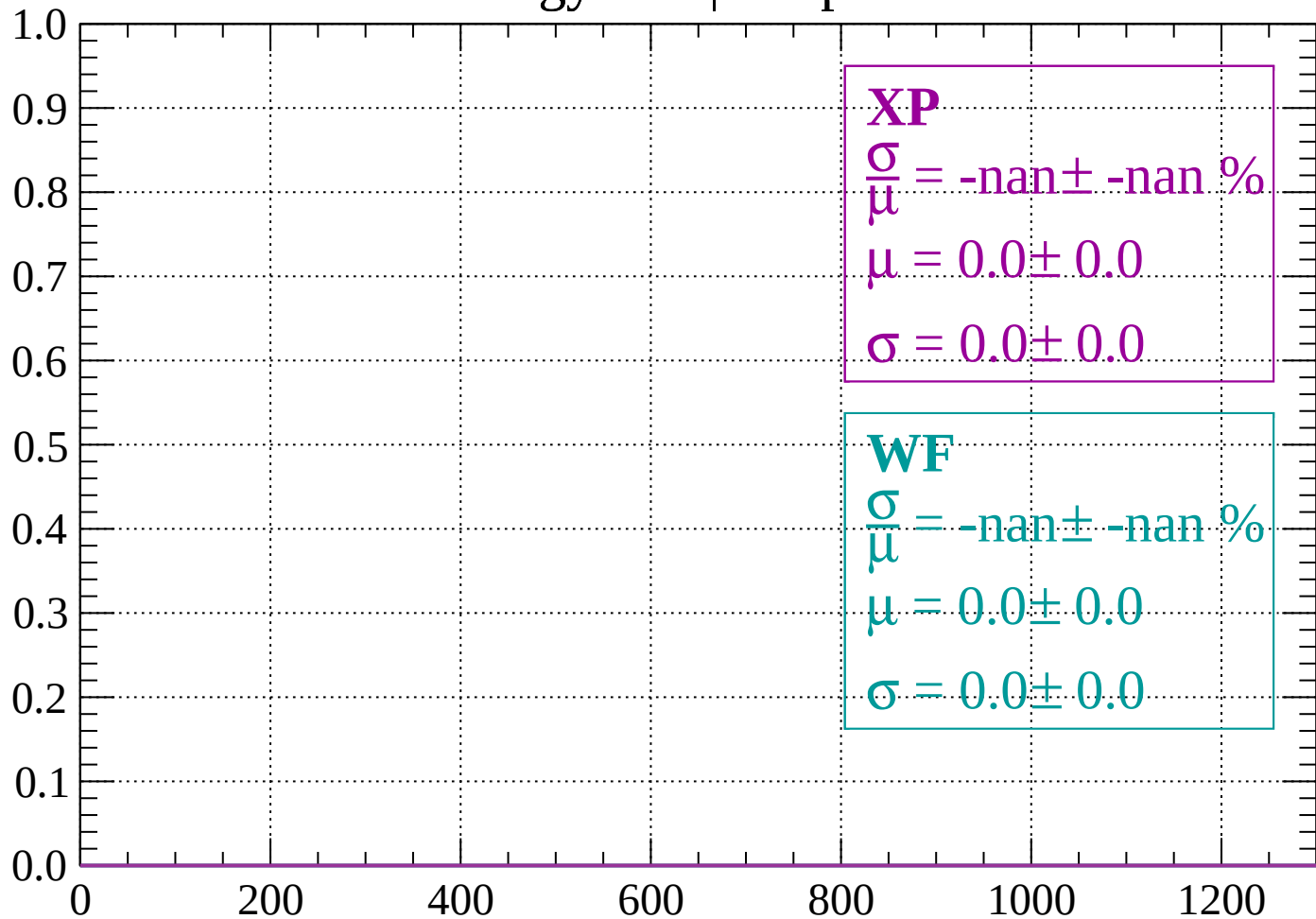
Energy loss | $-40 < p < 0$

Count



Energy loss | $0 < p < 40$

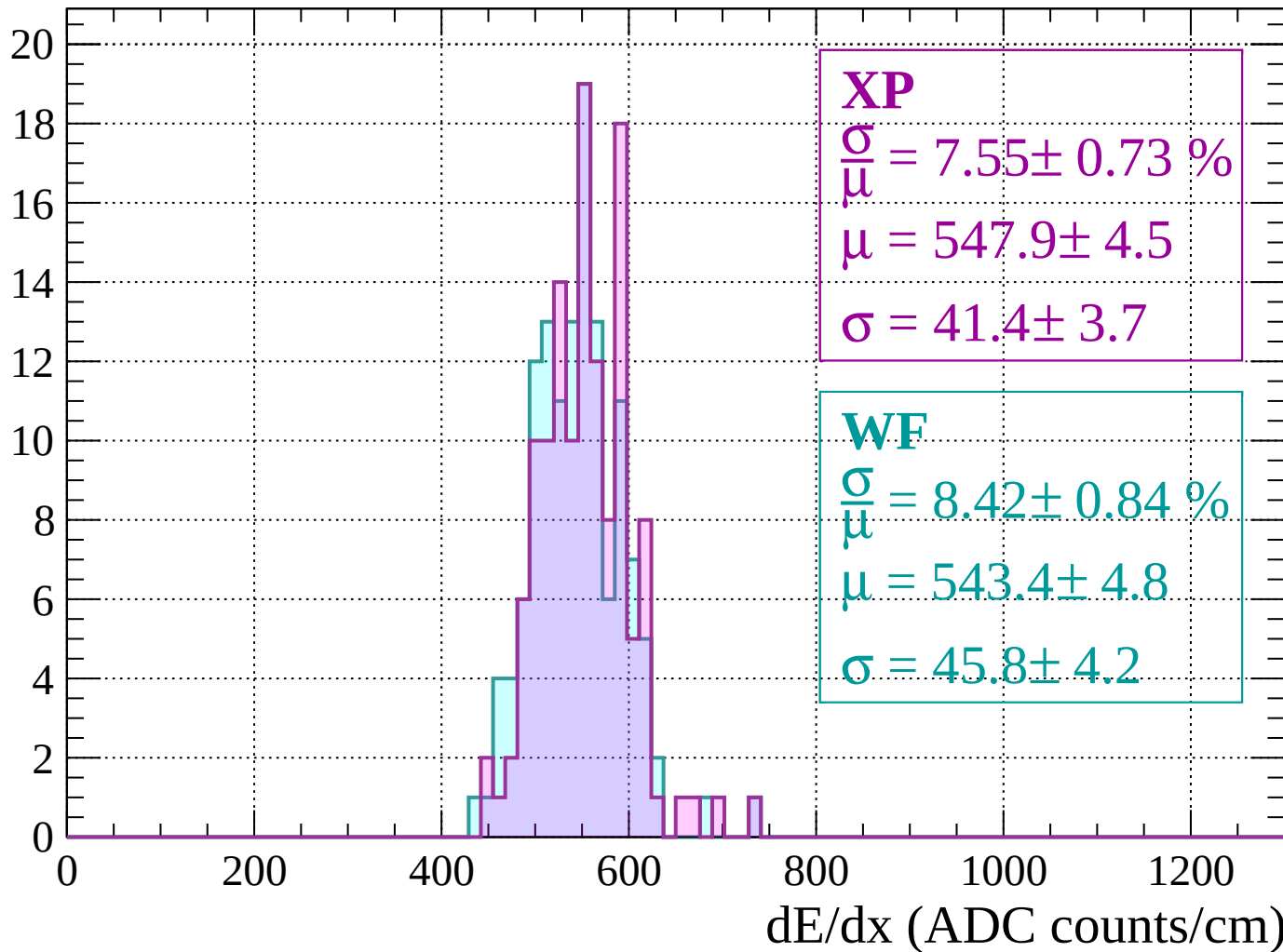
Count



dE/dx (ADC counts/cm)

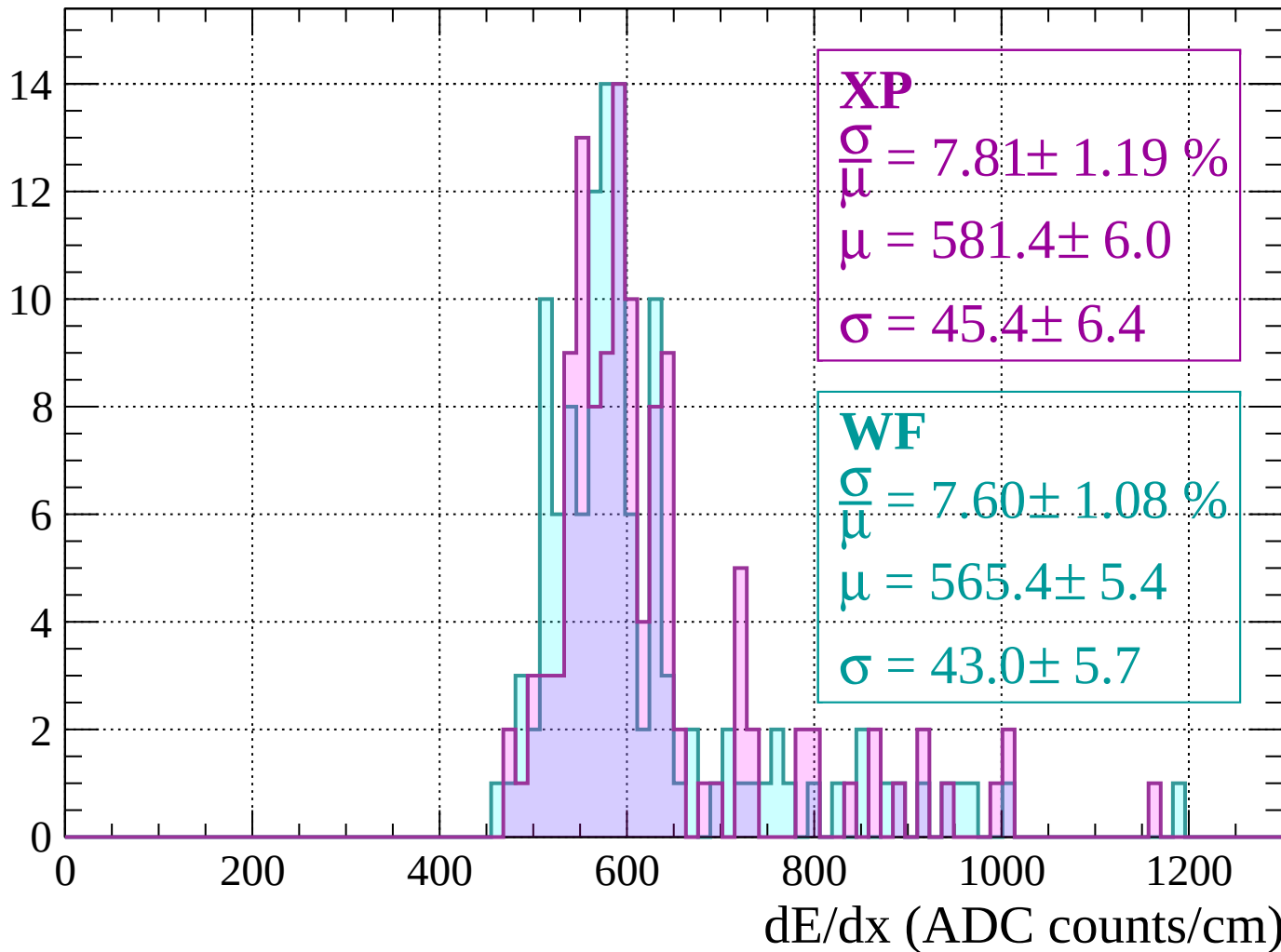
Energy loss | $40 < p < 80$

Count



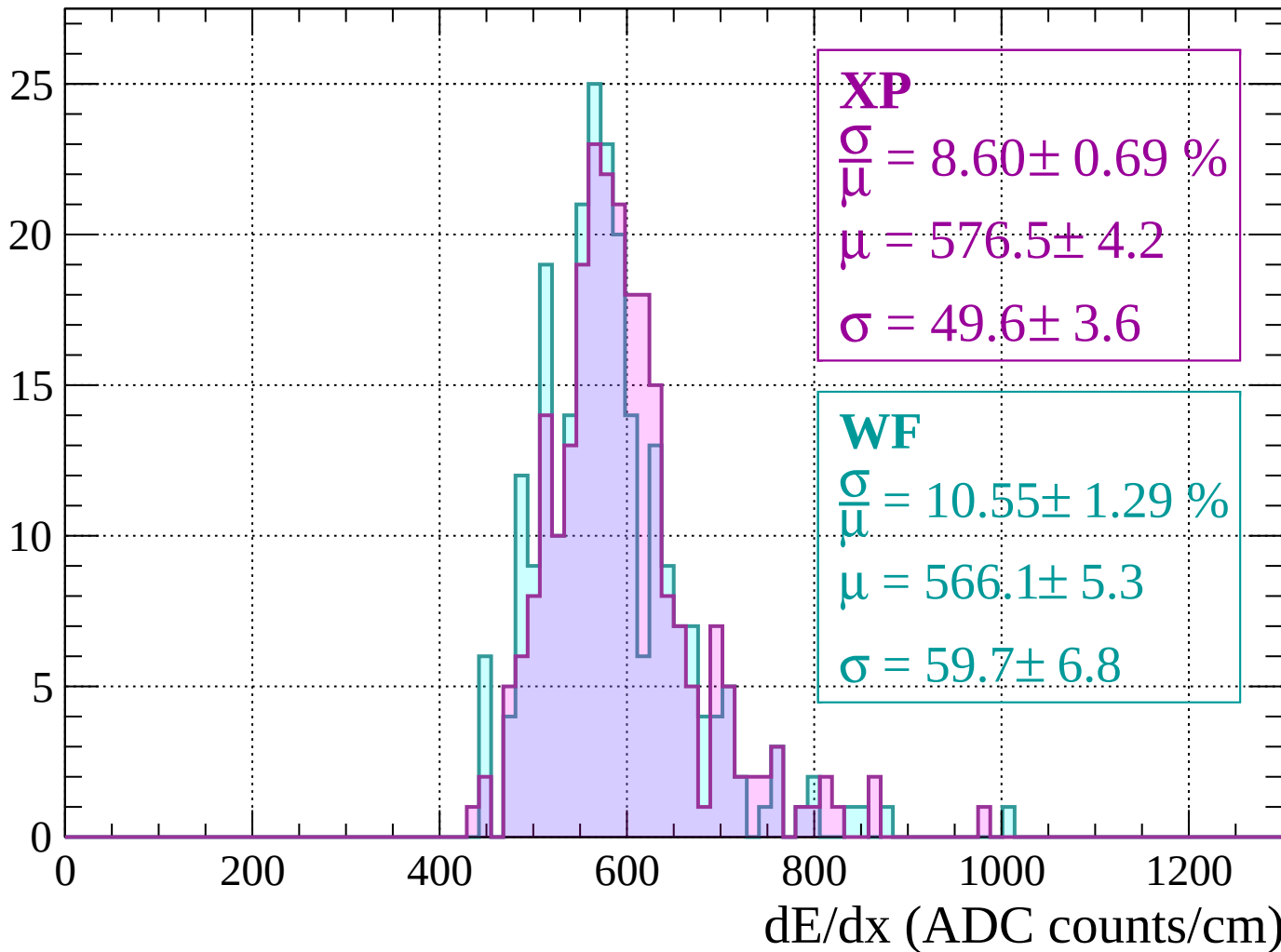
Energy loss | $80 < p < 120$

Count



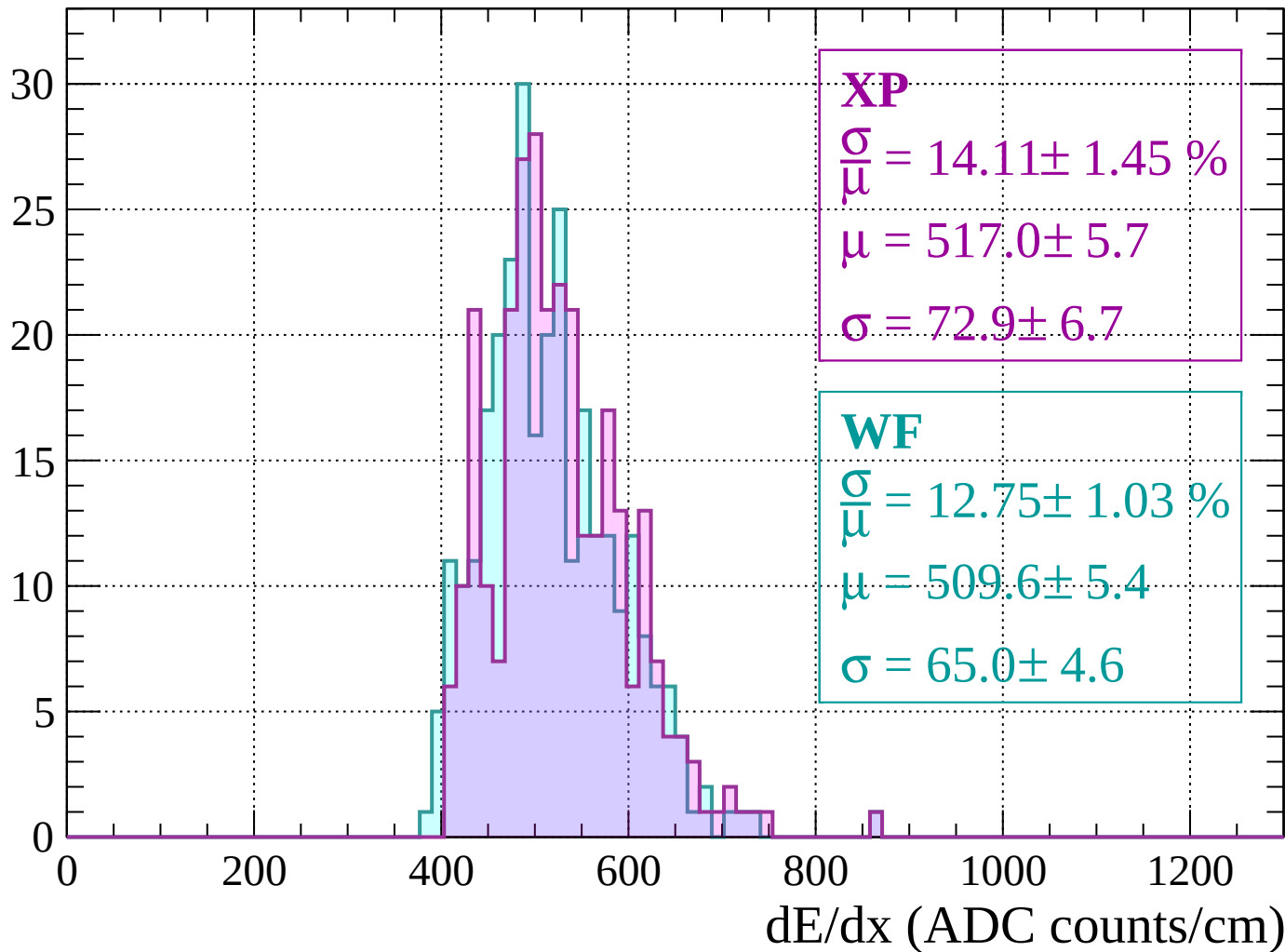
Energy loss | $120 < p < 160$

Count



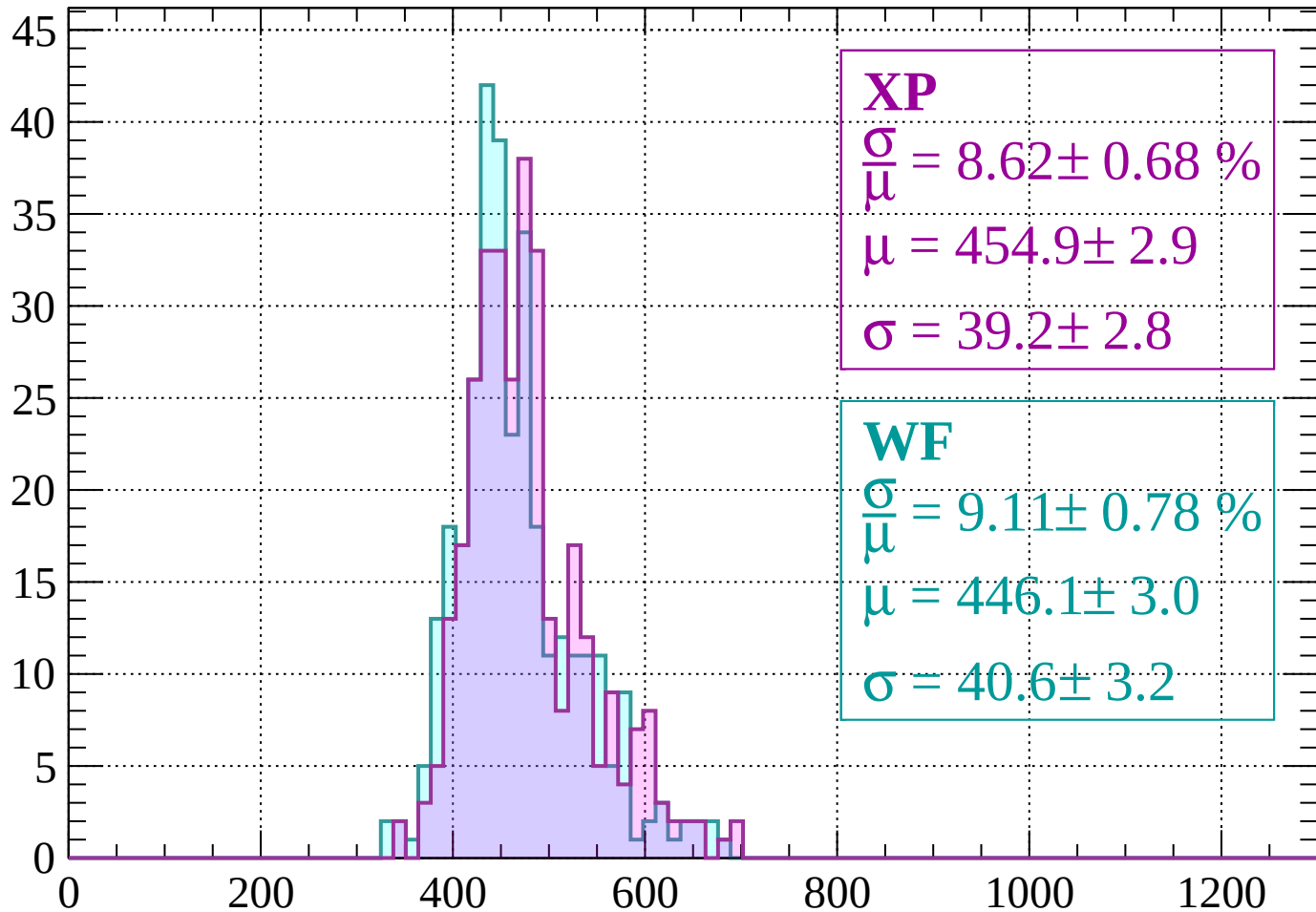
Energy loss | $160 < p < 200$

Count



Energy loss | $200 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 8.62 \pm 0.68 \%$$

$$\mu = 454.9 \pm 2.9$$

$$\sigma = 39.2 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 9.11 \pm 0.78 \%$$

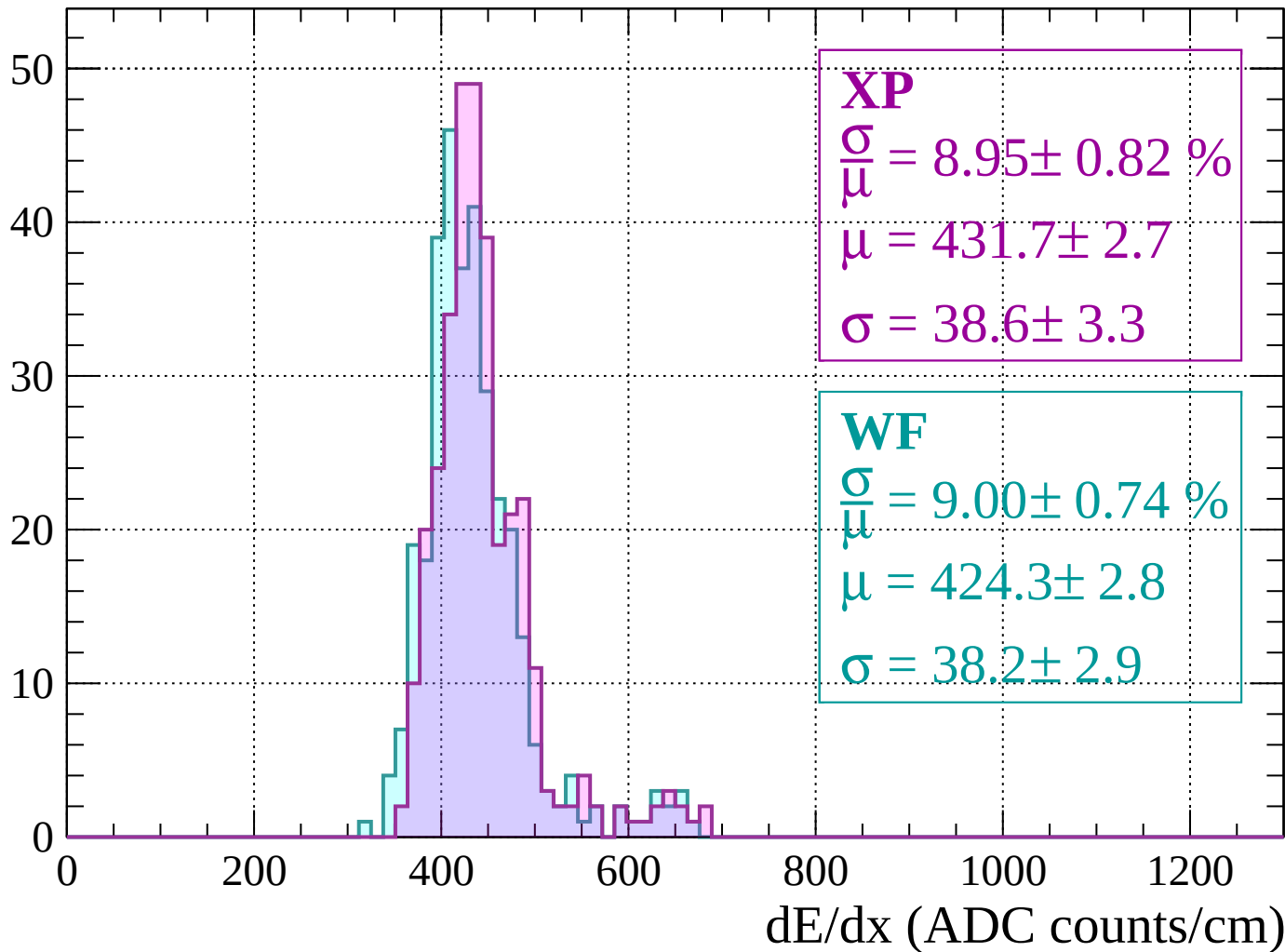
$$\mu = 446.1 \pm 3.0$$

$$\sigma = 40.6 \pm 3.2$$

dE/dx (ADC counts/cm)

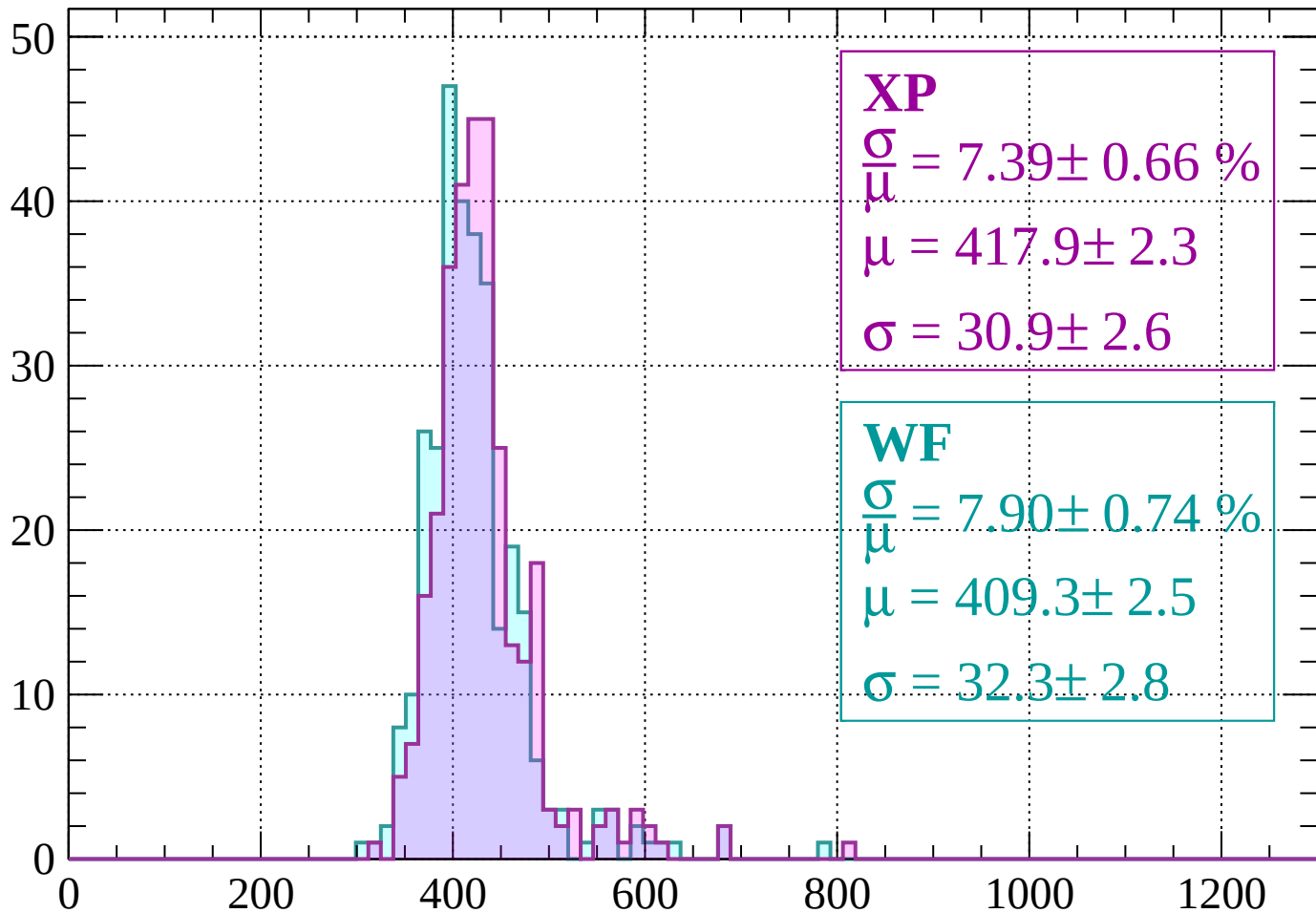
Energy loss | $240 < p < 280$

Count



Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 7.39 \pm 0.66 \%$$

$$\mu = 417.9 \pm 2.3$$

$$\sigma = 30.9 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 7.90 \pm 0.74 \%$$

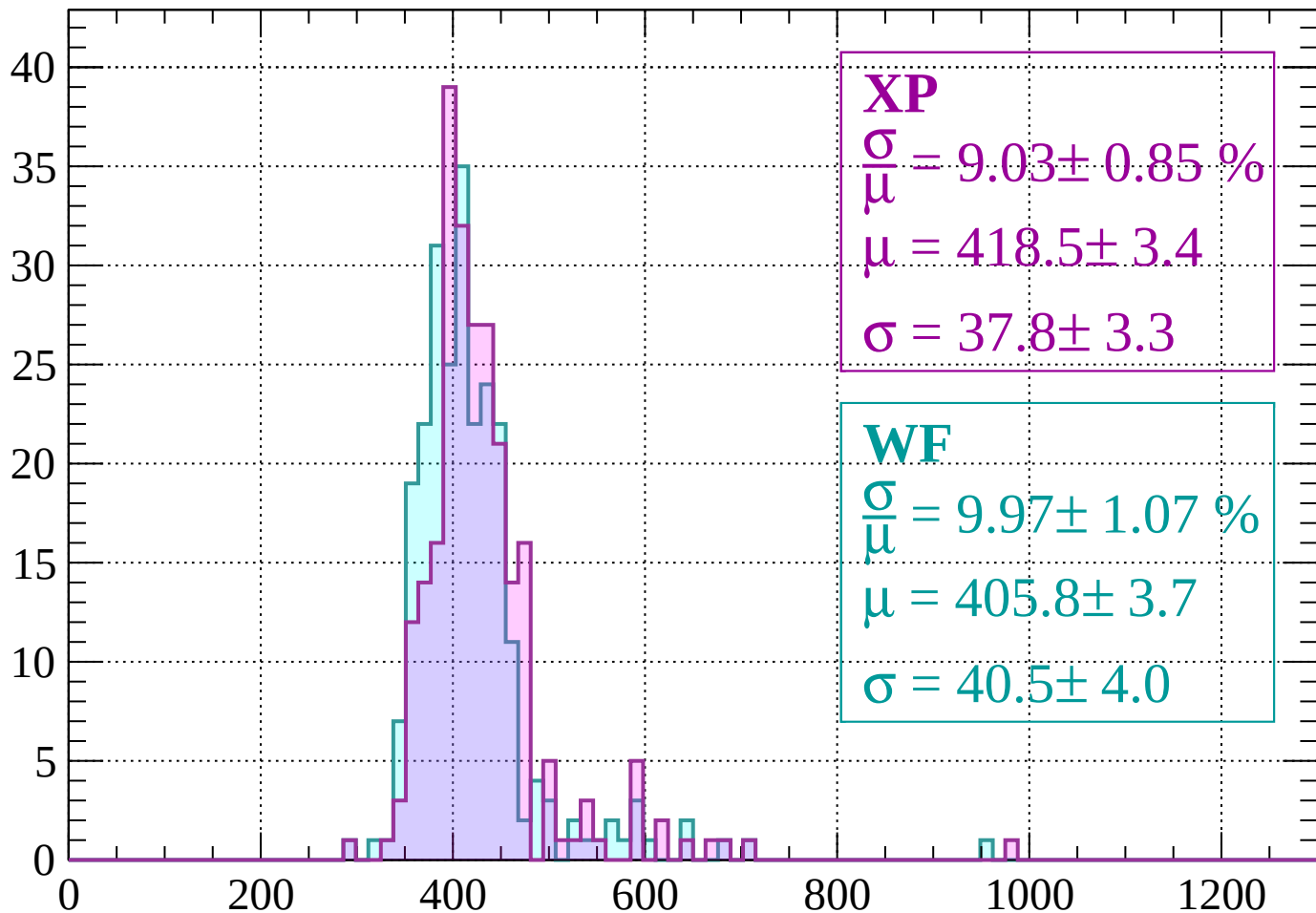
$$\mu = 409.3 \pm 2.5$$

$$\sigma = 32.3 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.03 \pm 0.85 \%$$

$$\mu = 418.5 \pm 3.4$$

$$\sigma = 37.8 \pm 3.3$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.97 \pm 1.07 \%$$

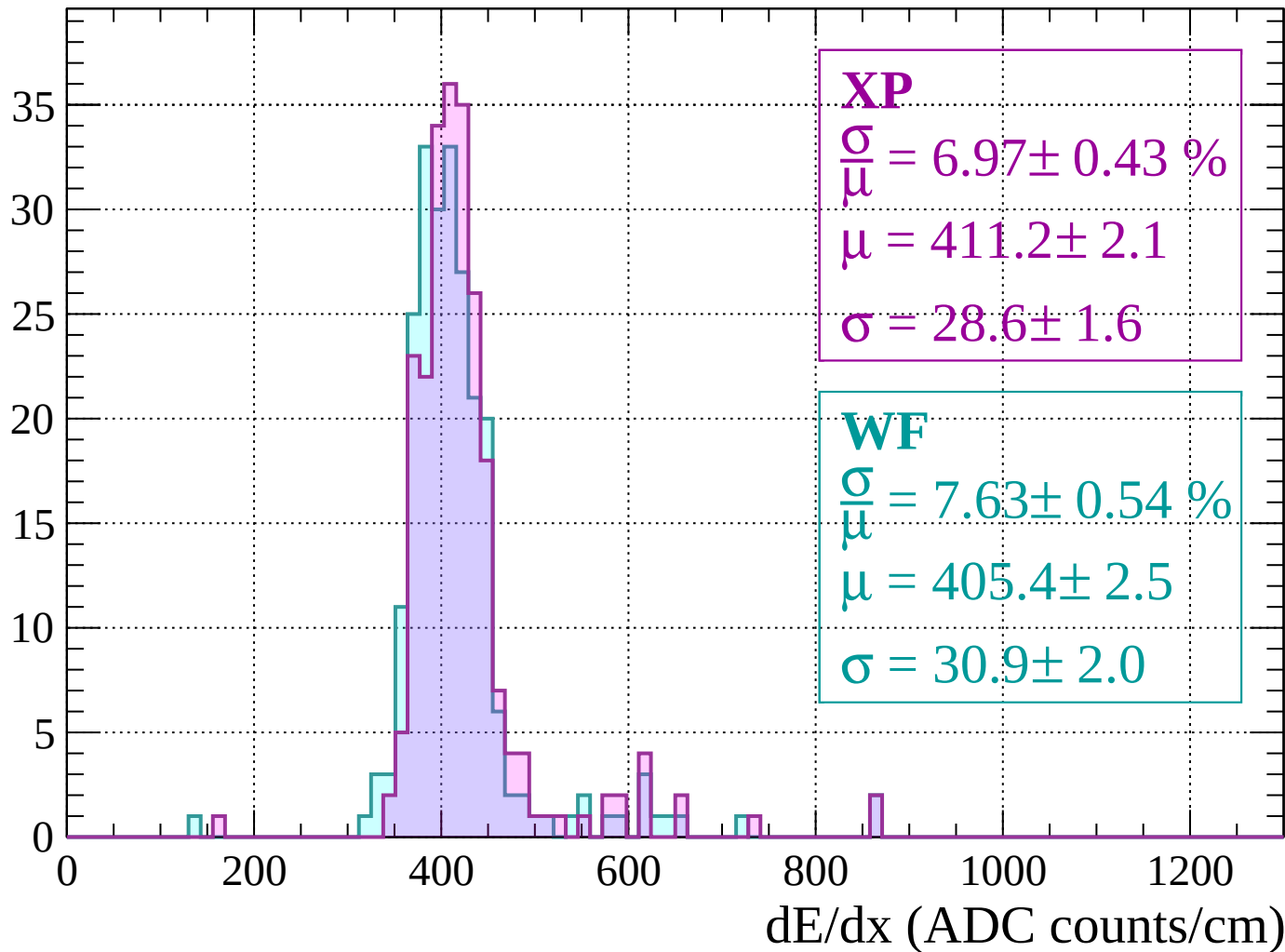
$$\mu = 405.8 \pm 3.7$$

$$\sigma = 40.5 \pm 4.0$$

dE/dx (ADC counts/cm)

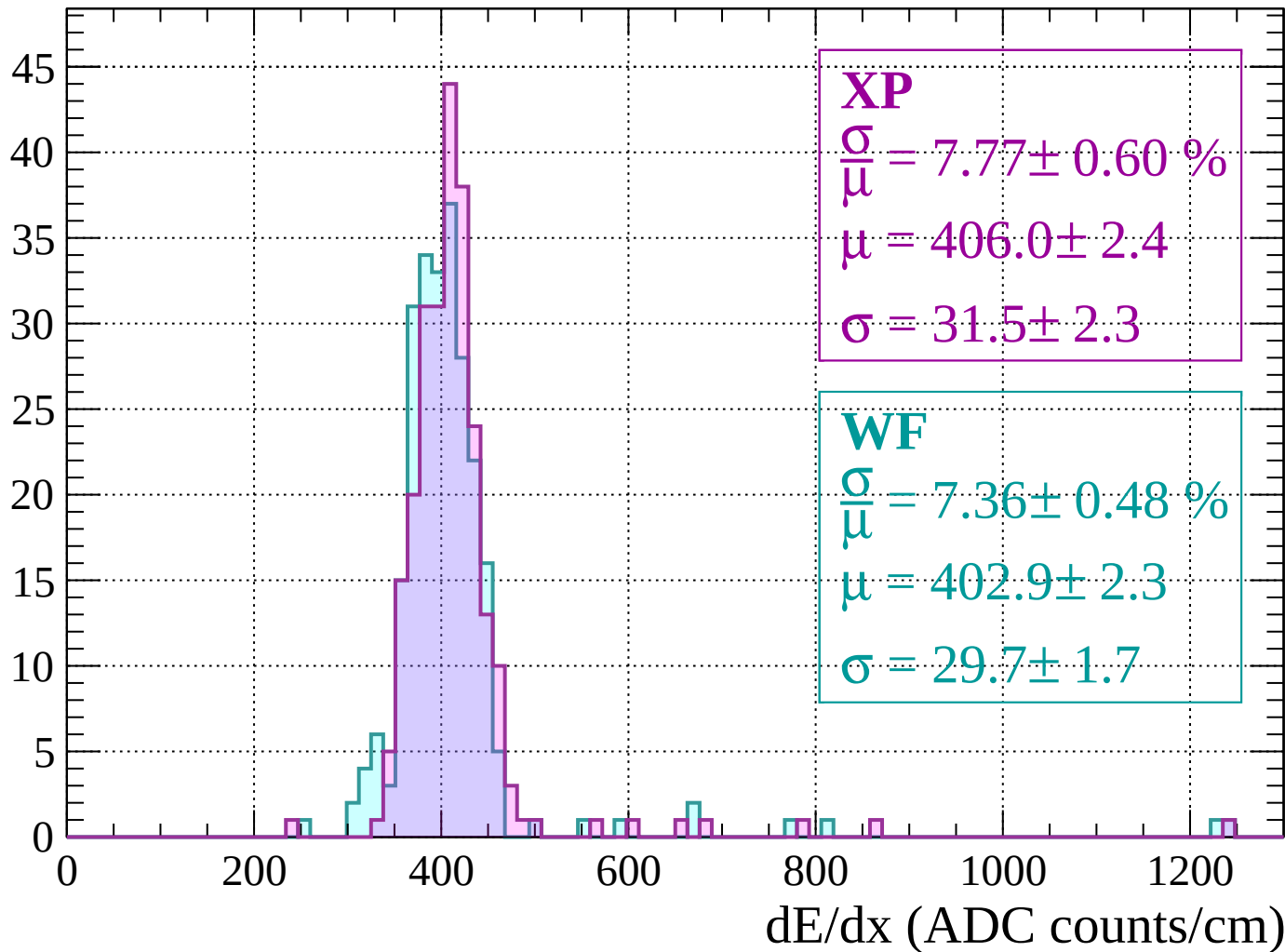
Energy loss | $360 < p < 400$

Count



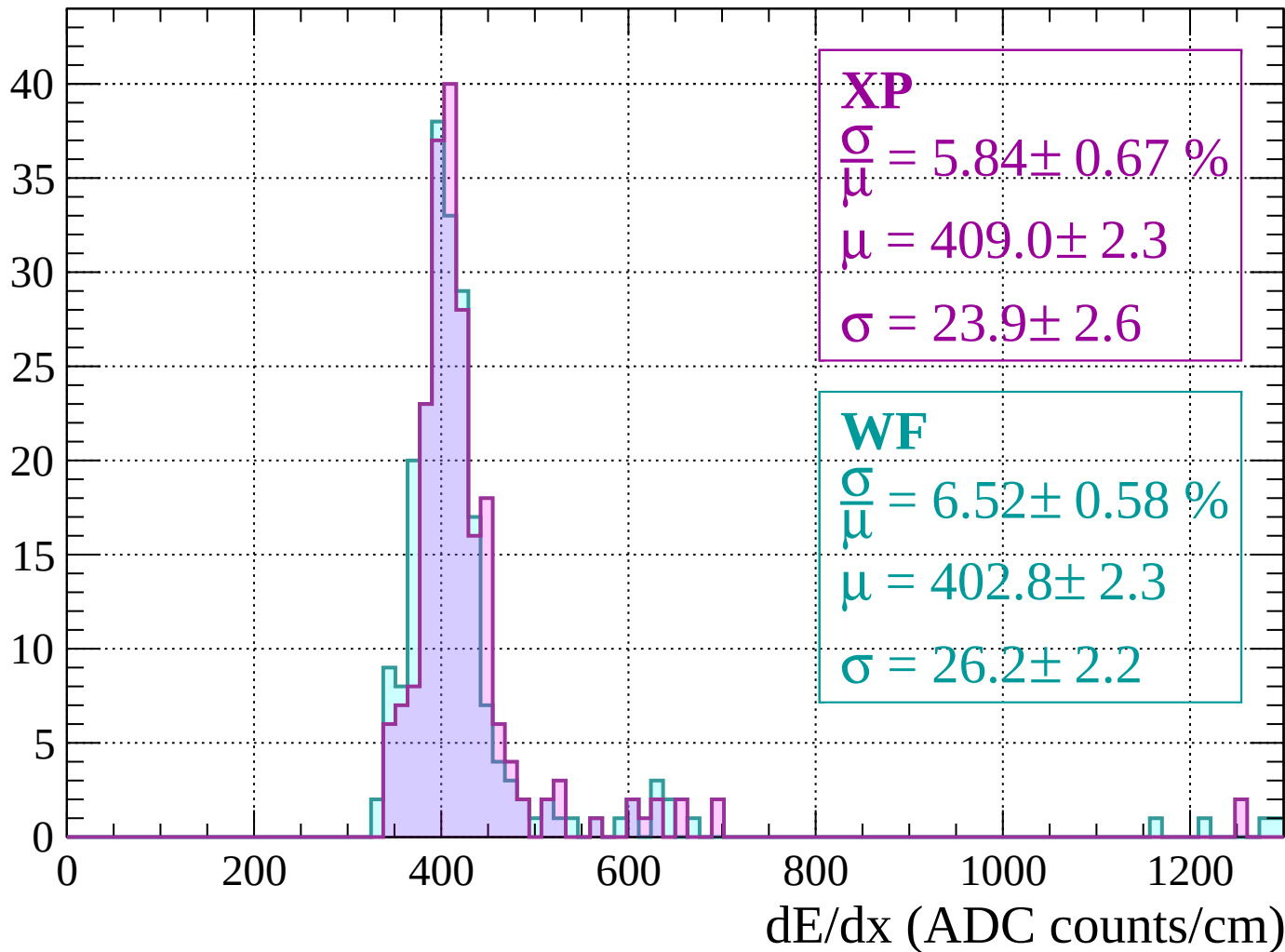
Energy loss | $400 < p < 440$

Count



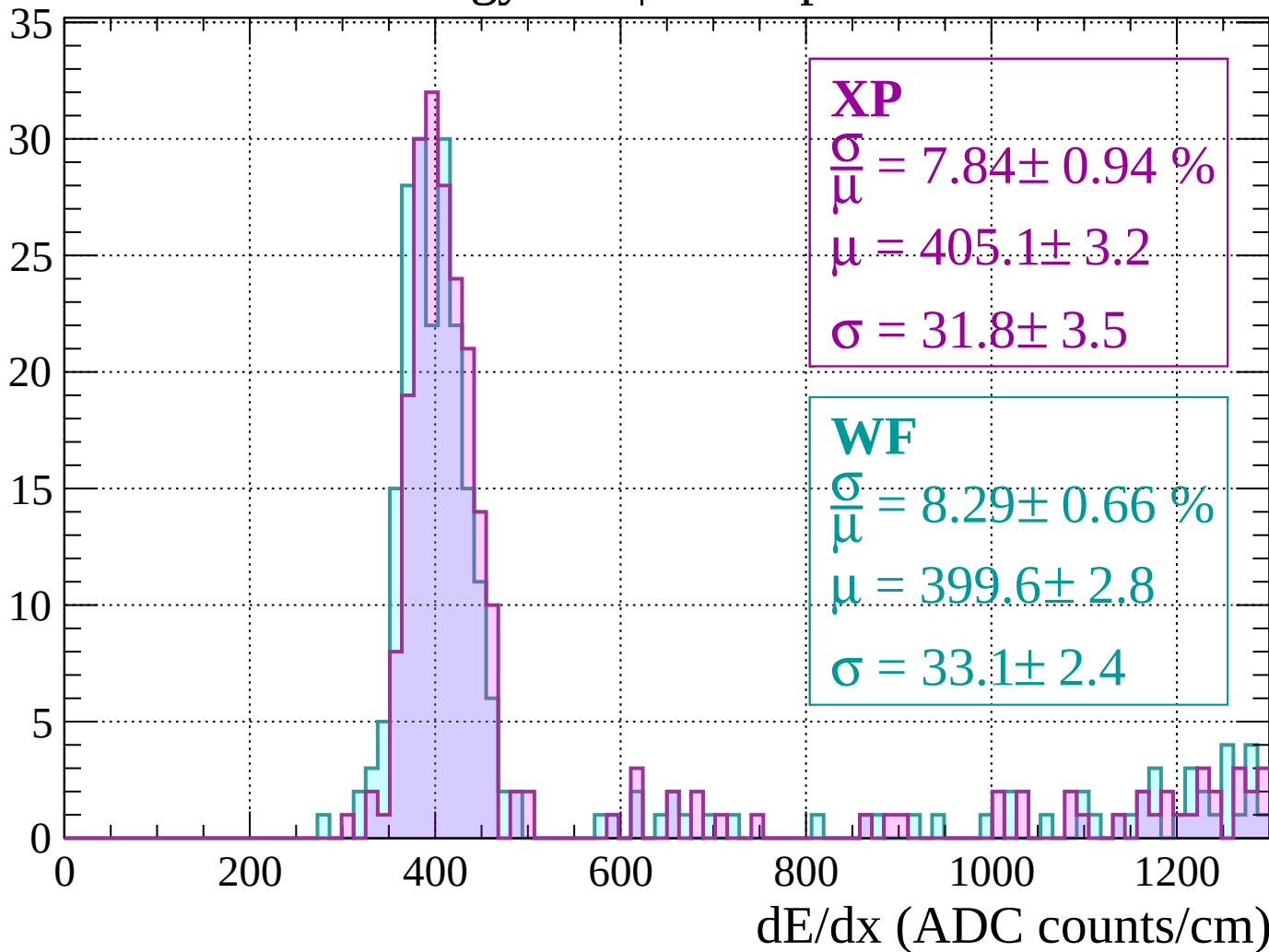
Energy loss | $440 < p < 480$

Count



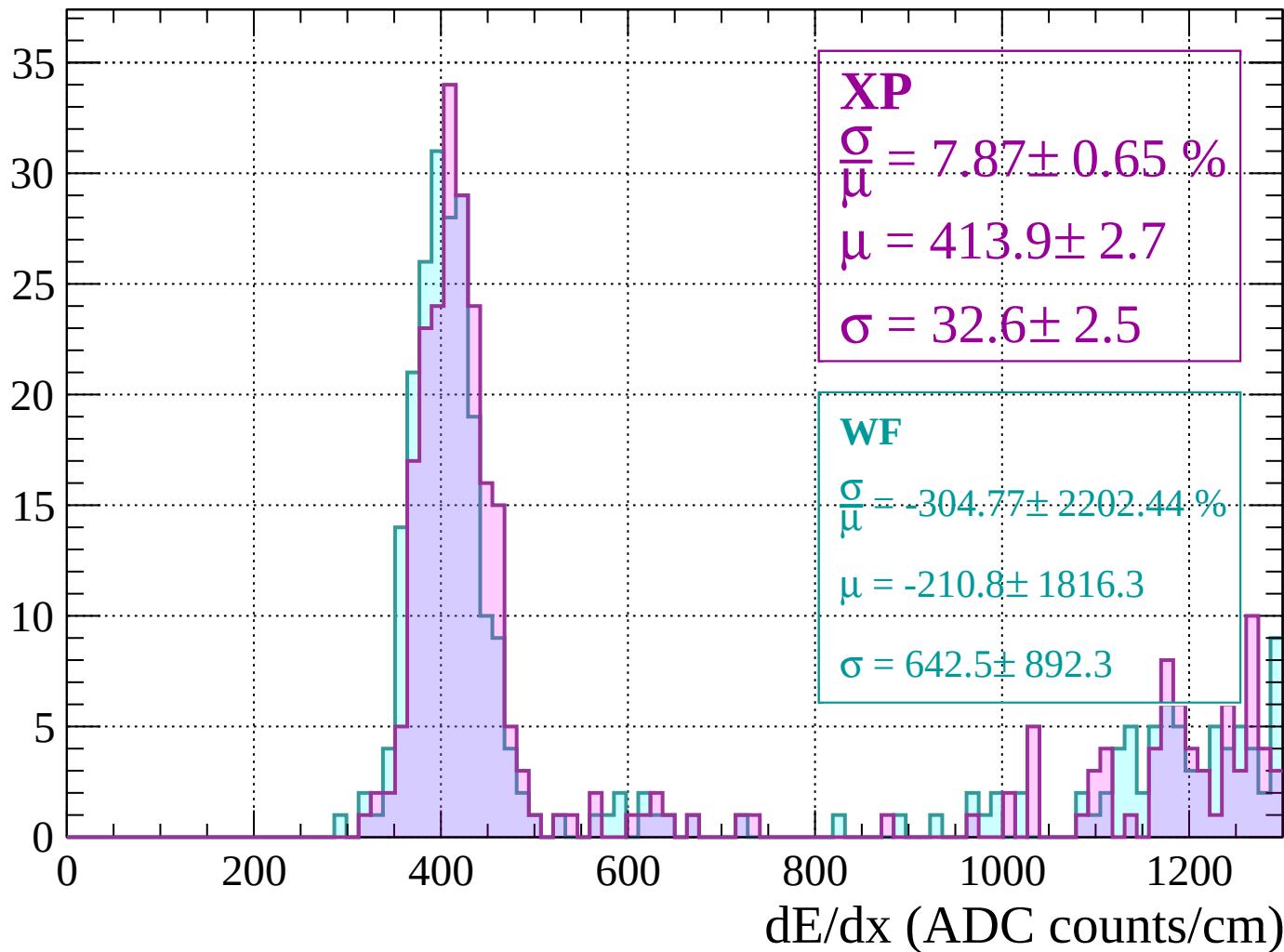
Energy loss | $480 < p < 520$

Count



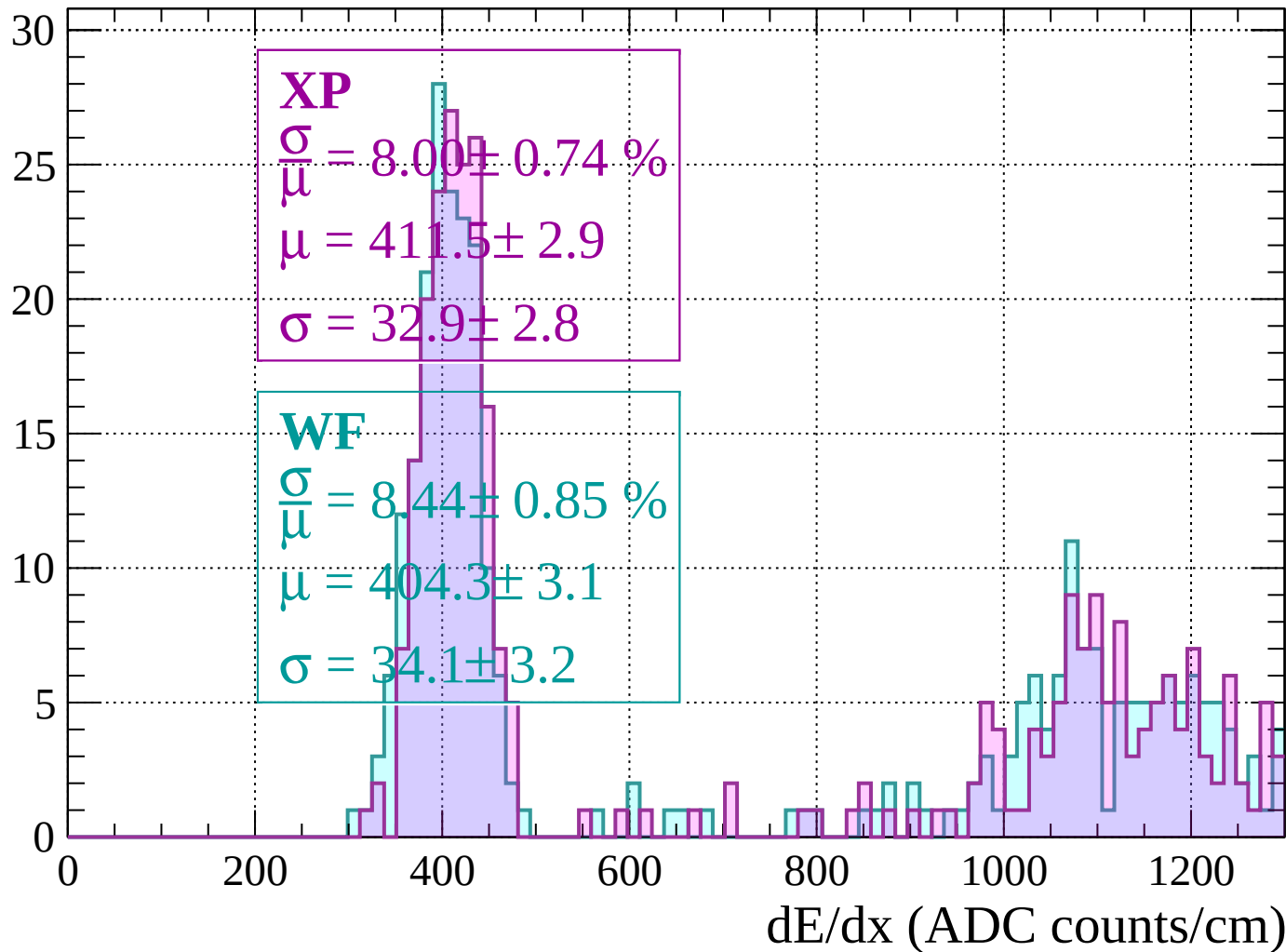
Energy loss | $520 < p < 560$

Count



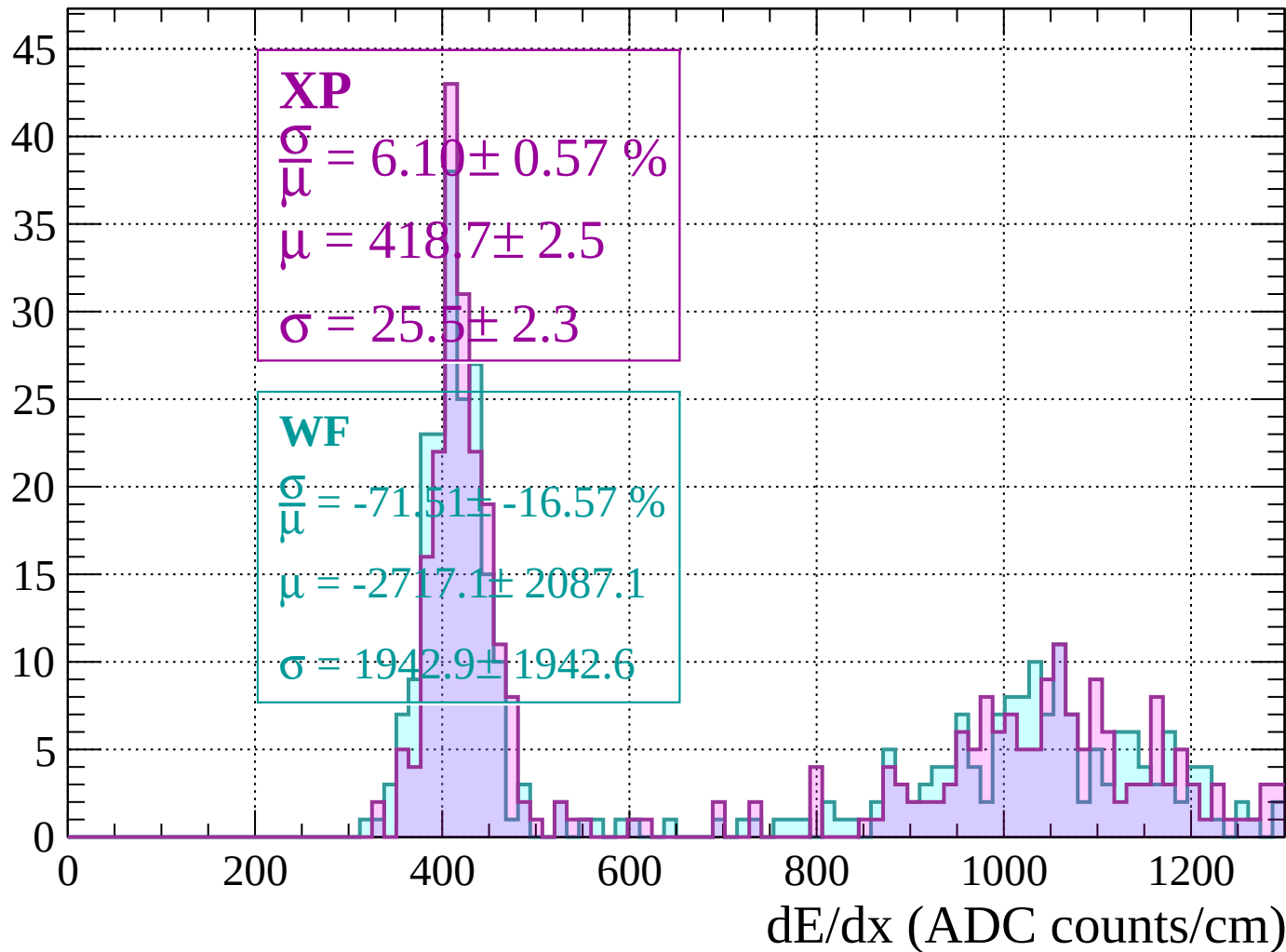
Energy loss | $560 < p < 600$

Count



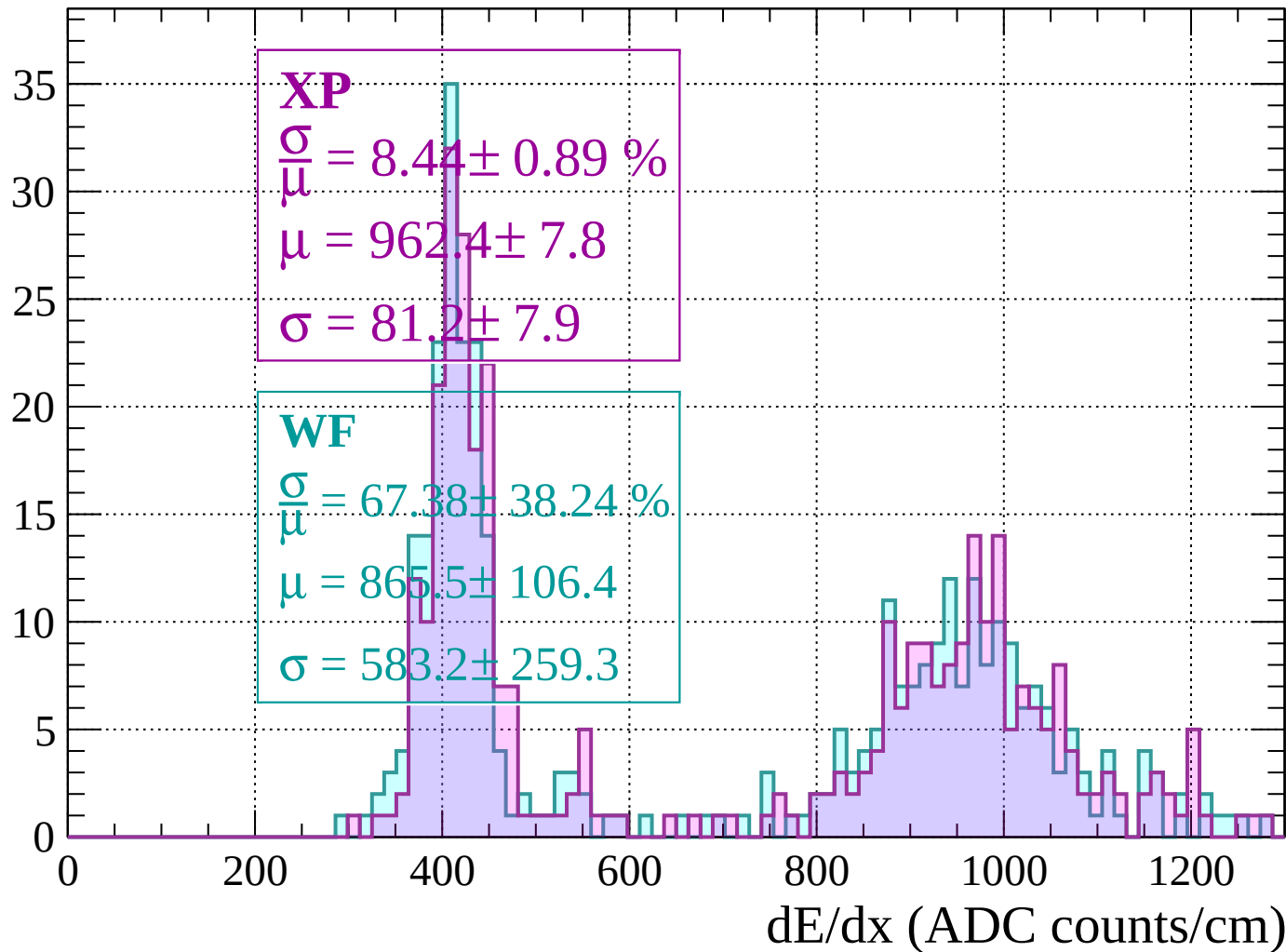
Energy loss | $600 < p < 640$

Count



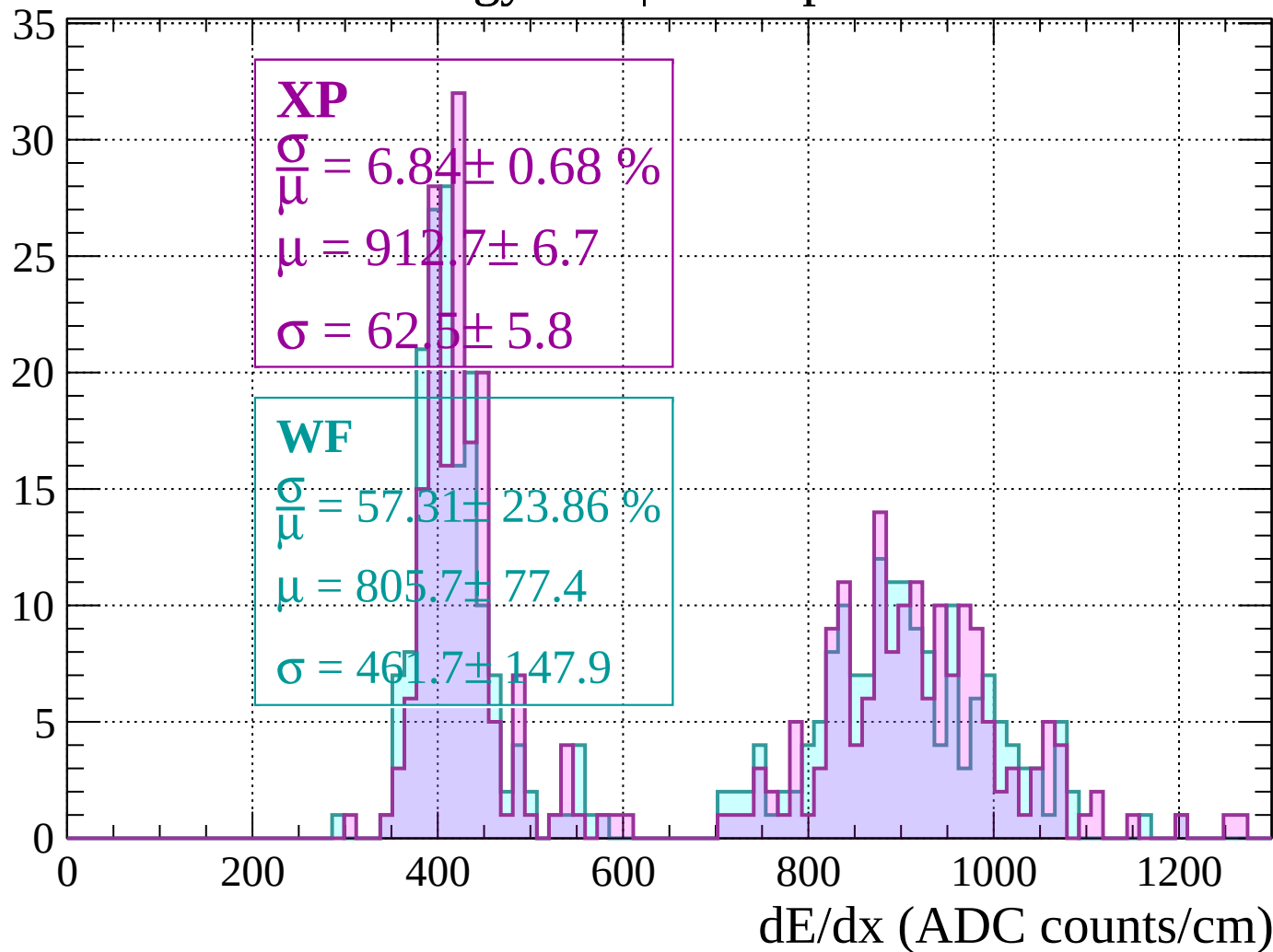
Energy loss | $640 < p < 680$

Count



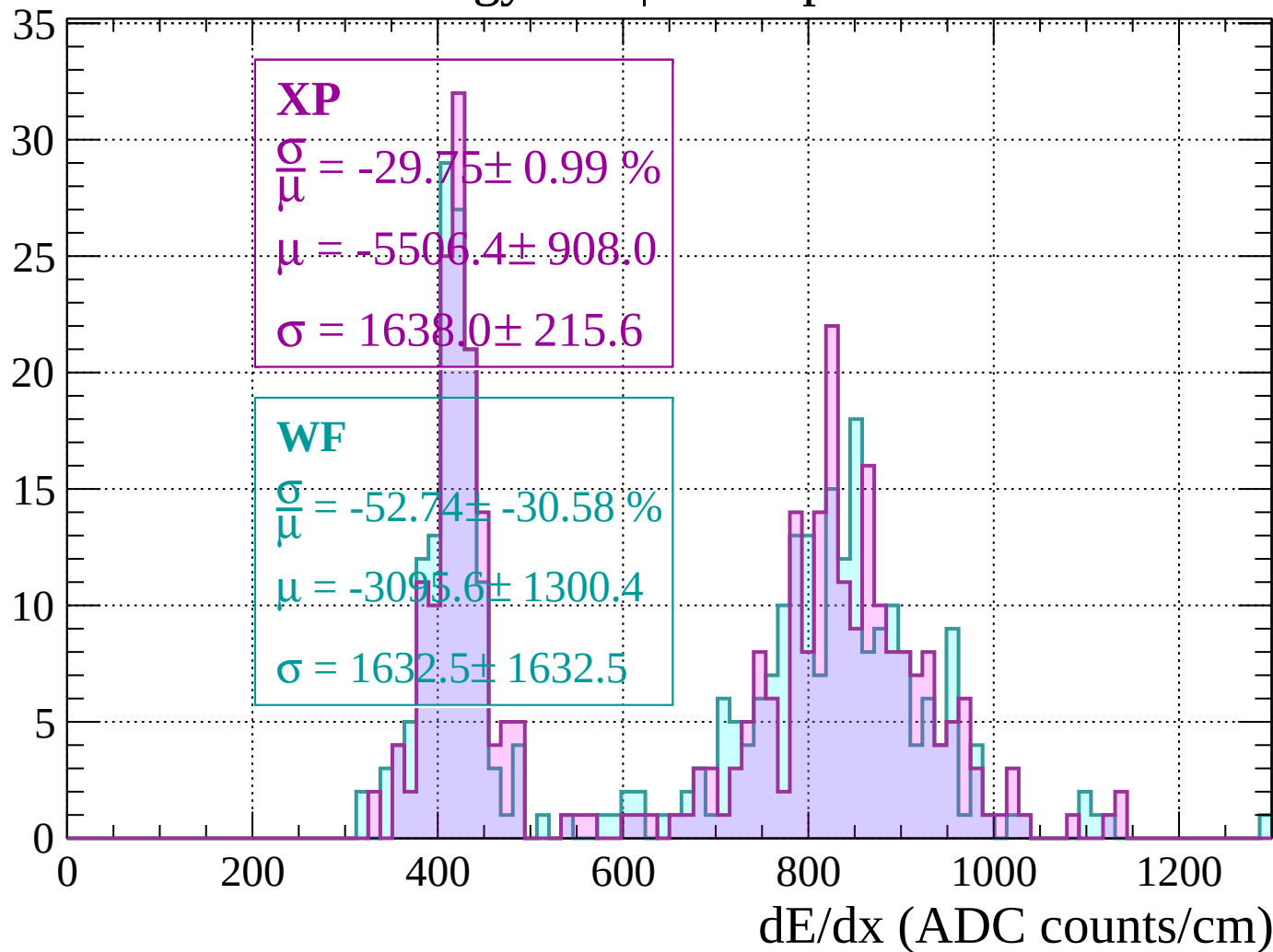
Energy loss | $680 < p < 720$

Count



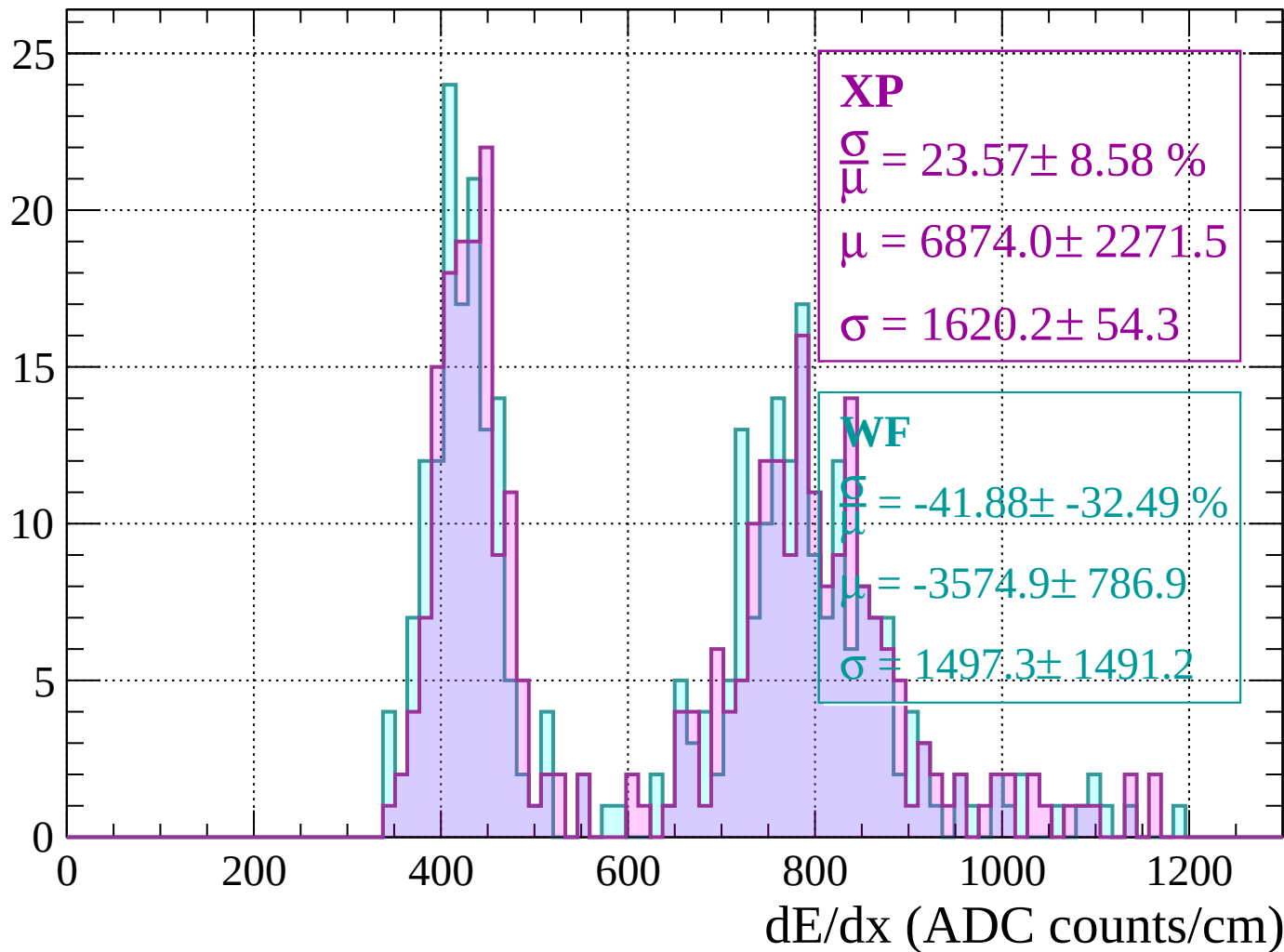
Energy loss | $720 < p < 760$

Count



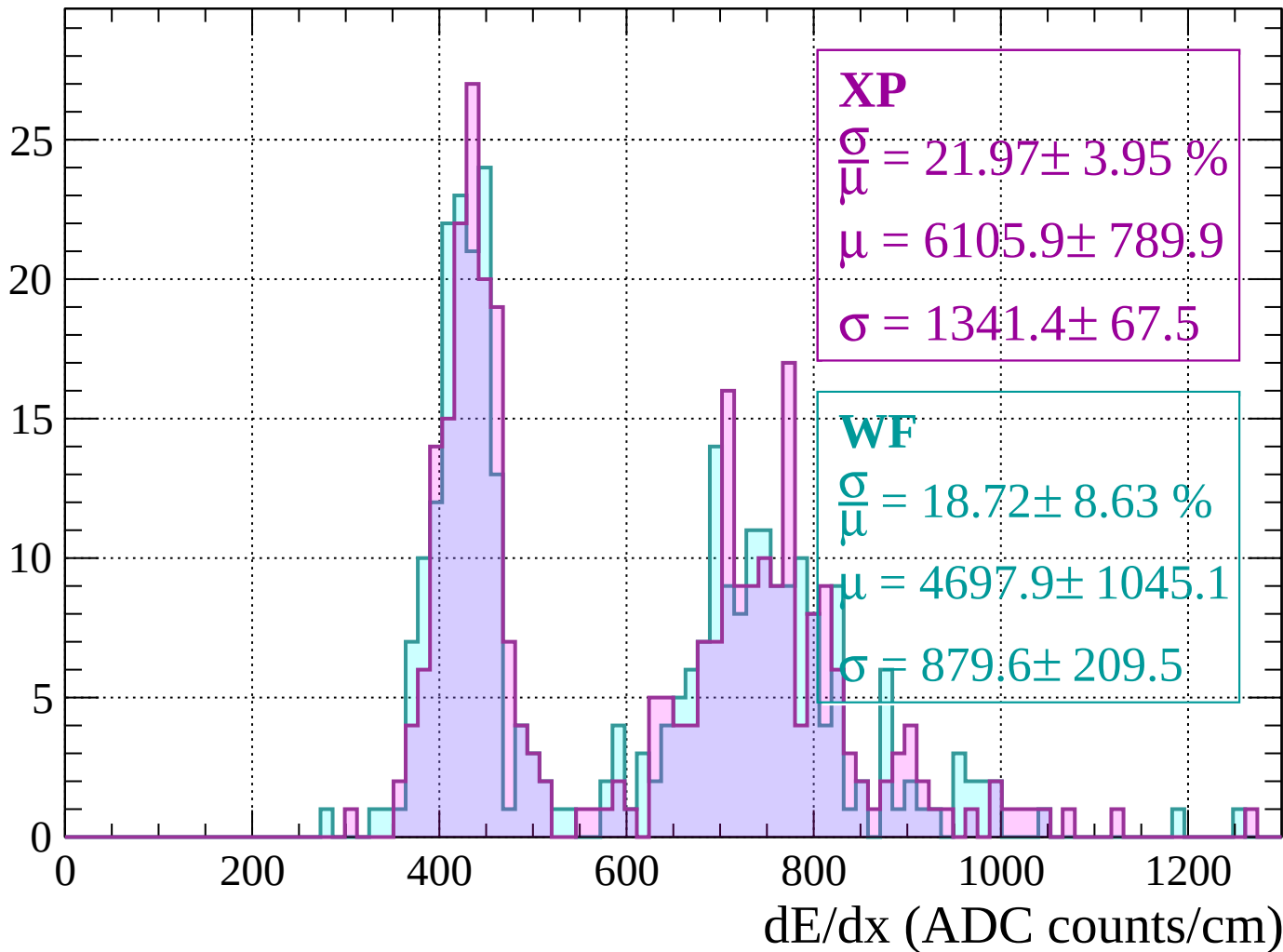
Energy loss | $760 < p < 800$

Count



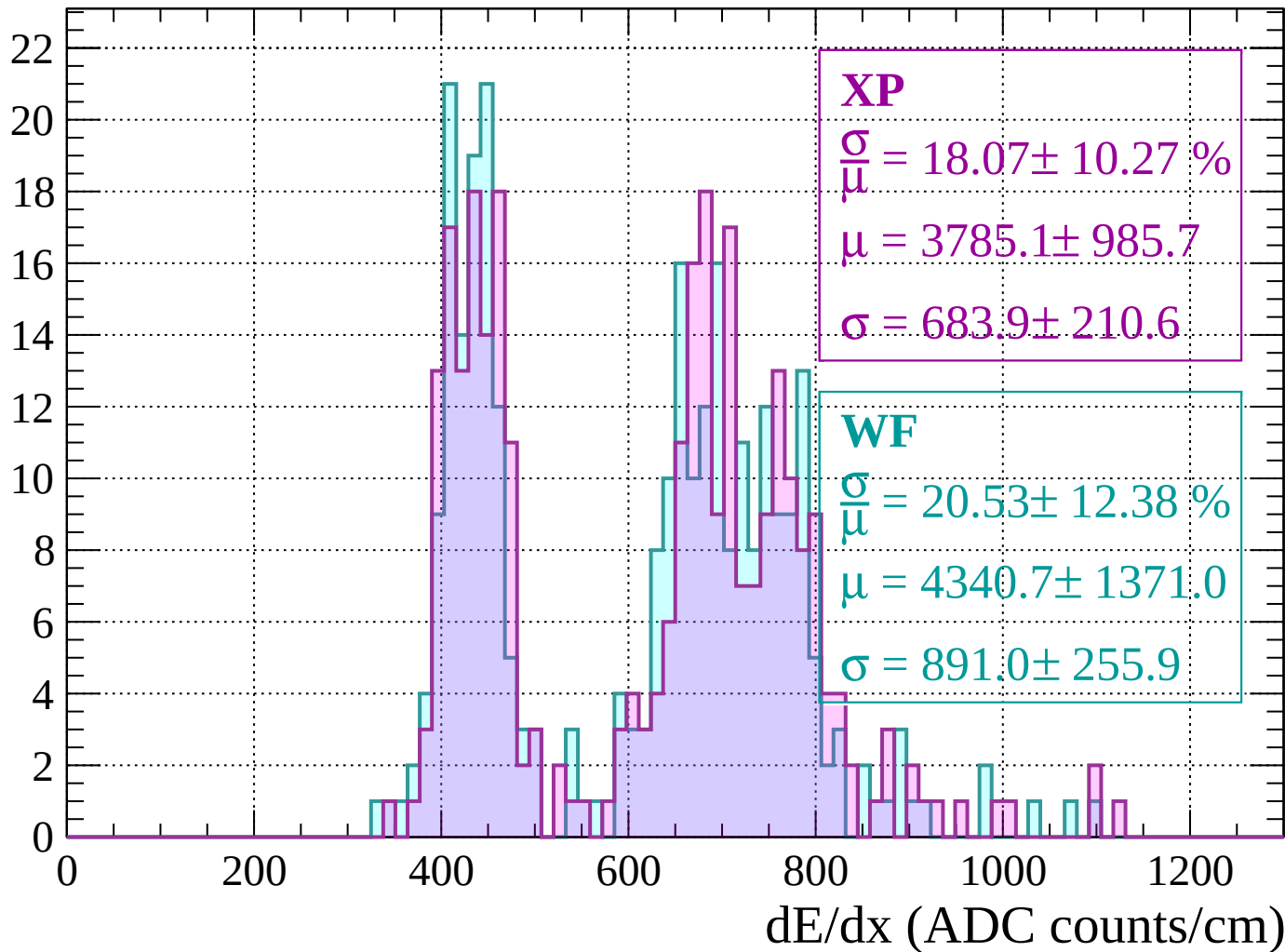
Energy loss | $800 < p < 840$

Count



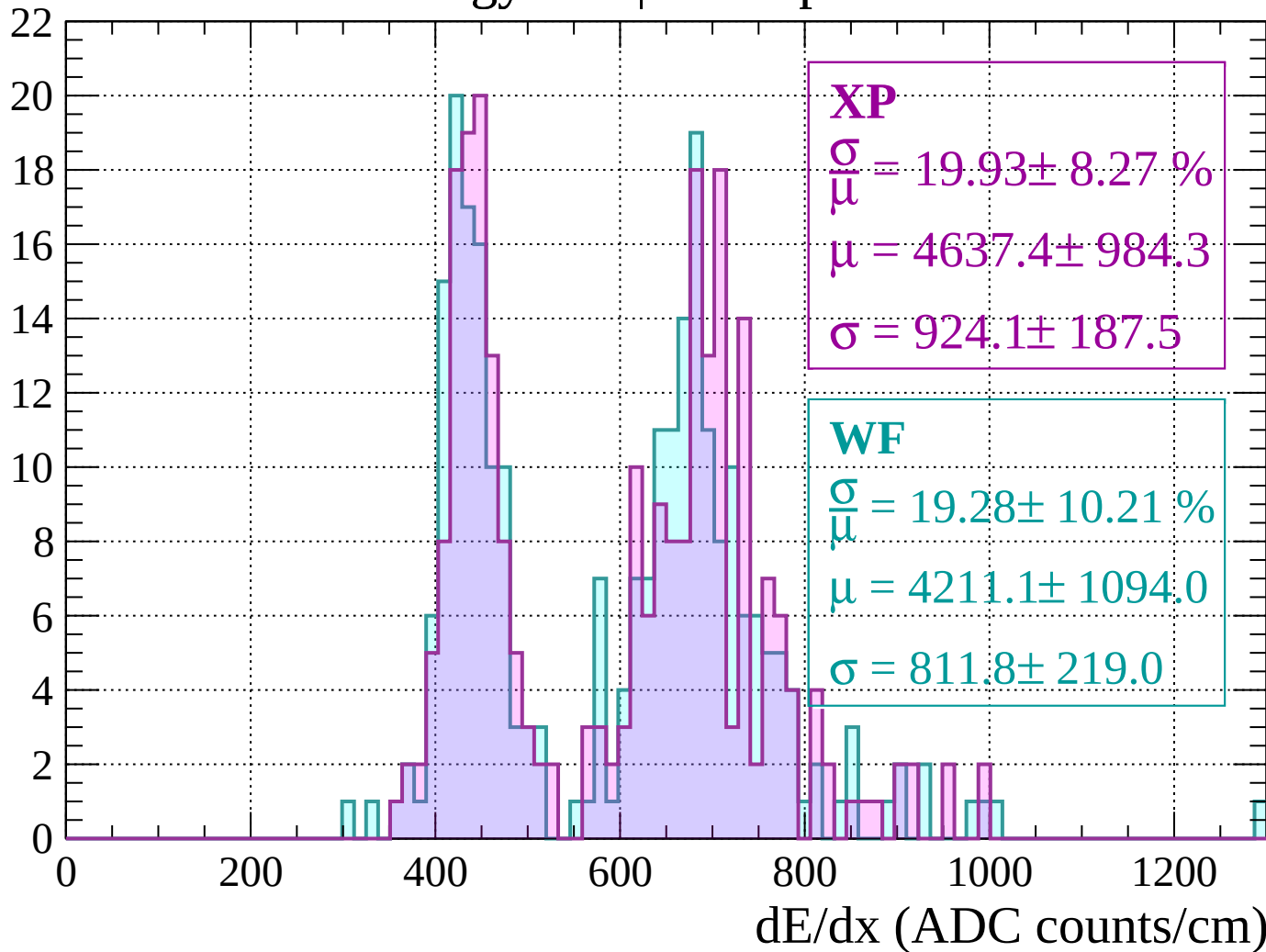
Energy loss | $840 < p < 880$

Count



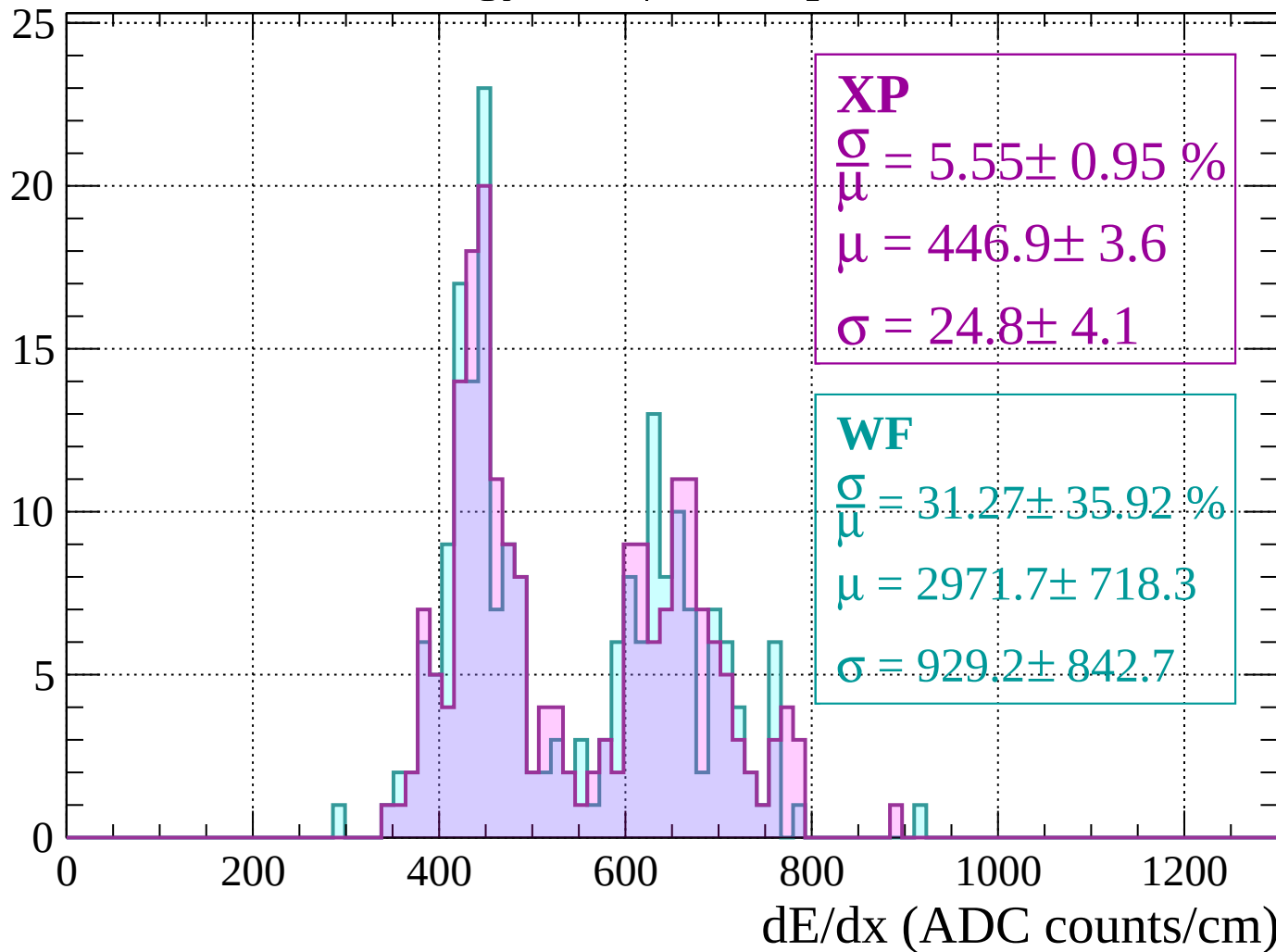
Energy loss | $880 < p < 920$

Count



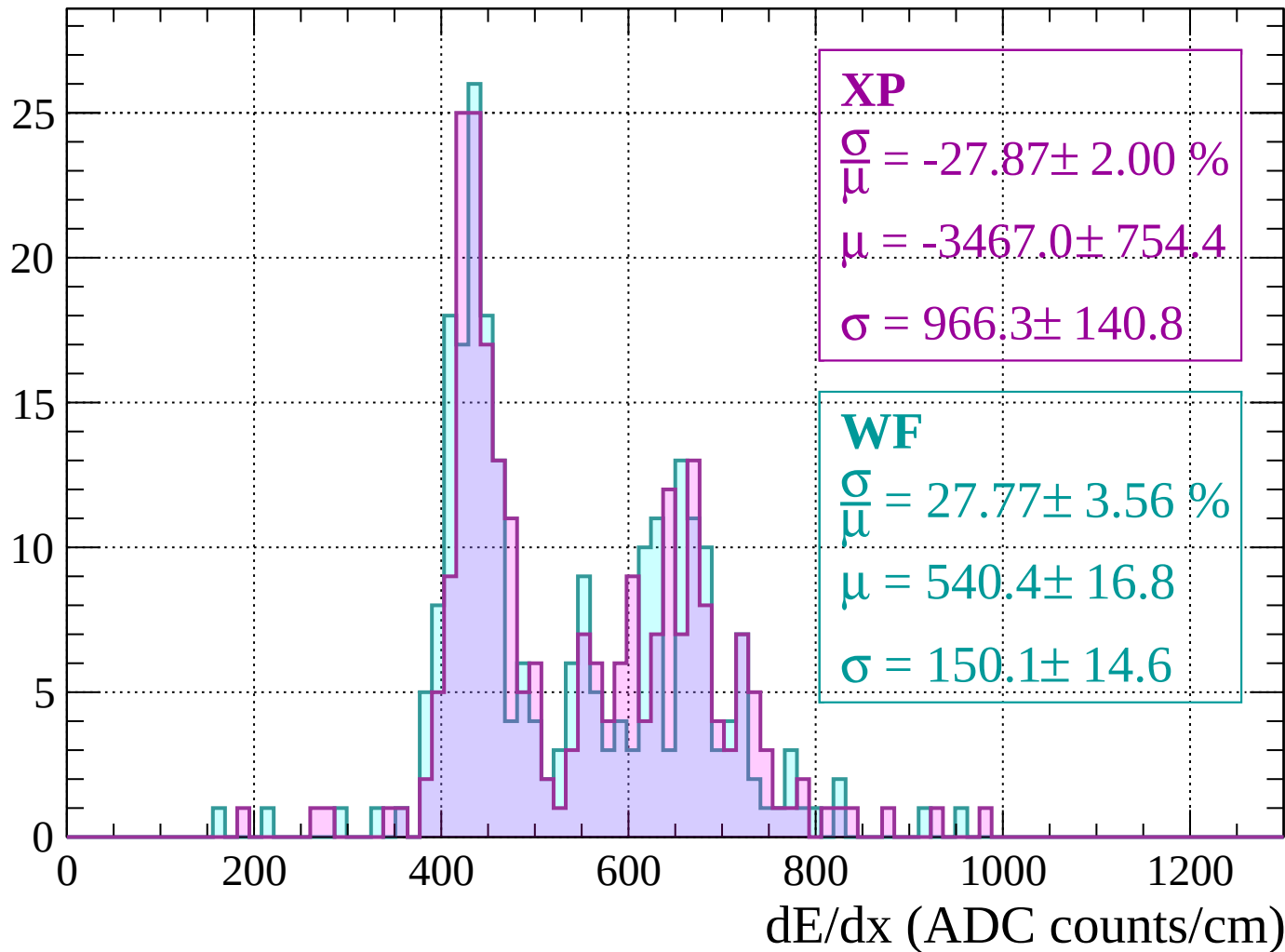
Energy loss | $920 < p < 960$

Count



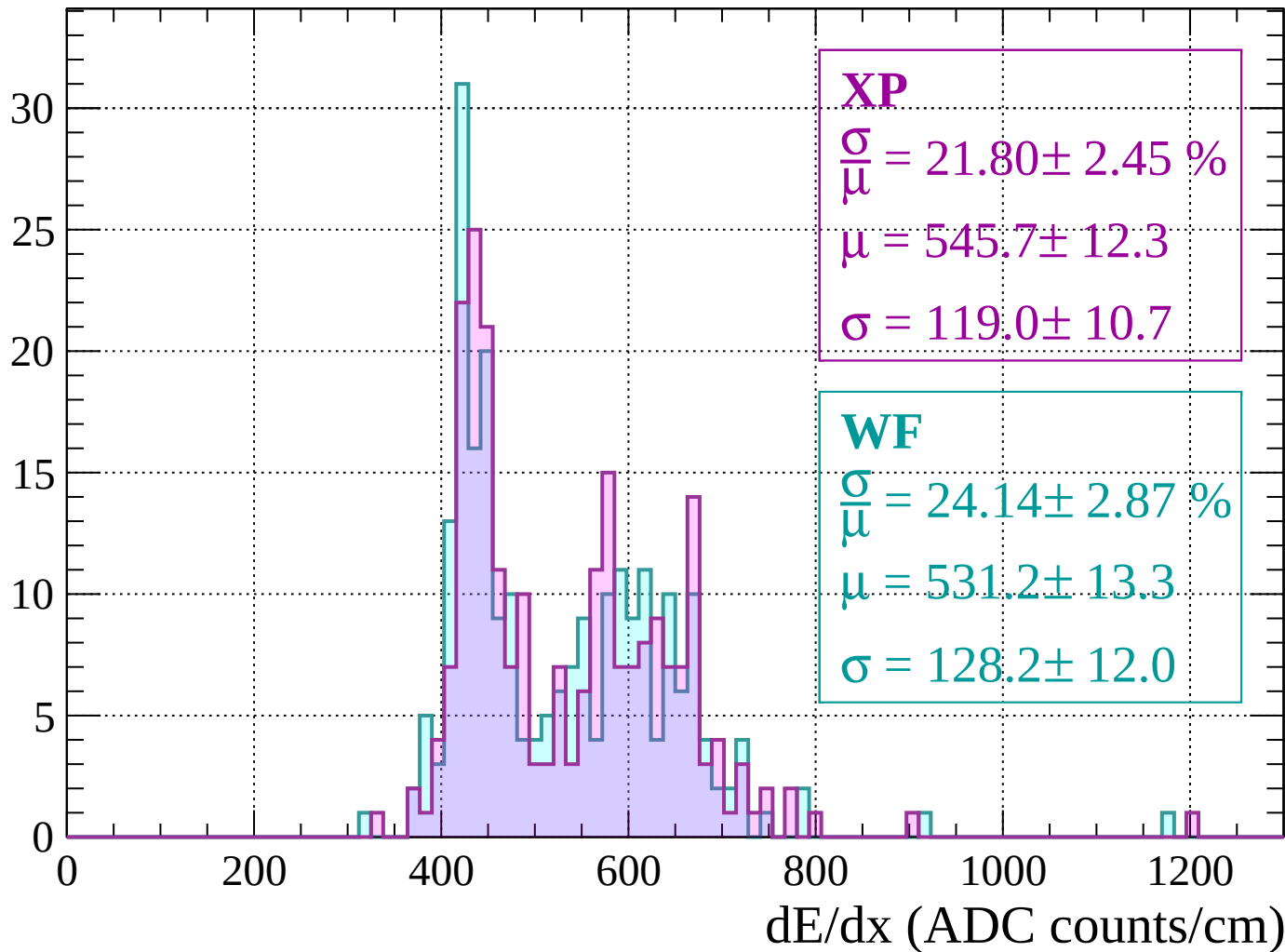
Energy loss | $960 < p < 1000$

Count



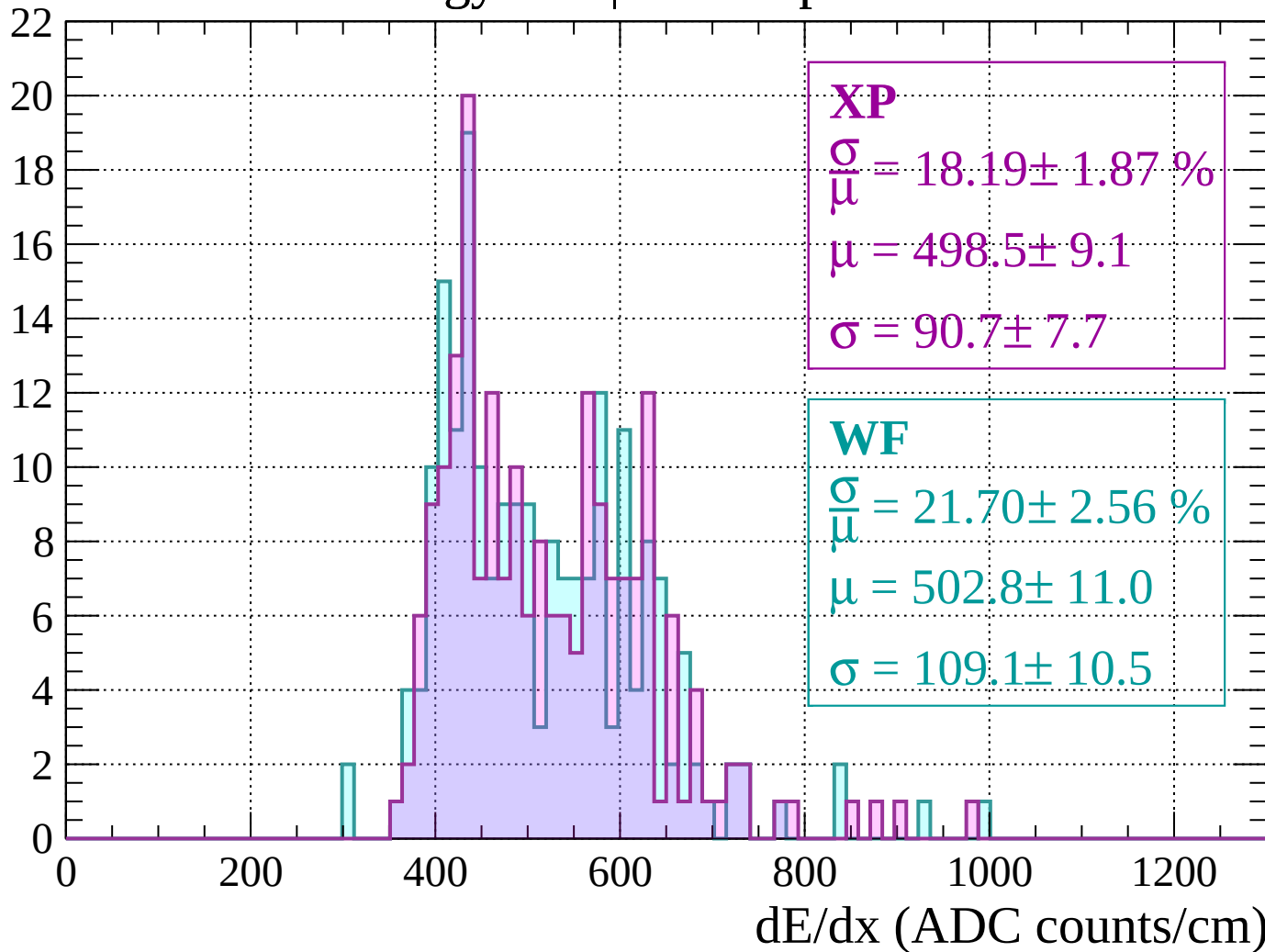
Energy loss | $1000 < p < 1040$

Count



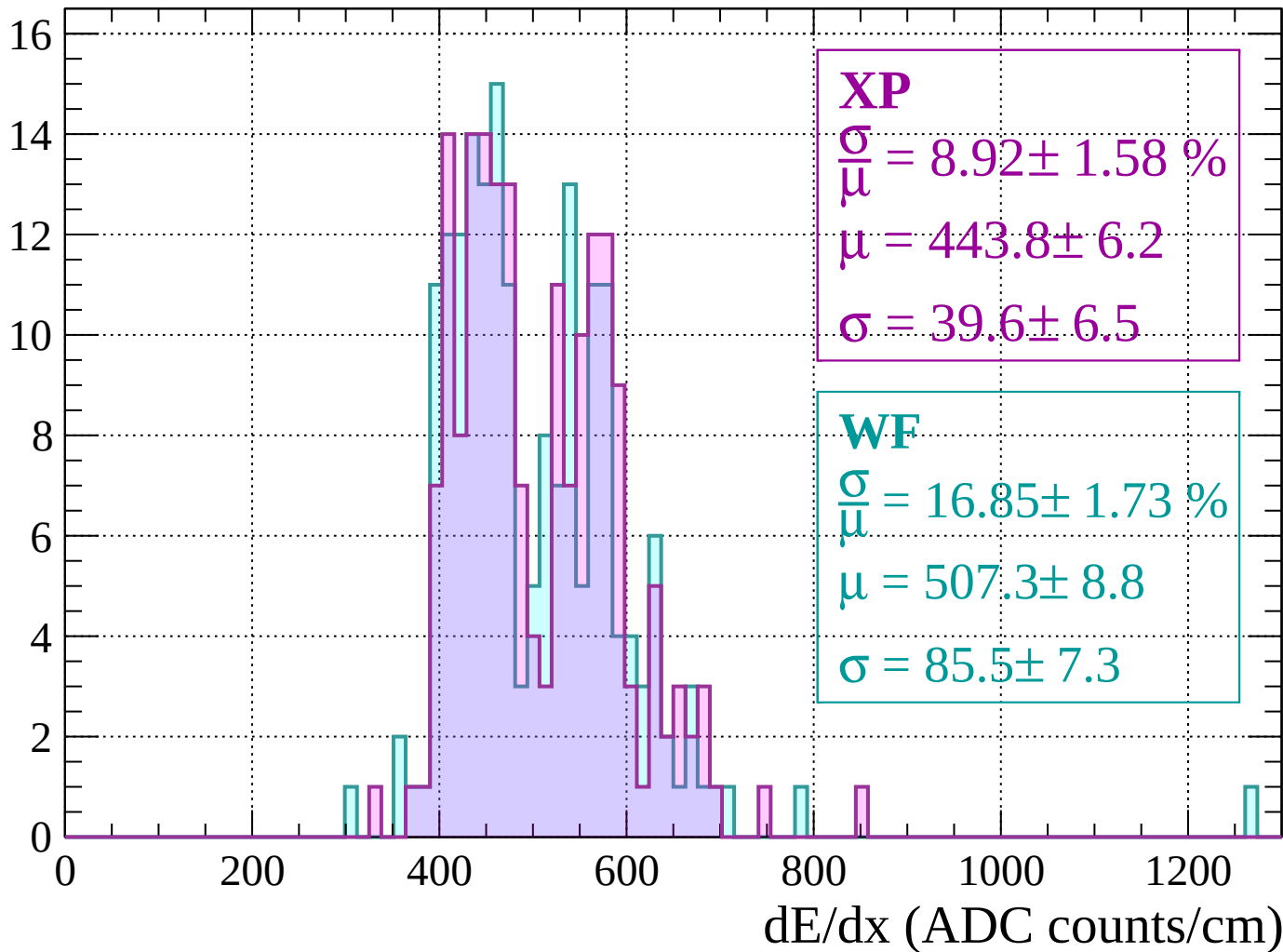
Energy loss | $1040 < p < 1080$

Count



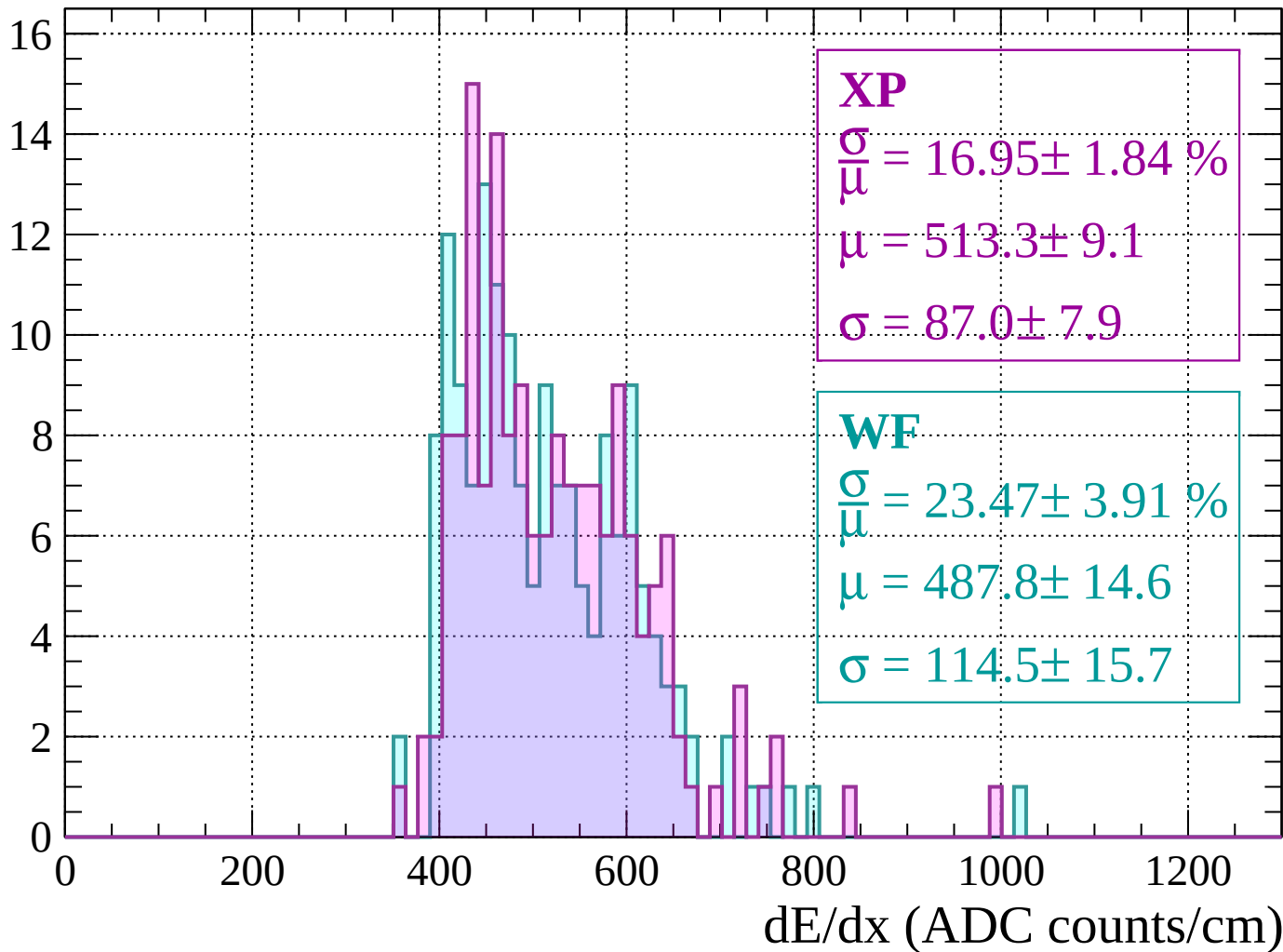
Energy loss | $1080 < p < 1120$

Count



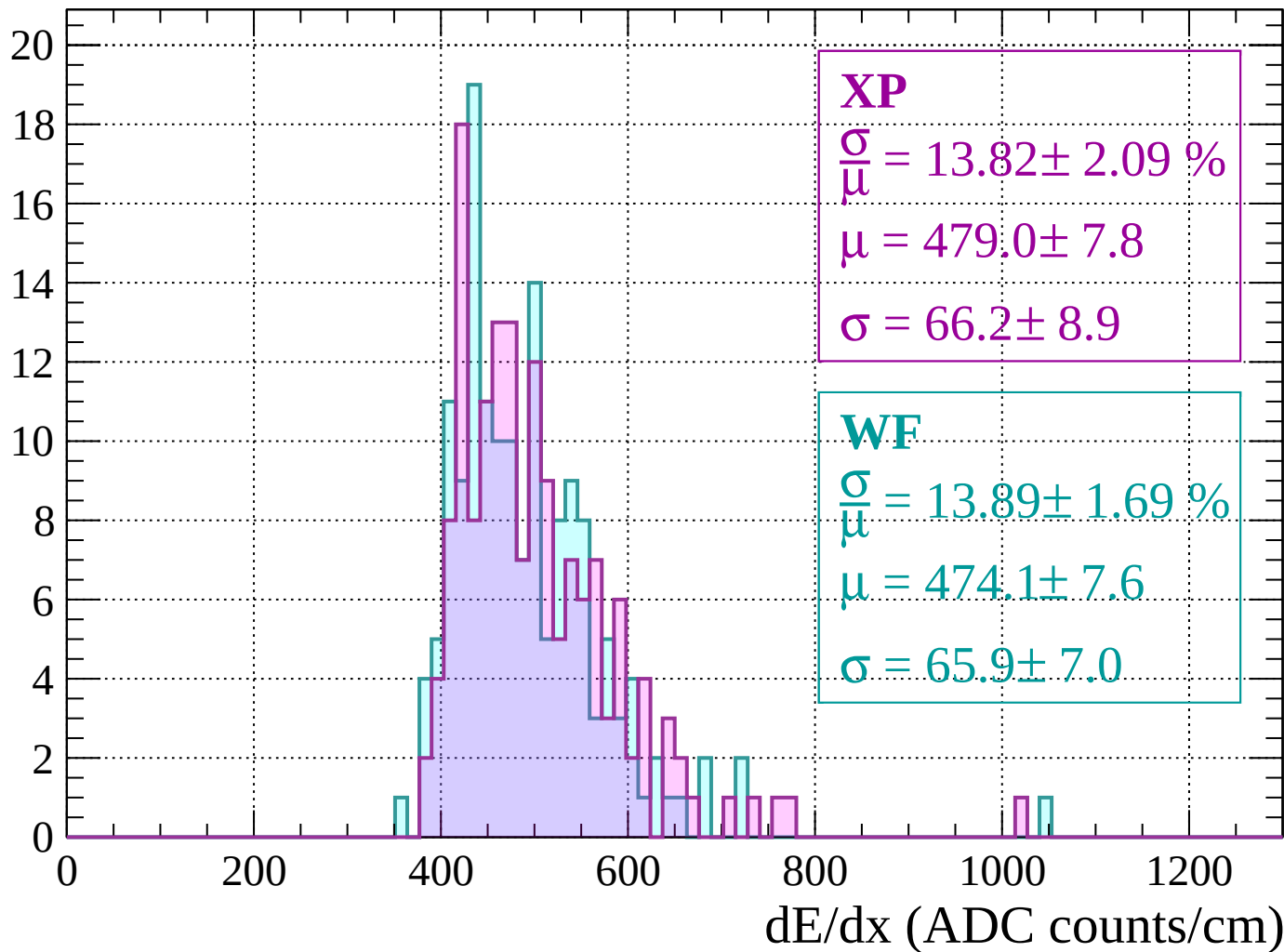
Energy loss | $1120 < p < 1160$

Count



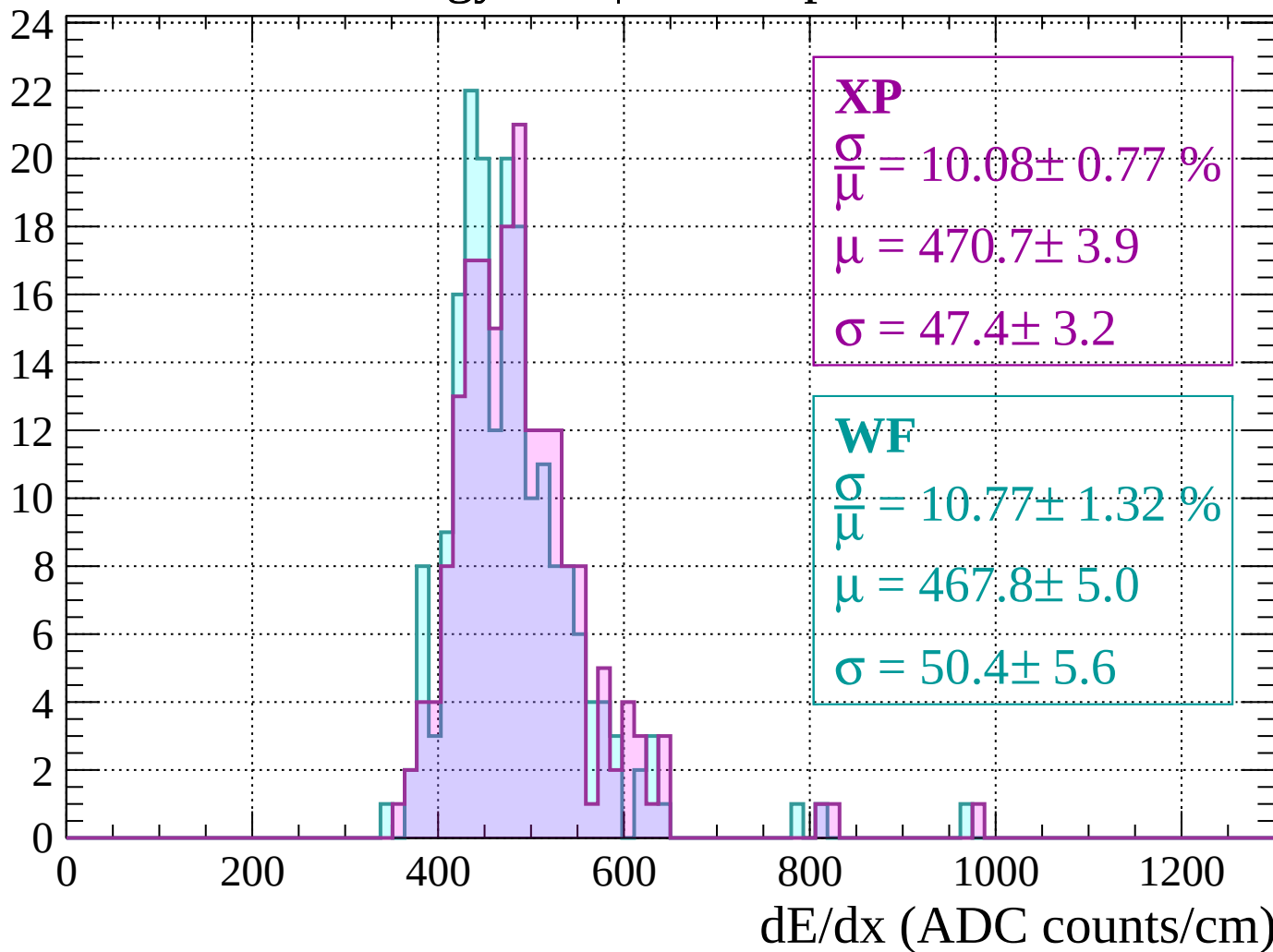
Energy loss | $1160 < p < 1200$

Count



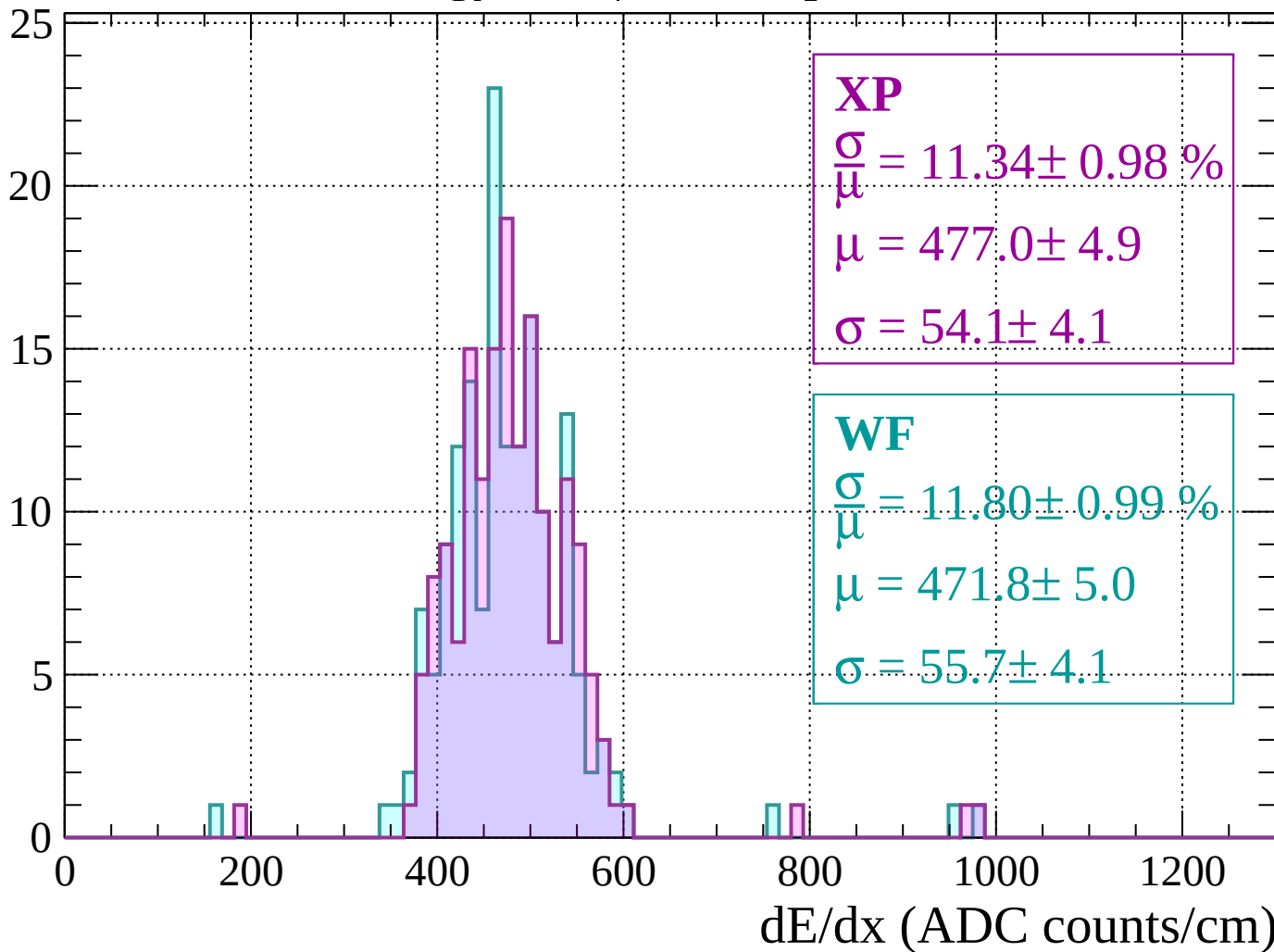
Energy loss | $1200 < p < 1240$

Count



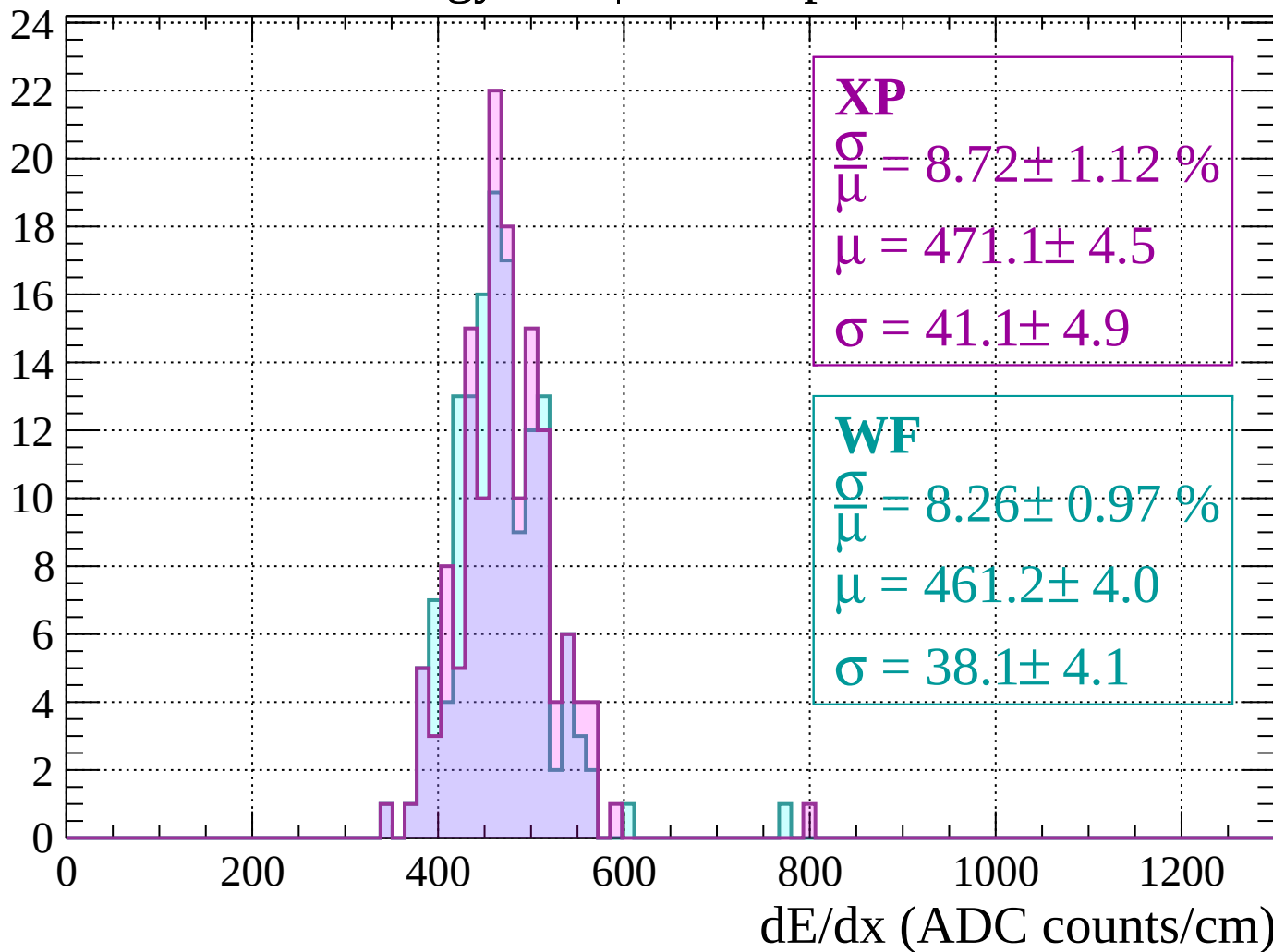
Energy loss | $1240 < p < 1280$

Count



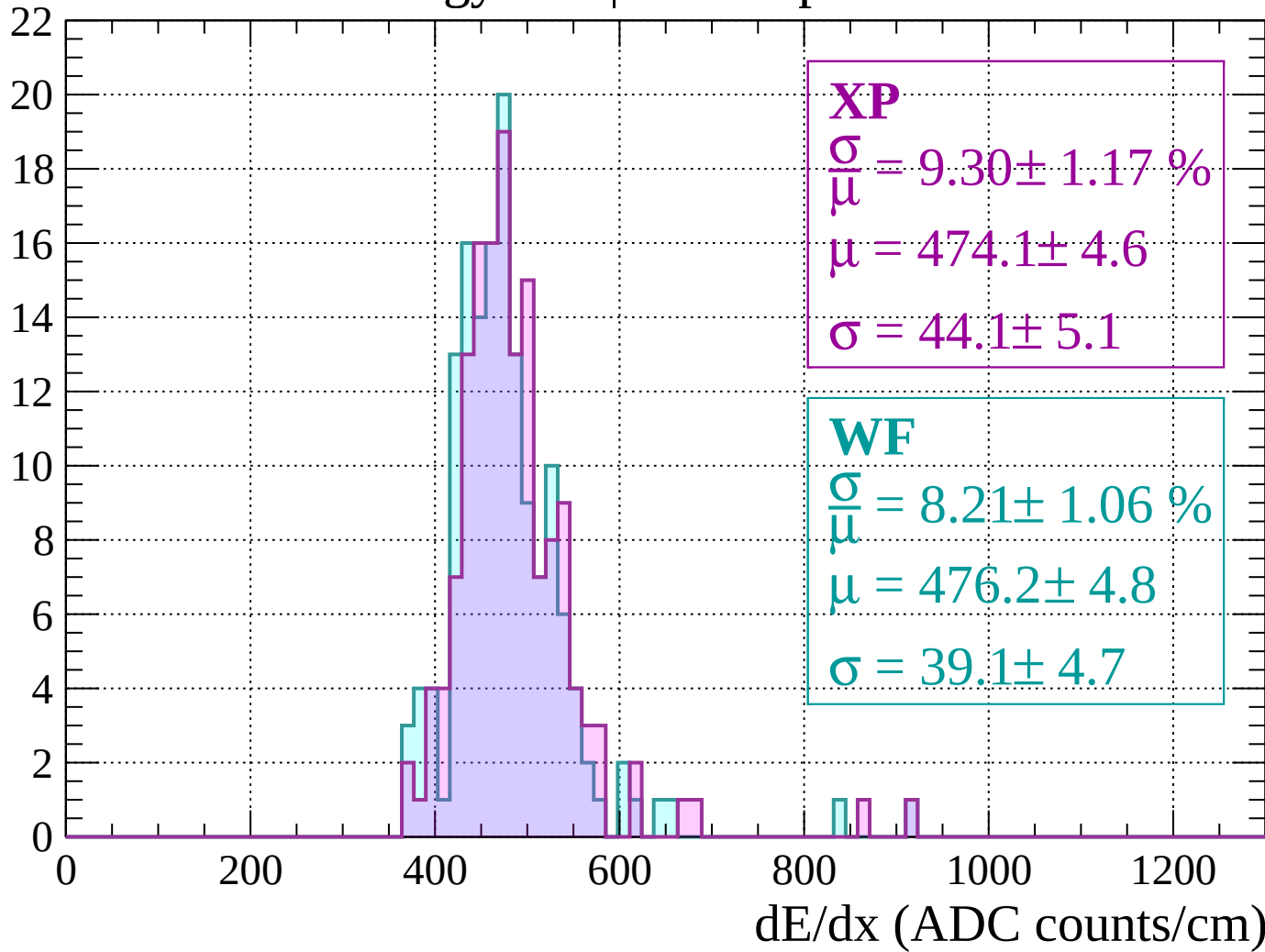
Energy loss | $1280 < p < 1320$

Count



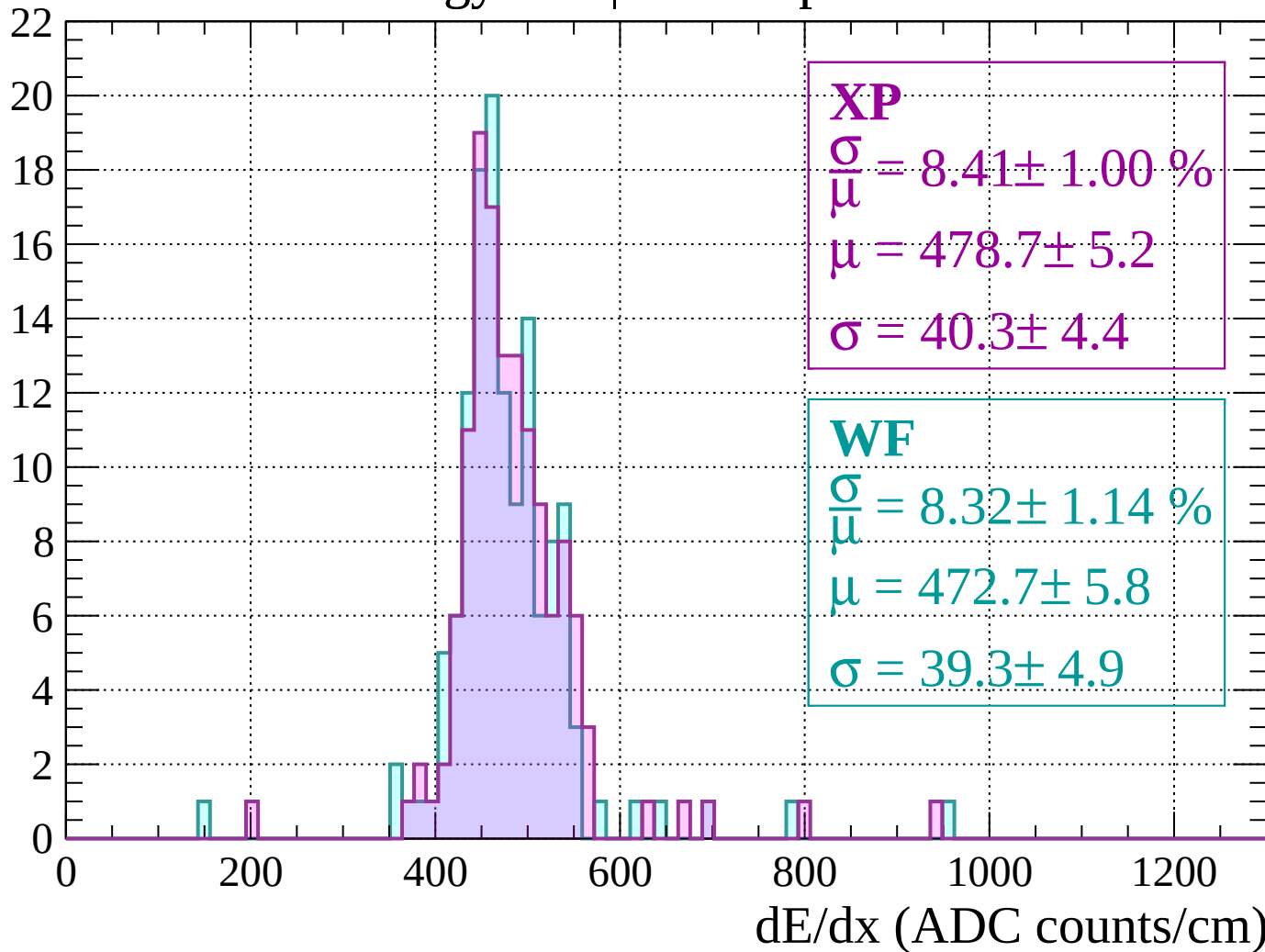
Energy loss | $1320 < p < 1360$

Count



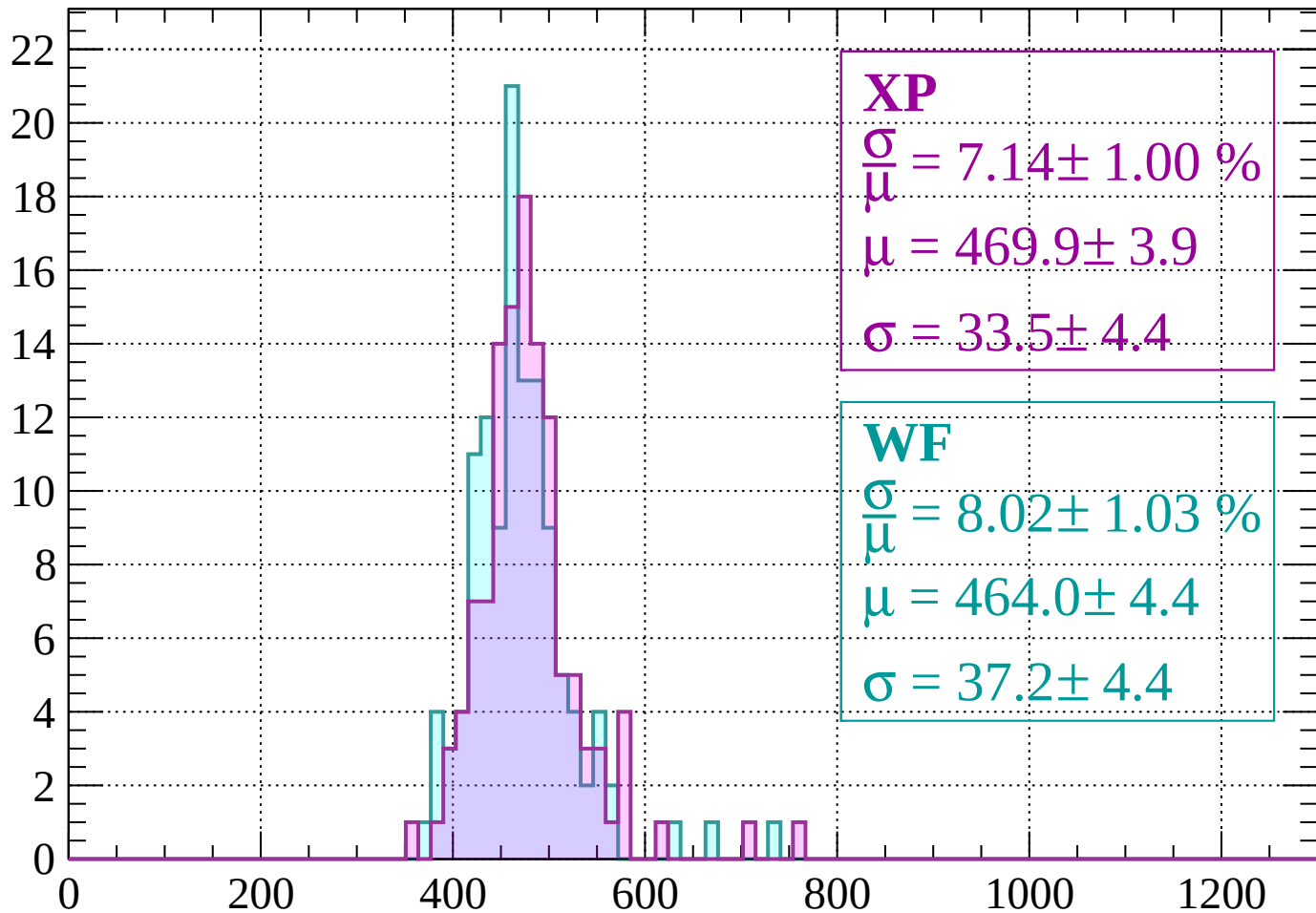
Energy loss | $1360 < p < 1400$

Count



Energy loss | $1400 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 7.14 \pm 1.00 \%$$

$$\mu = 469.9 \pm 3.9$$

$$\sigma = 33.5 \pm 4.4$$

WF

$$\frac{\sigma}{\mu} = 8.02 \pm 1.03 \%$$

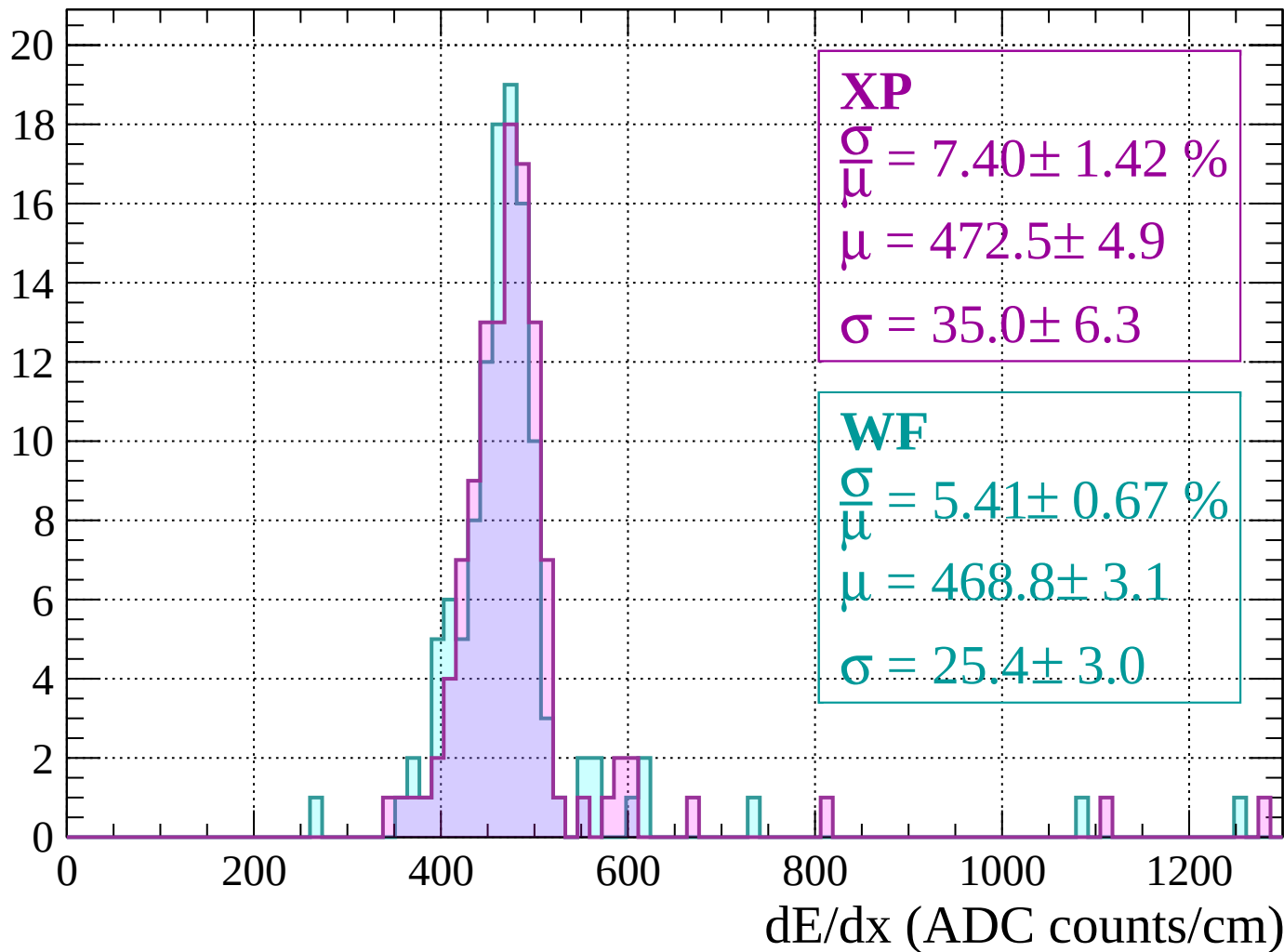
$$\mu = 464.0 \pm 4.4$$

$$\sigma = 37.2 \pm 4.4$$

dE/dx (ADC counts/cm)

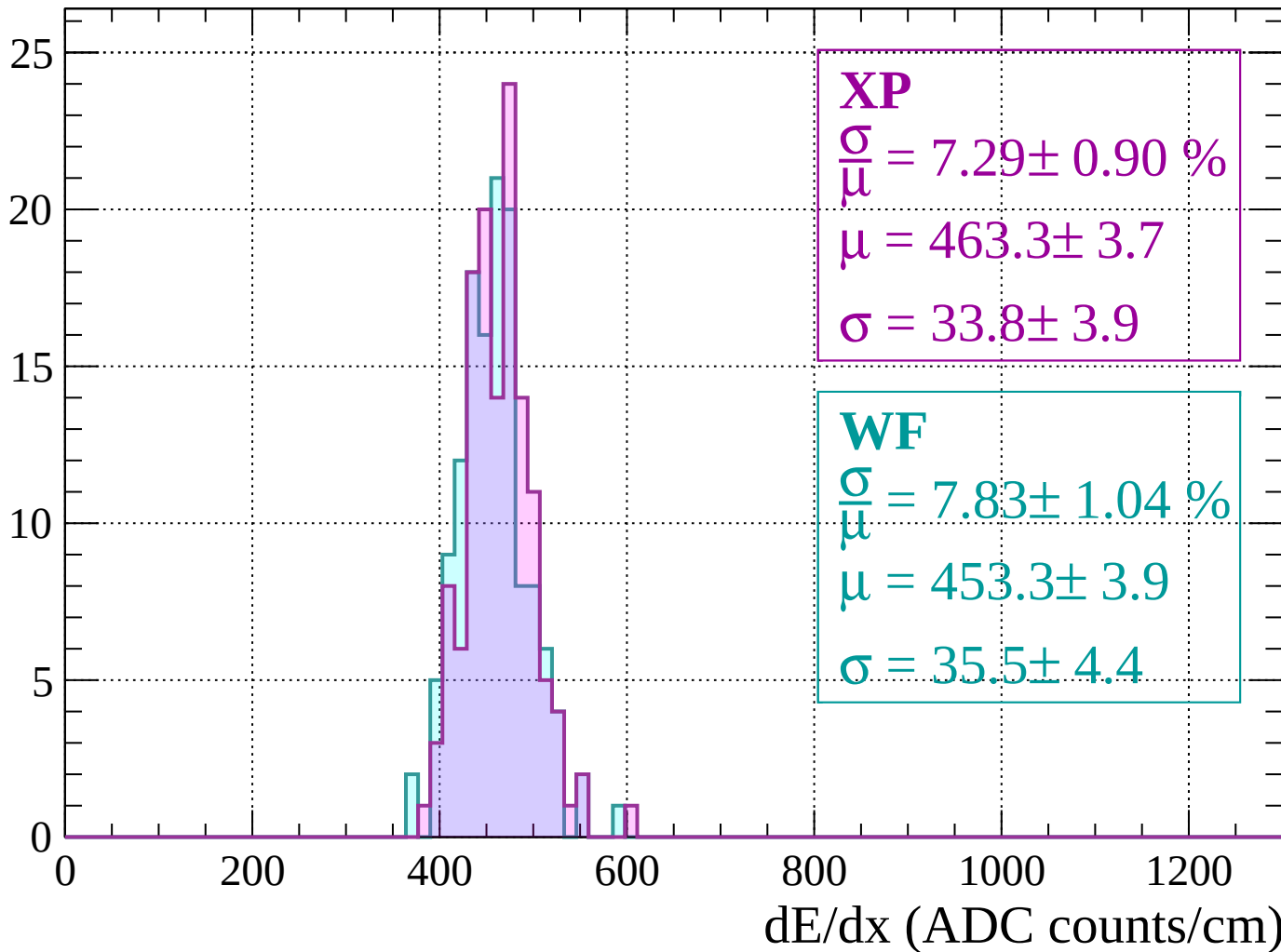
Energy loss | $1440 < p < 1480$

Count



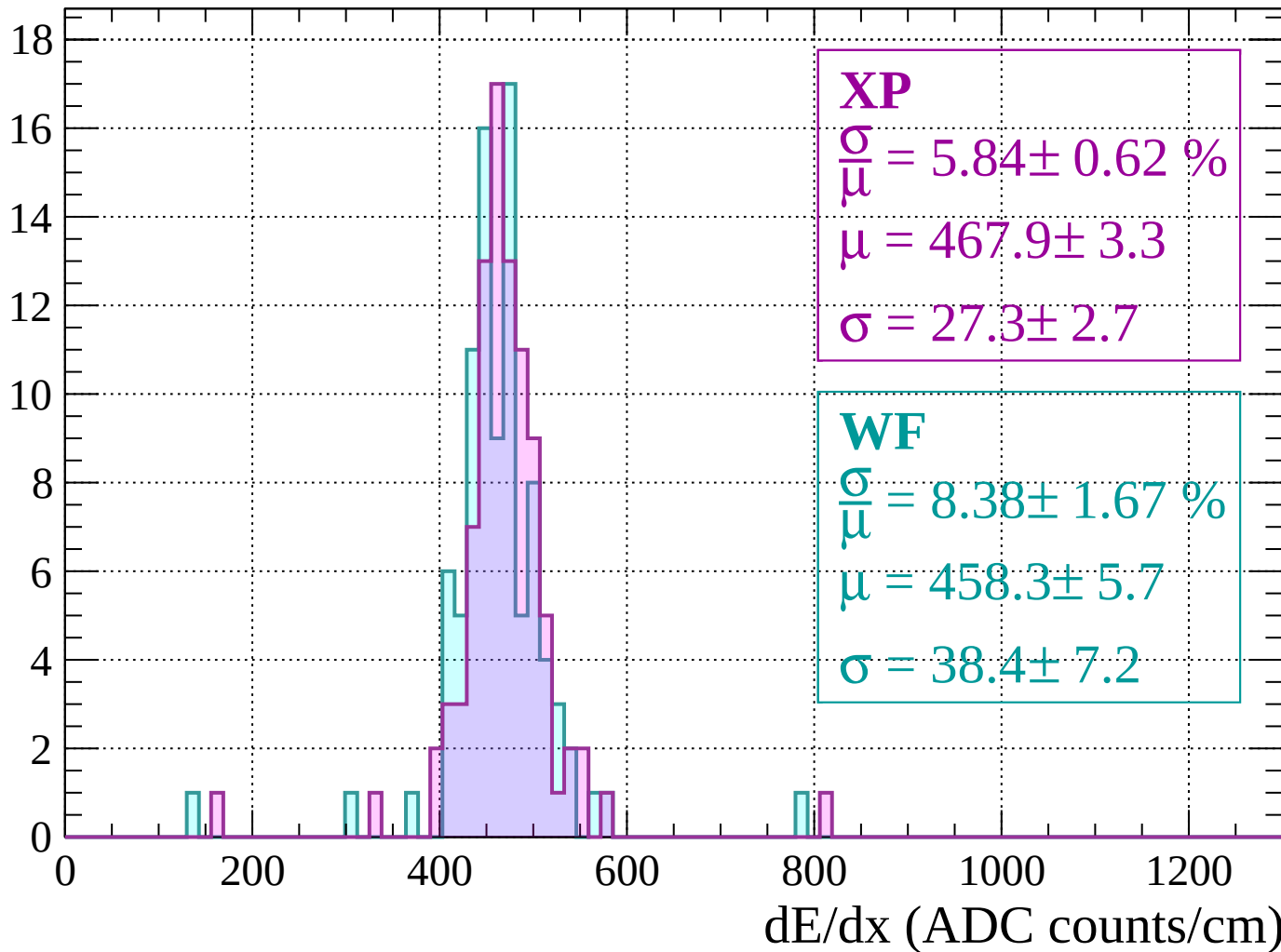
Energy loss | $1480 < p < 1520$

Count



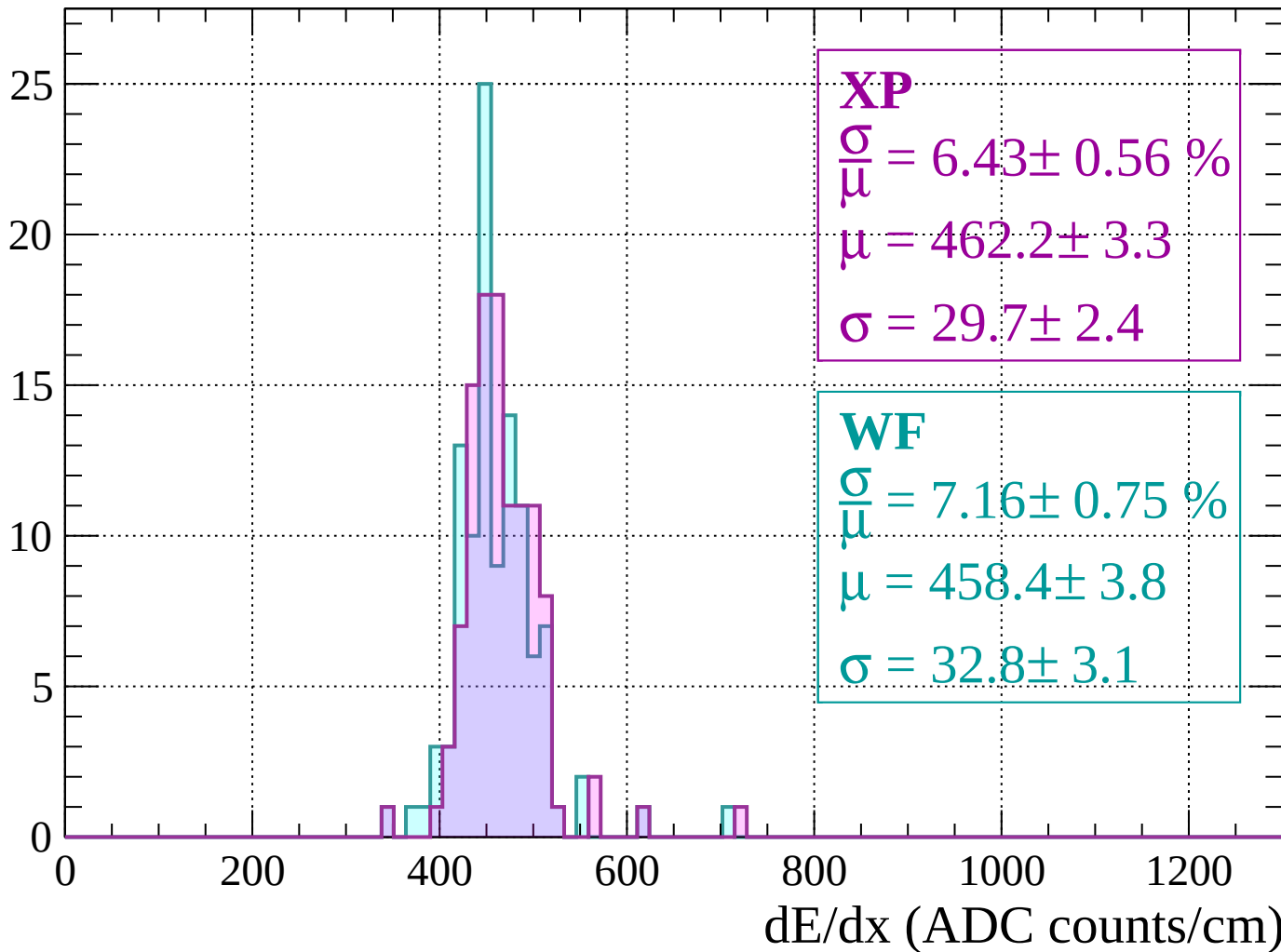
Energy loss | $1520 < p < 1560$

Count



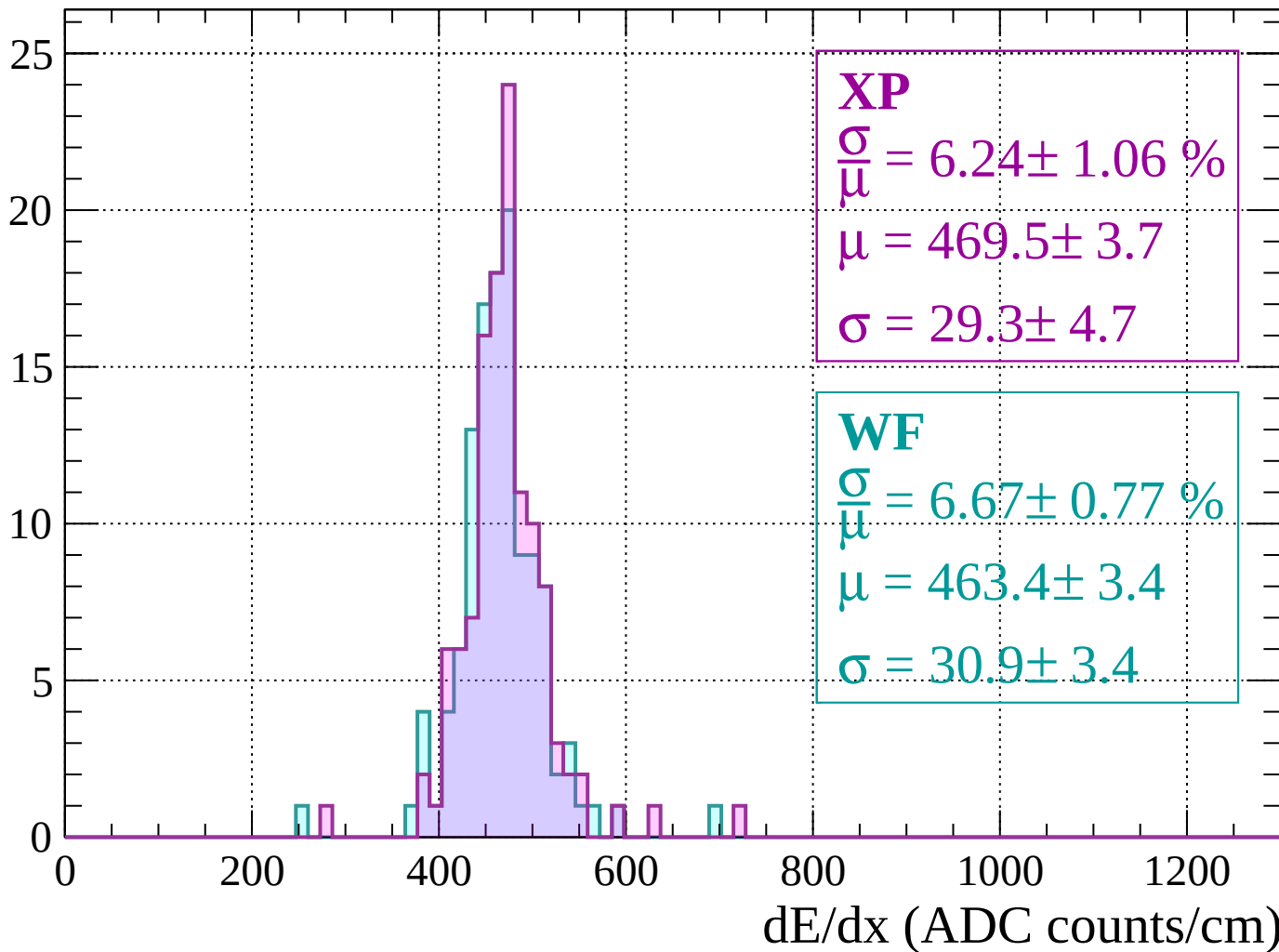
Energy loss | $1560 < p < 1600$

Count



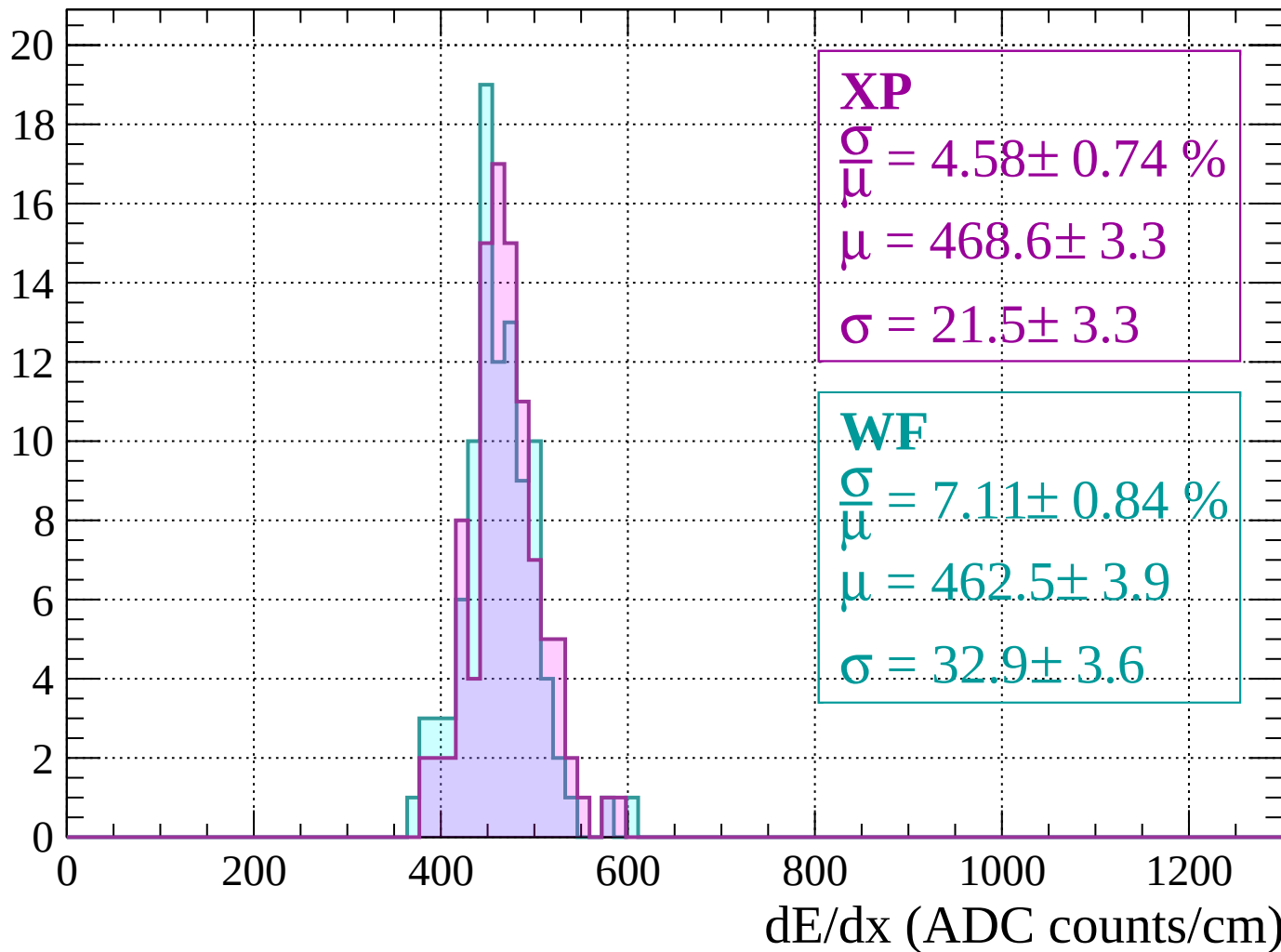
Energy loss | $1600 < p < 1640$

Count



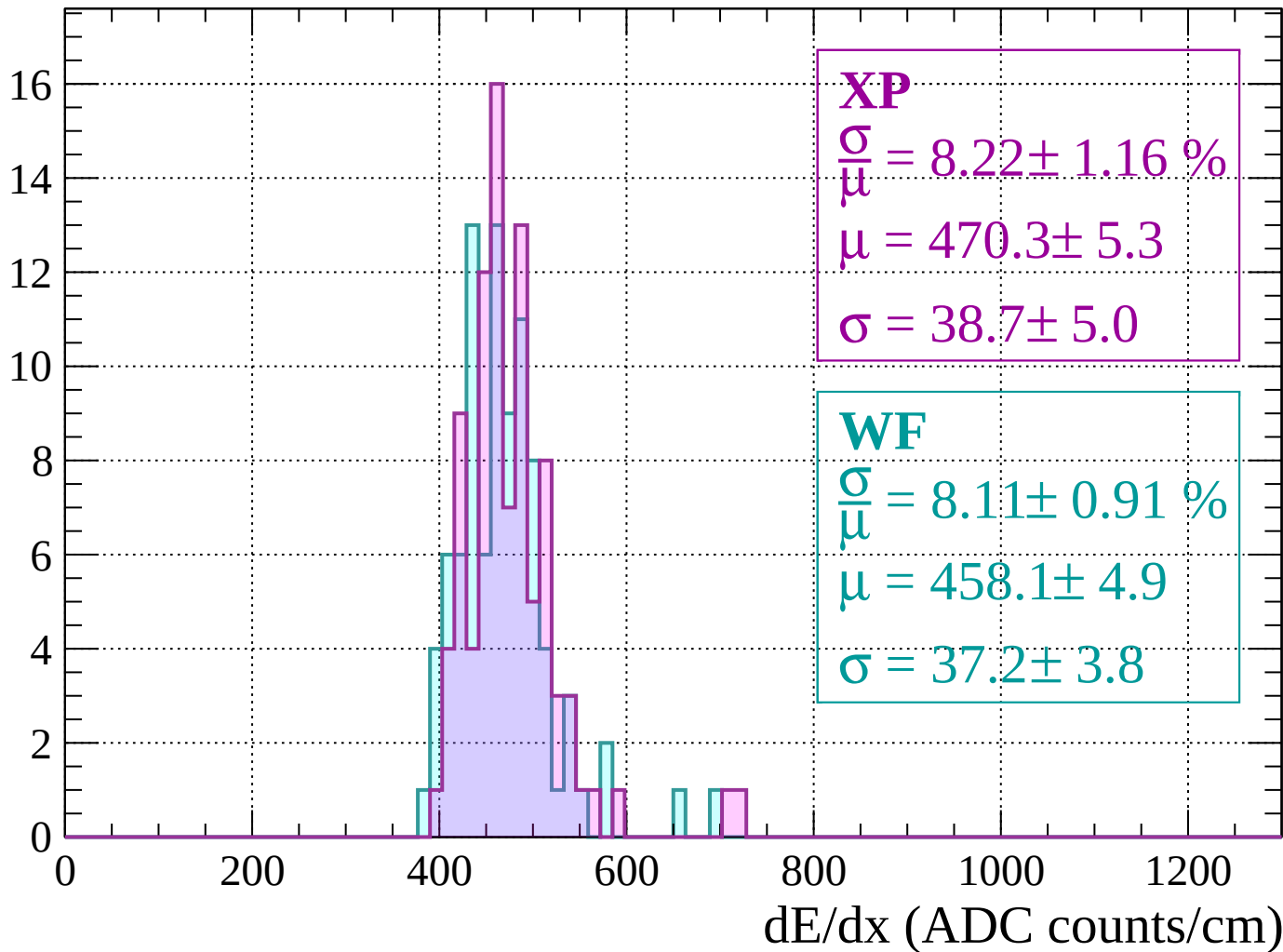
Energy loss | $1640 < p < 1680$

Count



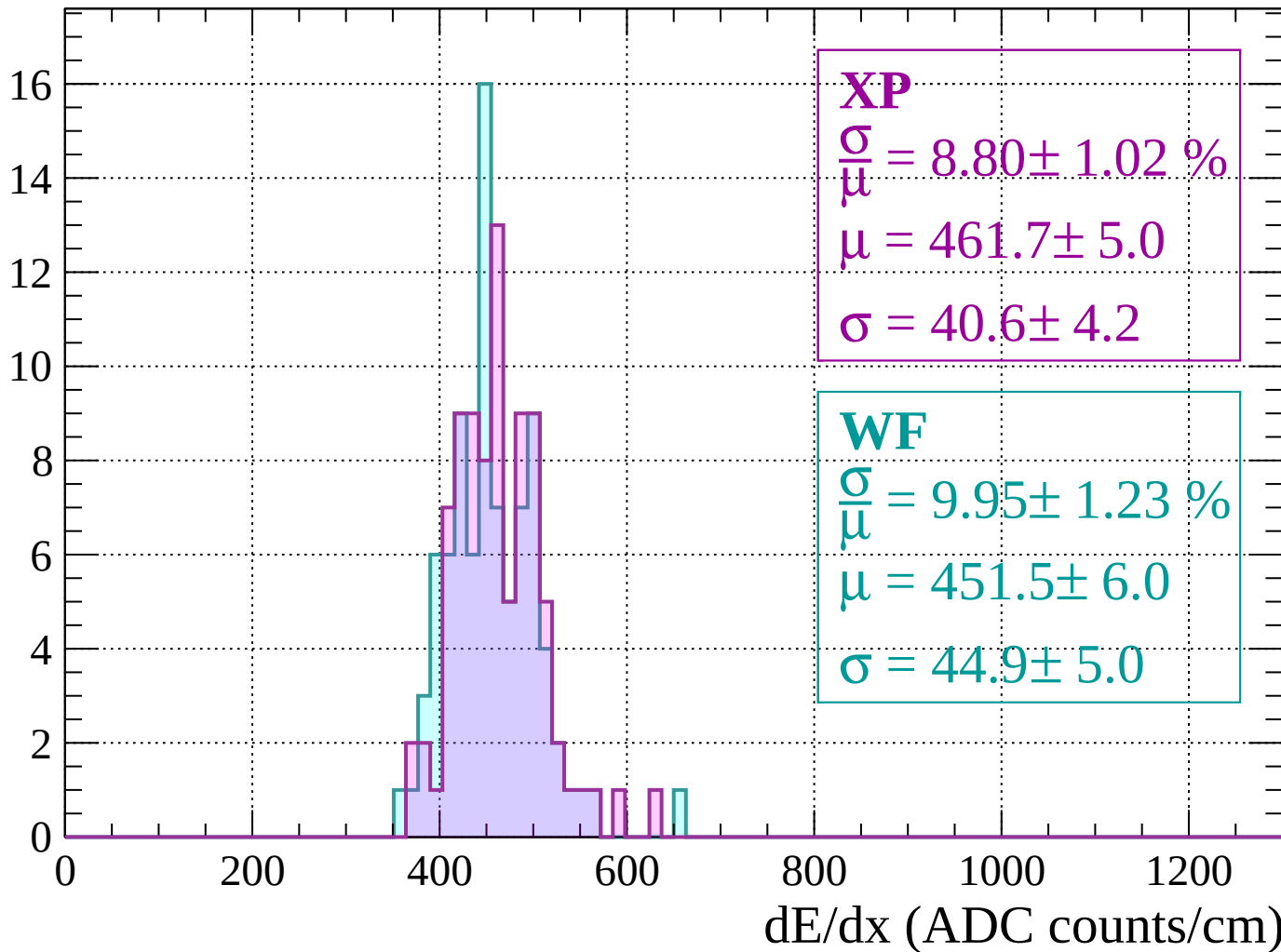
Energy loss | $1680 < p < 1720$

Count



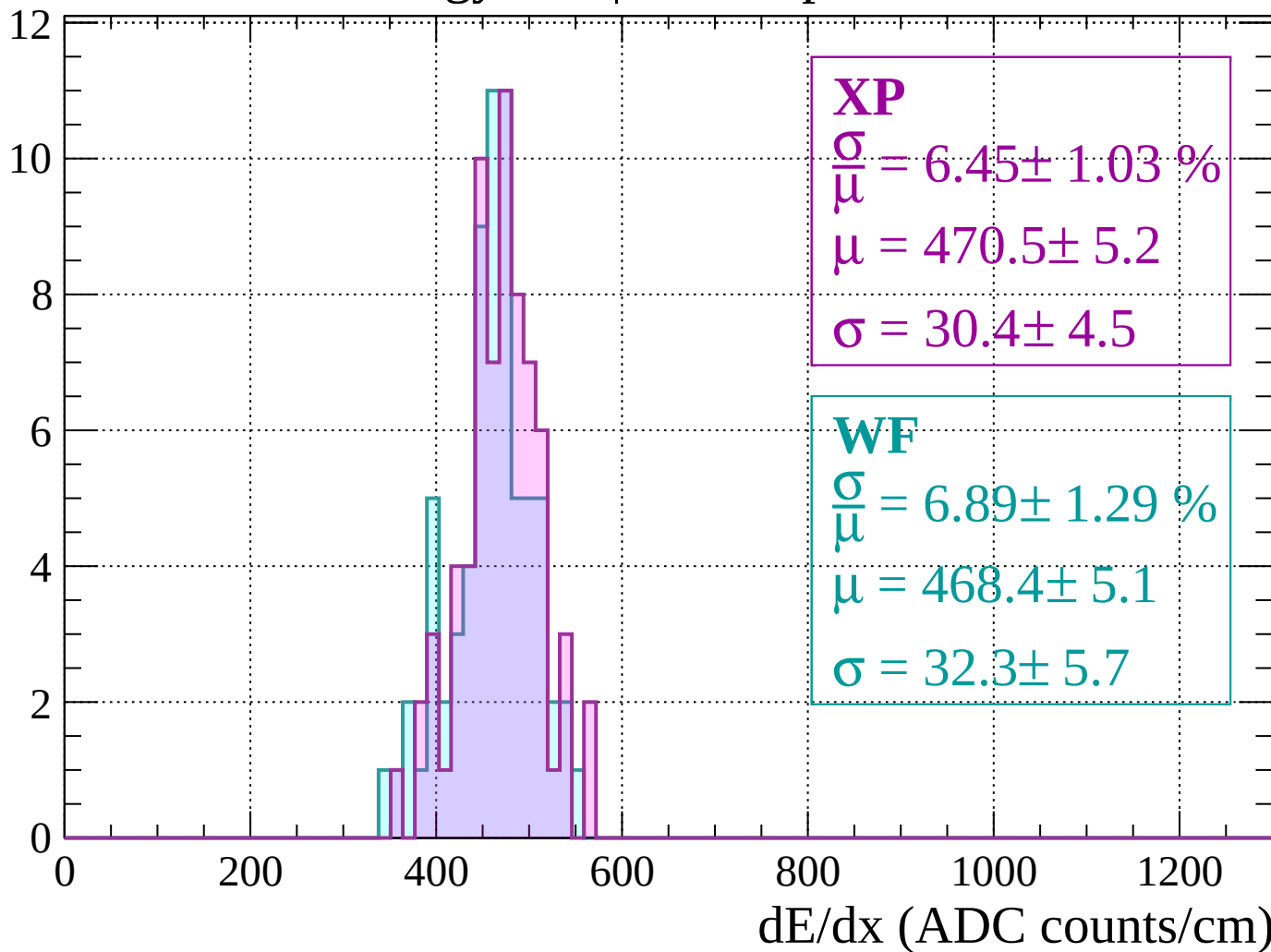
Energy loss | $1720 < p < 1760$

Count



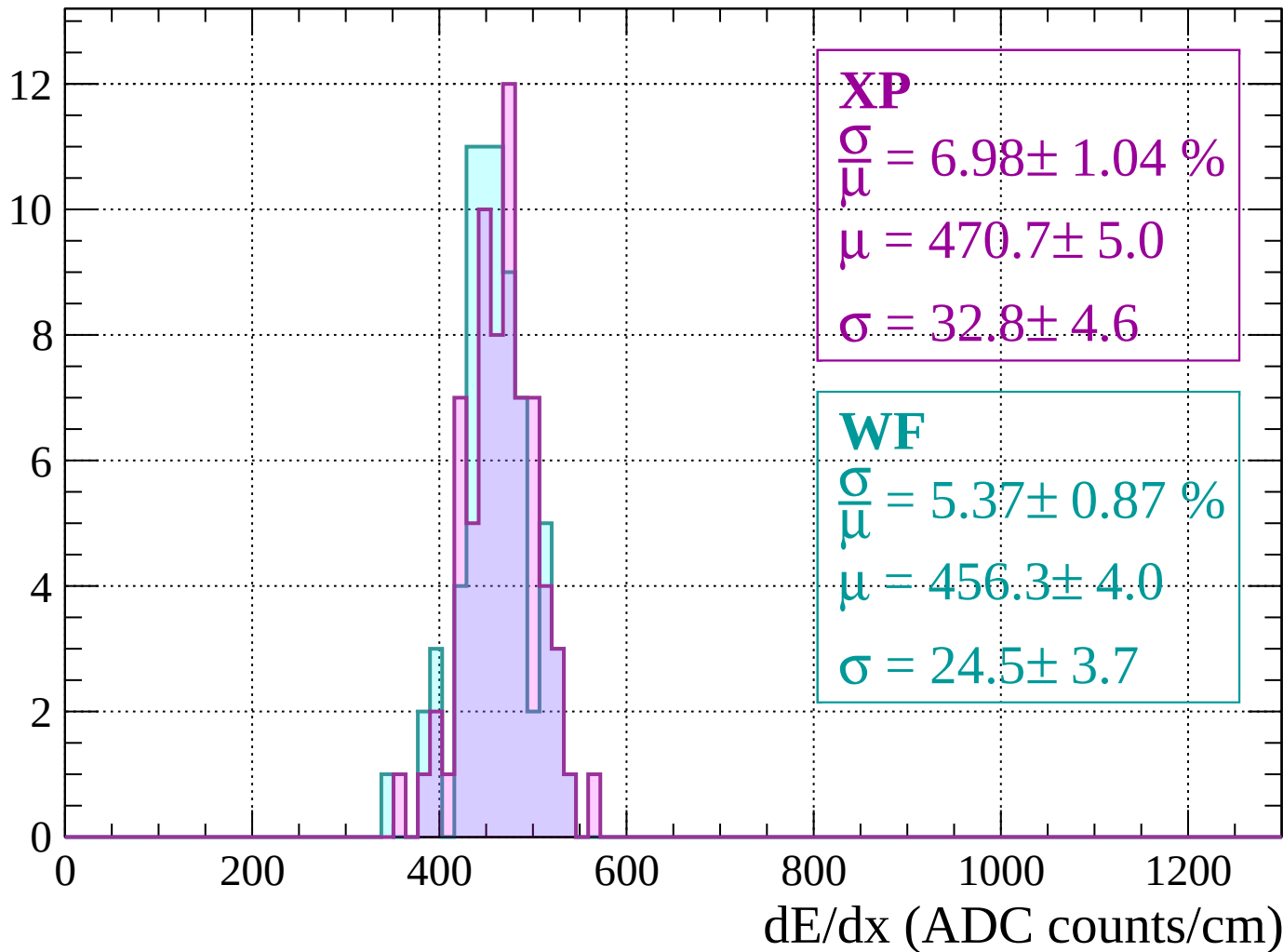
Energy loss | $1760 < p < 1800$

Count



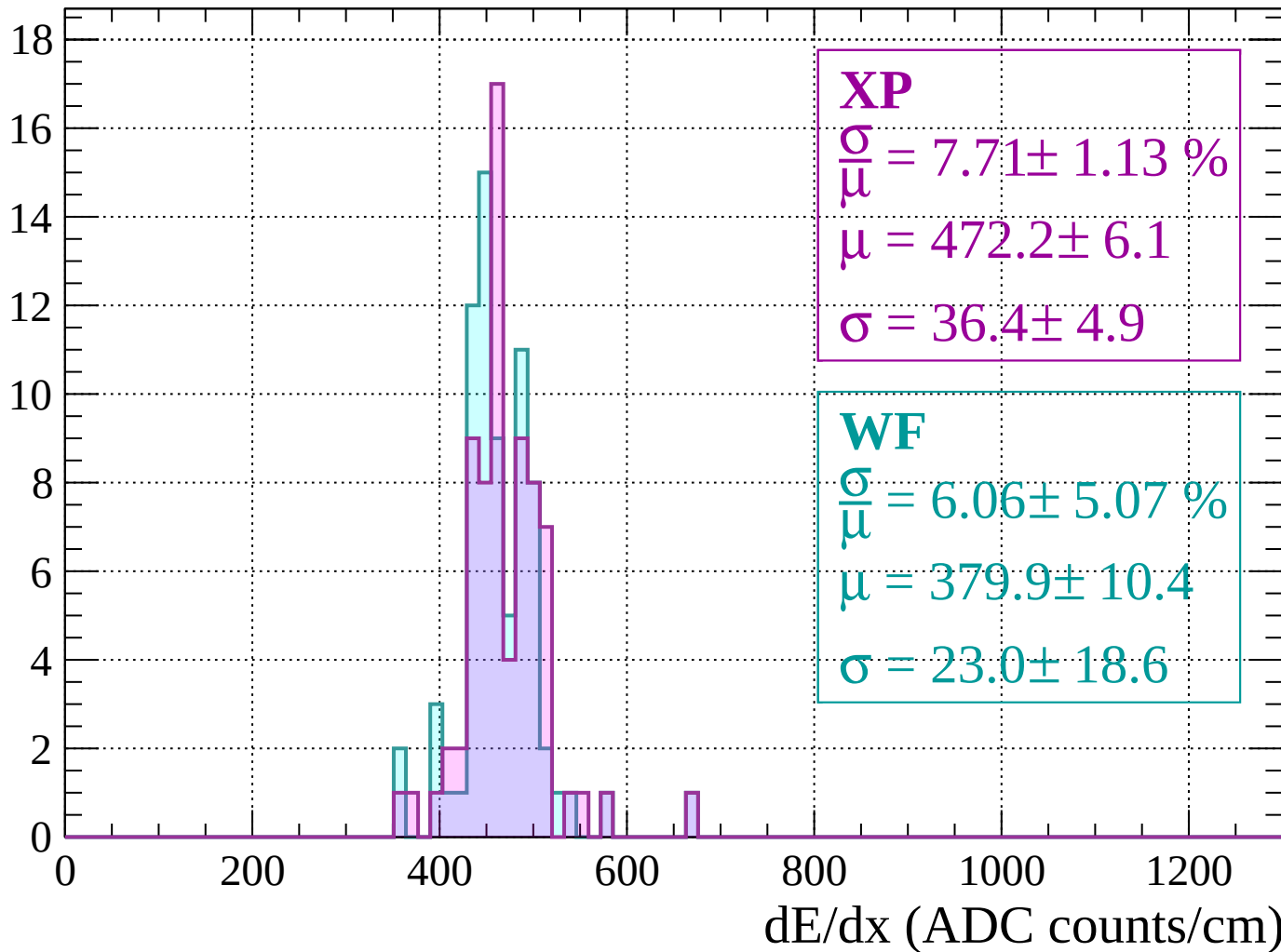
Energy loss | $1800 < p < 1840$

Count



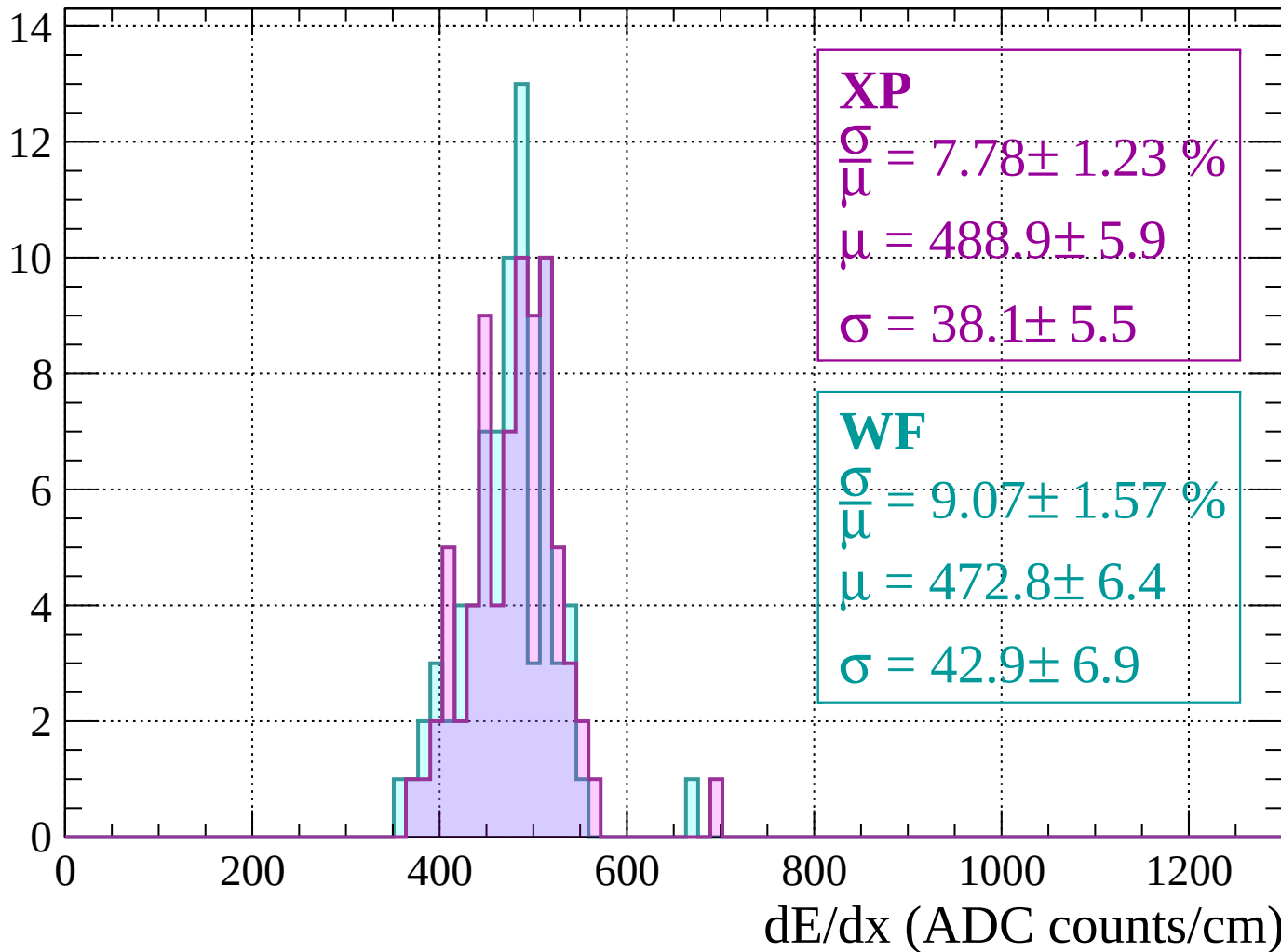
Energy loss | $1840 < p < 1880$

Count



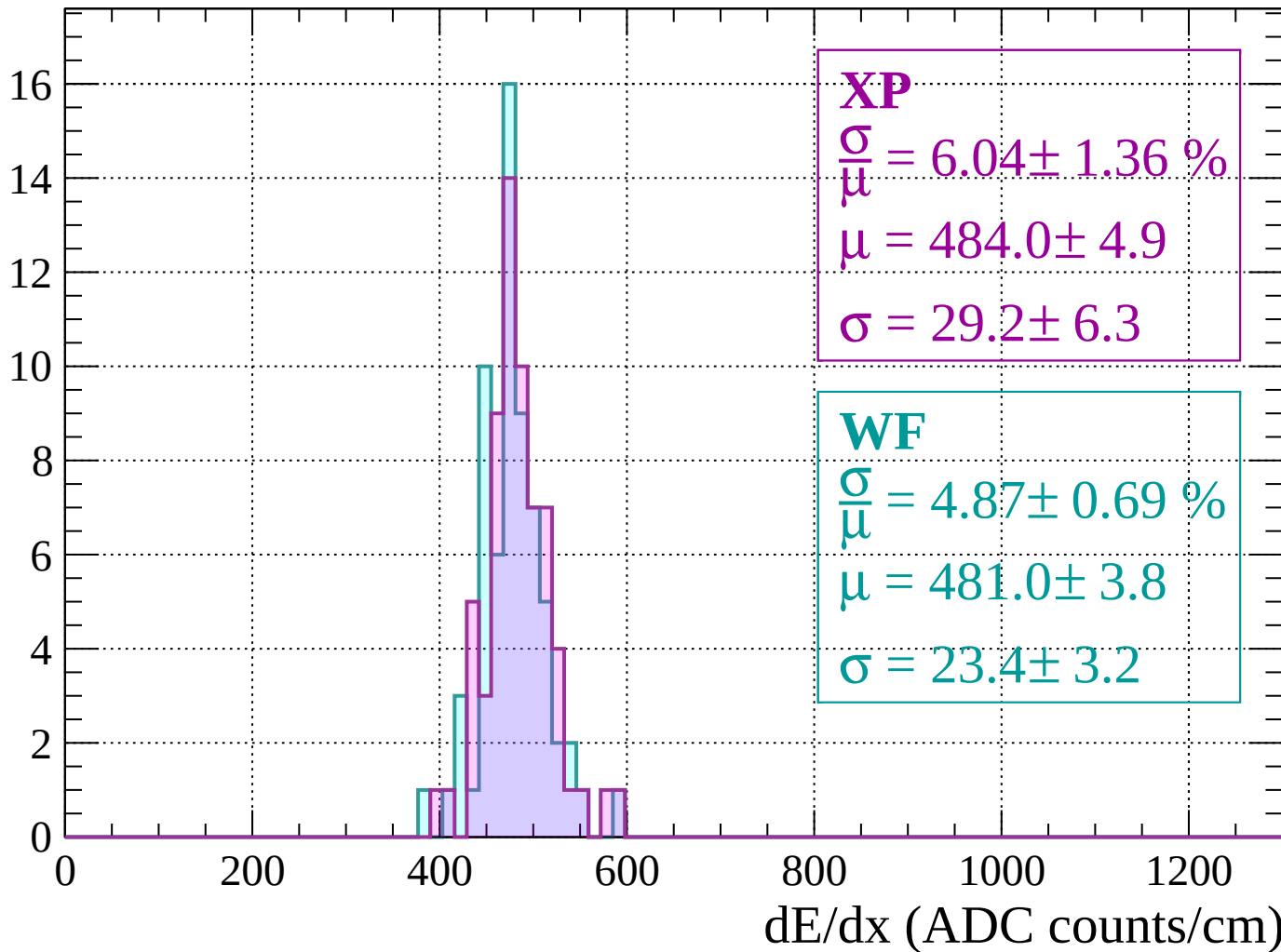
Energy loss | $1880 < p < 1920$

Count



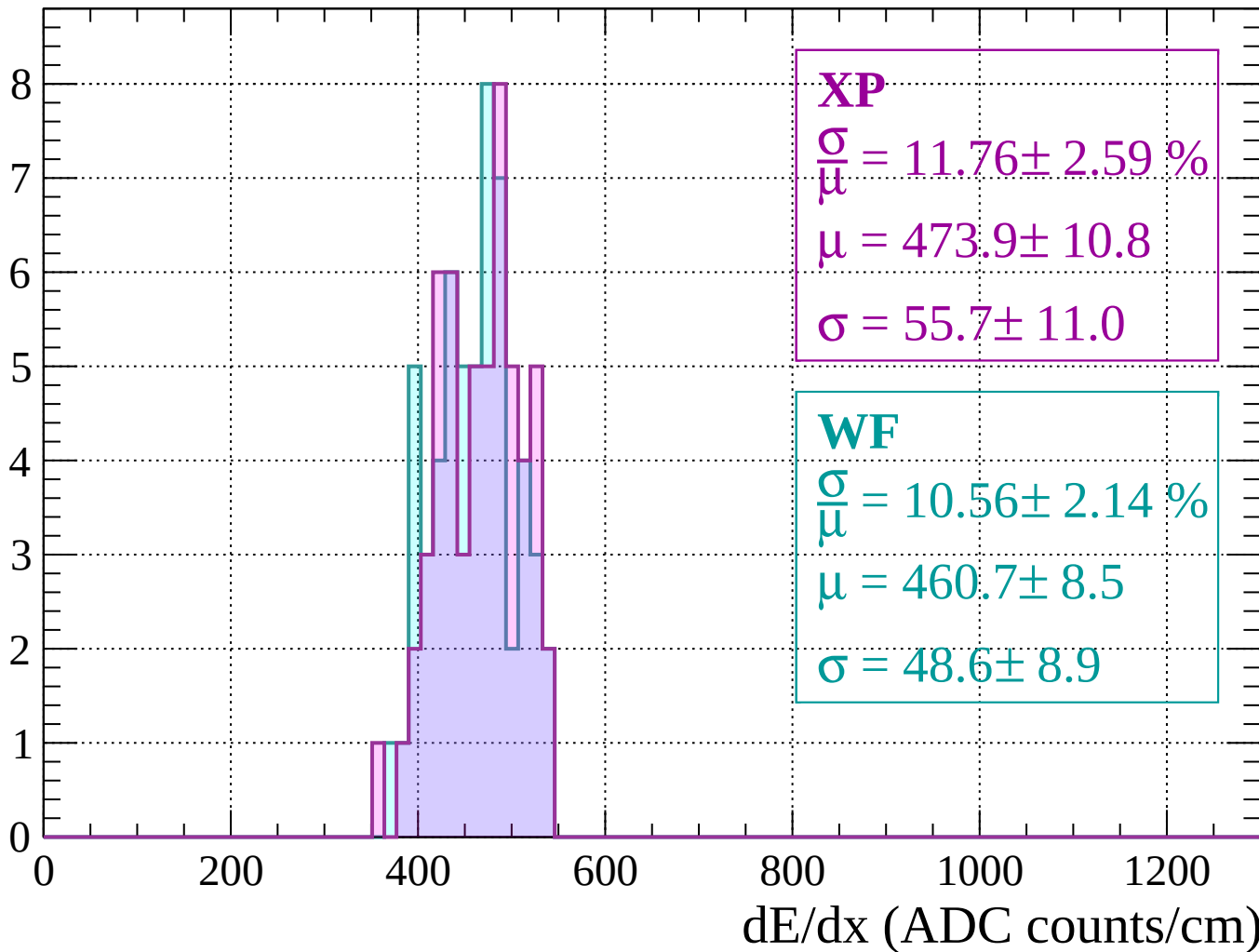
Energy loss | $1920 < p < 1960$

Count



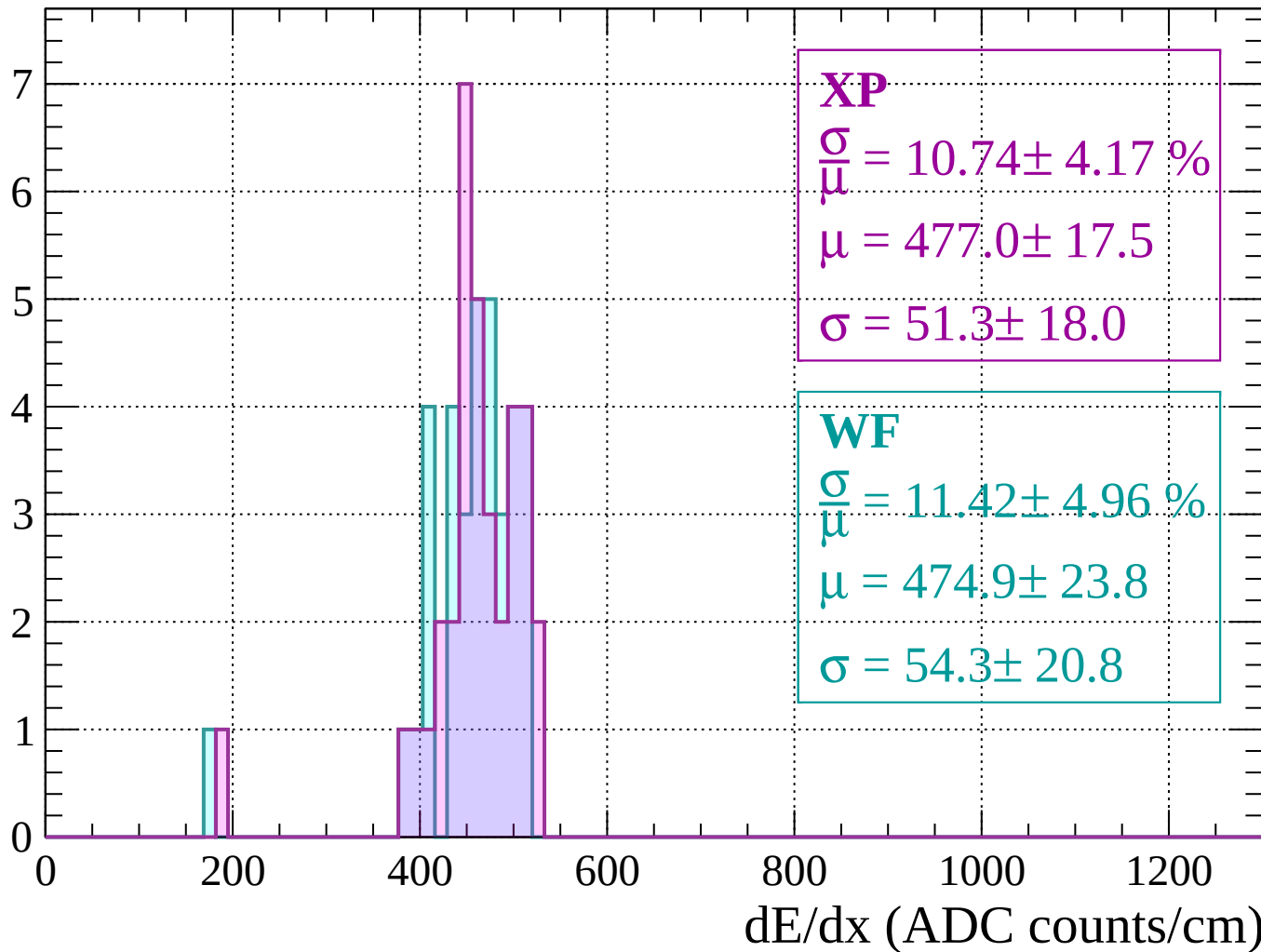
Energy loss | $1960 < p < 2000$

Count



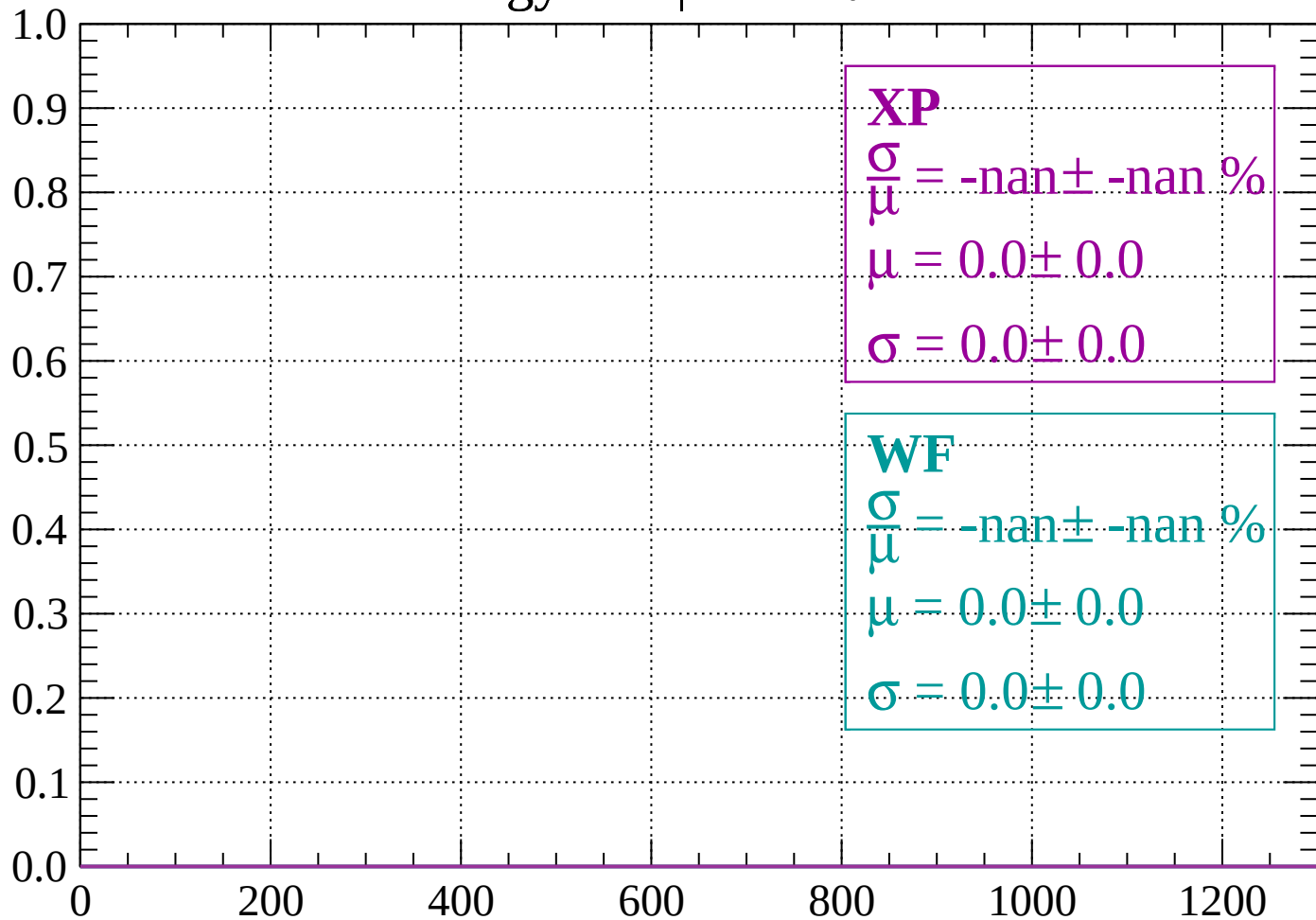
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

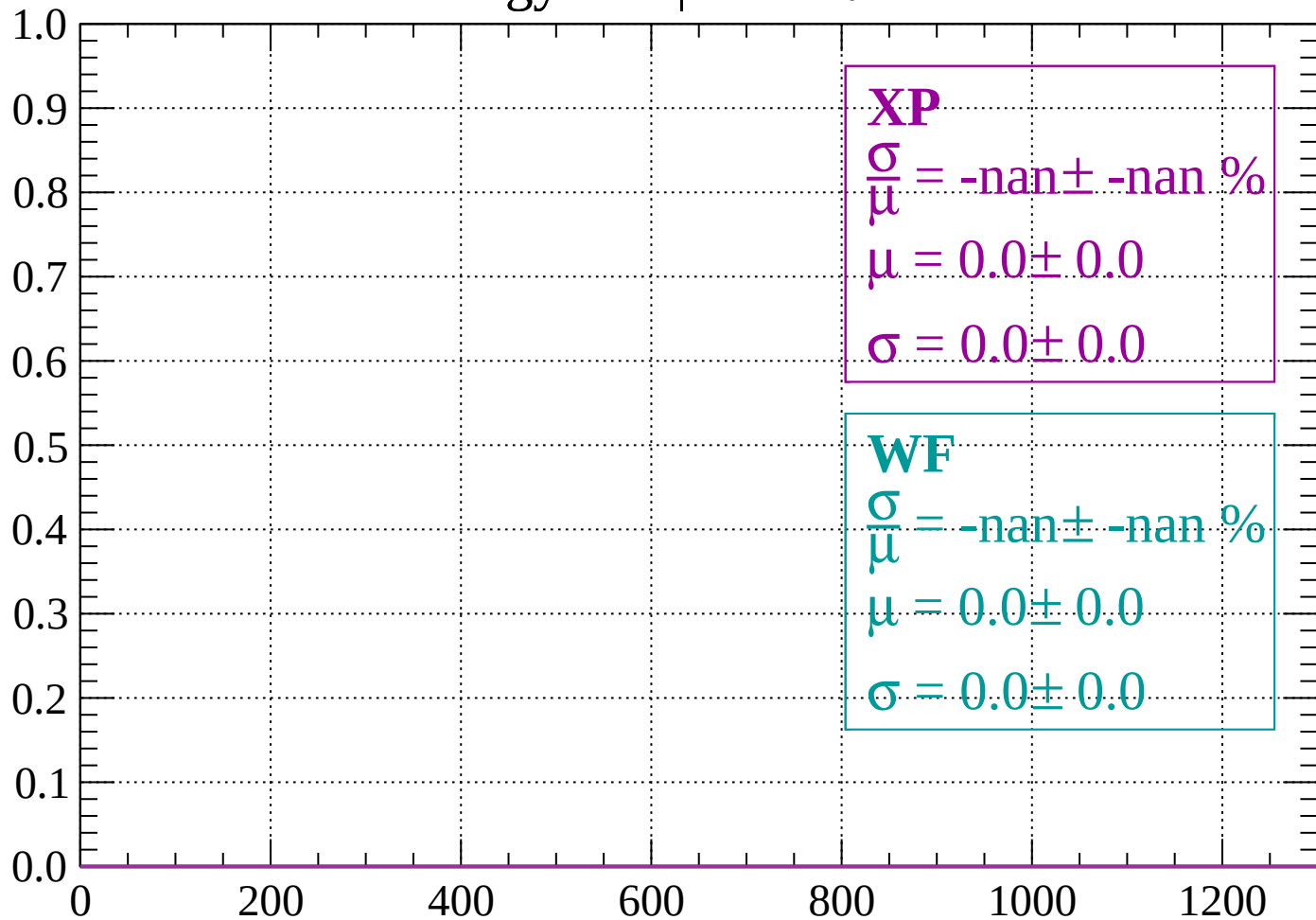
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

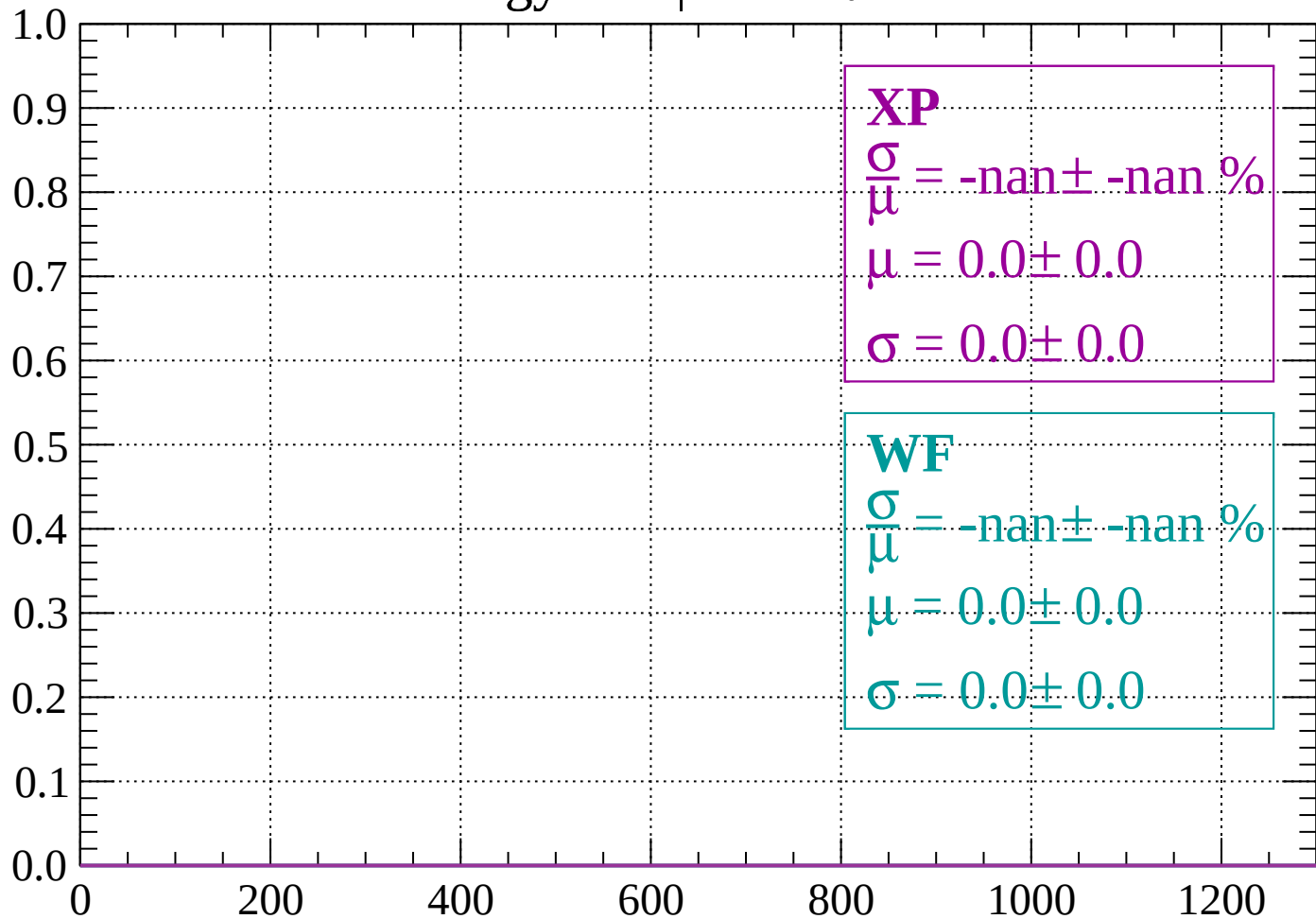
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

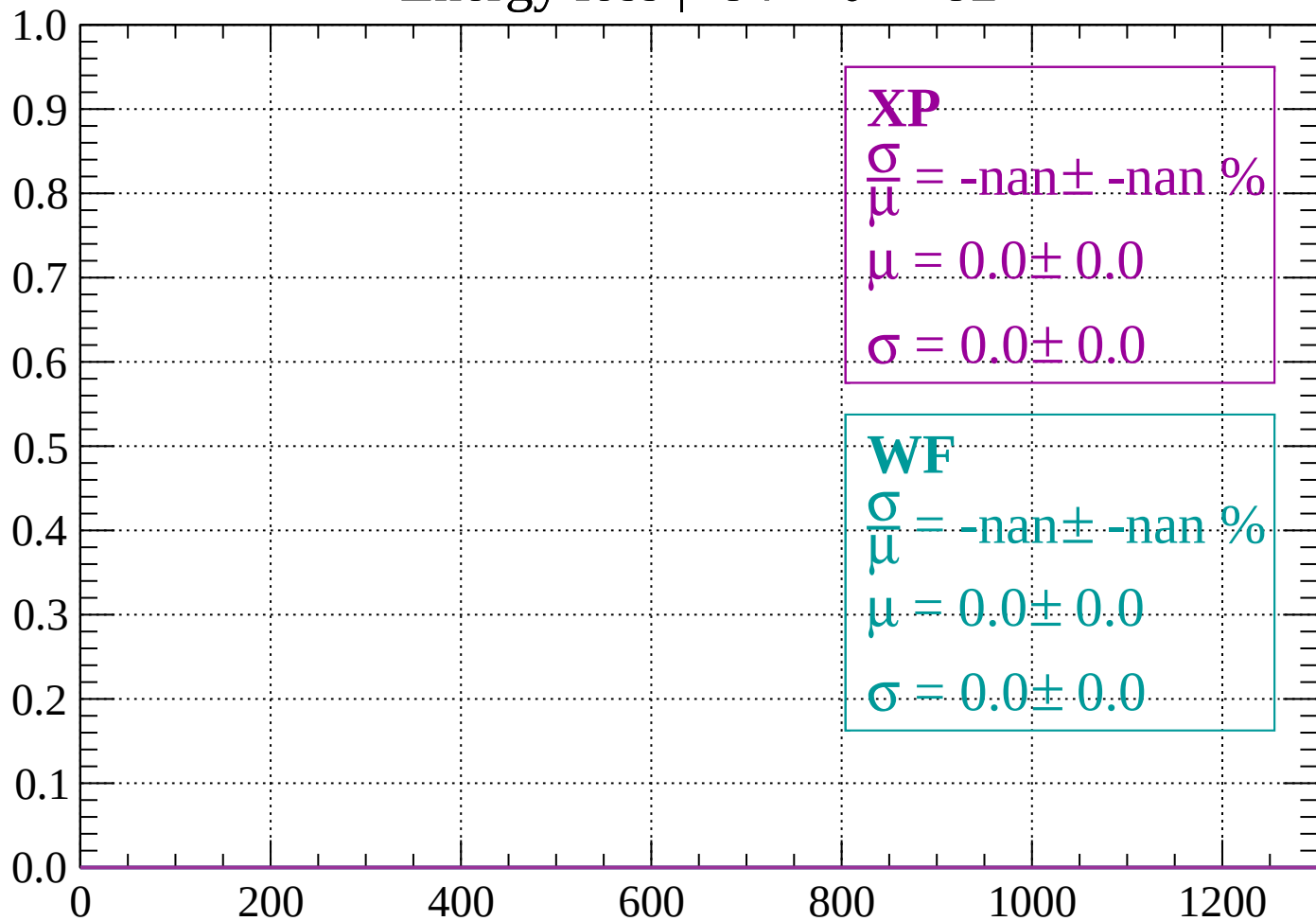
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

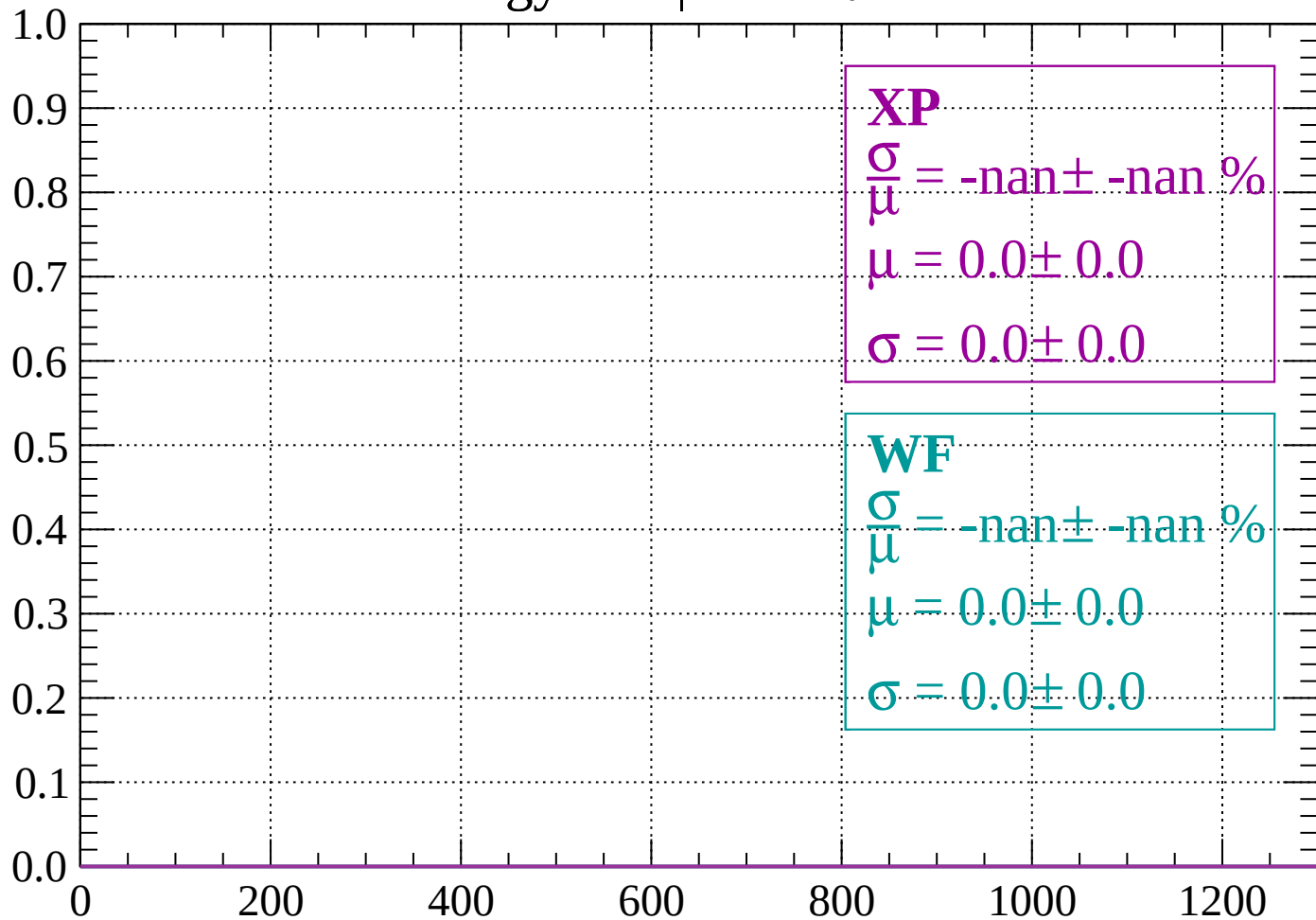
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

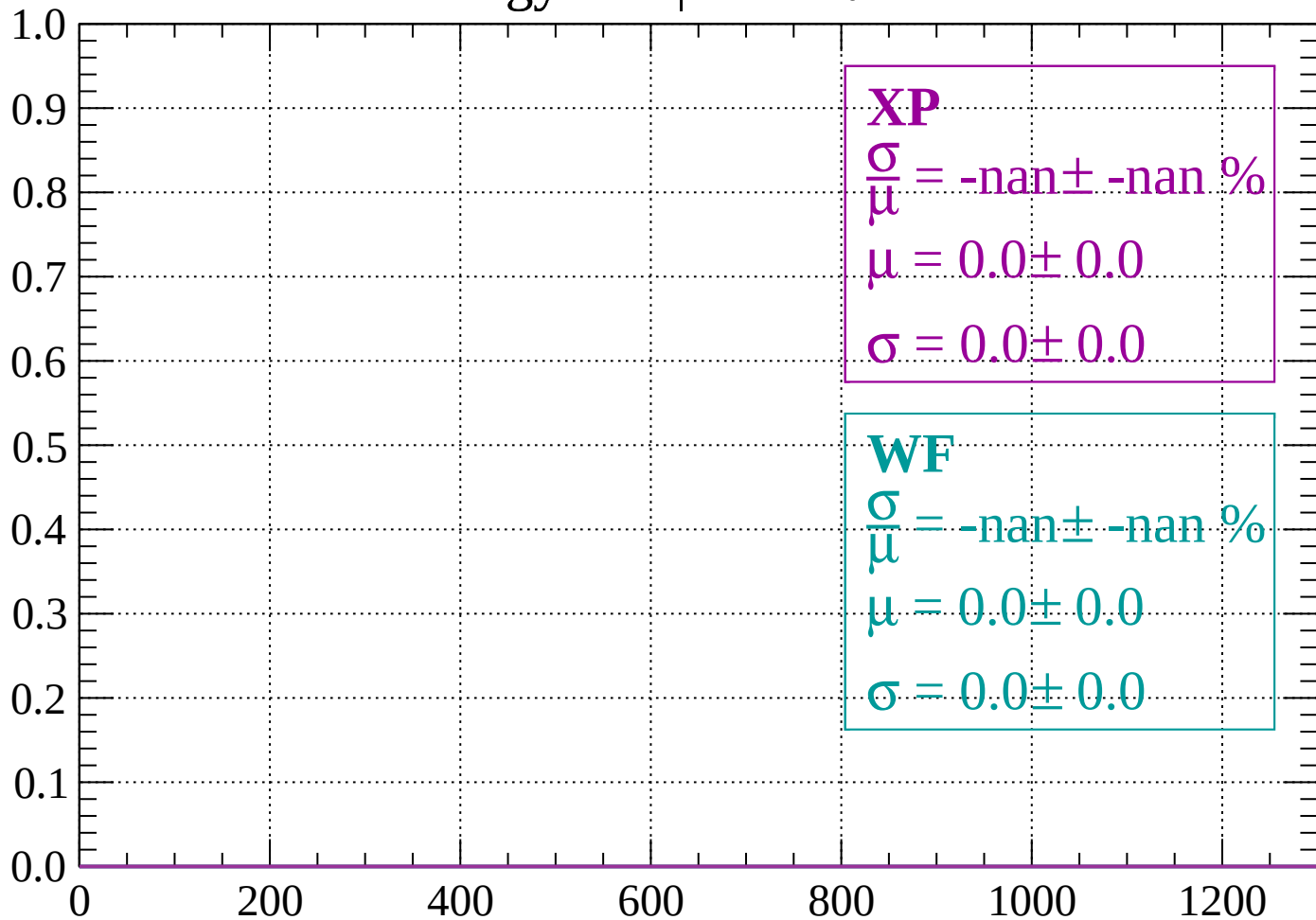
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

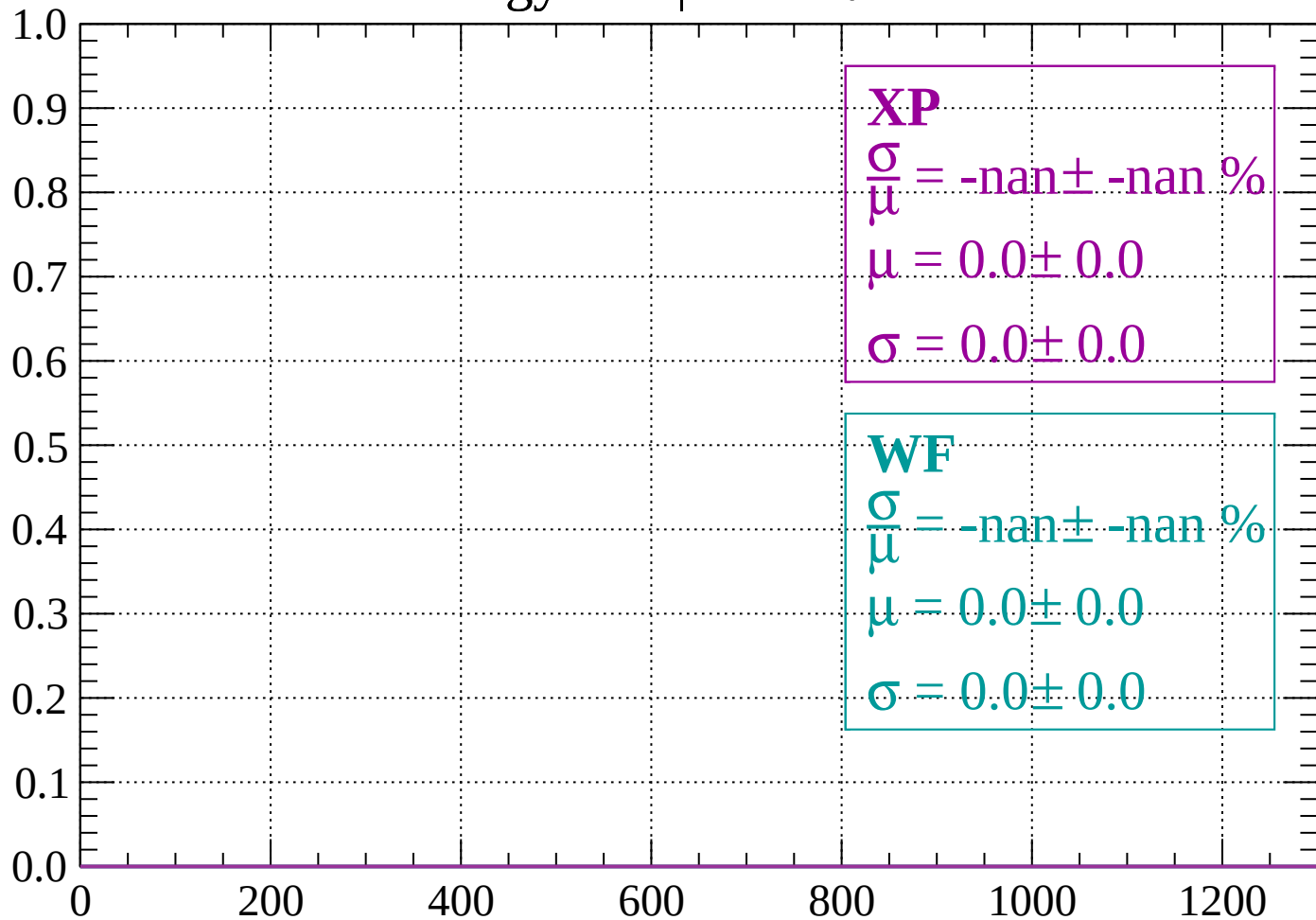
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

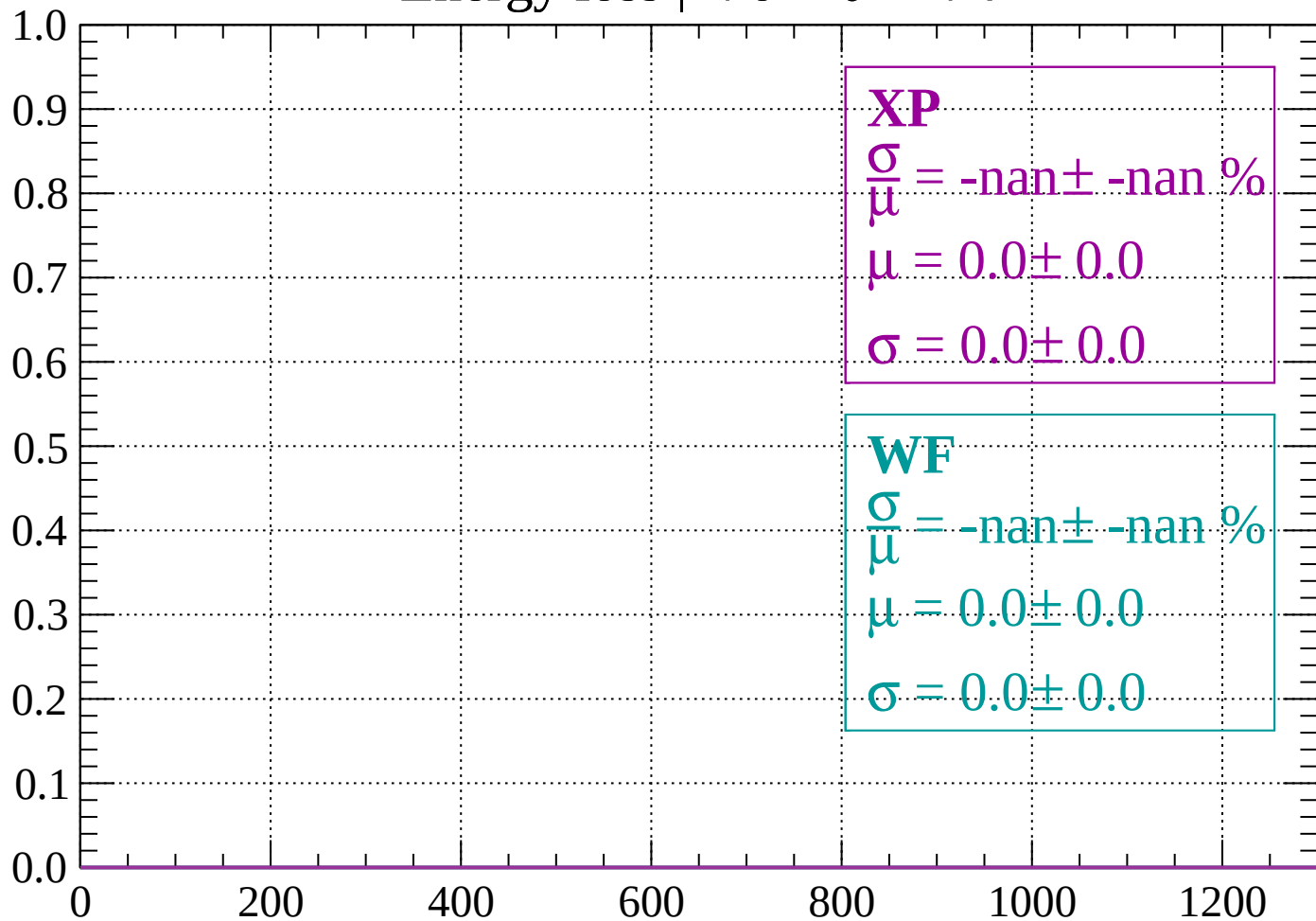
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

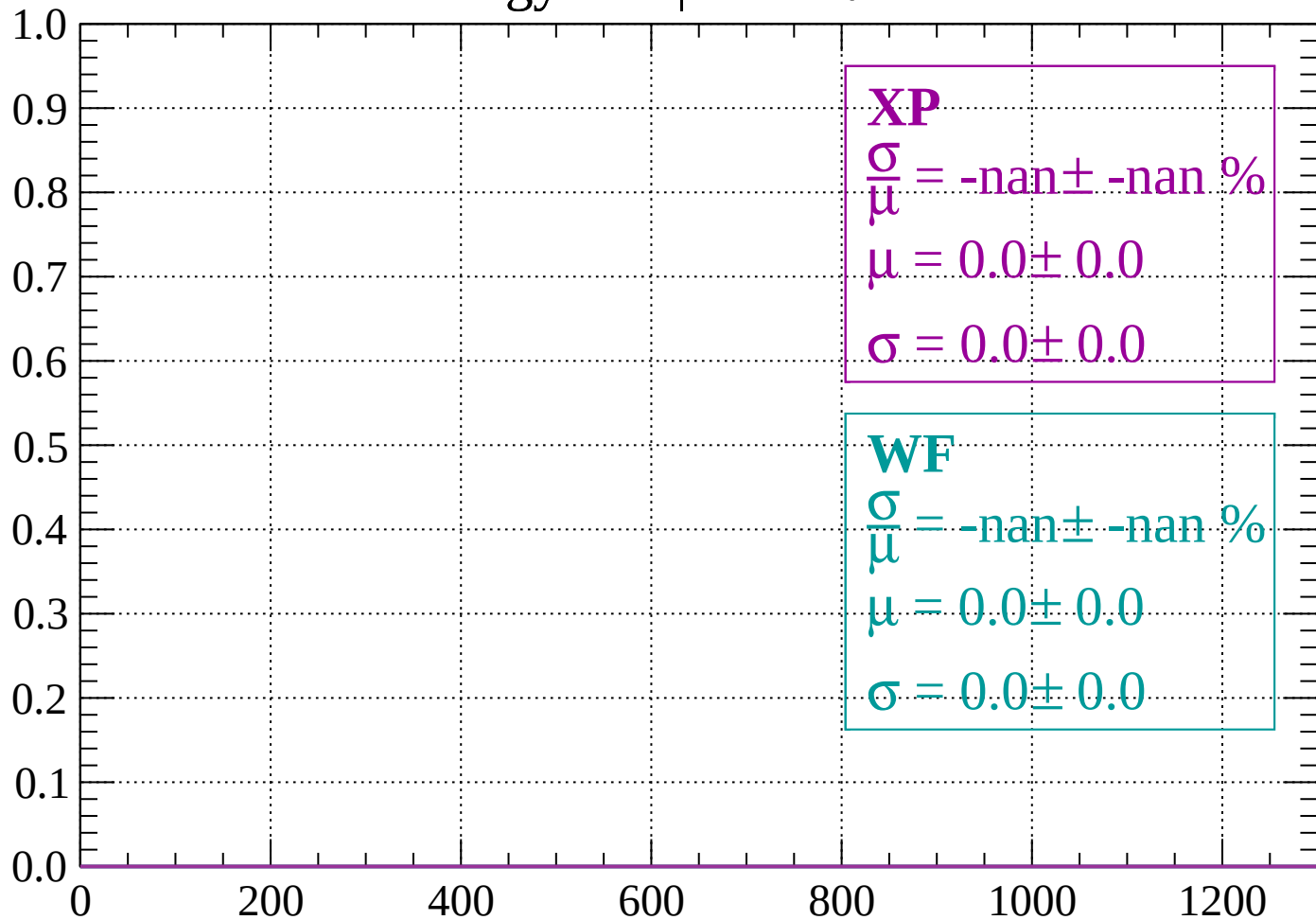
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

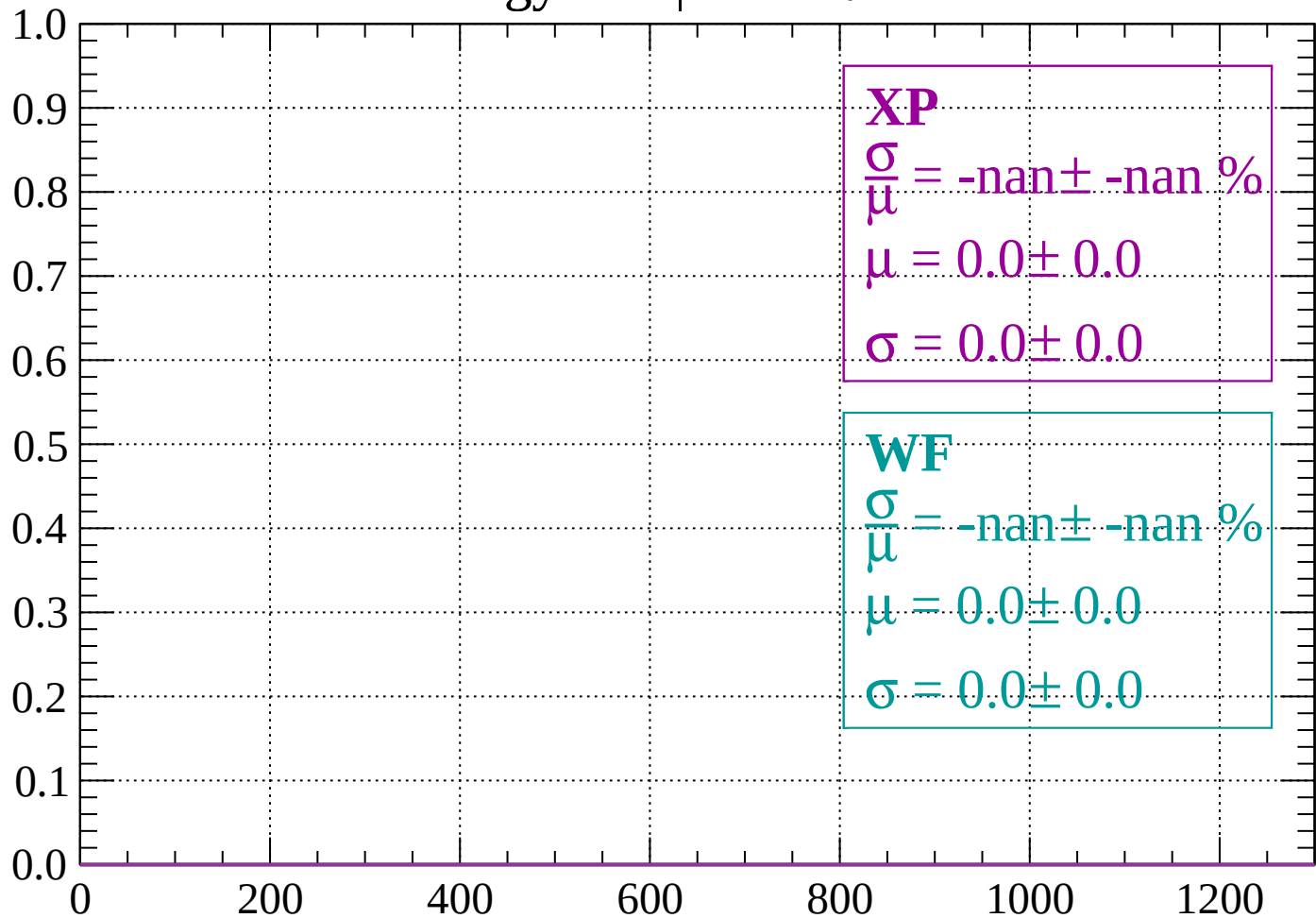
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

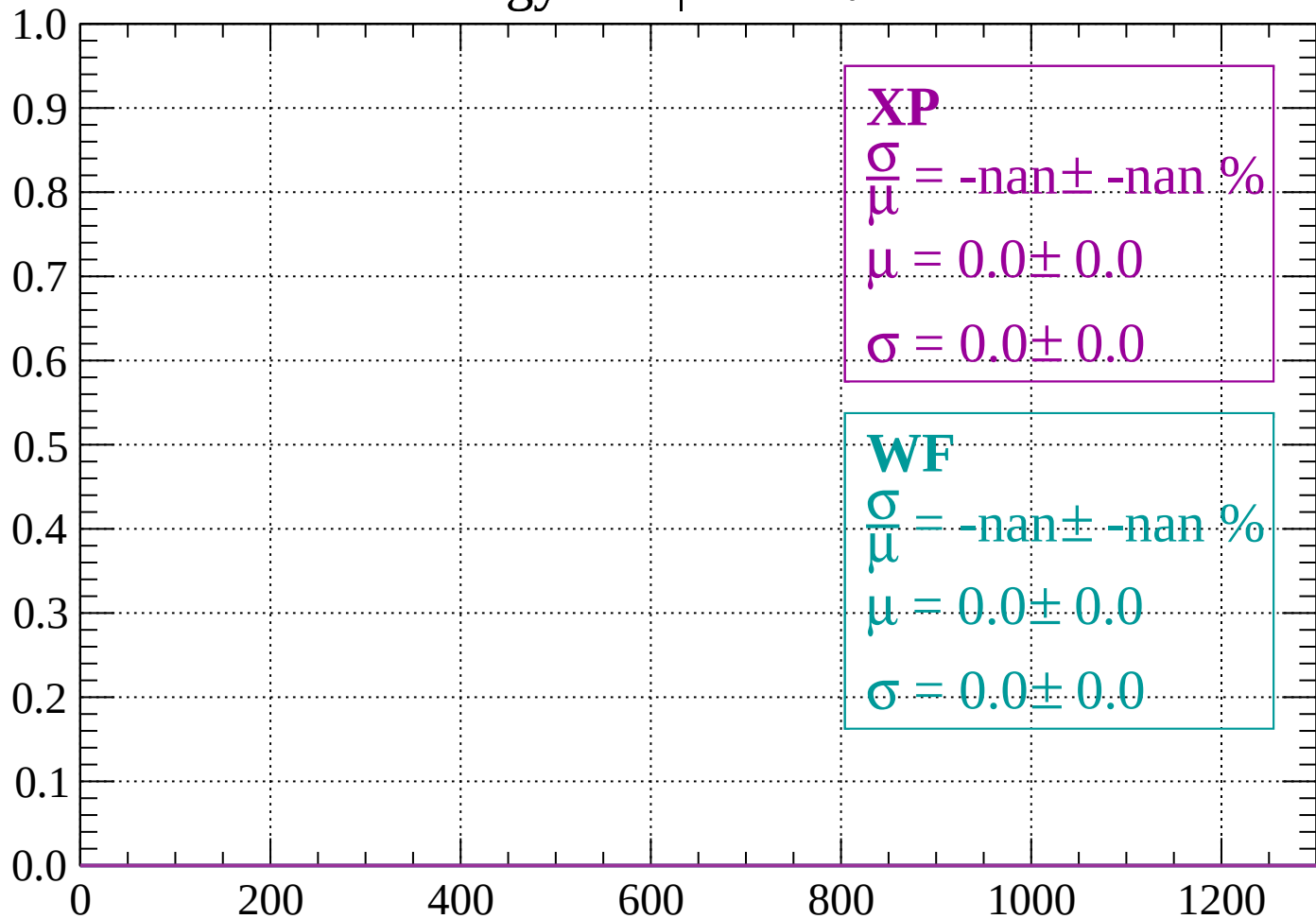
Count



dE/dx (ADC counts/cm)

Energy loss | $-70 < \theta < -68$

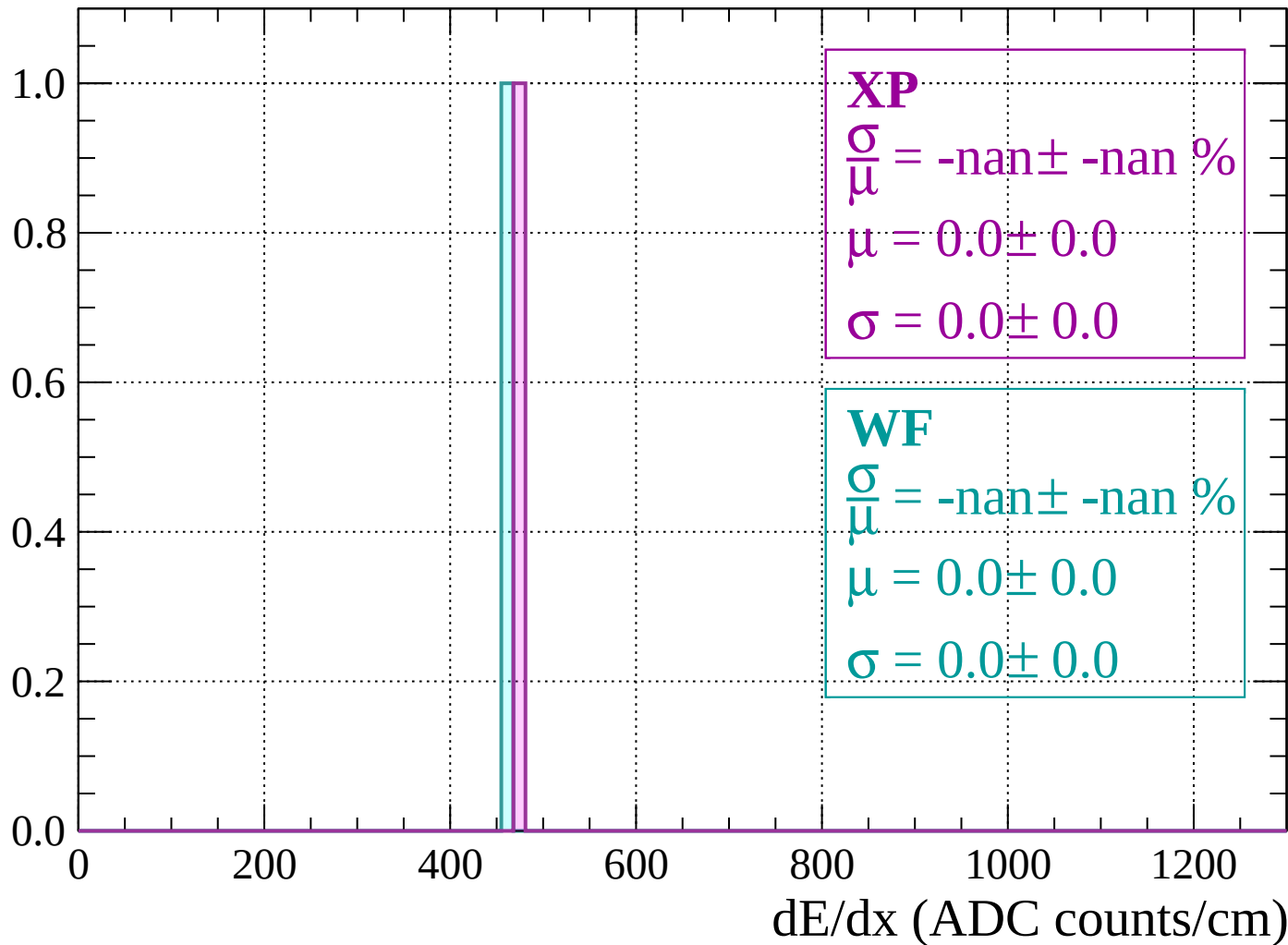
Count



dE/dx (ADC counts/cm)

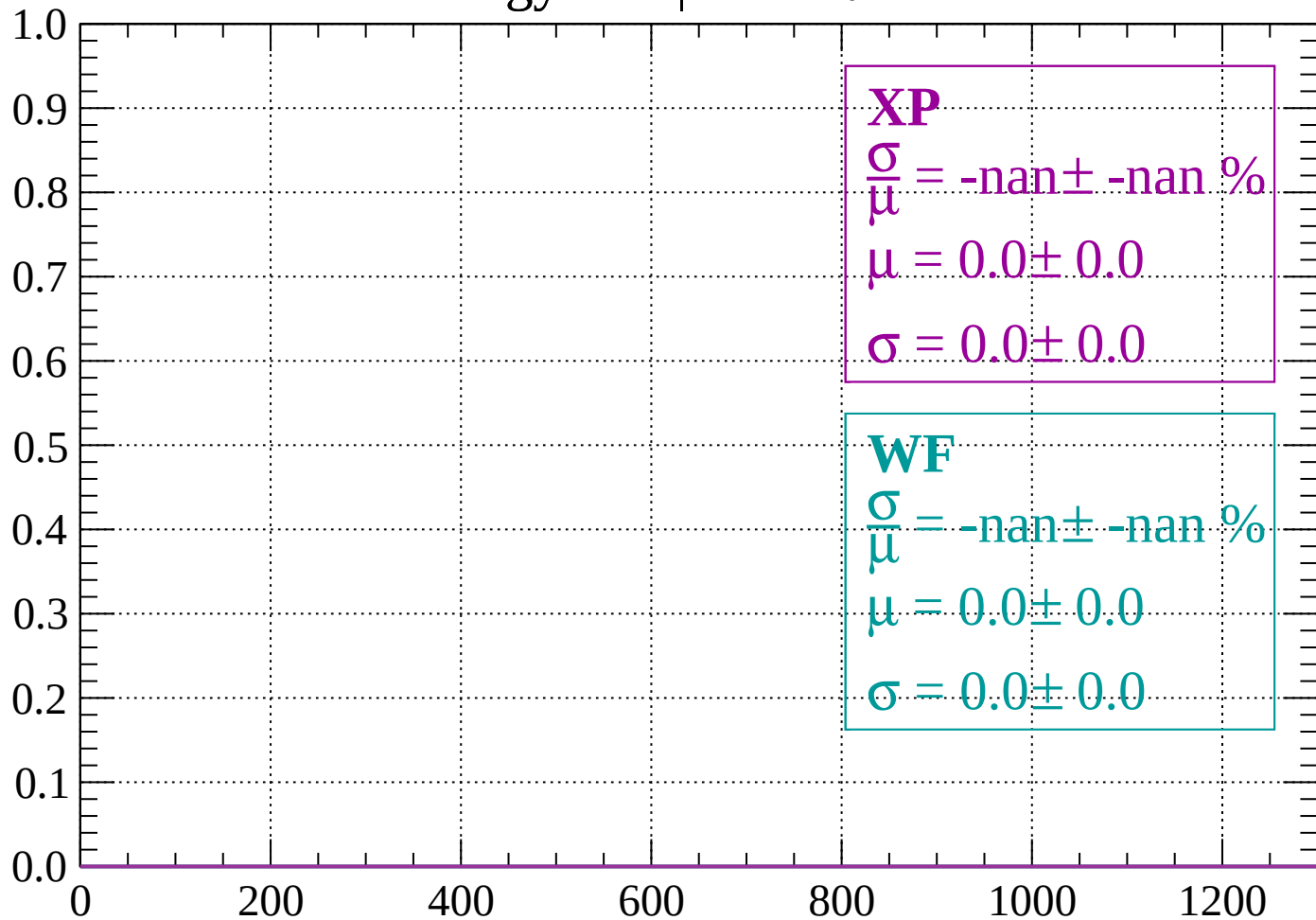
Energy loss | $-68 < \theta < -66$

Count



Energy loss | $-66 < \theta < -64$

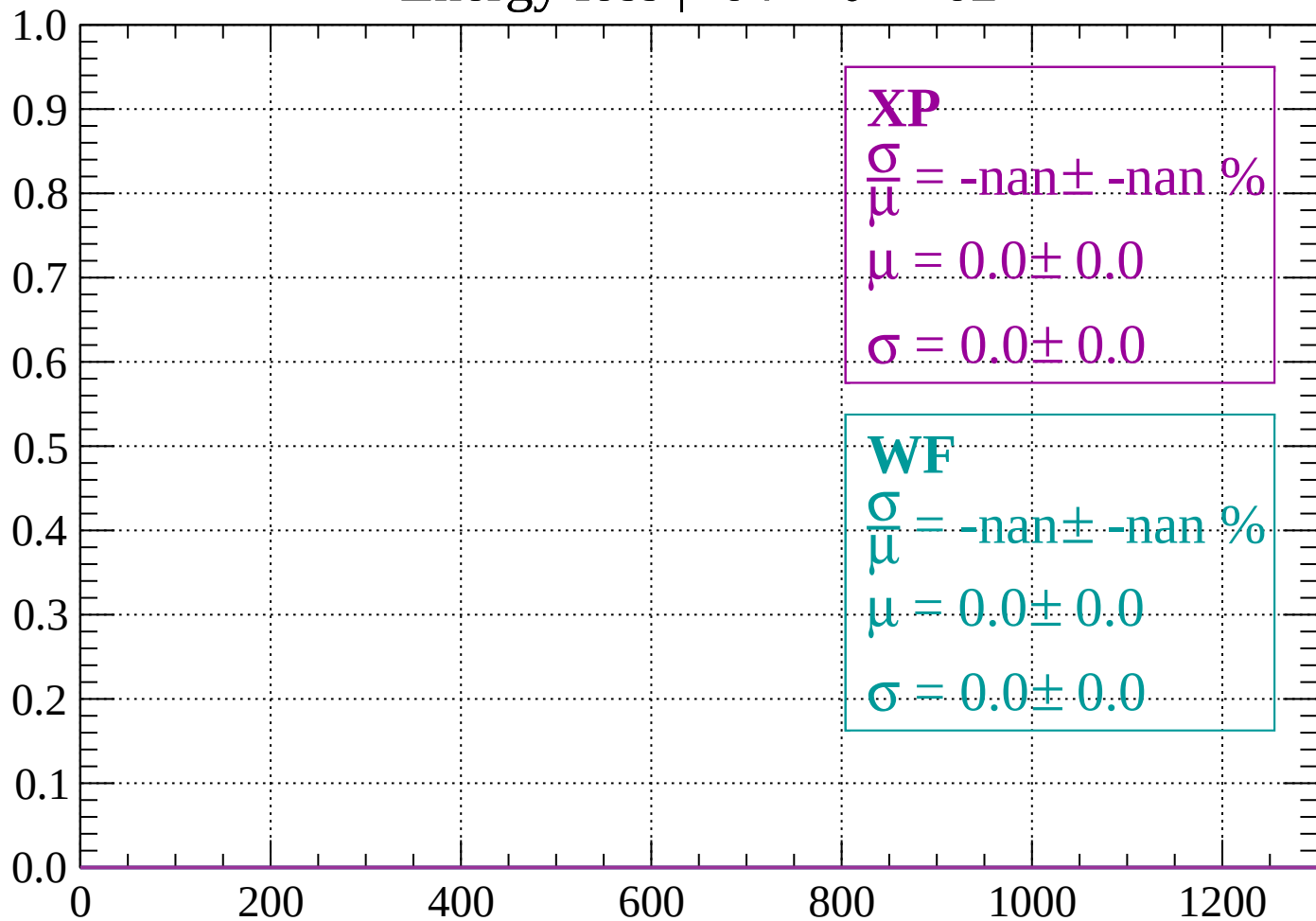
Count



dE/dx (ADC counts/cm)

Energy loss | $-64 < \theta < -62$

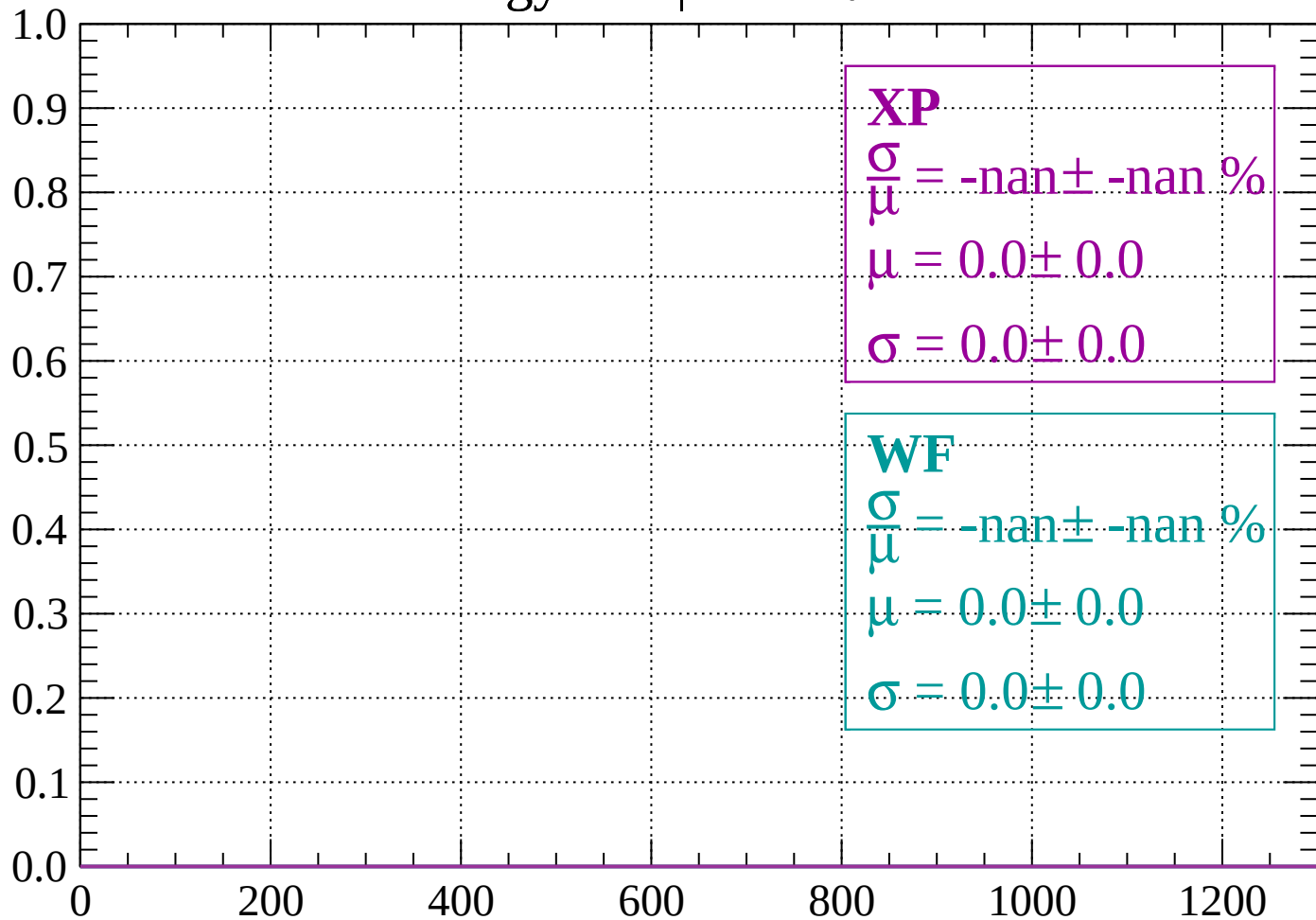
Count



dE/dx (ADC counts/cm)

Energy loss | $-62 < \theta < -60$

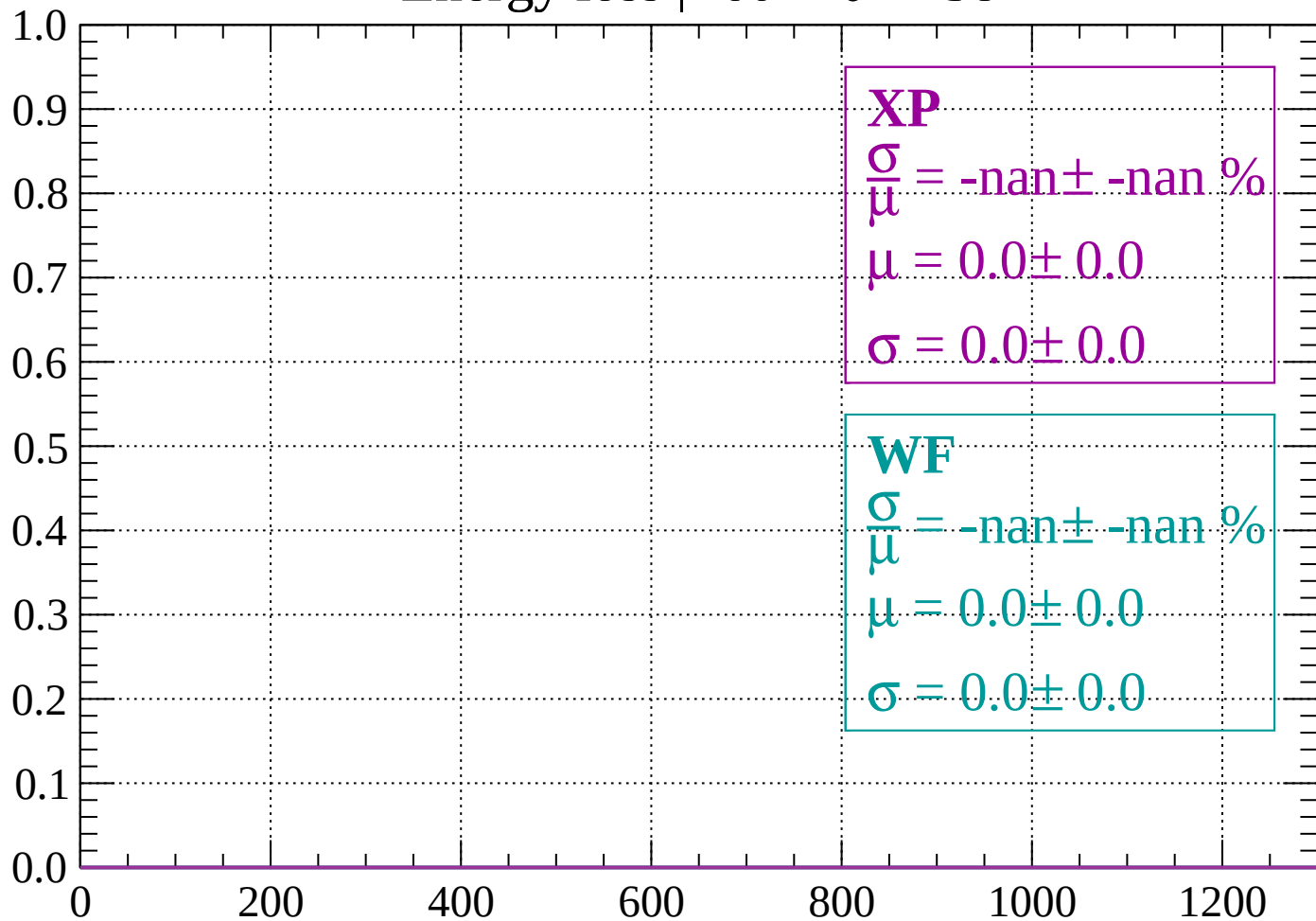
Count



dE/dx (ADC counts/cm)

Energy loss | $-60 < \theta < -58$

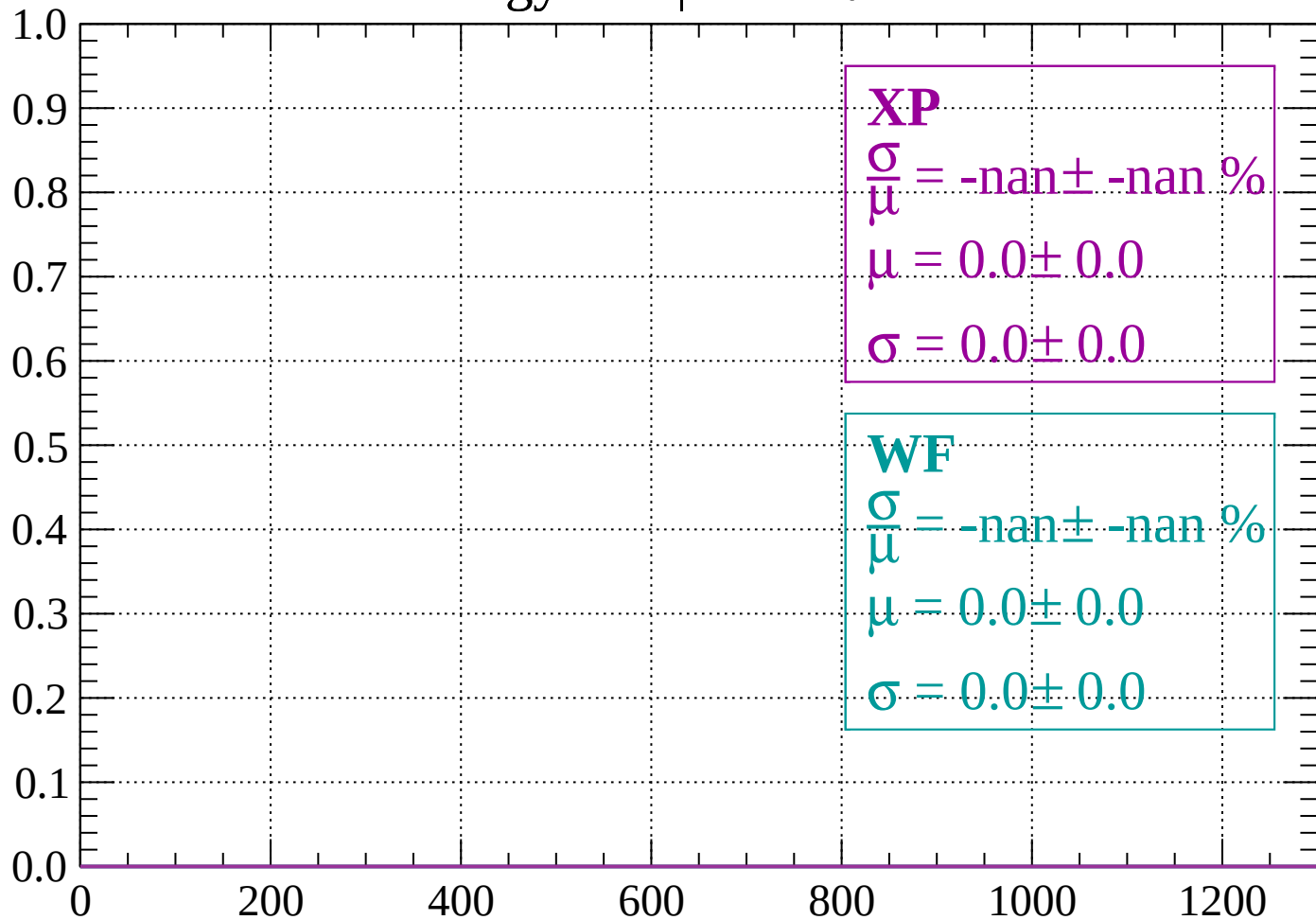
Count



dE/dx (ADC counts/cm)

Energy loss | $-58 < \theta < -56$

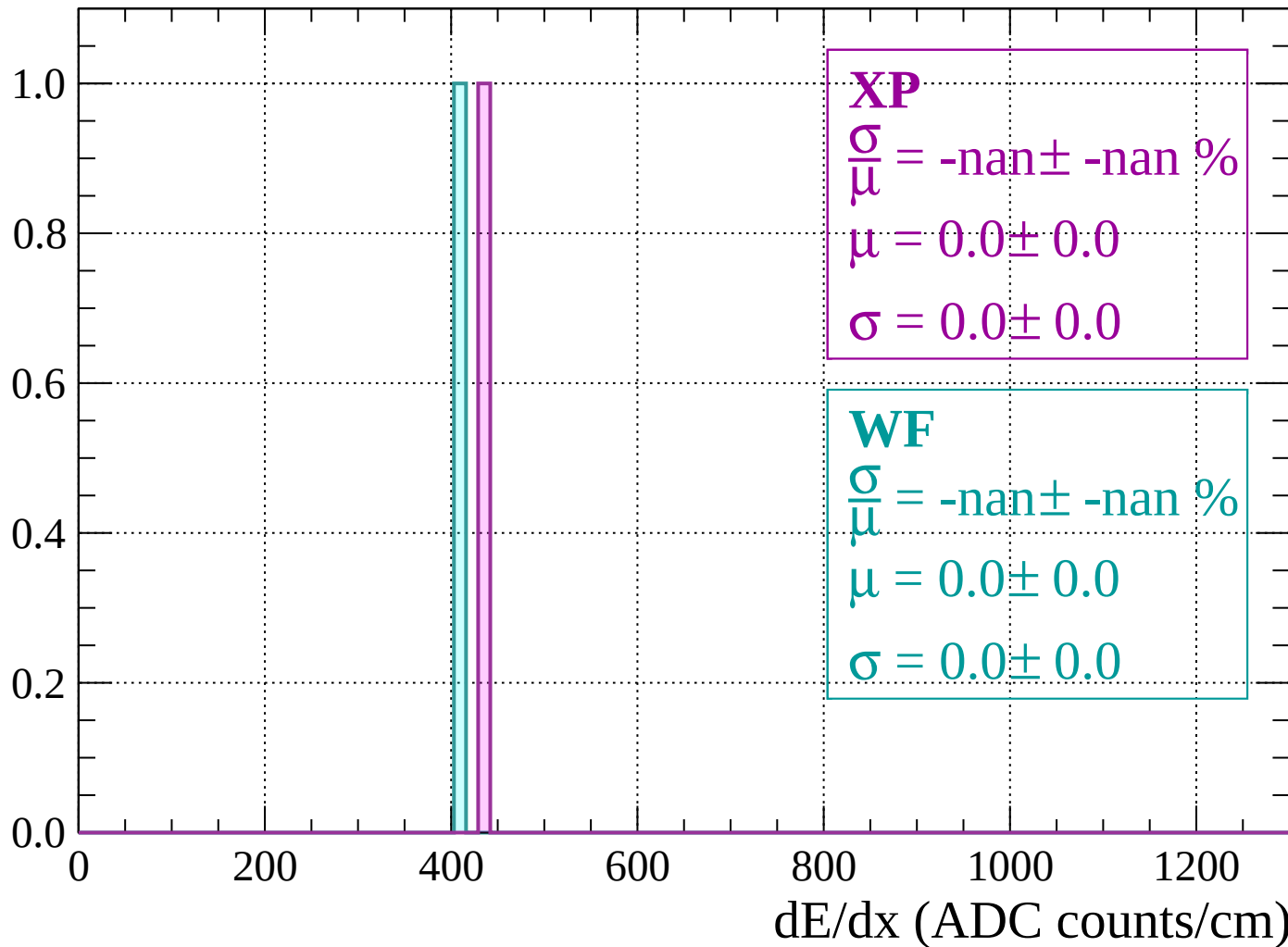
Count



dE/dx (ADC counts/cm)

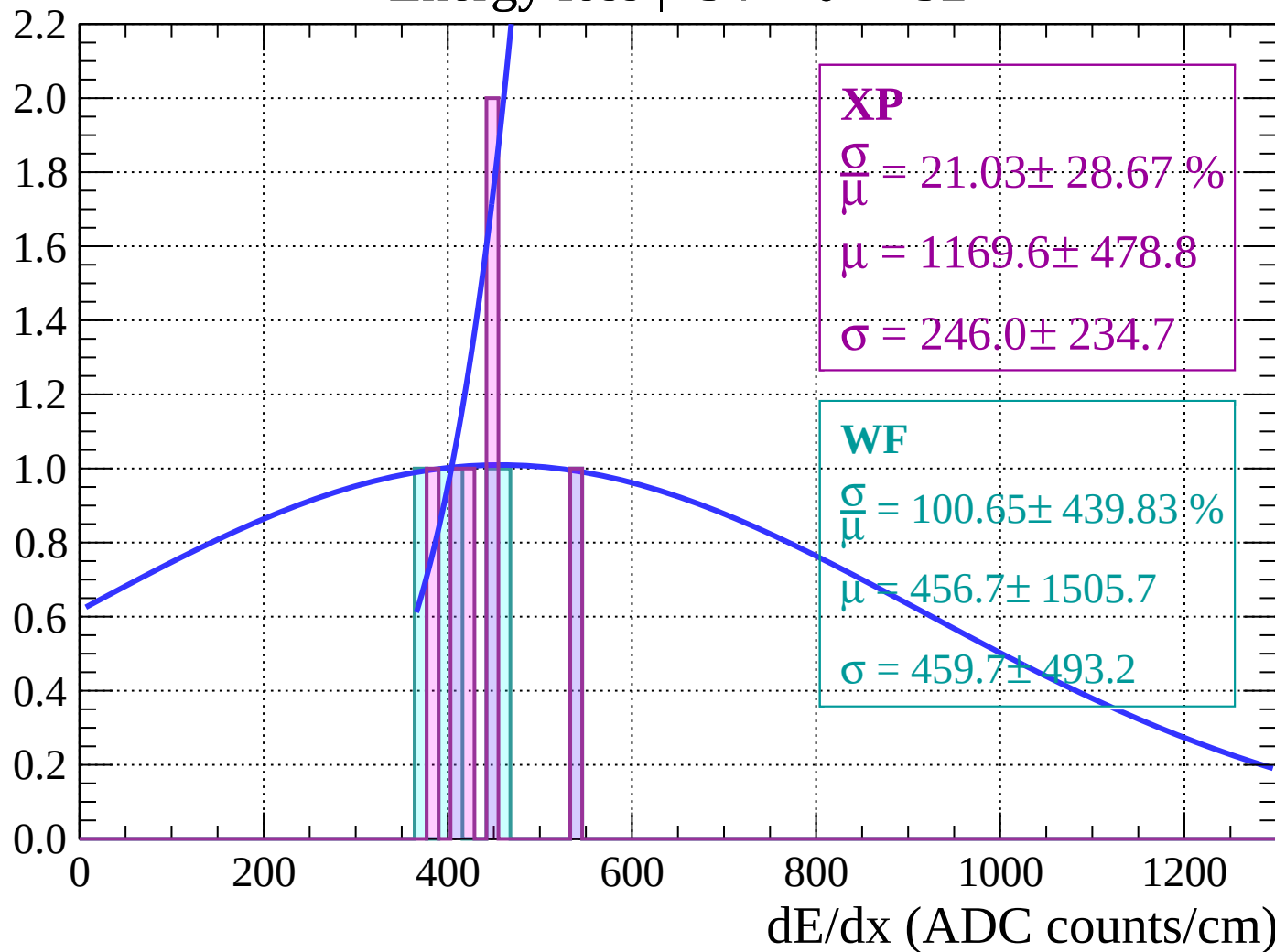
Energy loss | $-56 < \theta < -54$

Count



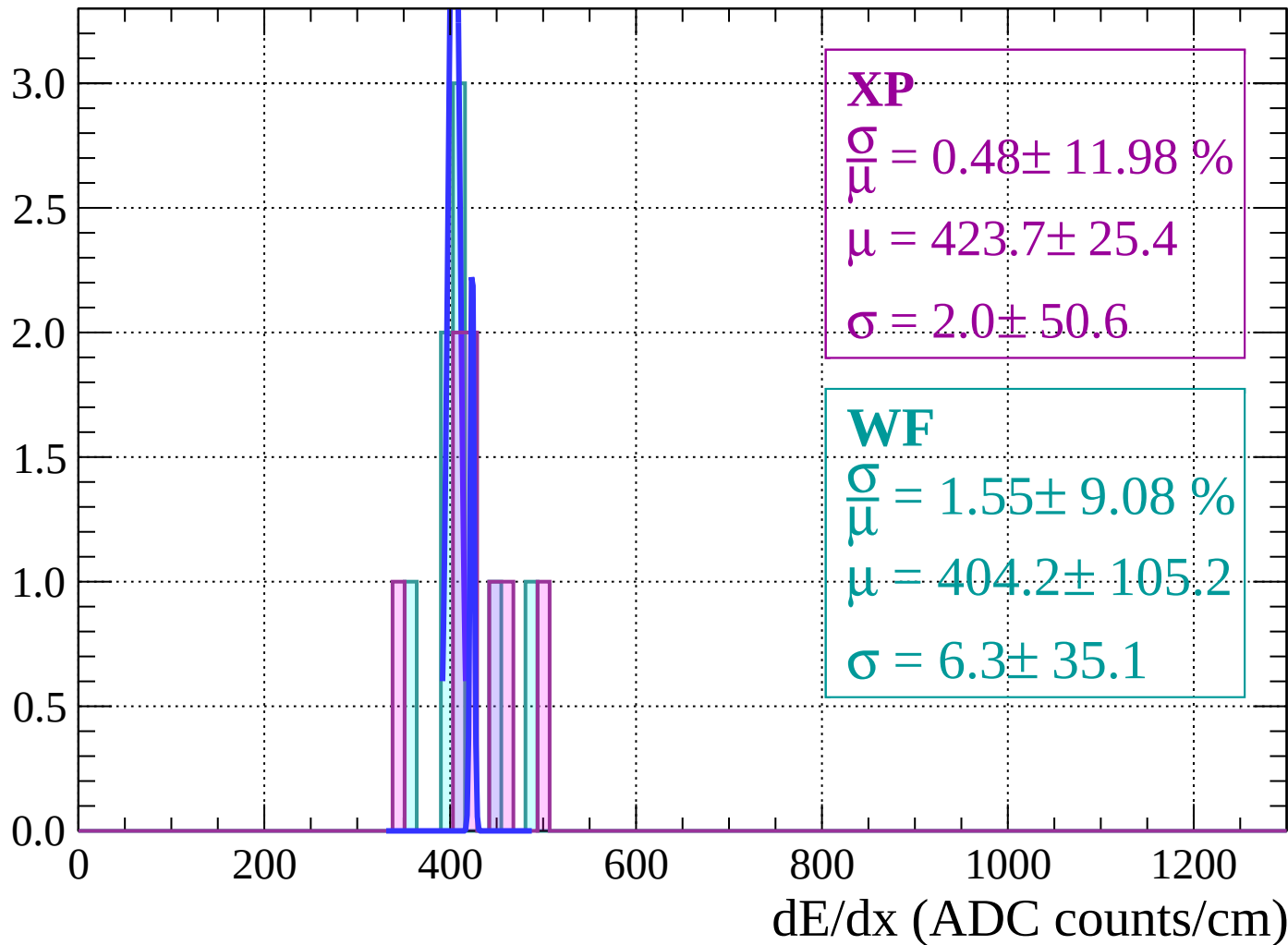
Energy loss $|-54 < \theta < -52$

Count



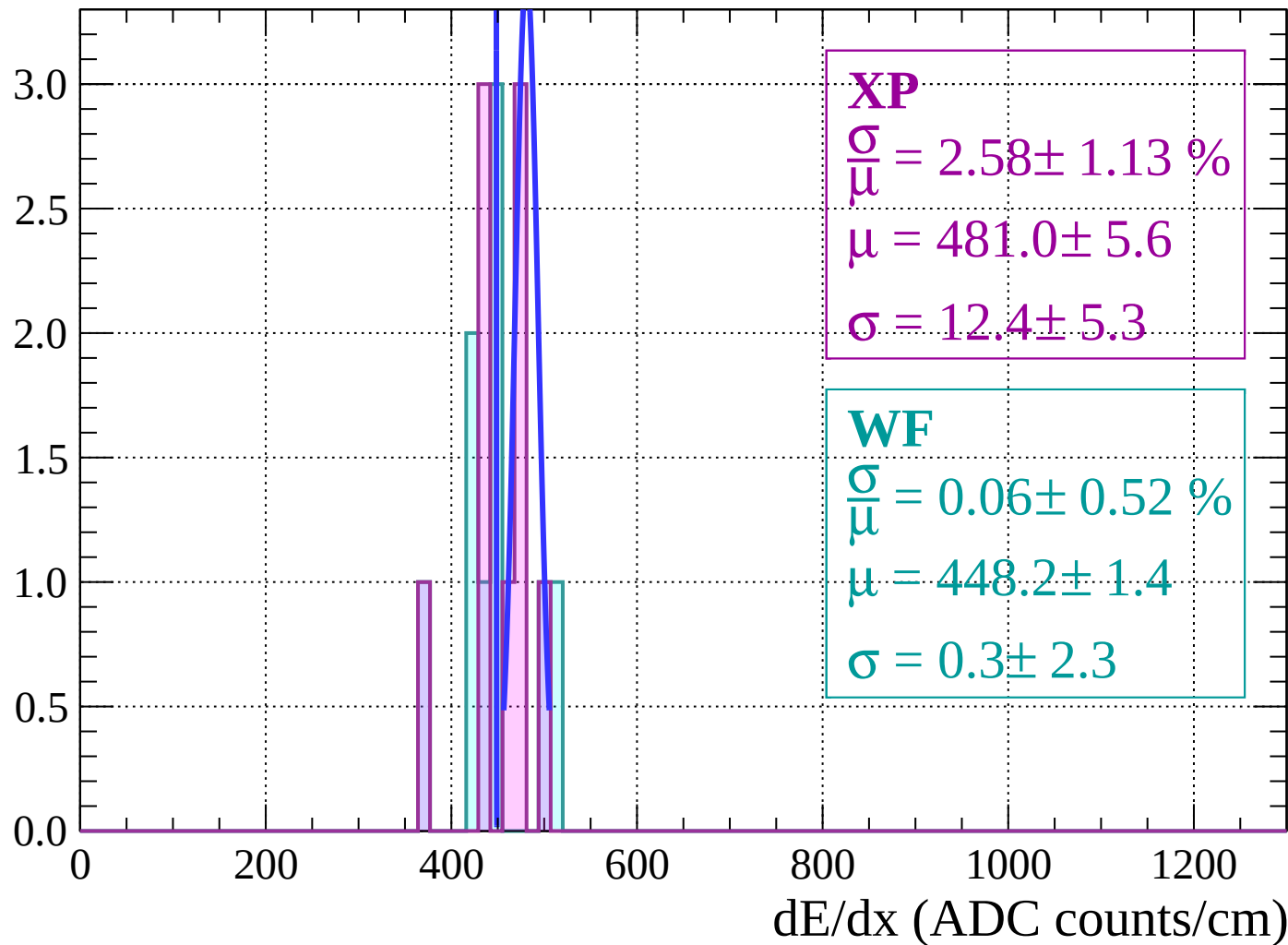
Energy loss | $-52 < \theta < -50$

Count



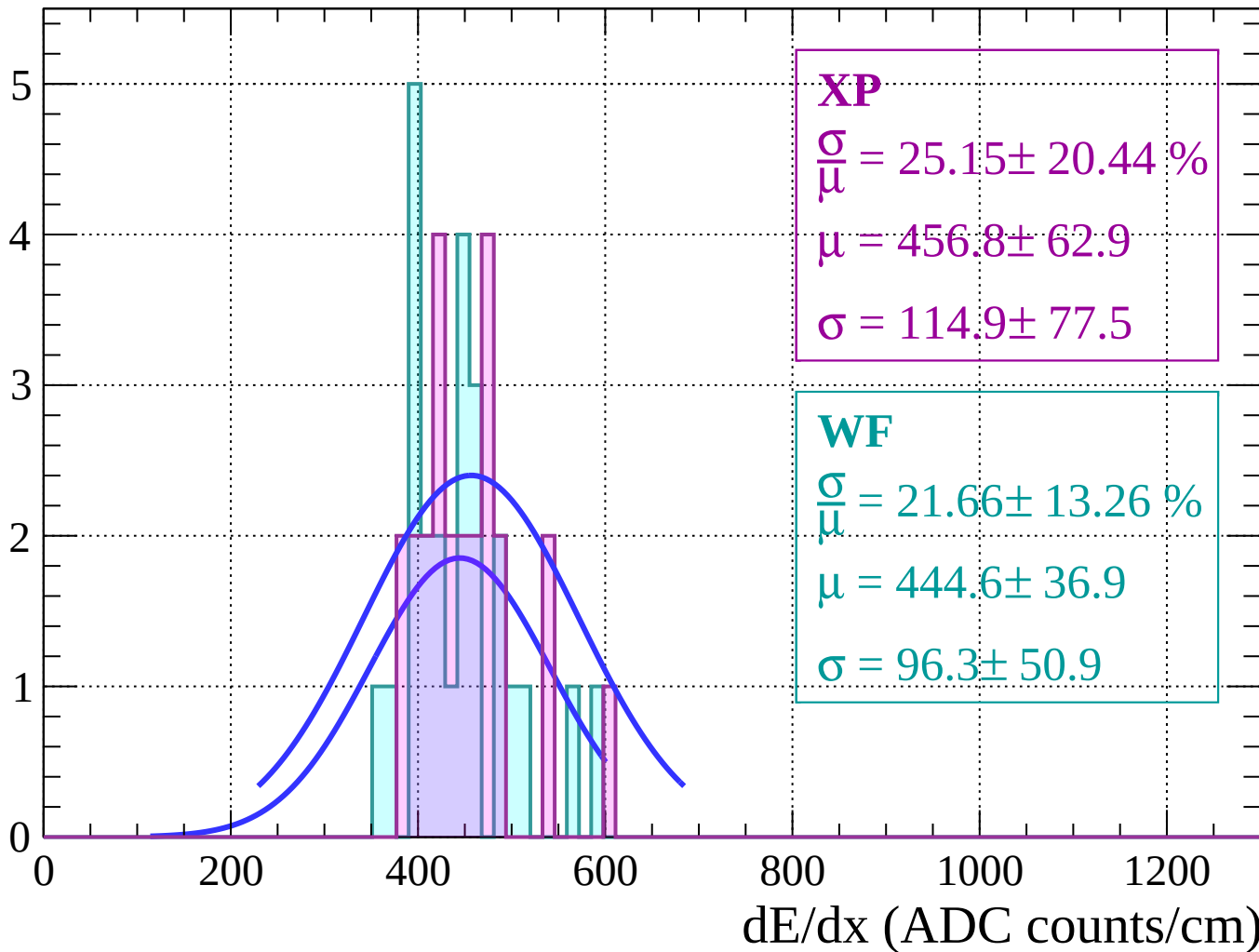
Energy loss | $-50 < \theta < -48$

Count



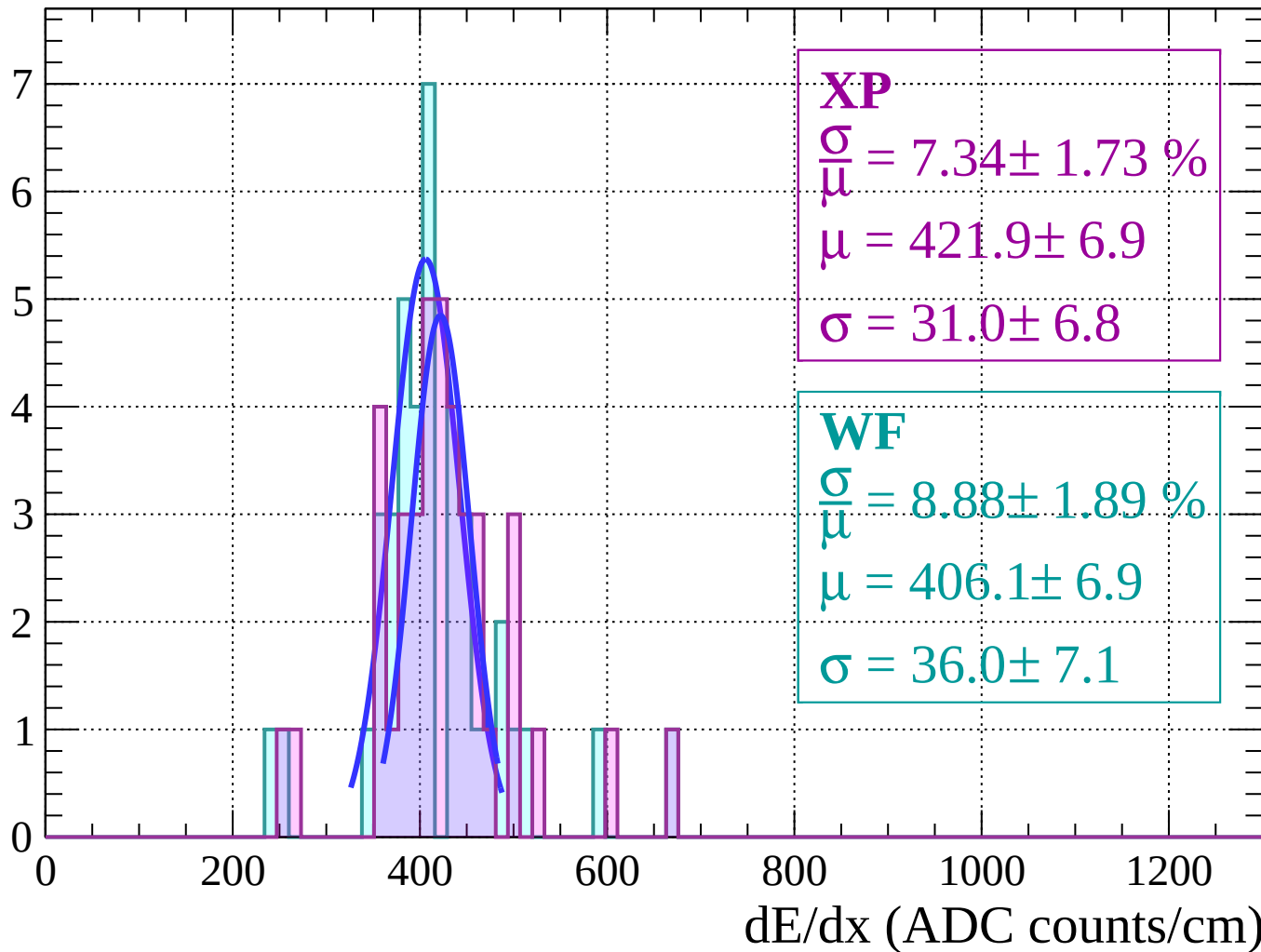
Energy loss | $-48 < \theta < -46$

Count



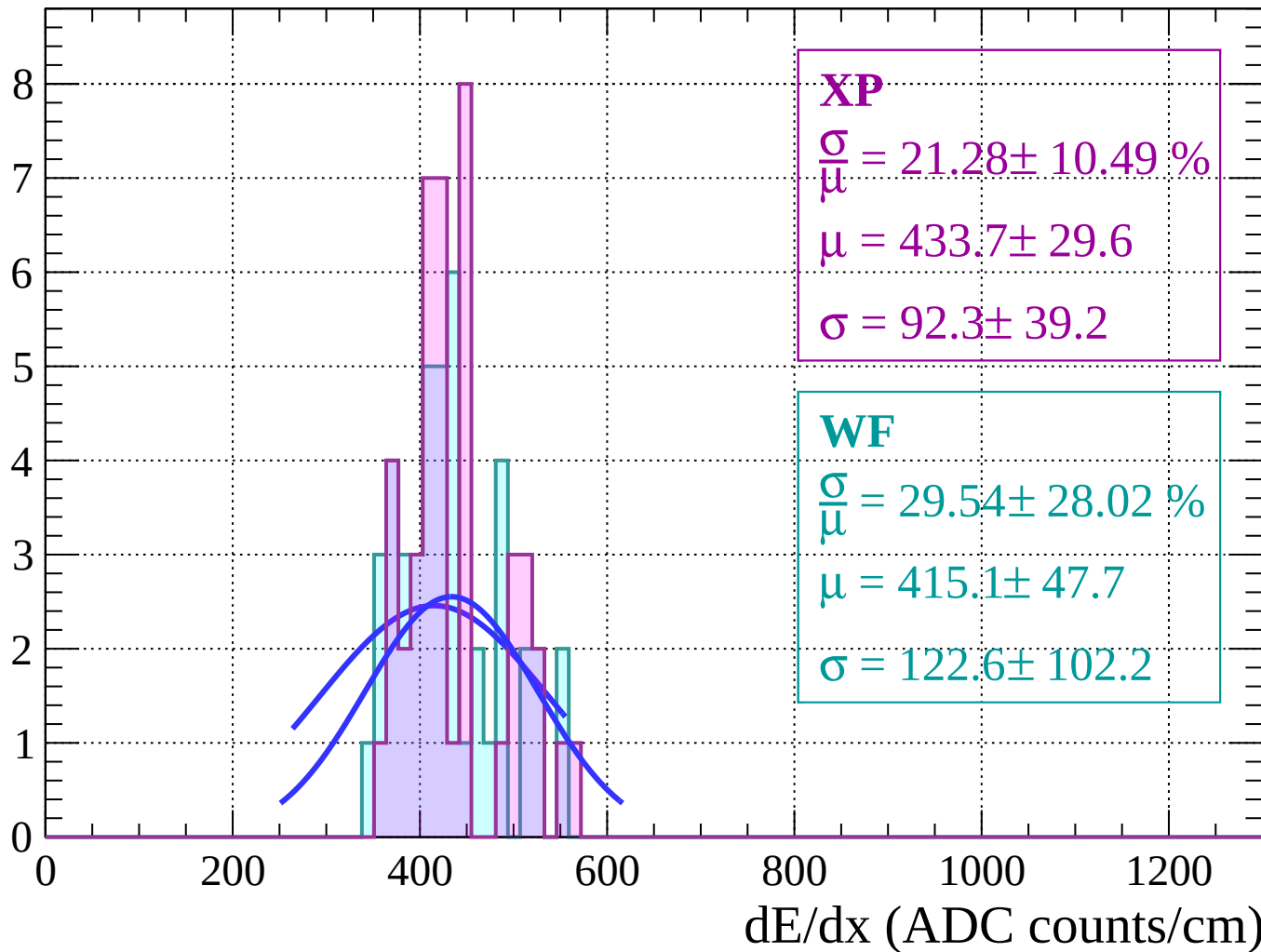
Energy loss $|-46 < \theta < -44$

Count



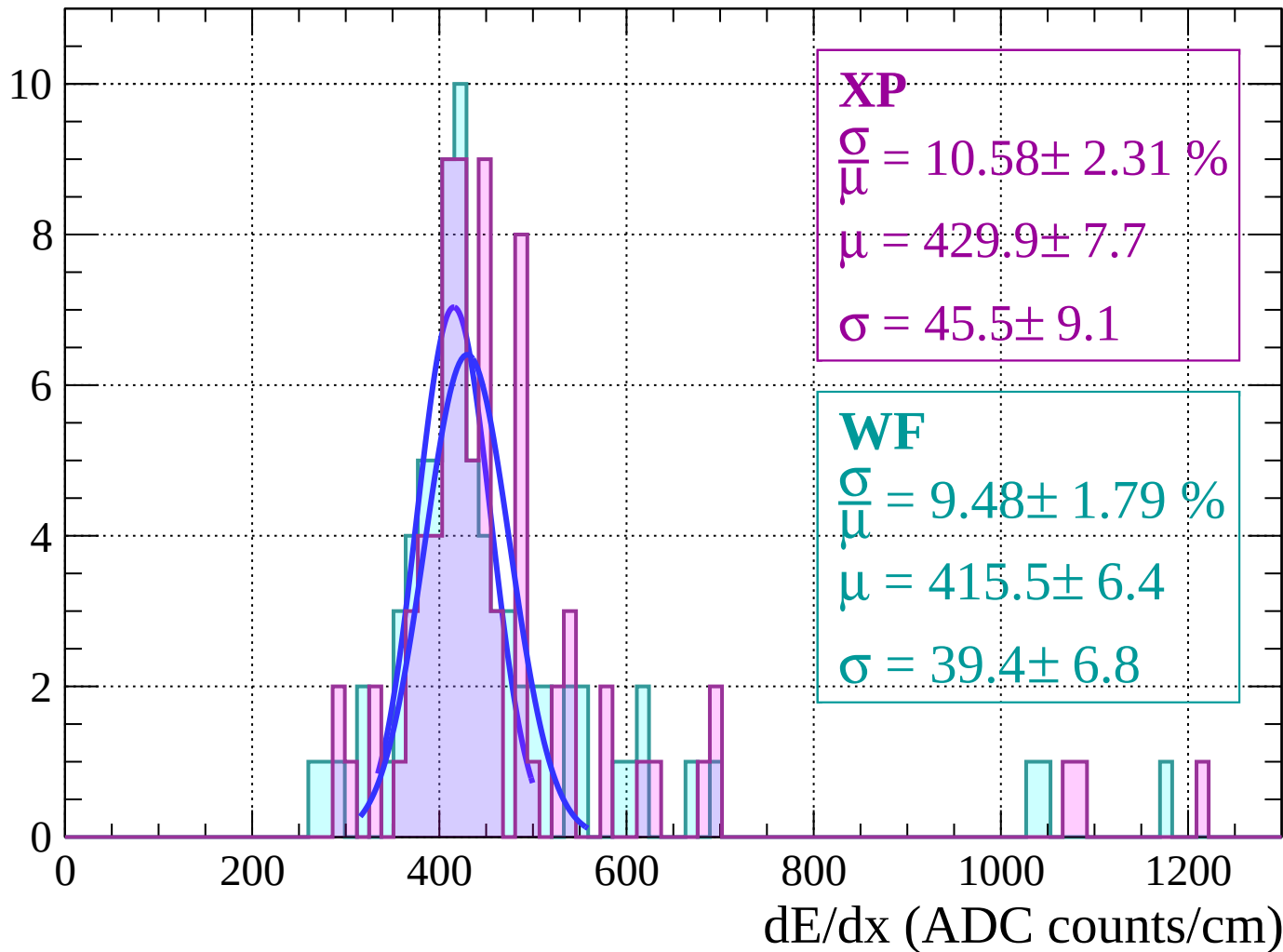
Energy loss | $-44 < \theta < -42$

Count



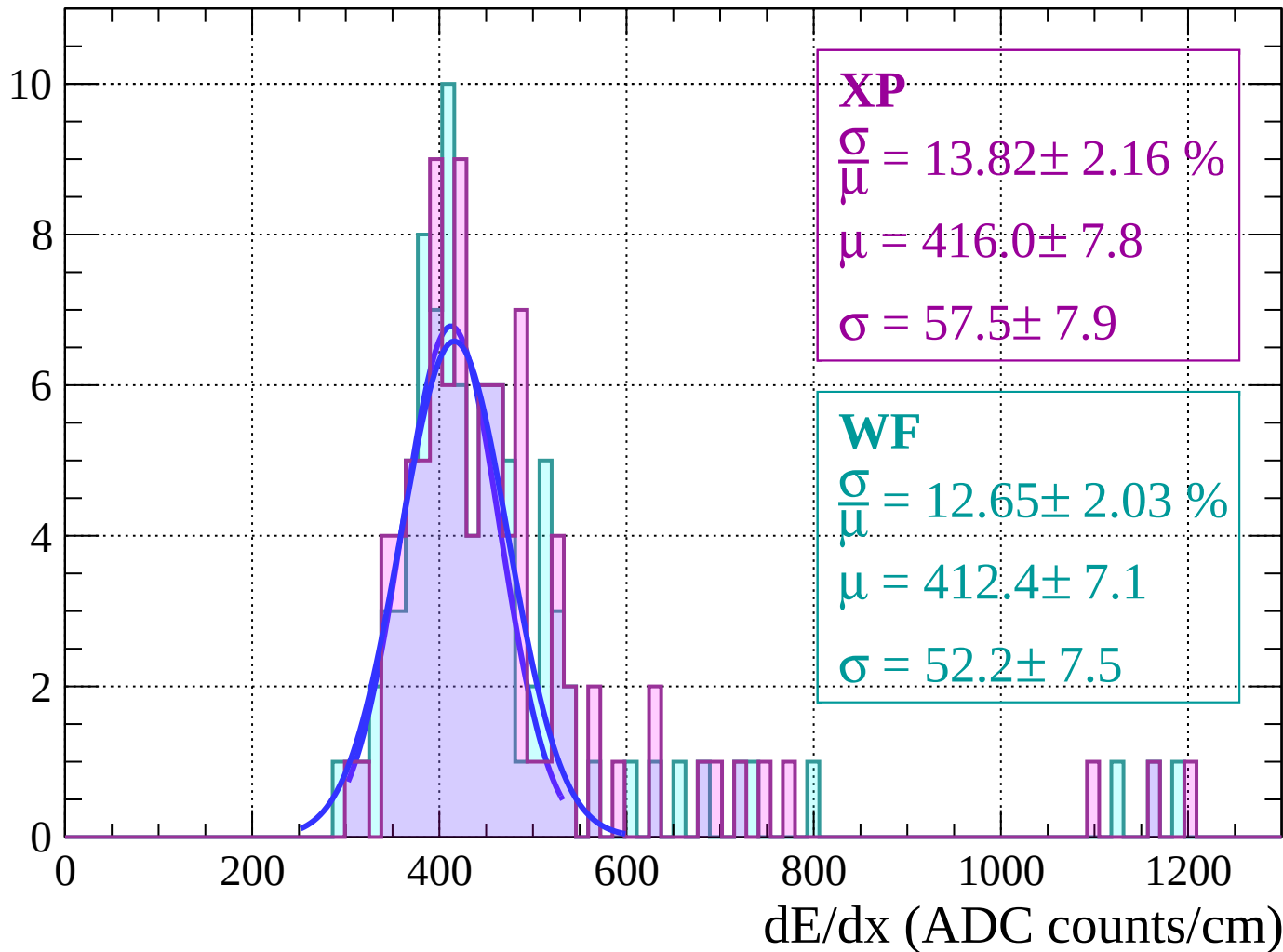
Energy loss | $-42 < \theta < -40$

Count



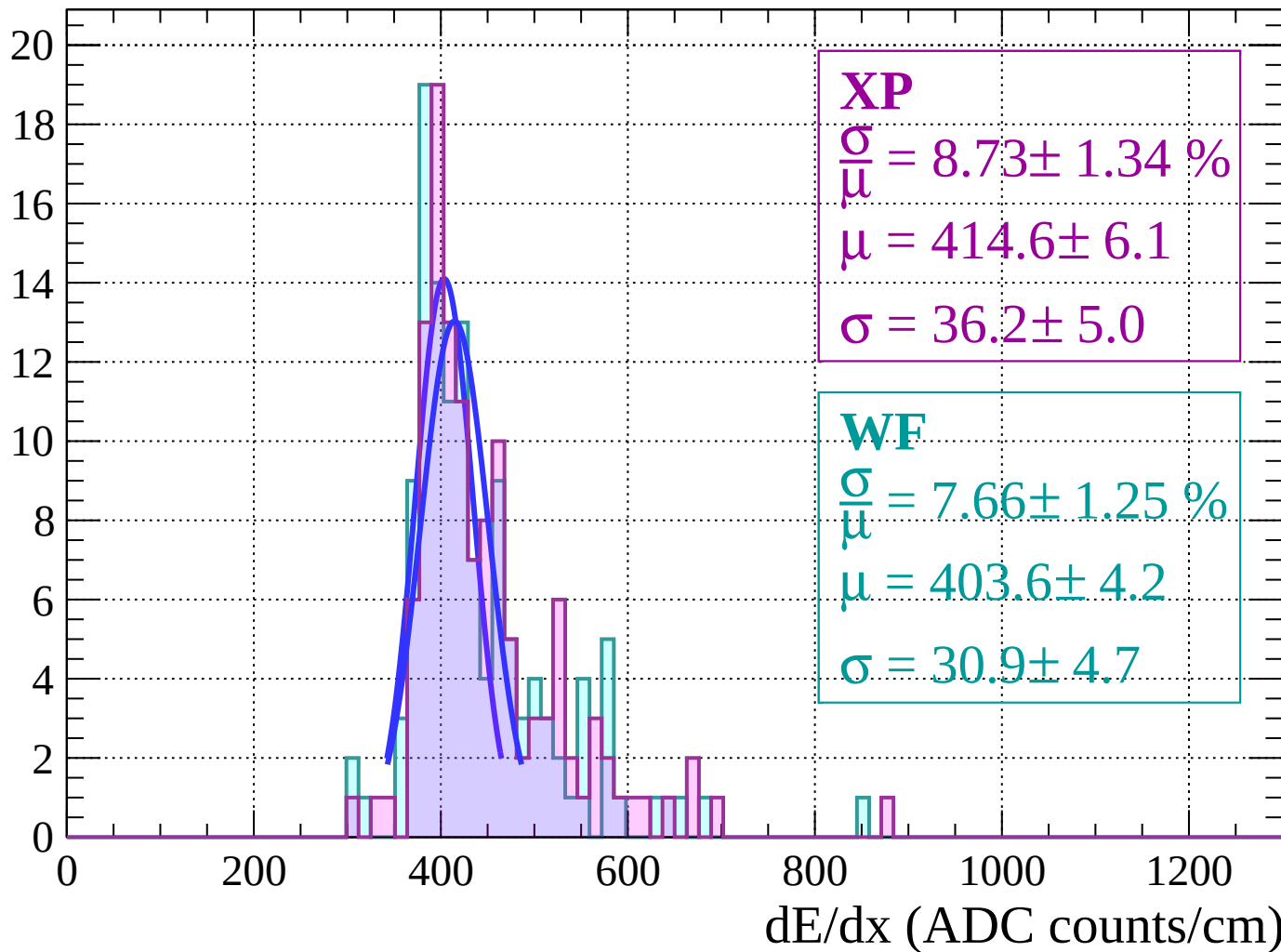
Energy loss | $-40 < \theta < -38$

Count



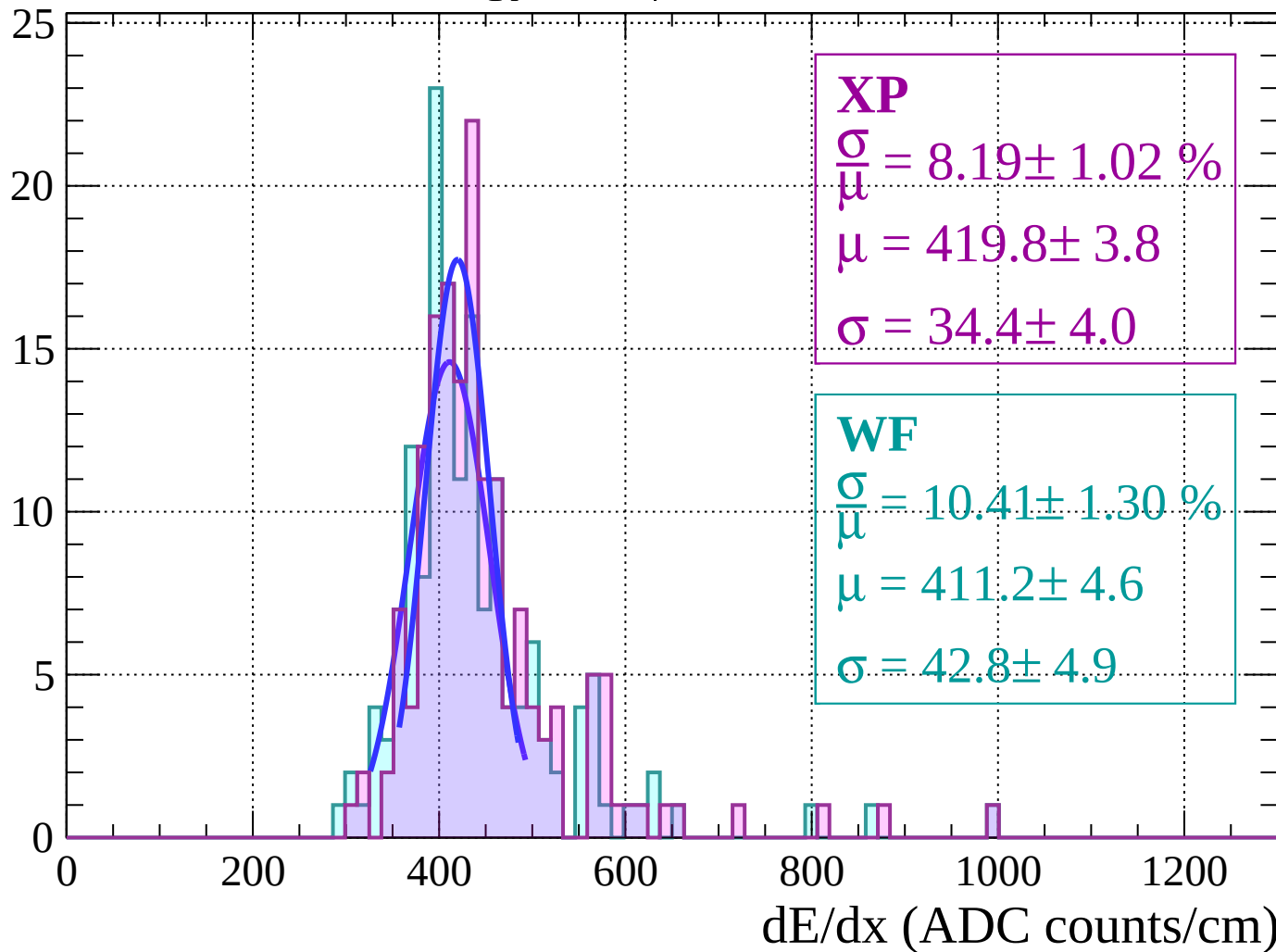
Energy loss | $-38 < \theta < -36$

Count



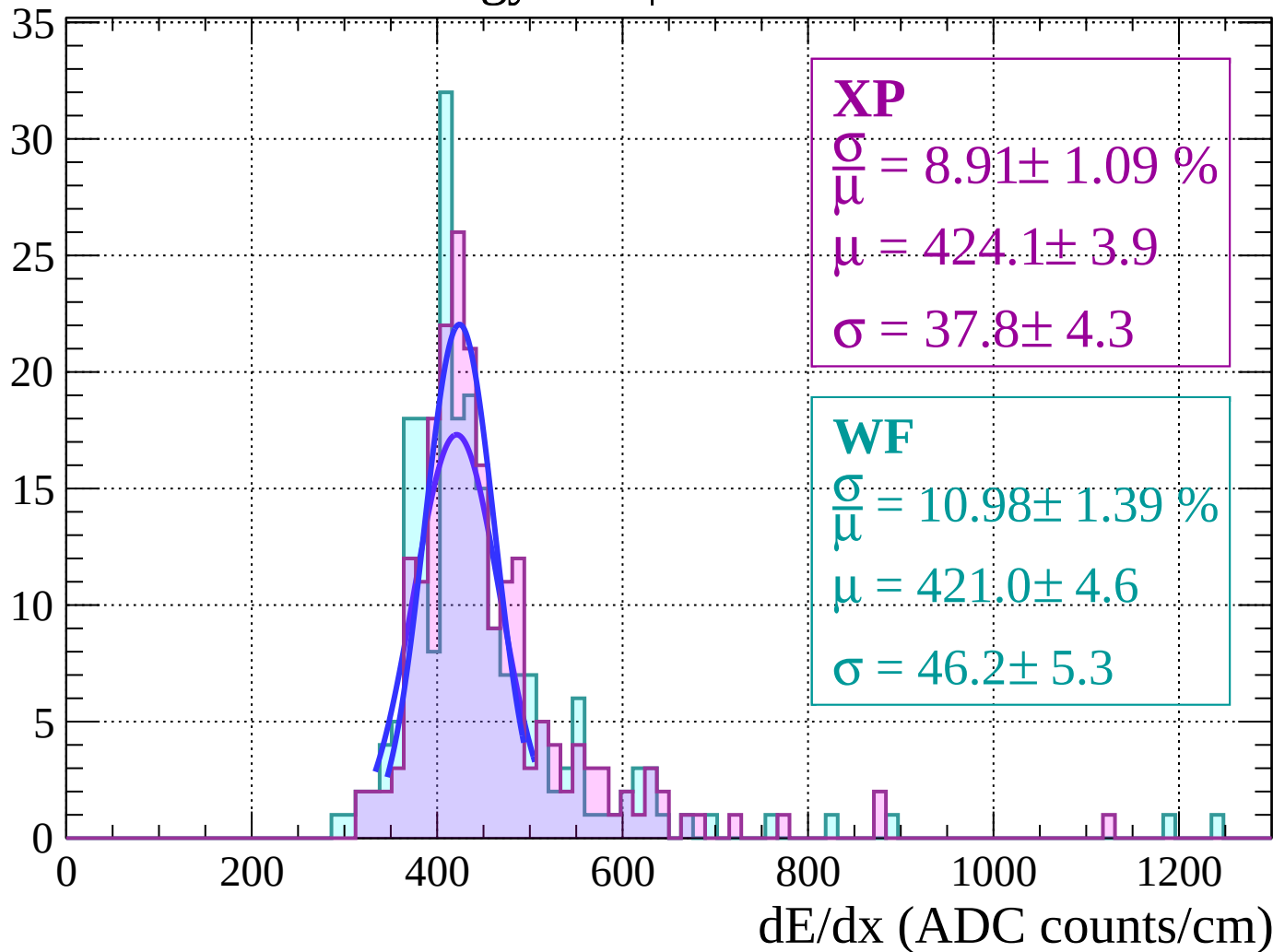
Energy loss | $-36 < \theta < -34$

Count



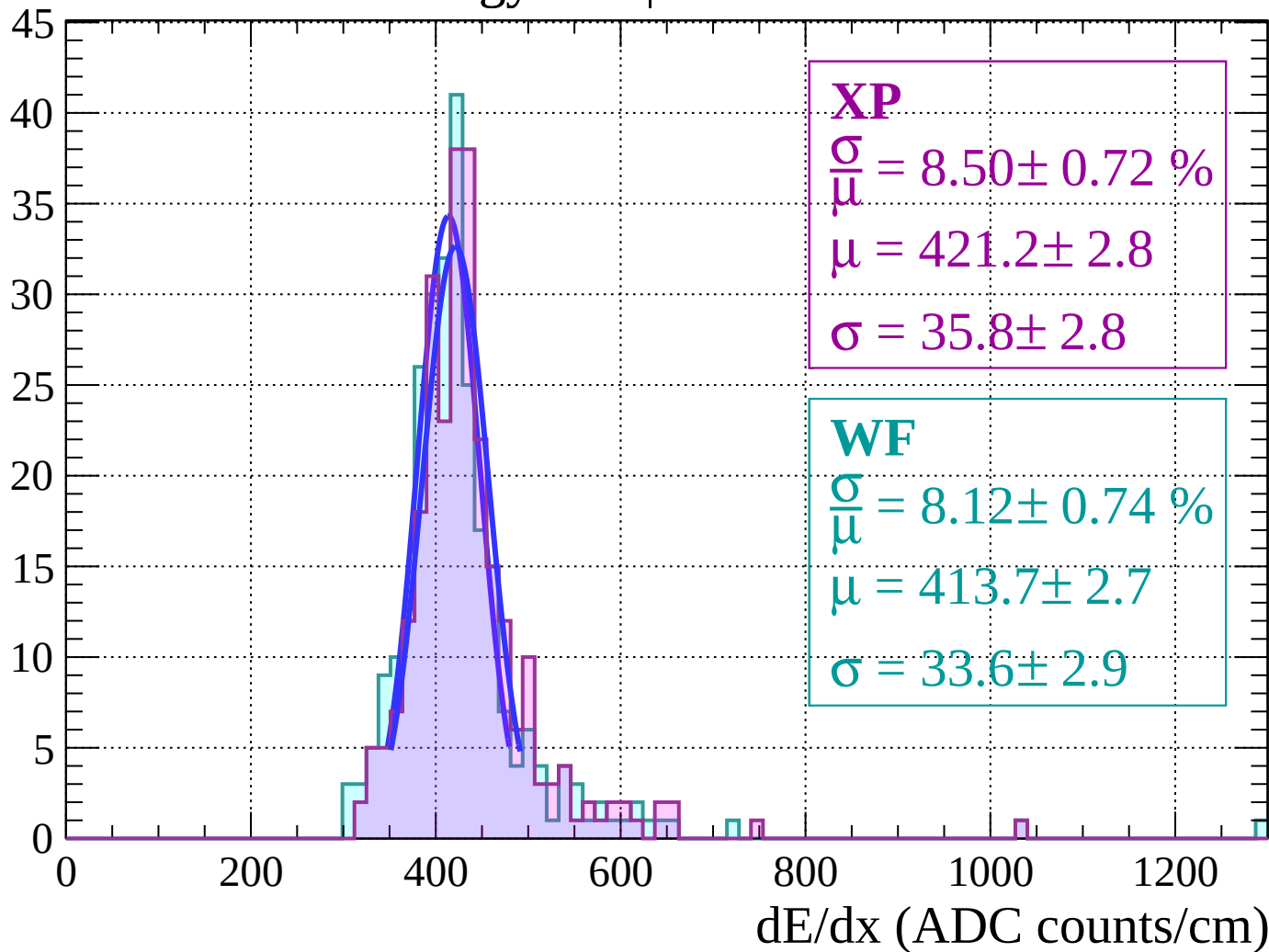
Energy loss | $-34 < \theta < -32$

Count



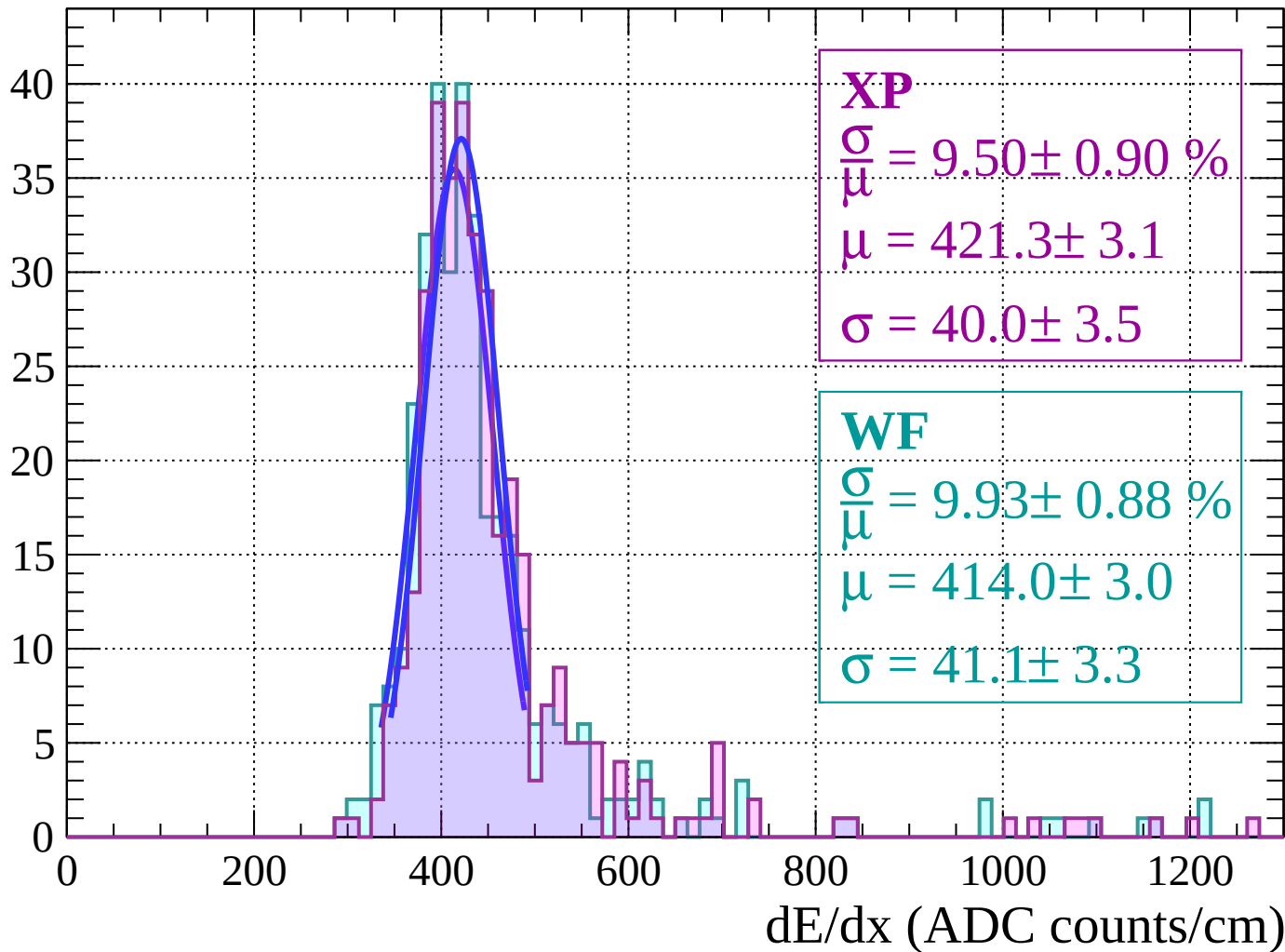
Energy loss | $-32 < \theta < -30$

Count



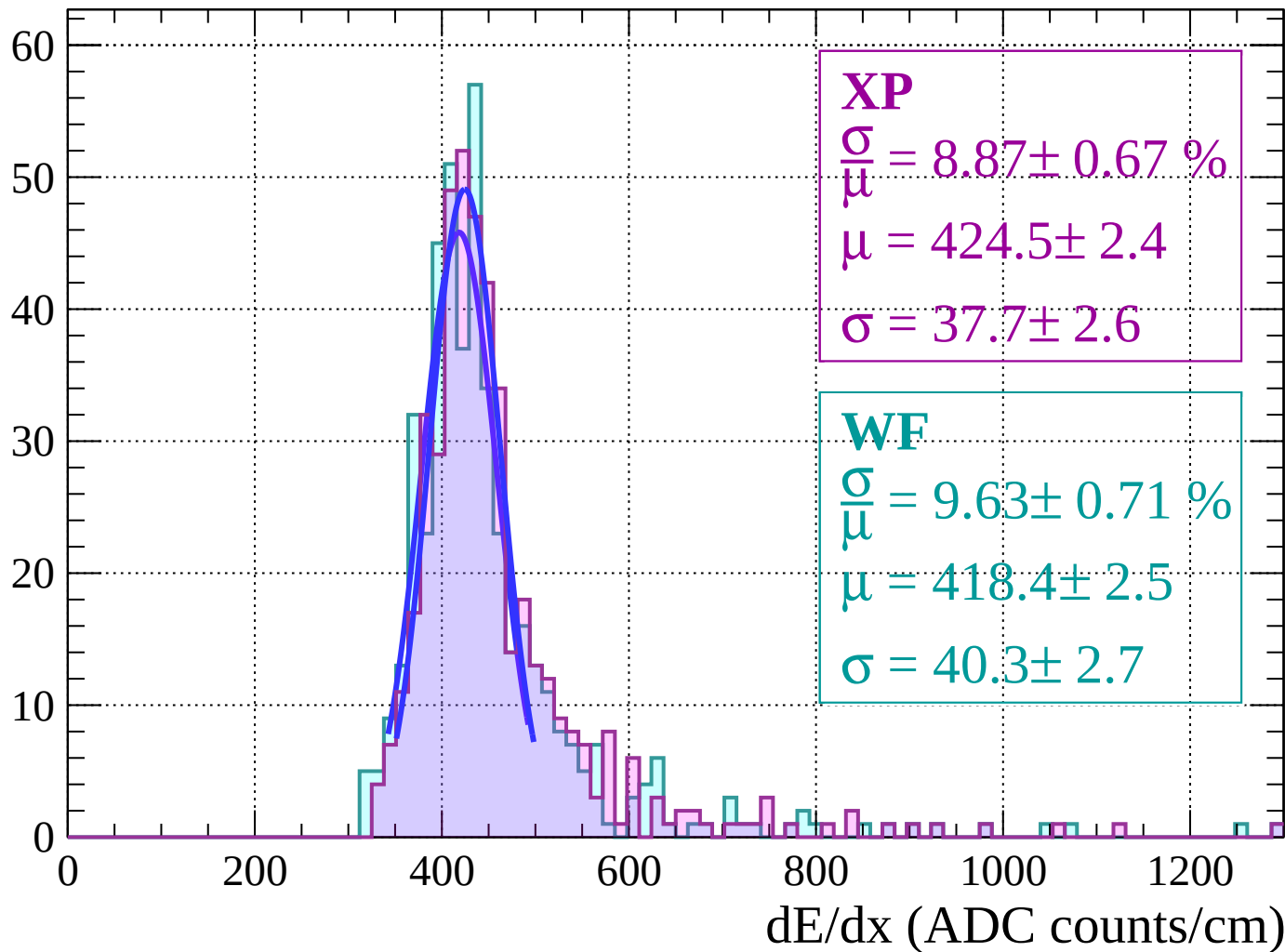
Energy loss | $-30 < \theta < -28$

Count



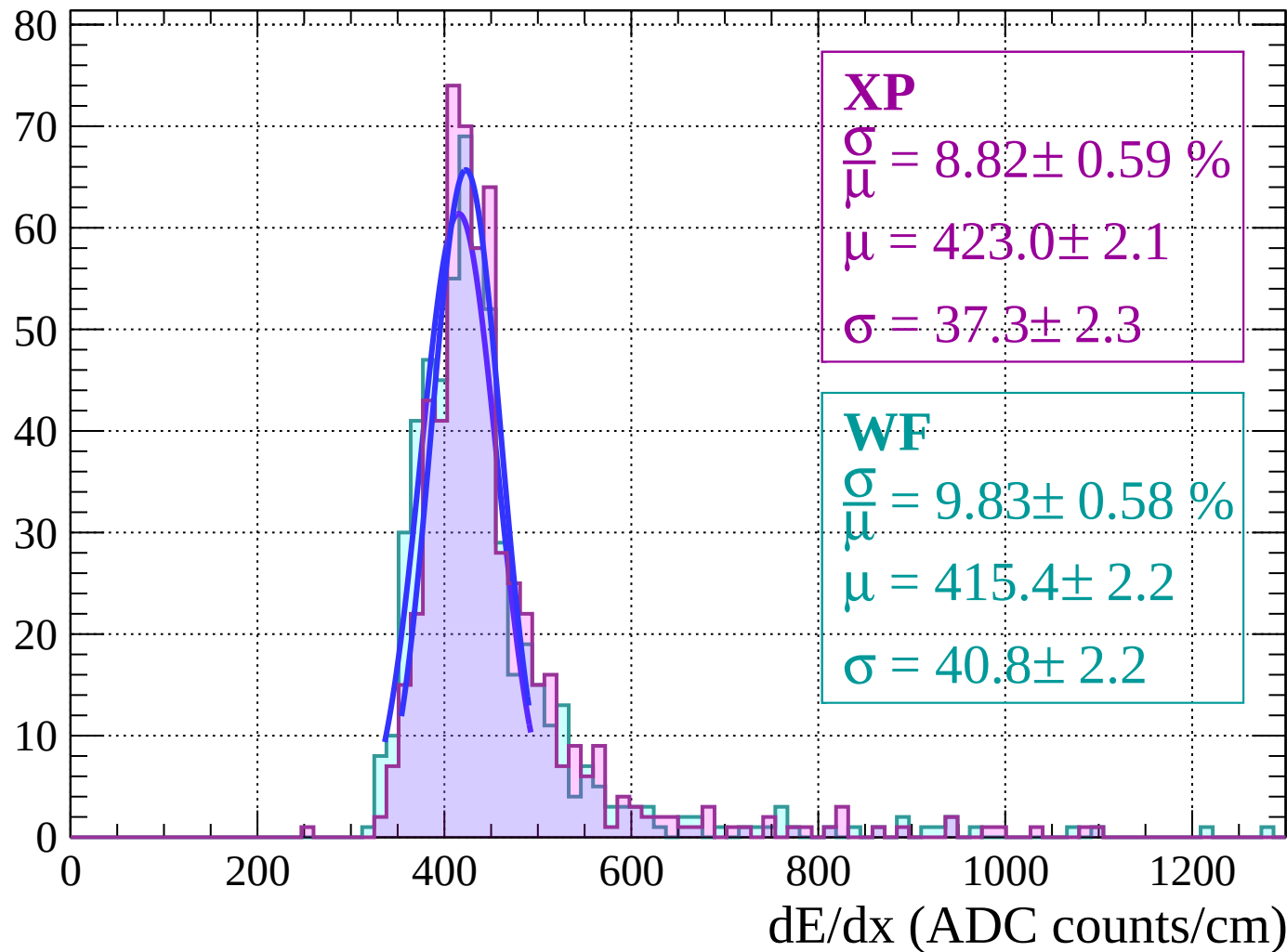
Energy loss | $-28 < \theta < -26$

Count



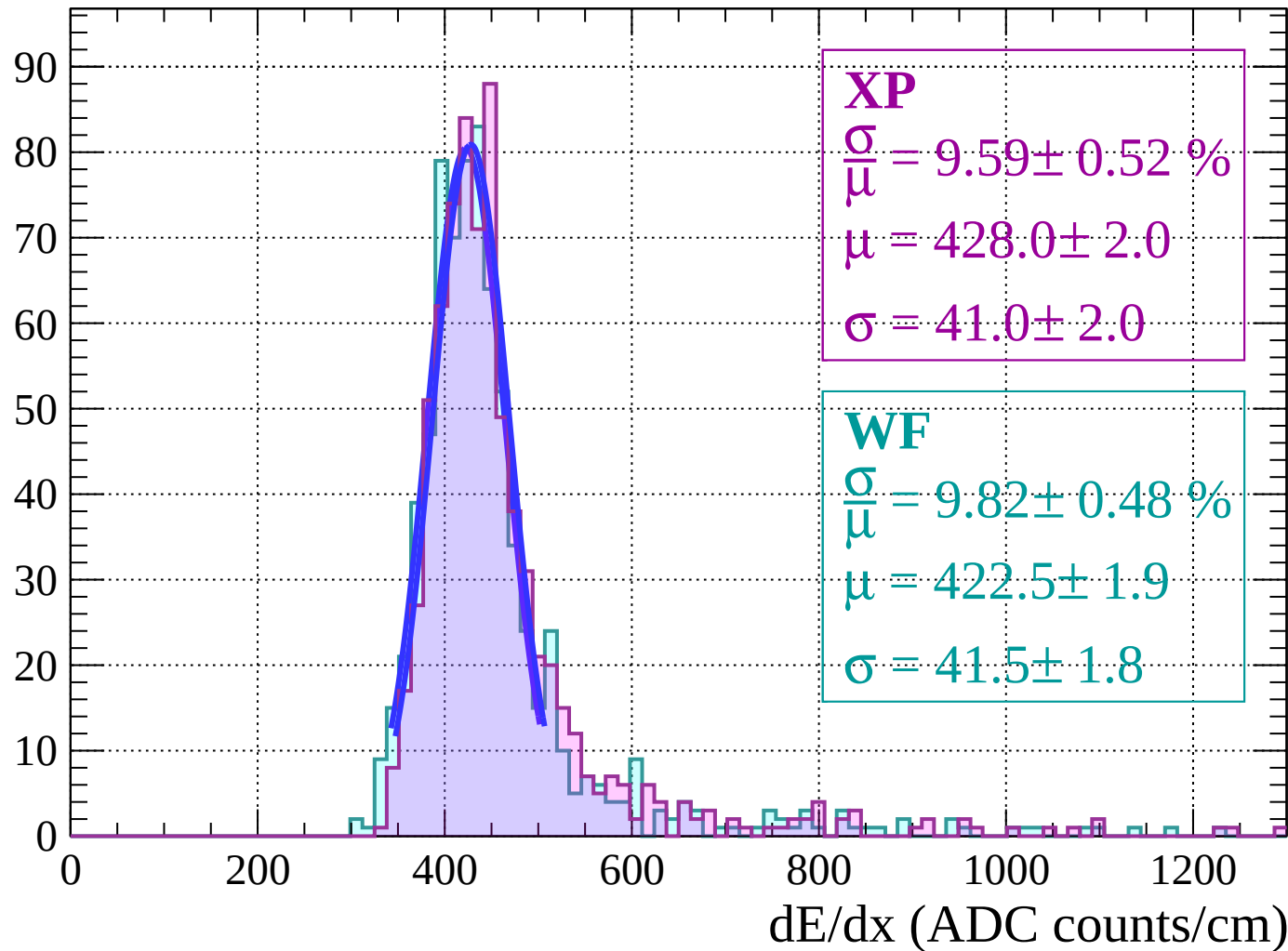
Energy loss | $-26 < \theta < -24$

Count



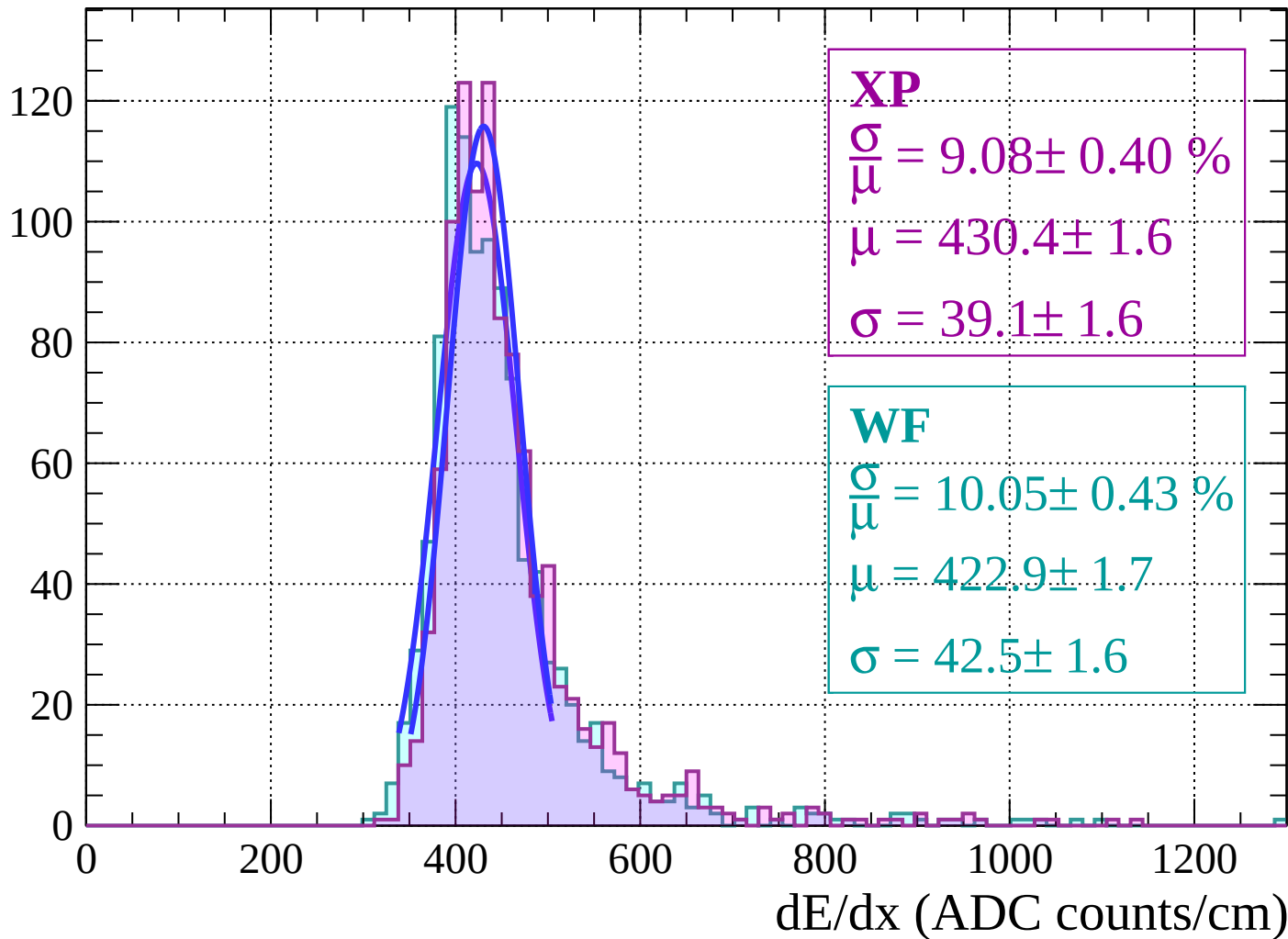
Energy loss | $-24 < \theta < -22$

Count



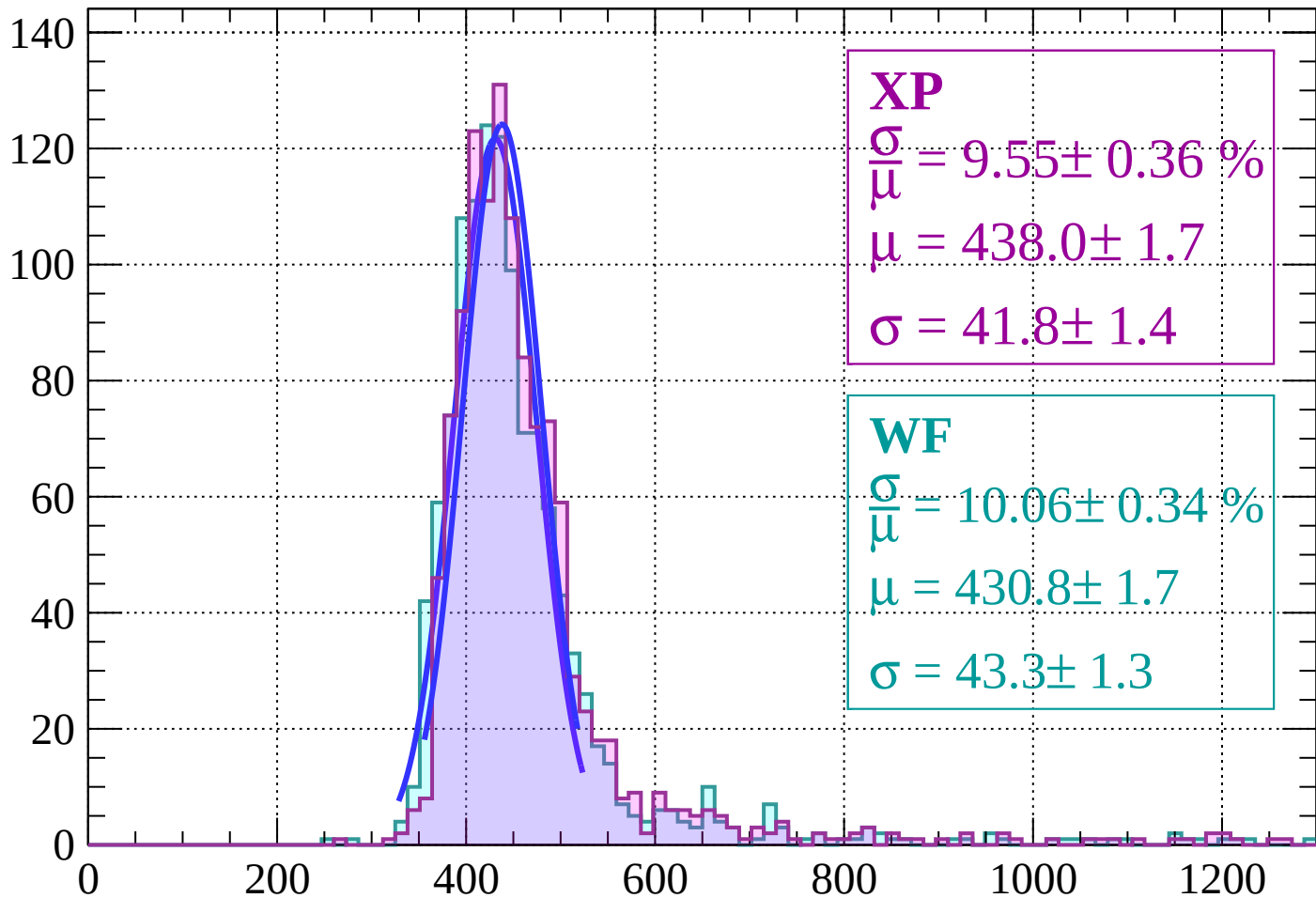
Energy loss | $-22 < \theta < -20$

Count



Energy loss | $-20 < \theta < -18$

Count



XP

$$\frac{\sigma}{\mu} = 9.55 \pm 0.36 \%$$

$$\mu = 438.0 \pm 1.7$$

$$\sigma = 41.8 \pm 1.4$$

WF

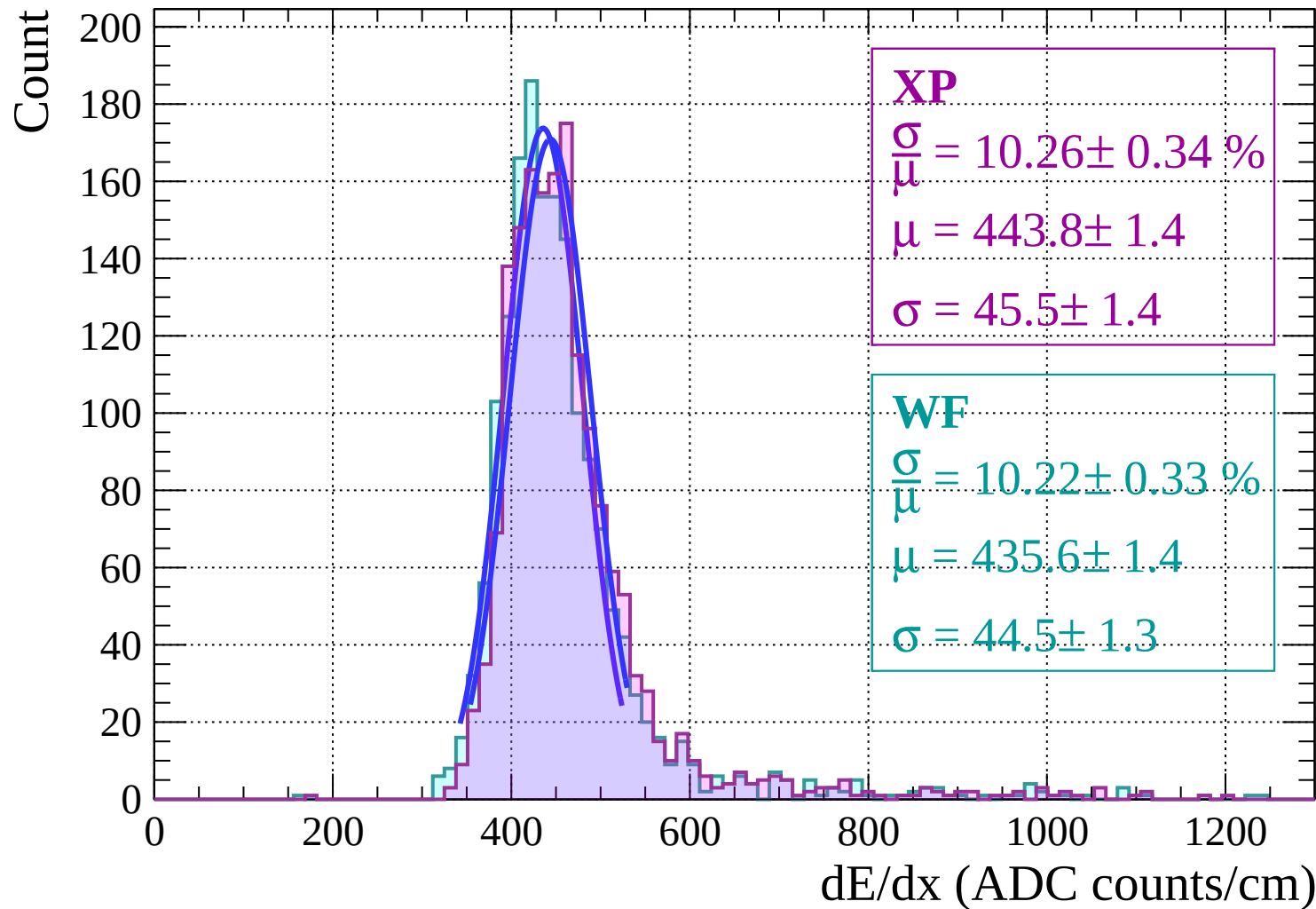
$$\frac{\sigma}{\mu} = 10.06 \pm 0.34 \%$$

$$\mu = 430.8 \pm 1.7$$

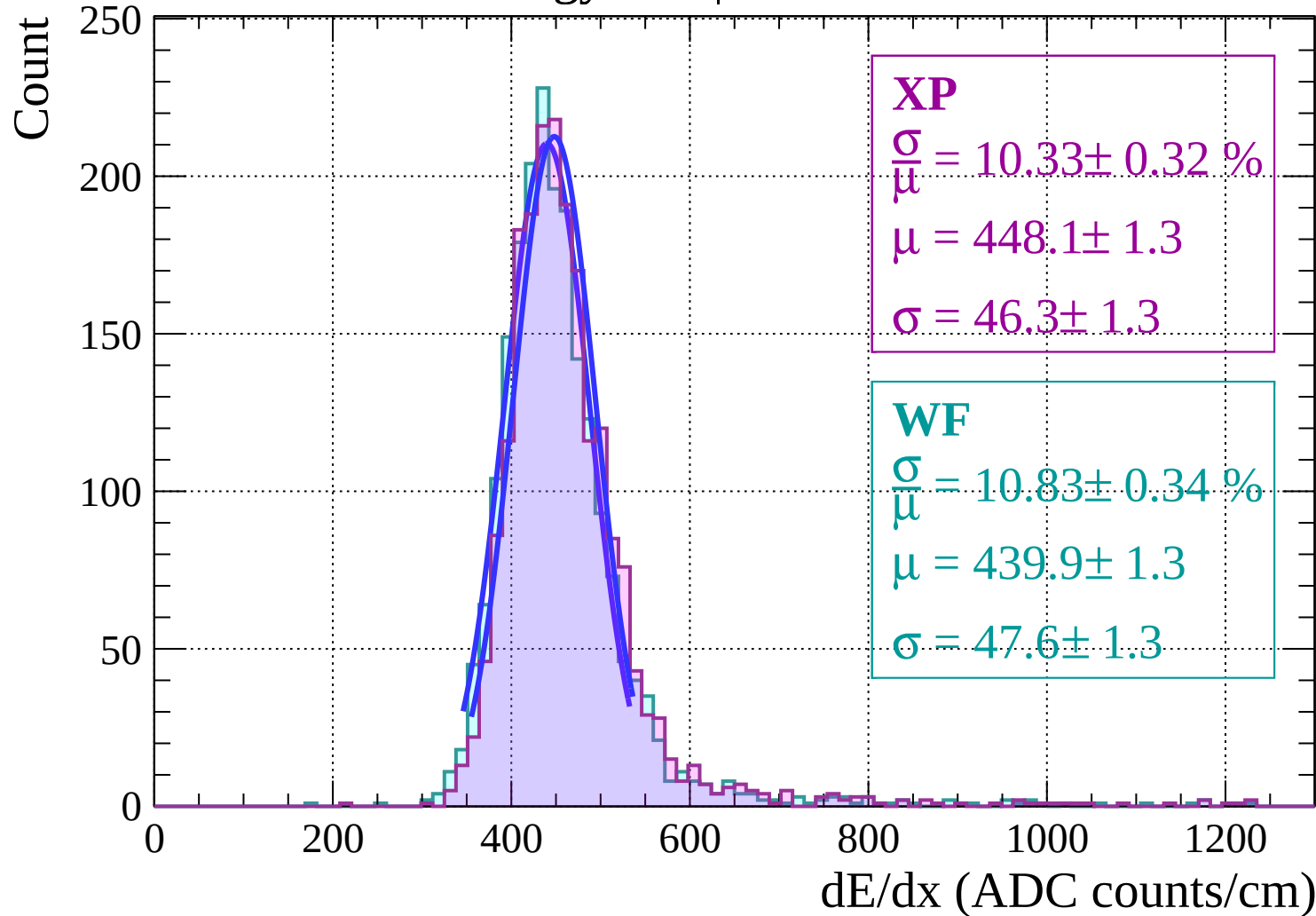
$$\sigma = 43.3 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-18 < \theta < -16$

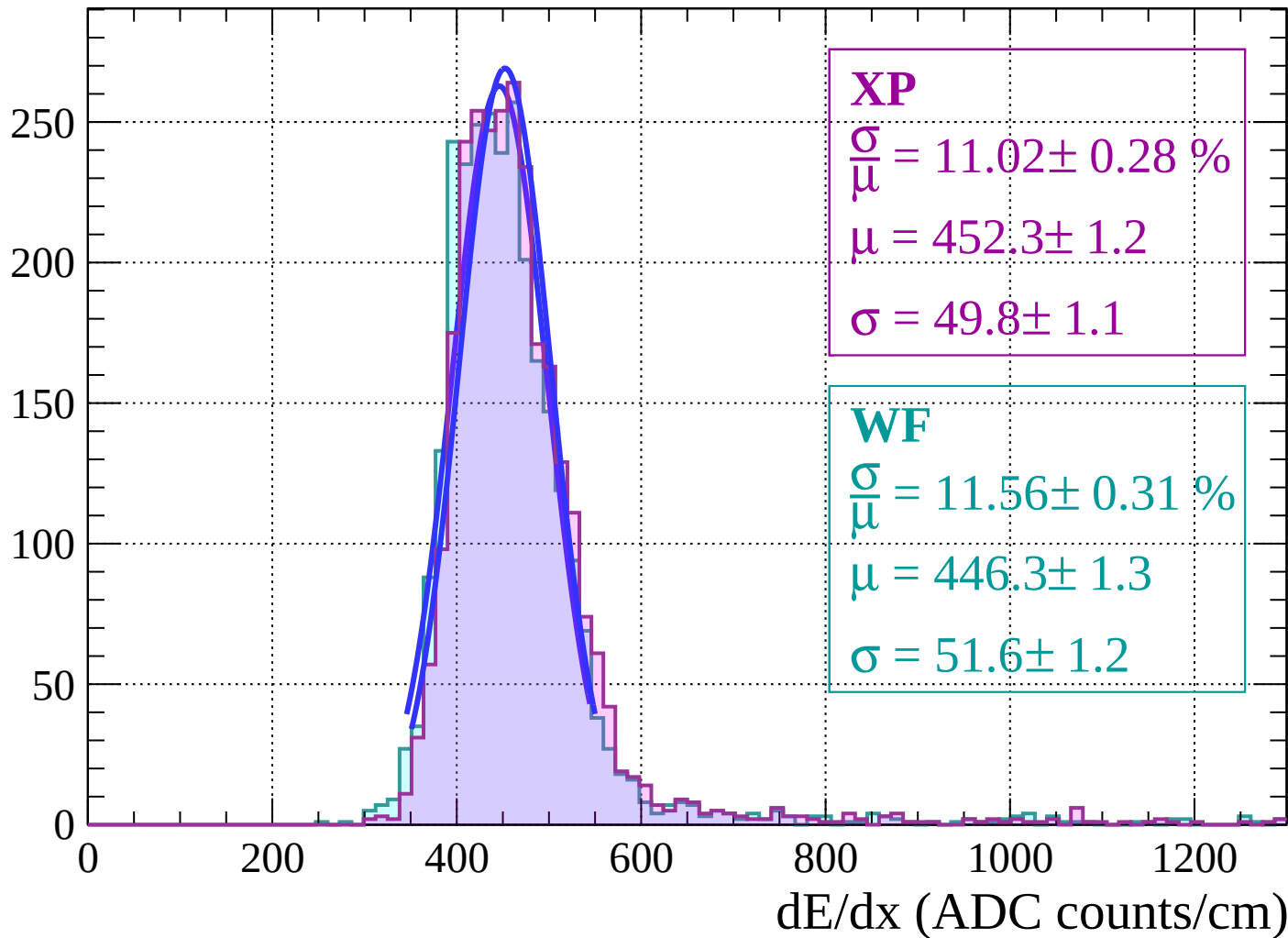


Energy loss | $-16 < \theta < -14$

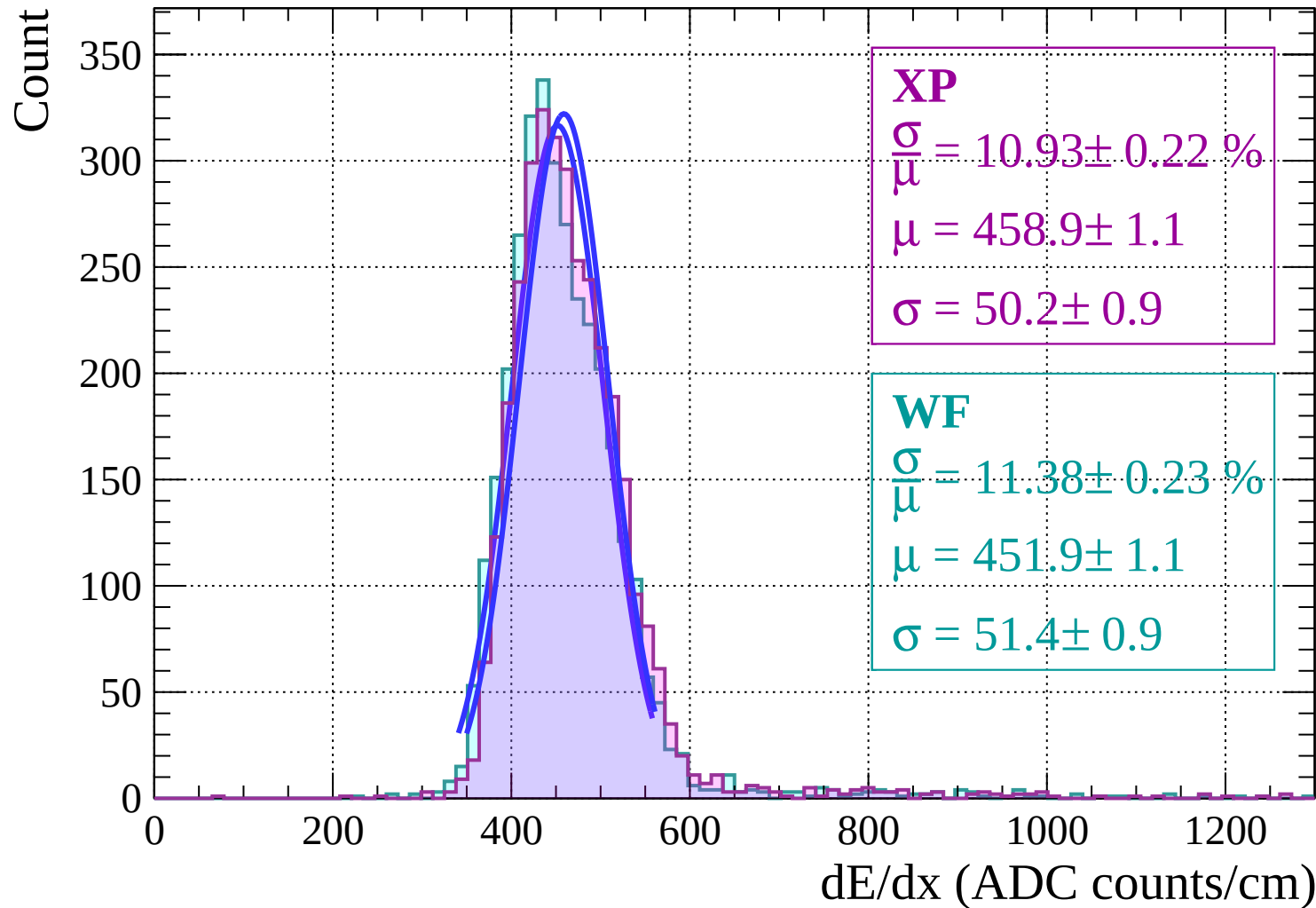


Energy loss $|-14 < \theta < -12$

Count

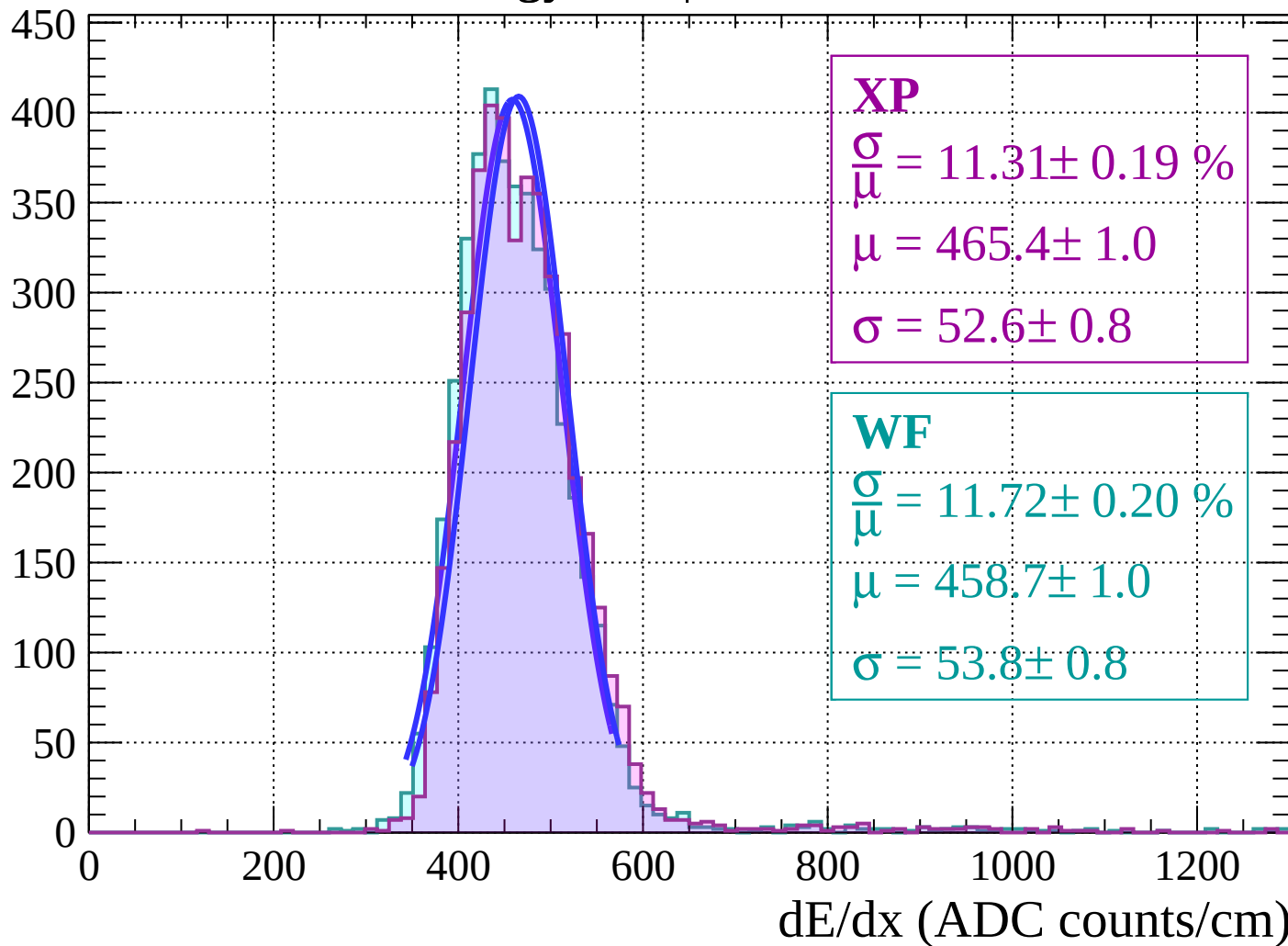


Energy loss | $-12 < \theta < -10$

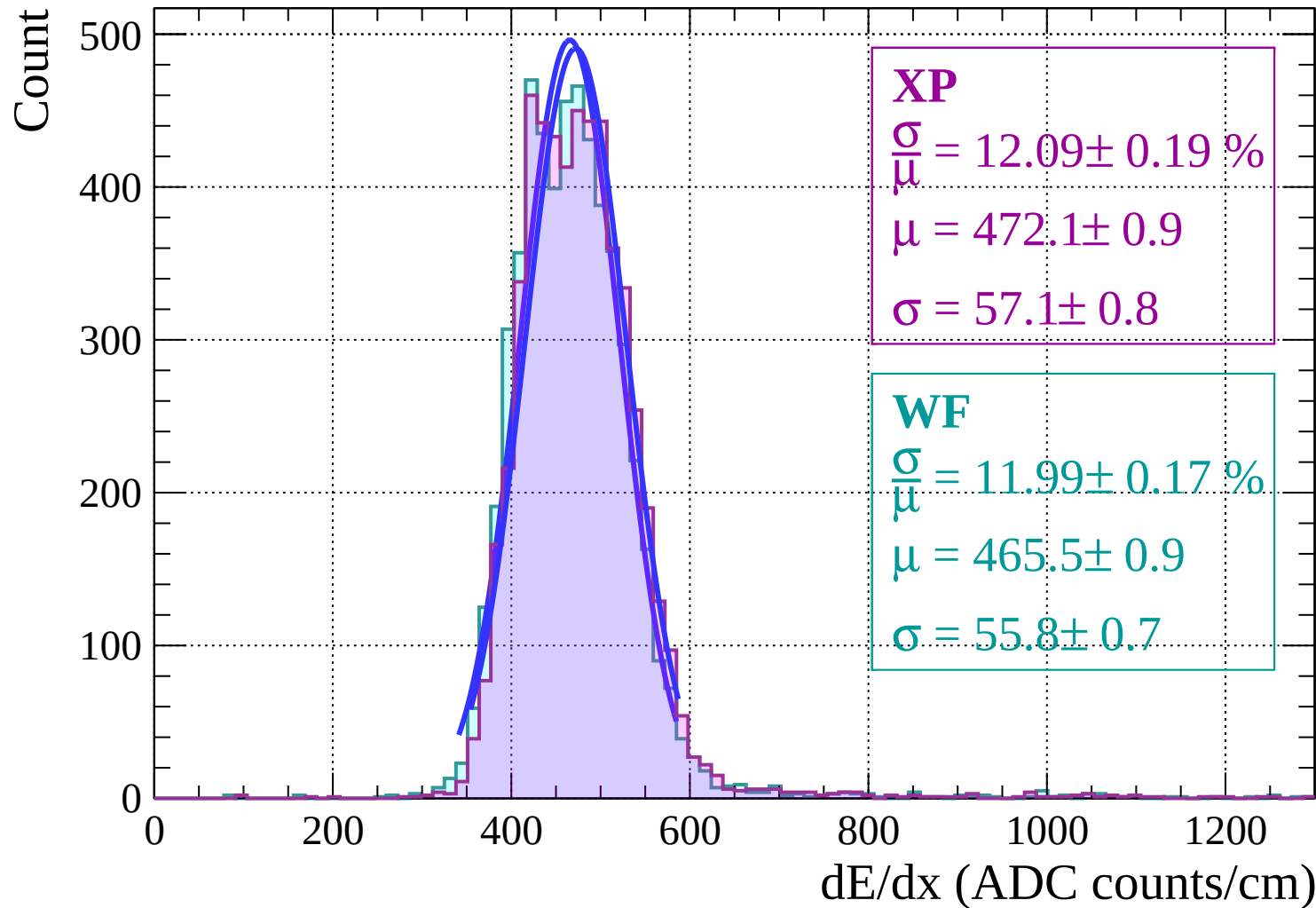


Energy loss | $-10 < \theta < -8$

Count

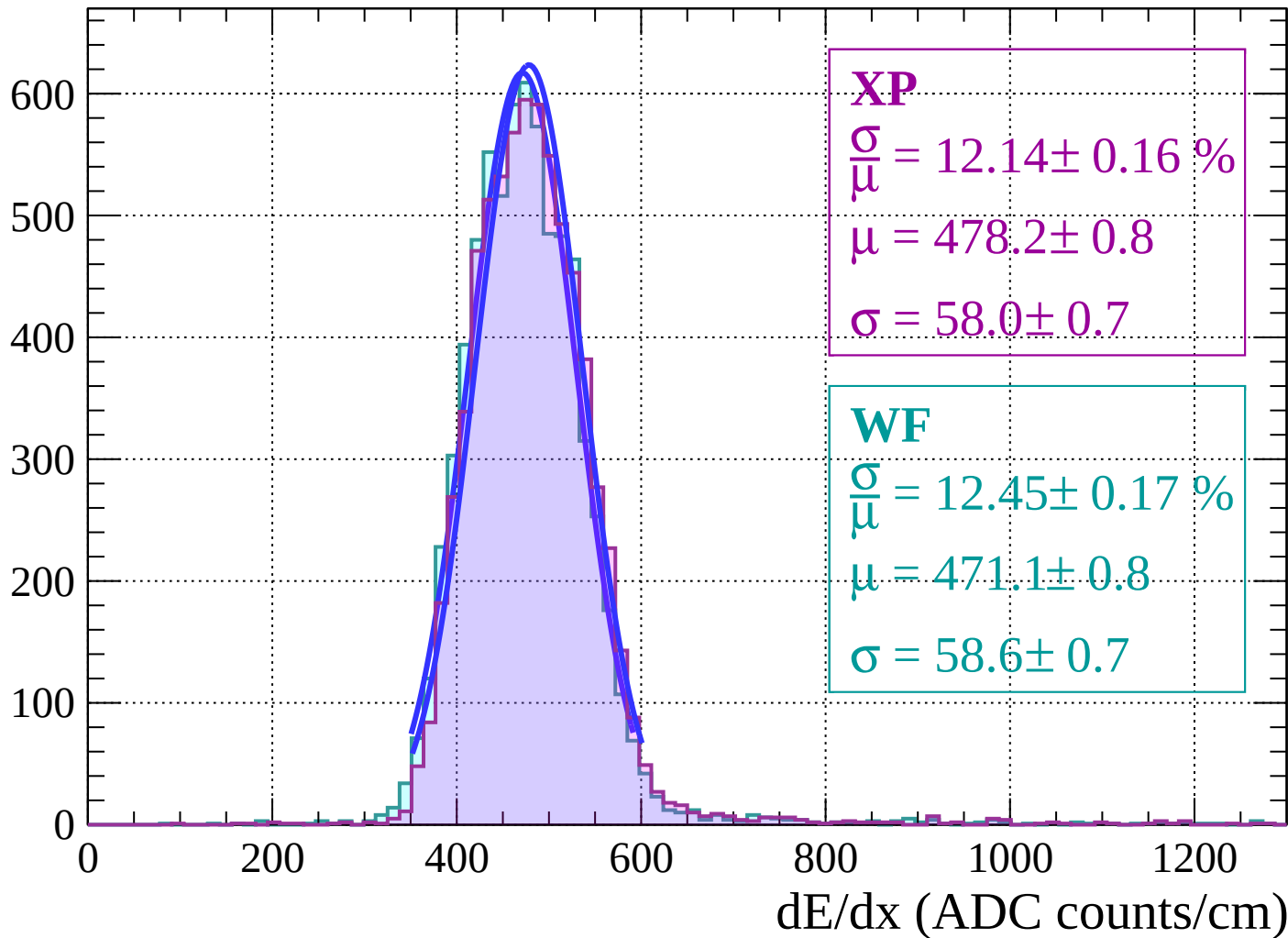


Energy loss | $-8 < \theta < -6$



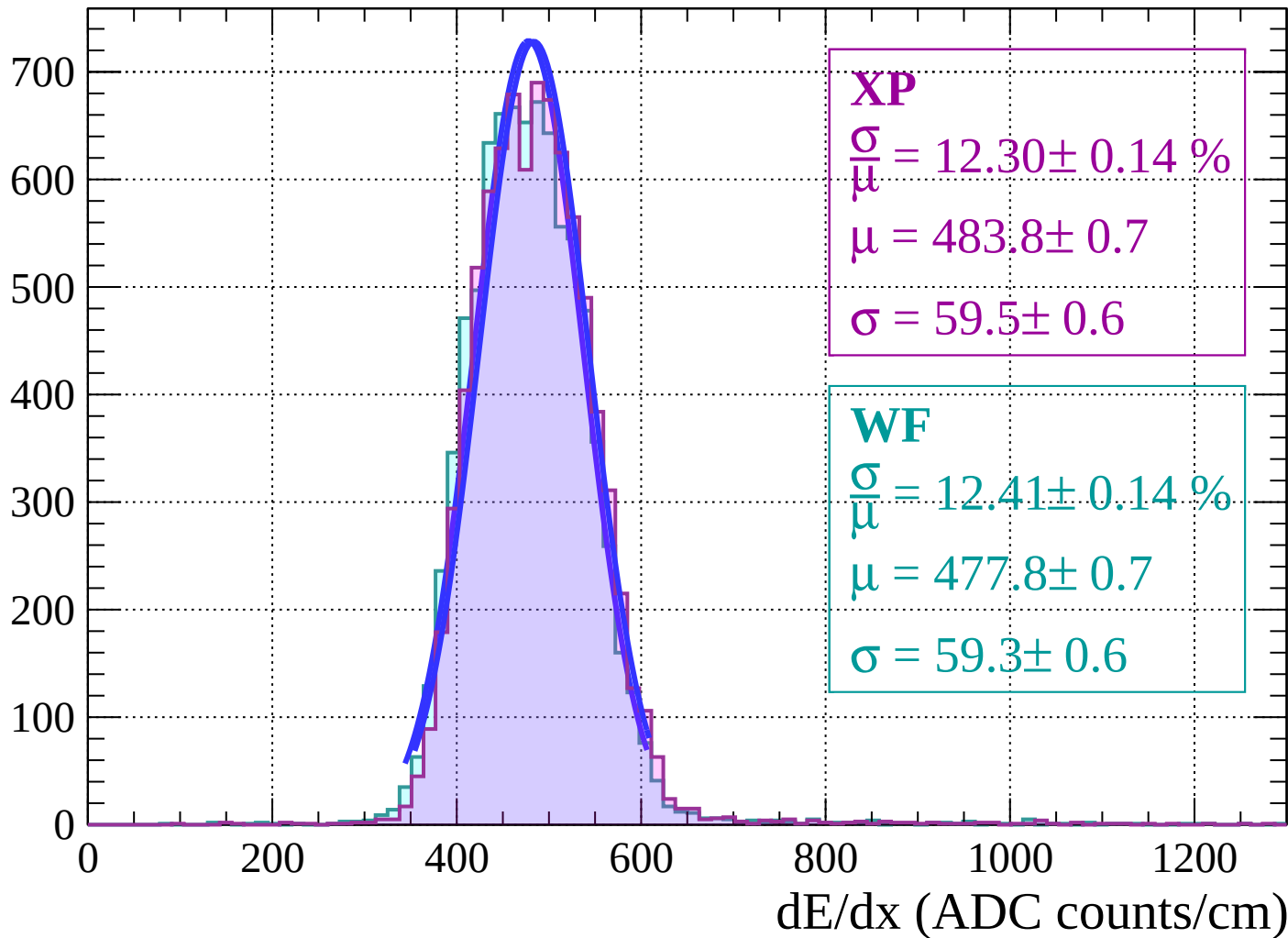
Energy loss | $-6 < \theta < -4$

Count

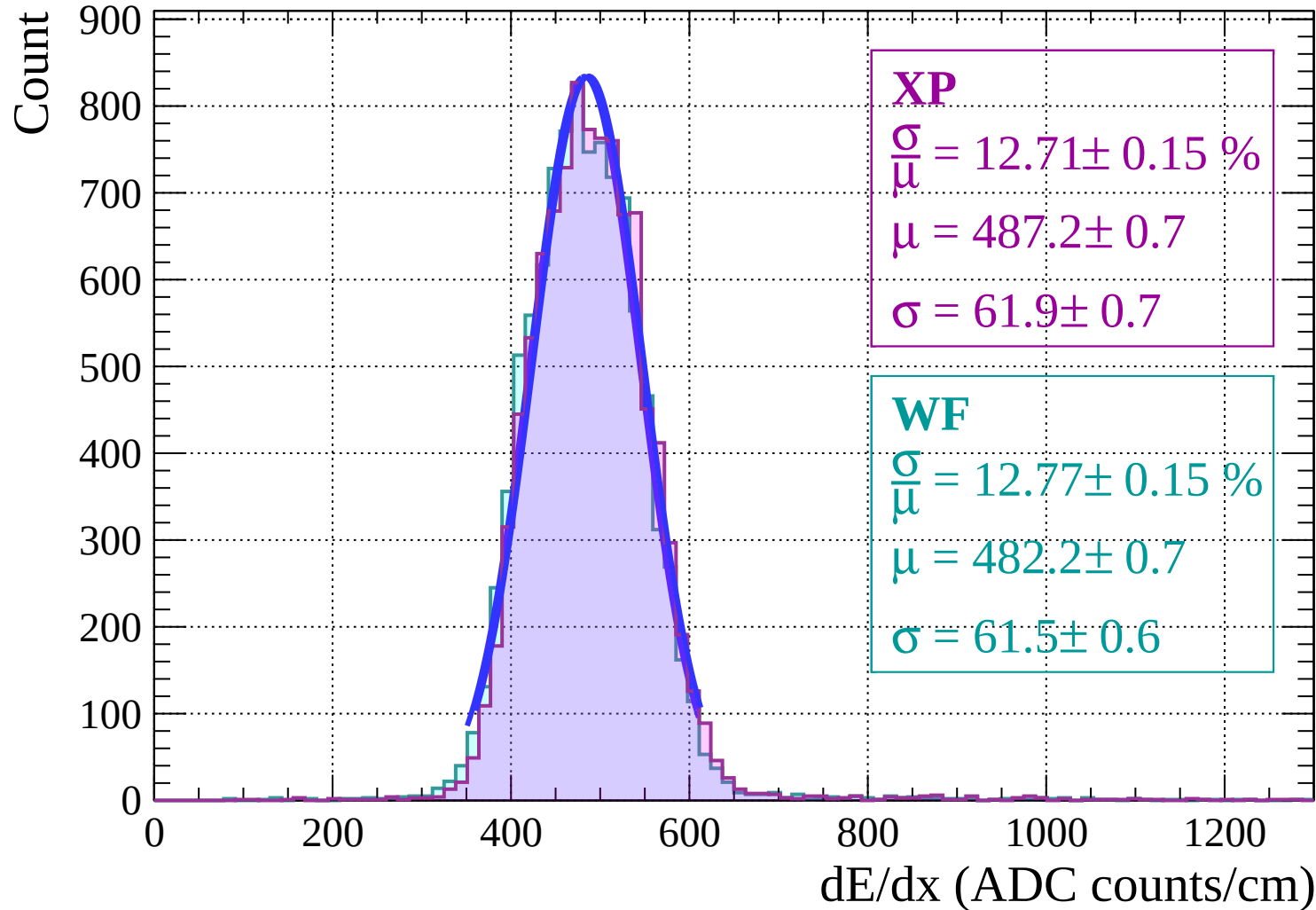


Energy loss $|-4 < \theta < -2$

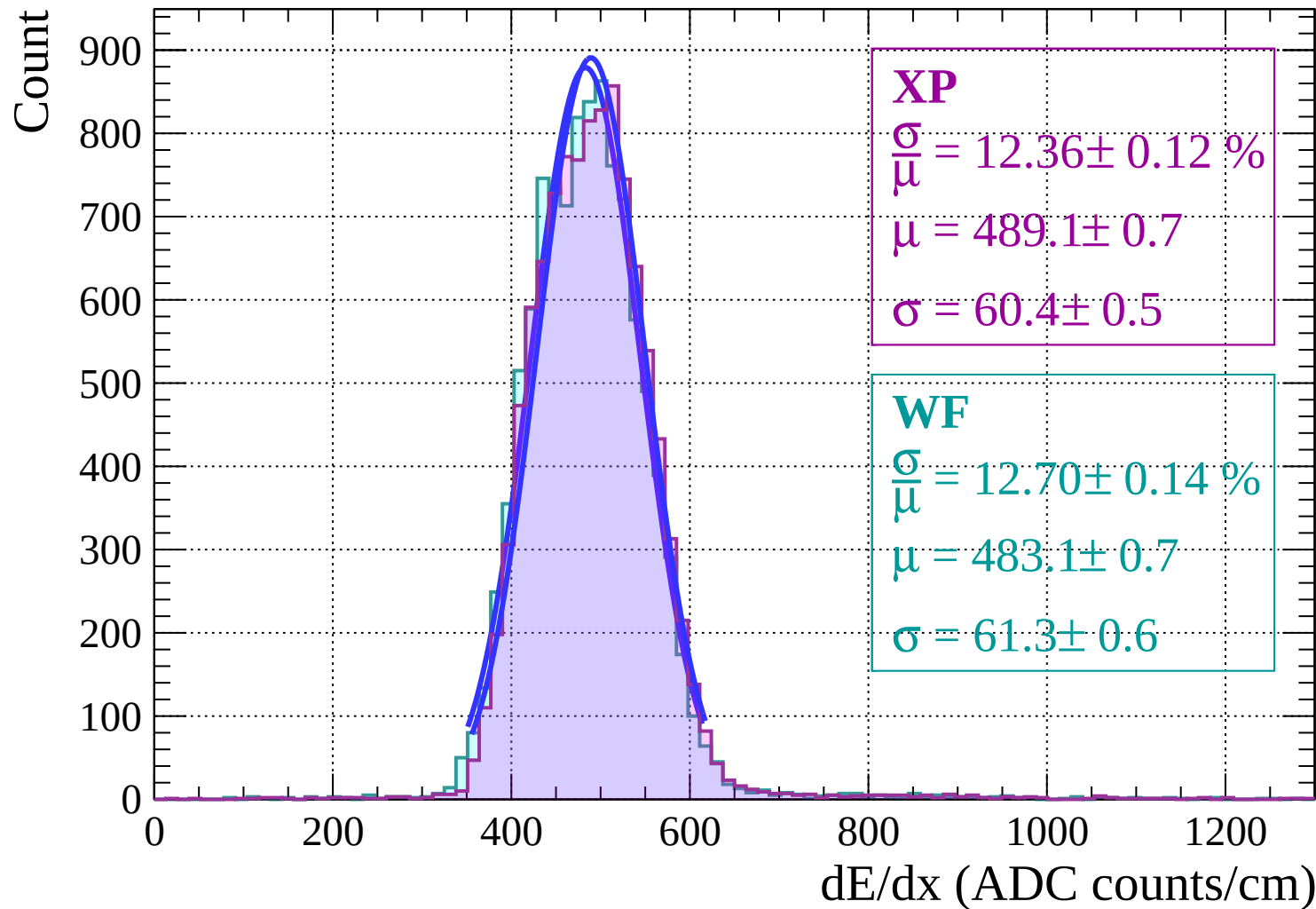
Count



Energy loss | $-2 < \theta < 0$

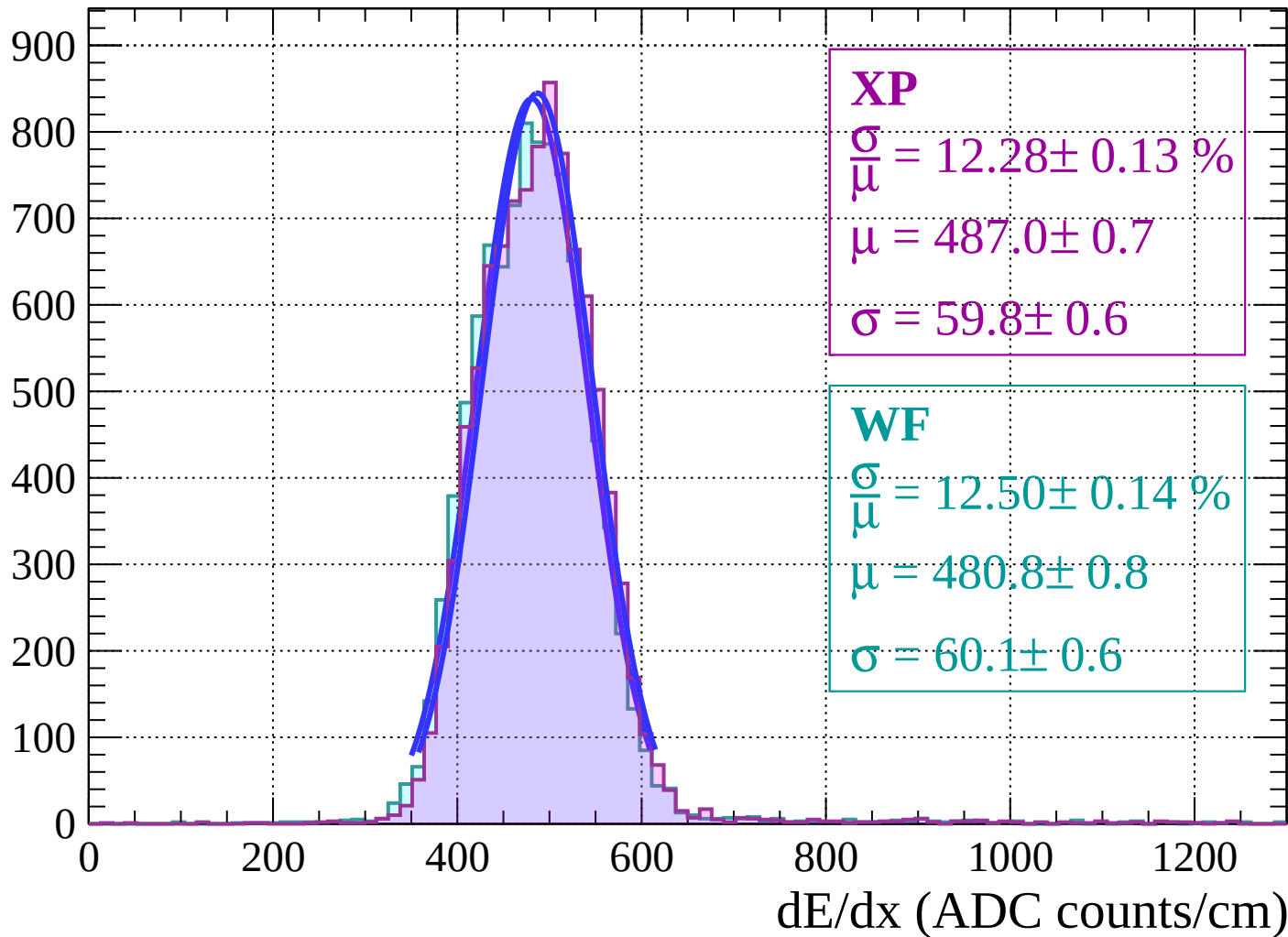


Energy loss | $0 < \theta < 2$



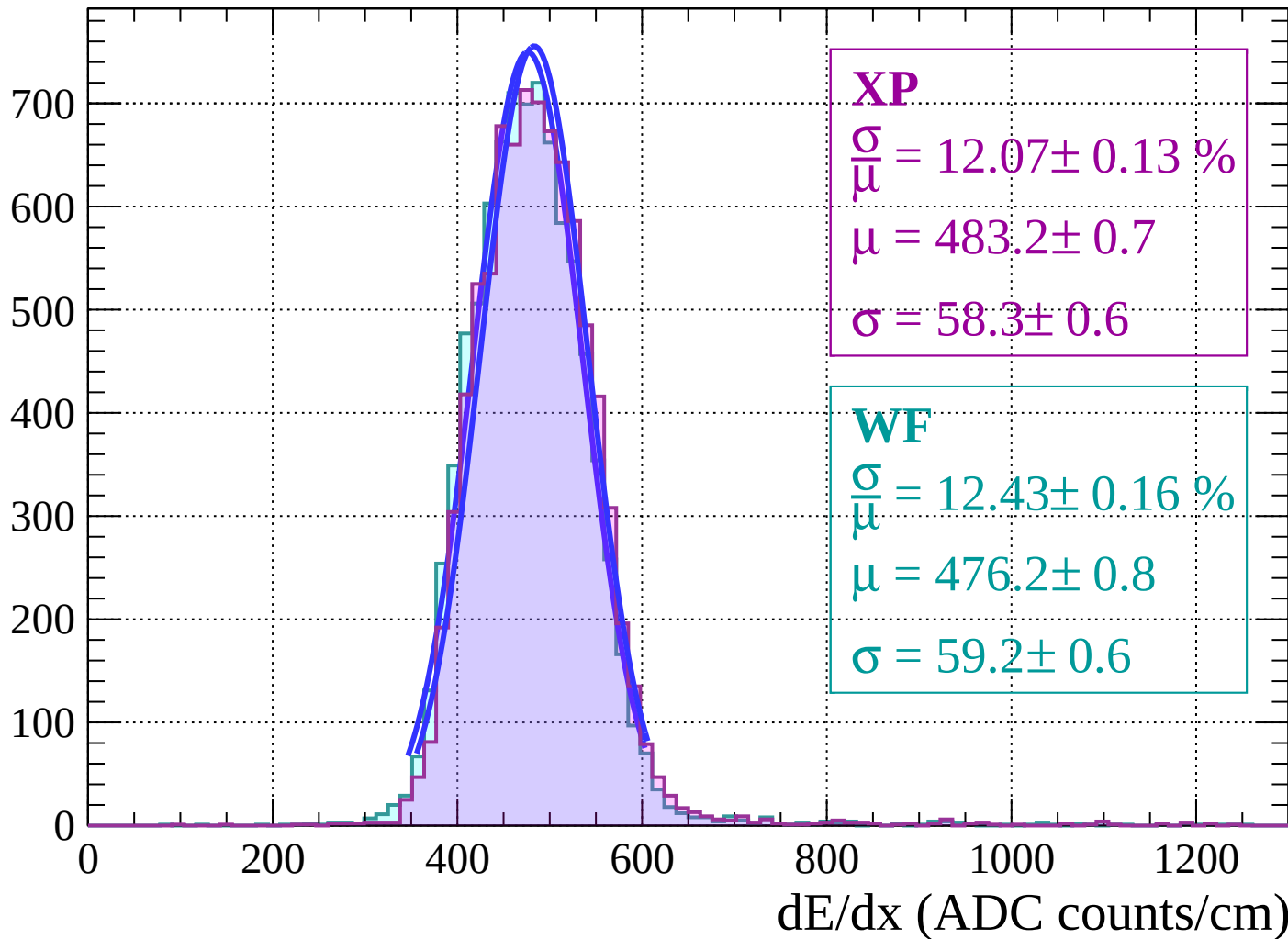
Energy loss | $2 < \theta < 4$

Count



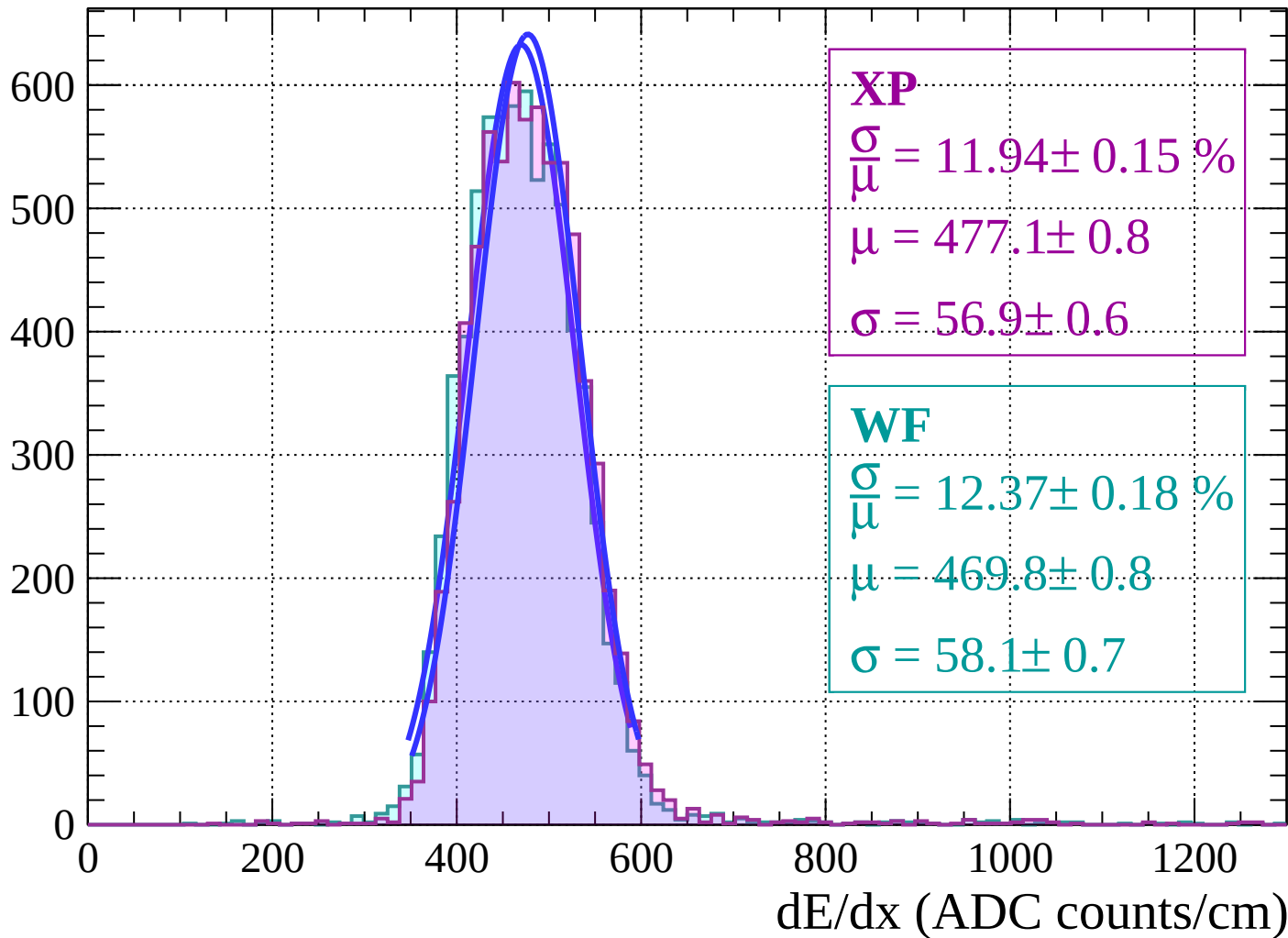
Energy loss | $4 < \theta < 6$

Count



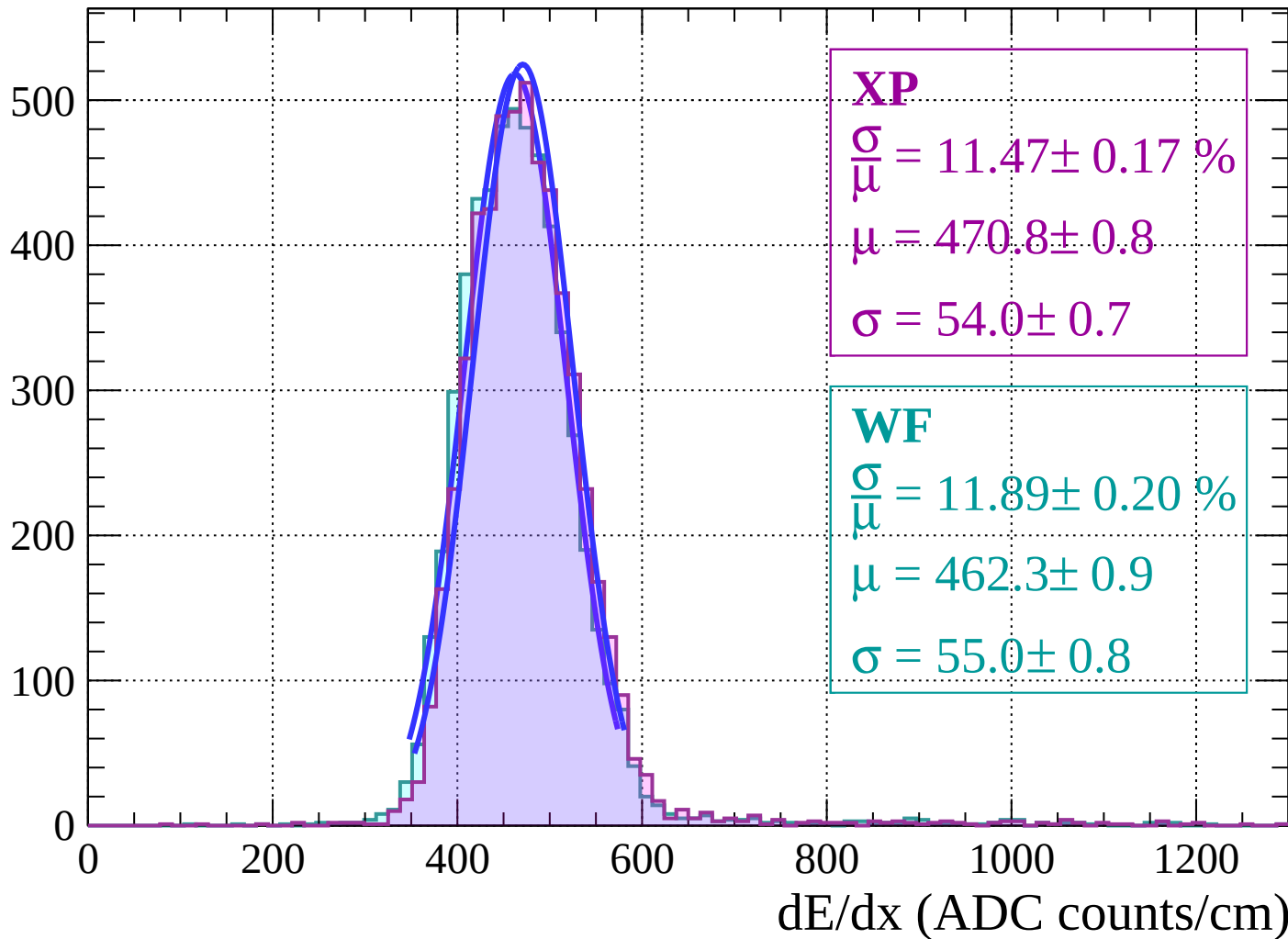
Energy loss | $6 < \theta < 8$

Count

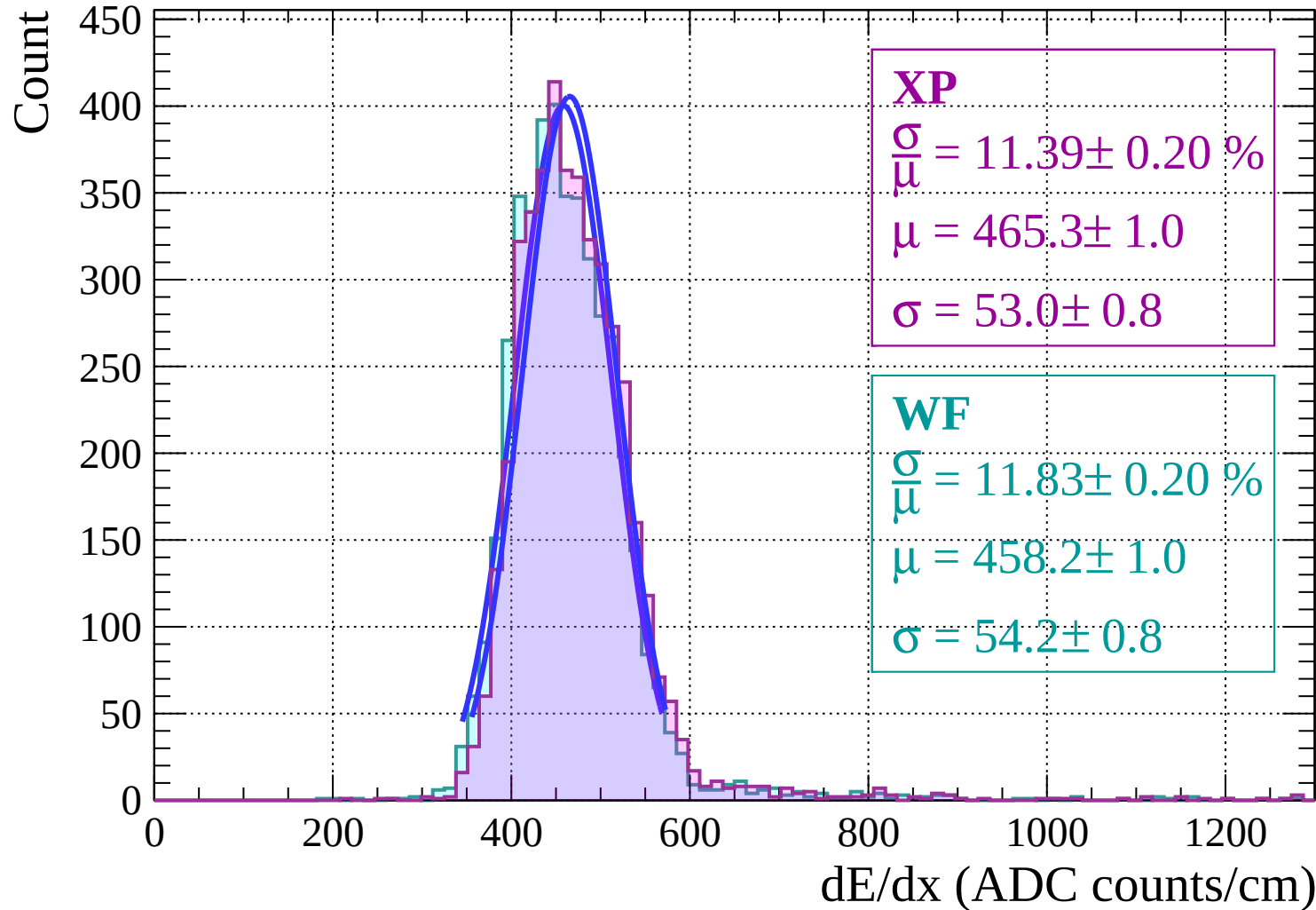


Energy loss | $8 < \theta < 10$

Count

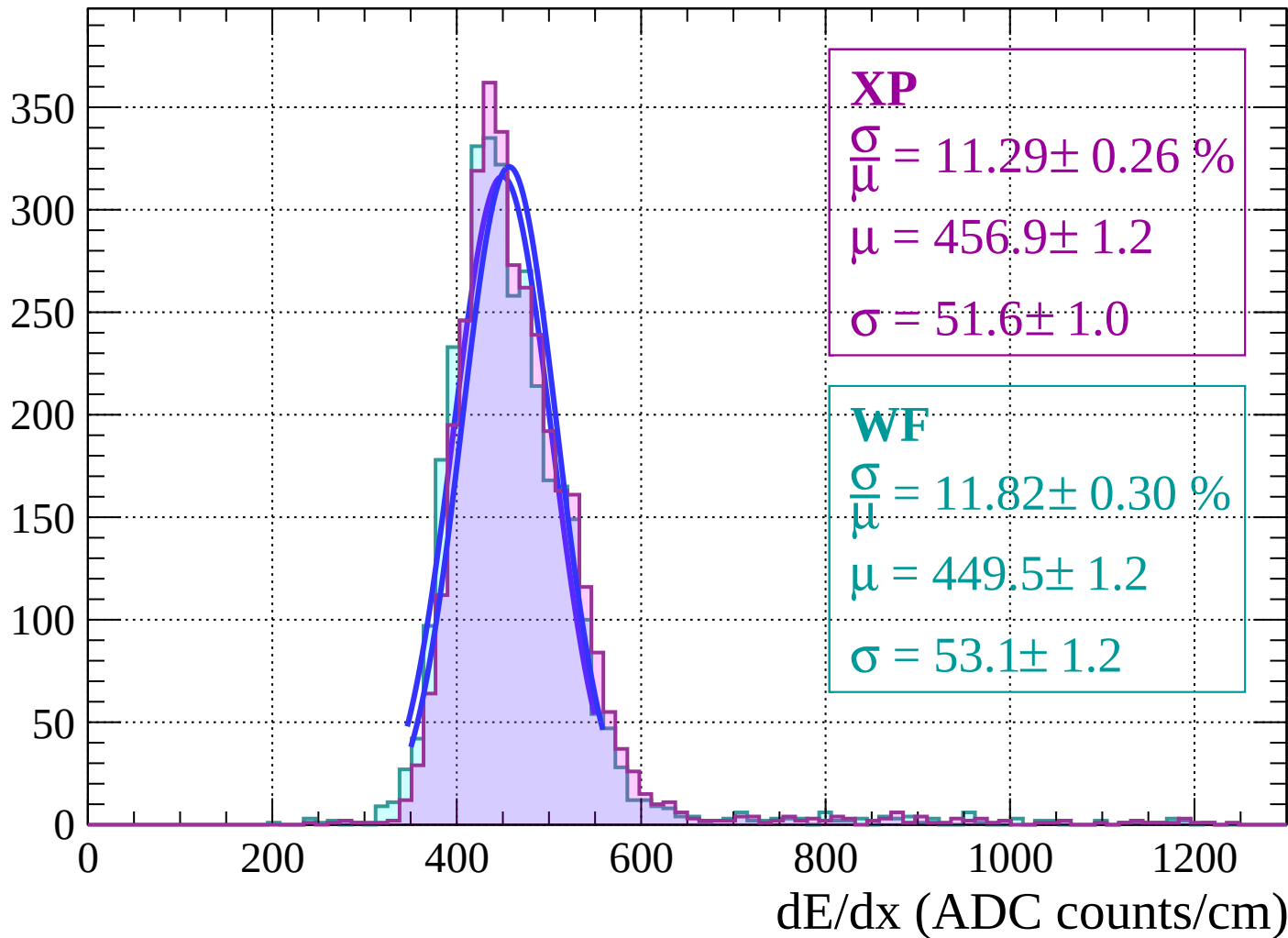


Energy loss | $10 < \theta < 12$



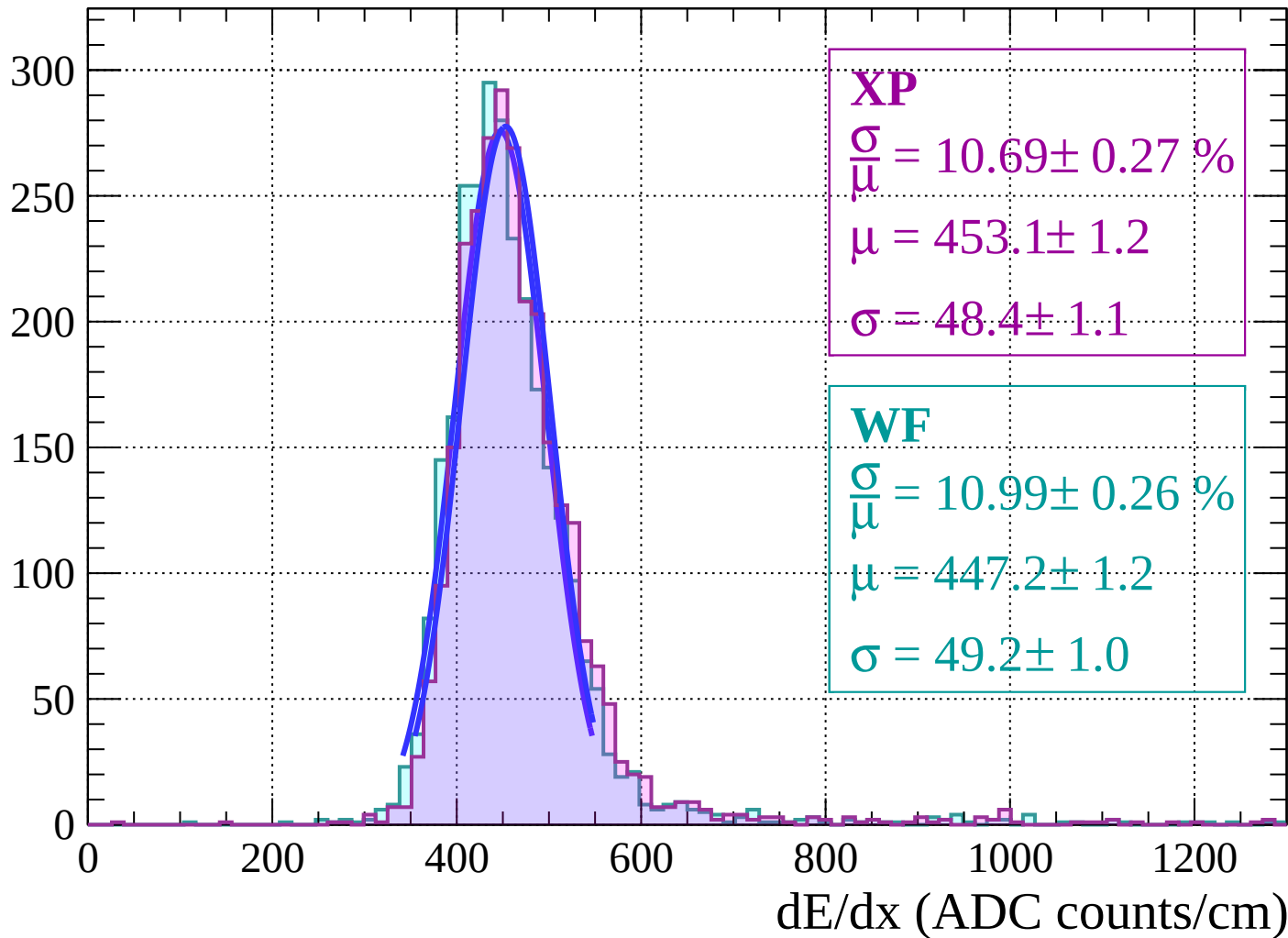
Energy loss | $12 < \theta < 14$

Count

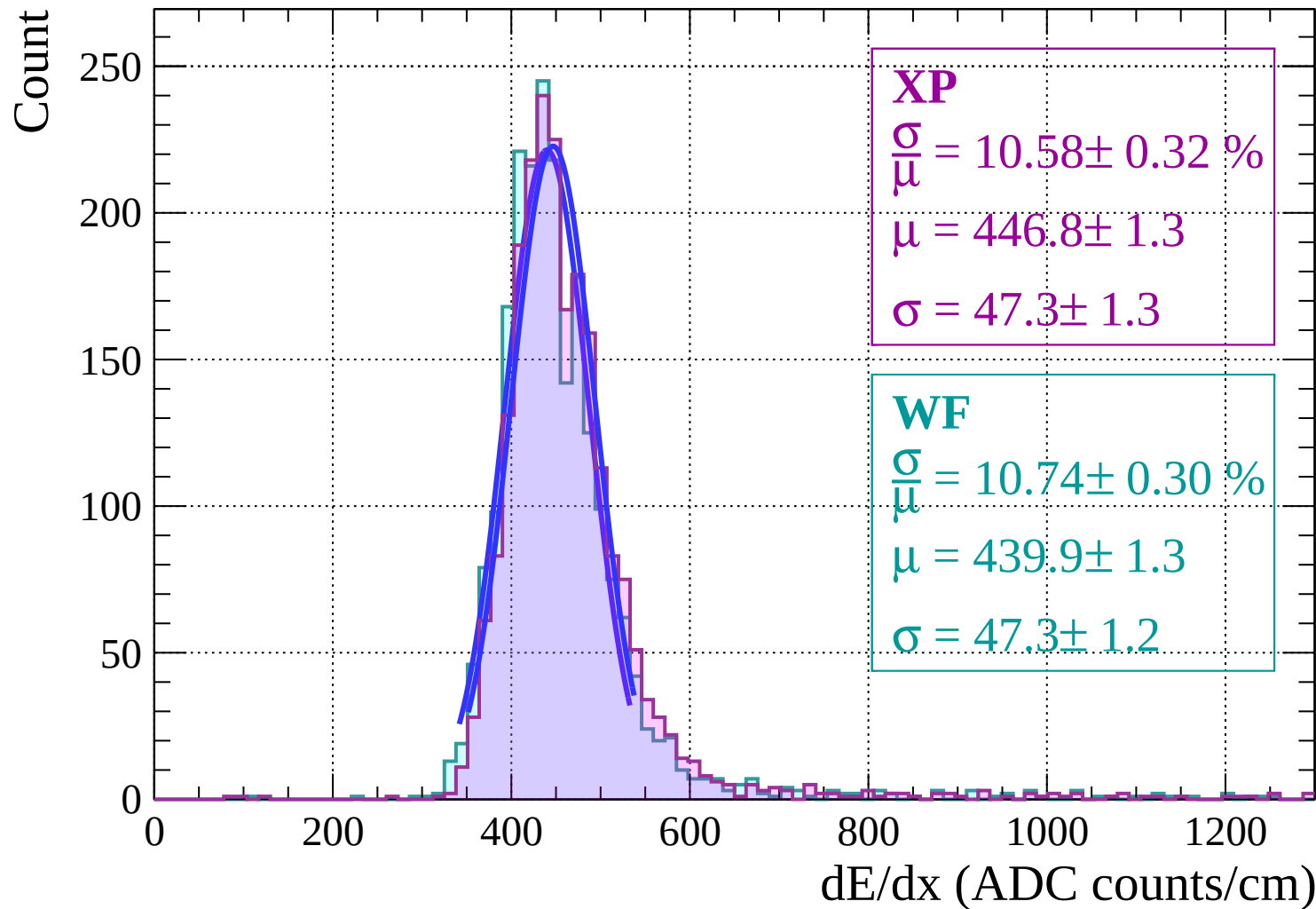


Energy loss | $14 < \theta < 16$

Count

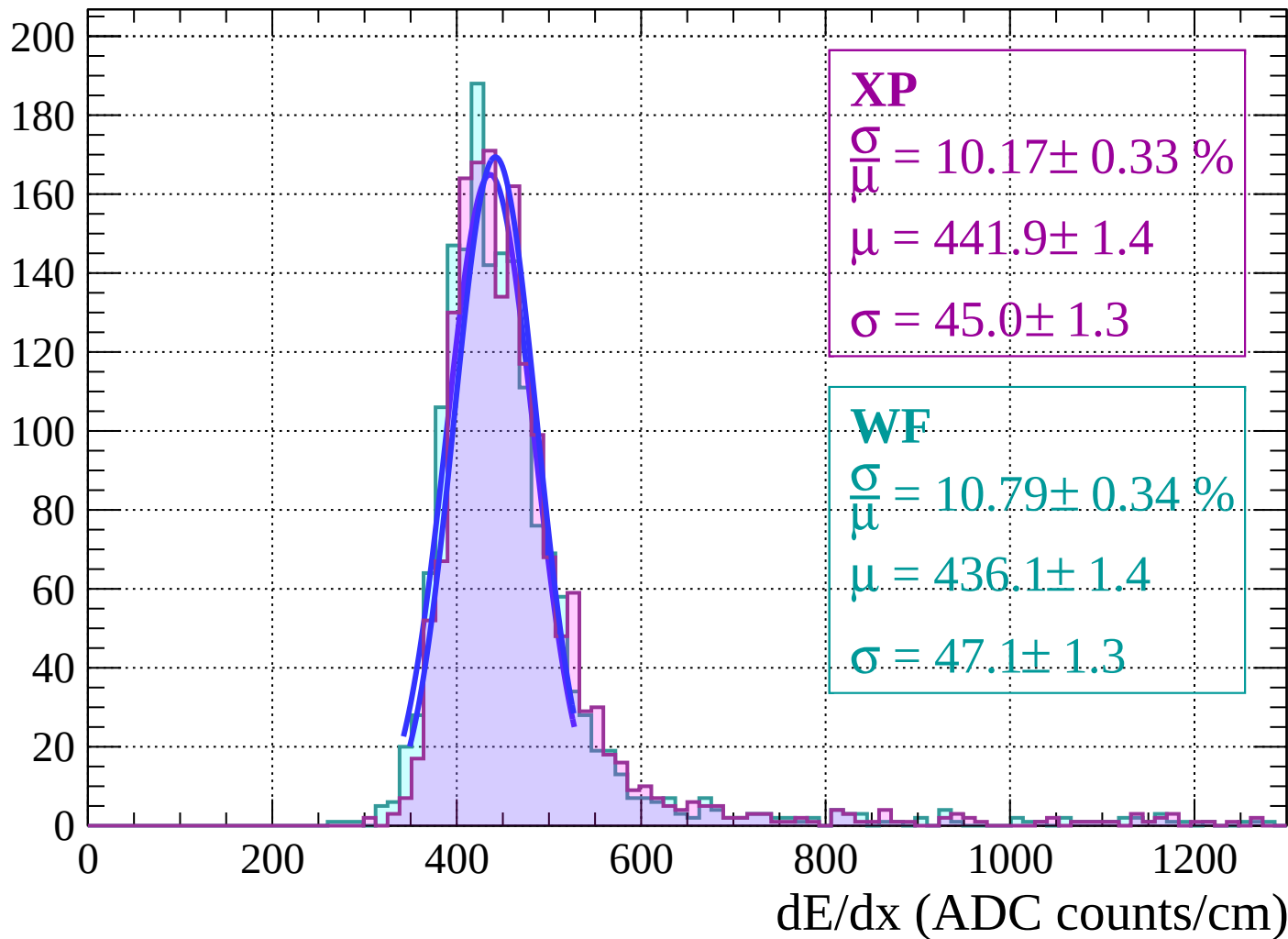


Energy loss | $16 < \theta < 18$



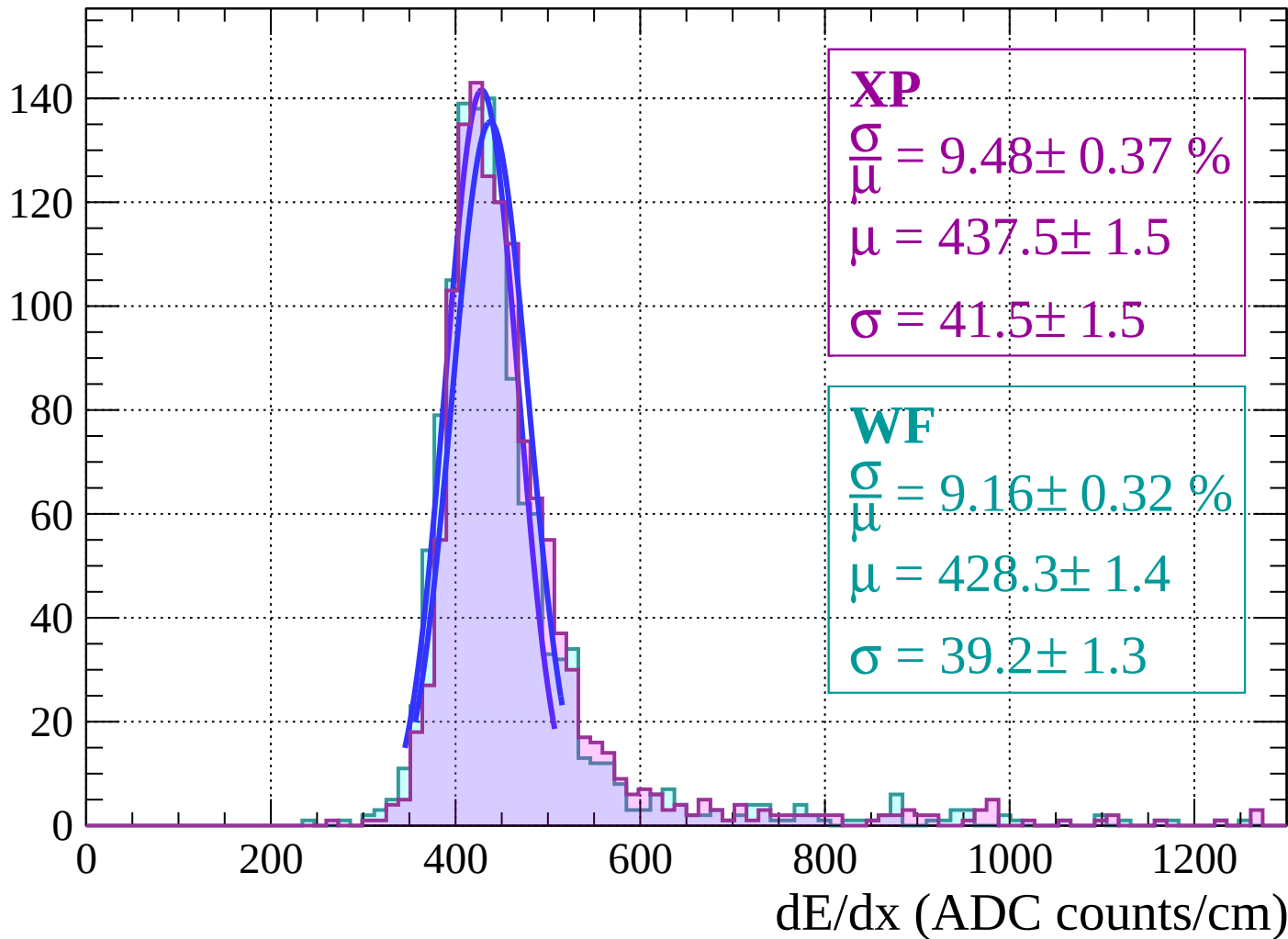
Energy loss | $18 < \theta < 20$

Count



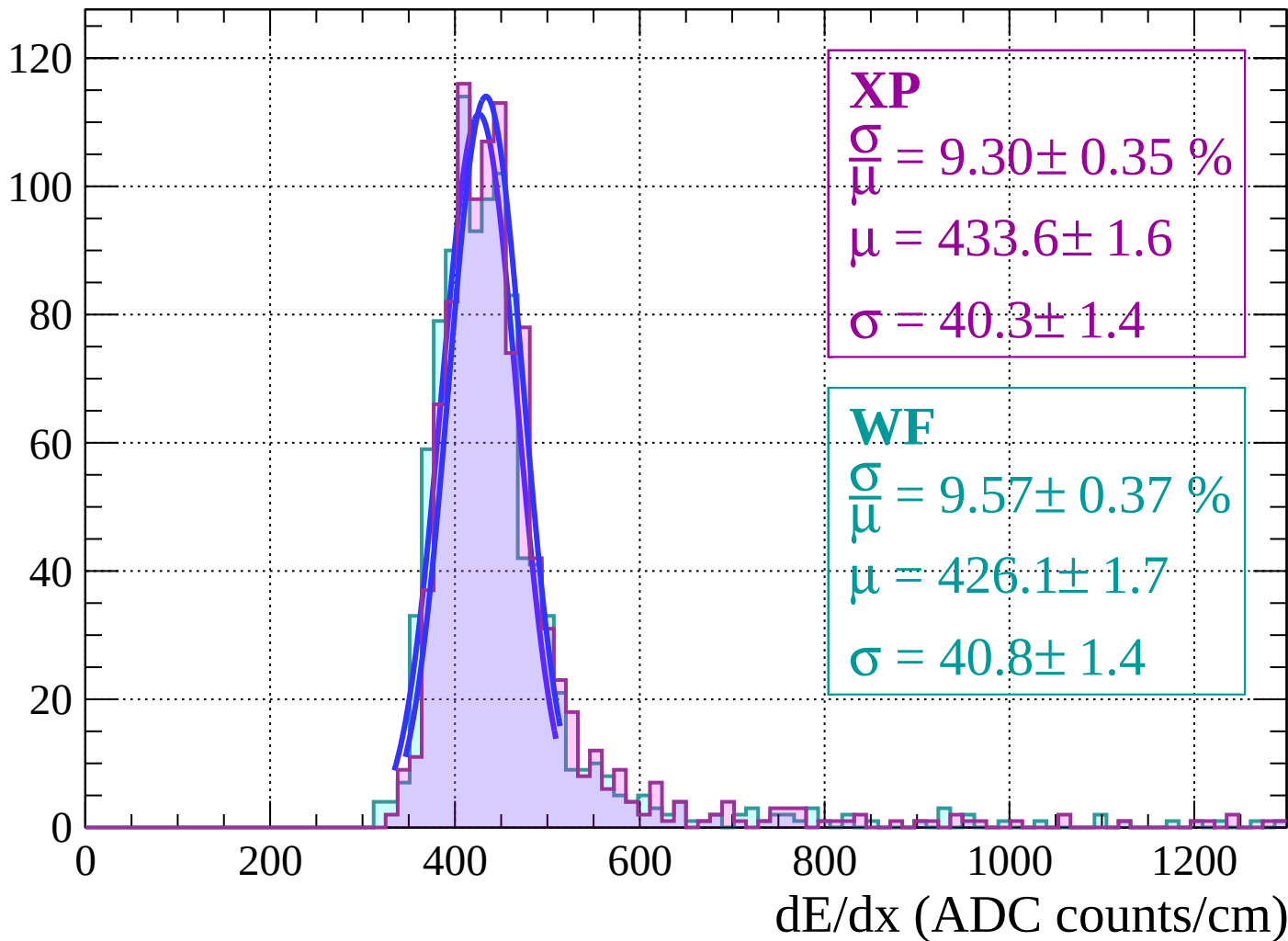
Energy loss | $20 < \theta < 22$

Count



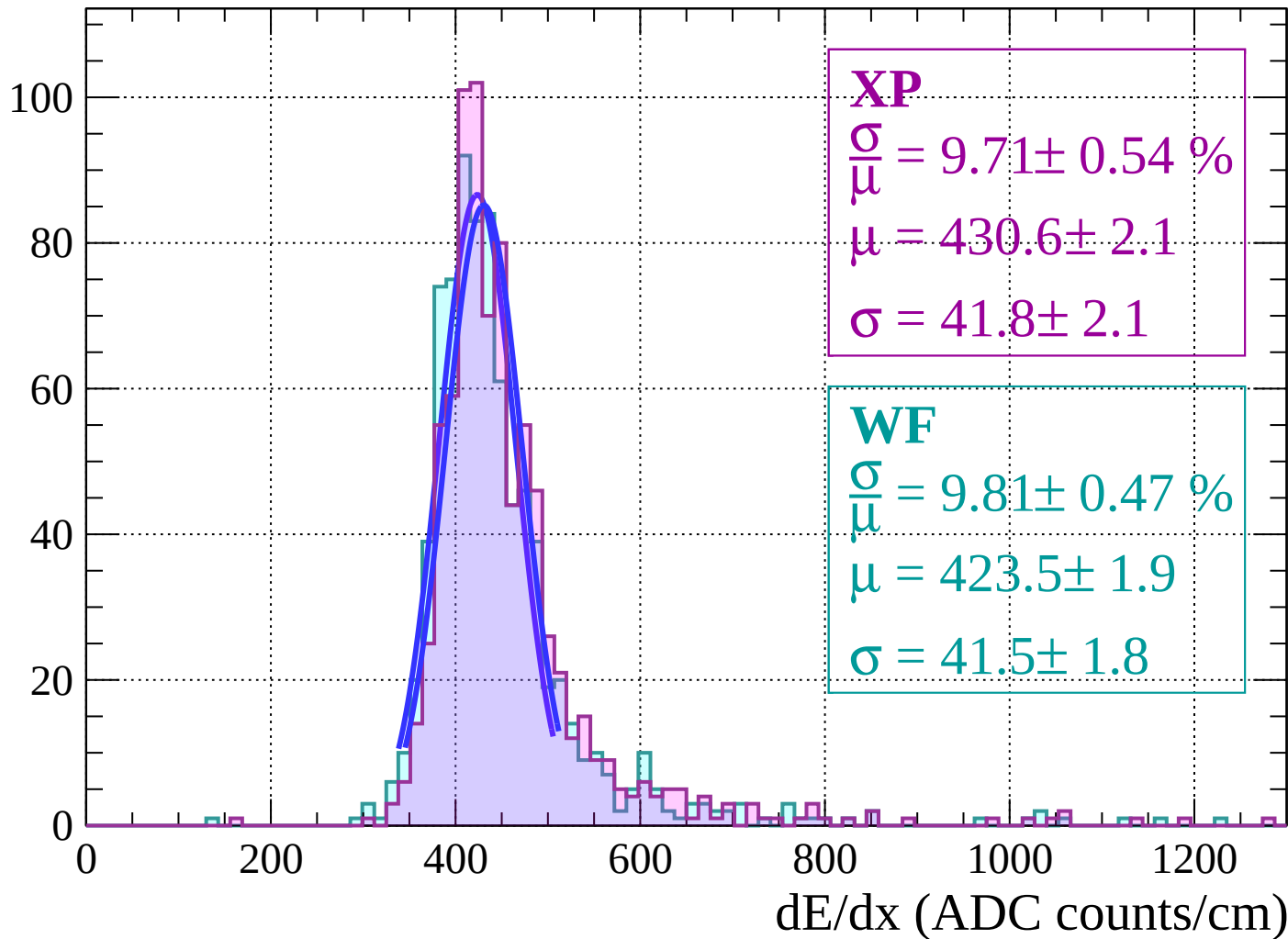
Energy loss | $22 < \theta < 24$

Count



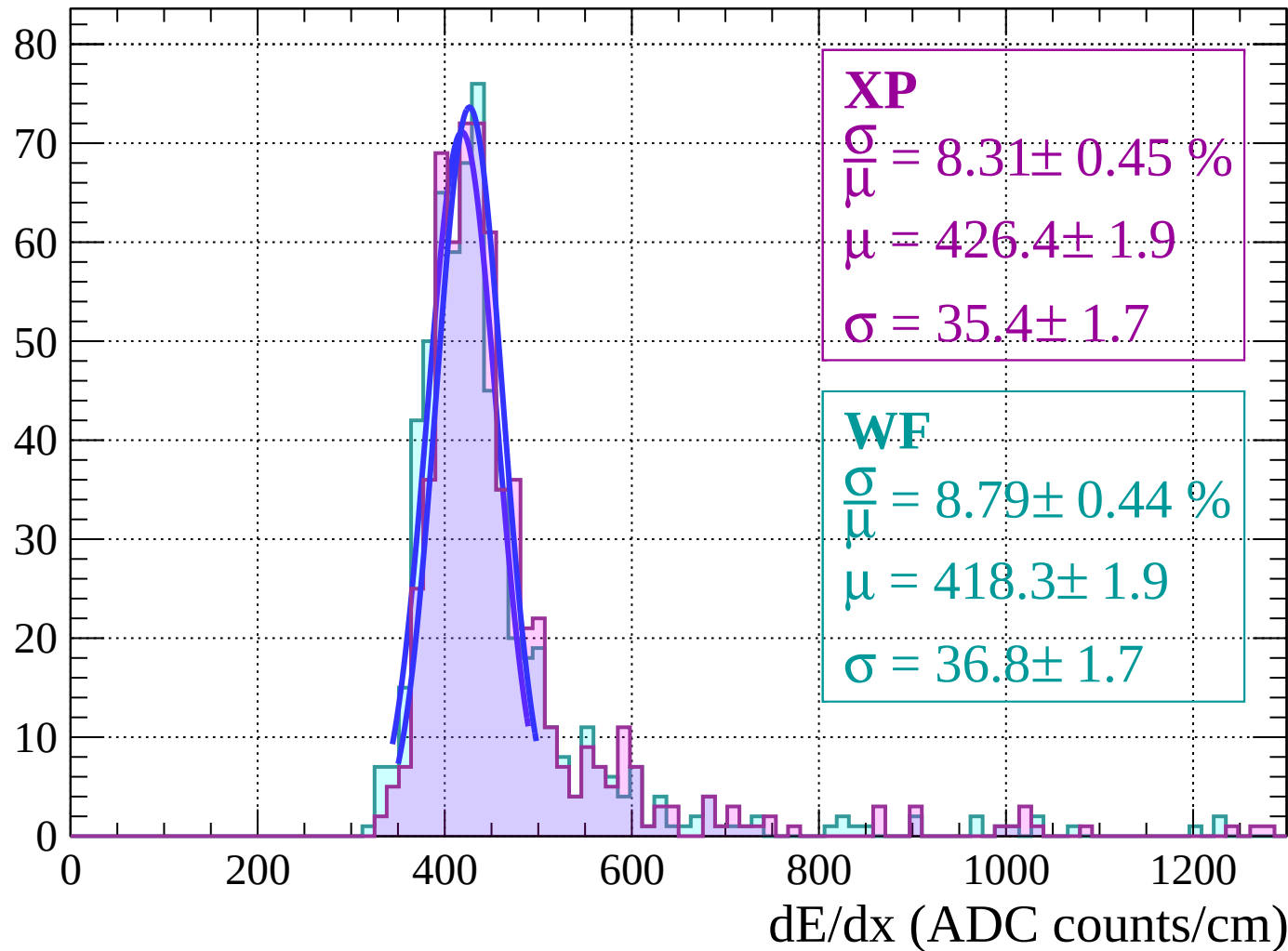
Energy loss | $24 < \theta < 26$

Count



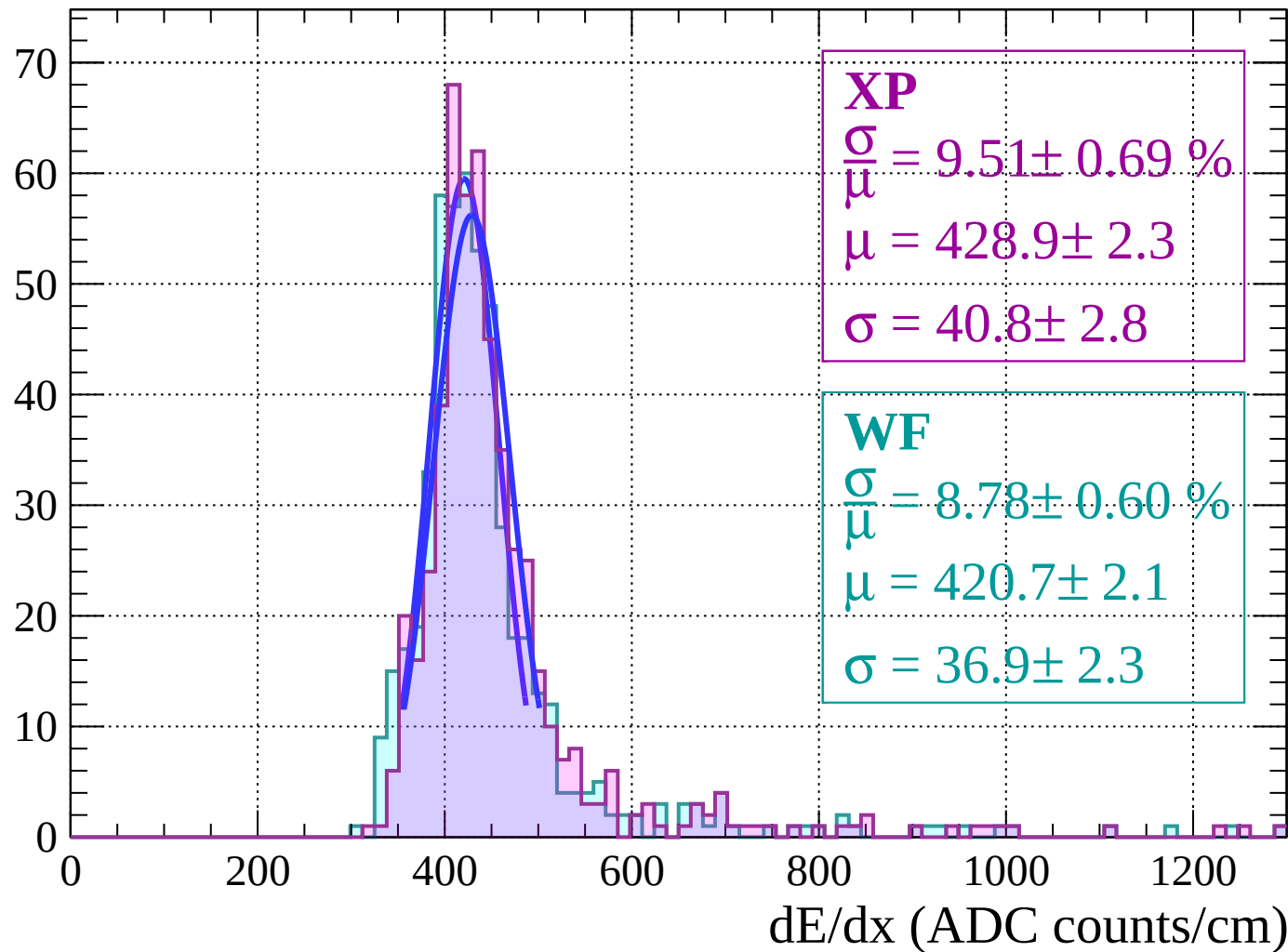
Energy loss | $26 < \theta < 28$

Count



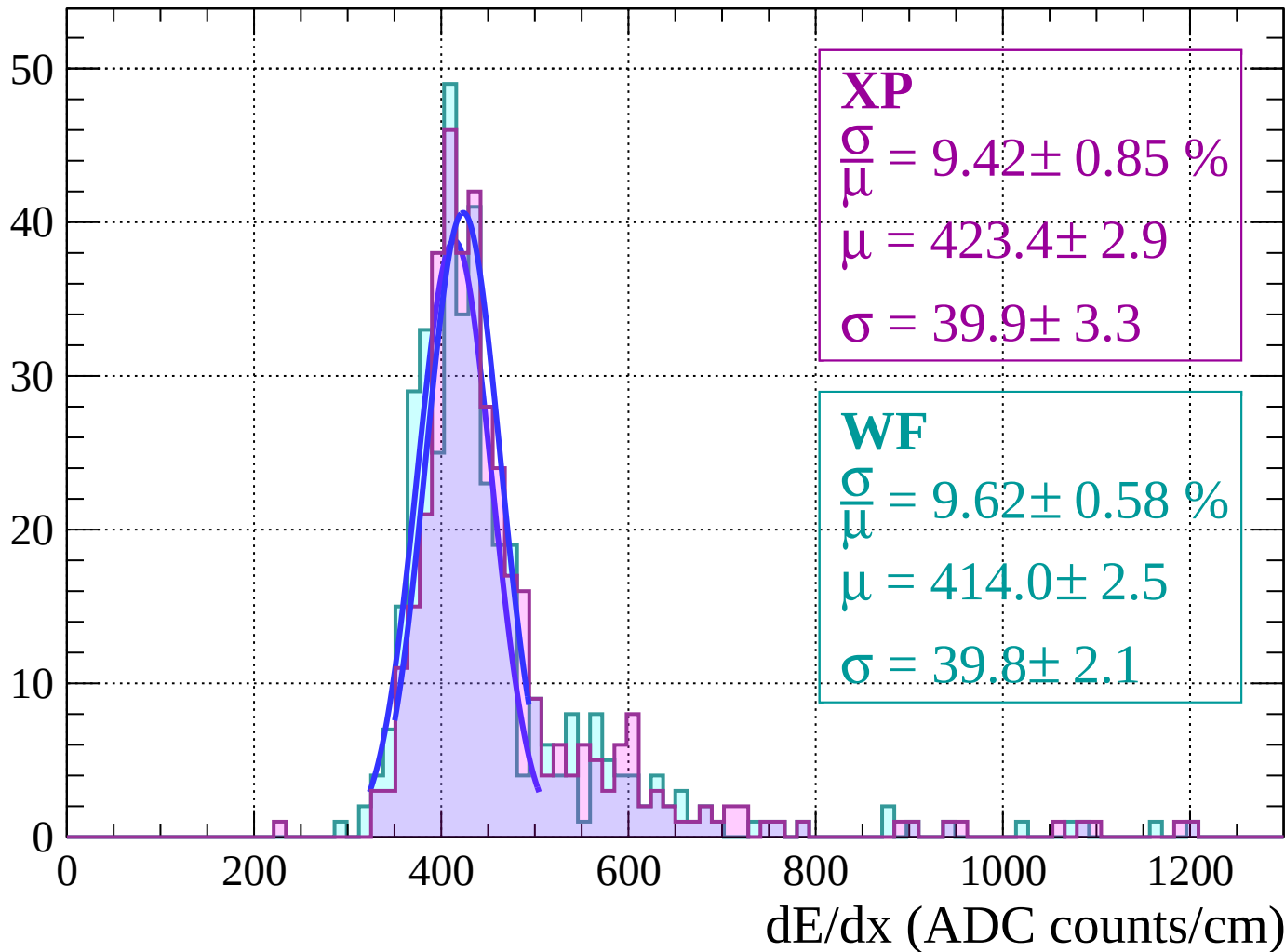
Energy loss | $28 < \theta < 30$

Count



Energy loss | $30 < \theta < 32$

Count



Energy loss | $32 < \theta < 34$

Count

XP

$$\frac{\sigma}{\mu} = 10.28 \pm 0.78 \%$$

$$\mu = 417.0 \pm 3.1$$

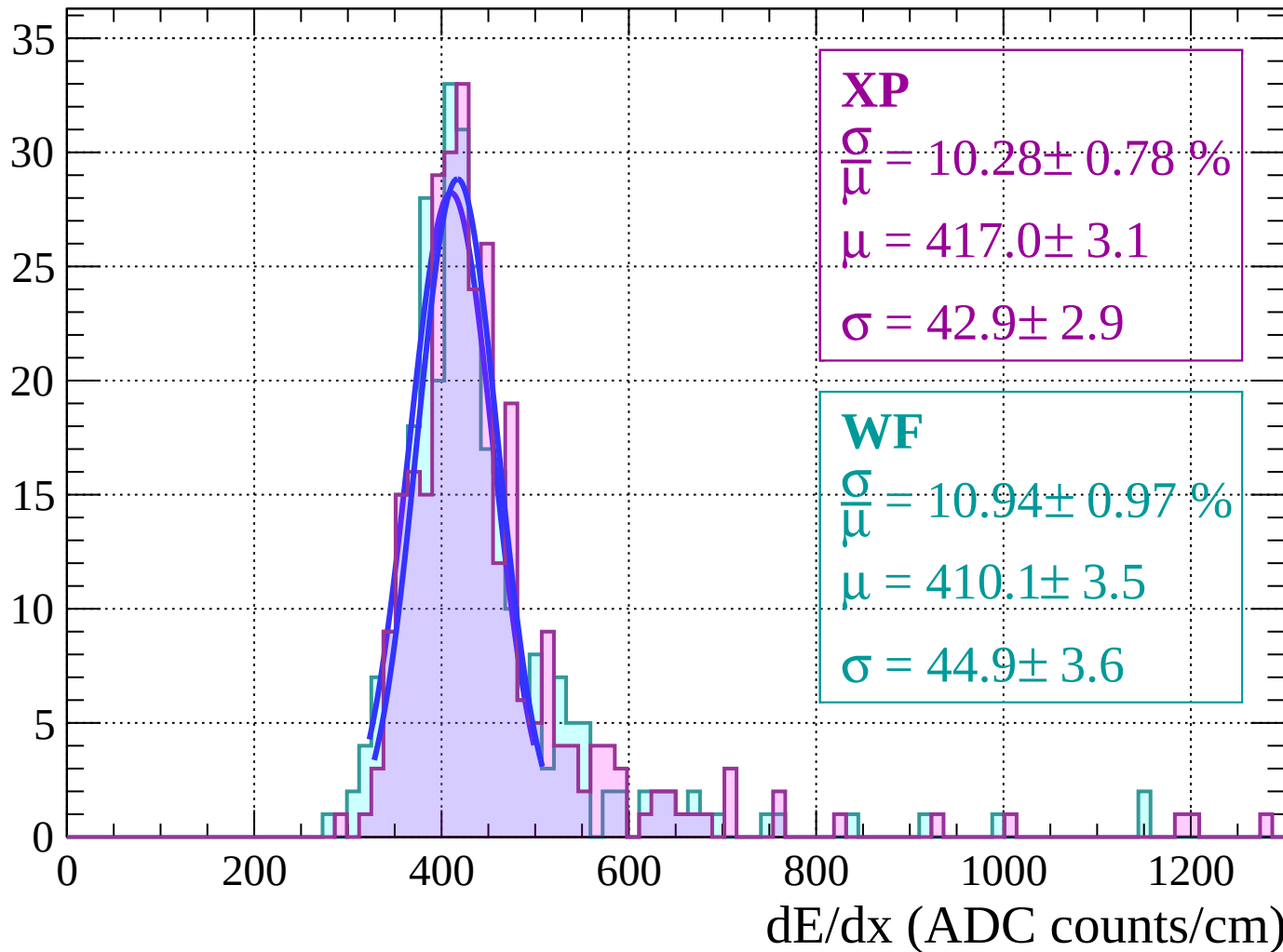
$$\sigma = 42.9 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 10.94 \pm 0.97 \%$$

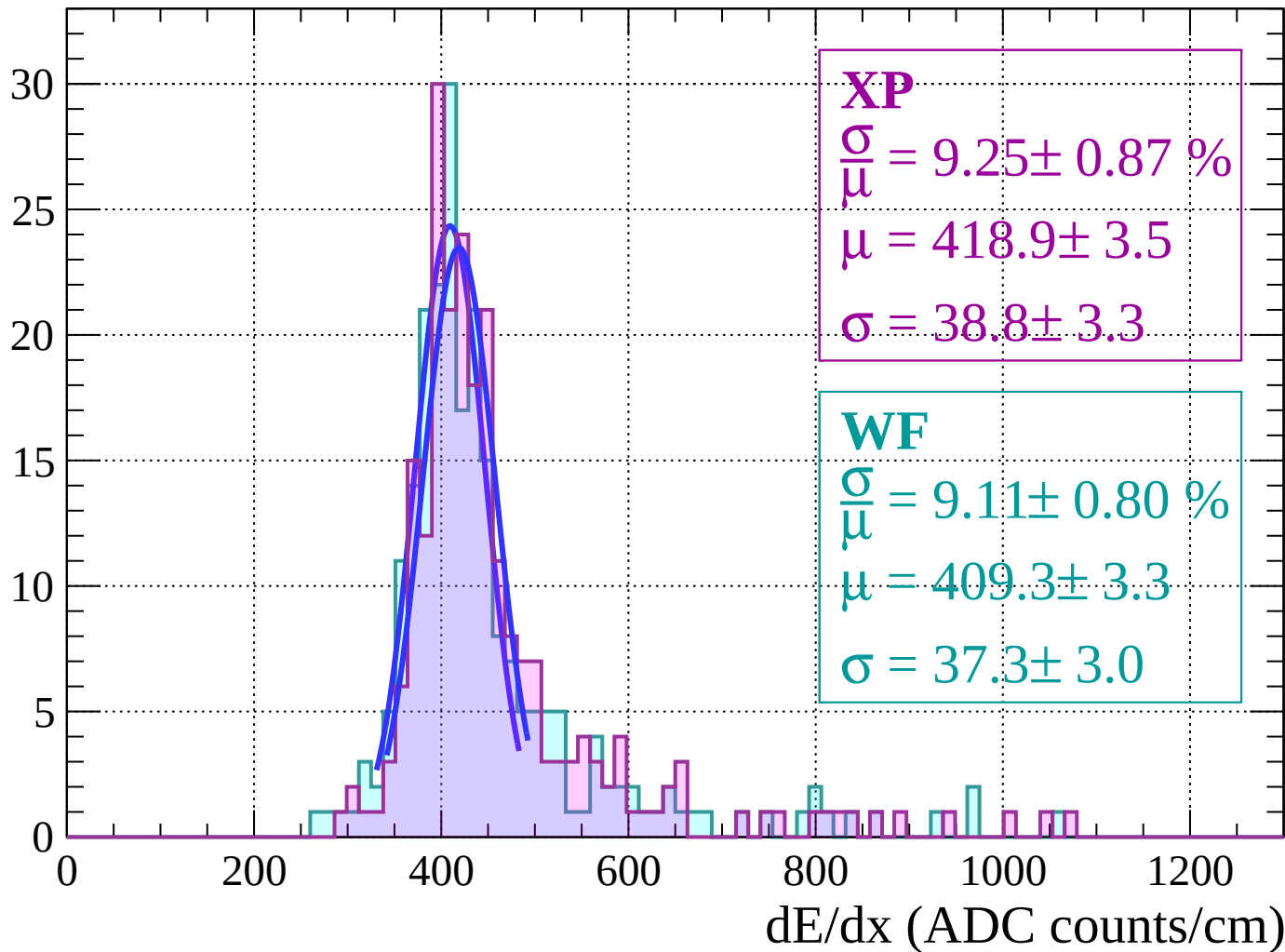
$$\mu = 410.1 \pm 3.5$$

$$\sigma = 44.9 \pm 3.6$$



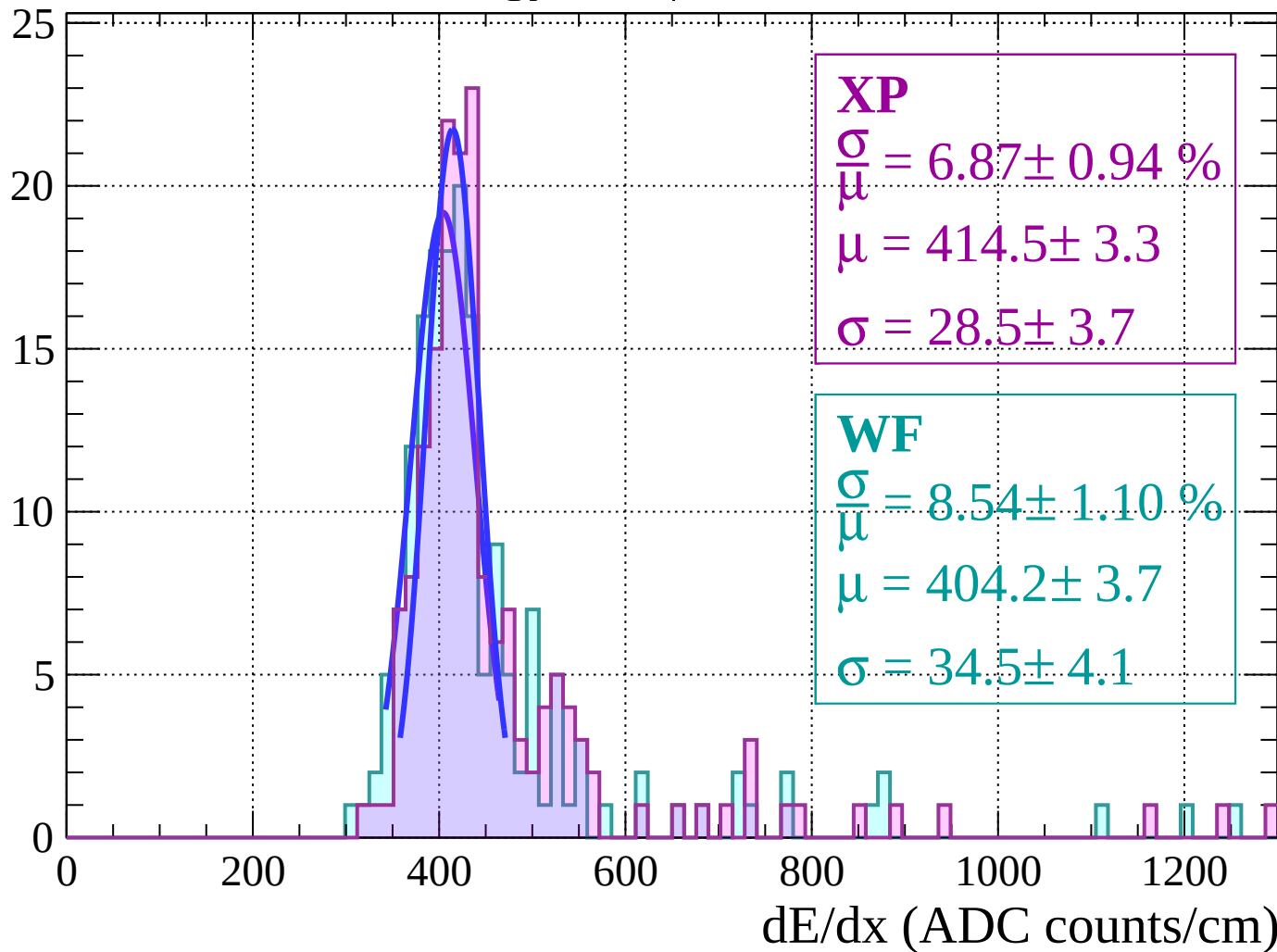
Energy loss | $34 < \theta < 36$

Count



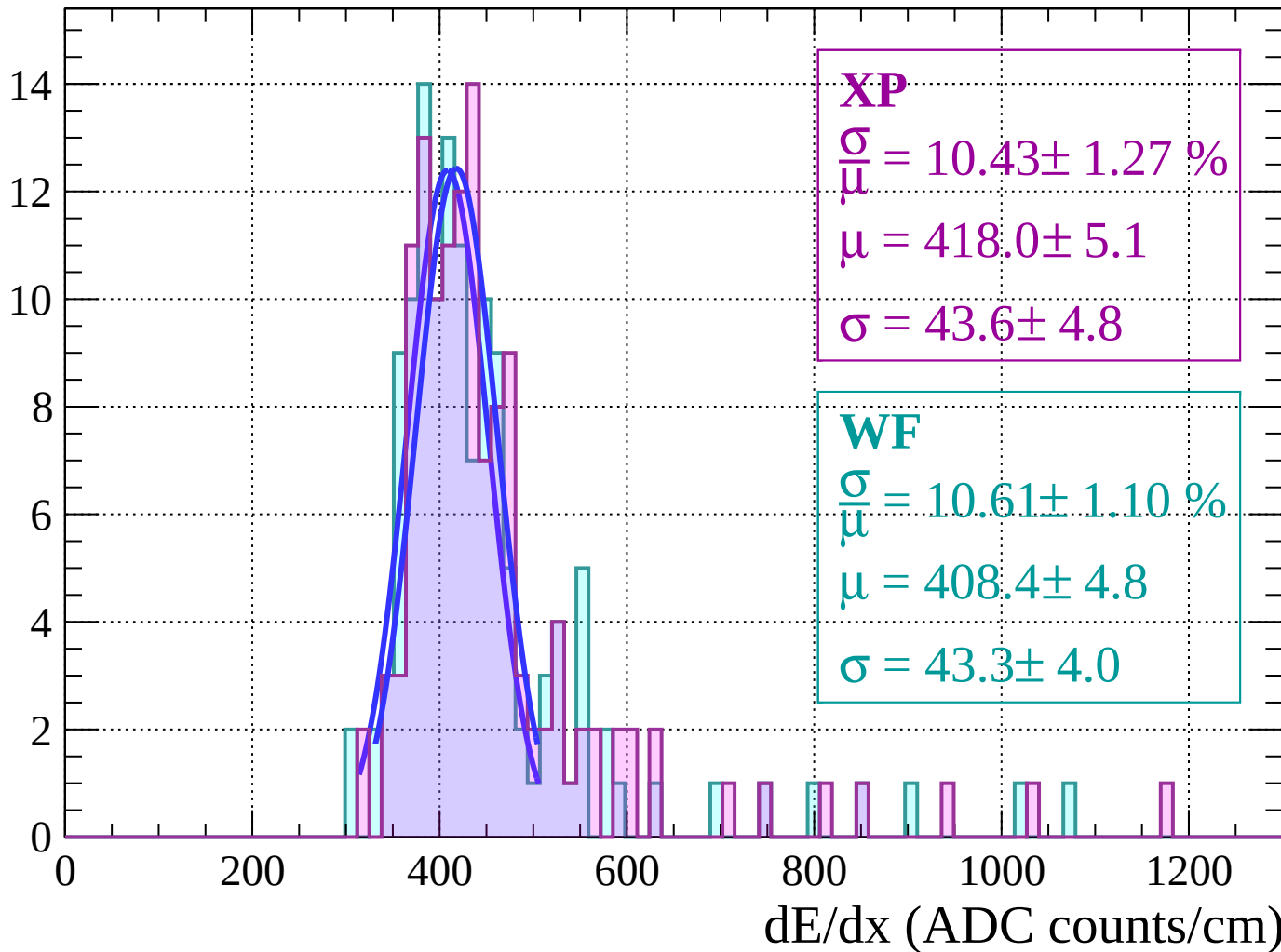
Energy loss | $36 < \theta < 38$

Count



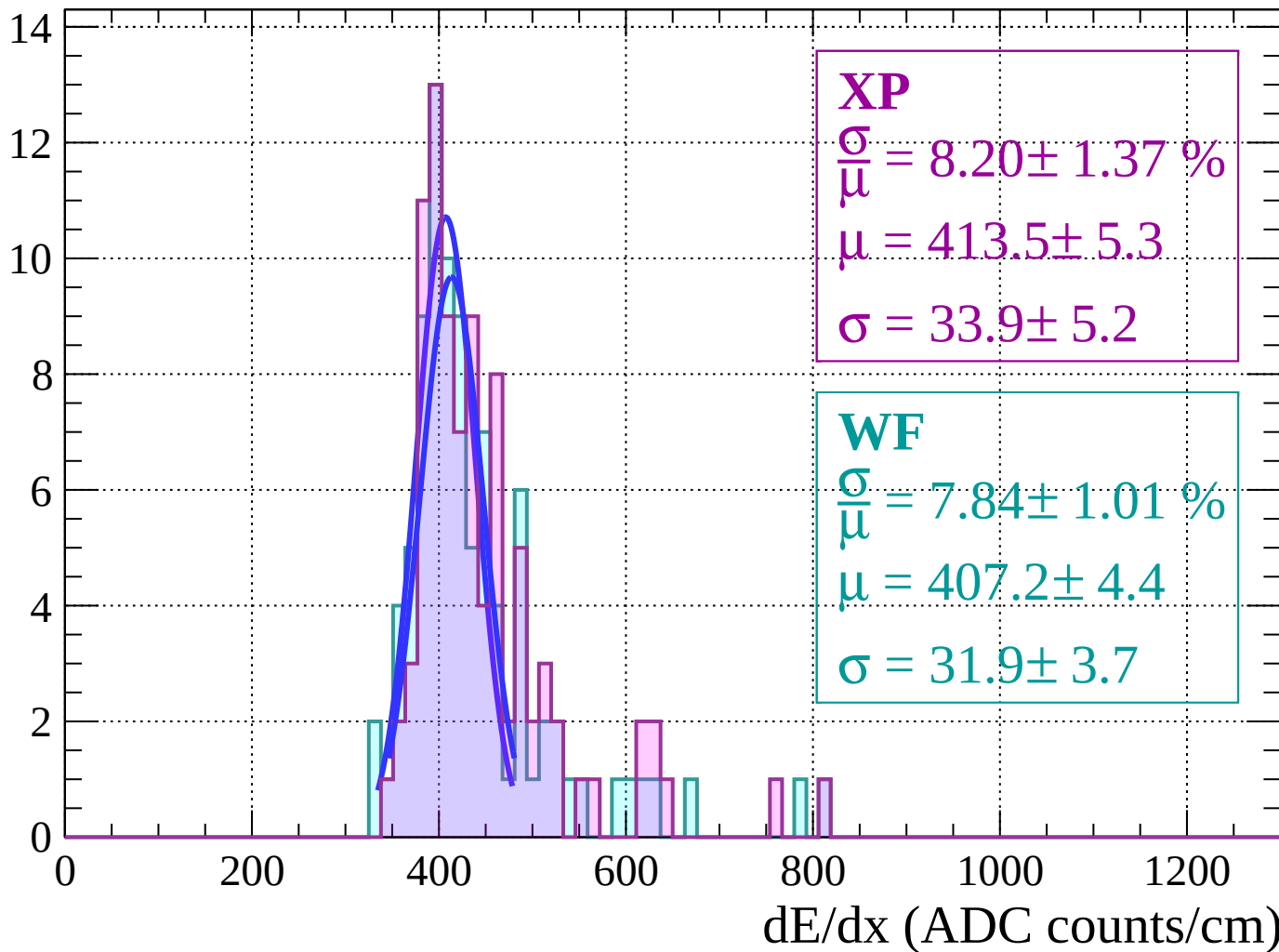
Energy loss | $38 < \theta < 40$

Count



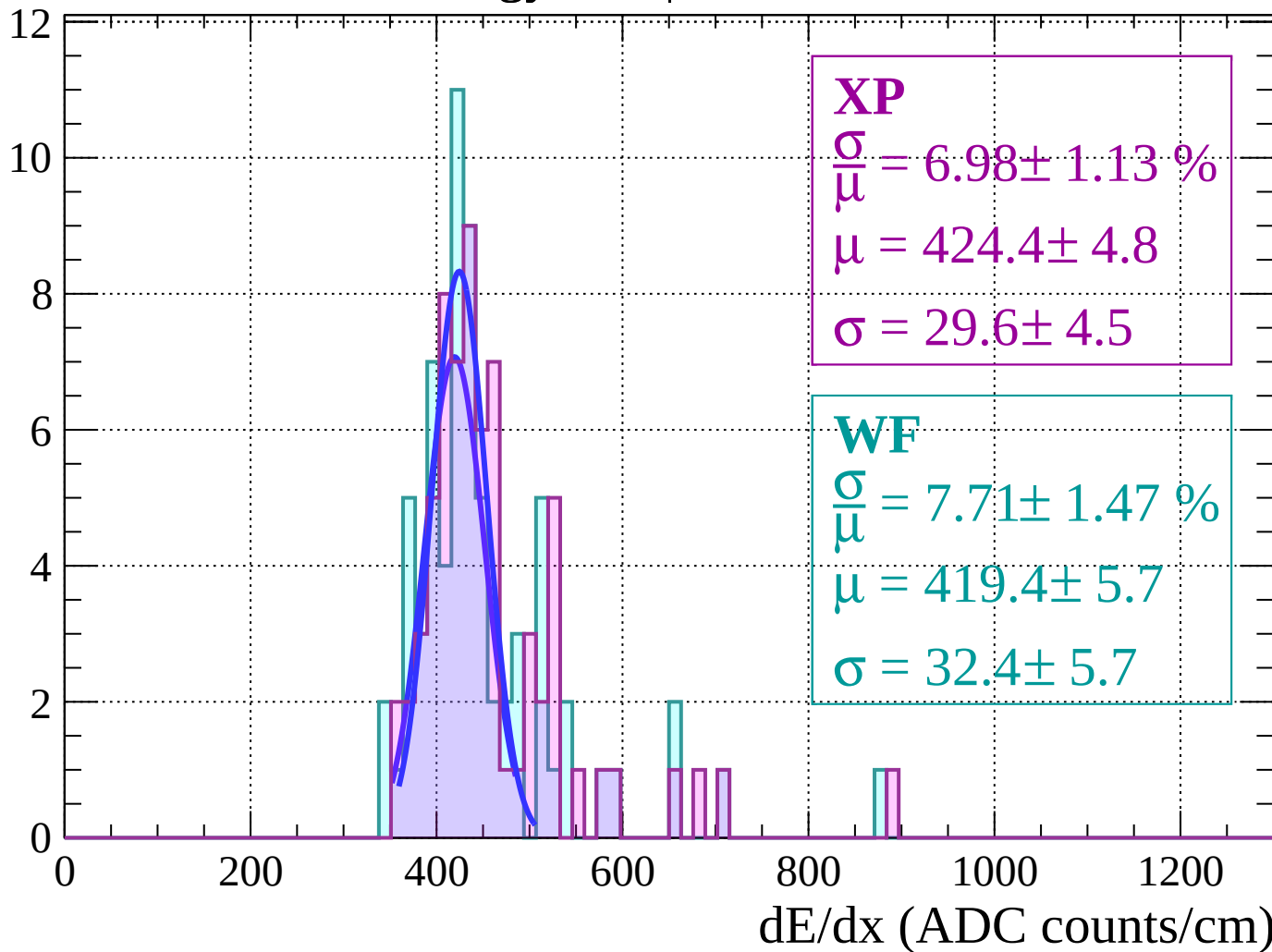
Energy loss | $40 < \theta < 42$

Count



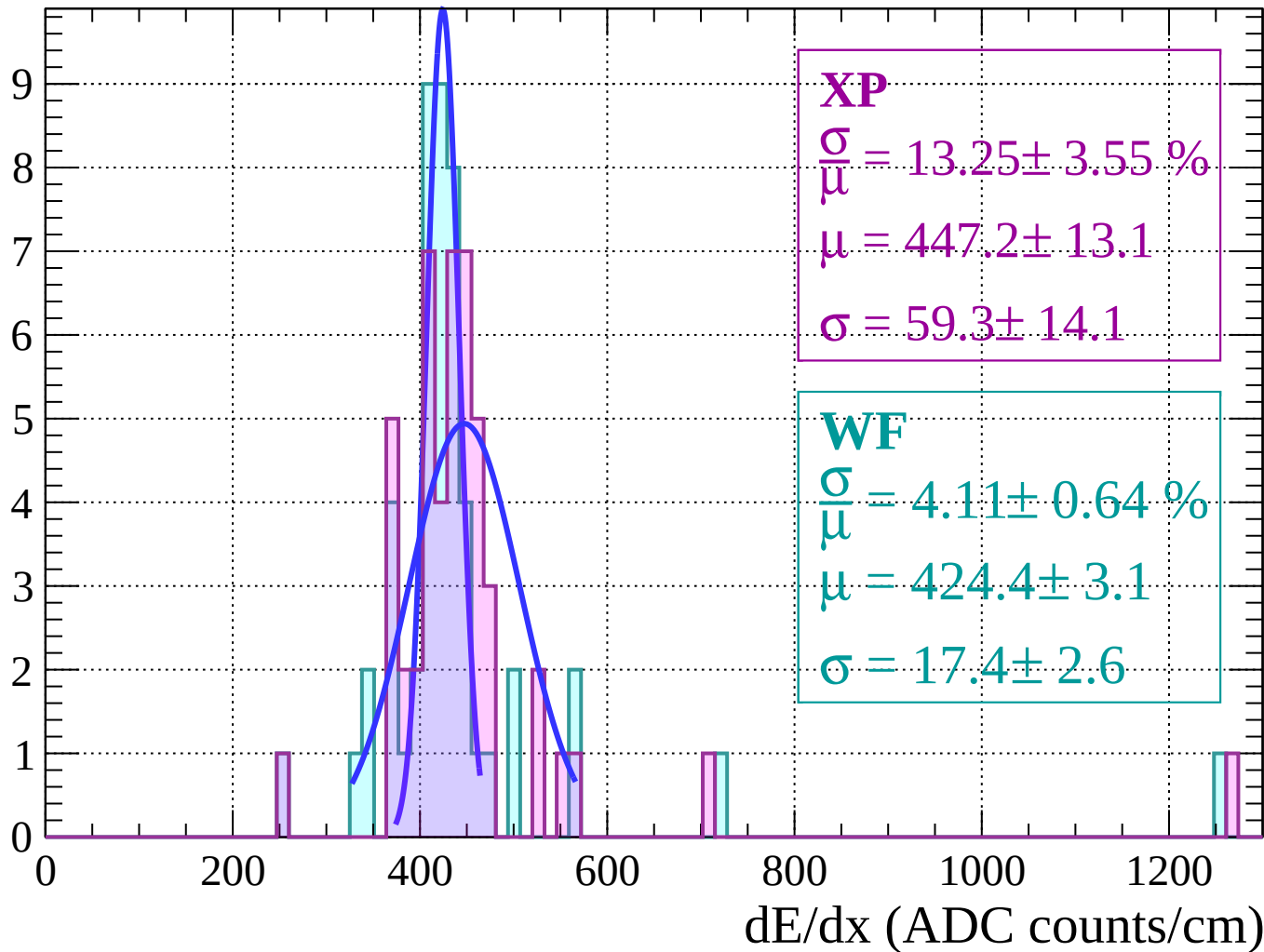
Energy loss | $42 < \theta < 44$

Count



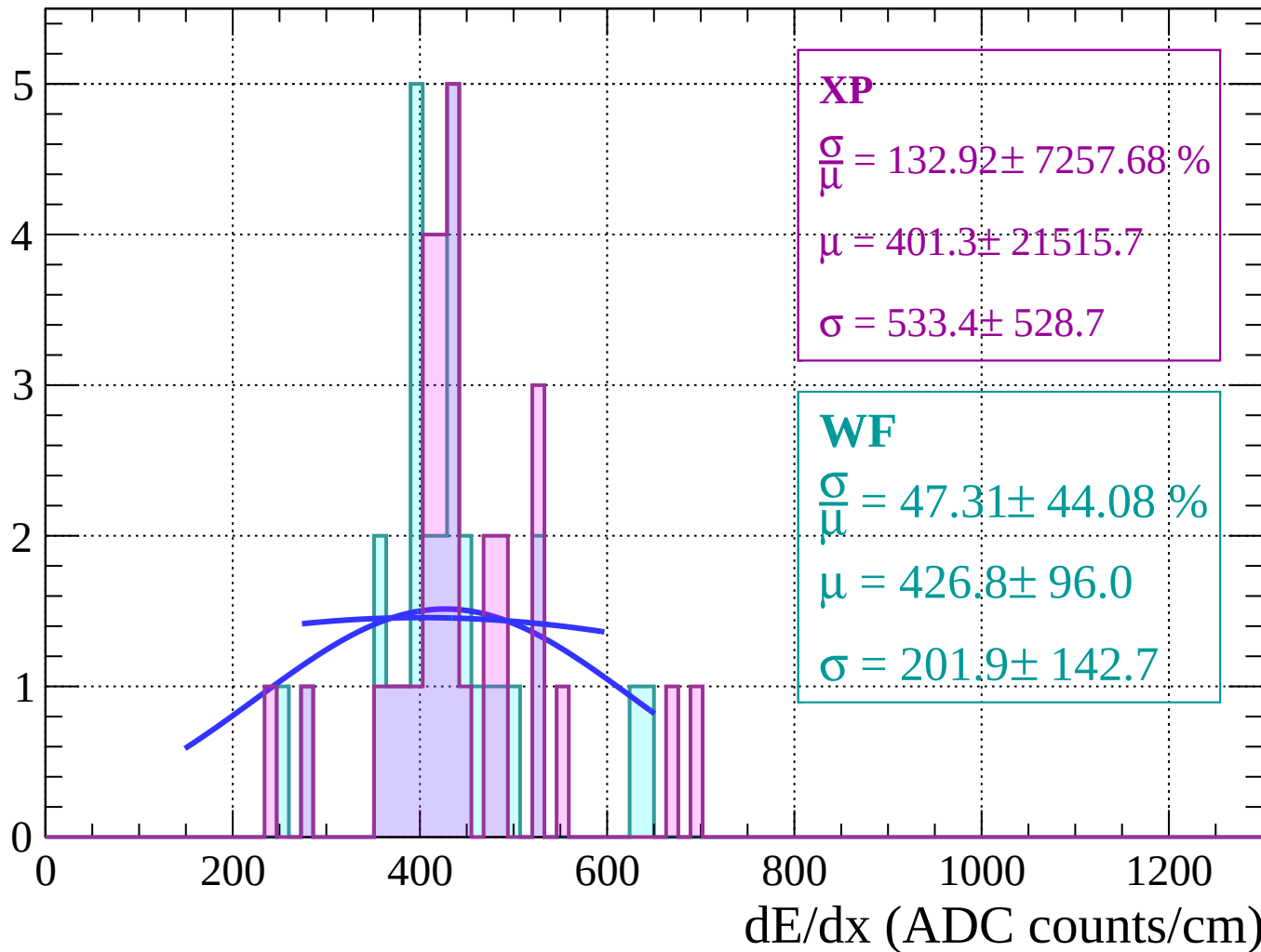
Energy loss | $44 < \theta < 46$

Count



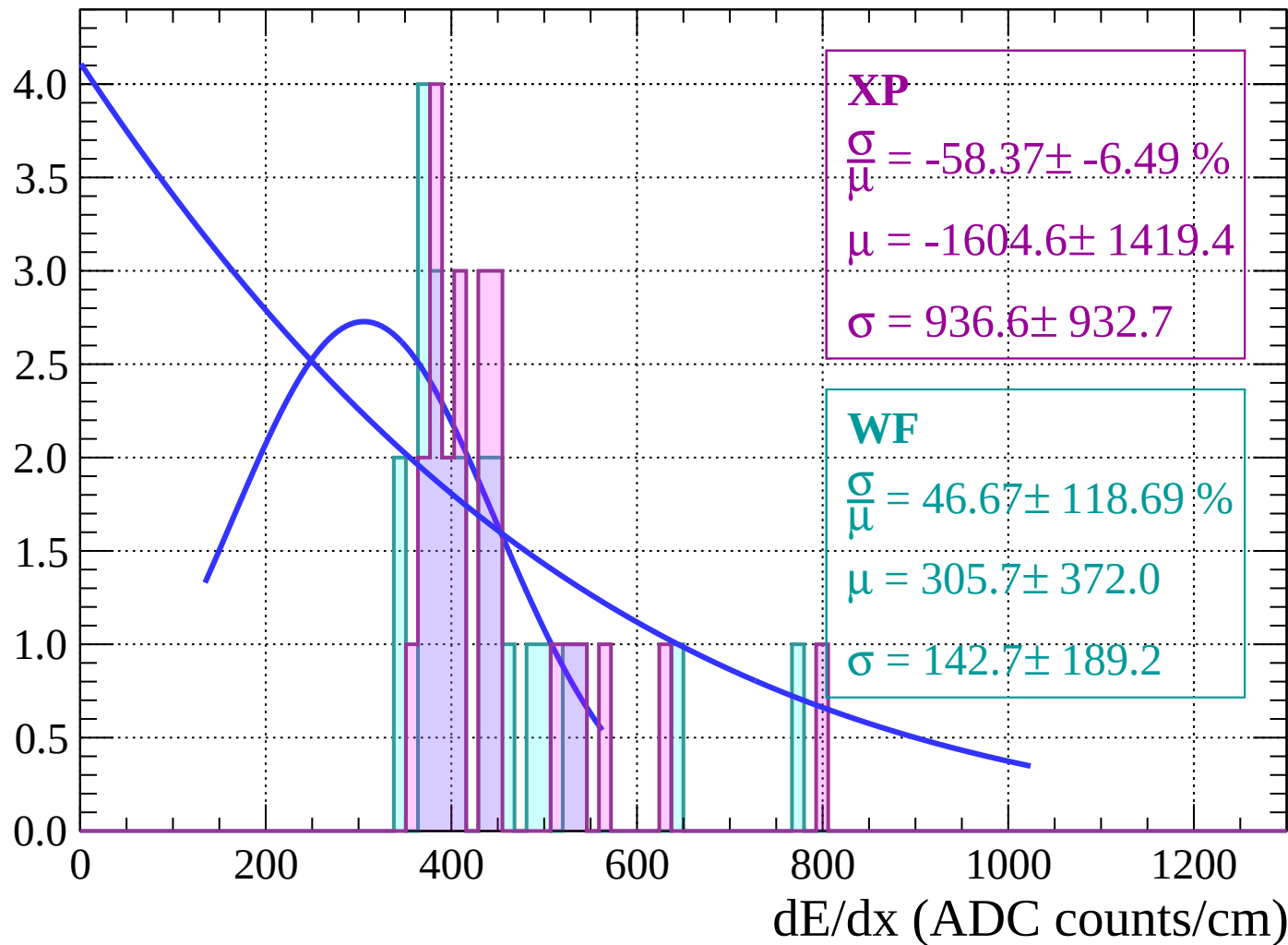
Energy loss | $46 < \theta < 48$

Count



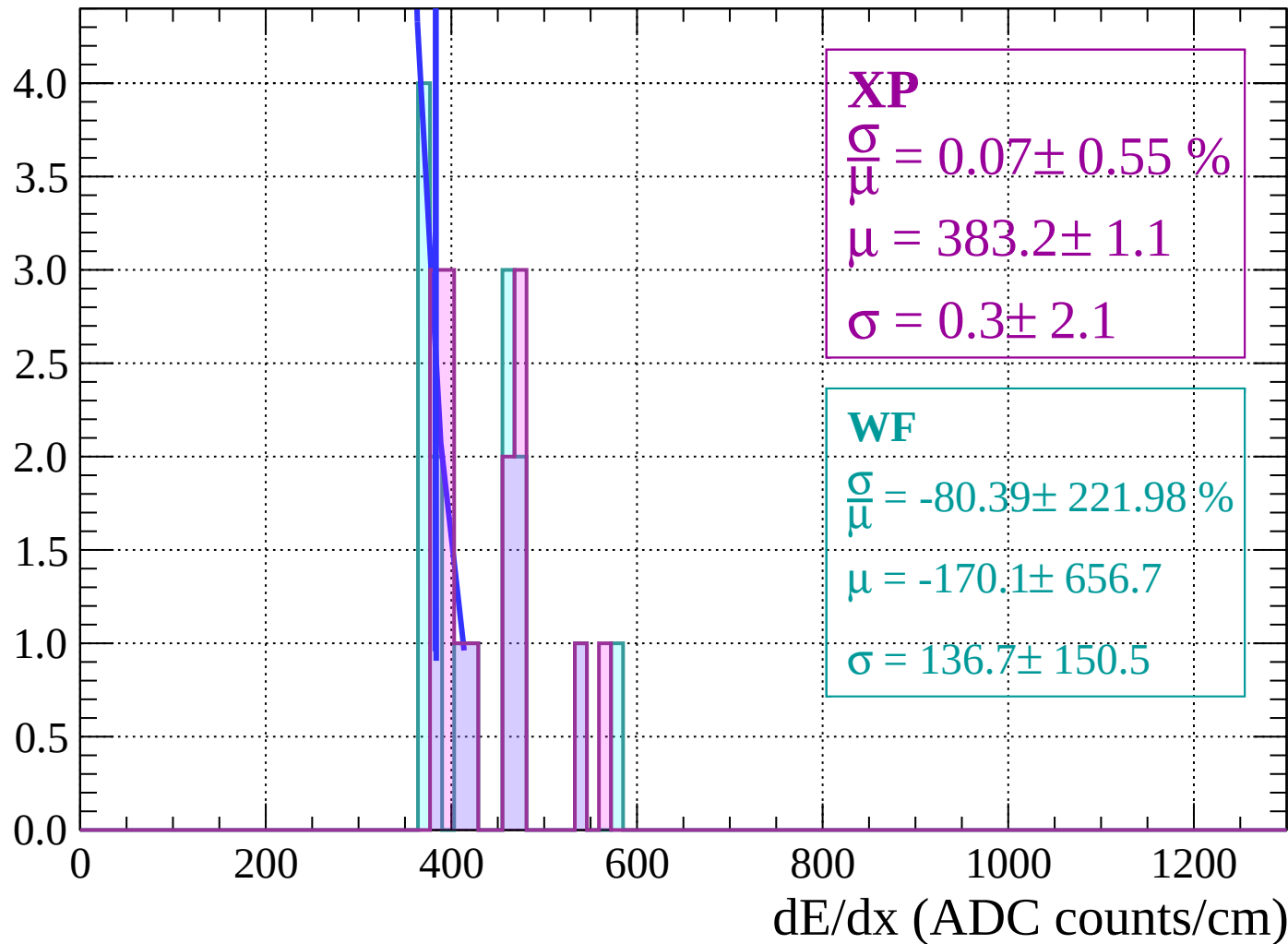
Energy loss | $48 < \theta < 50$

Count



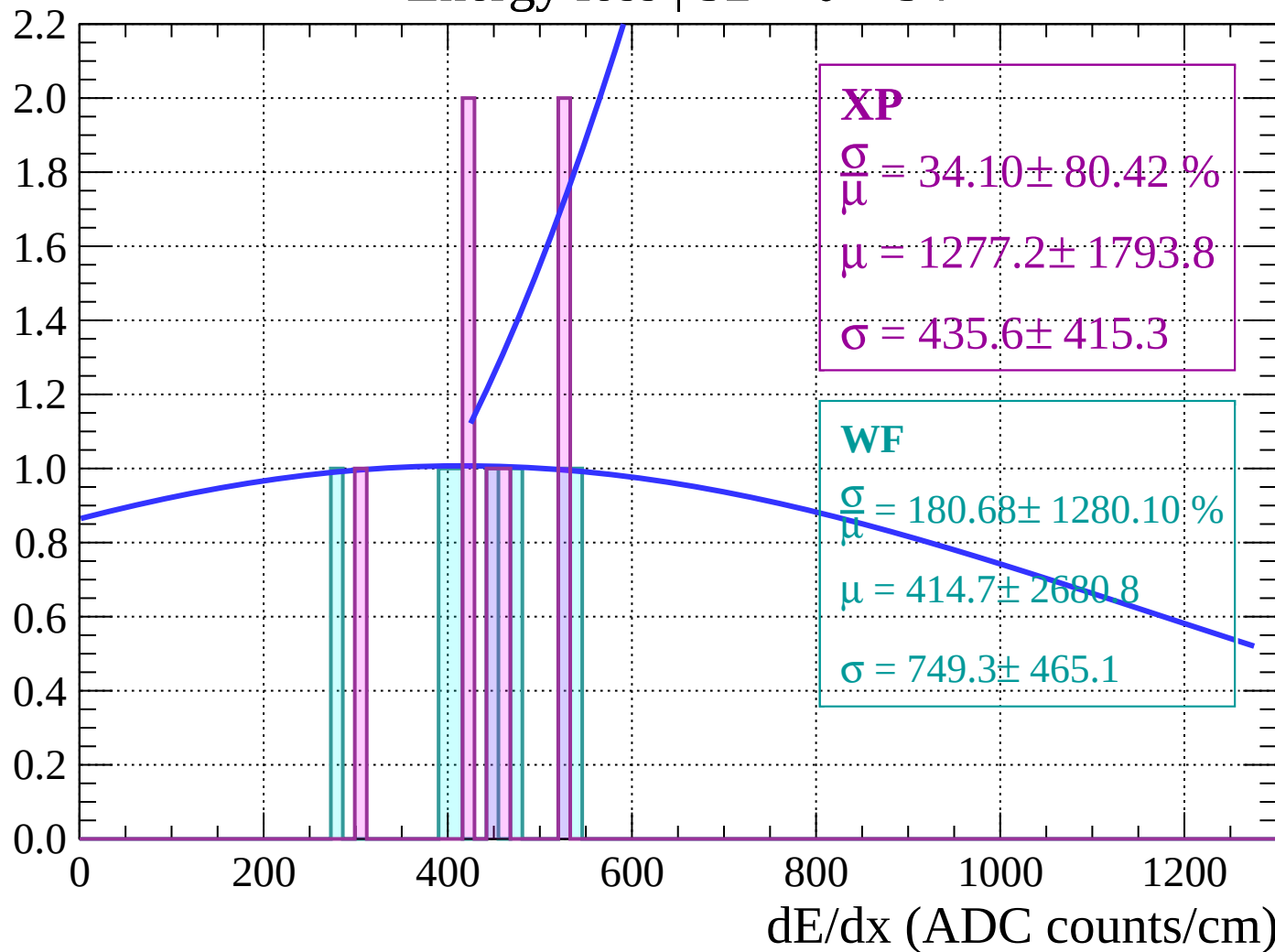
Energy loss | $50 < \theta < 52$

Count



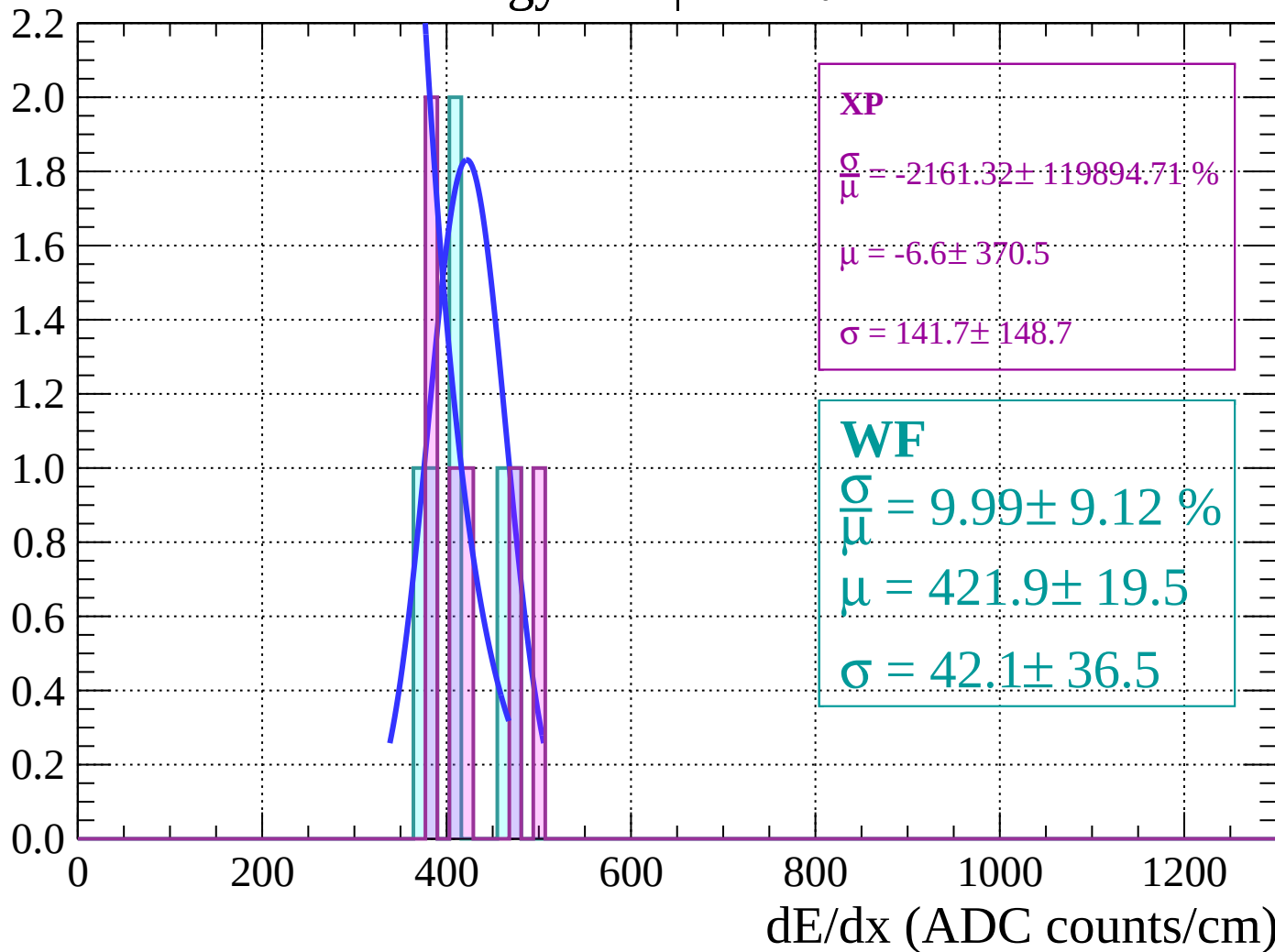
Energy loss | $52 < \theta < 54$

Count



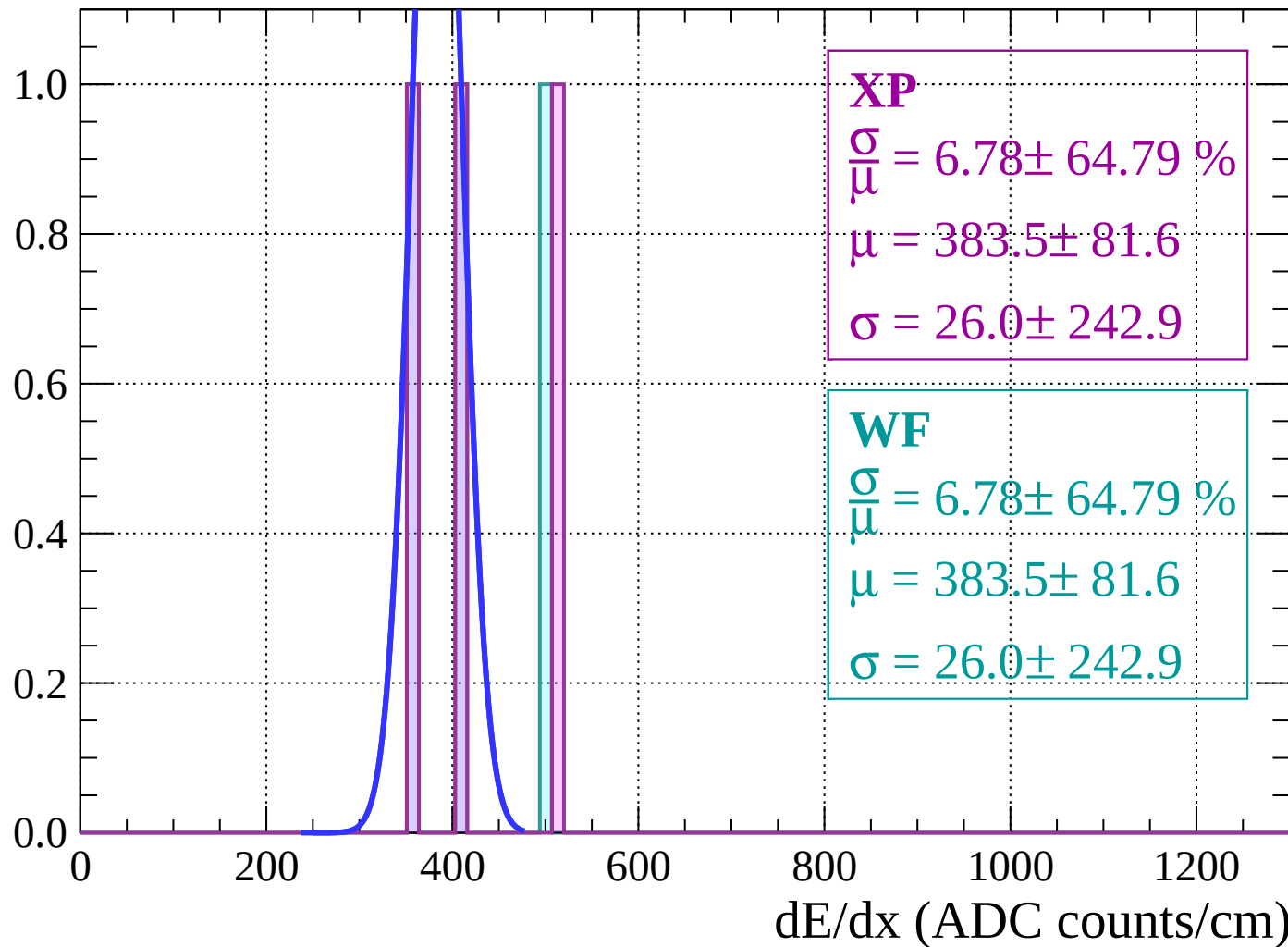
Energy loss | $54 < \theta < 56$

Count



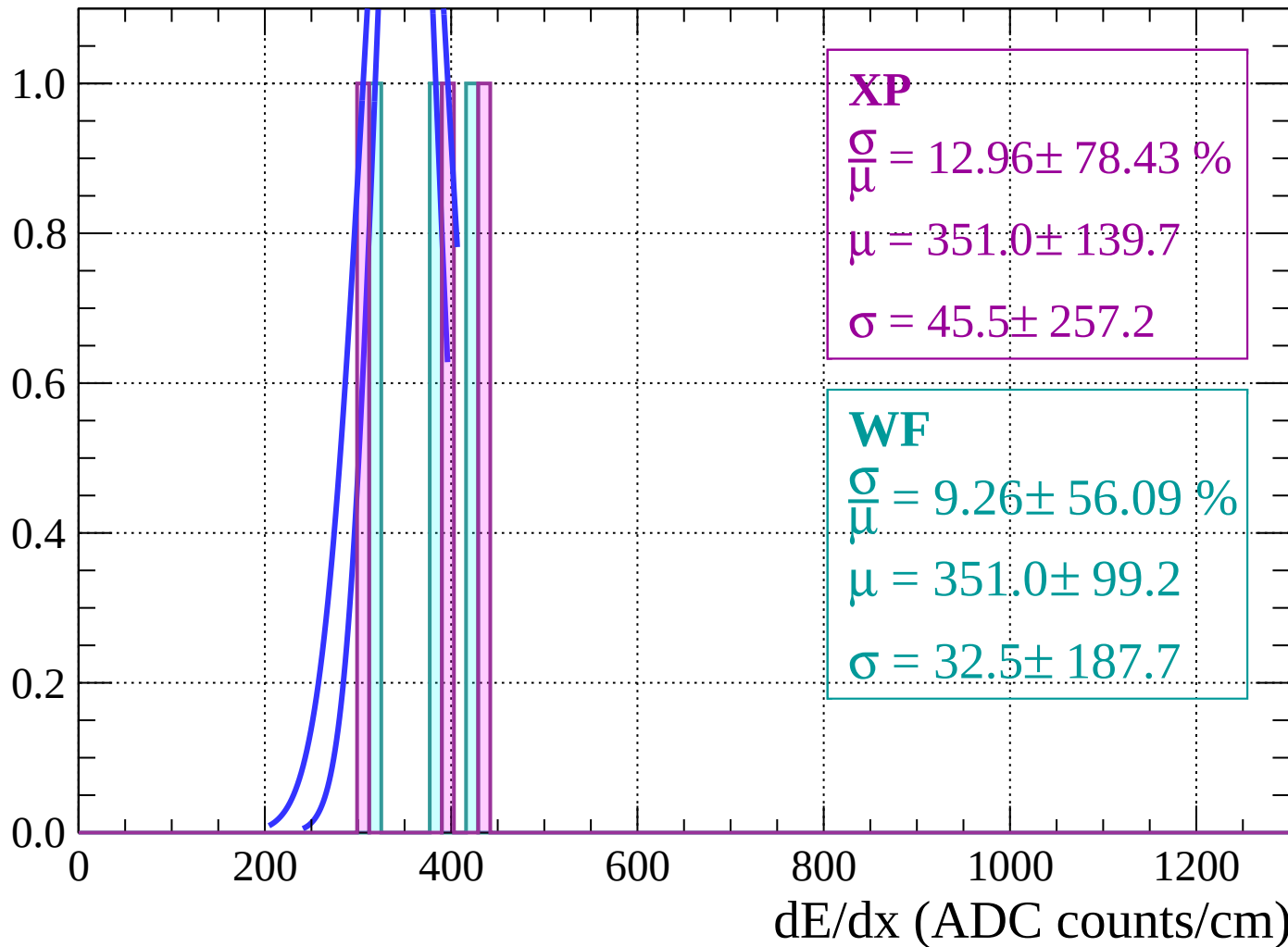
Energy loss | $56 < \theta < 58$

Count



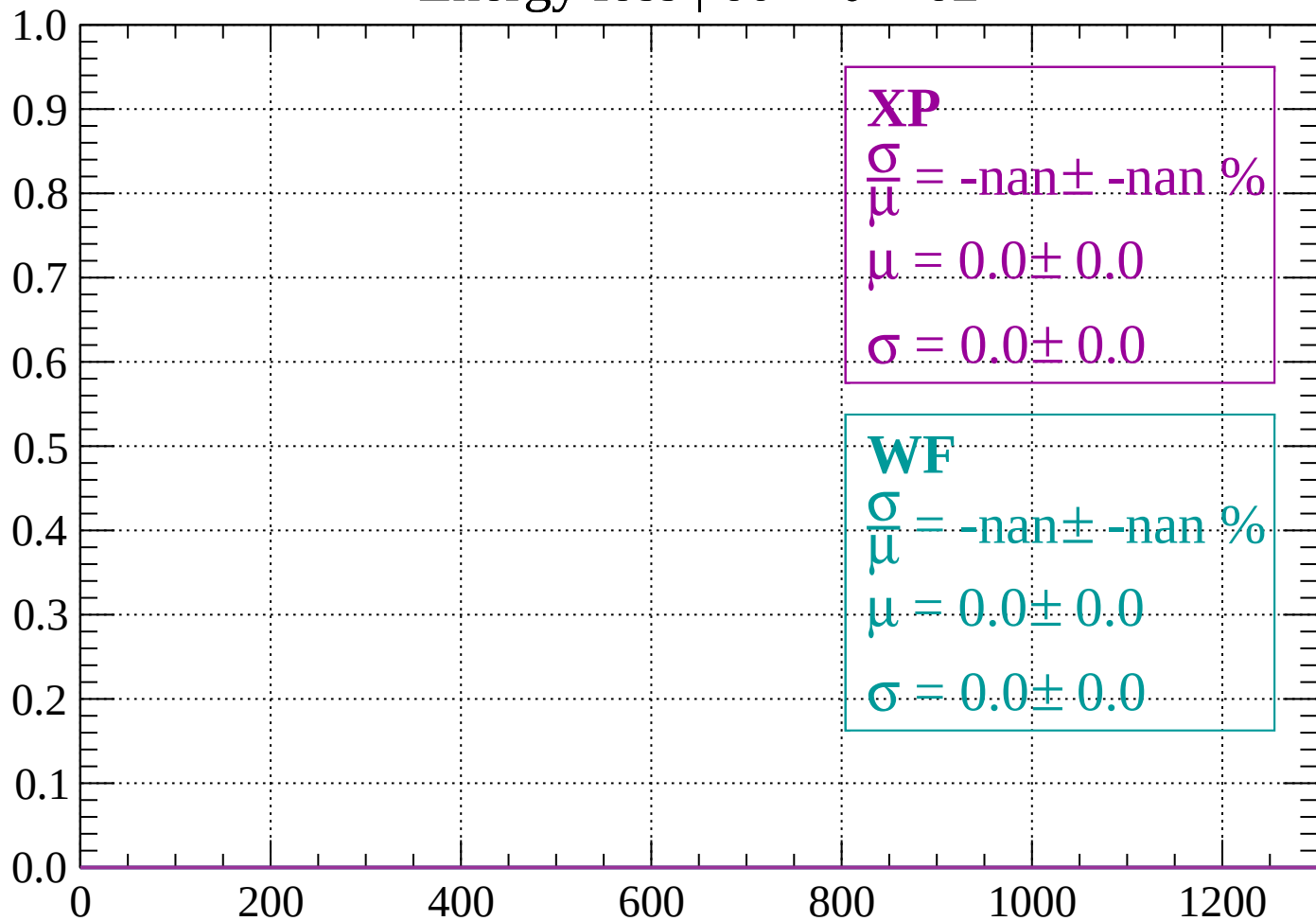
Energy loss | $58 < \theta < 60$

Count



Energy loss | $60 < \theta < 62$

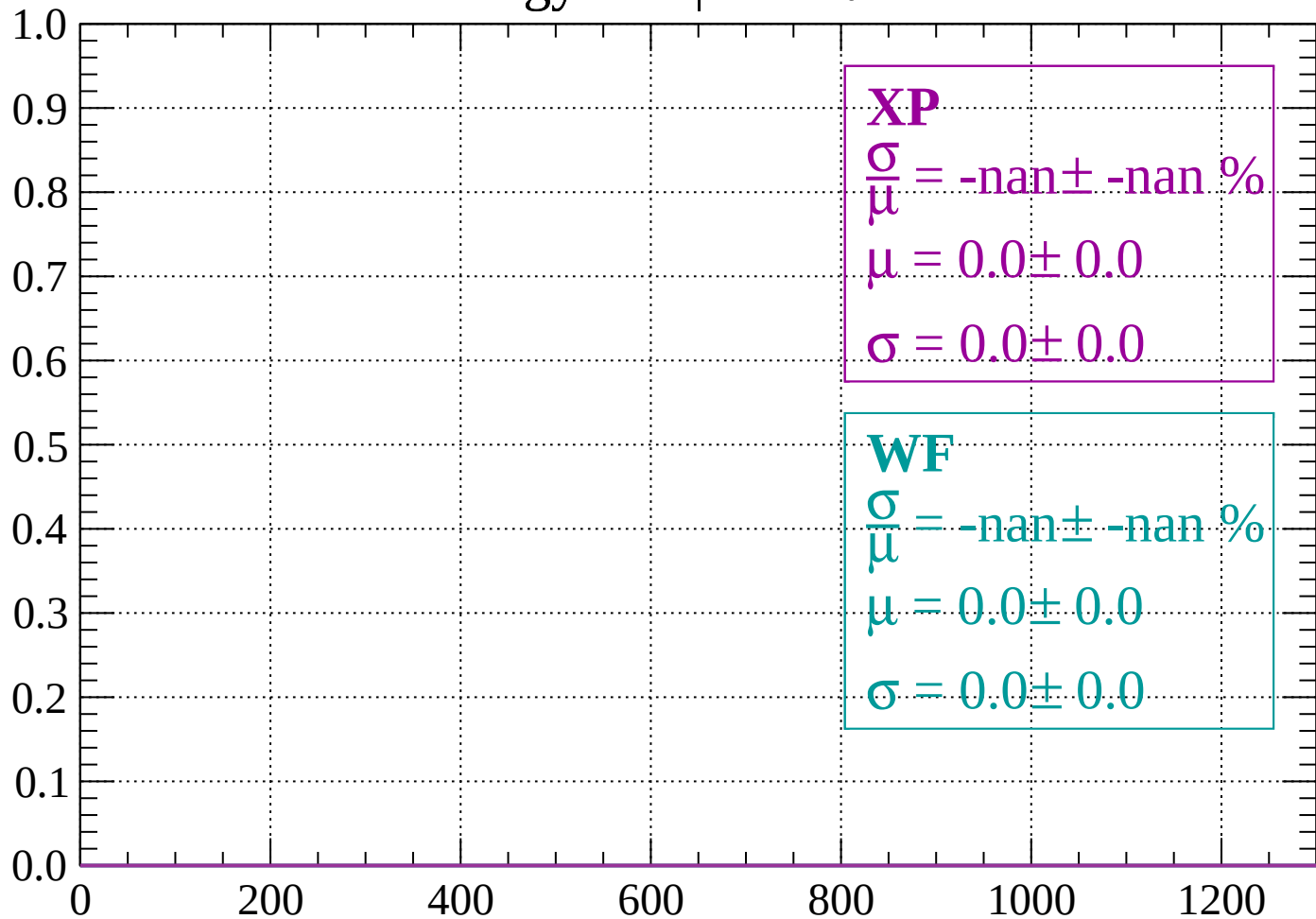
Count



dE/dx (ADC counts/cm)

Energy loss | $62 < \theta < 64$

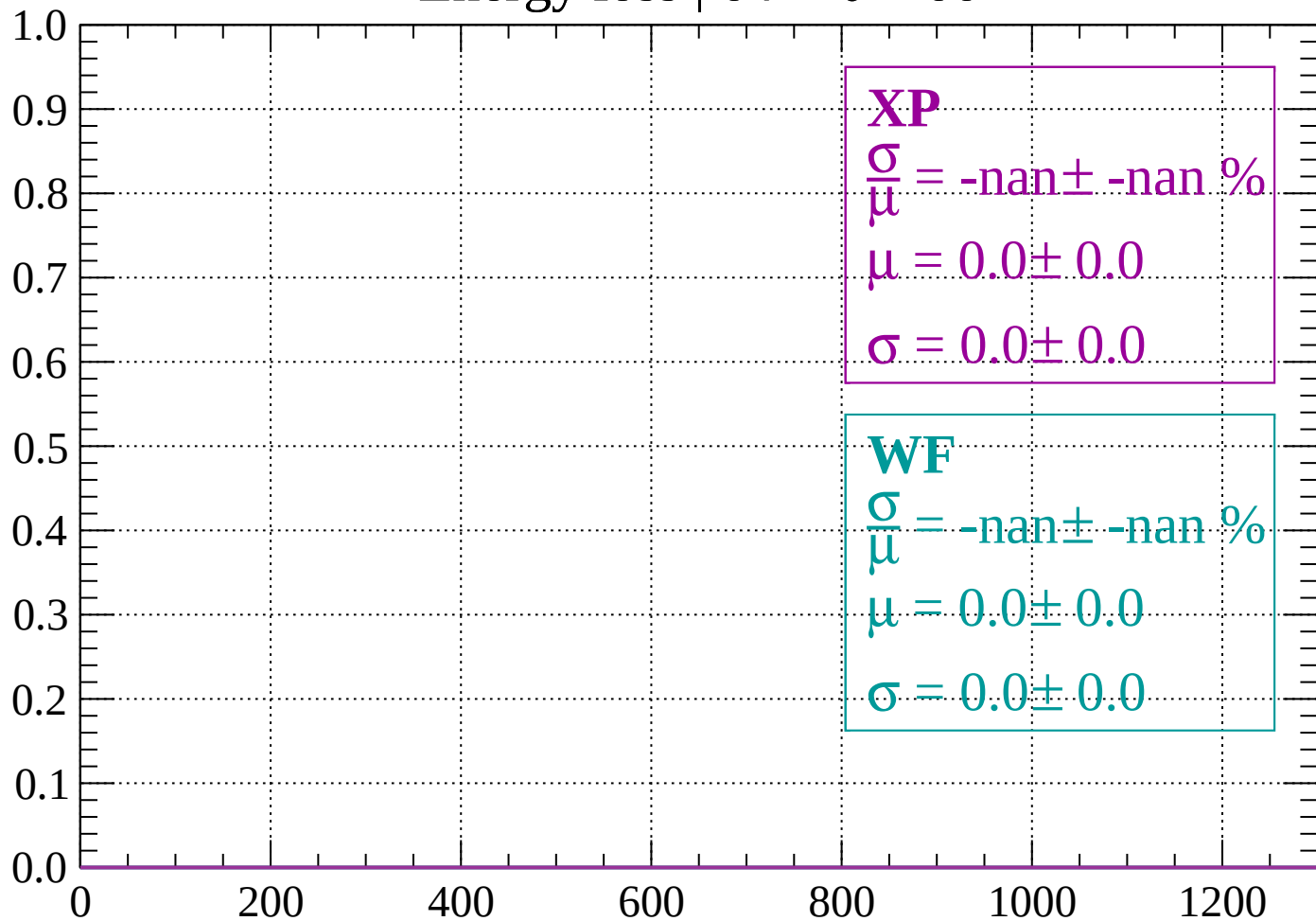
Count



dE/dx (ADC counts/cm)

Energy loss | $64 < \theta < 66$

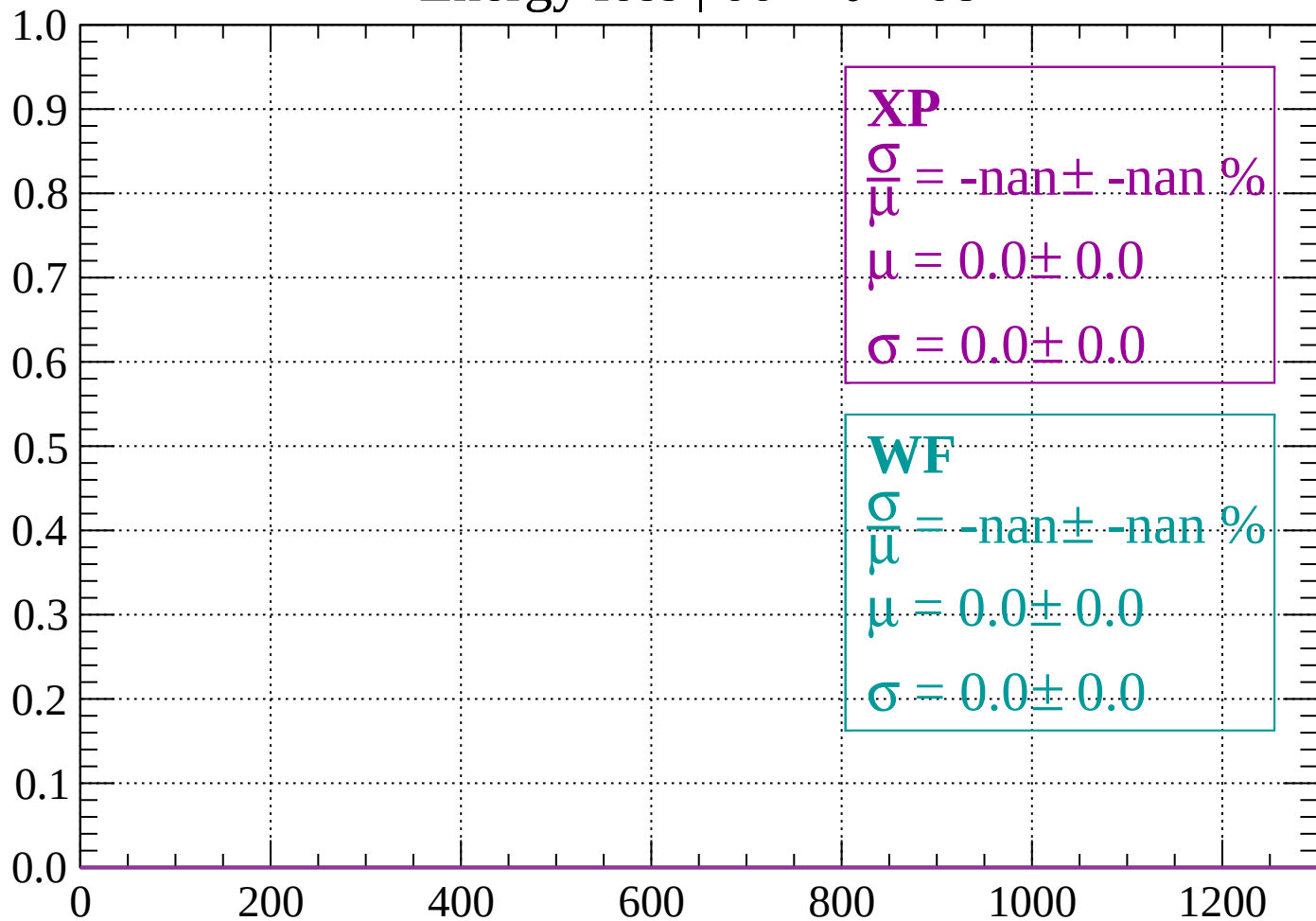
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 68$

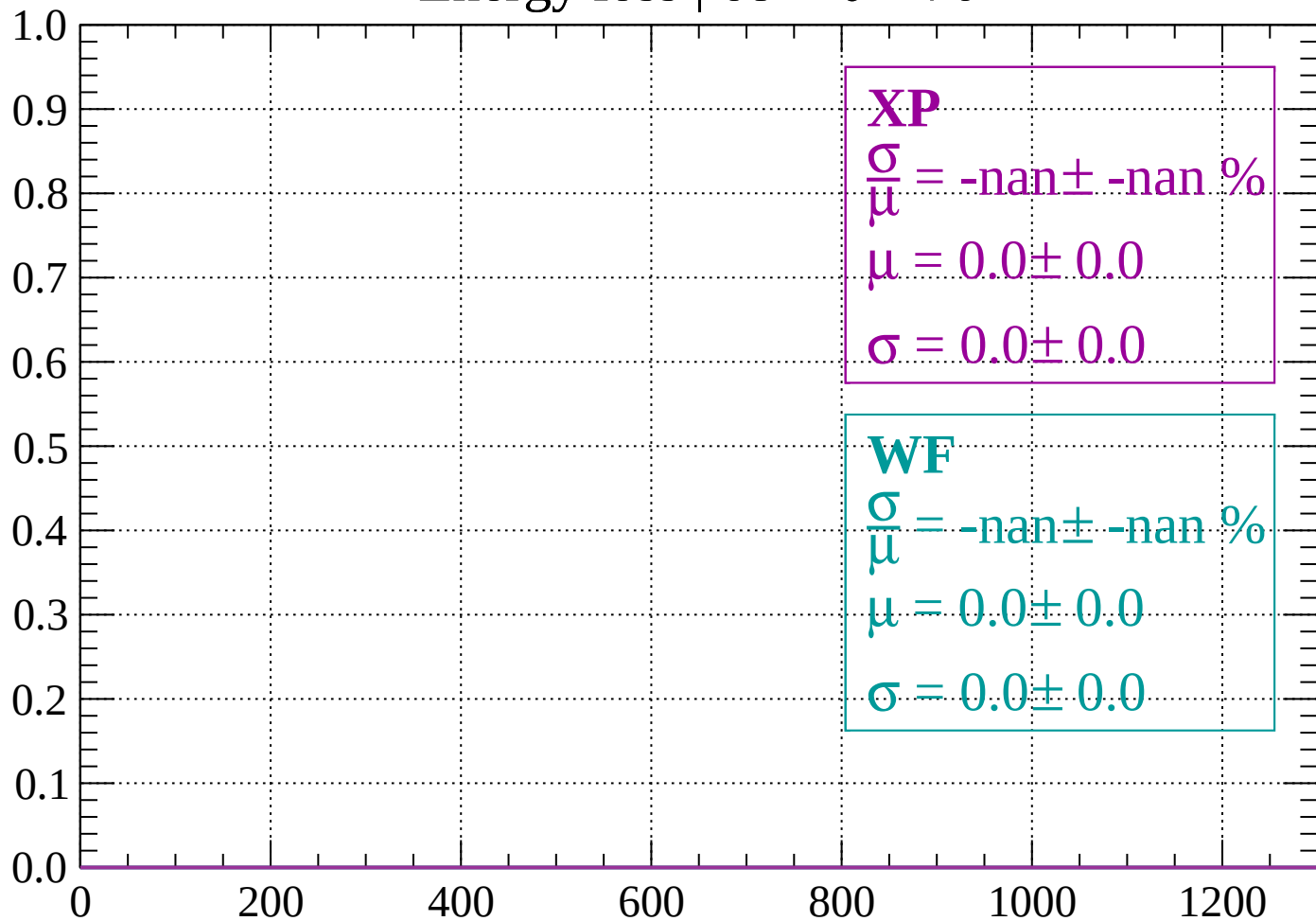
Count



dE/dx (ADC counts/cm)

Energy loss | $68 < \theta < 70$

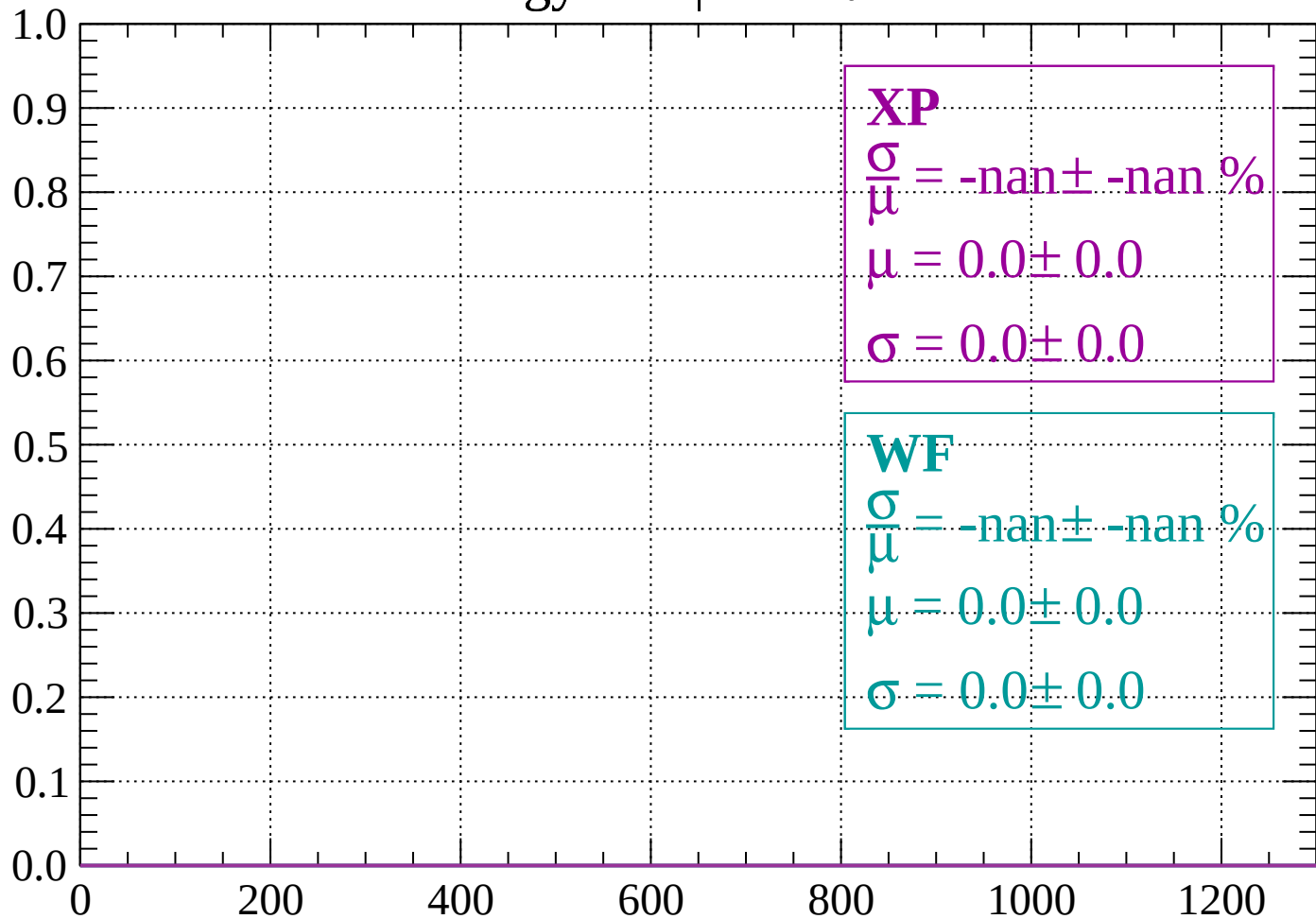
Count



dE/dx (ADC counts/cm)

Energy loss | $70 < \theta < 72$

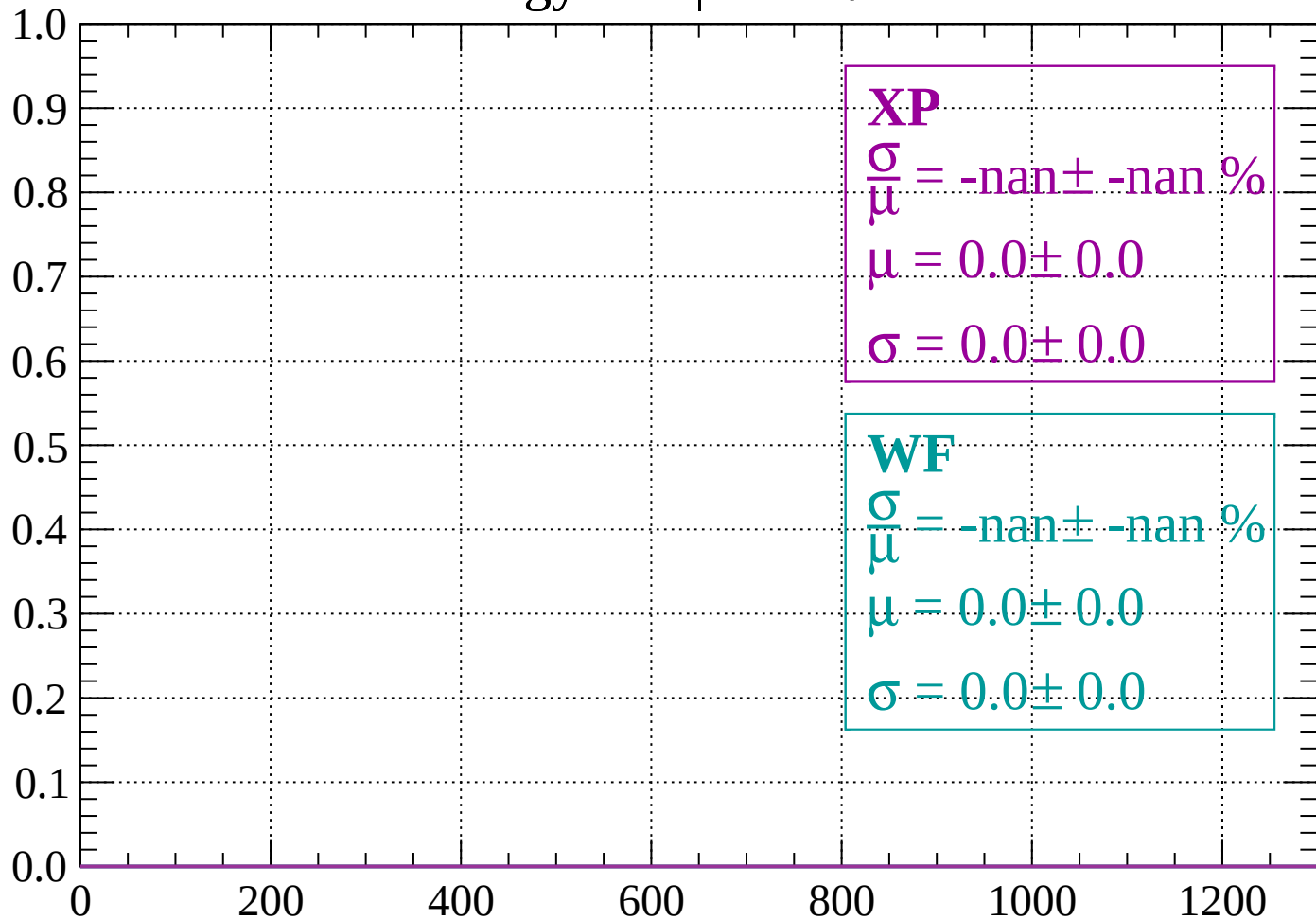
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 74$

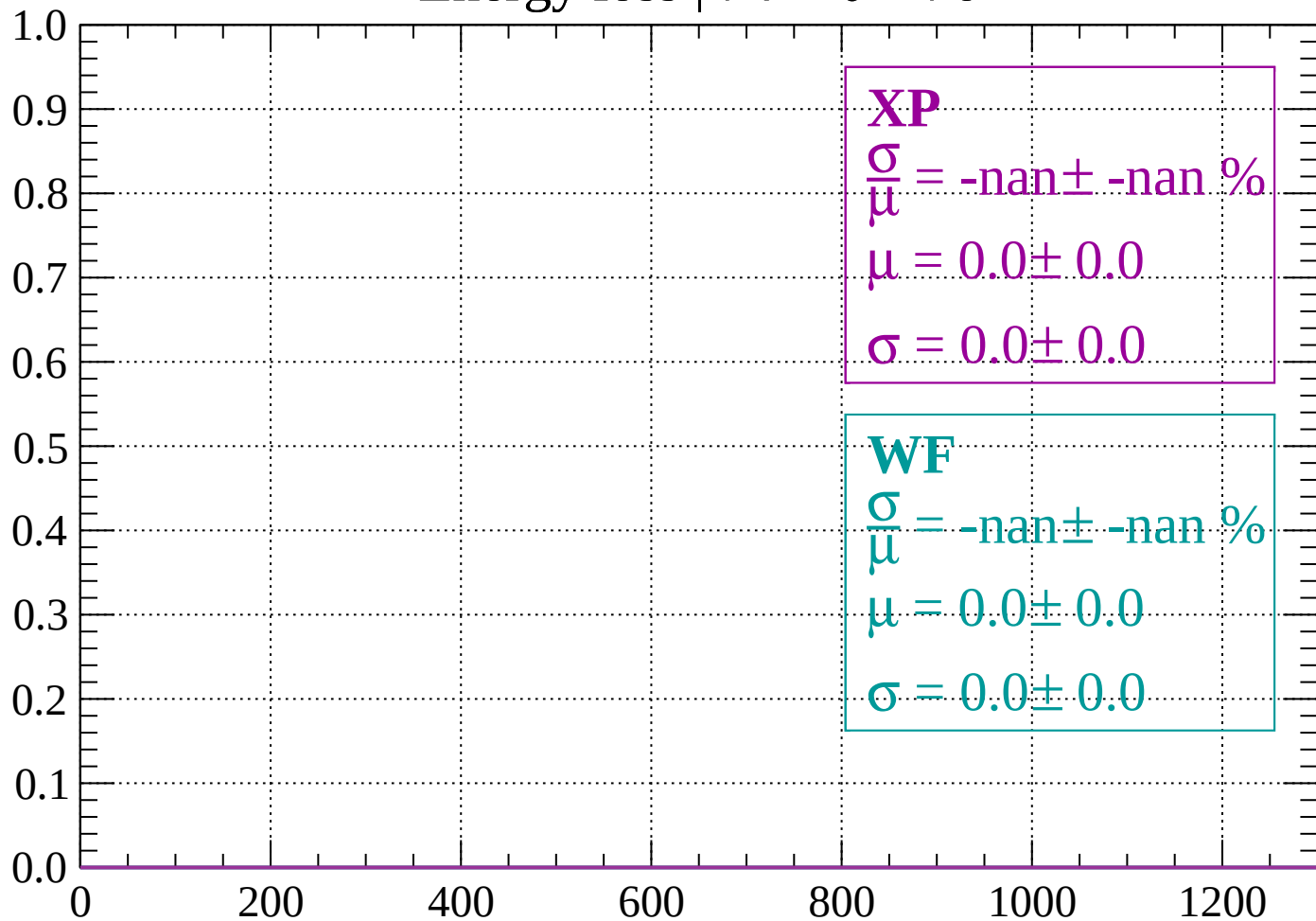
Count



dE/dx (ADC counts/cm)

Energy loss | $74 < \theta < 76$

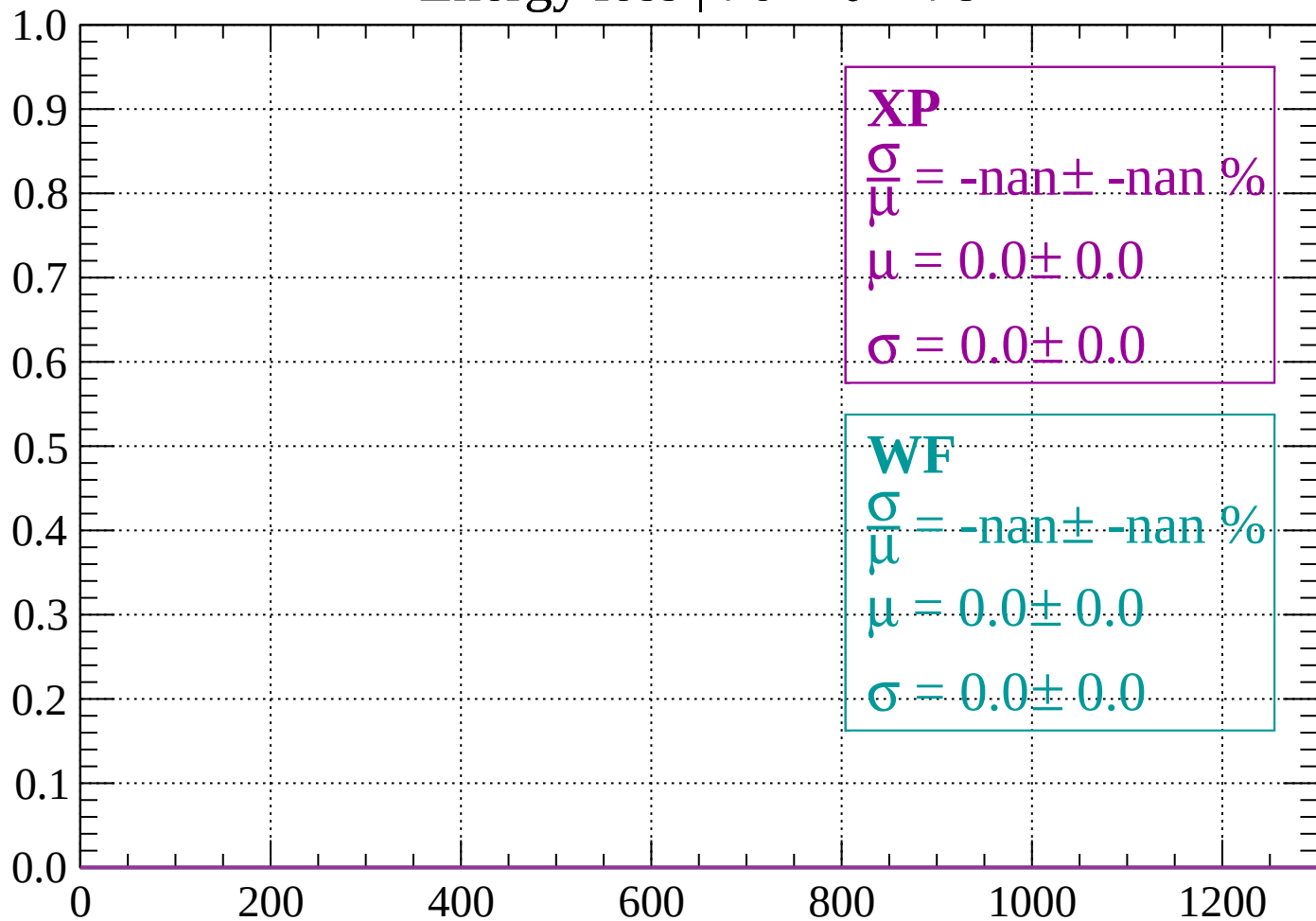
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

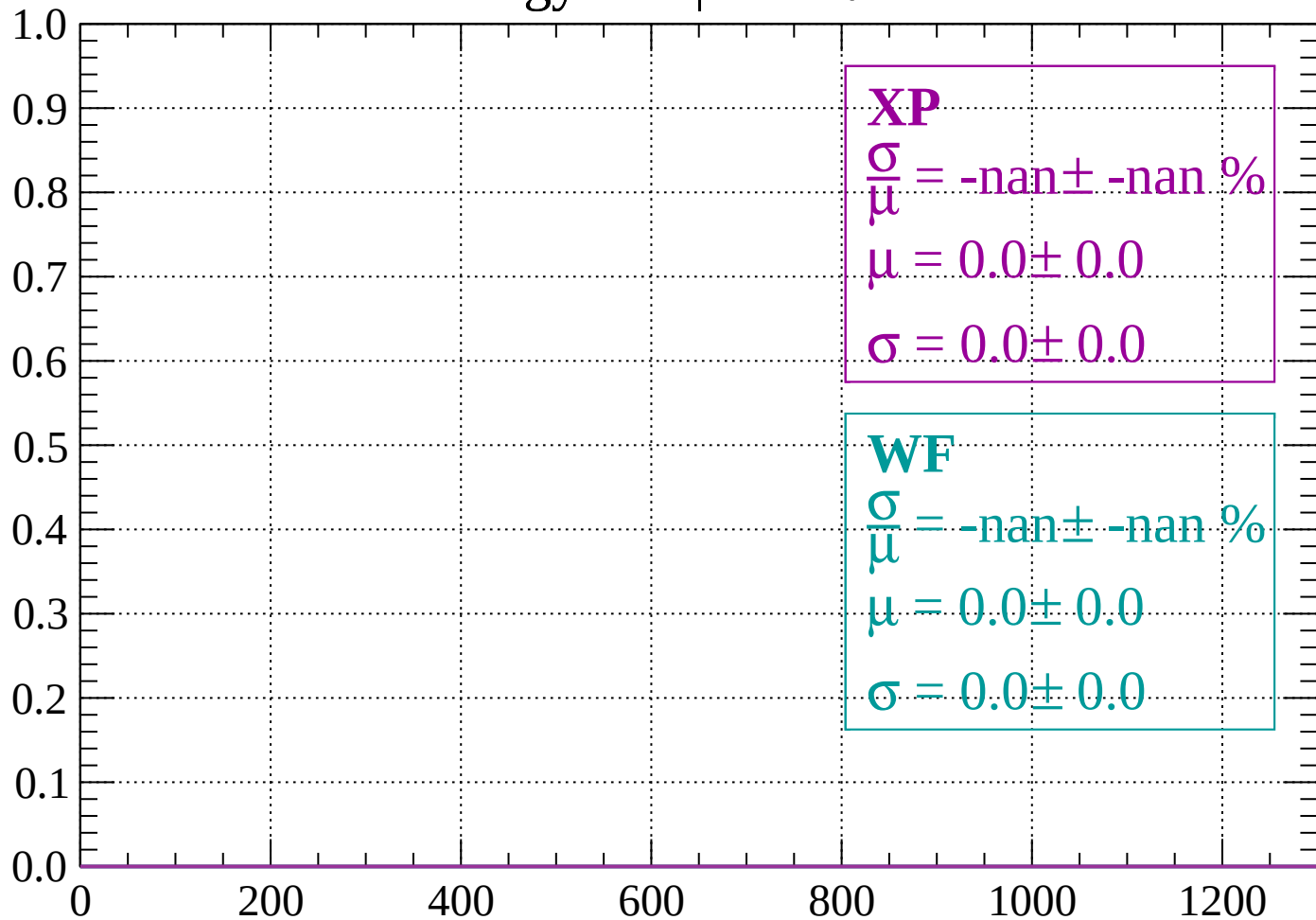
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

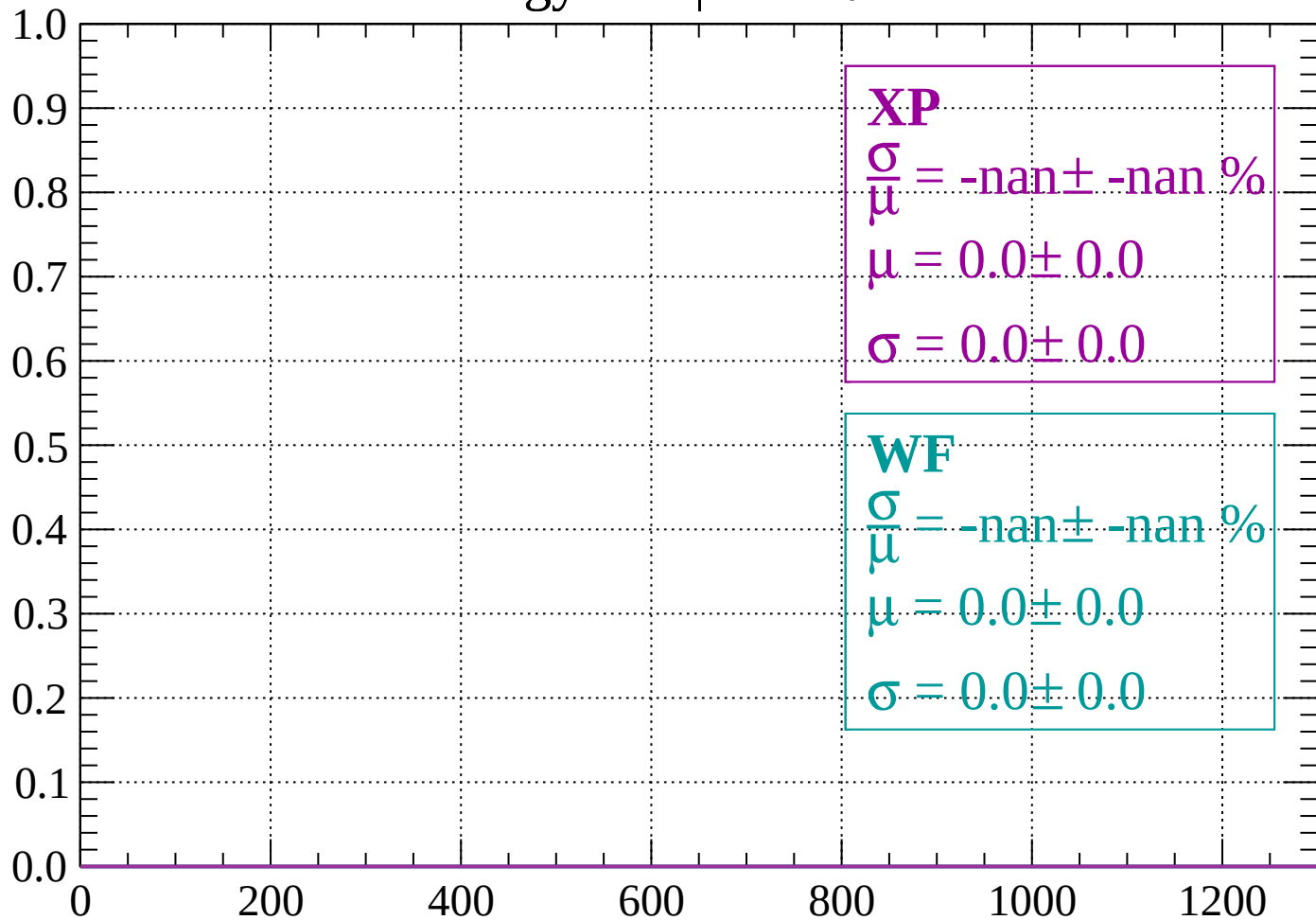
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

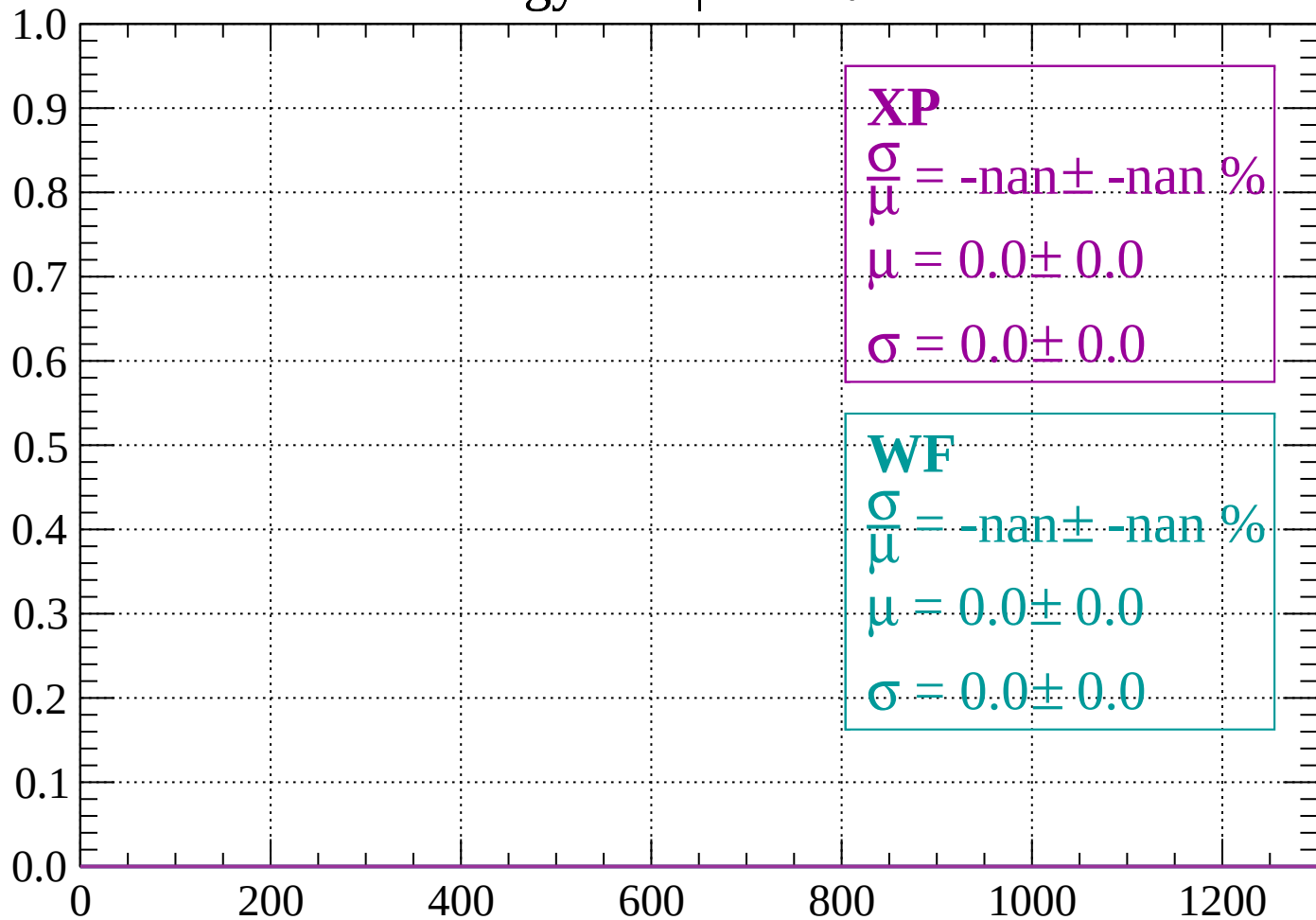
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

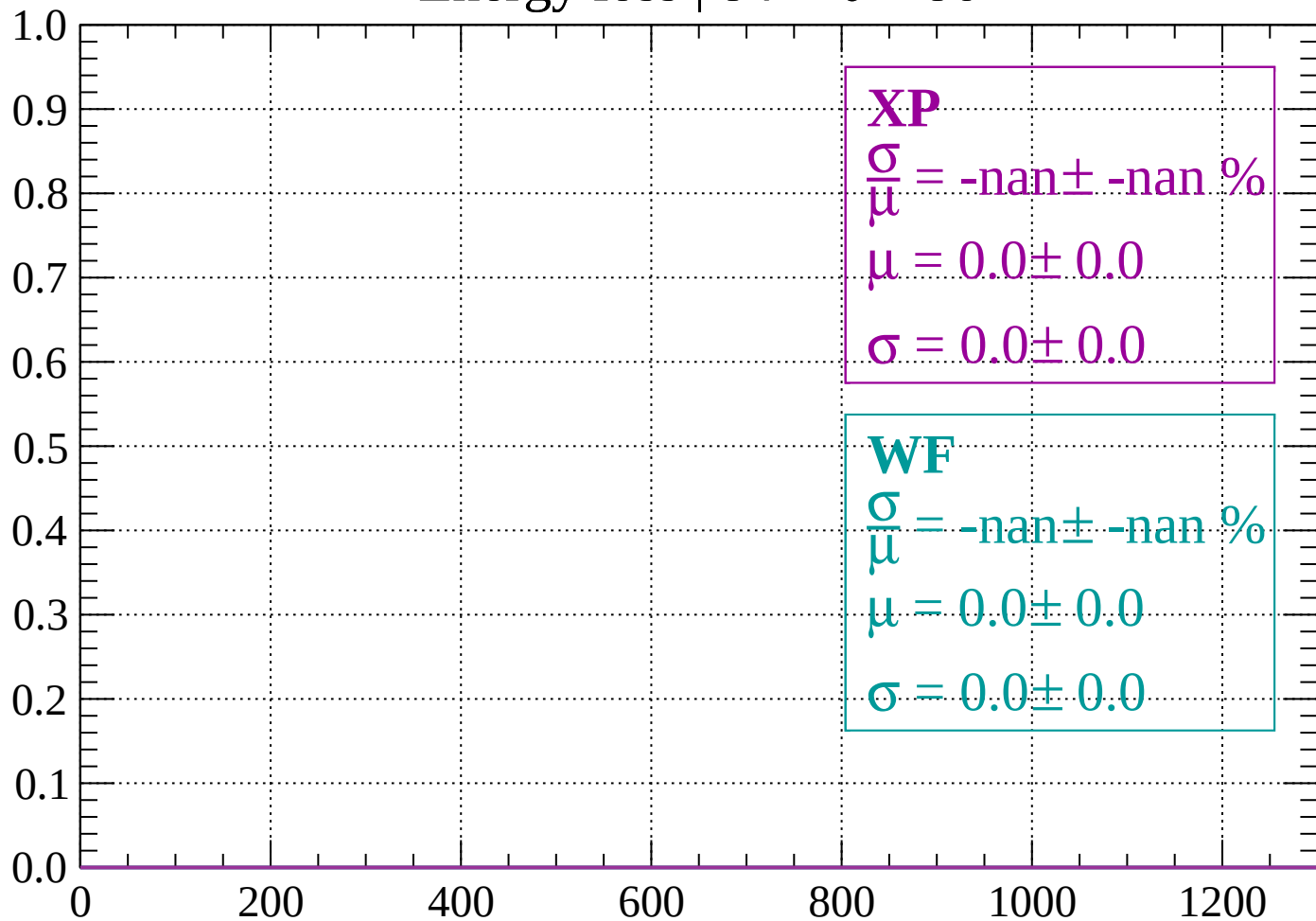
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

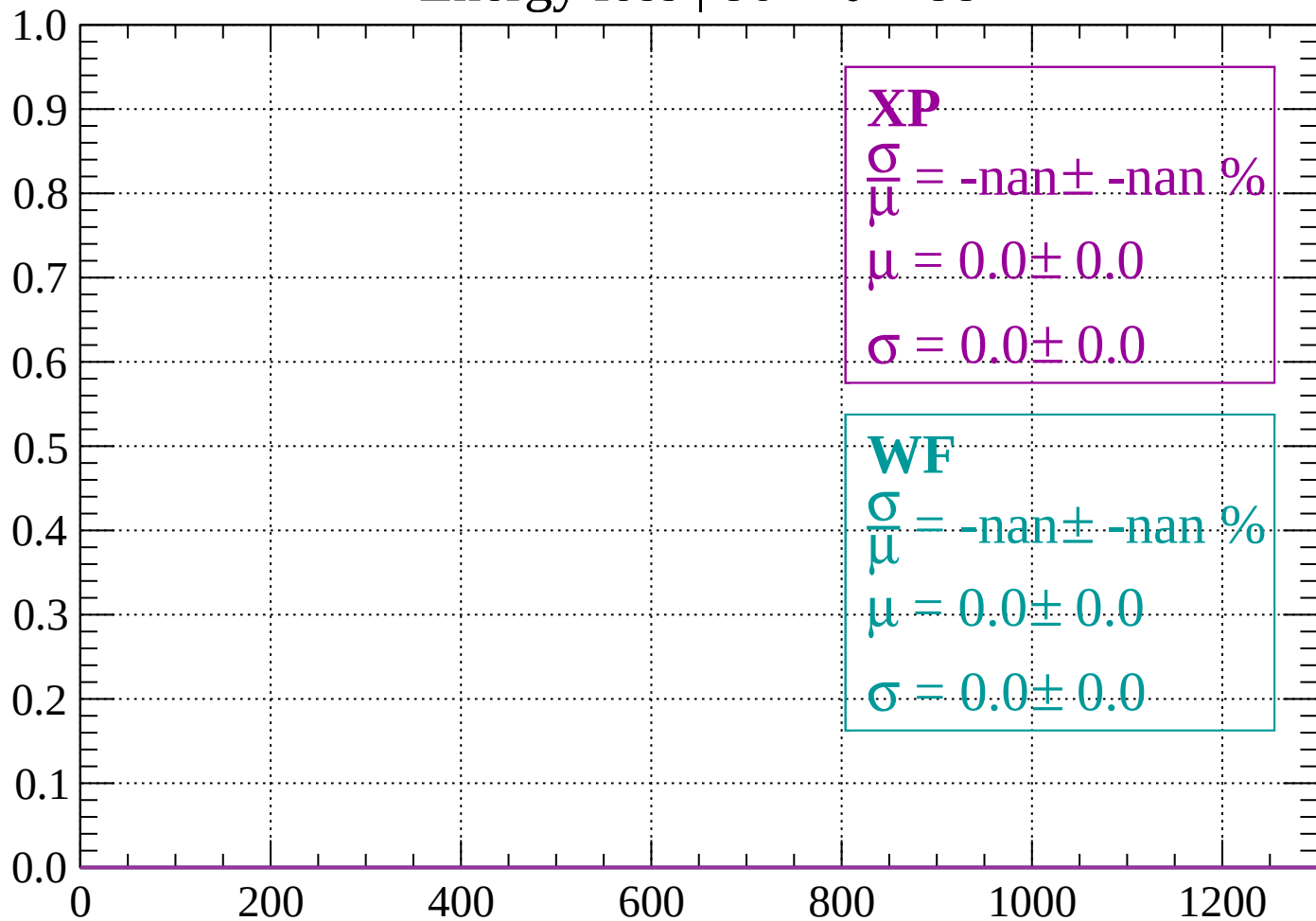
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

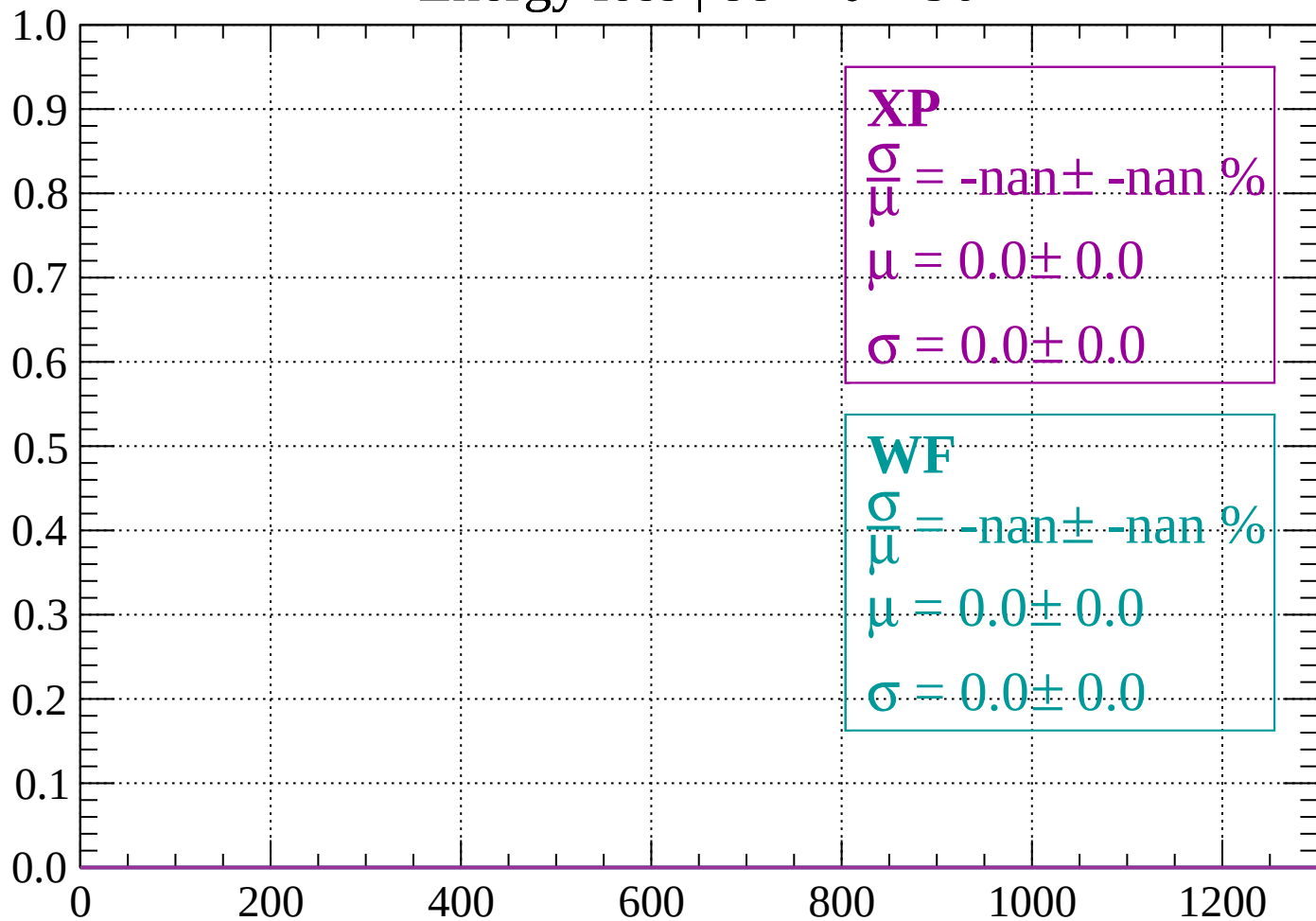
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

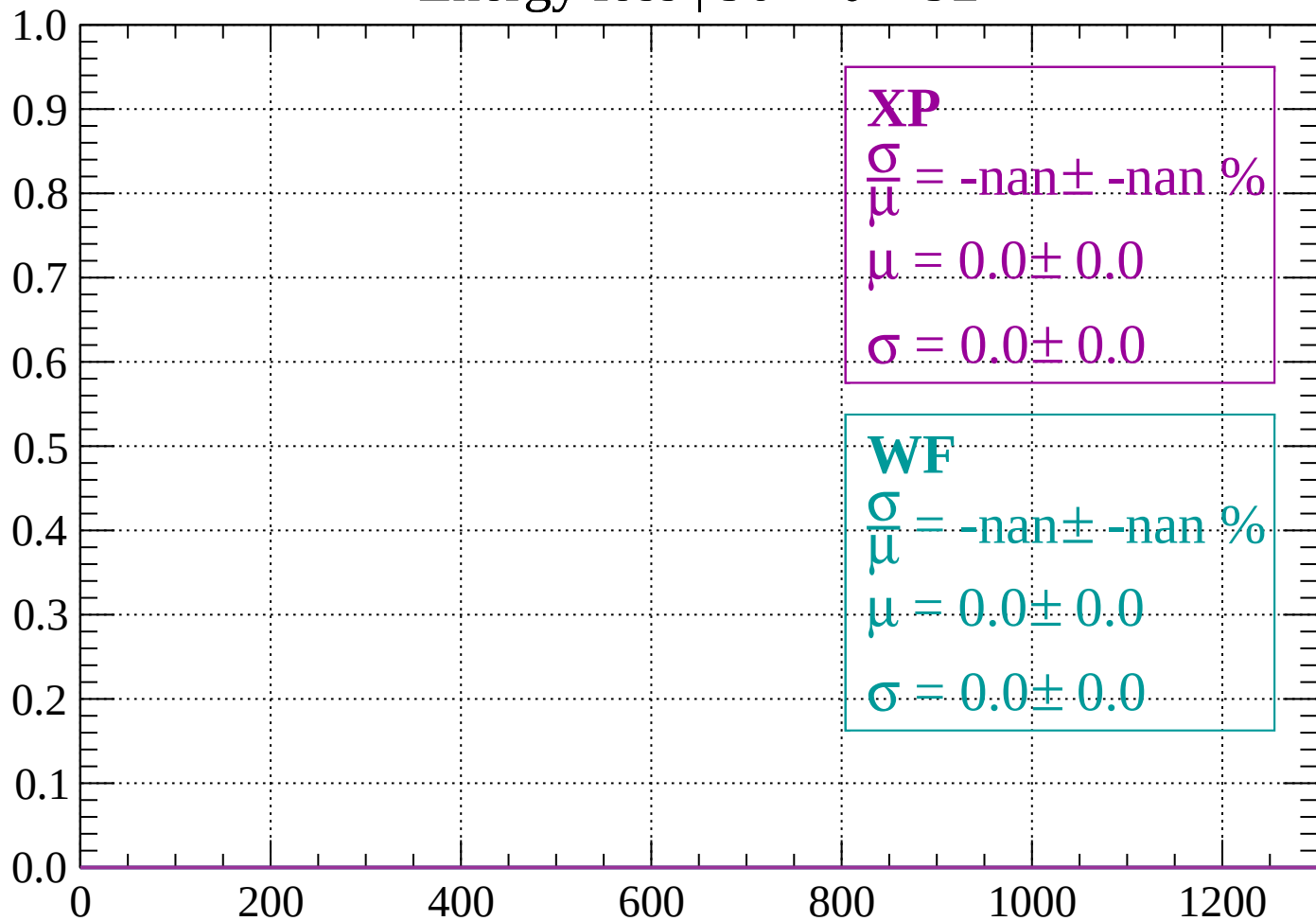
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

